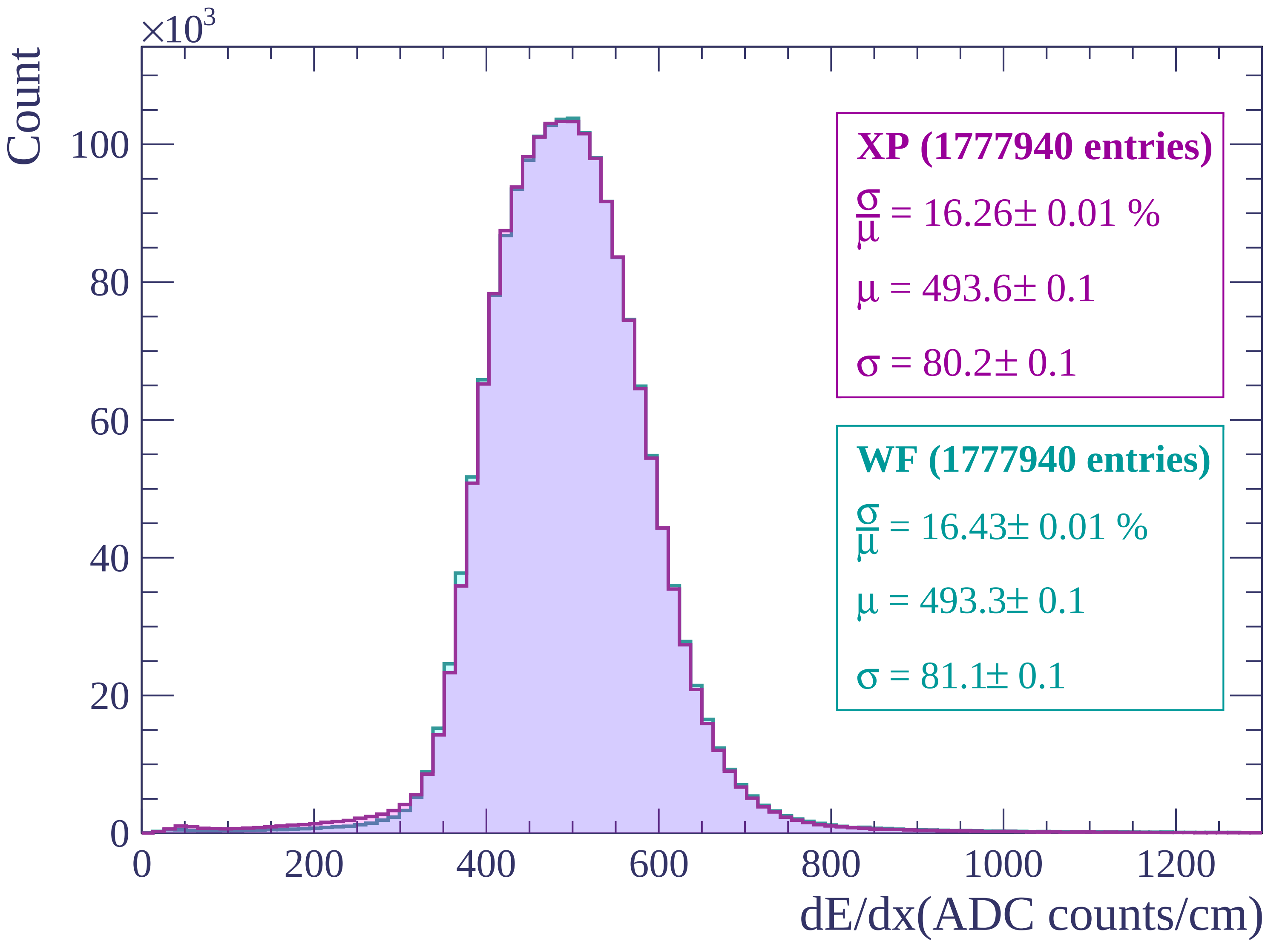
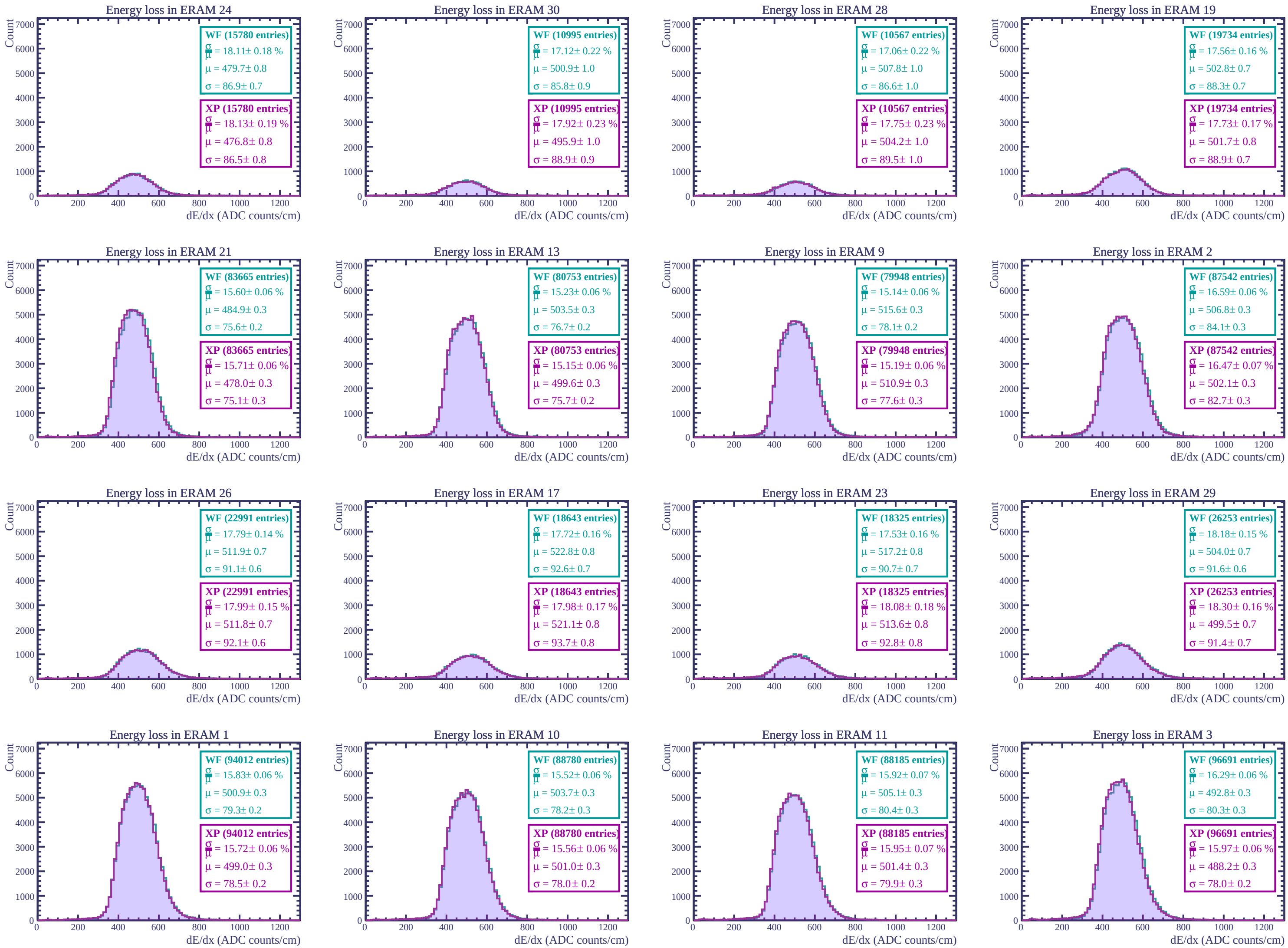
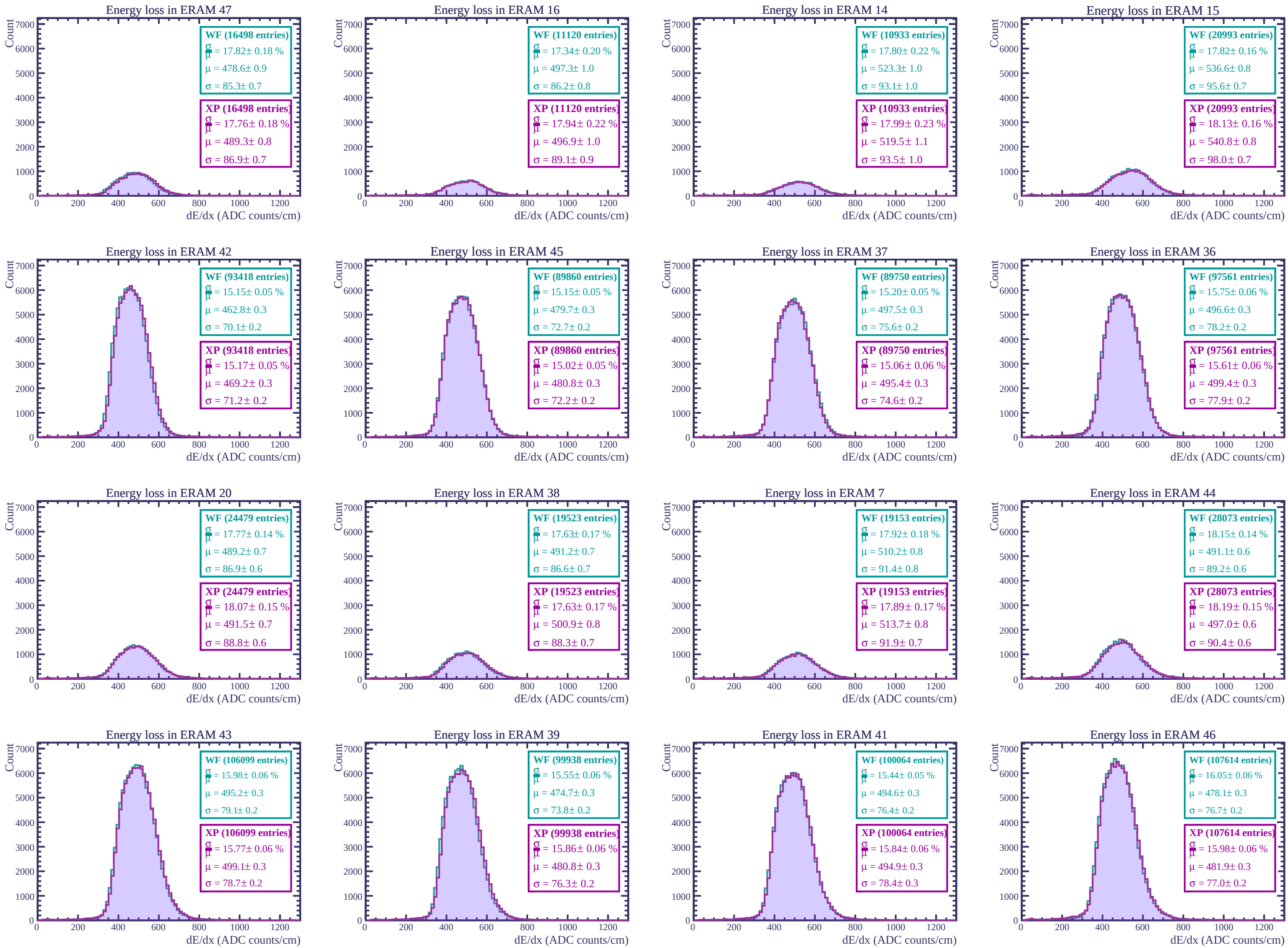
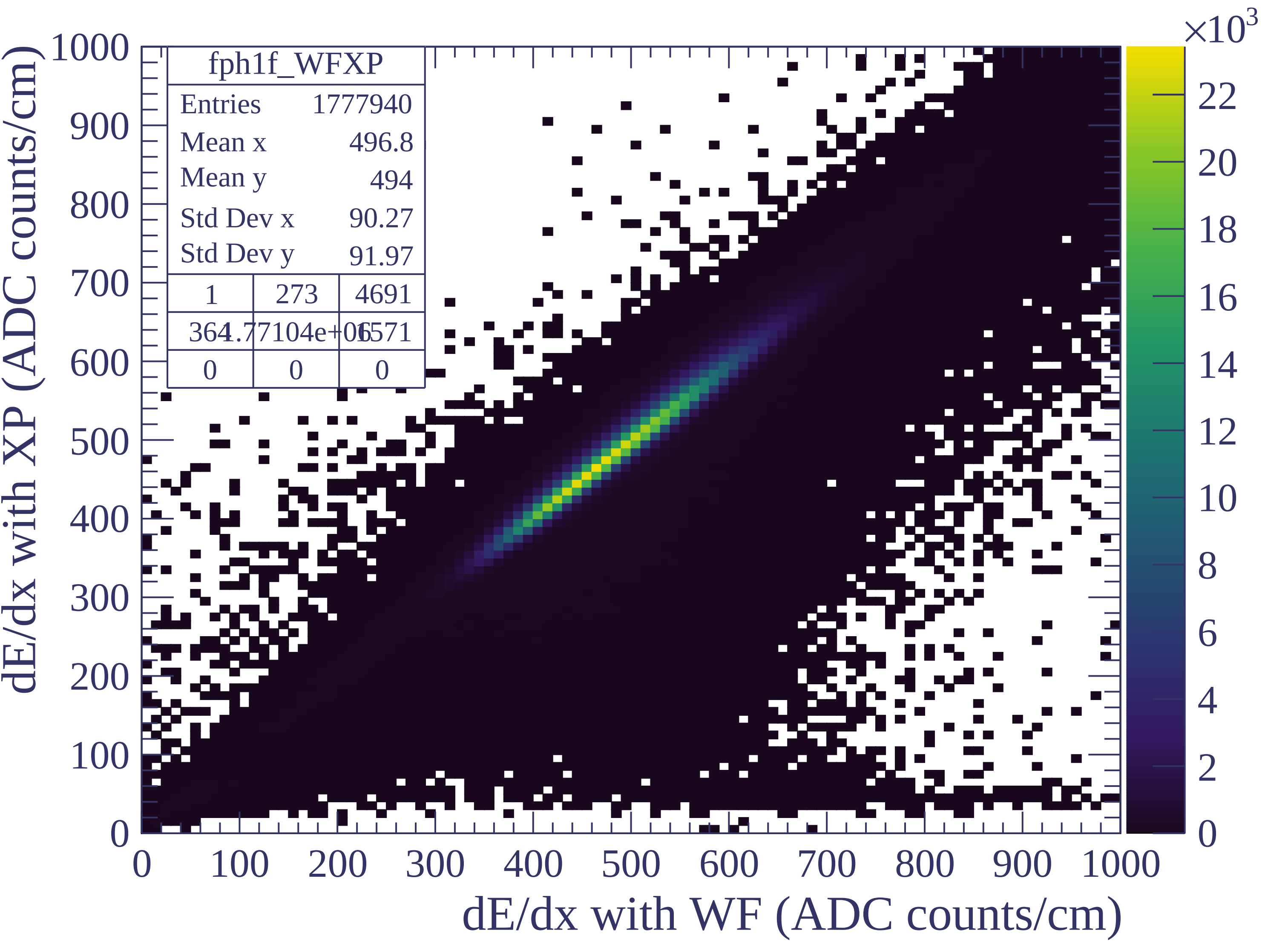
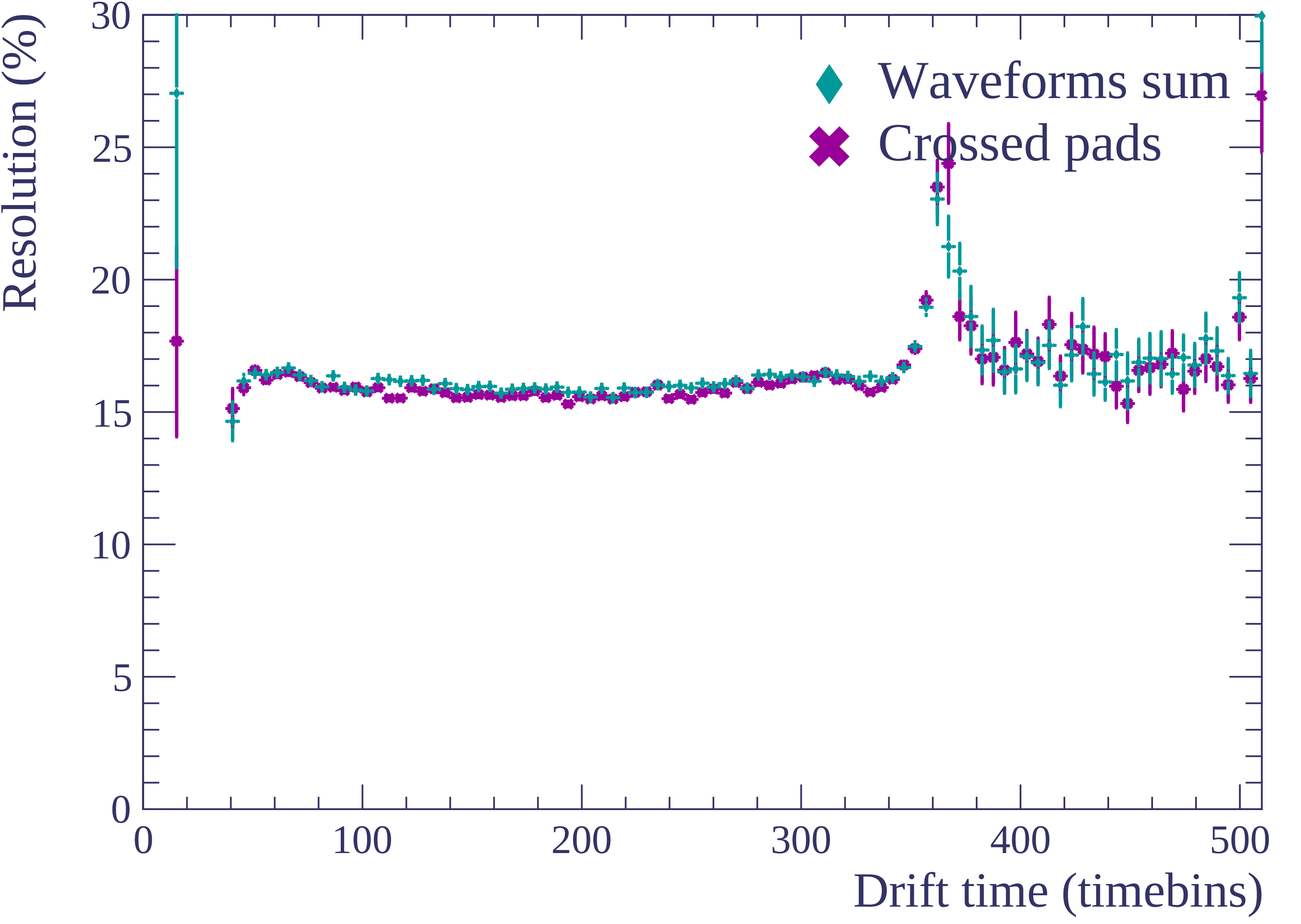


Count
$\times10^3$
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx(ADC counts/cm)
XP (1777940 entries)
$\frac{\sigma}{\mu}$ = 16.26± 0.01 %
μ = 493.6± 0.1
σ = 80.2± 0.1
WF (1777940 entries)
$\frac{\sigma}{\mu}$ = 16.43± 0.01 %
μ = 493.3± 0.1
σ = 81.1± 0.1

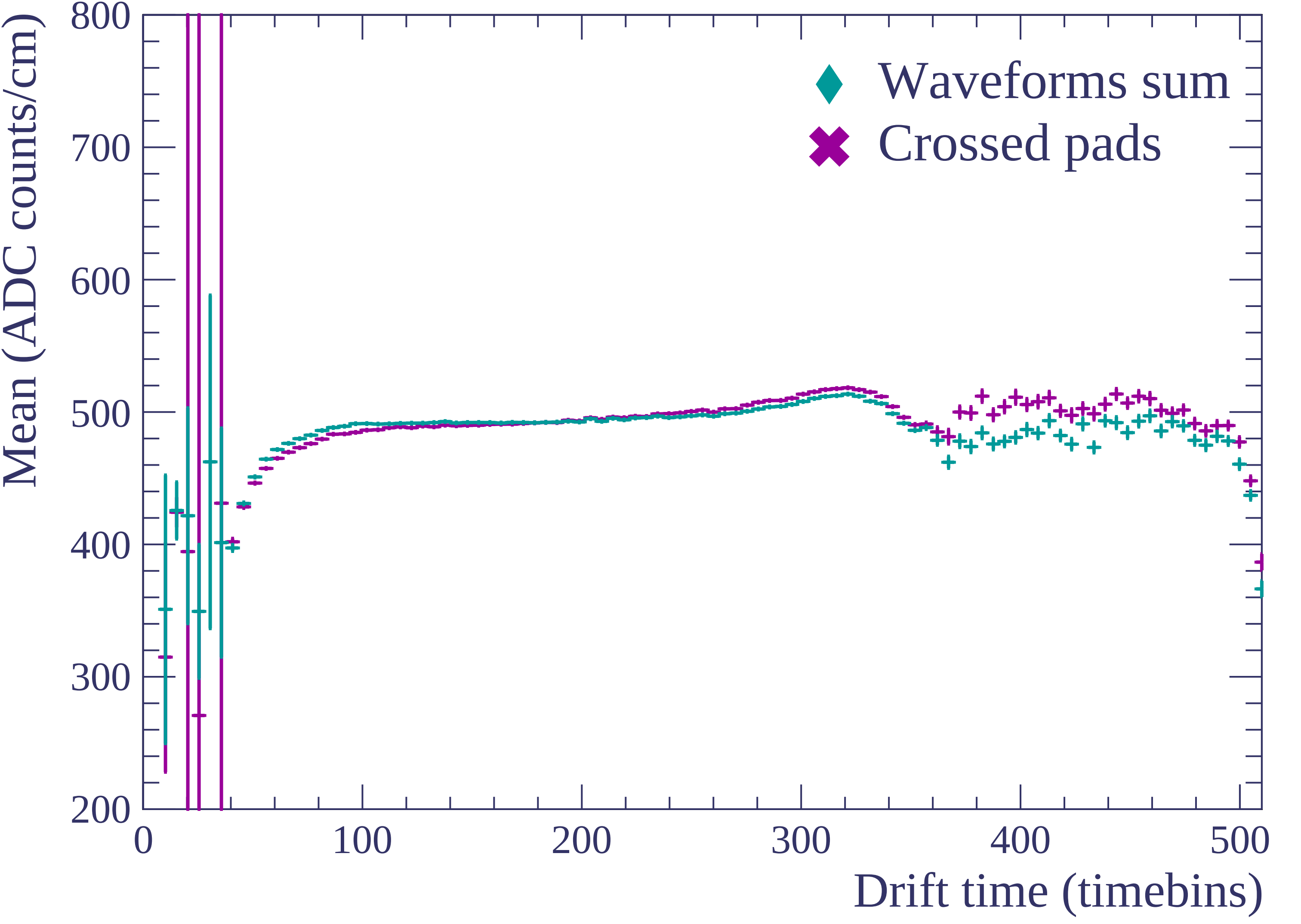

Energy loss in ERAM 24
WF (15780 entries)
σ/μ = 18.11± 0.18 %
μ = 479.7± 0.8
σ = 86.9± 0.7
XP (15780 entries)
σ/μ = 18.13± 0.19 %
μ = 476.8± 0.8
σ = 86.5± 0.8
Energy loss in ERAM 30
WF (10995 entries)
σ/μ = 17.12± 0.22 %
μ = 500.9± 1.0
σ = 85.8± 0.9
XP (10995 entries)
σ/μ = 17.92± 0.23 %
μ = 495.9± 1.0
σ = 88.9± 0.9
Energy loss in ERAM 28
WF (10567 entries)
σ/μ = 17.06± 0.22 %
μ = 507.8± 1.0
σ = 86.6± 1.0
XP (10567 entries)
σ/μ = 17.75± 0.23 %
μ = 504.2± 1.0
σ = 89.5± 1.0
Energy loss in ERAM 19
WF (19734 entries)
σ/μ = 17.56± 0.16 %
μ = 502.8± 0.7
σ = 88.3± 0.7
XP (19734 entries)
σ/μ = 17.73± 0.17 %
μ = 501.7± 0.8
σ = 88.9± 0.7
Energy loss in ERAM 21
WF (83665 entries)
σ/μ = 15.60± 0.06 %
μ = 484.9± 0.3
σ = 75.6± 0.2
XP (83665 entries)
σ/μ = 15.71± 0.06 %
μ = 478.0± 0.3
σ = 75.1± 0.3
Energy loss in ERAM 13
WF (80753 entries)
σ/μ = 15.23± 0.06 %
μ = 503.5± 0.3
σ = 76.7± 0.2
XP (80753 entries)
σ/μ = 15.15± 0.06 %
μ = 499.6± 0.3
σ = 75.7± 0.2
Energy loss in ERAM 9
WF (79948 entries)
σ/μ = 15.14± 0.06 %
μ = 515.6± 0.3
σ = 78.1± 0.2
XP (79948 entries)
σ/μ = 15.19± 0.06 %
μ = 510.9± 0.3
σ = 77.6± 0.3
Energy loss in ERAM 2
WF (87542 entries)
σ/μ = 16.59± 0.06 %
μ = 506.8± 0.3
σ = 84.1± 0.3
XP (87542 entries)
σ/μ = 16.47± 0.07 %
μ = 502.1± 0.3
σ = 82.7± 0.3
Energy loss in ERAM 26
WF (22991 entries)
σ/μ = 17.79± 0.14 %
μ = 511.9± 0.7
σ = 91.1± 0.6
XP (22991 entries)
σ/μ = 17.99± 0.15 %
μ = 511.8± 0.7
σ = 92.1± 0.6
Energy loss in ERAM 17
WF (18643 entries)
σ/μ = 17.72± 0.16 %
μ = 522.8± 0.8
σ = 92.6± 0.7
XP (18643 entries)
σ/μ = 17.98± 0.17 %
μ = 521.1± 0.8
σ = 93.7± 0.8
Energy loss in ERAM 23
WF (18325 entries)
σ/μ = 17.53± 0.16 %
μ = 517.2± 0.8
σ = 90.7± 0.7
XP (18325 entries)
σ/μ = 18.08± 0.18 %
μ = 513.6± 0.8
σ = 92.8± 0.8
Energy loss in ERAM 29
WF (26253 entries)
σ/μ = 18.18± 0.15 %
μ = 504.0± 0.7
σ = 91.6± 0.6
XP (26253 entries)
σ/μ = 18.30± 0.16 %
μ = 499.5± 0.7
σ = 91.4± 0.7
Energy loss in ERAM 1
WF (94012 entries)
σ/μ = 15.83± 0.06 %
μ = 500.9± 0.3
σ = 79.3± 0.2
XP (94012 entries)
σ/μ = 15.72± 0.06 %
μ = 499.0± 0.3
σ = 78.5± 0.2
Energy loss in ERAM 10
WF (88780 entries)
σ/μ = 15.52± 0.06 %
μ = 503.7± 0.3
σ = 78.2± 0.3
XP (88780 entries)
σ/μ = 15.56± 0.06 %
μ = 501.0± 0.3
σ = 78.0± 0.2
Energy loss in ERAM 11
WF (88185 entries)
σ/μ = 15.92± 0.07 %
μ = 505.1± 0.3
σ = 80.4± 0.3
XP (88185 entries)
σ/μ = 15.95± 0.07 %
μ = 501.4± 0.3
σ = 79.9± 0.3
Energy loss in ERAM 3
WF (96691 entries)
σ/μ = 16.29± 0.06 %
μ = 492.8± 0.3
σ = 80.3± 0.3
XP (96691 entries)
σ/μ = 15.97± 0.06 %
μ = 488.2± 0.3
σ = 78.0± 0.2
Count
dE/dx (ADC counts/cm)

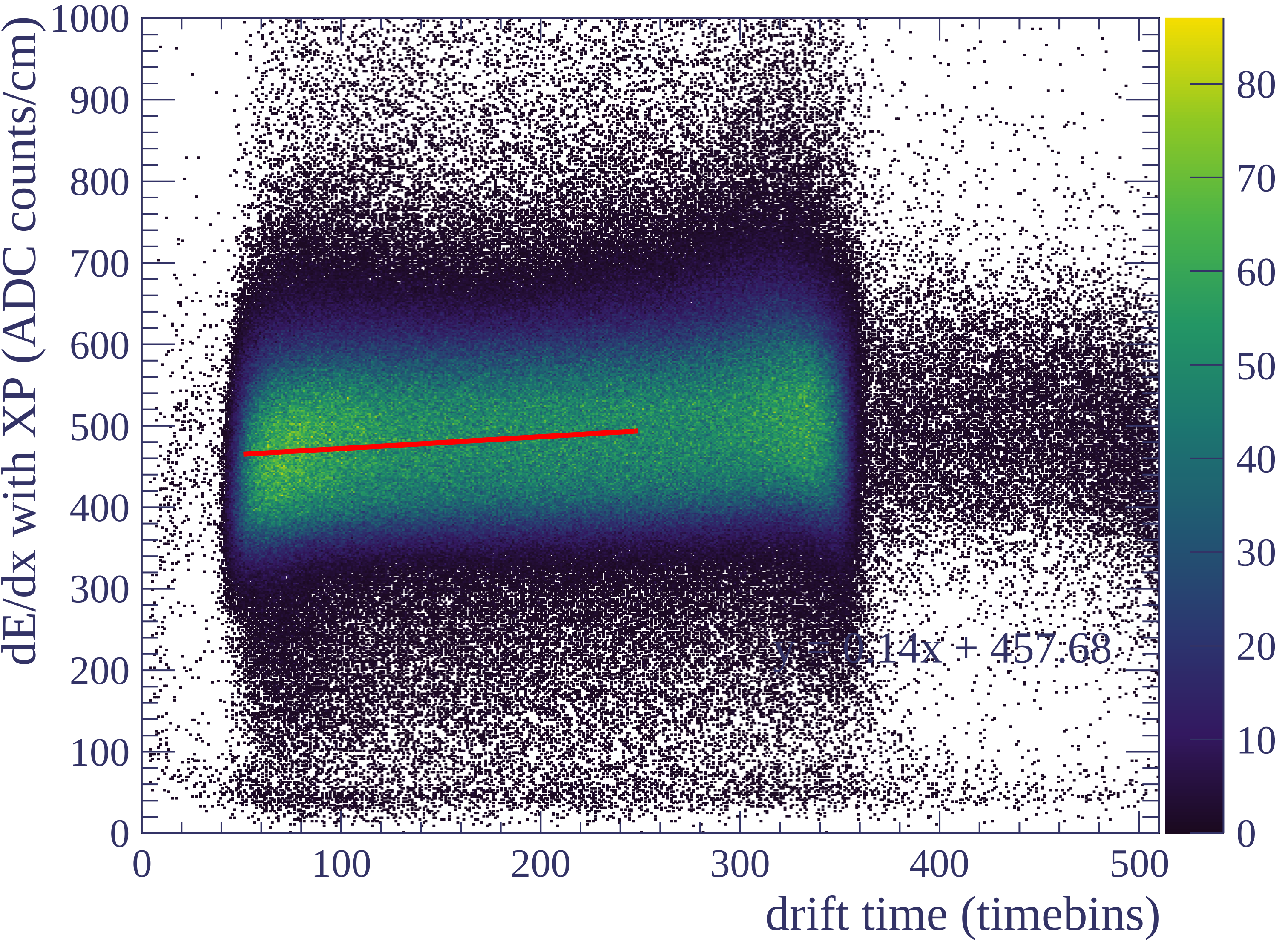

Energy loss in ERAM 47
WF (16498 entries)
σ/μ = 17.82± 0.18 %
μ = 478.6± 0.9
σ = 85.3± 0.7
XP (16498 entries)
σ/μ = 17.76± 0.18 %
μ = 489.3± 0.8
σ = 86.9± 0.7
Energy loss in ERAM 16
WF (11120 entries)
σ/μ = 17.34± 0.20 %
μ = 497.3± 1.0
σ = 86.2± 0.8
XP (11120 entries)
σ/μ = 17.94± 0.22 %
μ = 496.9± 1.0
σ = 89.1± 0.9
Energy loss in ERAM 14
WF (10933 entries)
σ/μ = 17.80± 0.22 %
μ = 523.3± 1.0
σ = 93.1± 1.0
XP (10933 entries)
σ/μ = 17.99± 0.23 %
μ = 519.5± 1.1
σ = 93.5± 1.0
Energy loss in ERAM 15
WF (20993 entries)
σ/μ = 17.82± 0.16 %
μ = 536.6± 0.8
σ = 95.6± 0.7
XP (20993 entries)
σ/μ = 18.13± 0.16 %
μ = 540.8± 0.8
σ = 98.0± 0.7
Energy loss in ERAM 42
WF (93418 entries)
σ/μ = 15.15± 0.05 %
μ = 462.8± 0.3
σ = 70.1± 0.2
XP (93418 entries)
σ/μ = 15.17± 0.05 %
μ = 469.2± 0.3
σ = 71.2± 0.2
Energy loss in ERAM 45
WF (89860 entries)
σ/μ = 15.15± 0.05 %
μ = 479.7± 0.3
σ = 72.7± 0.2
XP (89860 entries)
σ/μ = 15.02± 0.05 %
μ = 480.8± 0.3
σ = 72.2± 0.2
Energy loss in ERAM 37
WF (89750 entries)
σ/μ = 15.20± 0.05 %
μ = 497.5± 0.3
σ = 75.6± 0.2
XP (89750 entries)
σ/μ = 15.06± 0.06 %
μ = 495.4± 0.3
σ = 74.6± 0.2
Energy loss in ERAM 36
WF (97561 entries)
σ/μ = 15.75± 0.06 %
μ = 496.6± 0.3
σ = 78.2± 0.2
XP (97561 entries)
σ/μ = 15.61± 0.06 %
μ = 499.4± 0.3
σ = 77.9± 0.2
Energy loss in ERAM 20
WF (24479 entries)
σ/μ = 17.77± 0.14 %
μ = 489.2± 0.7
σ = 86.9± 0.6
XP (24479 entries)
σ/μ = 18.07± 0.15 %
μ = 491.5± 0.7
σ = 88.8± 0.6
Energy loss in ERAM 38
WF (19523 entries)
σ/μ = 17.63± 0.17 %
μ = 491.2± 0.7
σ = 86.6± 0.7
XP (19523 entries)
σ/μ = 17.63± 0.17 %
μ = 500.9± 0.8
σ = 88.3± 0.7
Energy loss in ERAM 7
WF (19153 entries)
σ/μ = 17.92± 0.18 %
μ = 510.2± 0.8
σ = 91.4± 0.8
XP (19153 entries)
σ/μ = 17.89± 0.17 %
μ = 513.7± 0.8
σ = 91.9± 0.7
Energy loss in ERAM 44
WF (28073 entries)
σ/μ = 18.15± 0.14 %
μ = 491.1± 0.6
σ = 89.2± 0.6
XP (28073 entries)
σ/μ = 18.19± 0.15 %
μ = 497.0± 0.6
σ = 90.4± 0.6
Energy loss in ERAM 43
WF (106099 entries)
σ/μ = 15.98± 0.06 %
μ = 495.2± 0.3
σ = 79.1± 0.2
XP (106099 entries)
σ/μ = 15.77± 0.06 %
μ = 499.1± 0.3
σ = 78.7± 0.2
Energy loss in ERAM 39
WF (99938 entries)
σ/μ = 15.55± 0.06 %
μ = 474.7± 0.3
σ = 73.8± 0.2
XP (99938 entries)
σ/μ = 15.86± 0.06 %
μ = 480.8± 0.3
σ = 76.3± 0.2
Energy loss in ERAM 41
WF (100064 entries)
σ/μ = 15.44± 0.05 %
μ = 494.6± 0.3
σ = 76.4± 0.2
XP (100064 entries)
σ/μ = 15.84± 0.06 %
μ = 494.9± 0.3
σ = 78.4± 0.3
Energy loss in ERAM 46
WF (107614 entries)
σ/μ = 16.05± 0.06 %
μ = 478.1± 0.3
σ = 76.7± 0.2
XP (107614 entries)
σ/μ = 15.98± 0.06 %
μ = 481.9± 0.3
σ = 77.0± 0.2
Count
dE/dx (ADC counts/cm)

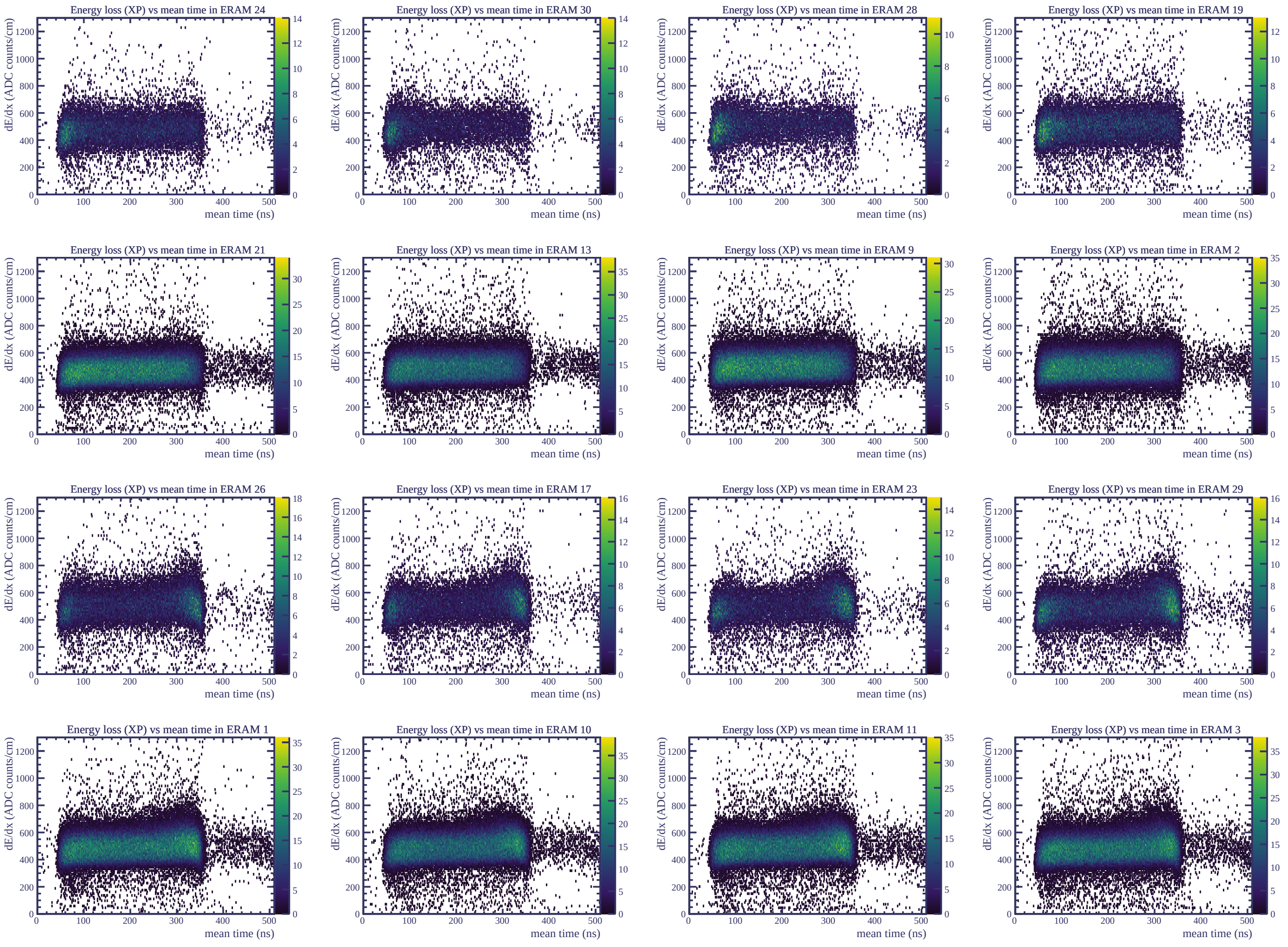

fph1f_WFXP
Entries 1777940
Mean x 496.8
Mean y 494
Std Dev x 90.27
Std Dev y 91.97
1 273 4691
0 0 0
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
×10^3
0 100 200 300 400 500 600 700 800 900 1000
0 2 4 6 8 10 12 14 16 18 20 22



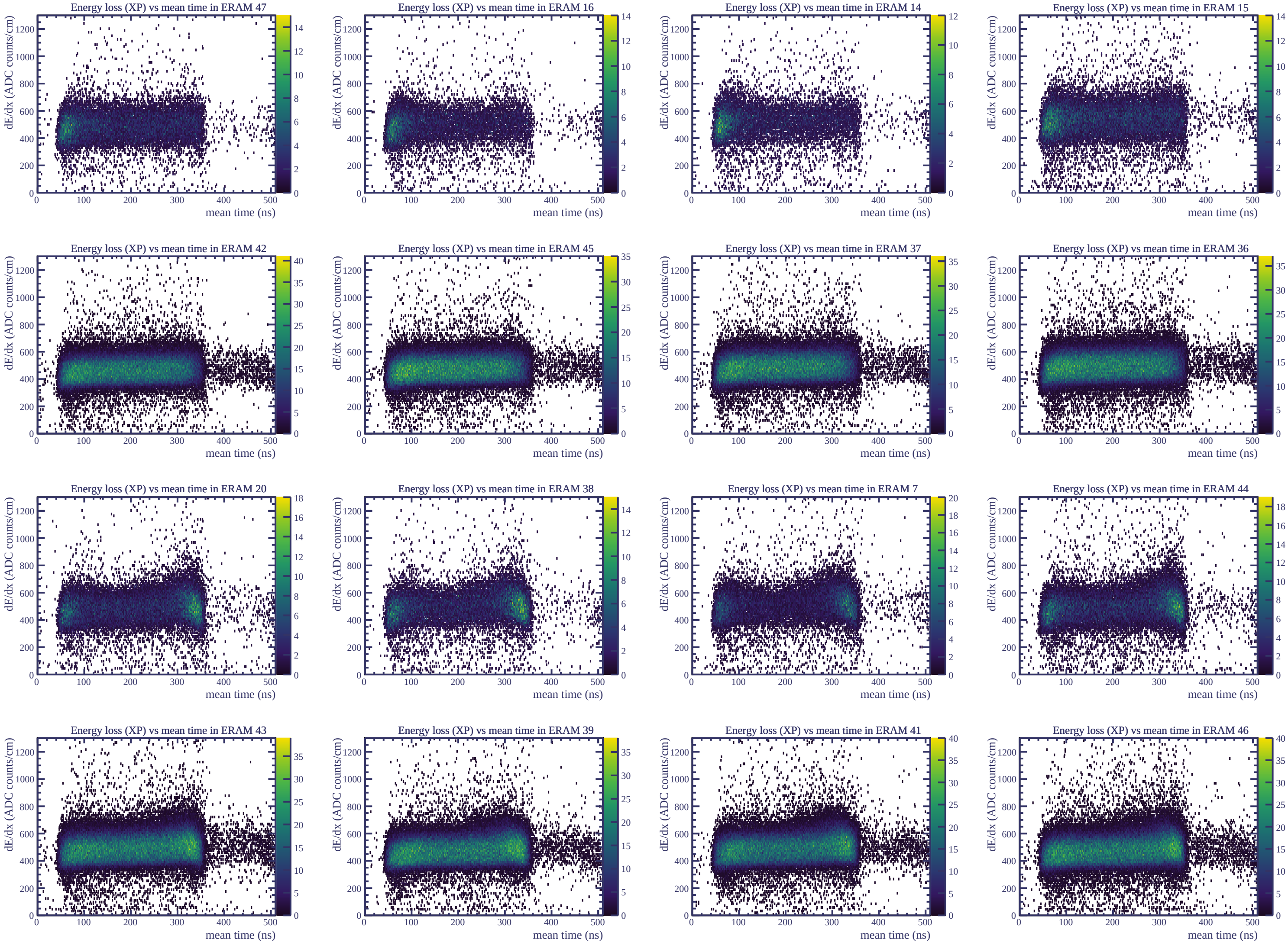
Resolution (%)
30
25
20
15
10
5
0
0
100
200
300
400
500
Drift time (timebins)
Waveforms sum
Crossed pads

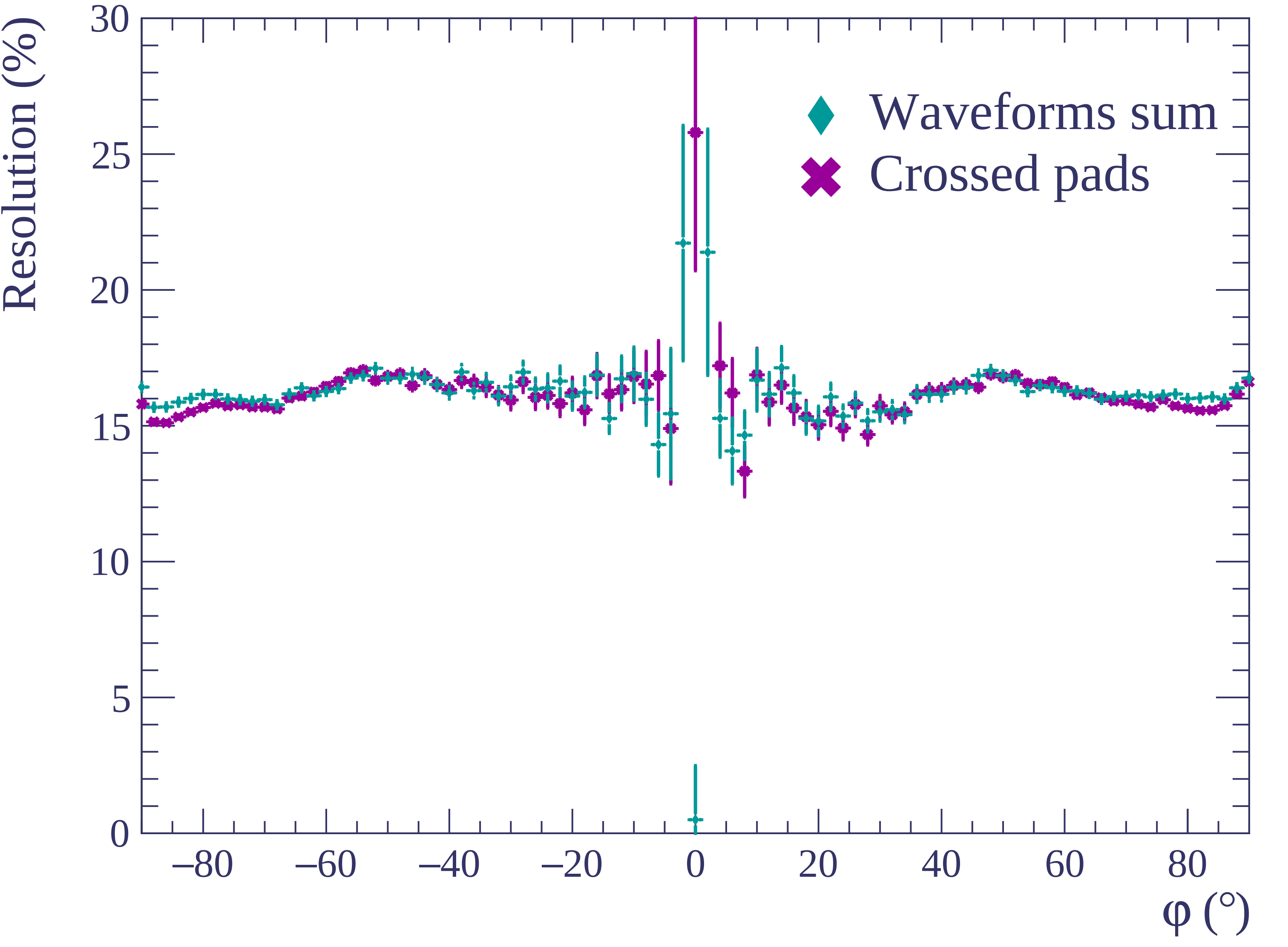

Mean (ADC counts/cm)
Drift time (timebins)
Waveforms sum
Crossed pads
800
700
600
500
400
300
200
0
100
200
300
400
500

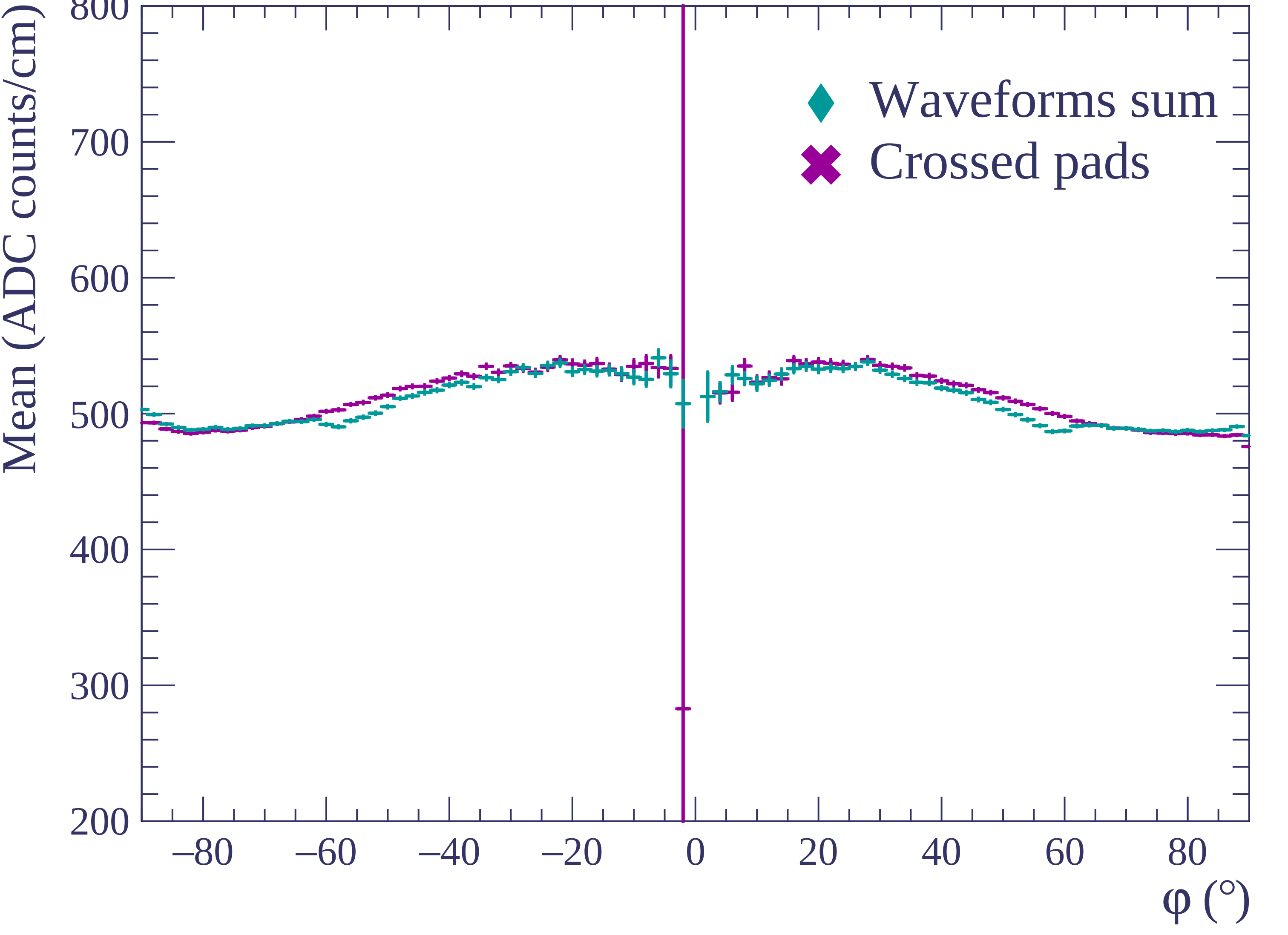

dE/dx with XP (ADC counts/cm)
drift time (timebins)
y = 0.14x + 457.68
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
10
20
30
40
50
60
70
80

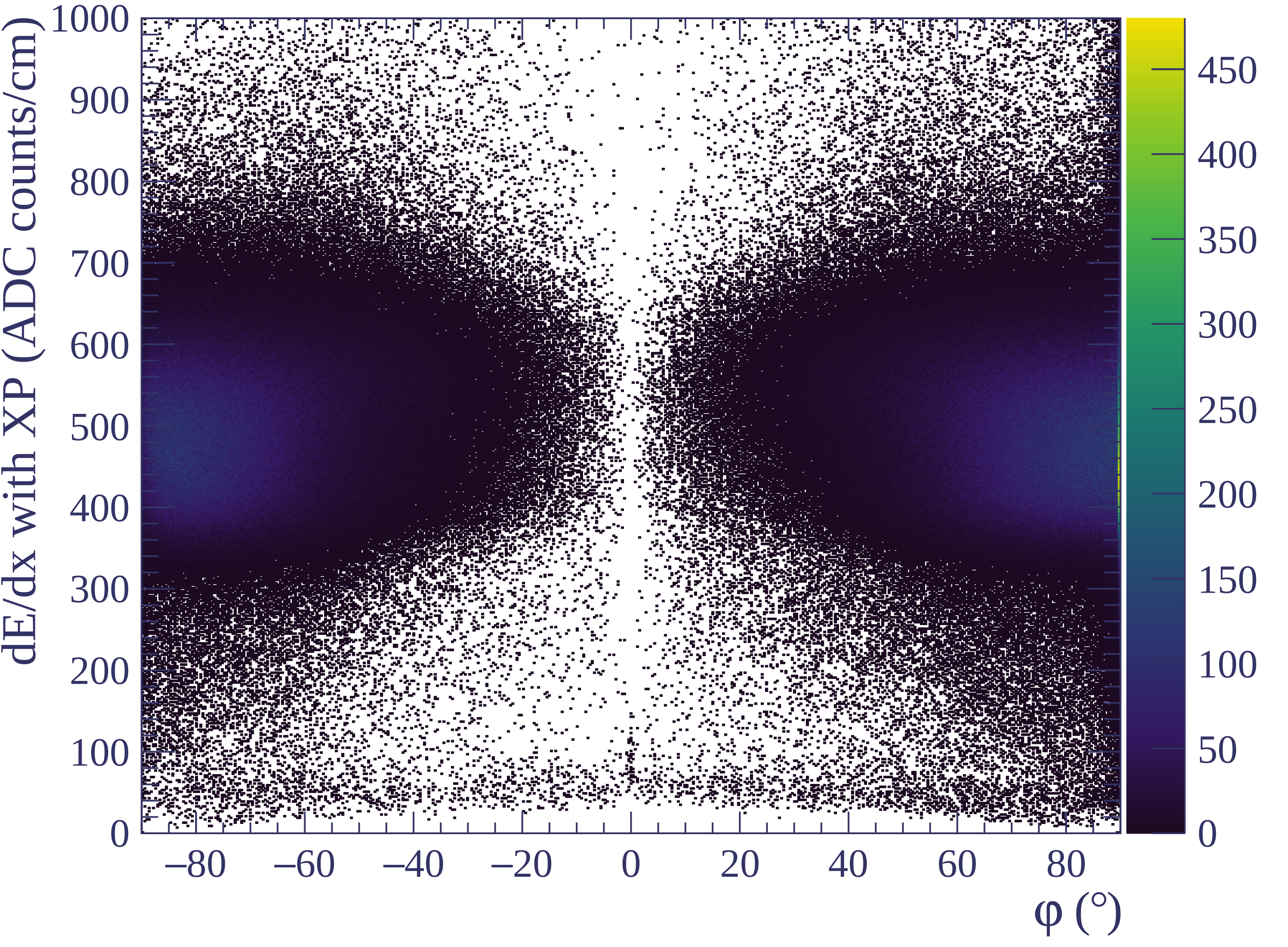


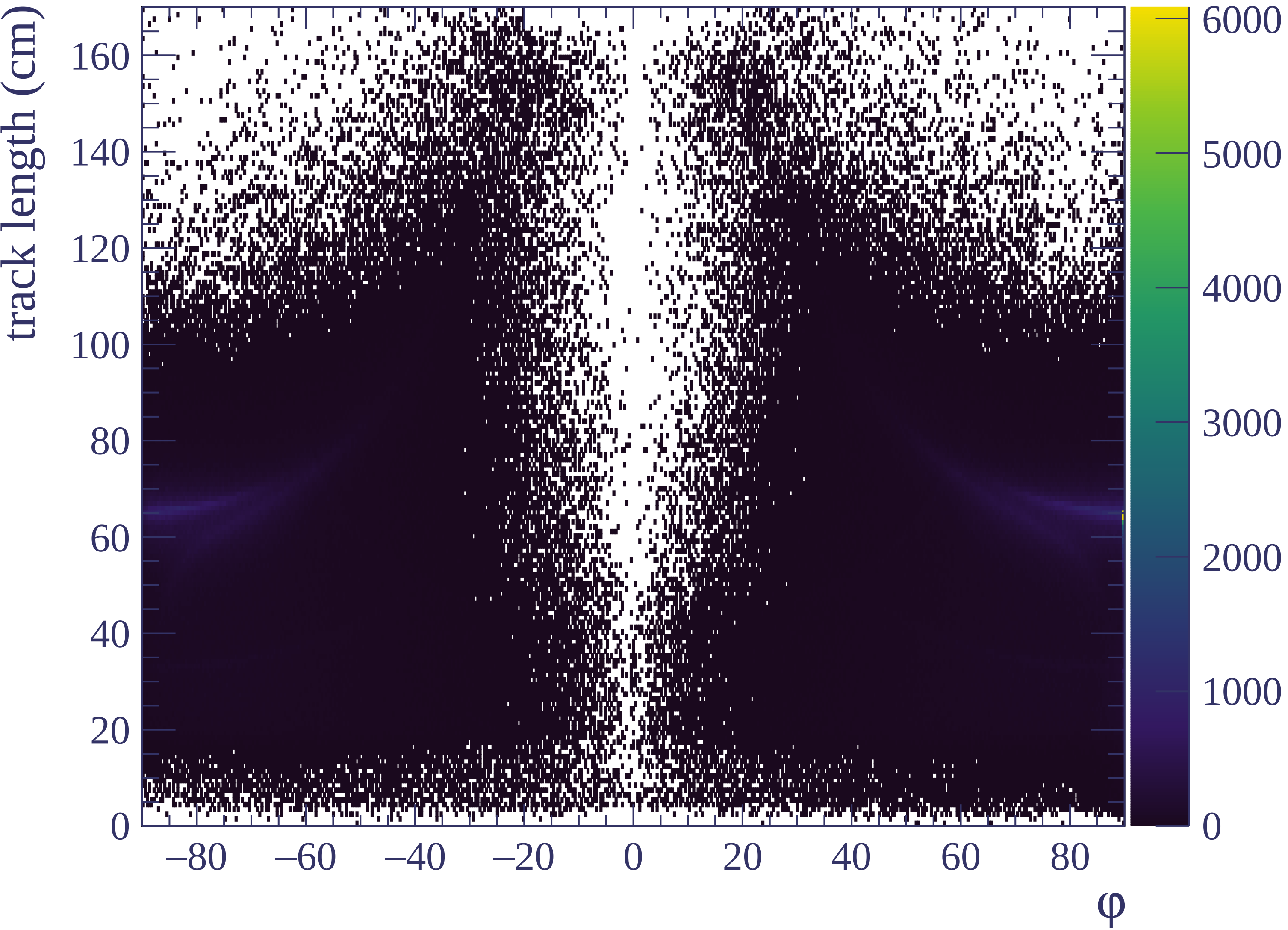
Energy loss (XP) vs mean time in ERAM 24
Energy loss (XP) vs mean time in ERAM 30
Energy loss (XP) vs mean time in ERAM 28
Energy loss (XP) vs mean time in ERAM 19
Energy loss (XP) vs mean time in ERAM 21
Energy loss (XP) vs mean time in ERAM 13
Energy loss (XP) vs mean time in ERAM 9
Energy loss (XP) vs mean time in ERAM 2
Energy loss (XP) vs mean time in ERAM 26
Energy loss (XP) vs mean time in ERAM 17
Energy loss (XP) vs mean time in ERAM 23
Energy loss (XP) vs mean time in ERAM 29
Energy loss (XP) vs mean time in ERAM 1
Energy loss (XP) vs mean time in ERAM 10
Energy loss (XP) vs mean time in ERAM 11
Energy loss (XP) vs mean time in ERAM 3
dE/dx (ADC counts/cm)
mean time (ns)


Energy loss (XP) vs mean time in ERAM 47
Energy loss (XP) vs mean time in ERAM 16
Energy loss (XP) vs mean time in ERAM 14
Energy loss (XP) vs mean time in ERAM 15
Energy loss (XP) vs mean time in ERAM 42
Energy loss (XP) vs mean time in ERAM 45
Energy loss (XP) vs mean time in ERAM 37
Energy loss (XP) vs mean time in ERAM 36
Energy loss (XP) vs mean time in ERAM 20
Energy loss (XP) vs mean time in ERAM 38
Energy loss (XP) vs mean time in ERAM 7
Energy loss (XP) vs mean time in ERAM 44
Energy loss (XP) vs mean time in ERAM 43
Energy loss (XP) vs mean time in ERAM 39
Energy loss (XP) vs mean time in ERAM 41
Energy loss (XP) vs mean time in ERAM 46
dE/dx (ADC counts/cm)
mean time (ns)

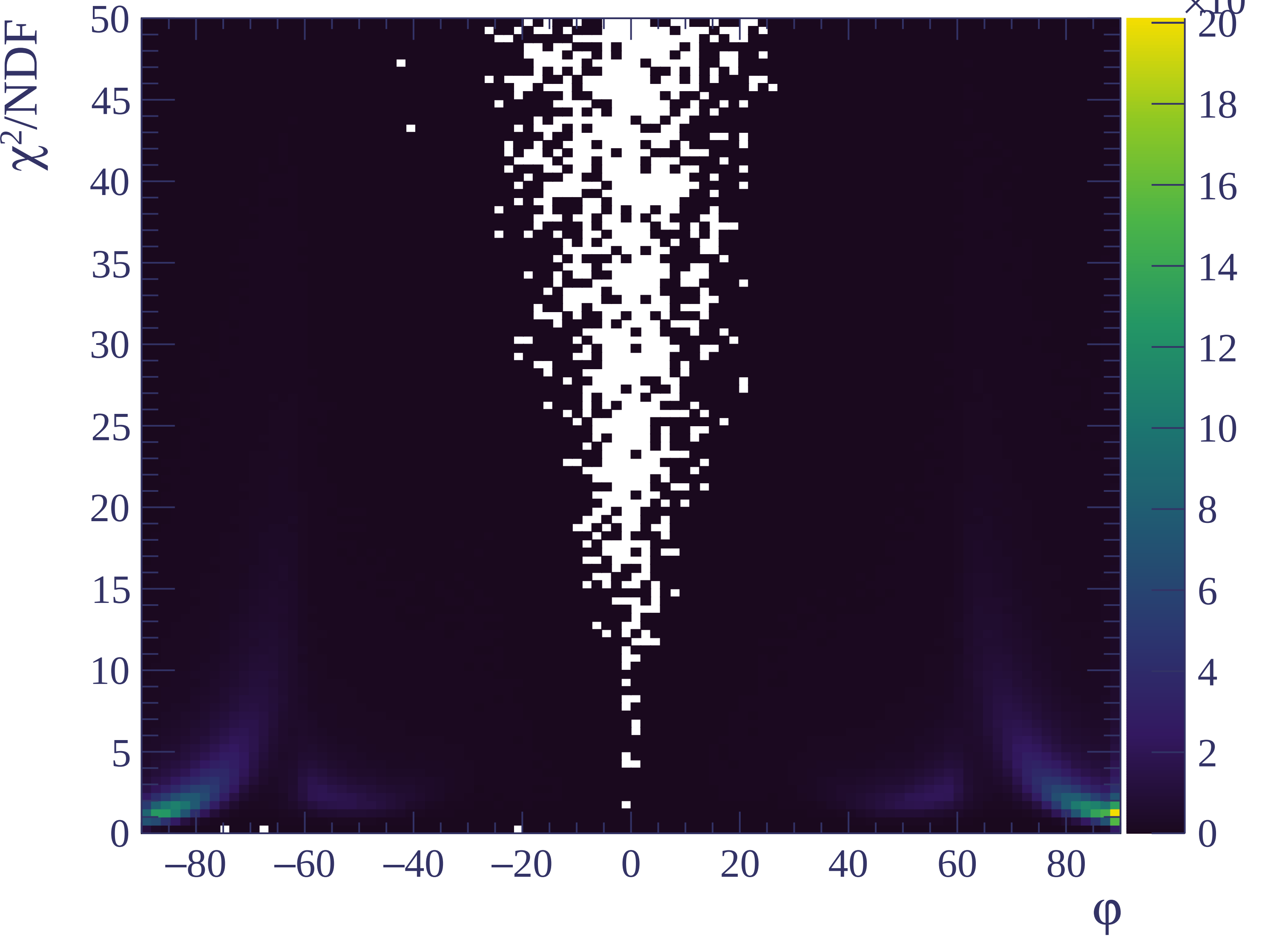

Resolution (%)
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ (°)
Waveforms sum
Crossed pads


Mean (ADC counts/cm)
800
700
600
500
400
300
200
−80
−60
−40
−20
0
20
40
60
80
φ (°)
Waveforms sum
Crossed pads

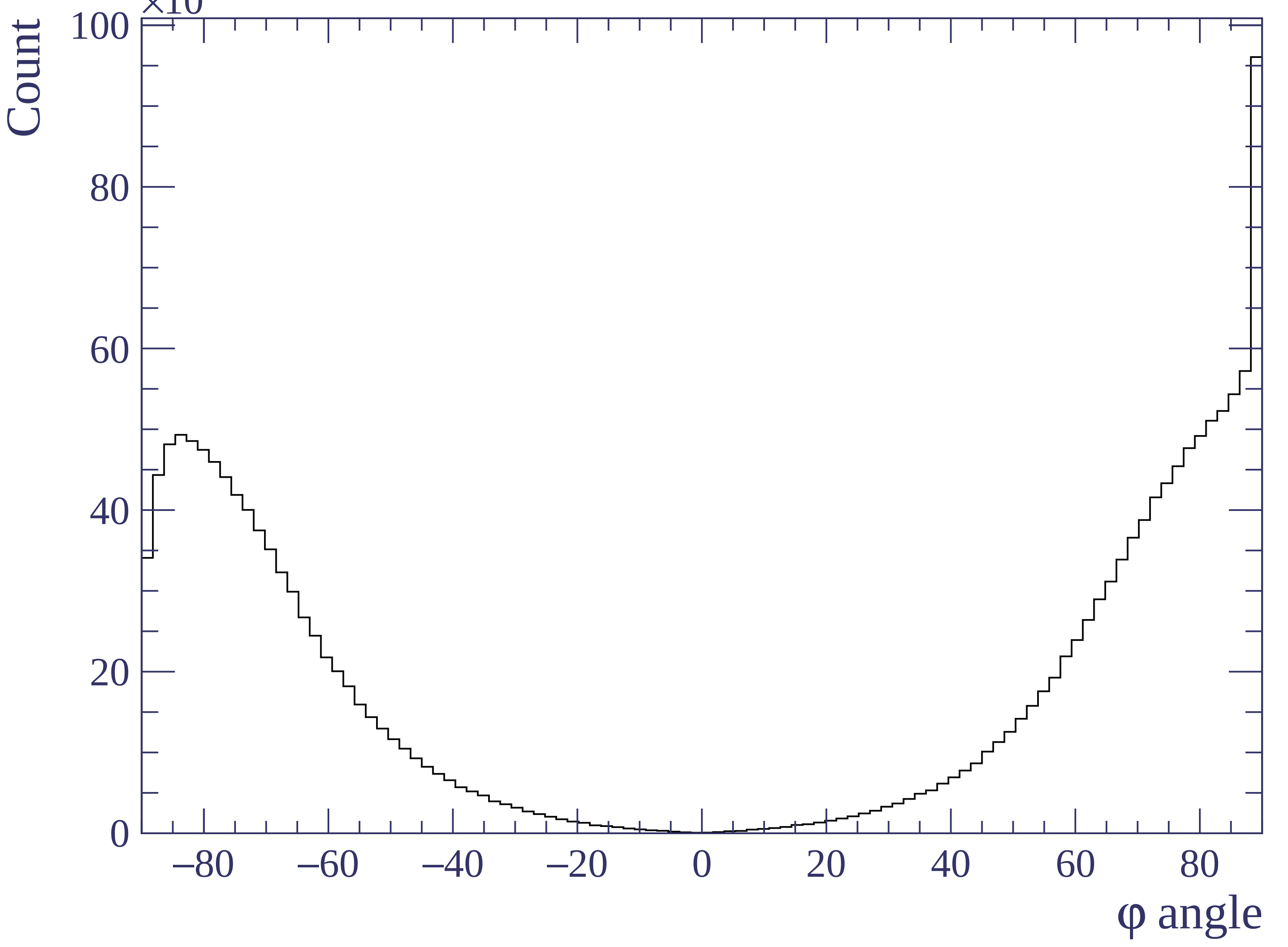

dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
φ (°)
450
400
350
300
250
200
150
100
50
0



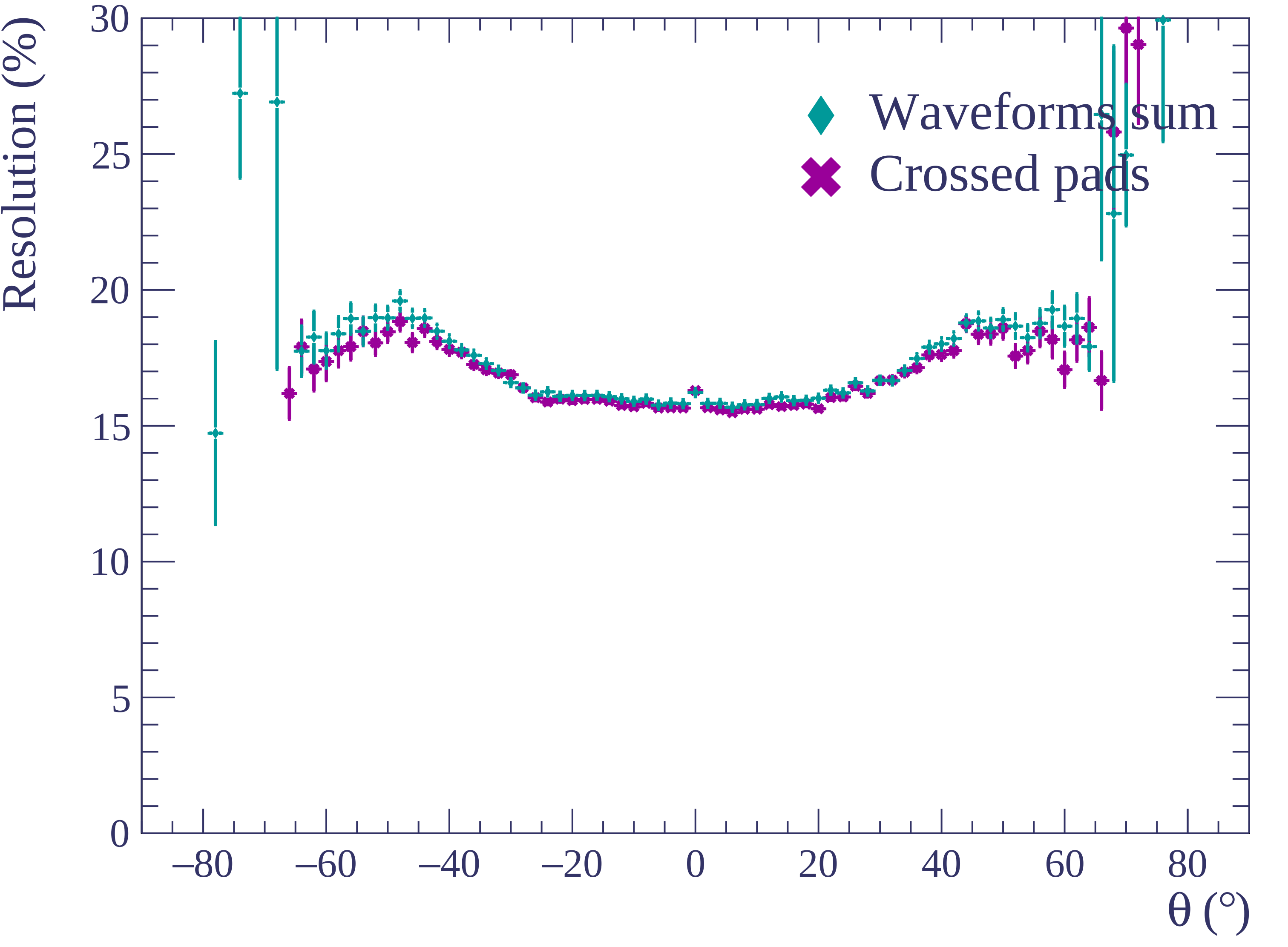
track length (cm)
0
20
40
60
80
100
120
140
160
−80
−60
−40
−20
0
20
40
60
80
φ
0
1000
2000
3000
4000
5000
6000

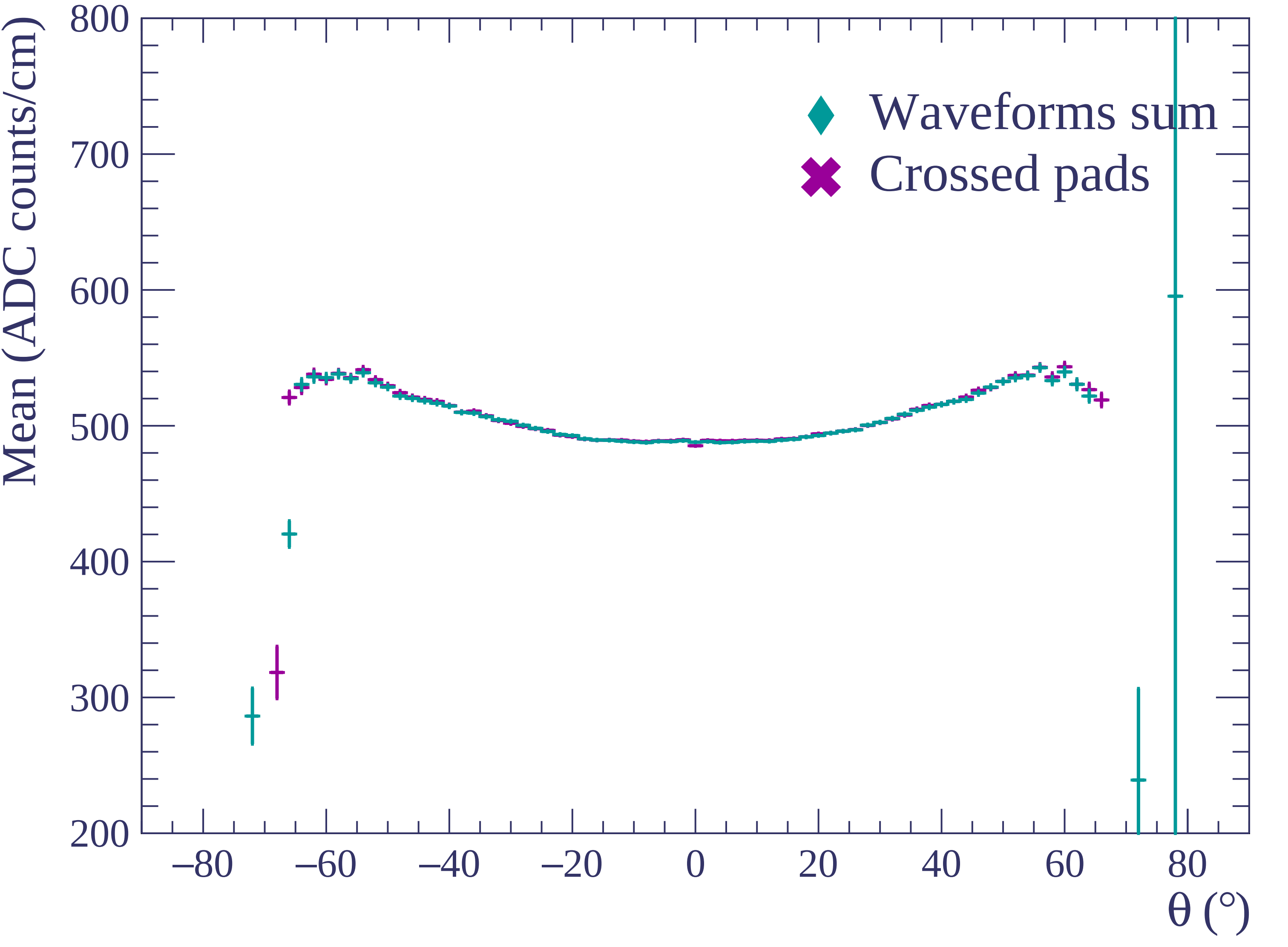


χ^2/NDF
φ
−80
−60
−40
−20
0
20
40
60
80
0
5
10
15
20
25
30
35
40
45
50
×10

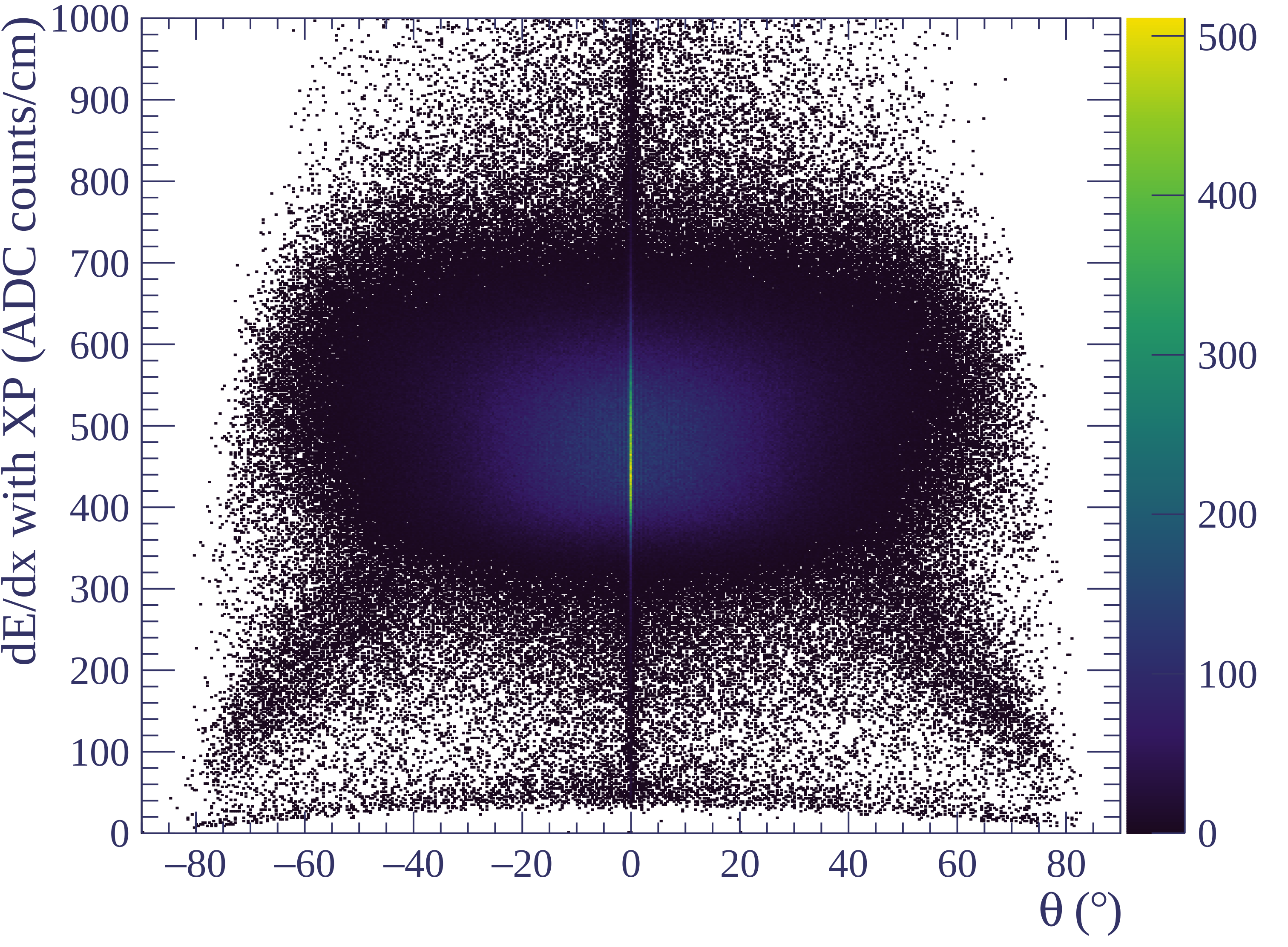


×10
Count
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
φ angle

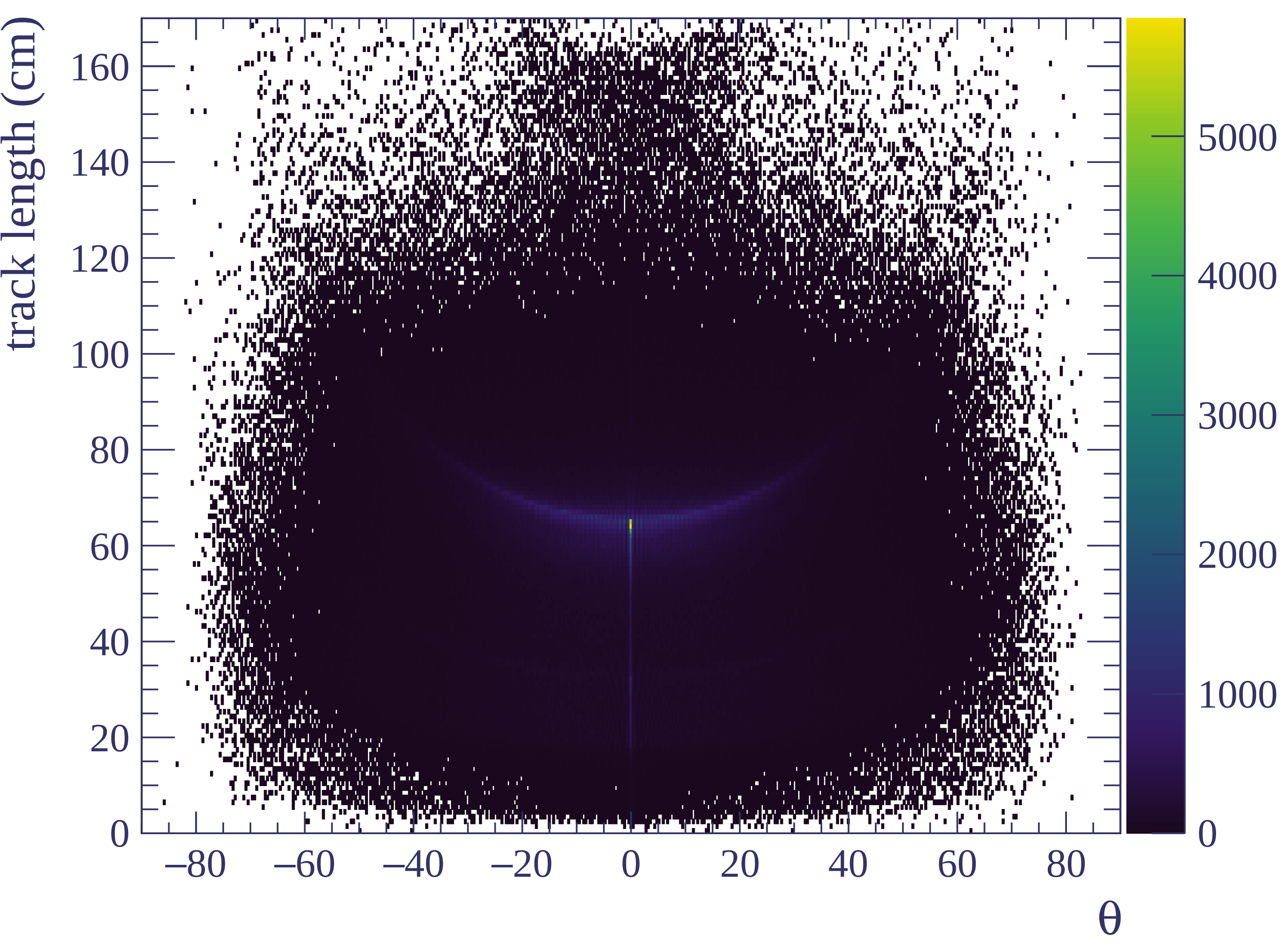

Resolution (%)
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
Waveforms sum
Crossed pads



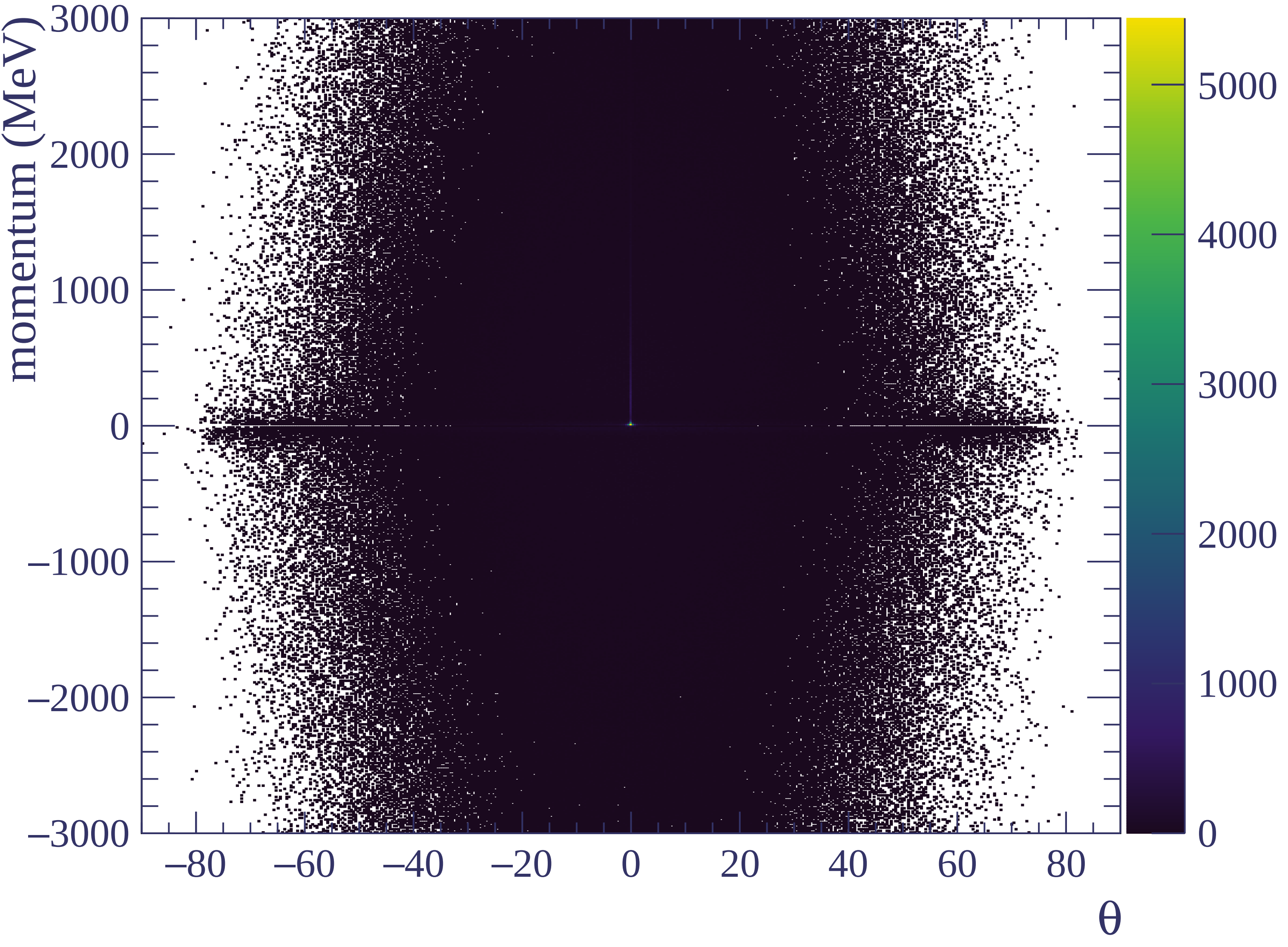
Waveforms sum
Crossed pads
Mean (ADC counts/cm)
θ (°)
800
700
600
500
400
300
200
−80
−60
−40
−20
0
20
40
60
80



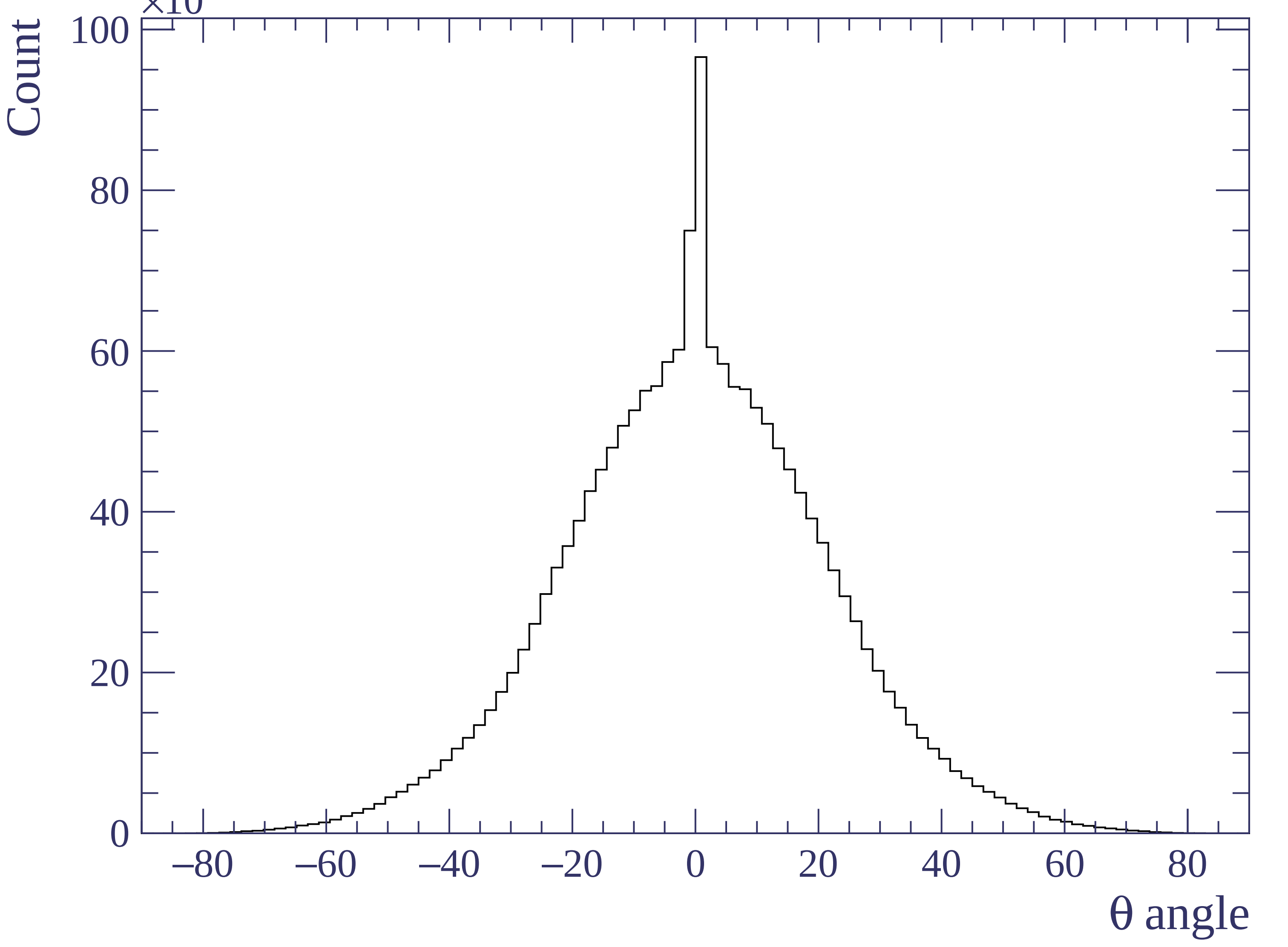
dE/dx with XP (ADC counts/cm)
θ (°)
0
100
200
300
400
500
600
700
800
900
1000
−80
−60
−40
−20
0
20
40
60
80
0
100
200
300
400
500



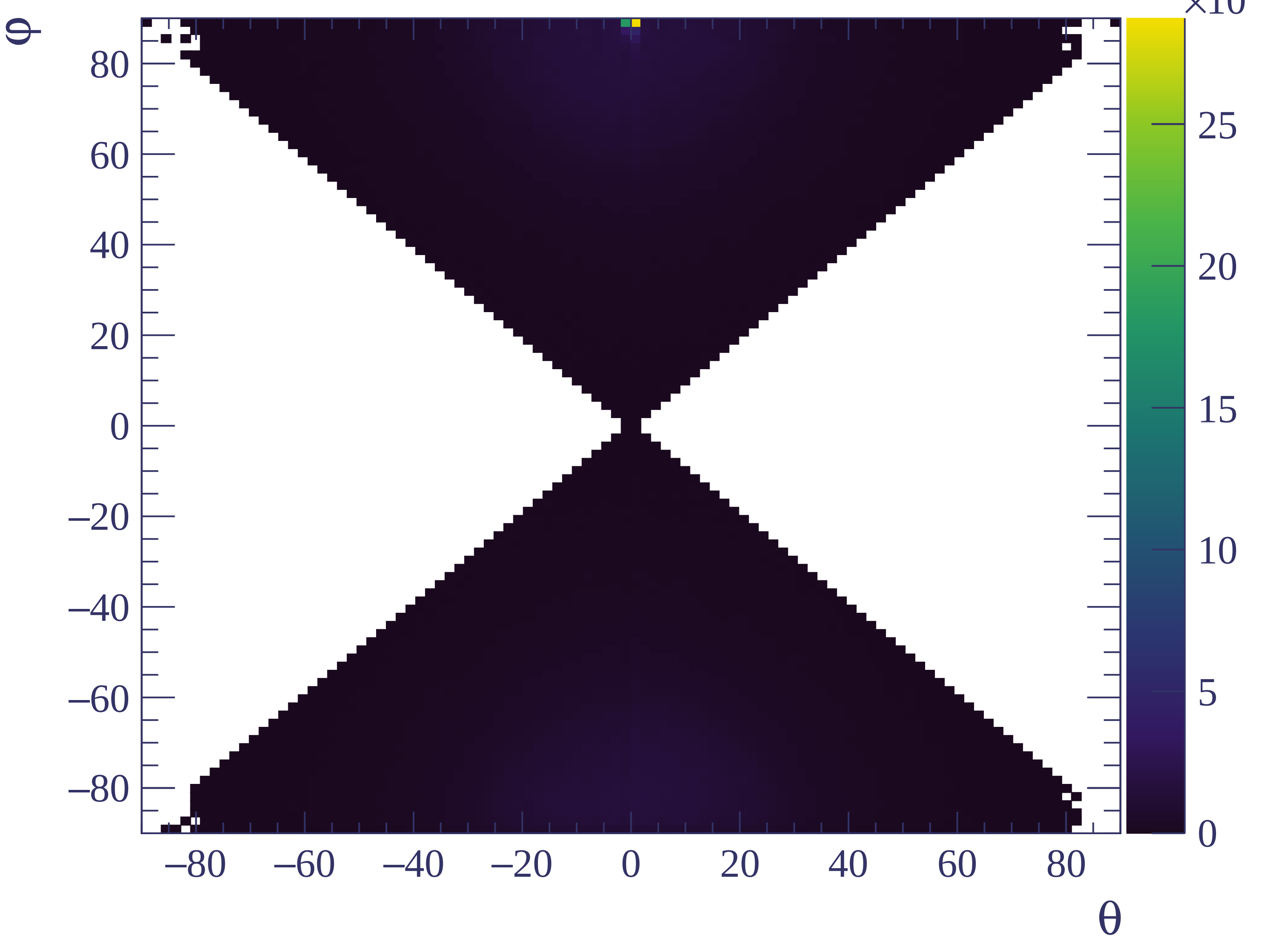
track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
θ
5000
4000
3000
2000
1000
0

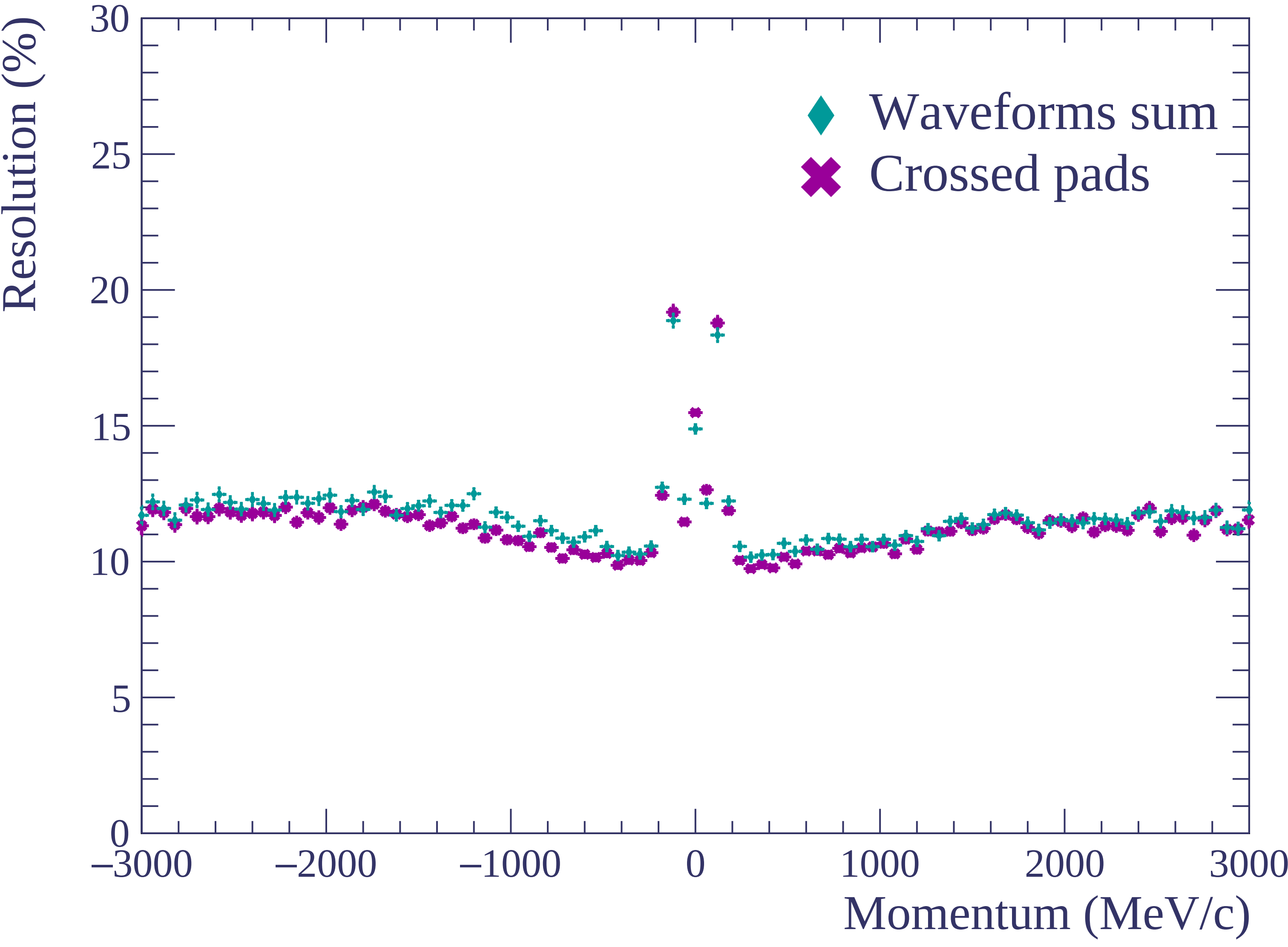


momentum (MeV)
3000
2000
1000
0
−1000
−2000
−3000
−80
−60
−40
−20
0
20
40
60
80
θ
5000
4000
3000
2000
1000
0

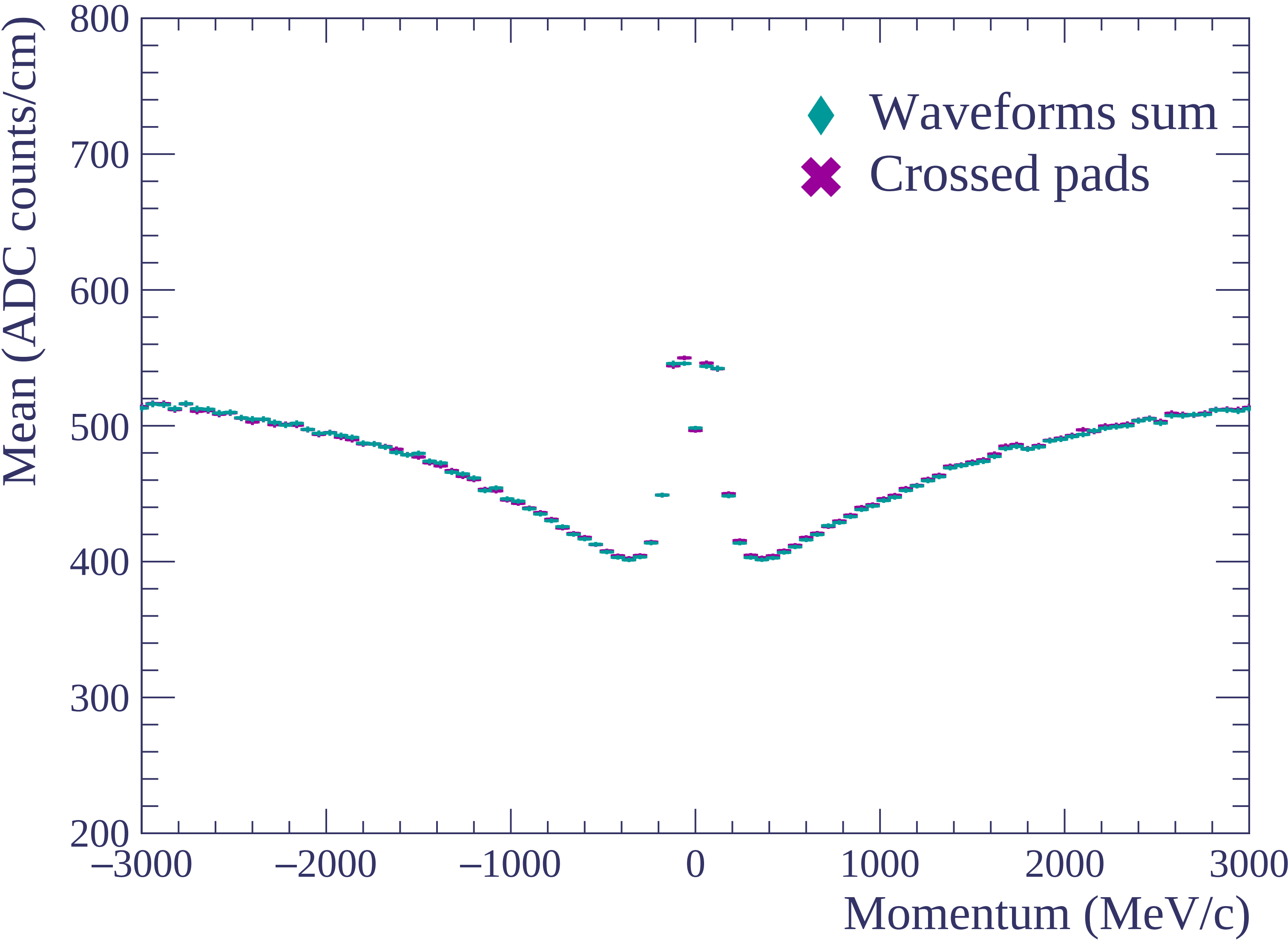


×10
Count
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
θ angle

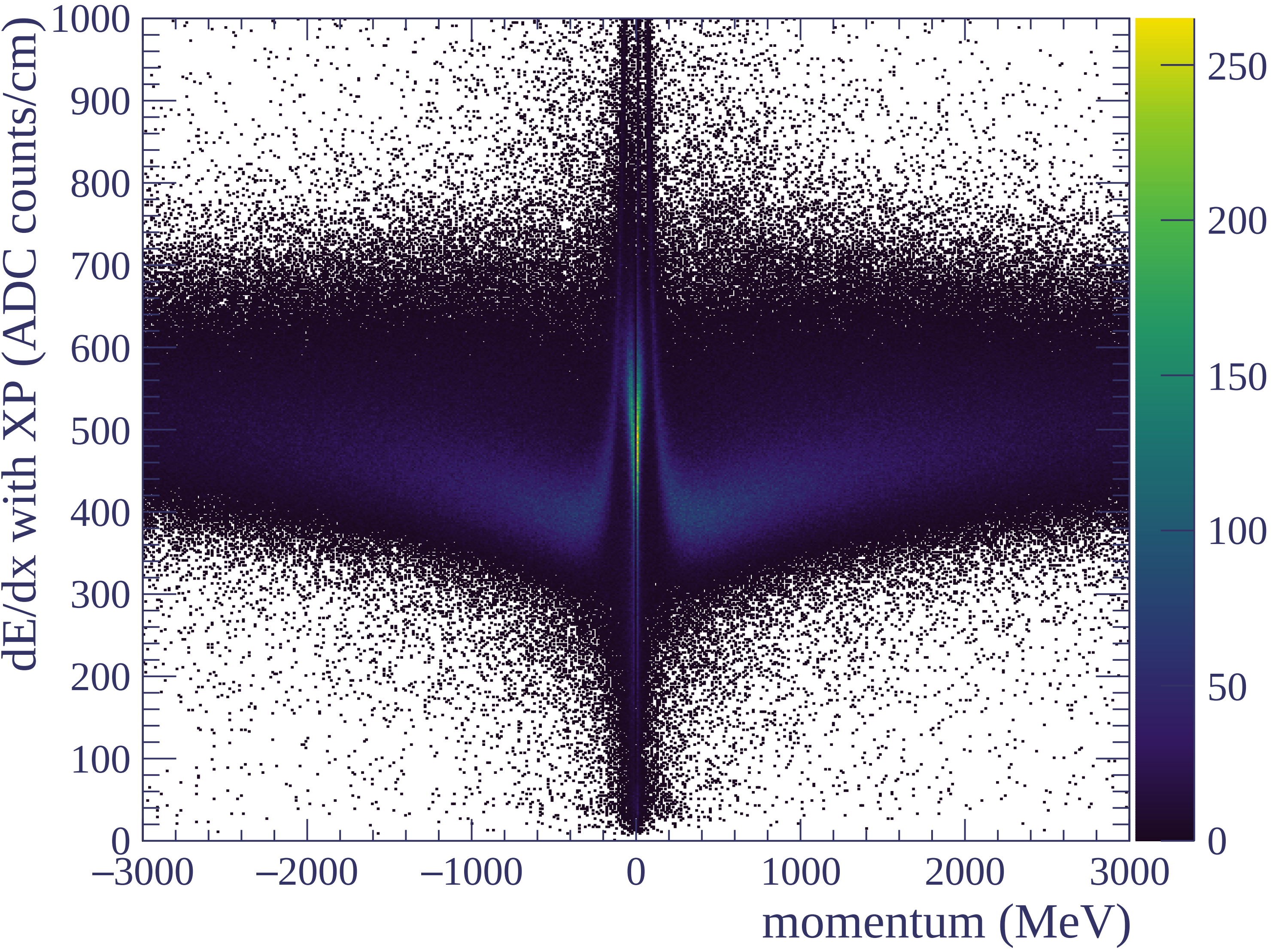

φ
80
60
40
20
0
−20
−40
−60
−80
−80
−60
−40
−20
0
20
40
60
80
θ
25
20
15
10
5
0

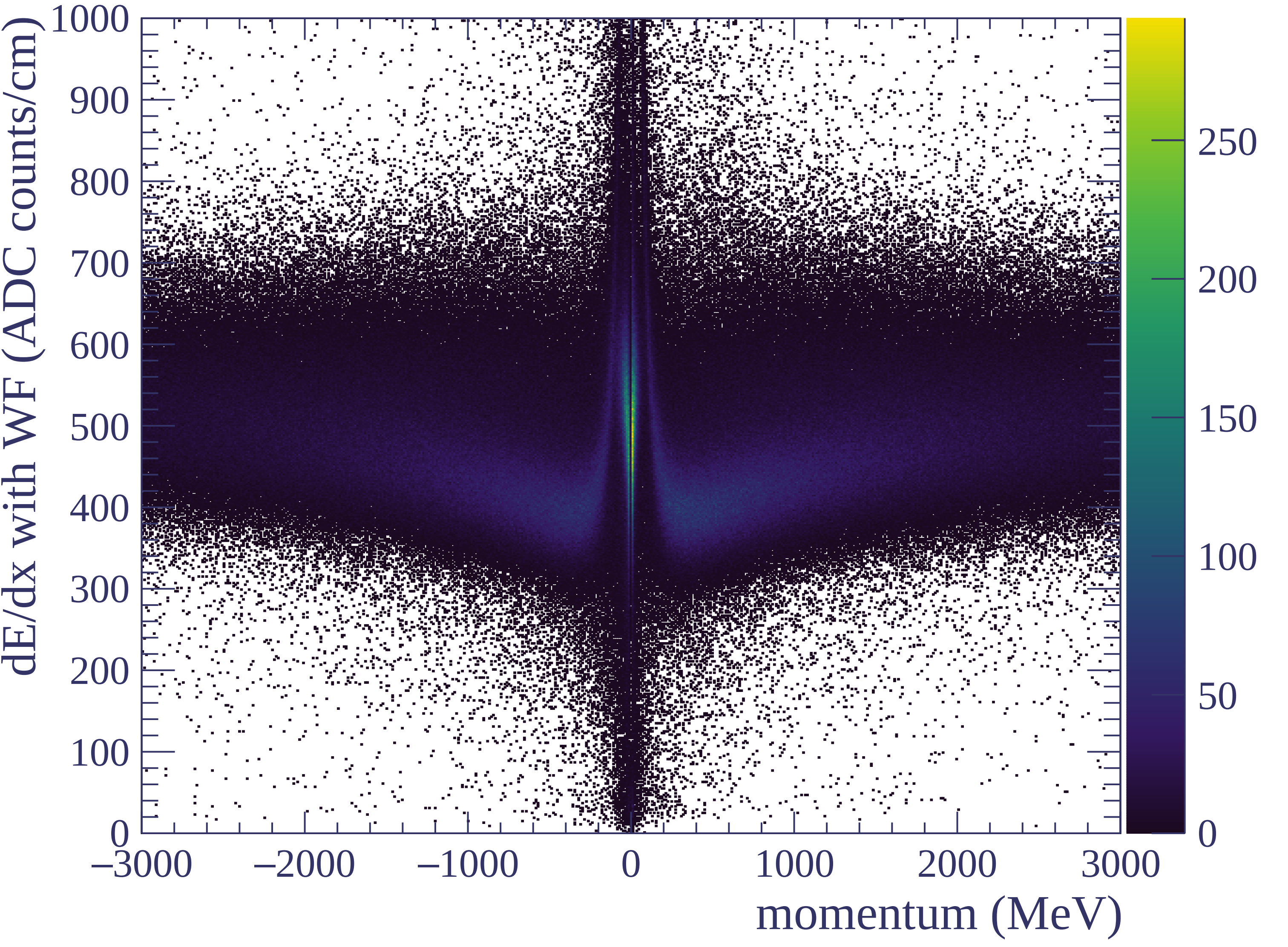


Resolution (%)
30
25
20
15
10
5
0
−3000
−2000
−1000
0
1000
2000
3000
Momentum (MeV/c)
Waveforms sum
Crossed pads

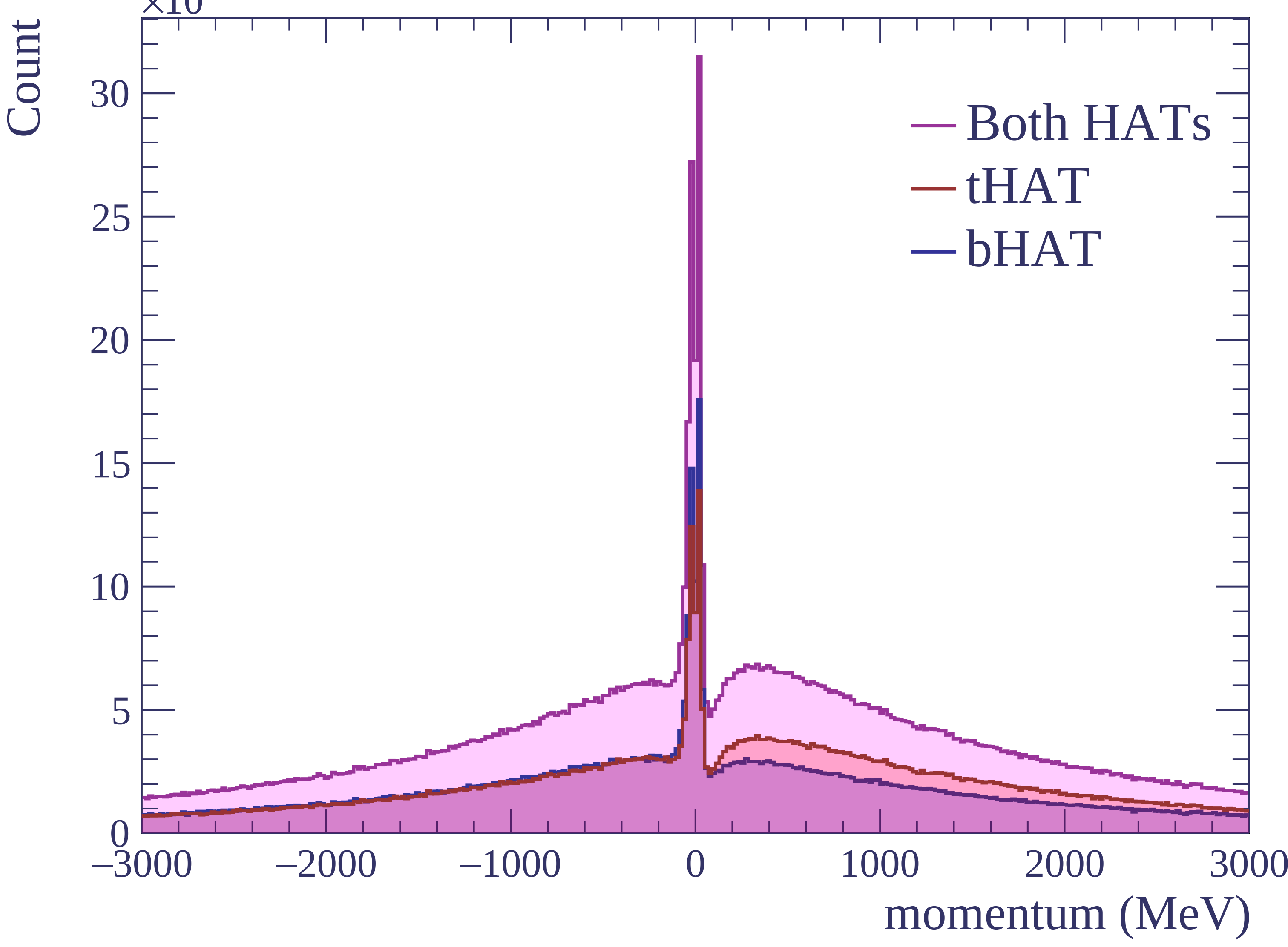


Mean (ADC counts/cm)
800
700
600
500
400
300
200
−3000
−2000
−1000
0
1000
2000
3000
Momentum (MeV/c)
Waveforms sum
Crossed pads

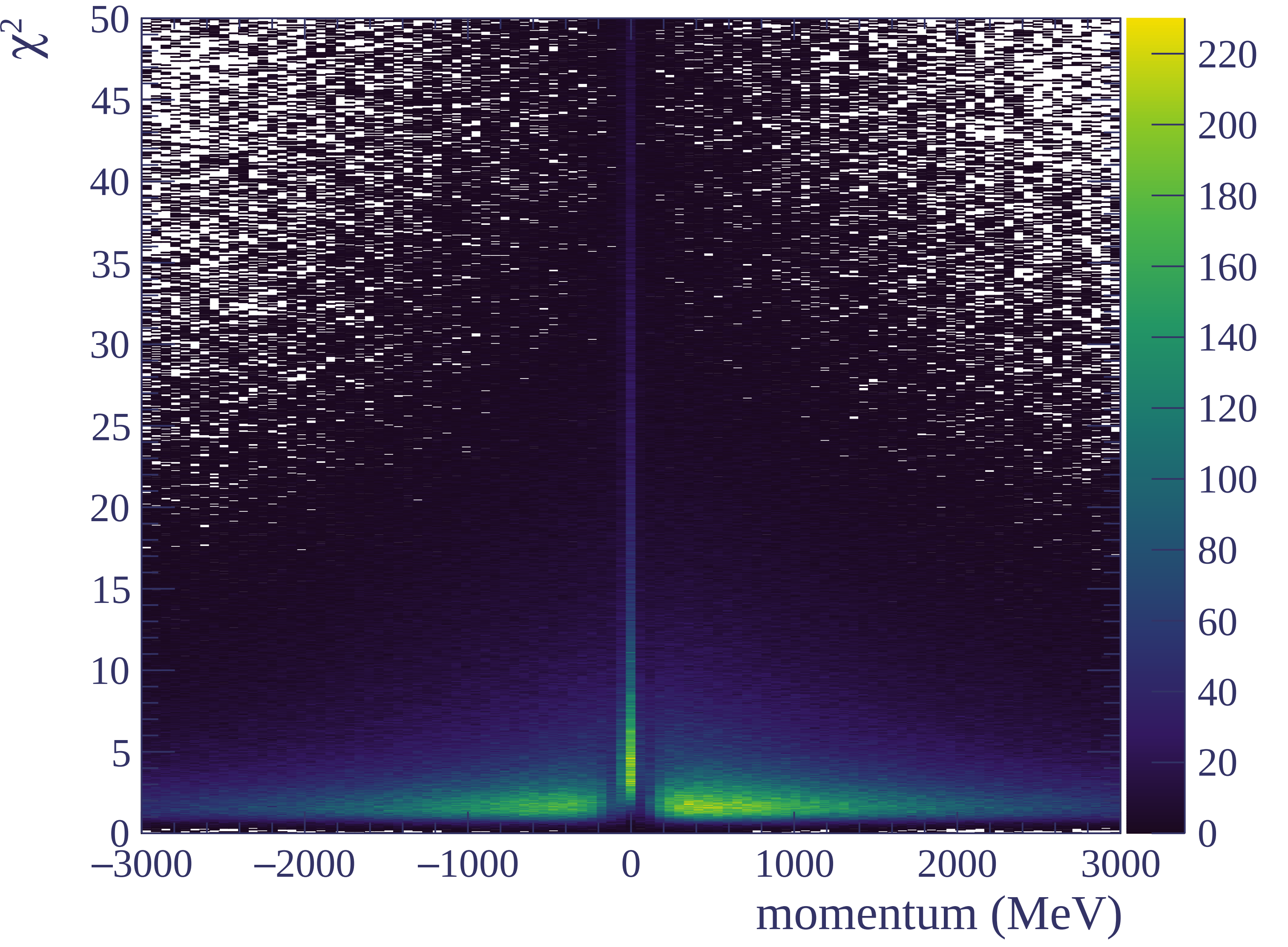

dE/dx with XP (ADC counts/cm)
momentum (MeV)
0
100
200
300
400
500
600
700
800
900
1000
−3000
−2000
−1000
0
1000
2000
3000
0
50
100
150
200
250



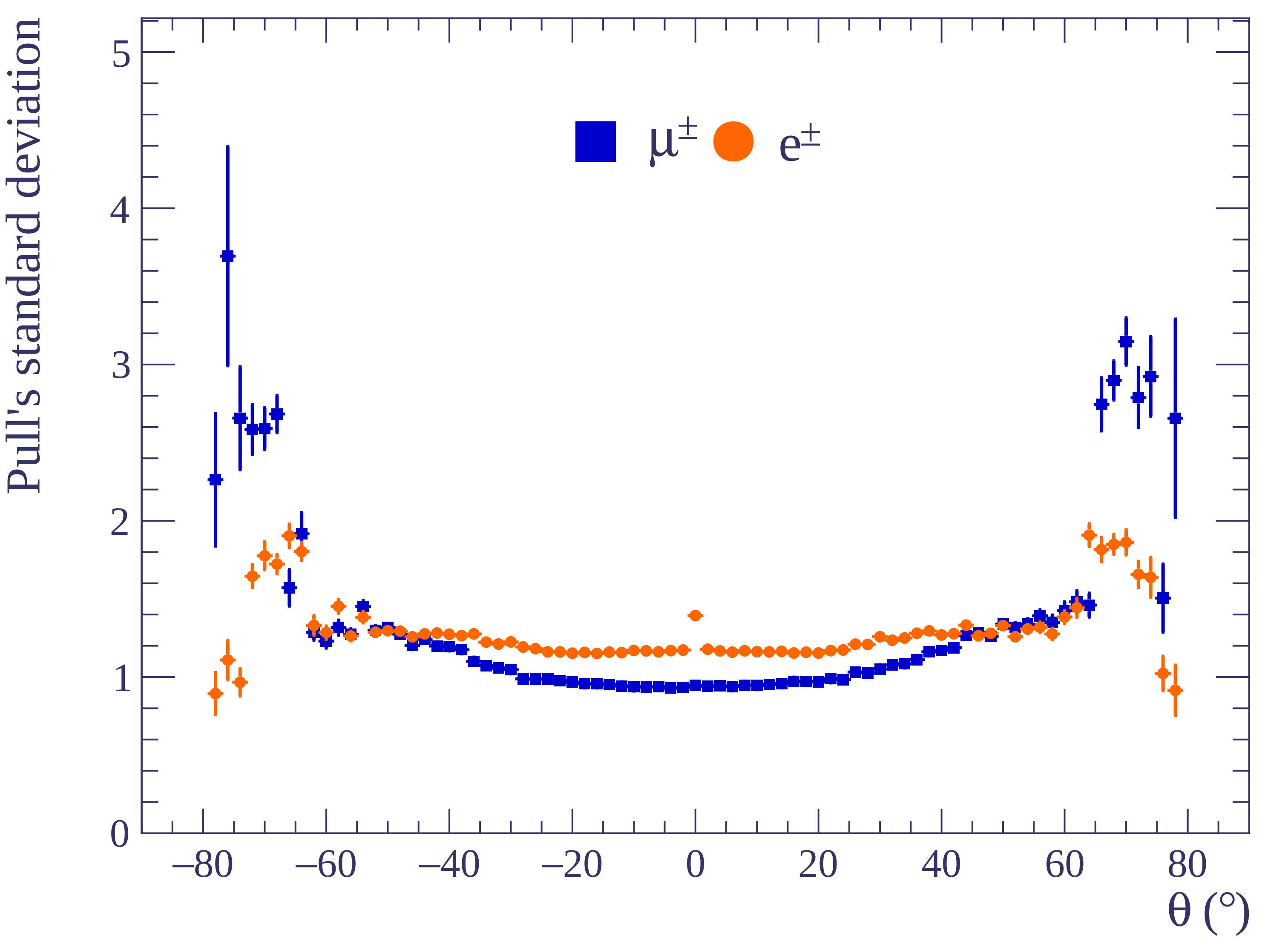
dE/dx with WF (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
250
200
150
100
50
0



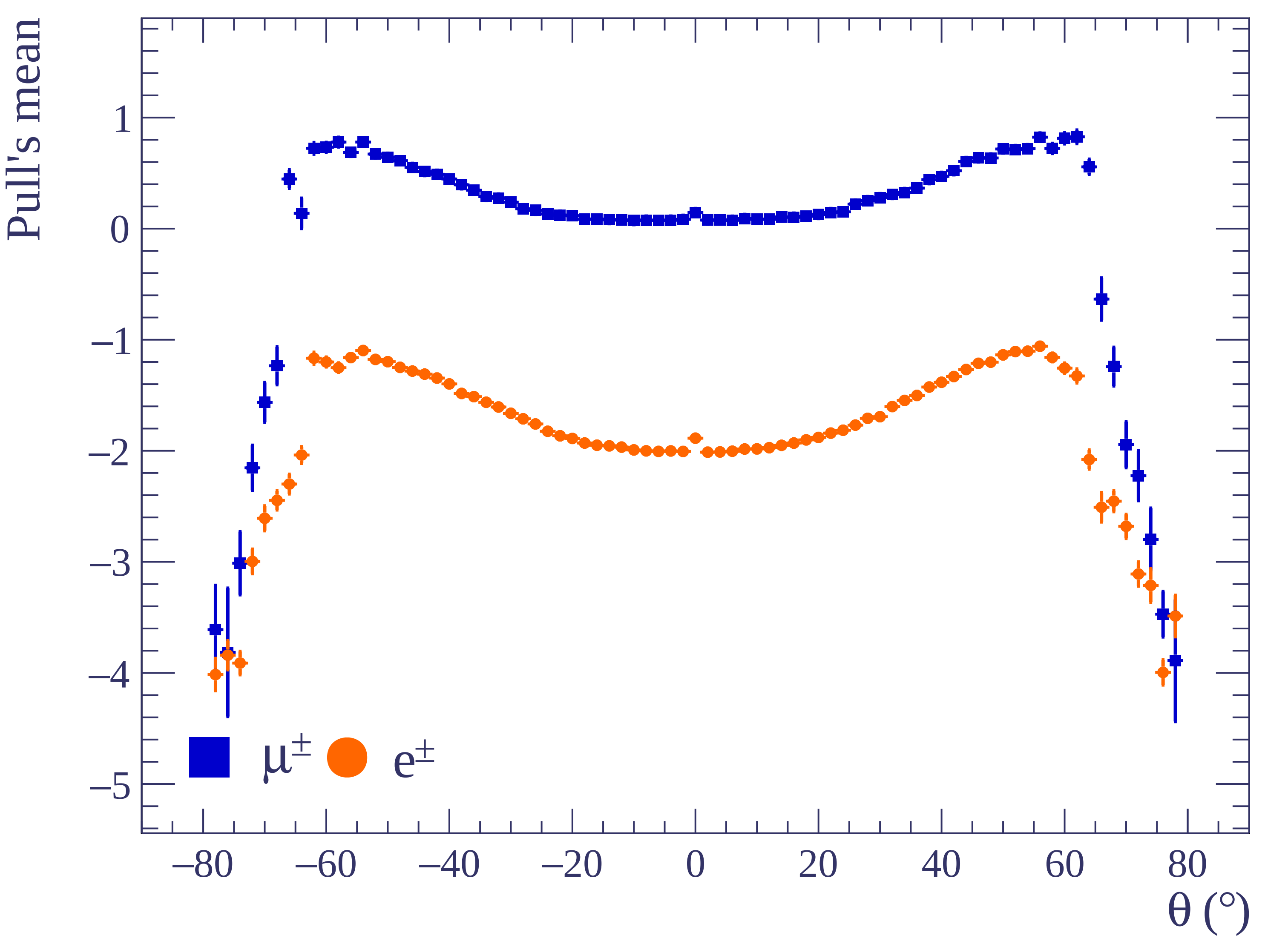
×10
Count
30
25
20
15
10
5
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
Both HATs
tHAT
bHAT



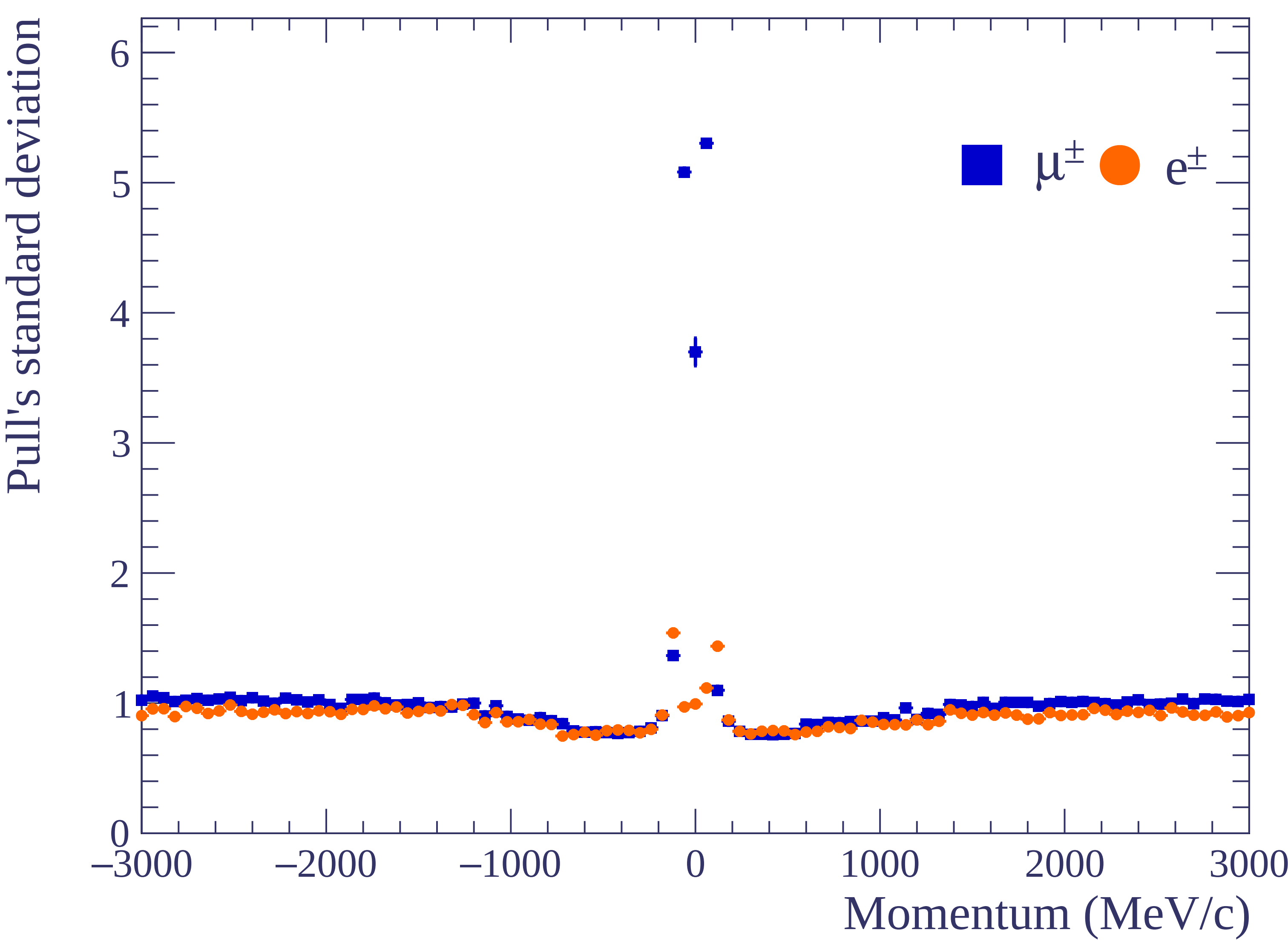
χ^2
50
45
40
35
30
25
20
15
10
5
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
220
200
180
160
140
120
100
80
60
40
20
0



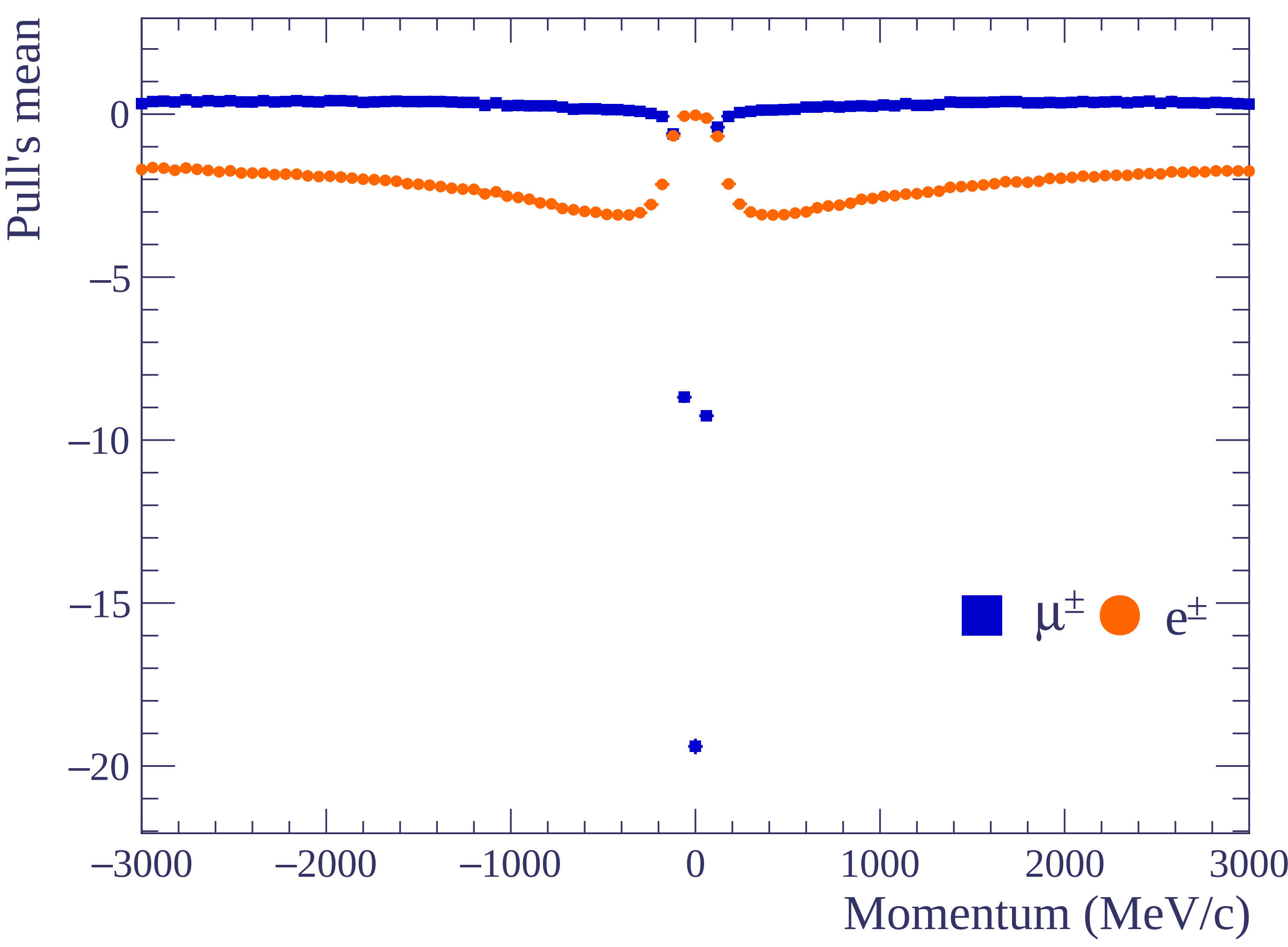
Pull's standard deviation
$\mu^{\pm}$
$e^{\pm}$
0
1
2
3
4
5
−80
−60
−40
−20
0
20
40
60
80
θ (°)



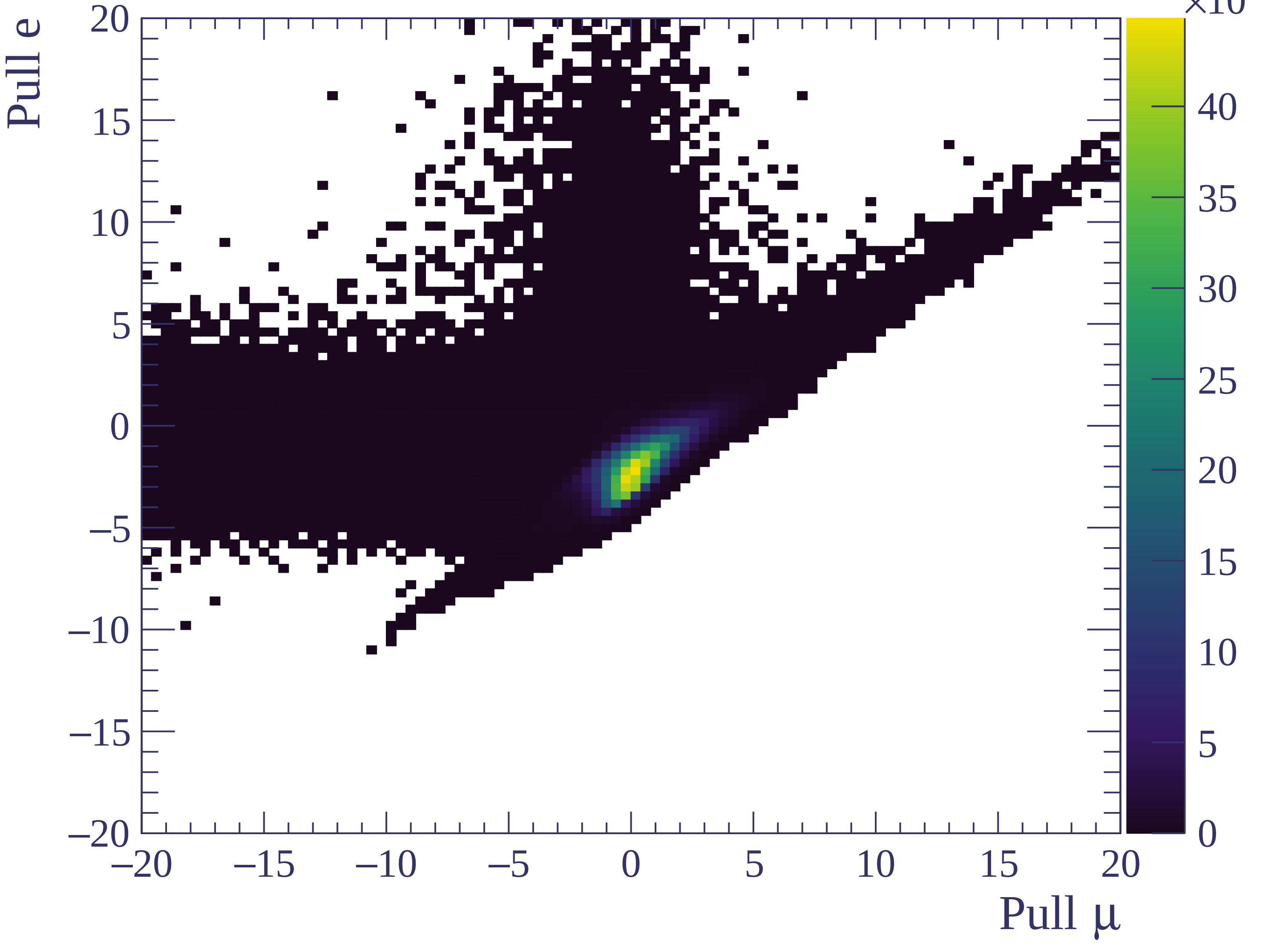
Pull's mean
1
0
−1
−2
−3
−4
−5
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±



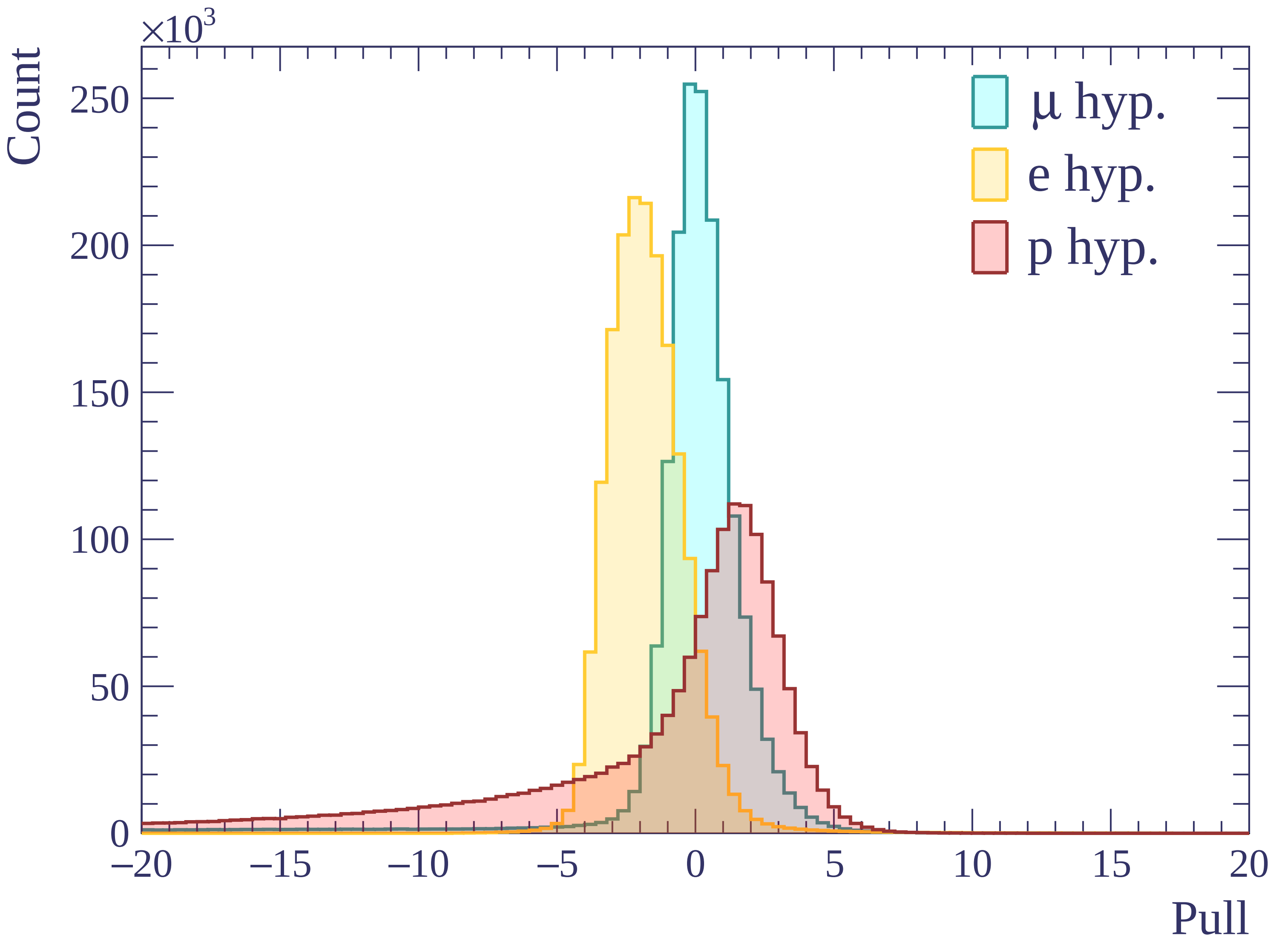
Pull's standard deviation
Momentum (MeV/c)
$\mu^{\pm}$
$e^{\pm}$
0
1
2
3
4
5
6
−3000
−2000
−1000
0
1000
2000
3000



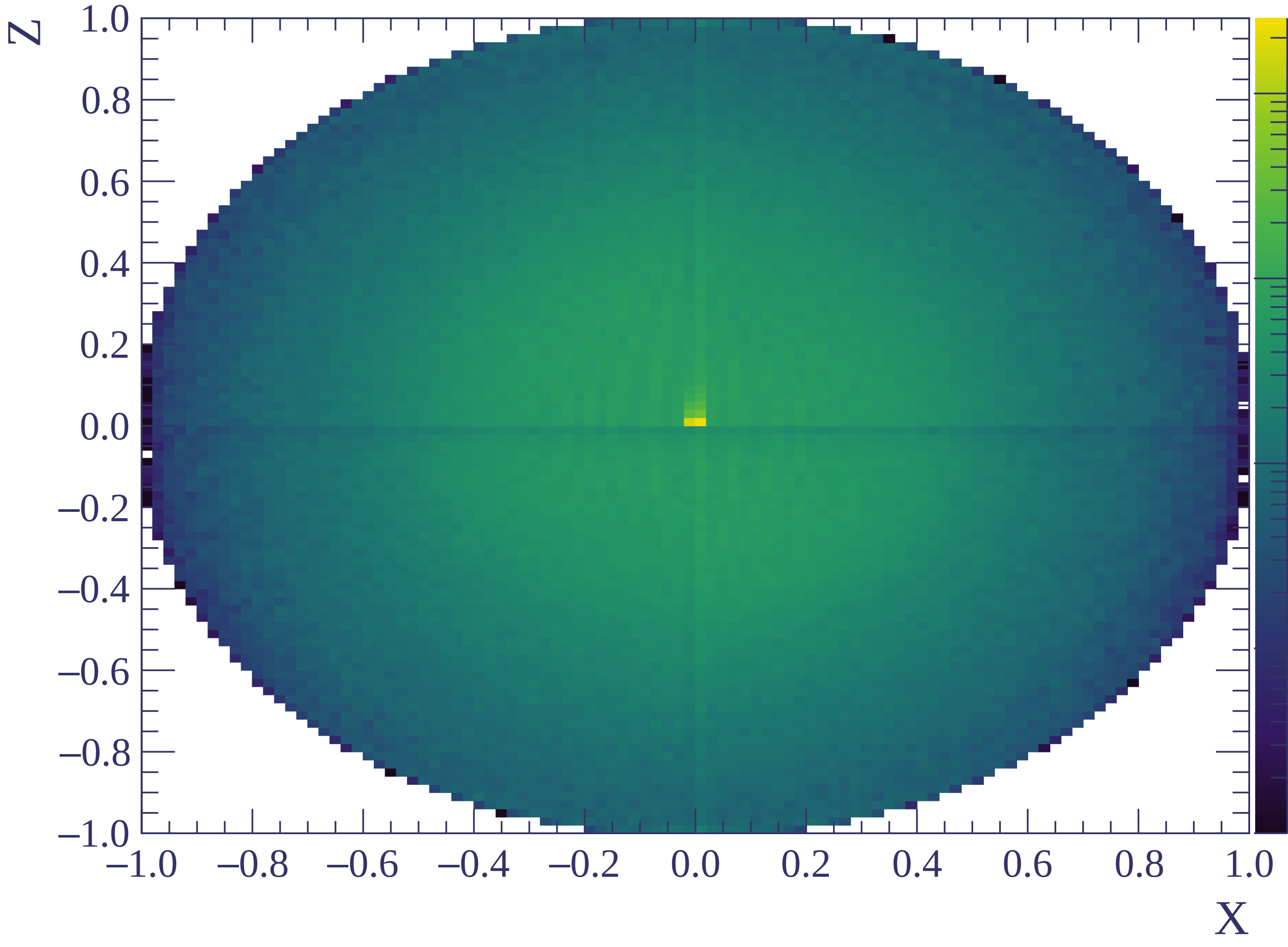
Pull's mean
0
−5
−10
−15
−20
−3000
−2000
−1000
0
1000
2000
3000
Momentum (MeV/c)
$\mu^{\pm}$
$e^{\pm}$



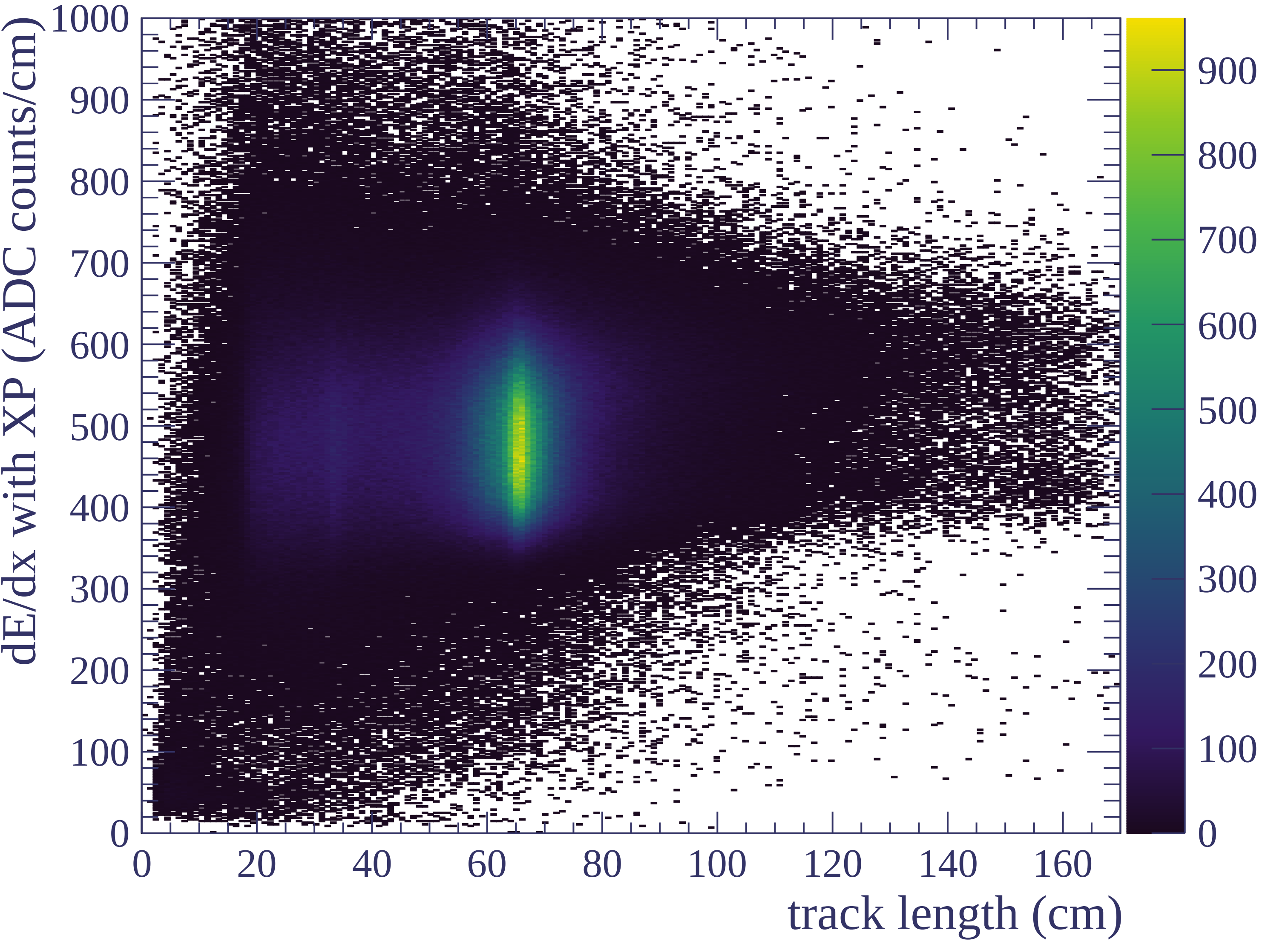
Pull e
Pull μ
×10
20
15
10
5
0
−5
−10
−15
−20
−20
−15
−10
−5
0
5
10
15
20
40
35
30
25
20
15
10
5
0



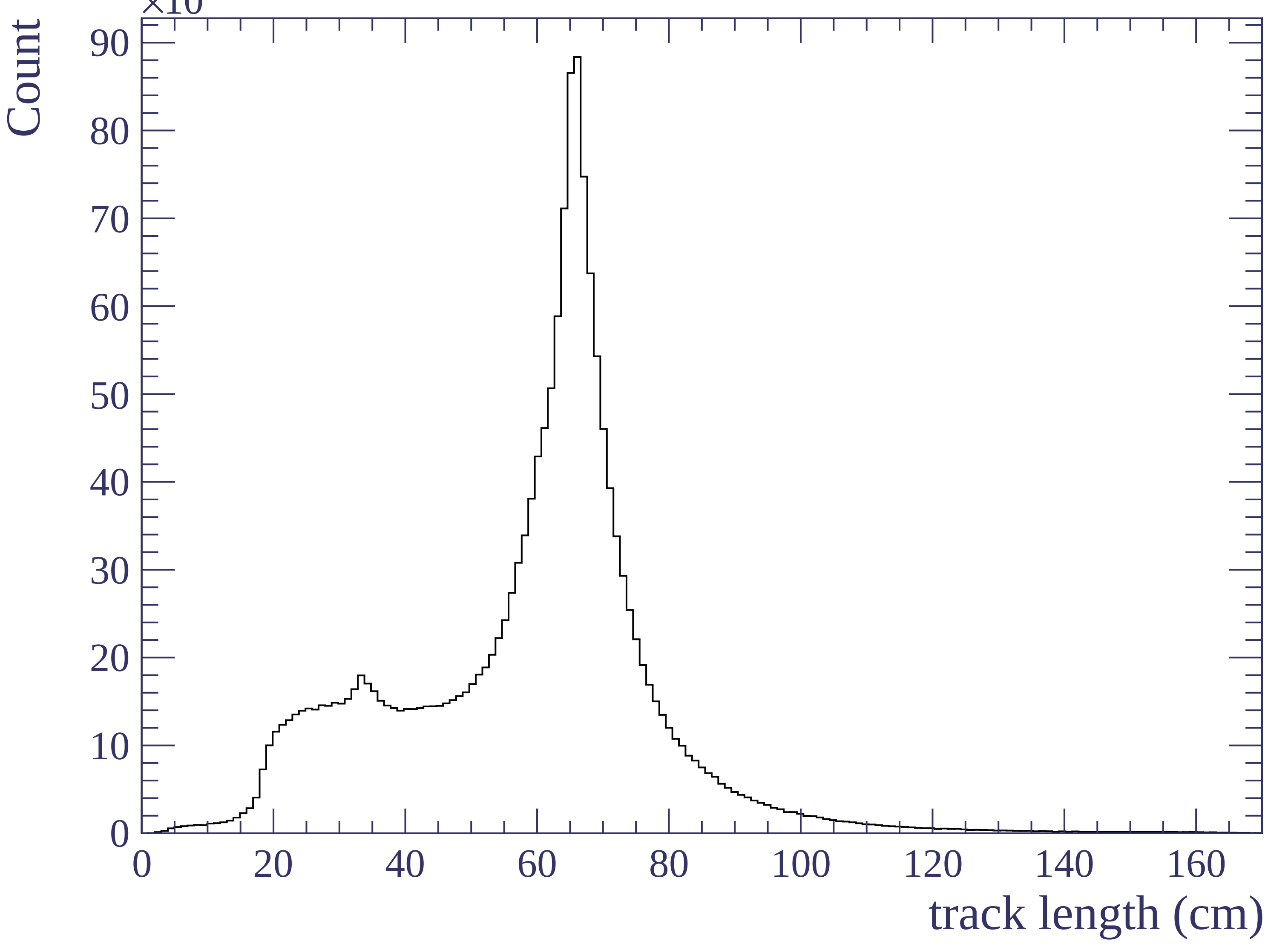
×10^3
Count
250
200
150
100
50
0
−20
−15
−10
−5
0
5
10
15
20
Pull
μ hyp.
e hyp.
p hyp.

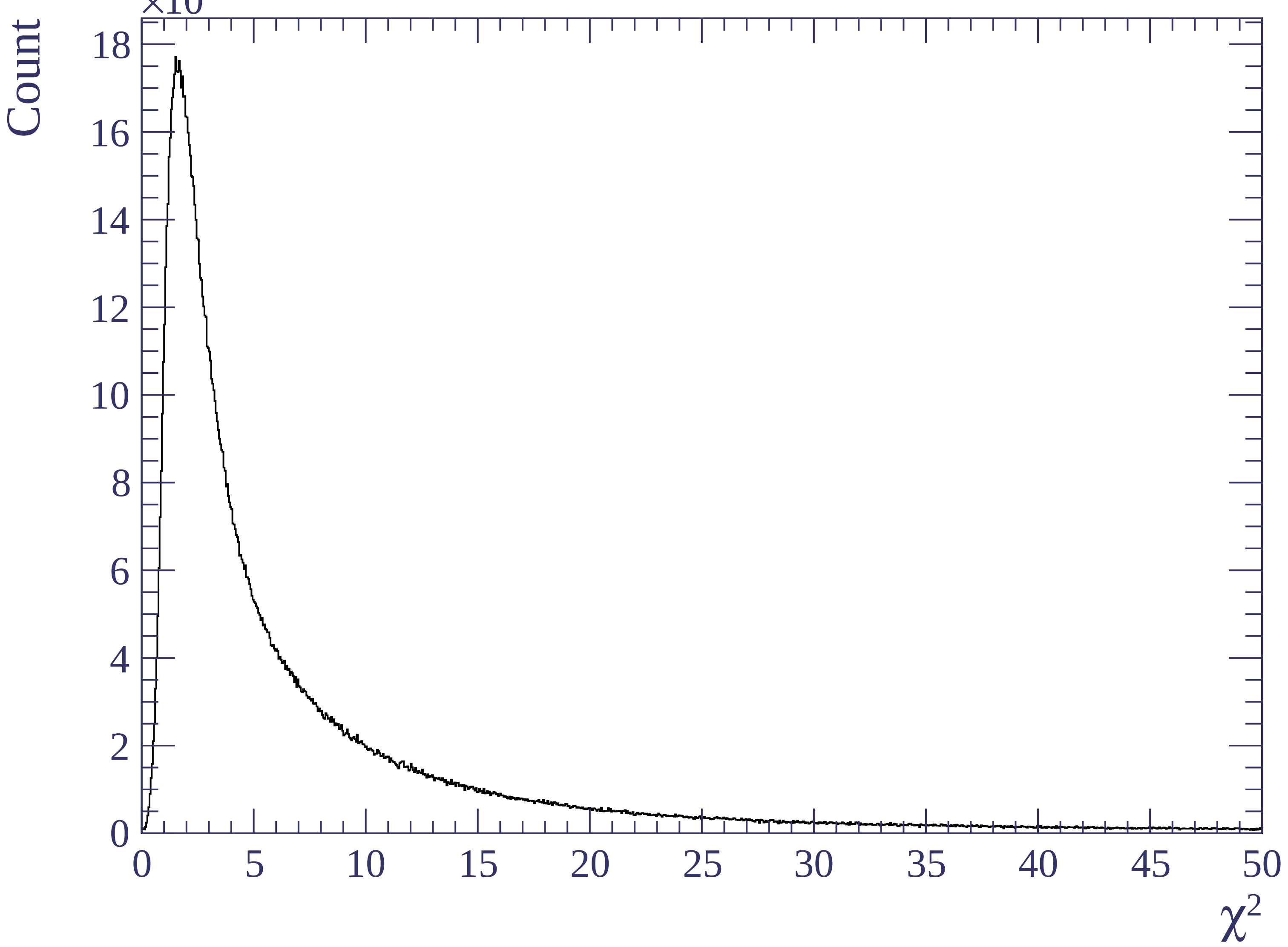


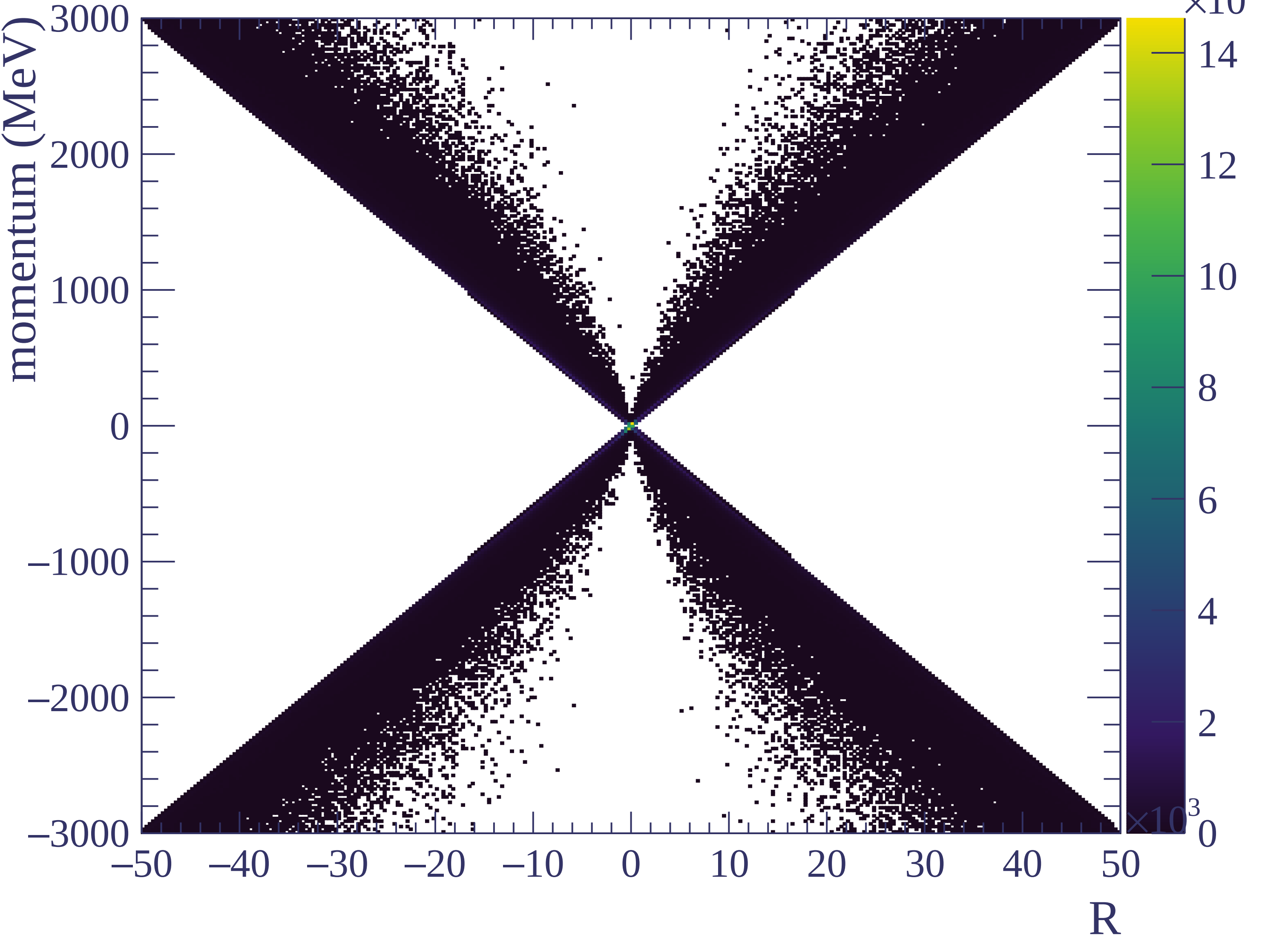
Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X

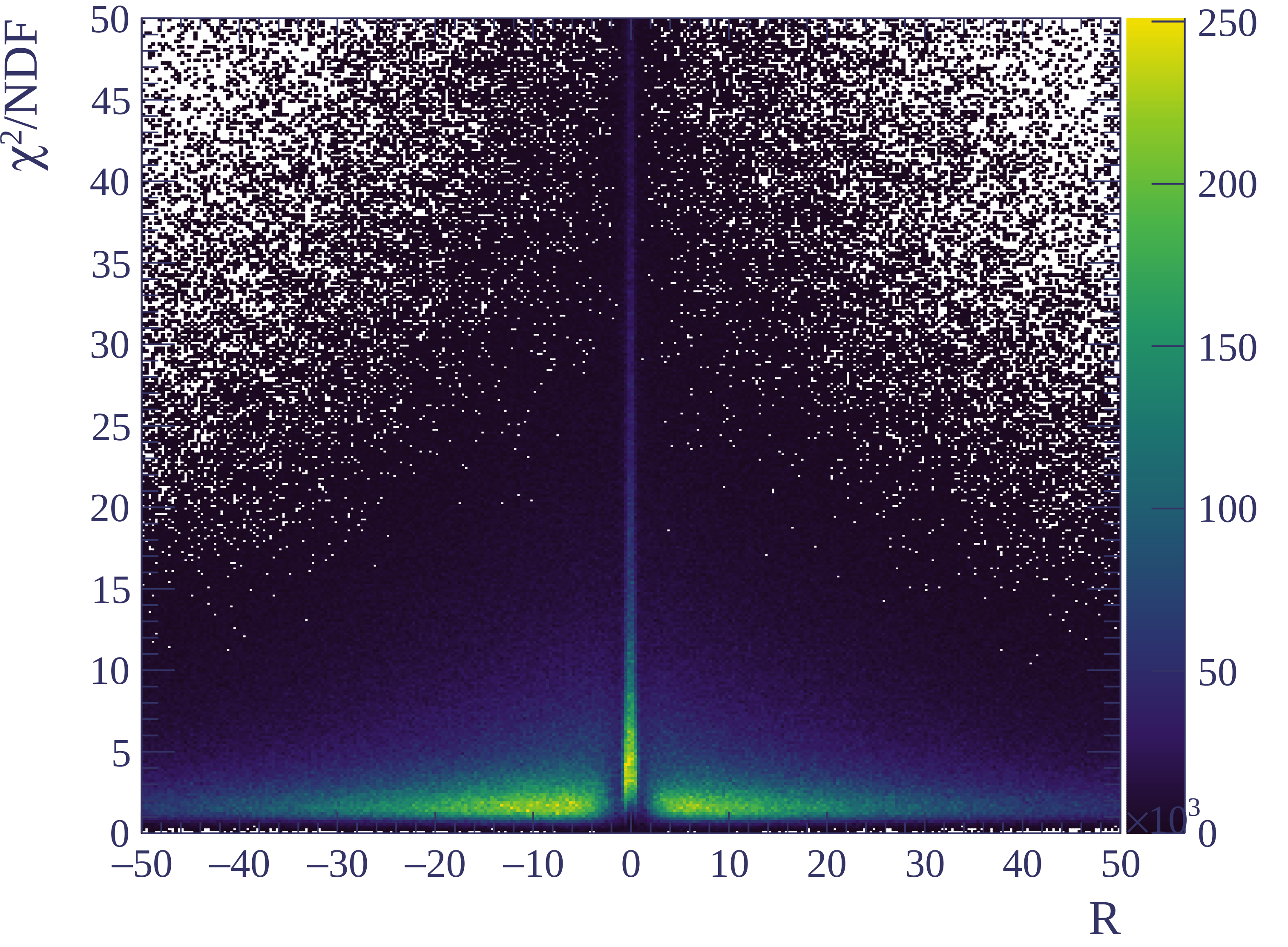


dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
0
20
40
60
80
100
120
140
160
track length (cm)
900
800
700
600
500
400
300
200
100
0

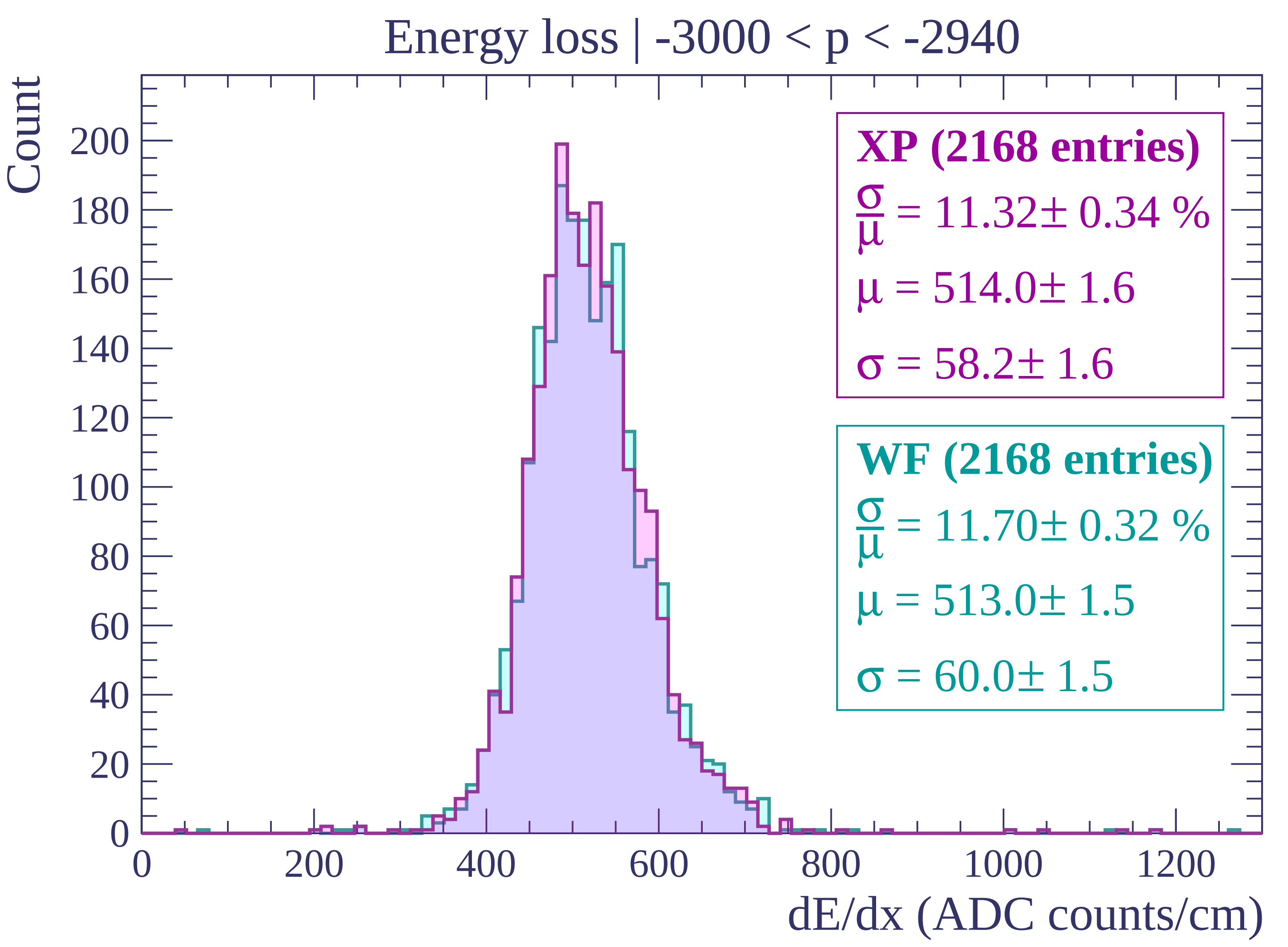

×10
Count
90
80
70
60
50
40
30
20
10
0
0
20
40
60
80
100
120
140
160
track length (cm)

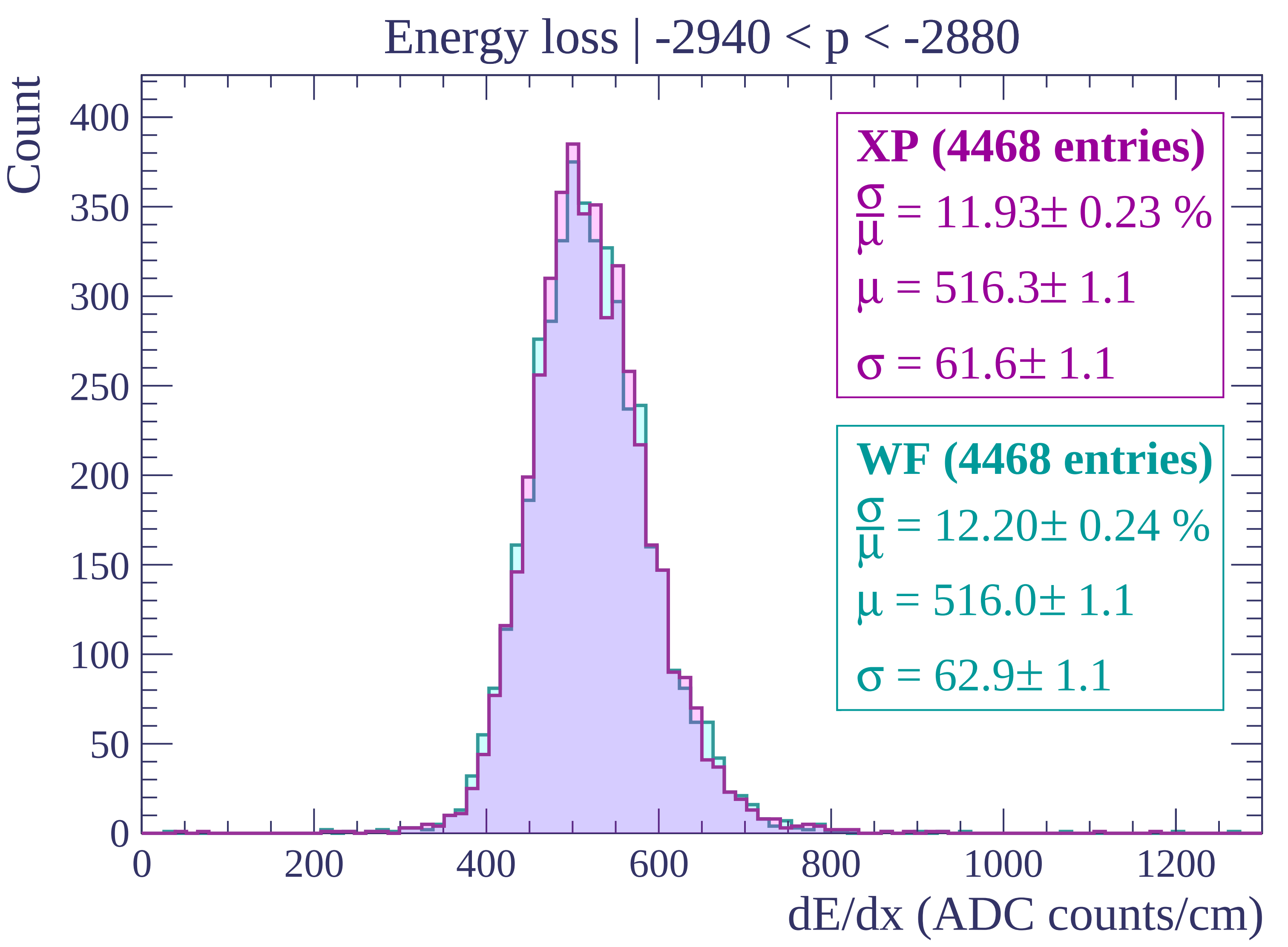

×10
Count
18
16
14
12
10
8
6
4
2
0
0
5
10
15
20
25
30
35
40
45
50
χ^2

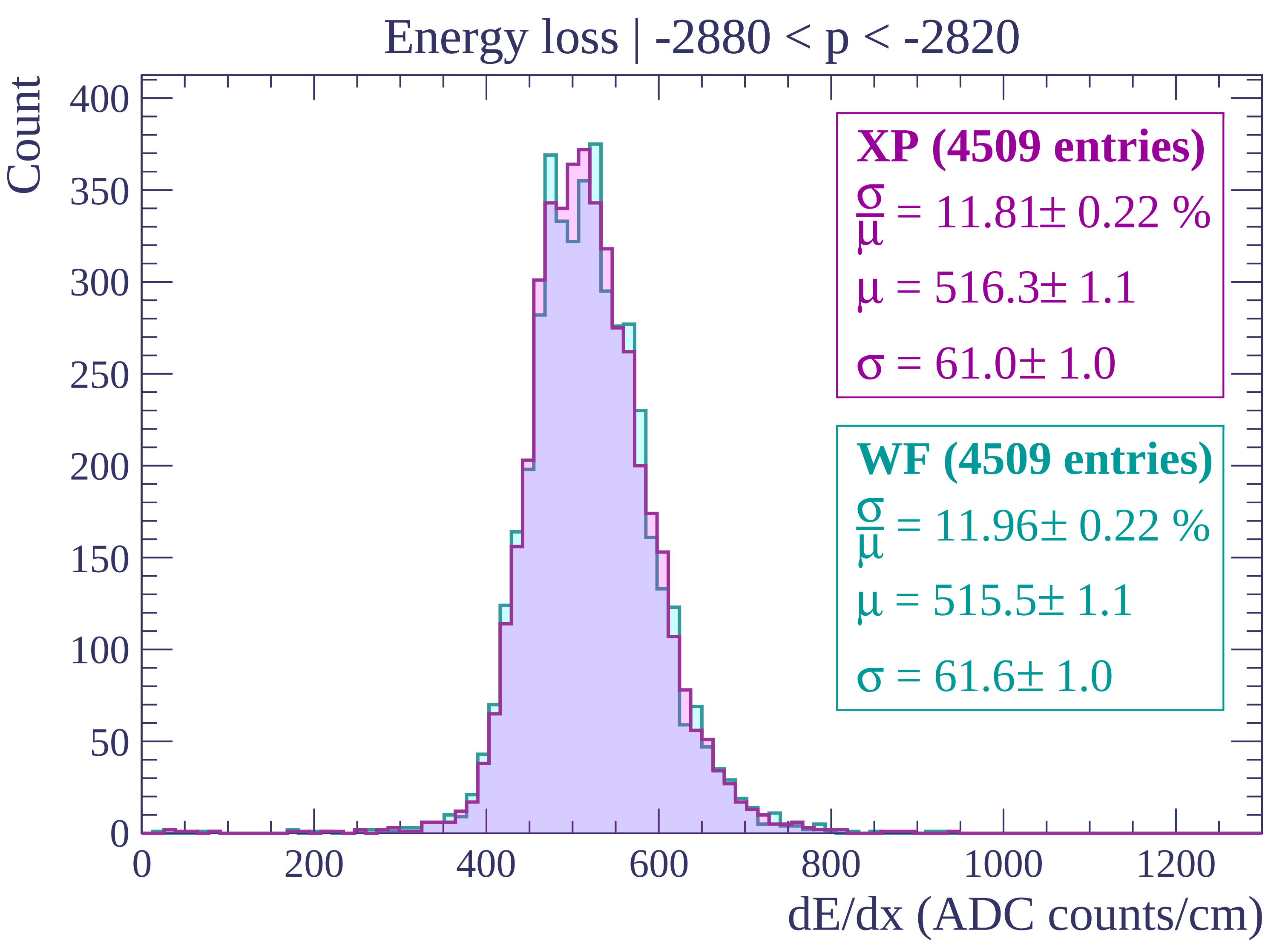

momentum (MeV)
3000
2000
1000
0
−1000
−2000
−3000
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
×10³
14
12
10
8
6
4
2
0



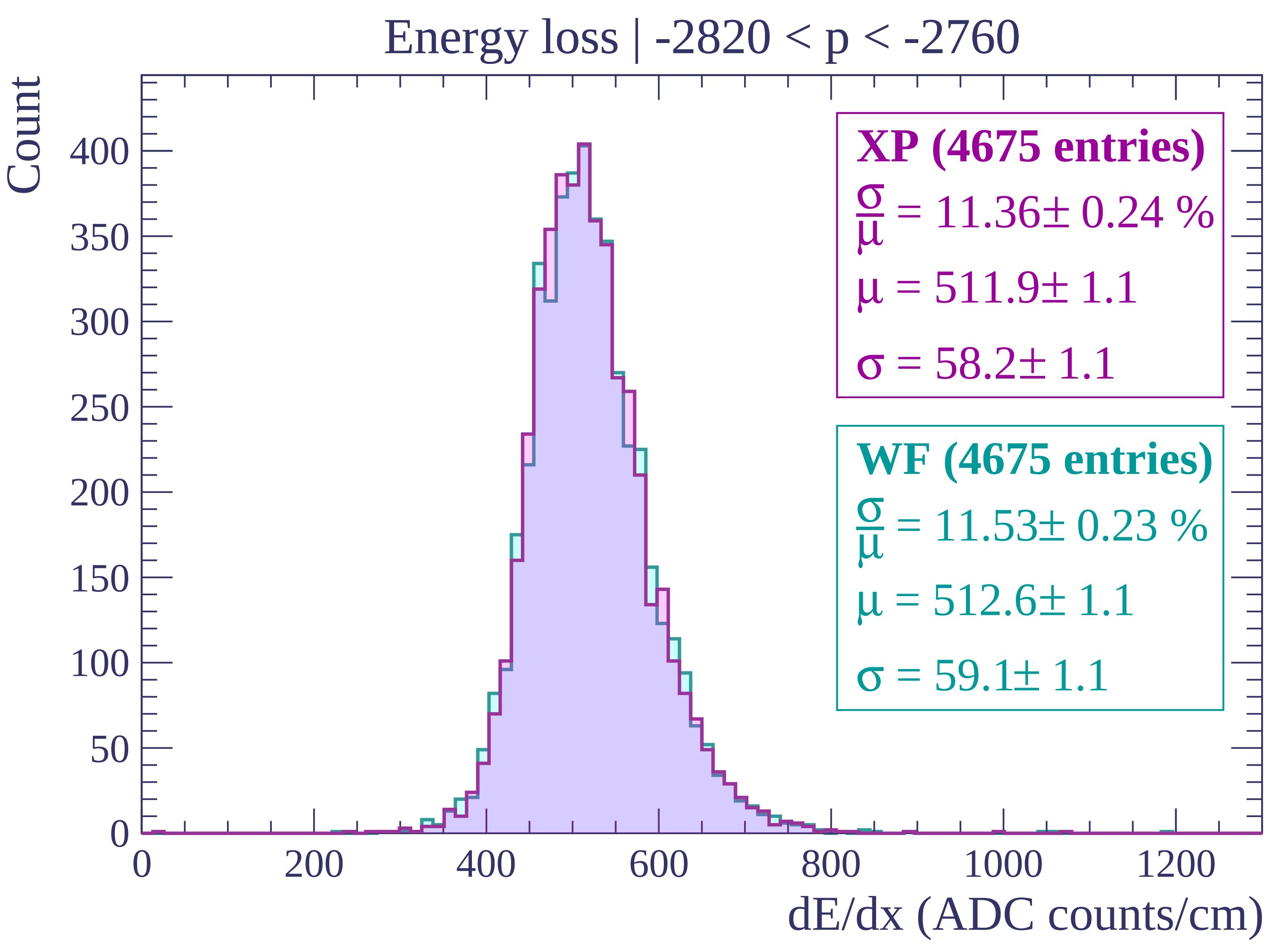
χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
250
200
150
100
50
0
$\times 10^3$


Energy loss | -3000 < p < -2940
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
XP (2168 entries)
σ/μ = 11.32± 0.34 %
μ = 514.0± 1.6
σ = 58.2± 1.6
WF (2168 entries)
σ/μ = 11.70± 0.32 %
μ = 513.0± 1.5
σ = 60.0± 1.5

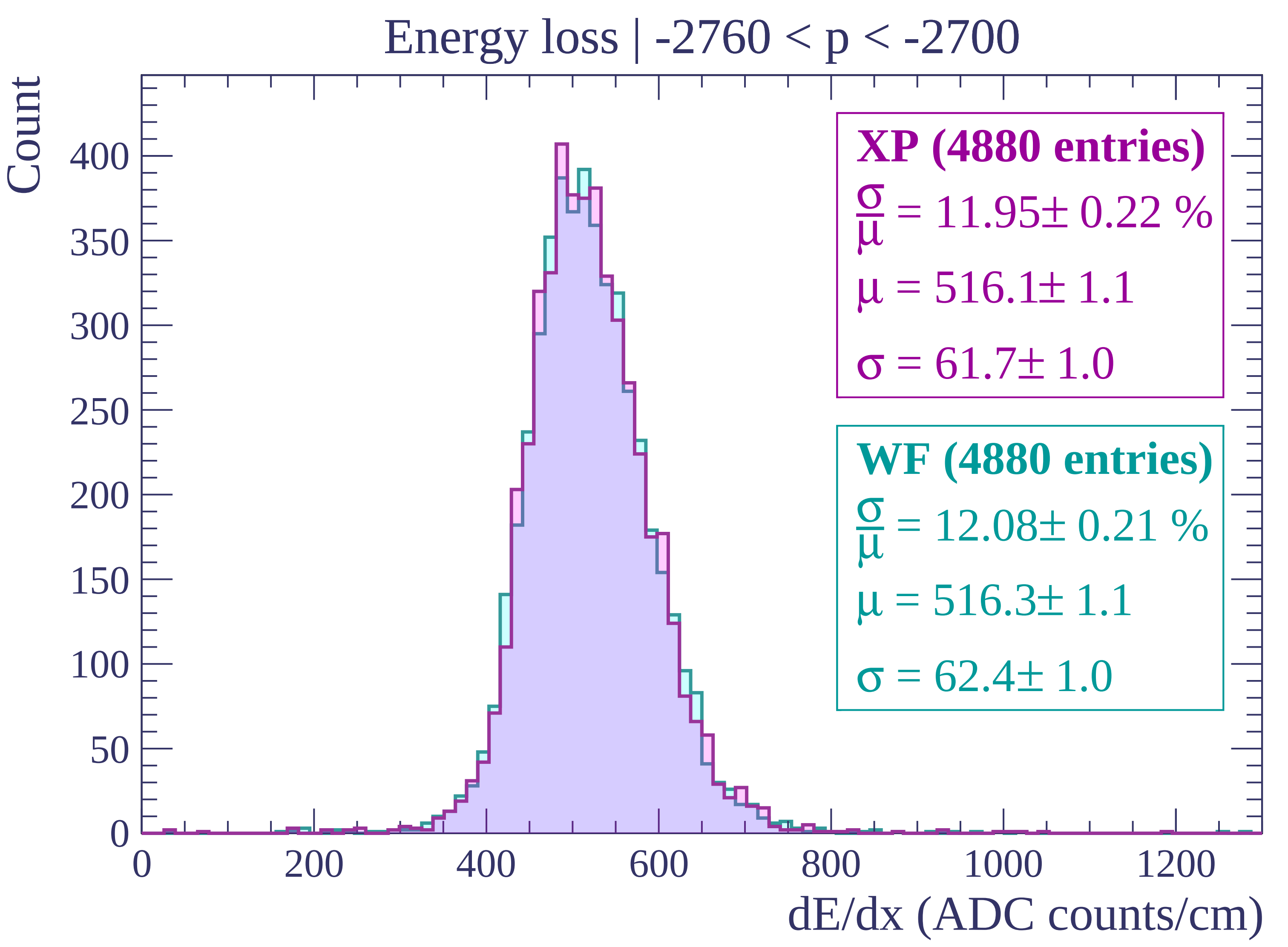

Energy loss | -2940 < p < -2880
Count
dE/dx (ADC counts/cm)
XP (4468 entries)
$\frac{\sigma}{\mu}$ = 11.93± 0.23 %
μ = 516.3± 1.1
σ = 61.6± 1.1
WF (4468 entries)
$\frac{\sigma}{\mu}$ = 12.20± 0.24 %
μ = 516.0± 1.1
σ = 62.9± 1.1
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200



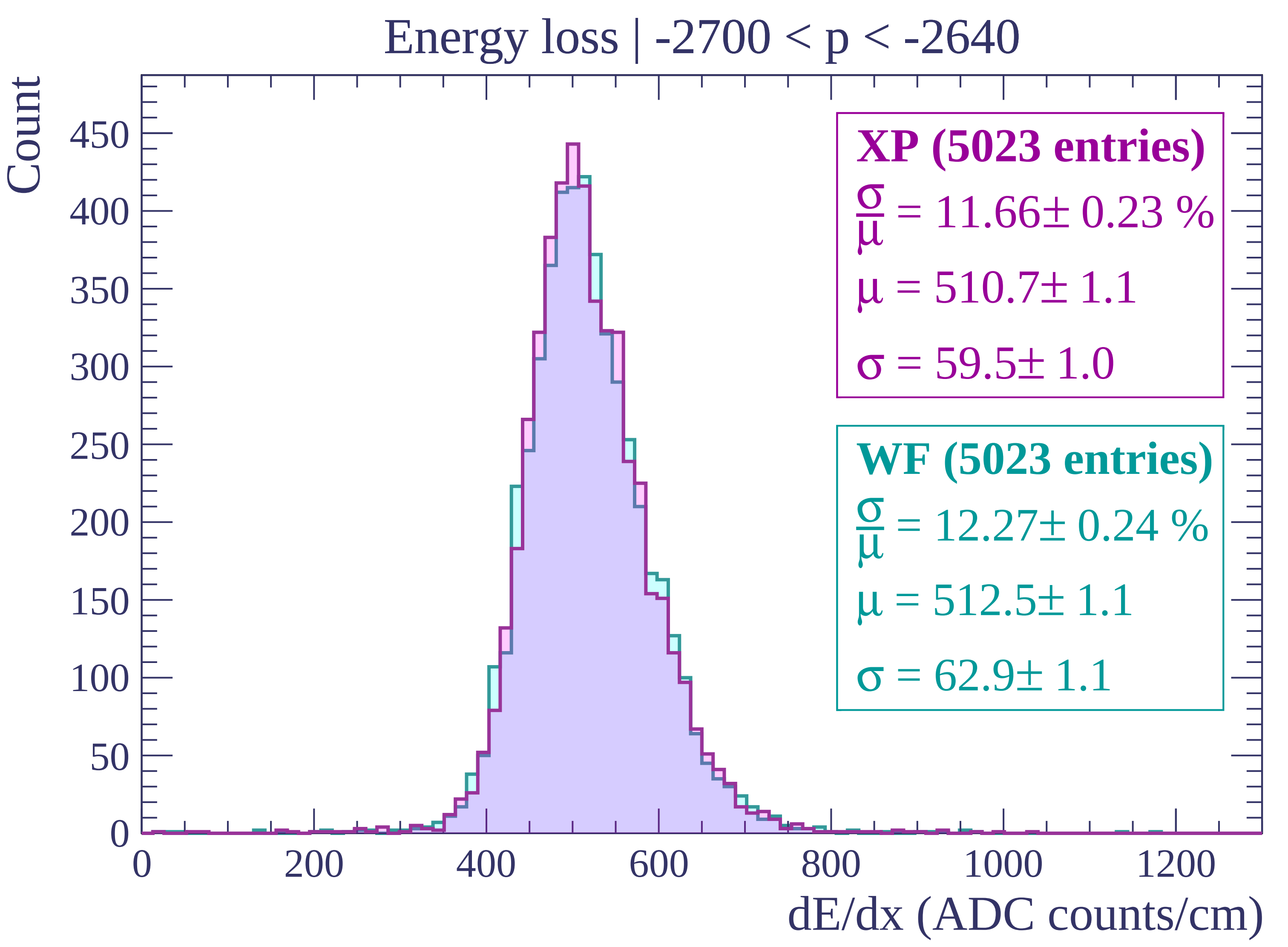
Energy loss | -2880 < p < -2820
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP (4509 entries)
σ/μ = 11.81± 0.22 %
μ = 516.3± 1.1
σ = 61.0± 1.0
WF (4509 entries)
σ/μ = 11.96± 0.22 %
μ = 515.5± 1.1
σ = 61.6± 1.0



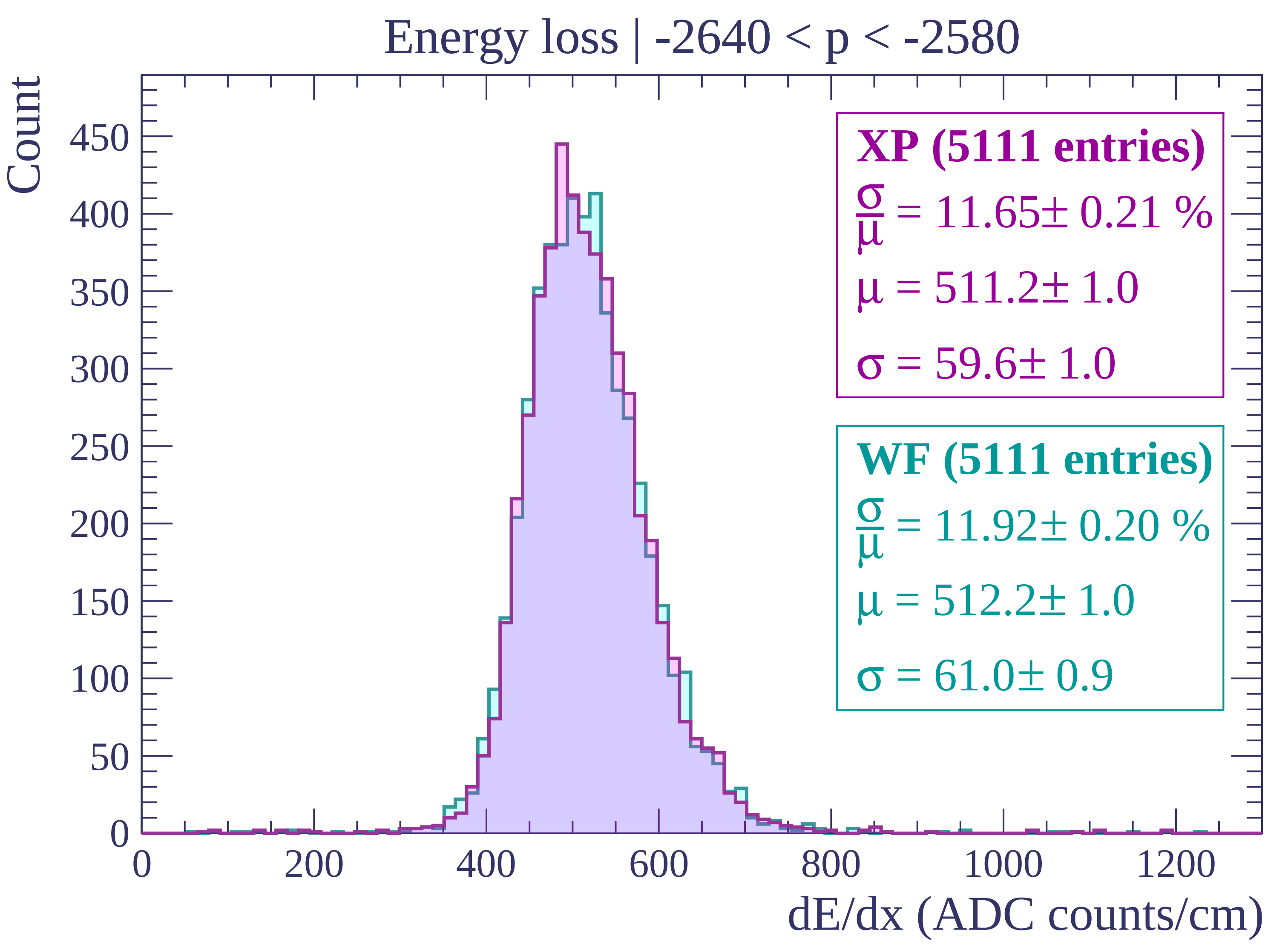
Energy loss | -2820 < p < -2760
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP (4675 entries)
$\frac{\sigma}{\mu}$ = 11.36± 0.24 %
μ = 511.9± 1.1
σ = 58.2± 1.1
WF (4675 entries)
$\frac{\sigma}{\mu}$ = 11.53± 0.23 %
μ = 512.6± 1.1
σ = 59.1± 1.1



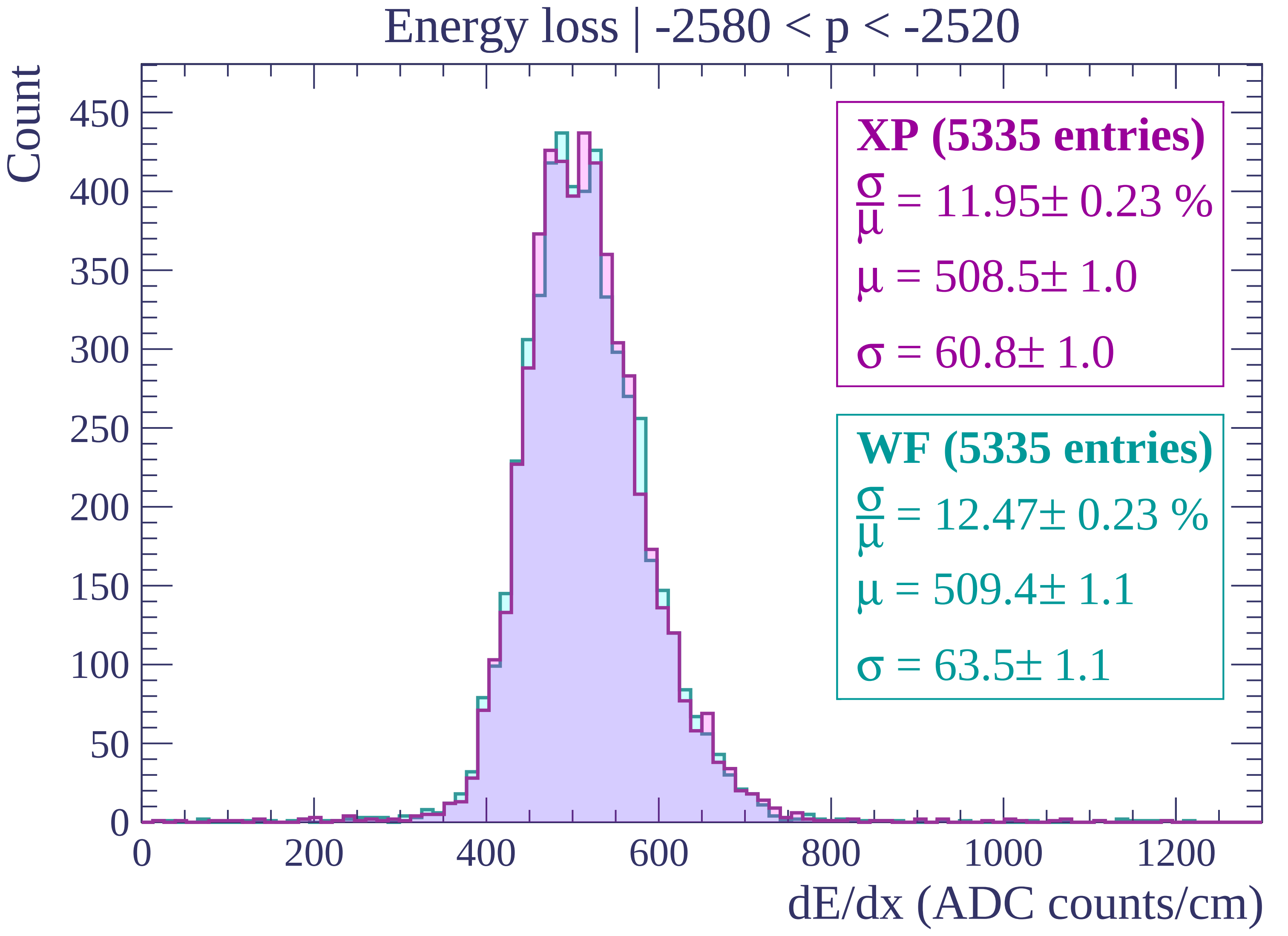
Energy loss | -2760 < p < -2700
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
400
XP (4880 entries)
$\frac{\sigma}{\mu}$ = 11.95± 0.22 %
μ = 516.1± 1.1
σ = 61.7± 1.0
WF (4880 entries)
$\frac{\sigma}{\mu}$ = 12.08± 0.21 %
μ = 516.3± 1.1
σ = 62.4± 1.0



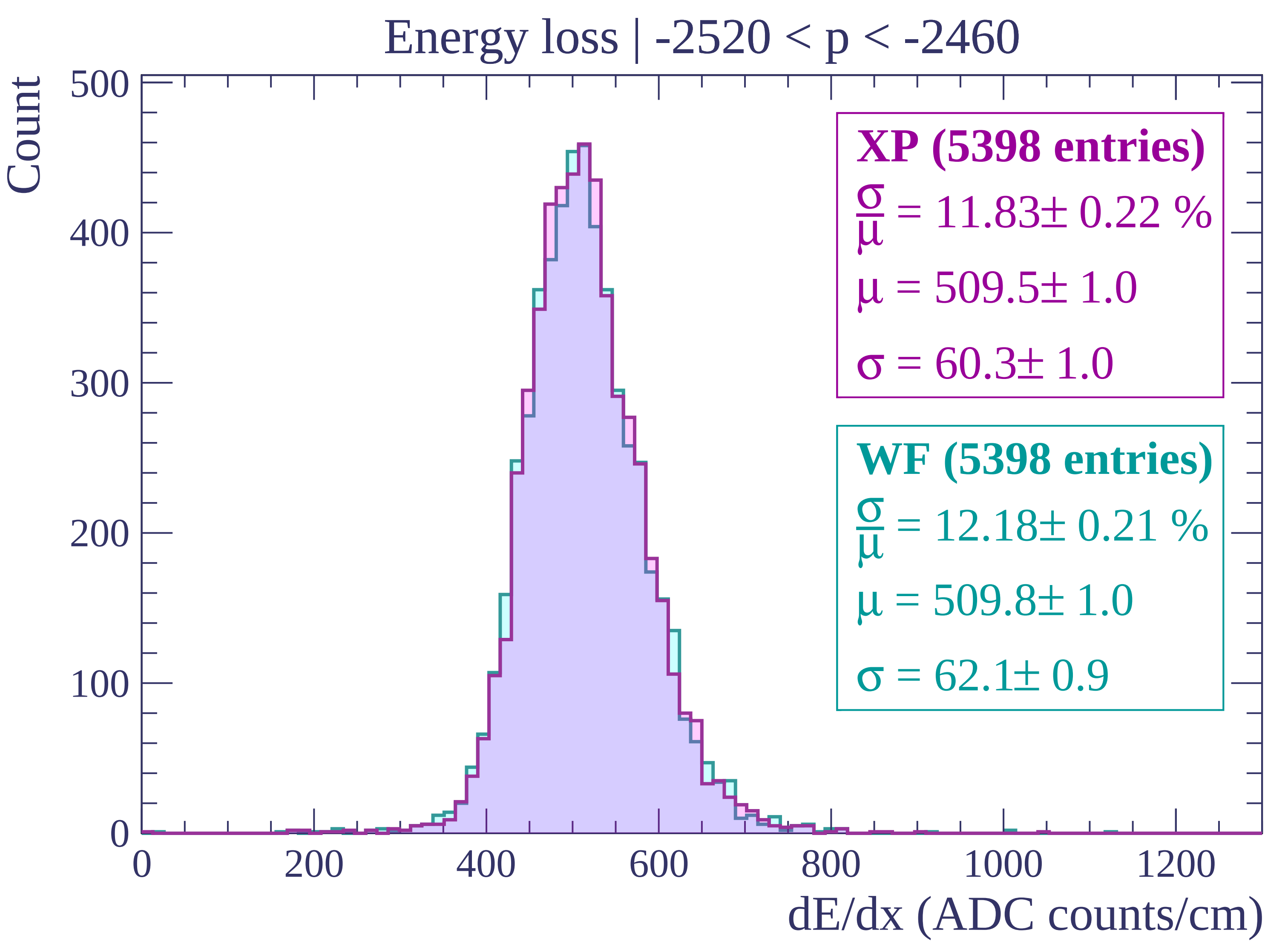
Energy loss | -2700 < p < -2640
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
400
450
XP (5023 entries)
σ/μ = 11.66± 0.23 %
μ = 510.7± 1.1
σ = 59.5± 1.0
WF (5023 entries)
σ/μ = 12.27± 0.24 %
μ = 512.5± 1.1
σ = 62.9± 1.1



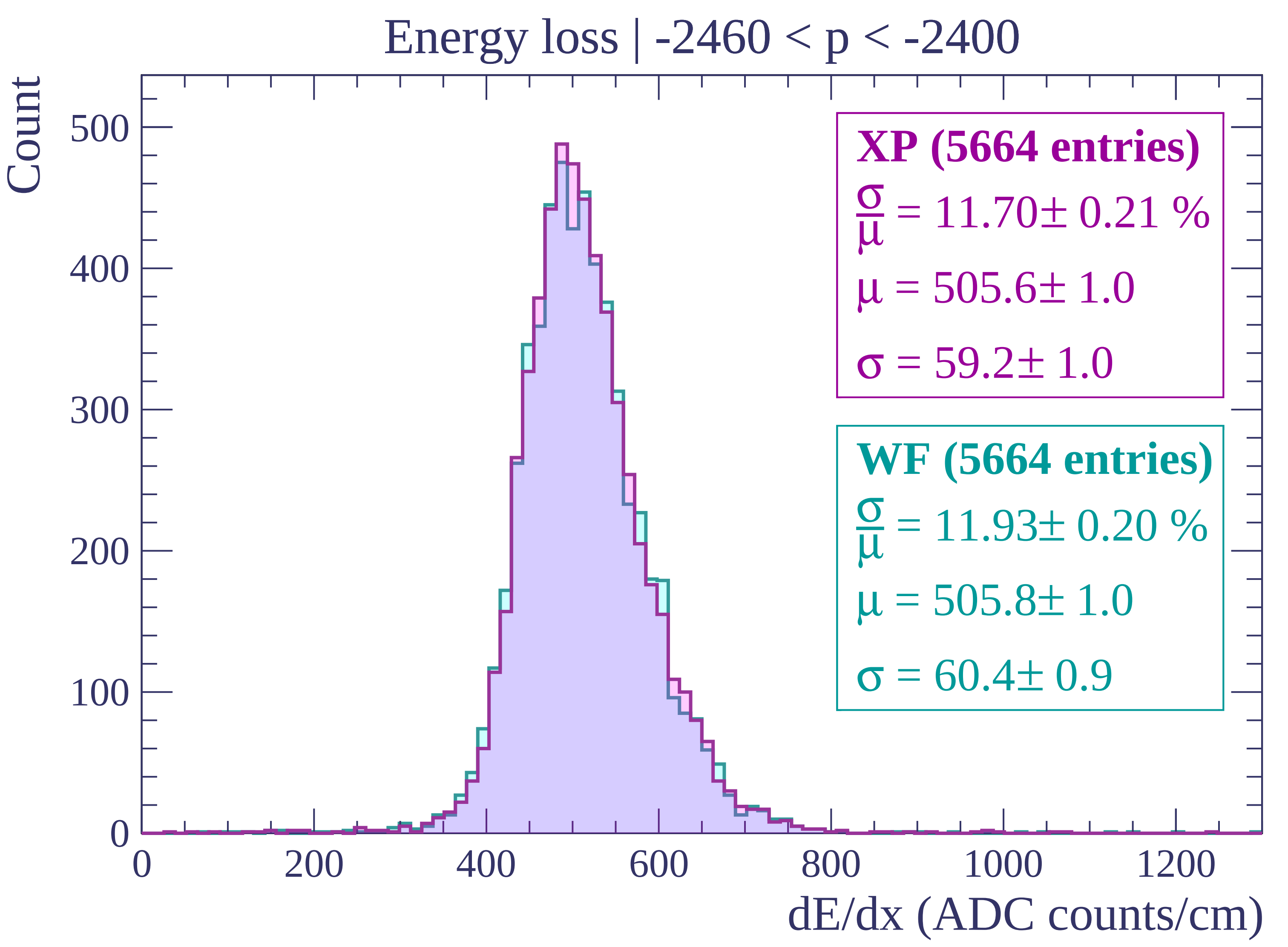
Energy loss | -2640 < p < -2580
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (5111 entries)
σ/μ = 11.65 ± 0.21 %
μ = 511.2 ± 1.0
σ = 59.6 ± 1.0
WF (5111 entries)
σ/μ = 11.92 ± 0.20 %
μ = 512.2 ± 1.0
σ = 61.0 ± 0.9

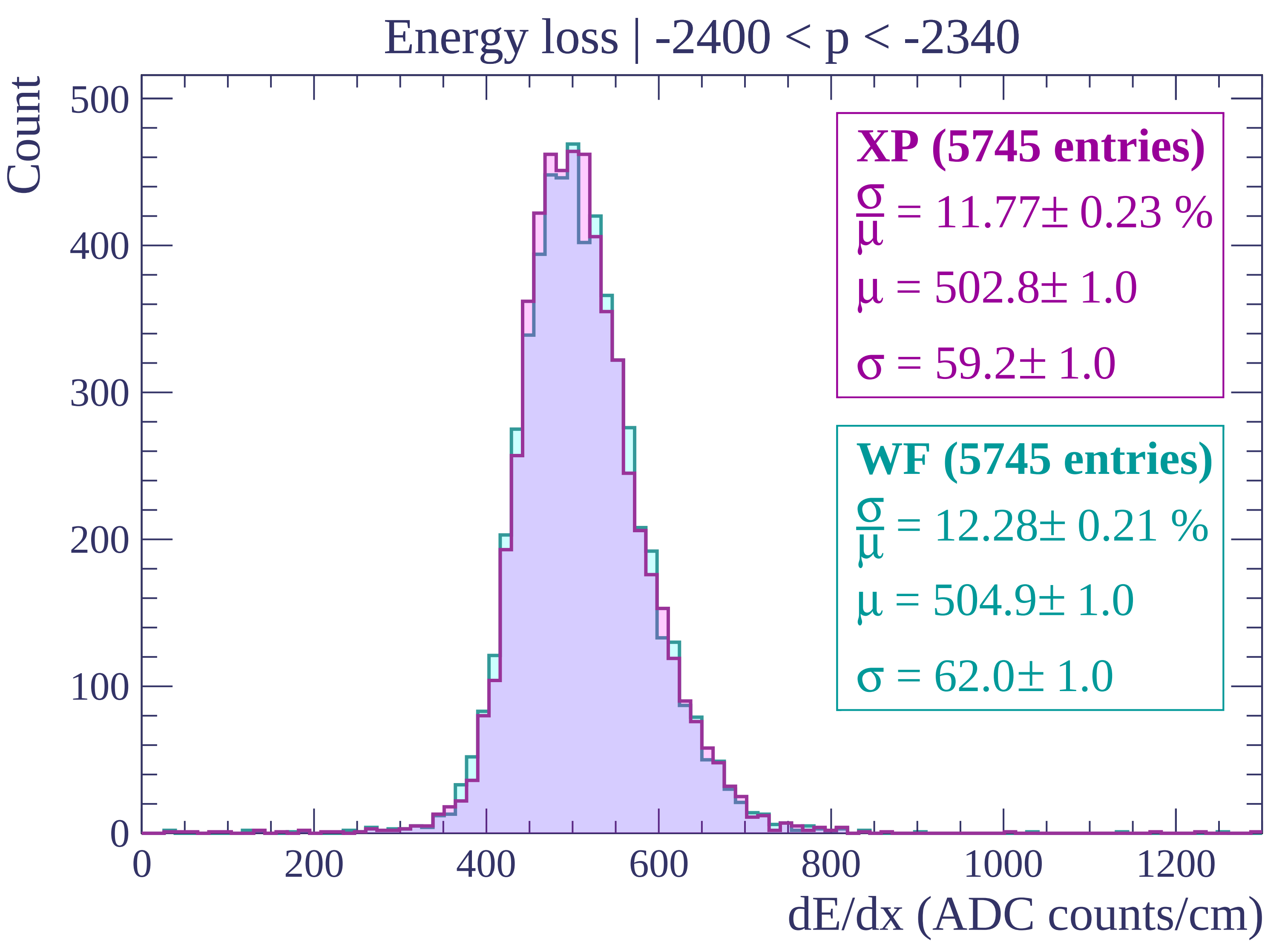


Energy loss | -2580 < p < -2520
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (5335 entries)
σ/μ = 11.95± 0.23 %
μ = 508.5± 1.0
σ = 60.8± 1.0
WF (5335 entries)
σ/μ = 12.47± 0.23 %
μ = 509.4± 1.1
σ = 63.5± 1.1

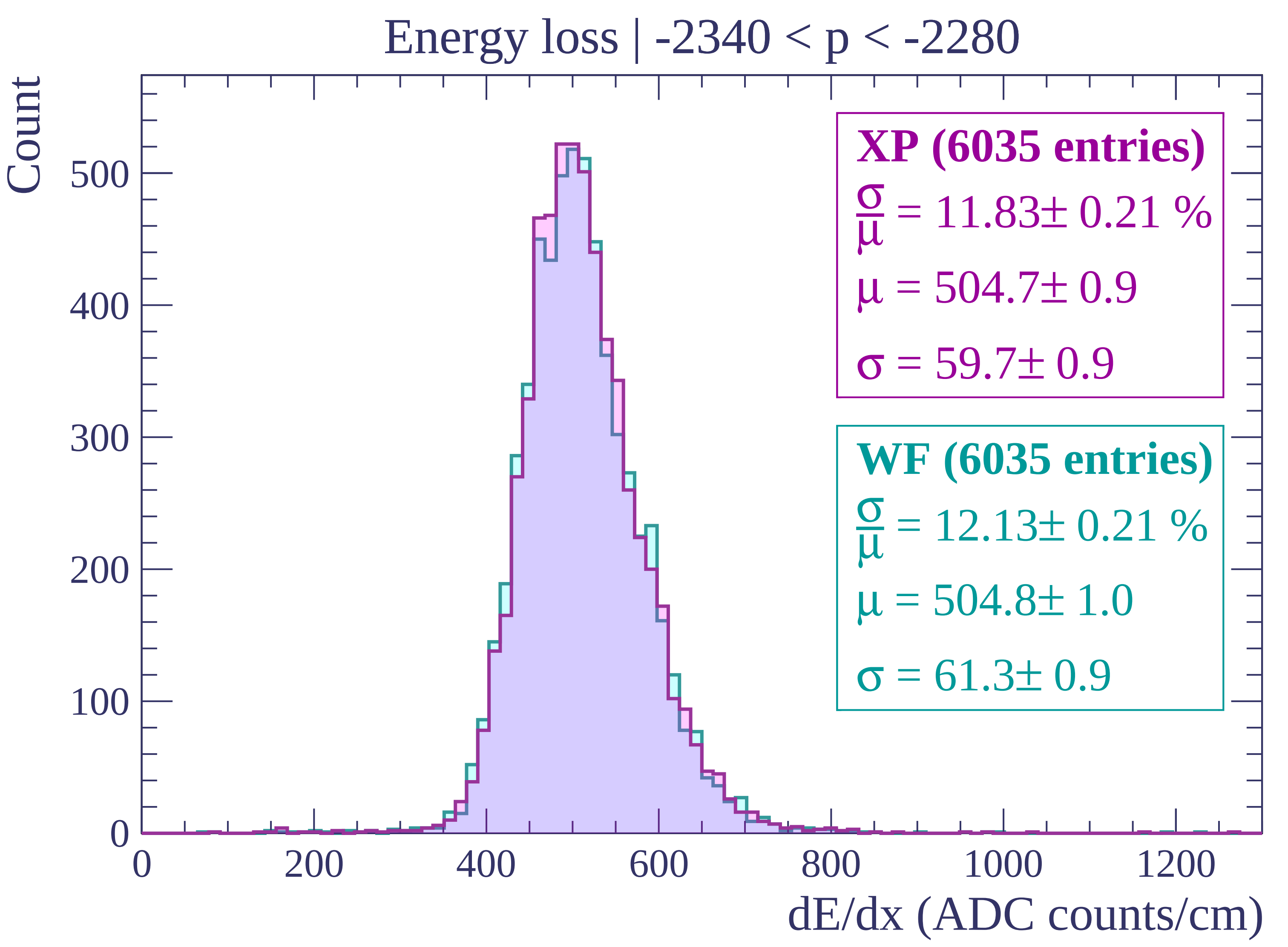


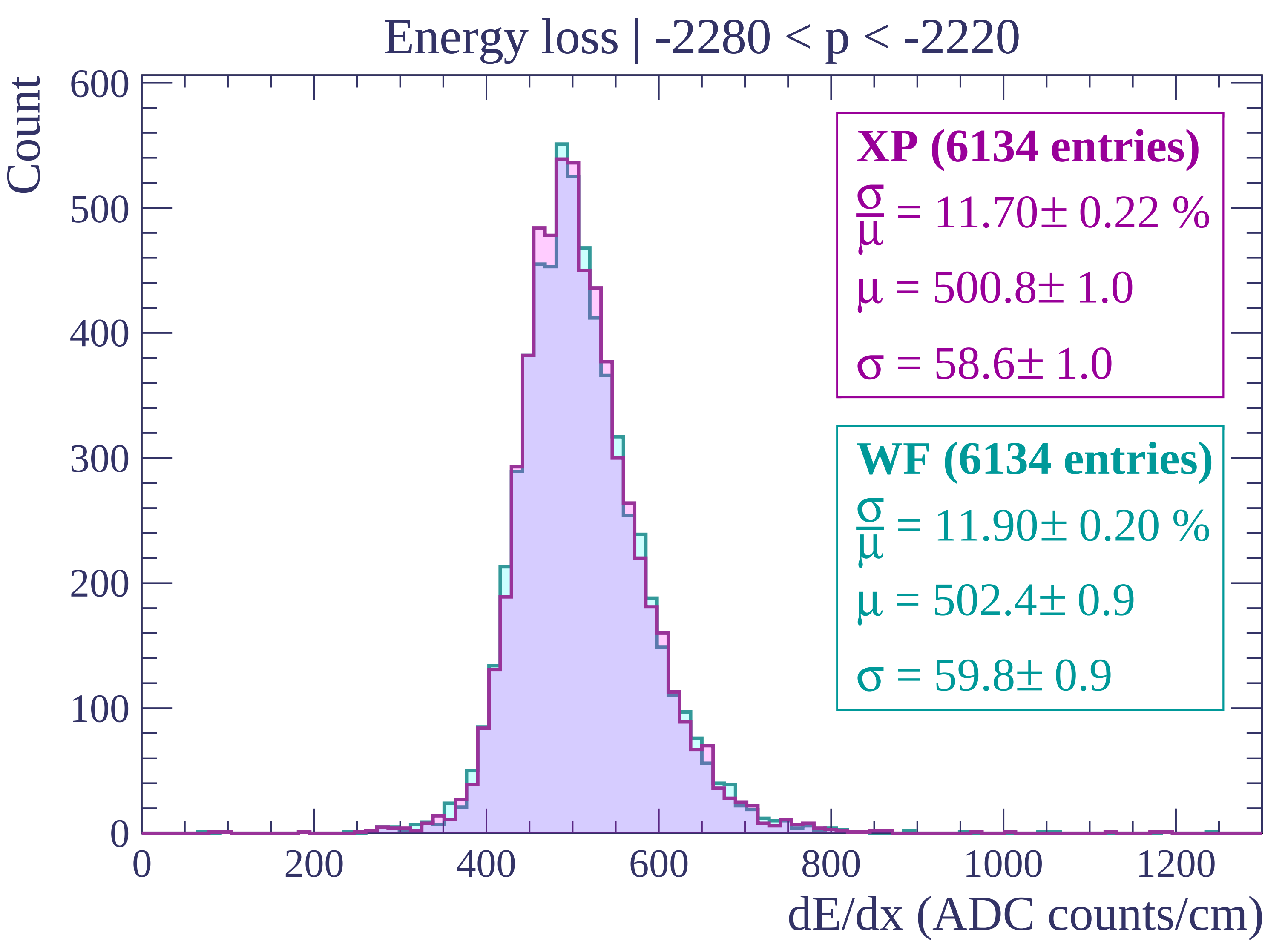
Energy loss | -2520 < p < -2460
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (5398 entries)
σ/μ = 11.83± 0.22 %
μ = 509.5± 1.0
σ = 60.3± 1.0
WF (5398 entries)
σ/μ = 12.18± 0.21 %
μ = 509.8± 1.0
σ = 62.1± 0.9


Energy loss | -2460 < p < -2400
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (5664 entries)
$\frac{\sigma}{\mu}$ = 11.70± 0.21 %
μ = 505.6± 1.0
σ = 59.2± 1.0
WF (5664 entries)
$\frac{\sigma}{\mu}$ = 11.93± 0.20 %
μ = 505.8± 1.0
σ = 60.4± 0.9

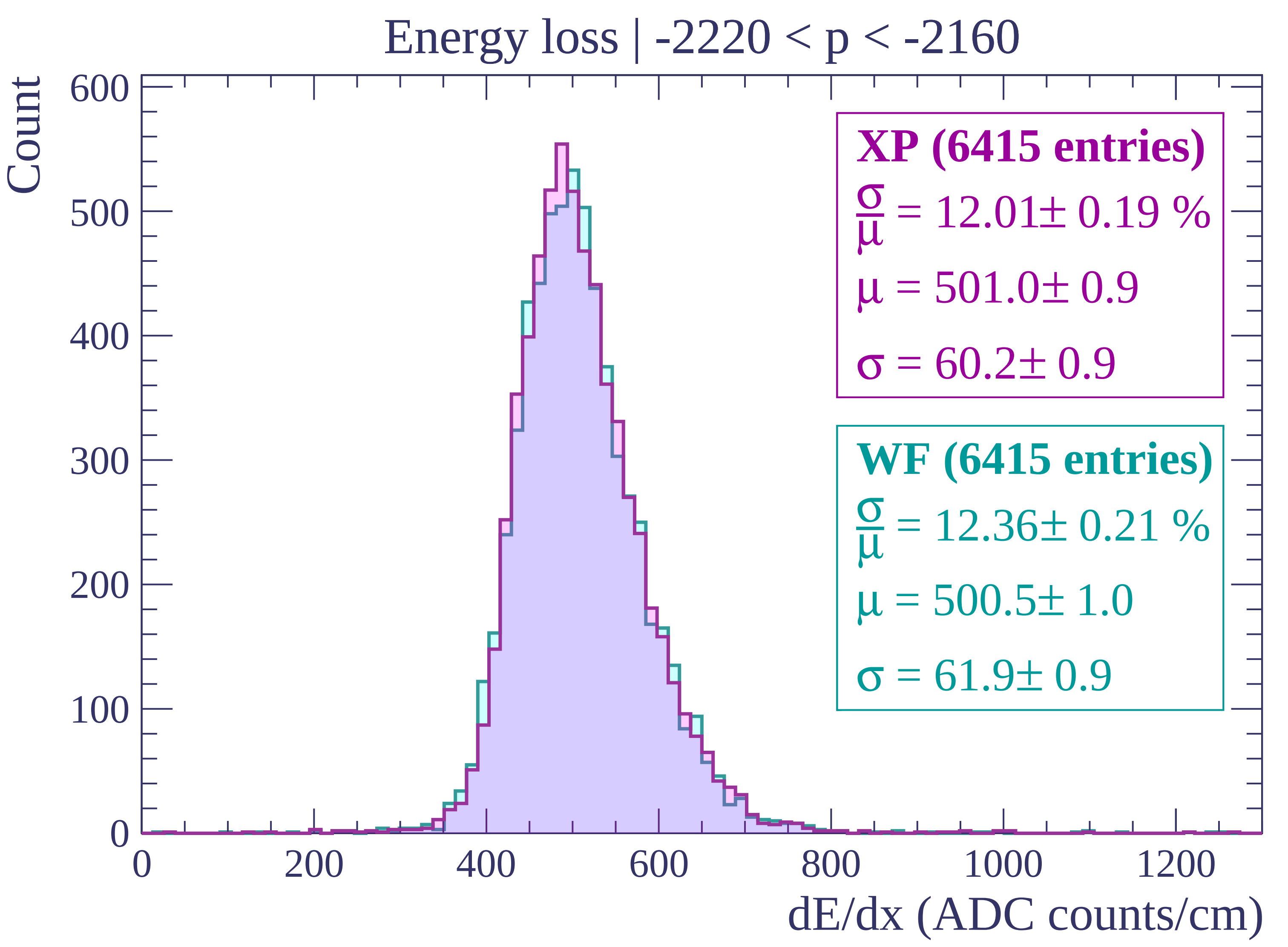


Energy loss | -2400 < p < -2340
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (5745 entries)
σ/μ = 11.77± 0.23 %
μ = 502.8± 1.0
σ = 59.2± 1.0
WF (5745 entries)
σ/μ = 12.28± 0.21 %
μ = 504.9± 1.0
σ = 62.0± 1.0

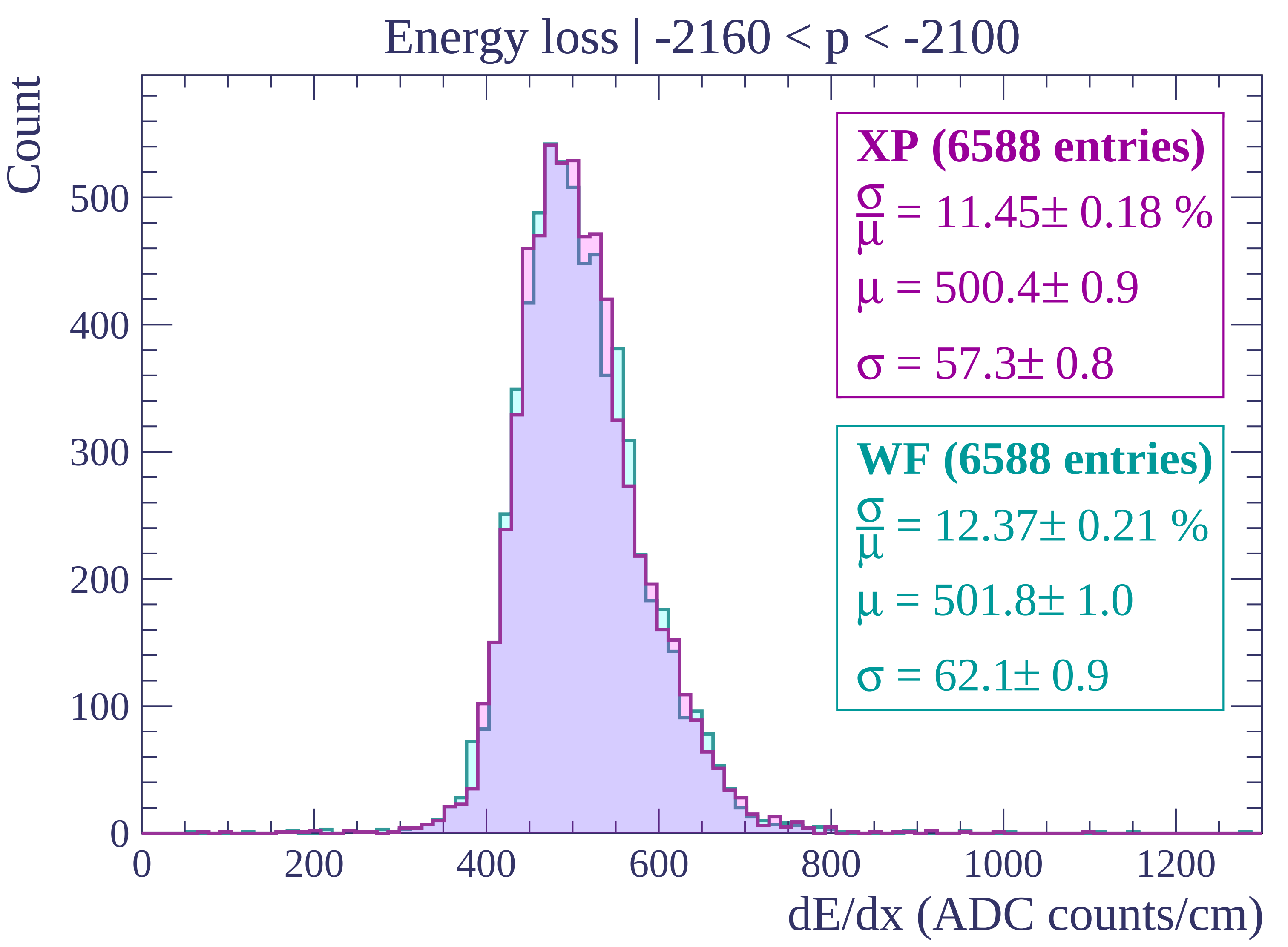

Energy loss | -2340 < p < -2280
Count
dE/dx (ADC counts/cm)
XP (6035 entries)
σ/μ = 11.83± 0.21 %
μ = 504.7± 0.9
σ = 59.7± 0.9
WF (6035 entries)
σ/μ = 12.13± 0.21 %
μ = 504.8± 1.0
σ = 61.3± 0.9

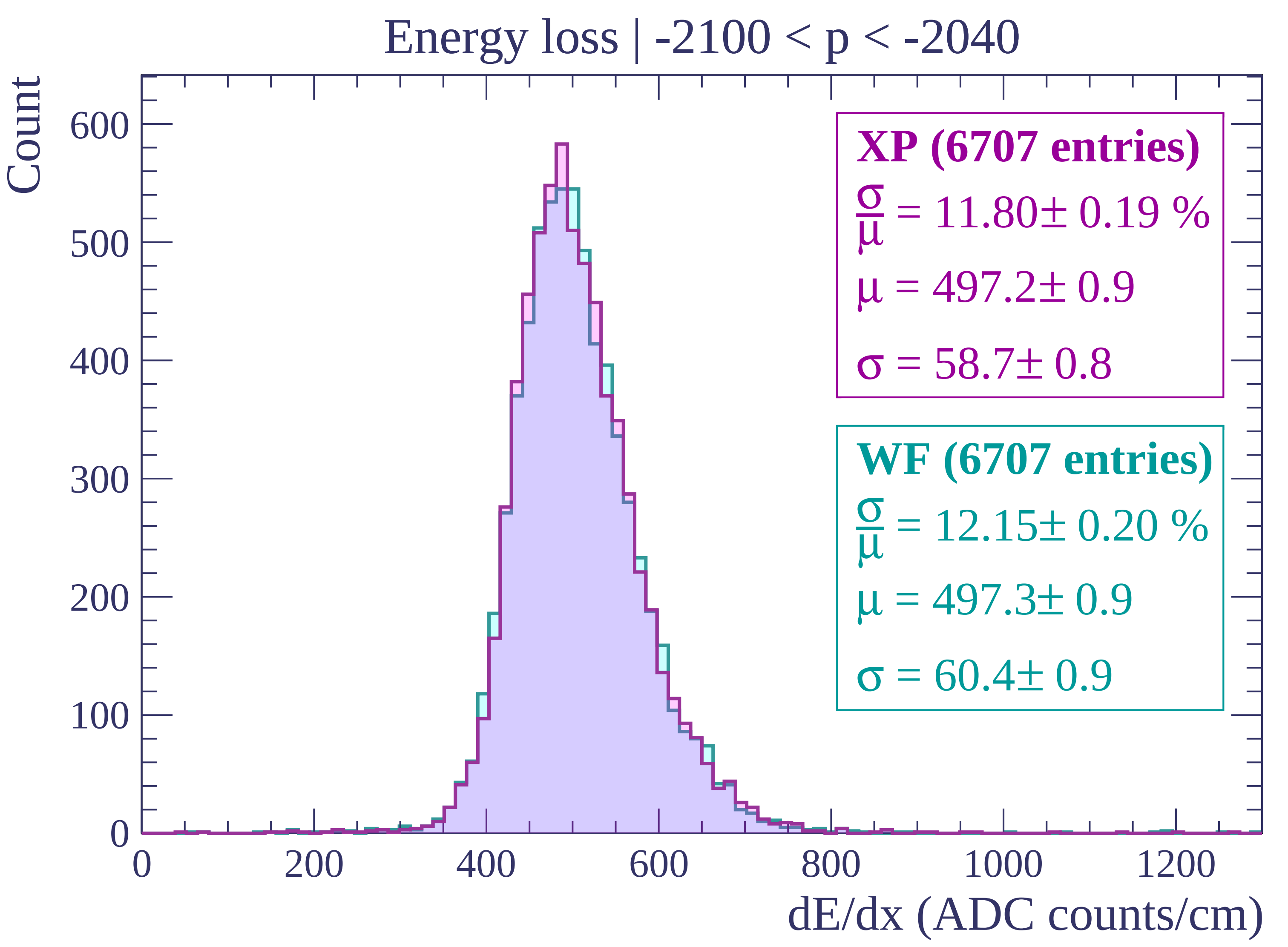


Energy loss | -2280 < p < -2220
Count
dE/dx (ADC counts/cm)
XP (6134 entries)
σ/μ = 11.70± 0.22 %
μ = 500.8± 1.0
σ = 58.6± 1.0
WF (6134 entries)
σ/μ = 11.90± 0.20 %
μ = 502.4± 0.9
σ = 59.8± 0.9
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200

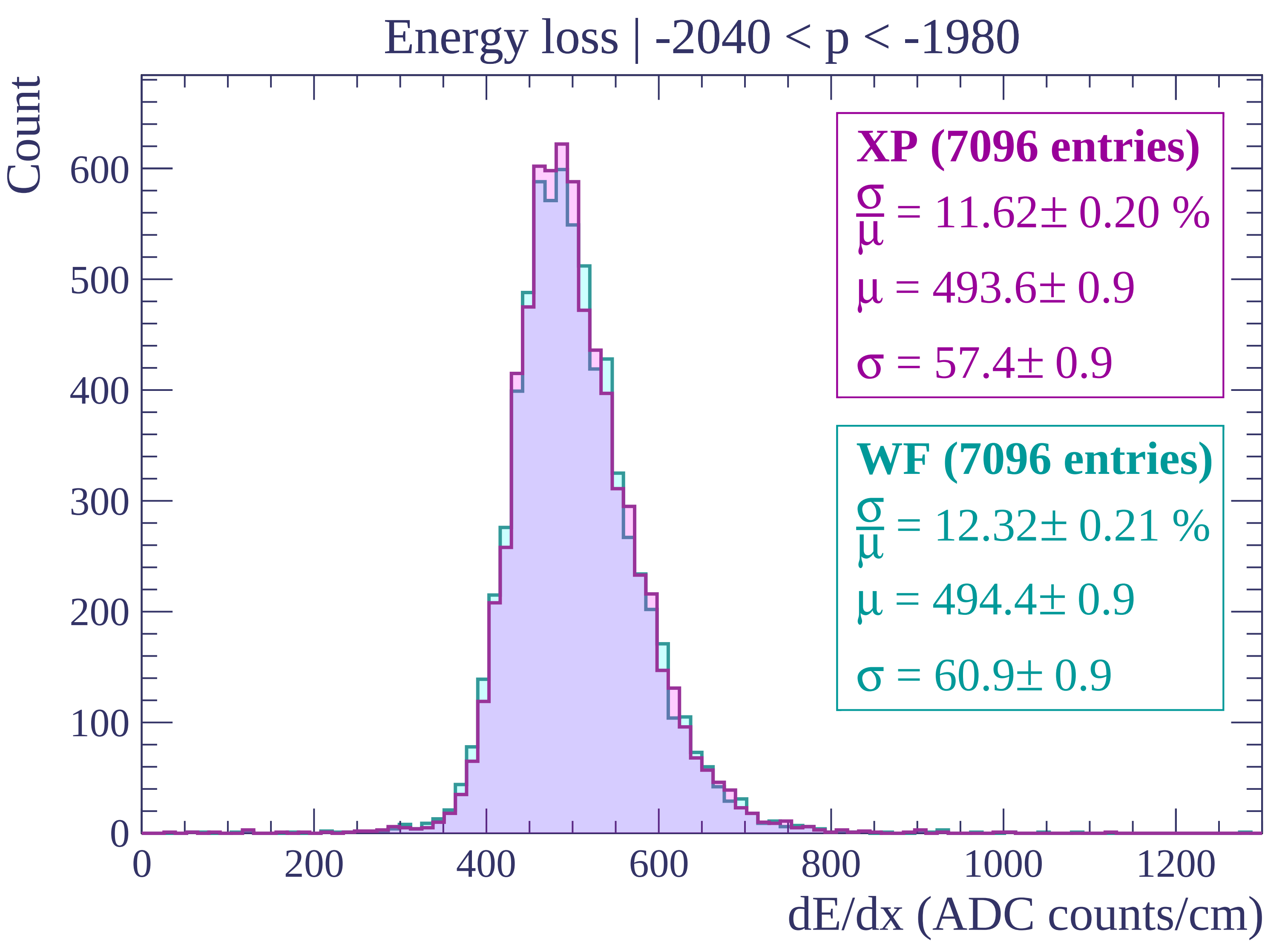


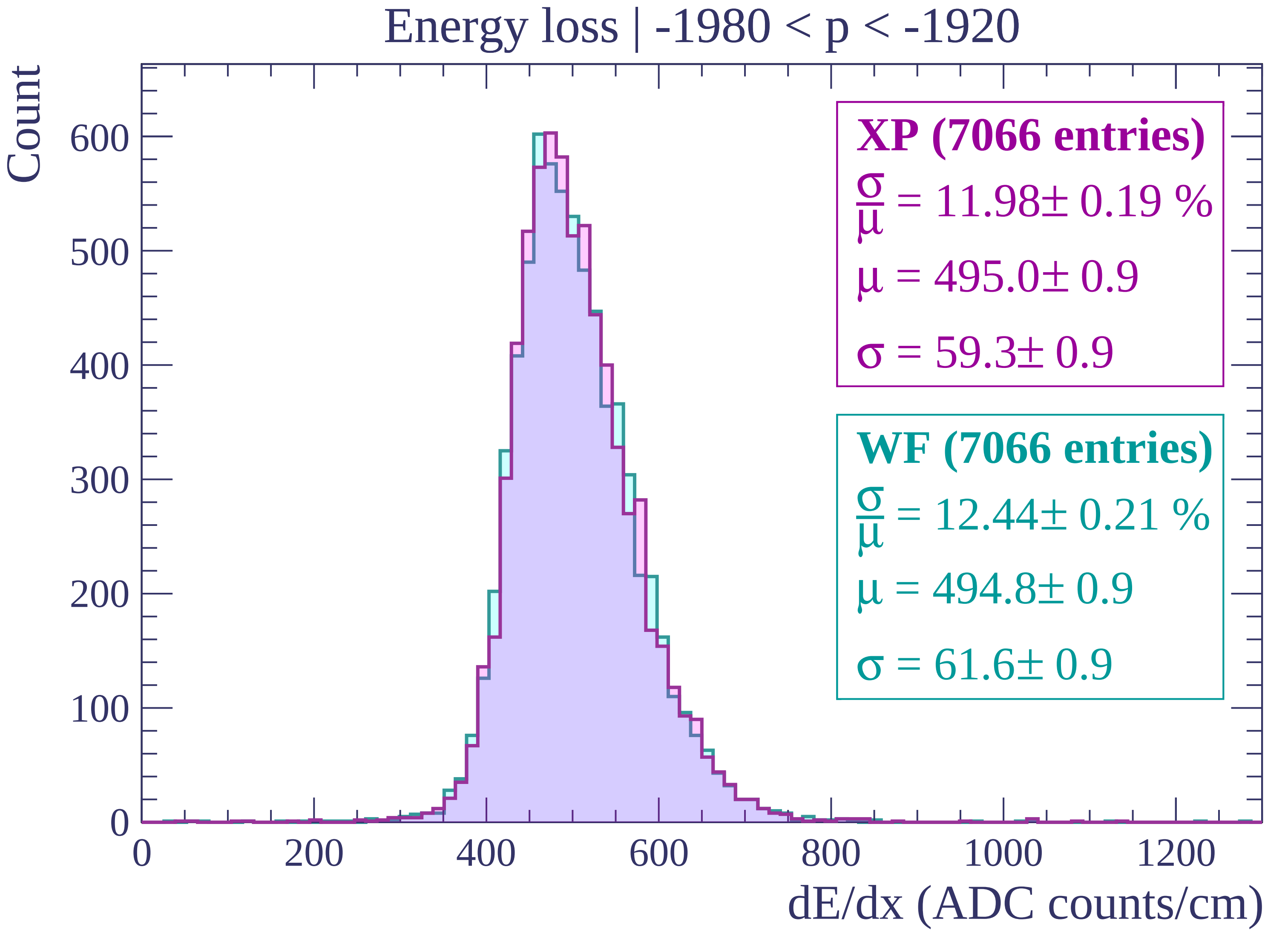
Energy loss | -2220 < p < -2160
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6415 entries)
σ/μ = 12.01± 0.19 %
μ = 501.0± 0.9
σ = 60.2± 0.9
WF (6415 entries)
σ/μ = 12.36± 0.21 %
μ = 500.5± 1.0
σ = 61.9± 0.9

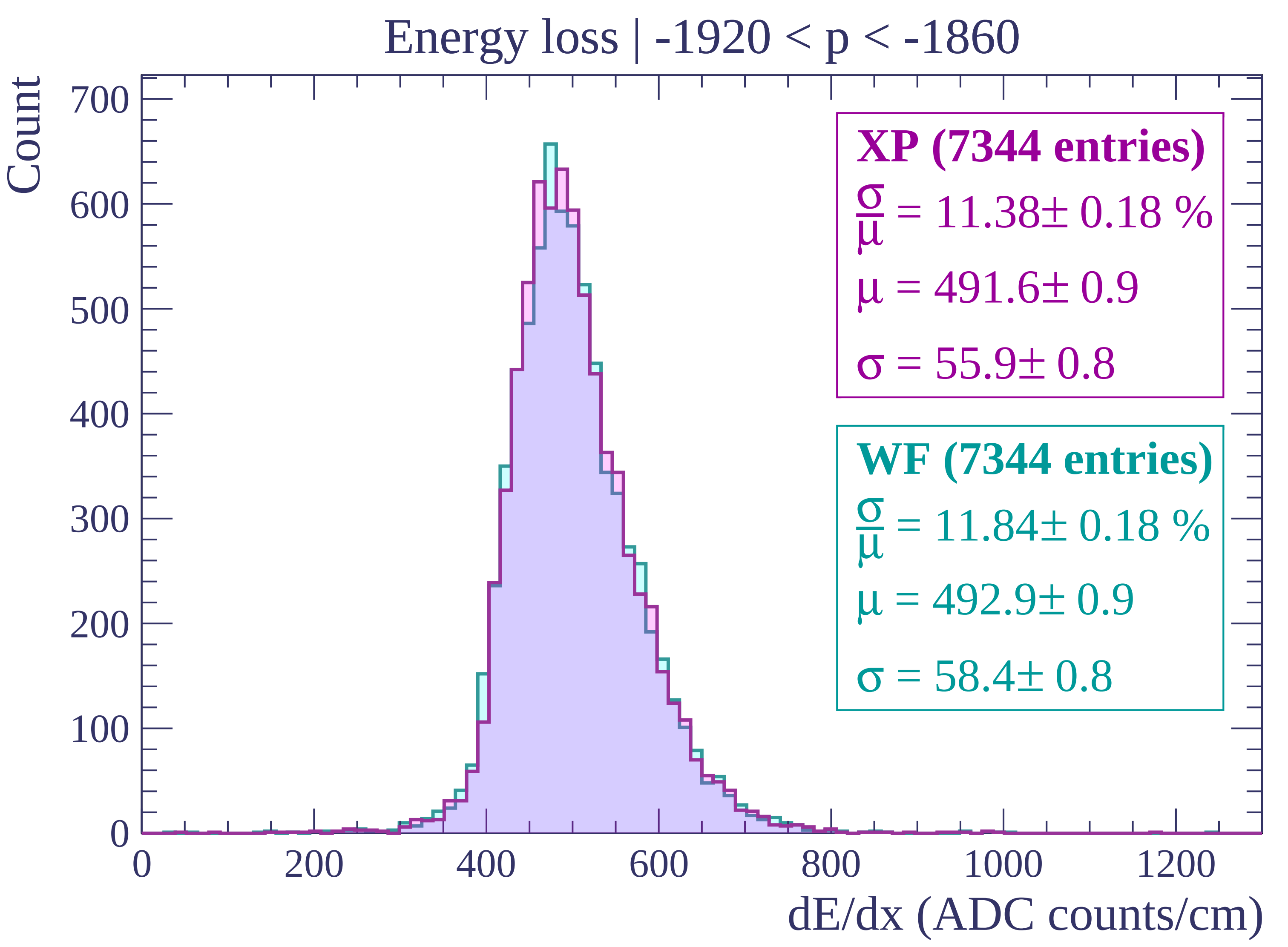

Energy loss | -2160 < p < -2100
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6588 entries)
σ/μ = 11.45± 0.18 %
μ = 500.4± 0.9
σ = 57.3± 0.8
WF (6588 entries)
σ/μ = 12.37± 0.21 %
μ = 501.8± 1.0
σ = 62.1± 0.9



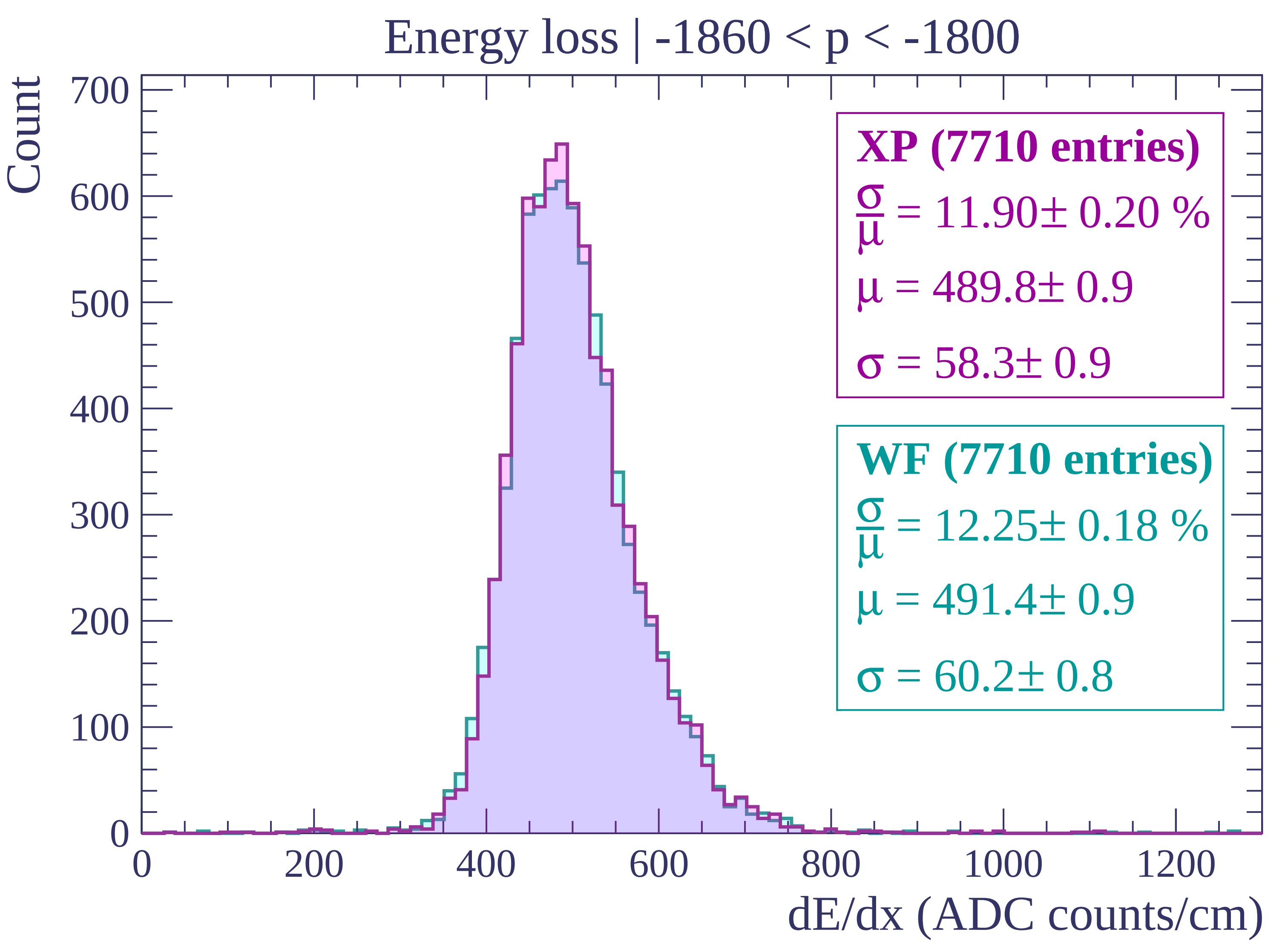
Energy loss | -2100 < p < -2040
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6707 entries)
$\frac{\sigma}{\mu}$ = 11.80± 0.19 %
μ = 497.2± 0.9
σ = 58.7± 0.8
WF (6707 entries)
$\frac{\sigma}{\mu}$ = 12.15± 0.20 %
μ = 497.3± 0.9
σ = 60.4± 0.9

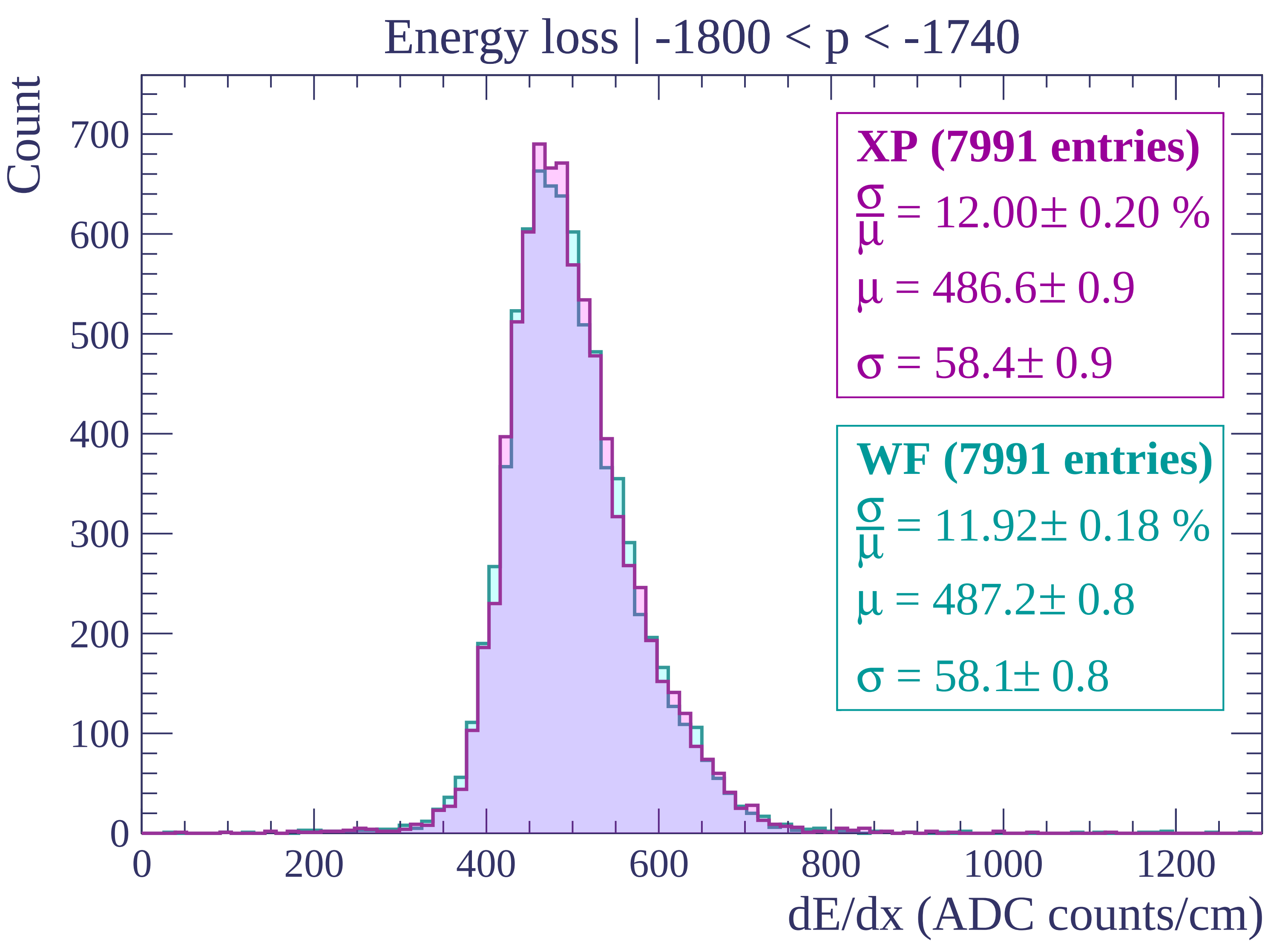

Energy loss | -2040 < p < -1980
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (7096 entries)
σ/μ = 11.62± 0.20 %
μ = 493.6± 0.9
σ = 57.4± 0.9
WF (7096 entries)
σ/μ = 12.32± 0.21 %
μ = 494.4± 0.9
σ = 60.9± 0.9


Energy loss | -1980 < p < -1920
Count
dE/dx (ADC counts/cm)
XP (7066 entries)
σ/μ = 11.98± 0.19 %
μ = 495.0± 0.9
σ = 59.3± 0.9
WF (7066 entries)
σ/μ = 12.44± 0.21 %
μ = 494.8± 0.9
σ = 61.6± 0.9
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200

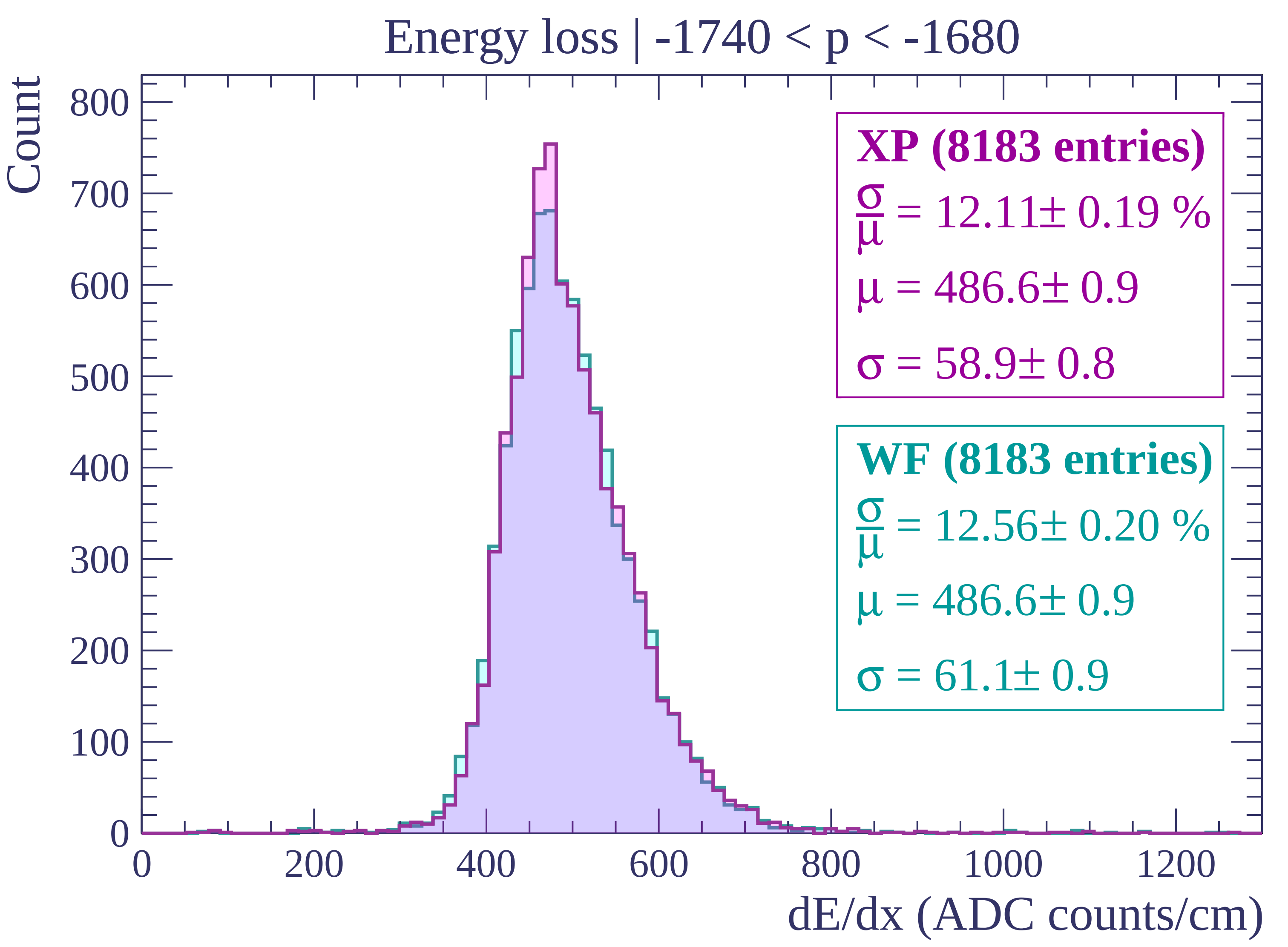


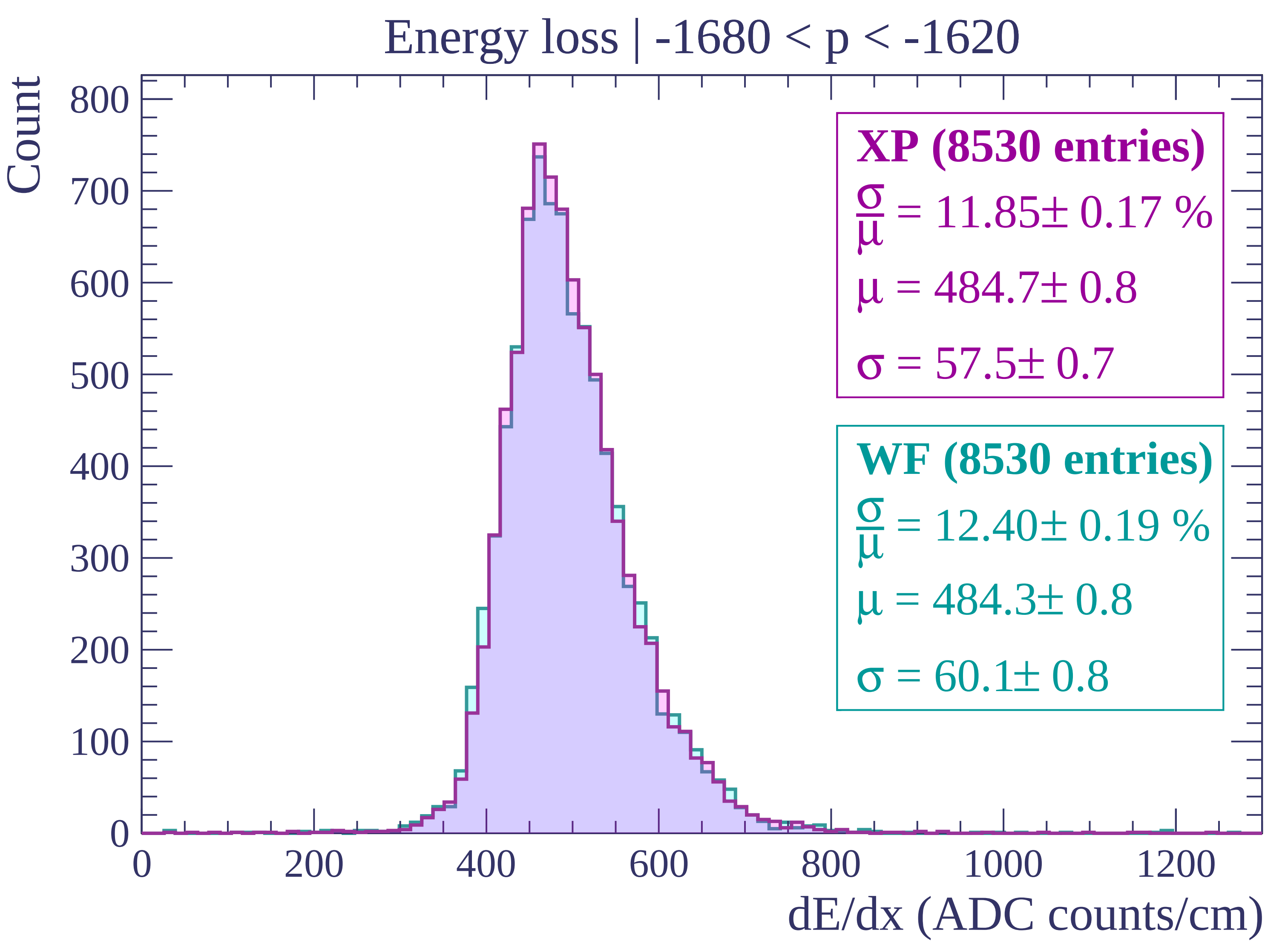
Energy loss | -1920 < p < -1860
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (7344 entries)
$\frac{\sigma}{\mu}$ = 11.38± 0.18 %
μ = 491.6± 0.9
σ = 55.9± 0.8
WF (7344 entries)
$\frac{\sigma}{\mu}$ = 11.84± 0.18 %
μ = 492.9± 0.9
σ = 58.4± 0.8


Energy loss | -1860 < p < -1800
Count
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (7710 entries)
σ/μ = 11.90± 0.20 %
μ = 489.8± 0.9
σ = 58.3± 0.9
WF (7710 entries)
σ/μ = 12.25± 0.18 %
μ = 491.4± 0.9
σ = 60.2± 0.8

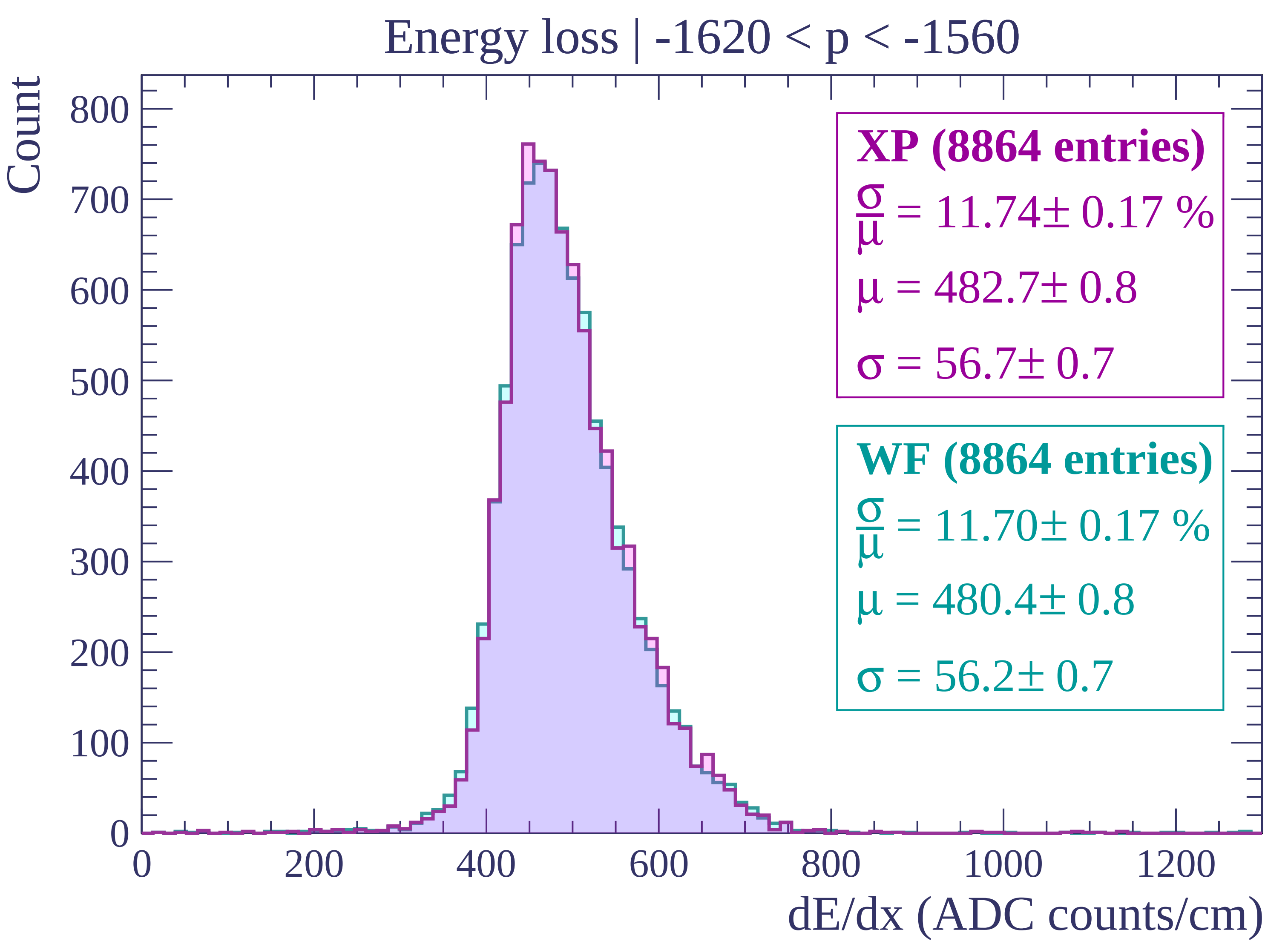


Energy loss | -1800 < p < -1740
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (7991 entries)
$\frac{\sigma}{\mu}$ = 12.00± 0.20 %
μ = 486.6± 0.9
σ = 58.4± 0.9
WF (7991 entries)
$\frac{\sigma}{\mu}$ = 11.92± 0.18 %
μ = 487.2± 0.8
σ = 58.1± 0.8

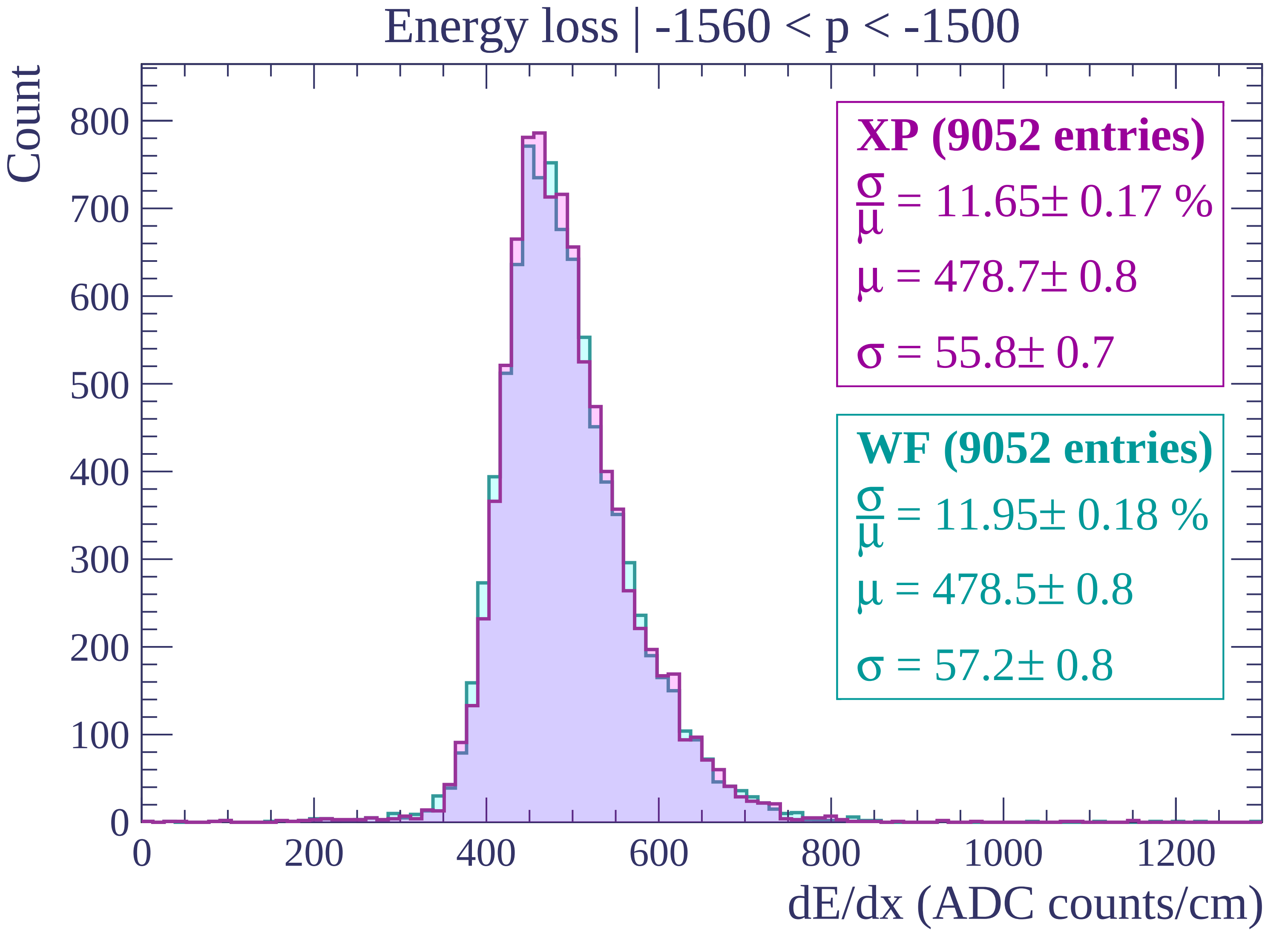

Energy loss | -1740 < p < -1680
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
700
XP (8183 entries)
$\frac{\sigma}{\mu}$ = 12.11± 0.19 %
μ = 486.6± 0.9
σ = 58.9± 0.8
WF (8183 entries)
$\frac{\sigma}{\mu}$ = 12.56± 0.20 %
μ = 486.6± 0.9
σ = 61.1± 0.9



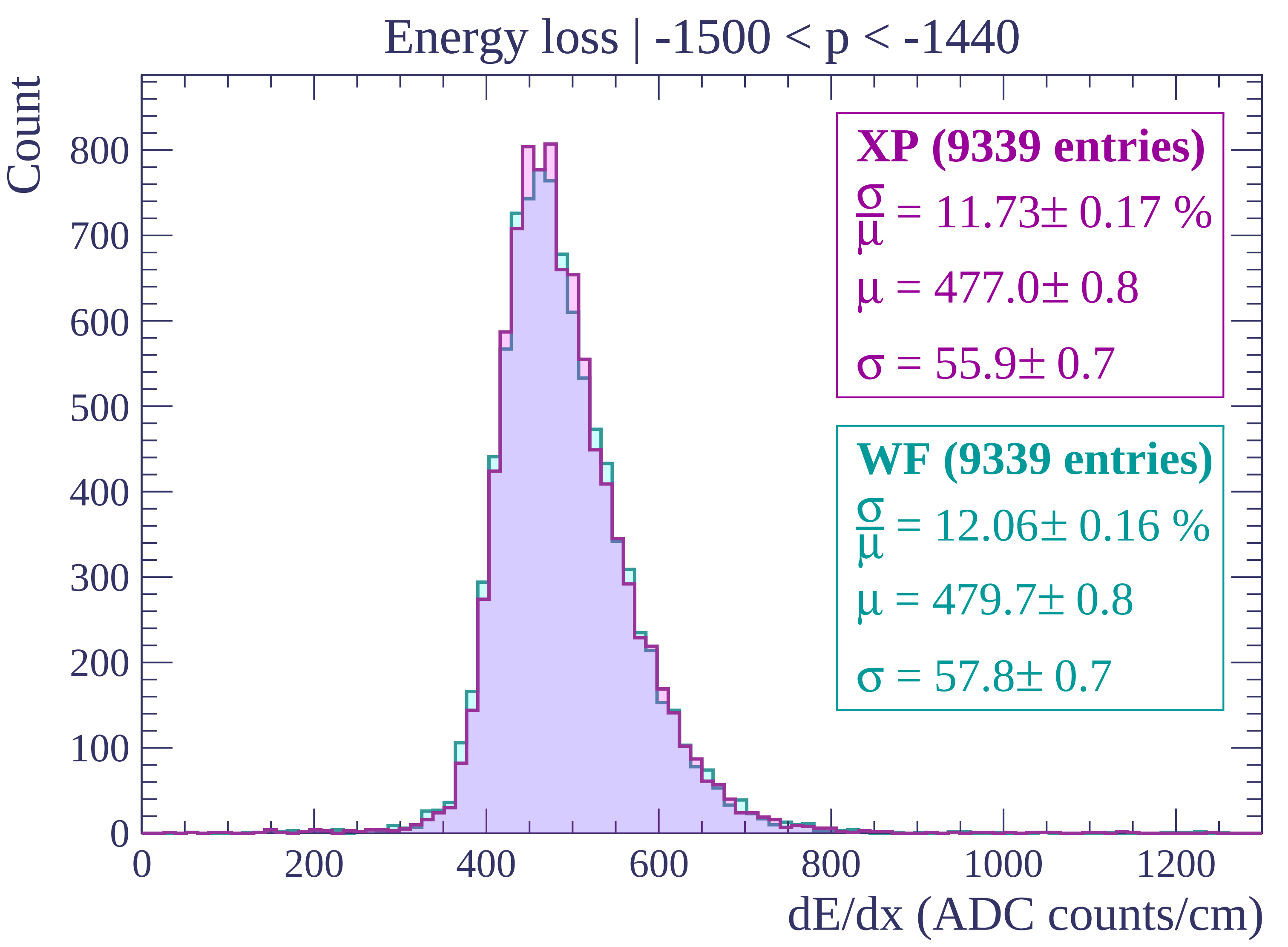
Energy loss | -1680 < p < -1620
Count
dE/dx (ADC counts/cm)
XP (8530 entries)
$\frac{\sigma}{\mu}$ = 11.85± 0.17 %
μ = 484.7± 0.8
σ = 57.5± 0.7
WF (8530 entries)
$\frac{\sigma}{\mu}$ = 12.40± 0.19 %
μ = 484.3± 0.8
σ = 60.1± 0.8
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200



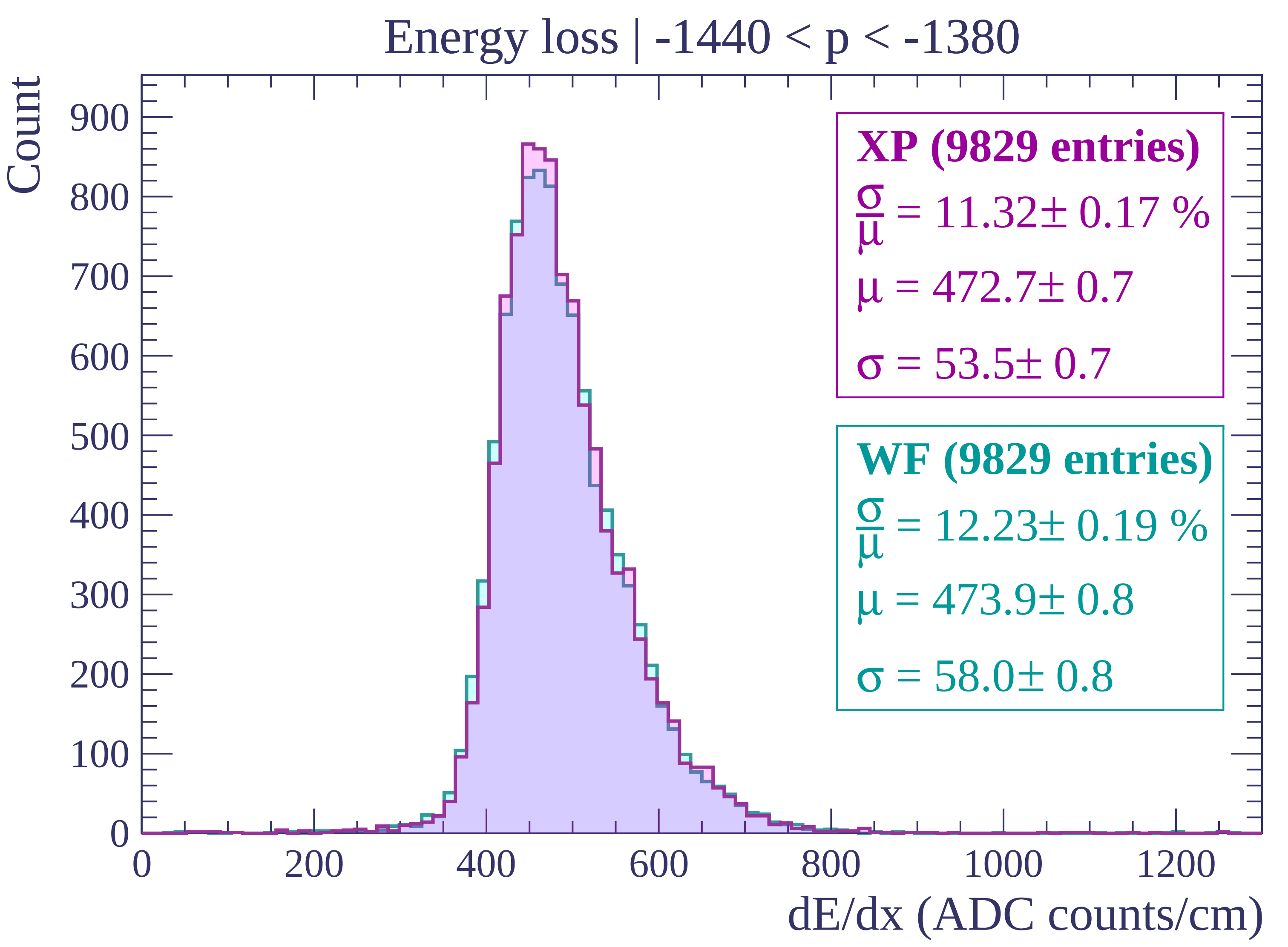
Energy loss | -1620 < p < -1560
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP (8864 entries)
σ/μ = 11.74± 0.17 %
μ = 482.7± 0.8
σ = 56.7± 0.7
WF (8864 entries)
σ/μ = 11.70± 0.17 %
μ = 480.4± 0.8
σ = 56.2± 0.7



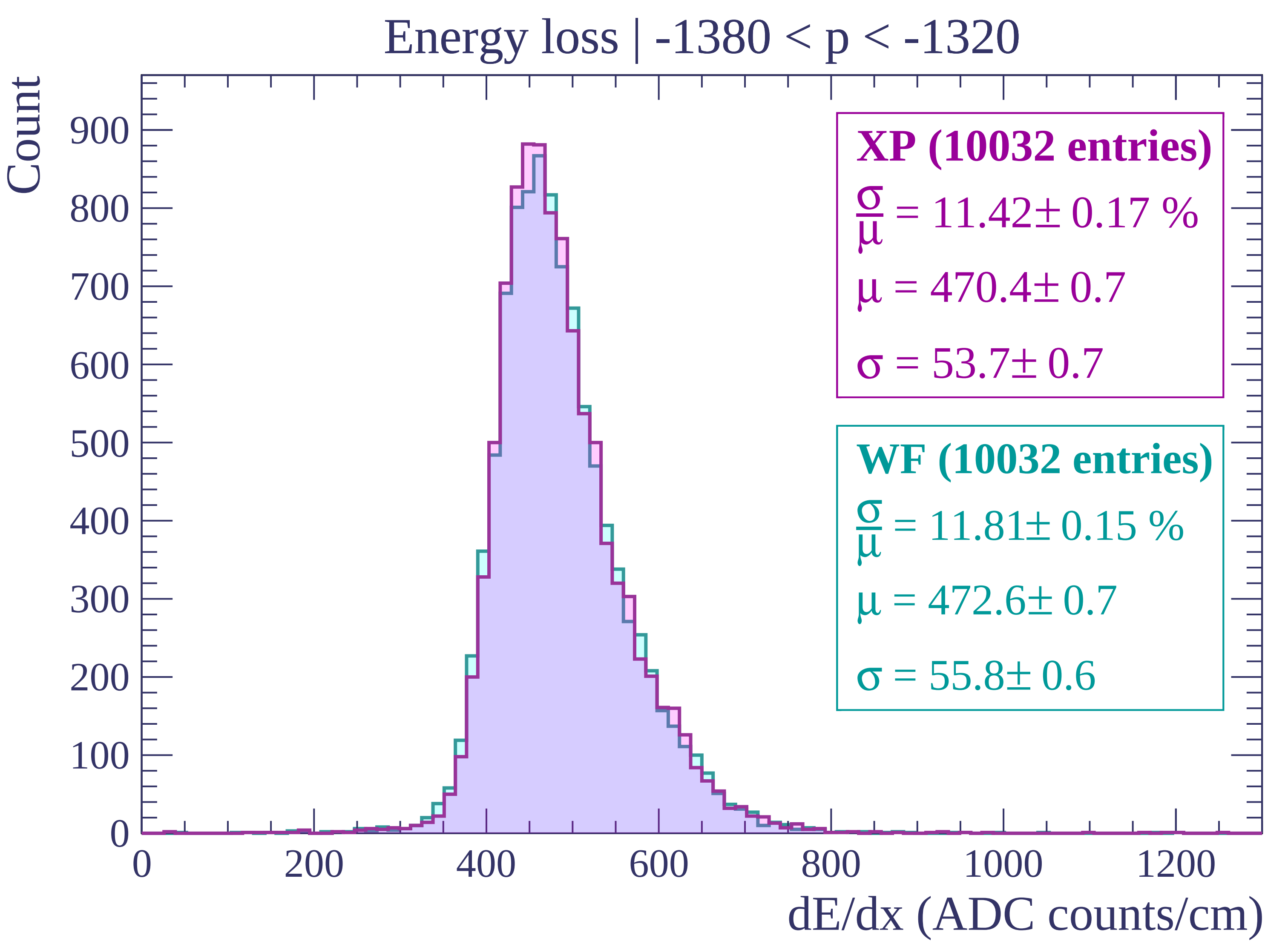
Energy loss | -1560 < p < -1500
Count
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (9052 entries)
σ/μ = 11.65± 0.17 %
μ = 478.7± 0.8
σ = 55.8± 0.7
WF (9052 entries)
σ/μ = 11.95± 0.18 %
μ = 478.5± 0.8
σ = 57.2± 0.8



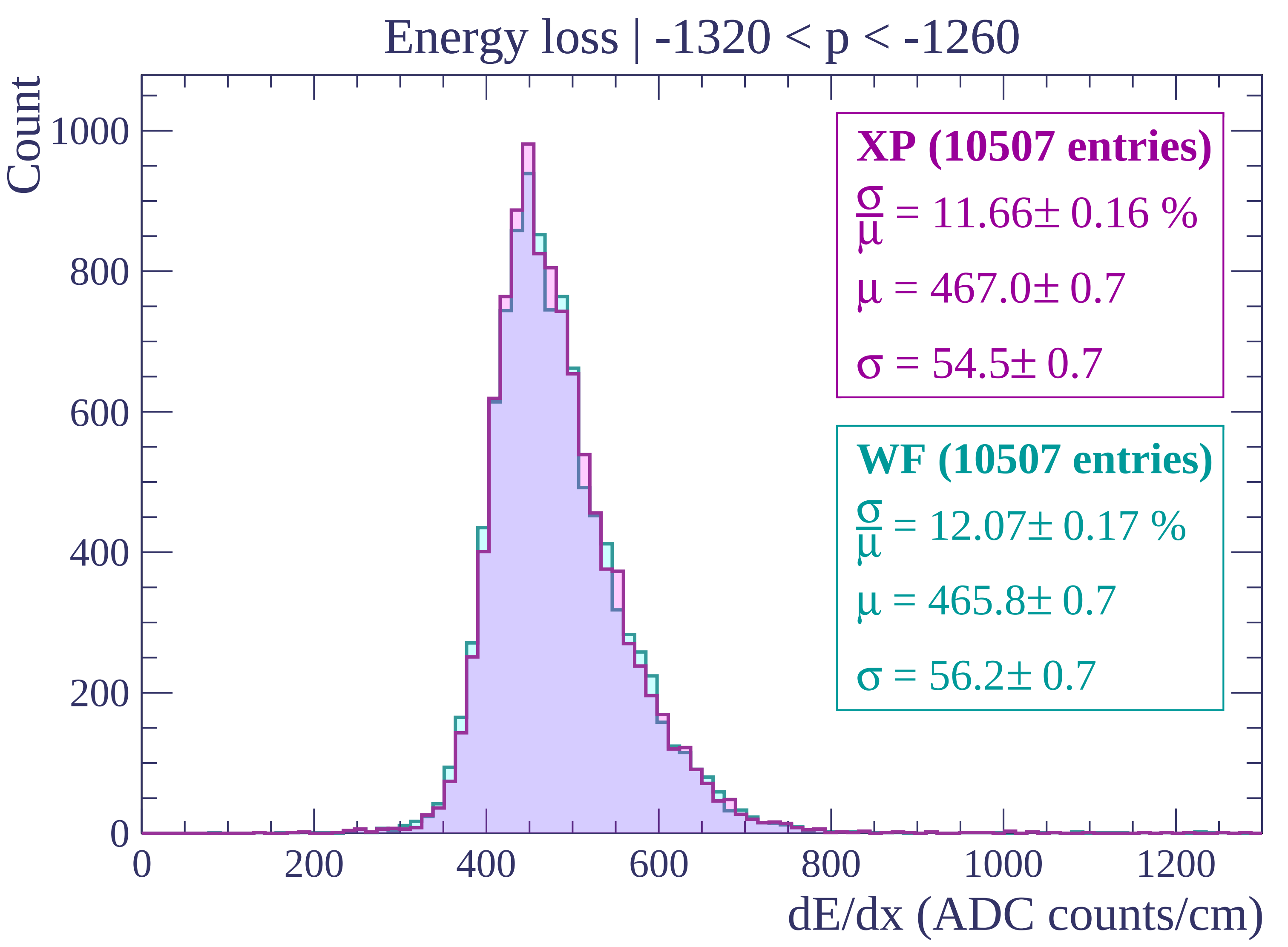
Energy loss | -1500 < p < -1440
Count
dE/dx (ADC counts/cm)
XP (9339 entries)
σ/μ = 11.73± 0.17 %
μ = 477.0± 0.8
σ = 55.9± 0.7
WF (9339 entries)
σ/μ = 12.06± 0.16 %
μ = 479.7± 0.8
σ = 57.8± 0.7



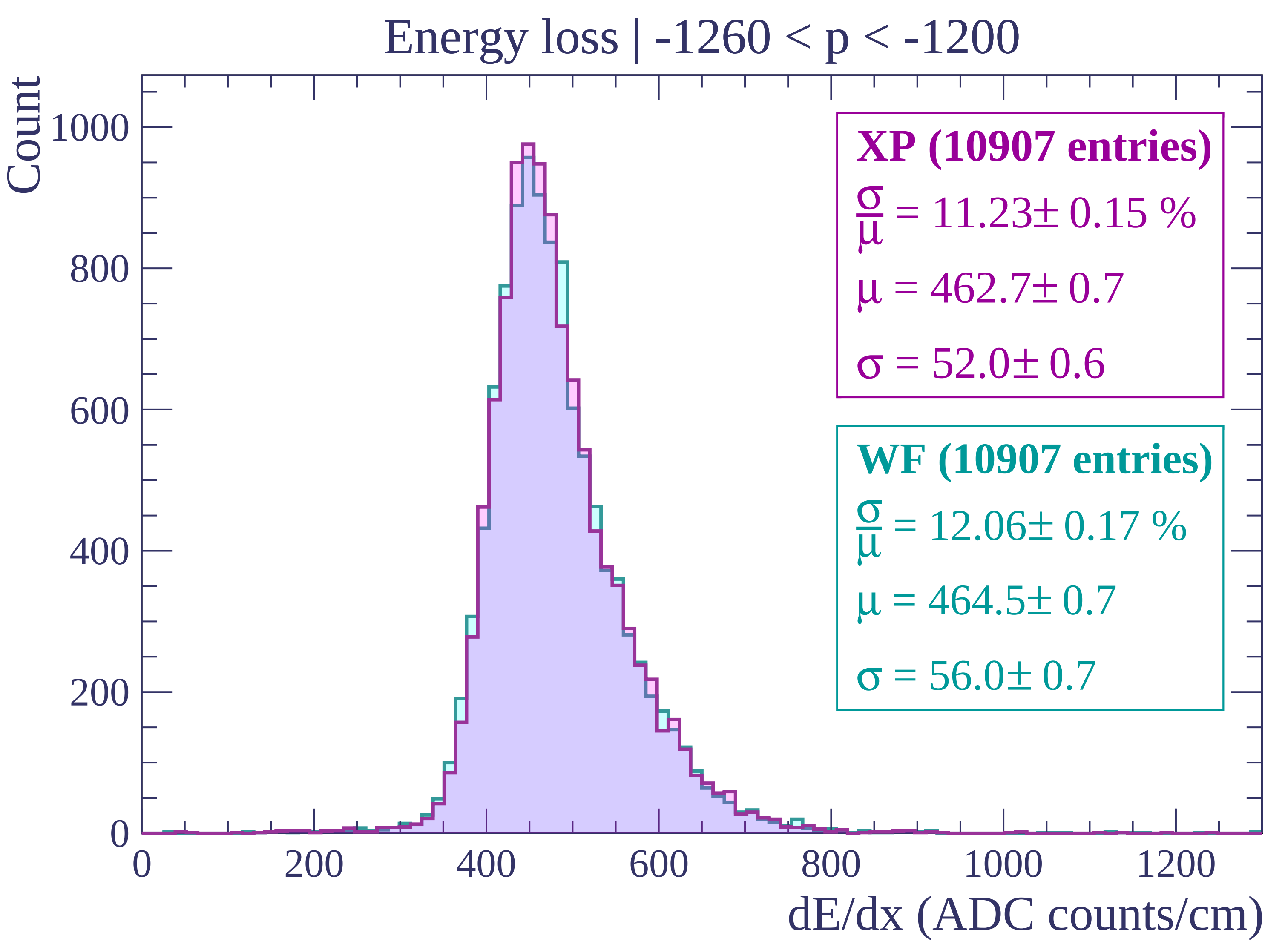
Energy loss | -1440 < p < -1380
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP (9829 entries)
σ/μ = 11.32± 0.17 %
μ = 472.7± 0.7
σ = 53.5± 0.7
WF (9829 entries)
σ/μ = 12.23± 0.19 %
μ = 473.9± 0.8
σ = 58.0± 0.8

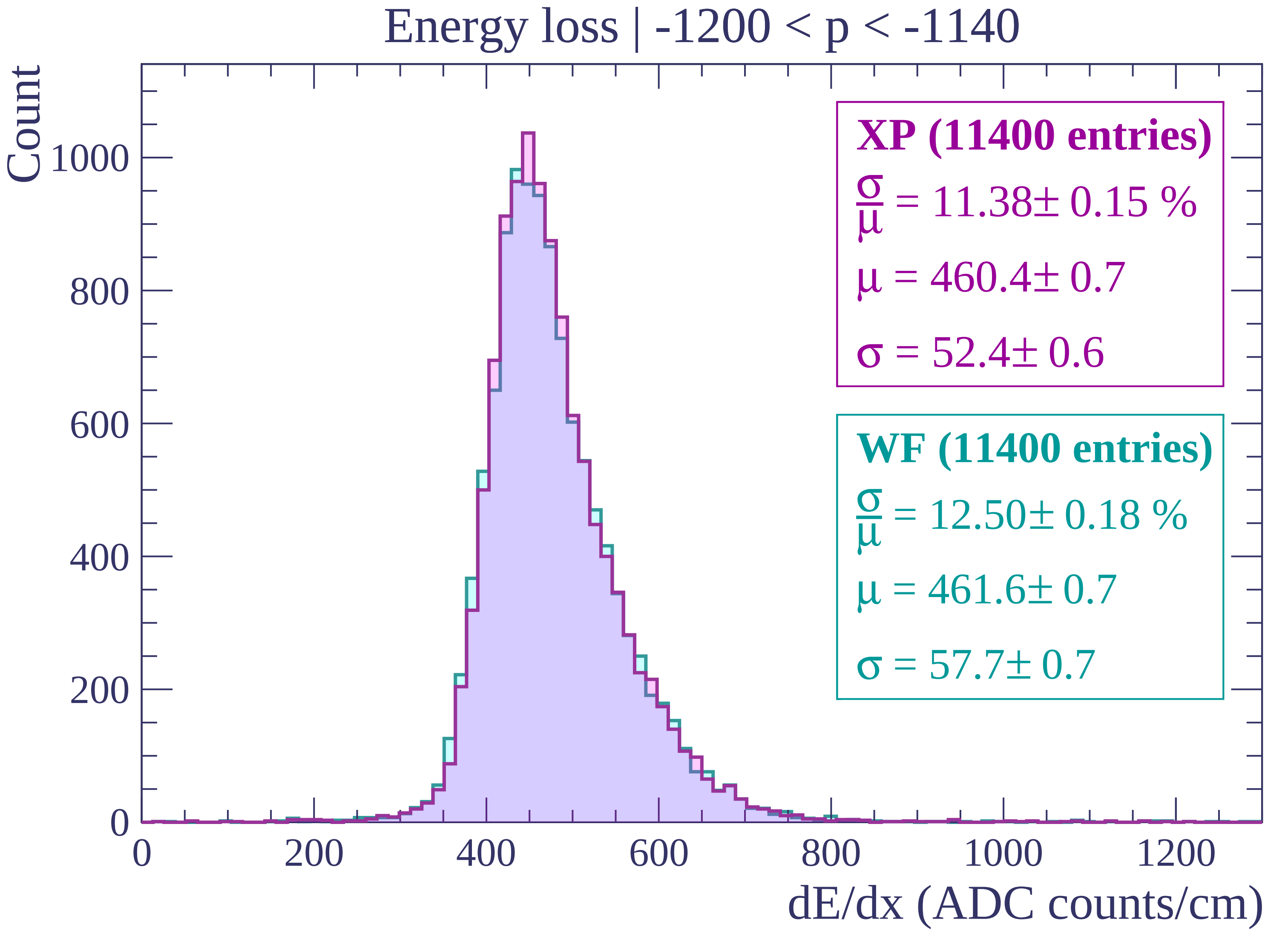


Energy loss | -1380 < p < -1320
Count
dE/dx (ADC counts/cm)
XP (10032 entries)
$\frac{\sigma}{\mu}$ = 11.42± 0.17 %
μ = 470.4± 0.7
σ = 53.7± 0.7
WF (10032 entries)
$\frac{\sigma}{\mu}$ = 11.81± 0.15 %
μ = 472.6± 0.7
σ = 55.8± 0.6
0
100
200
300
400
500
600
700
800
900
1000
1200

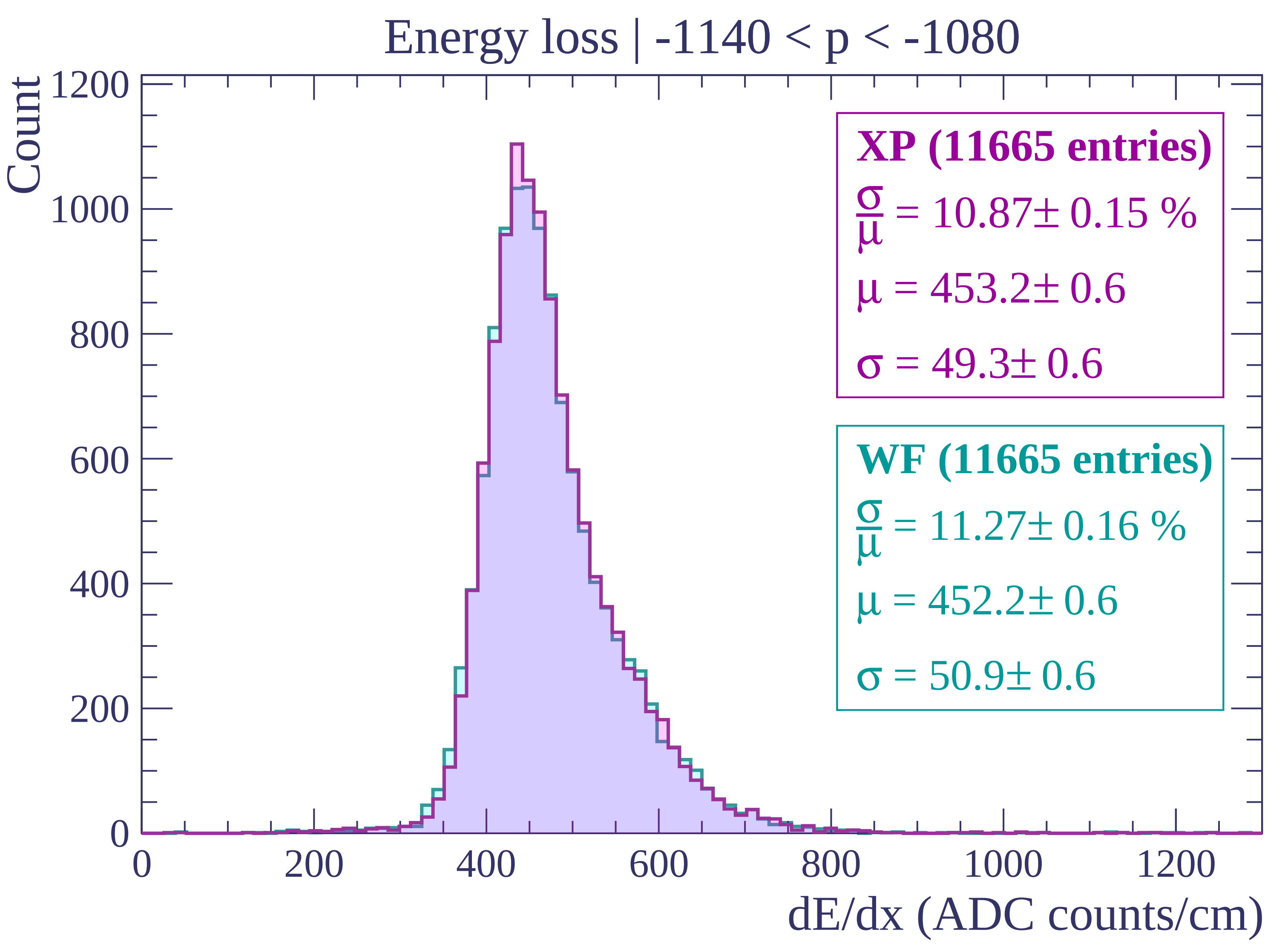


Energy loss | -1320 < p < -1260
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (10507 entries)
σ/μ = 11.66± 0.16 %
μ = 467.0± 0.7
σ = 54.5± 0.7
WF (10507 entries)
σ/μ = 12.07± 0.17 %
μ = 465.8± 0.7
σ = 56.2± 0.7

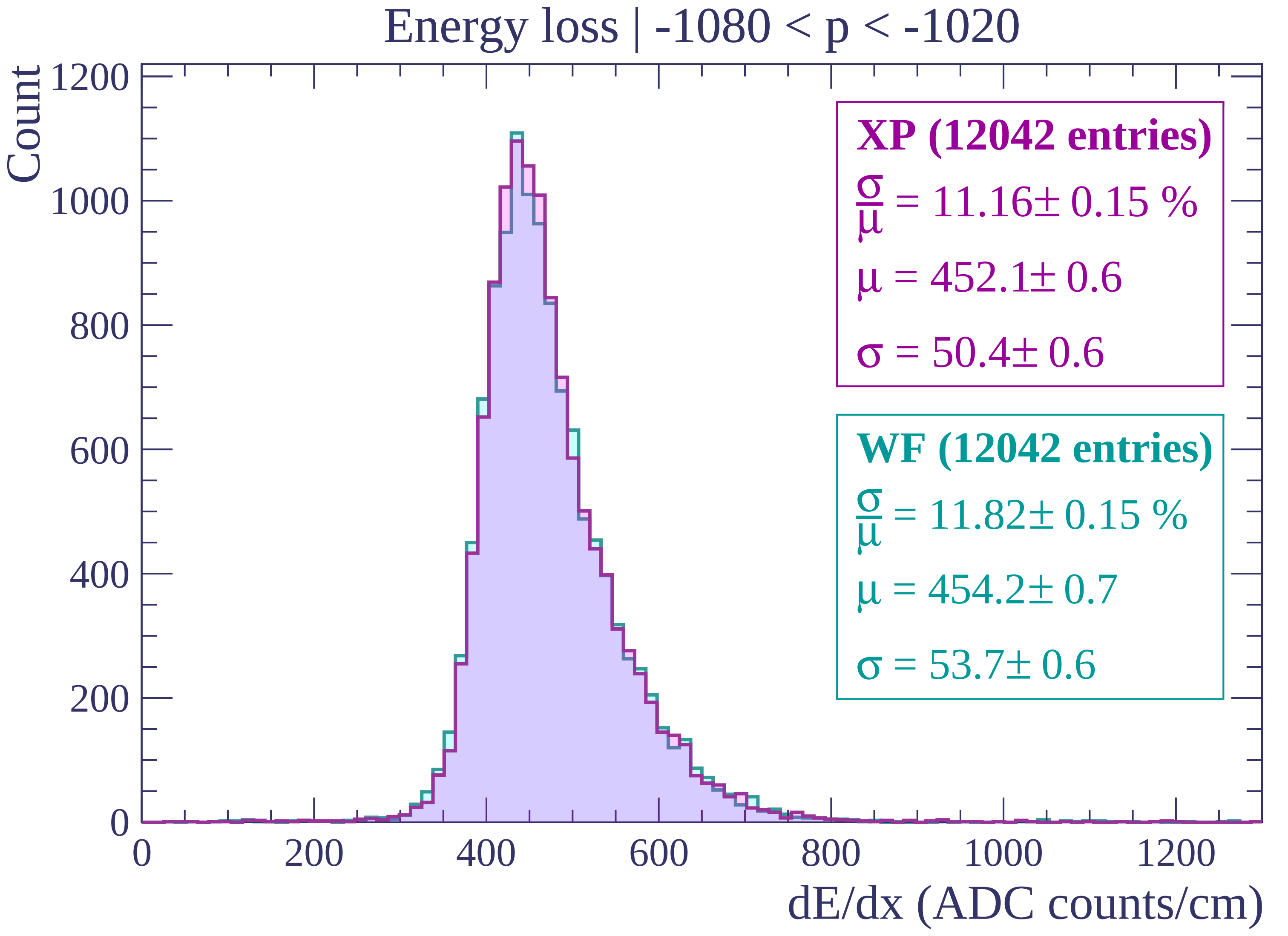

Energy loss | -1260 < p < -1200
Count
dE/dx (ADC counts/cm)
XP (10907 entries)
σ/μ = 11.23± 0.15 %
μ = 462.7± 0.7
σ = 52.0± 0.6
WF (10907 entries)
σ/μ = 12.06± 0.17 %
μ = 464.5± 0.7
σ = 56.0± 0.7
0
200
400
600
800
1000
1200

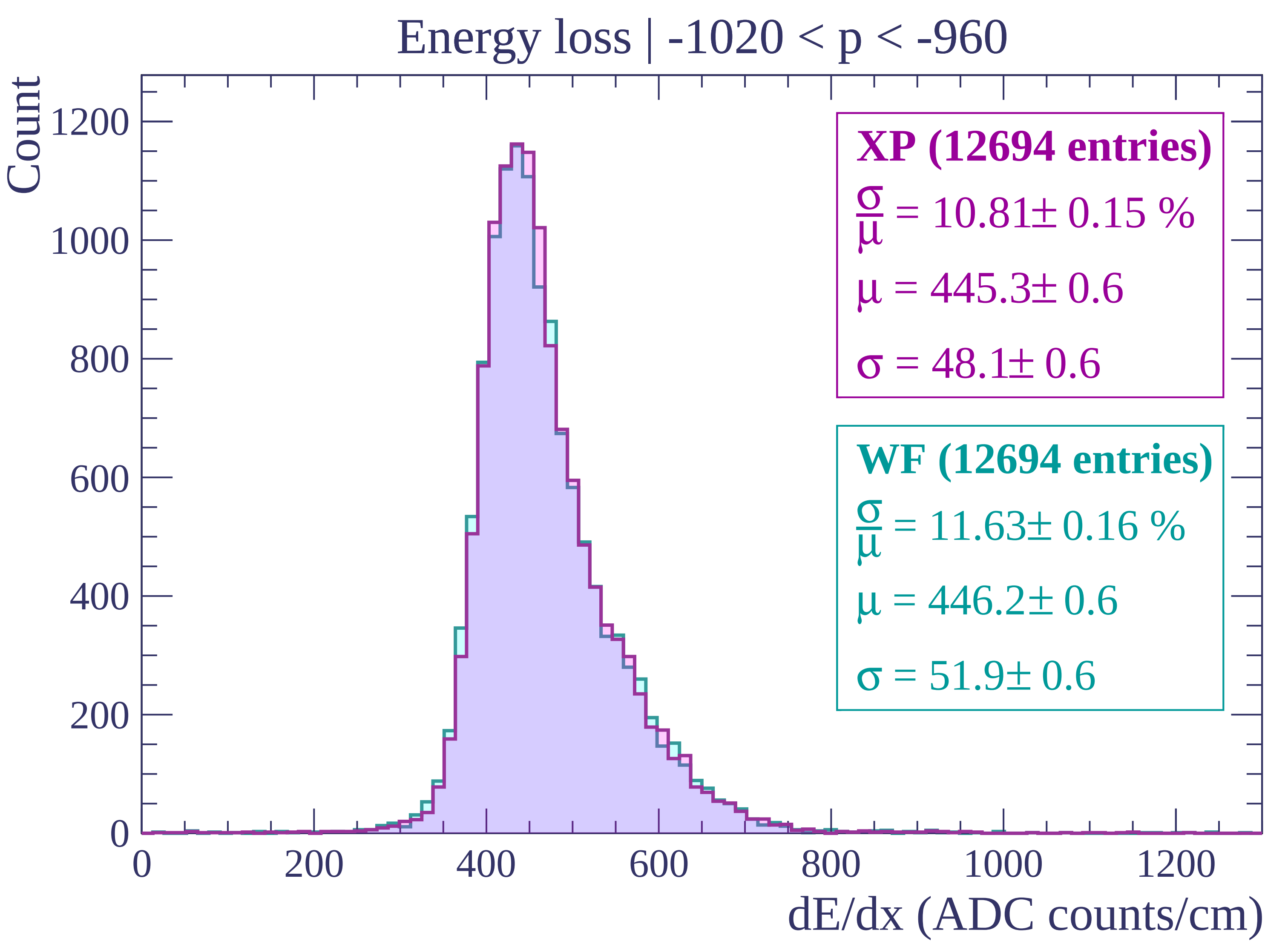


Energy loss | -1200 < p < -1140
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (11400 entries)
$\frac{\sigma}{\mu}$ = 11.38± 0.15 %
μ = 460.4± 0.7
σ = 52.4± 0.6
WF (11400 entries)
$\frac{\sigma}{\mu}$ = 12.50± 0.18 %
μ = 461.6± 0.7
σ = 57.7± 0.7

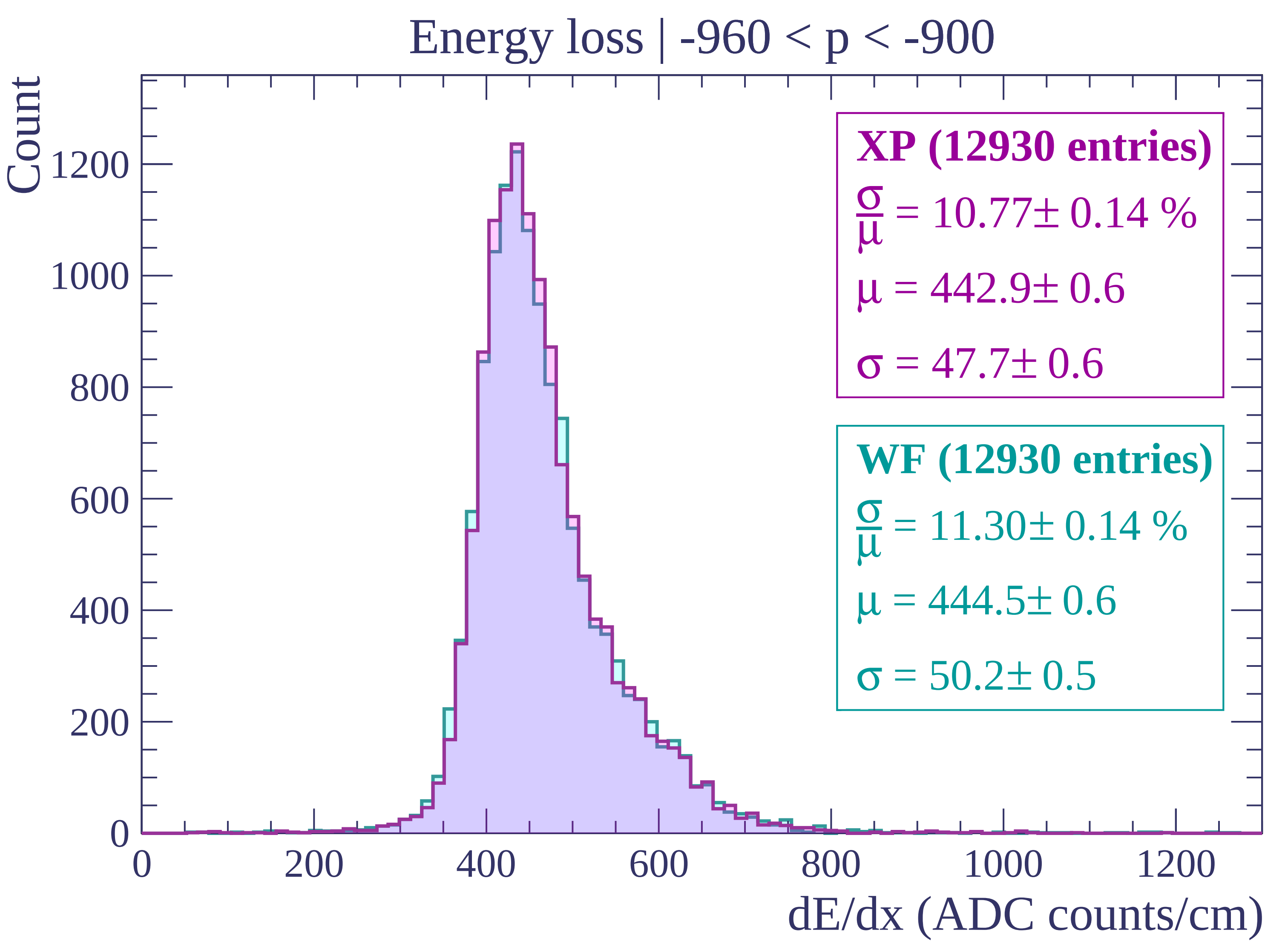


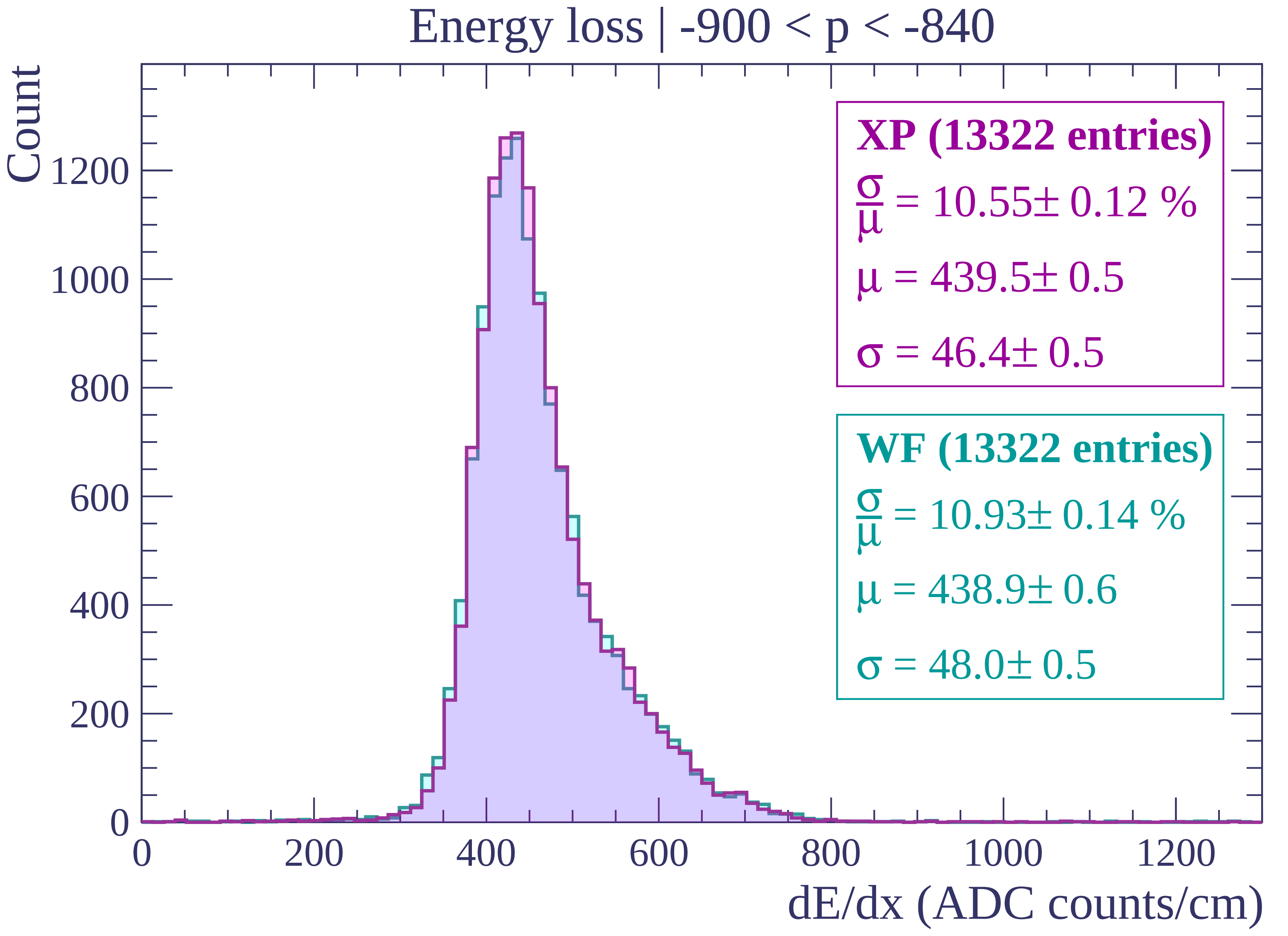
Energy loss | -1140 < p < -1080
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (11665 entries)
$\frac{\sigma}{\mu}$ = 10.87± 0.15 %
μ = 453.2± 0.6
σ = 49.3± 0.6
WF (11665 entries)
$\frac{\sigma}{\mu}$ = 11.27± 0.16 %
μ = 452.2± 0.6
σ = 50.9± 0.6

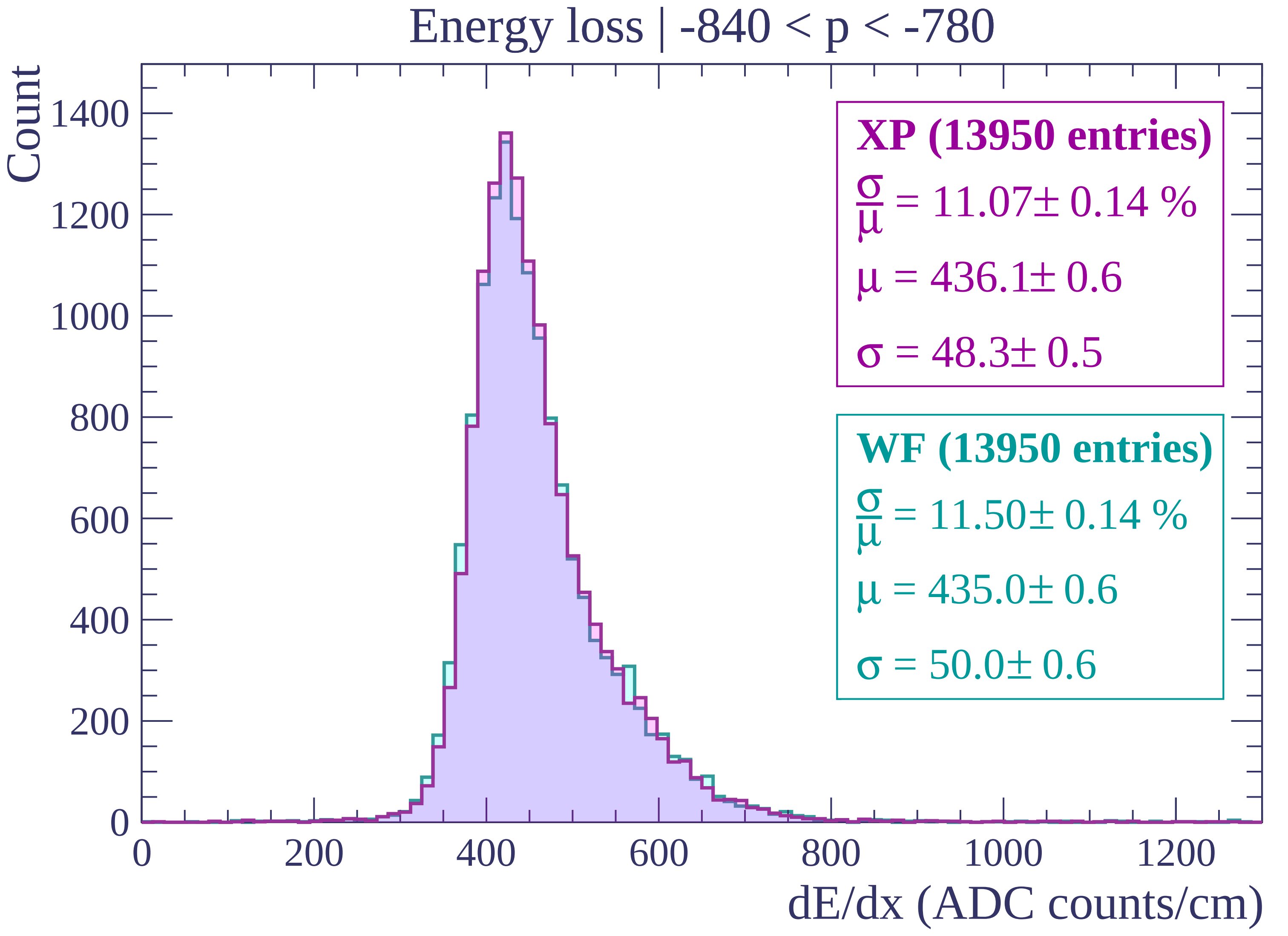

Energy loss | -1080 < p < -1020
Count
0
200
400
600
800
1000
1200
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (12042 entries)
$\frac{\sigma}{\mu}$ = 11.16± 0.15 %
μ = 452.1± 0.6
σ = 50.4± 0.6
WF (12042 entries)
$\frac{\sigma}{\mu}$ = 11.82± 0.15 %
μ = 454.2± 0.7
σ = 53.7± 0.6

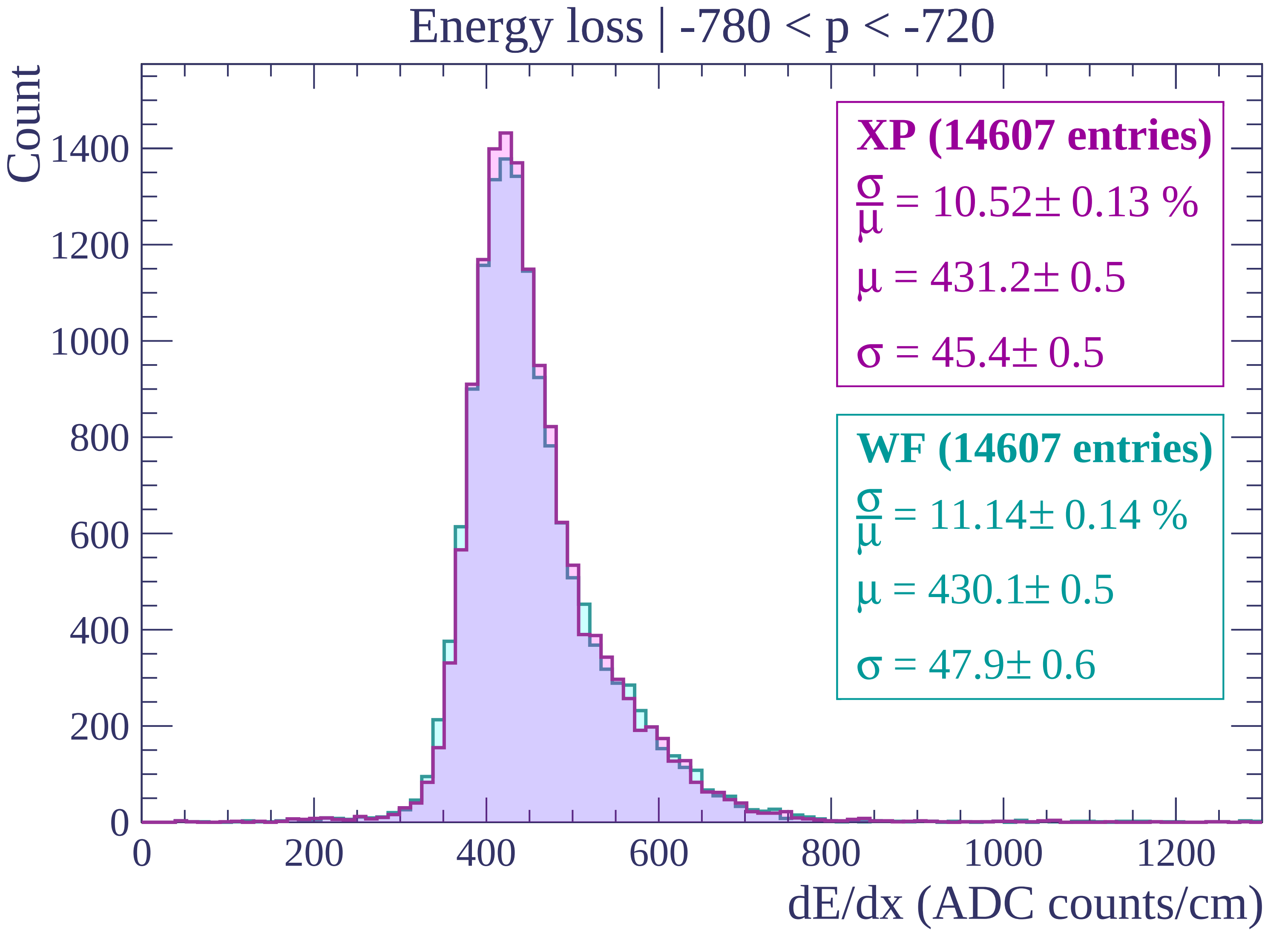


Energy loss | -1020 < p < -960
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (12694 entries)
σ/μ = 10.81± 0.15 %
μ = 445.3± 0.6
σ = 48.1± 0.6
WF (12694 entries)
σ/μ = 11.63± 0.16 %
μ = 446.2± 0.6
σ = 51.9± 0.6

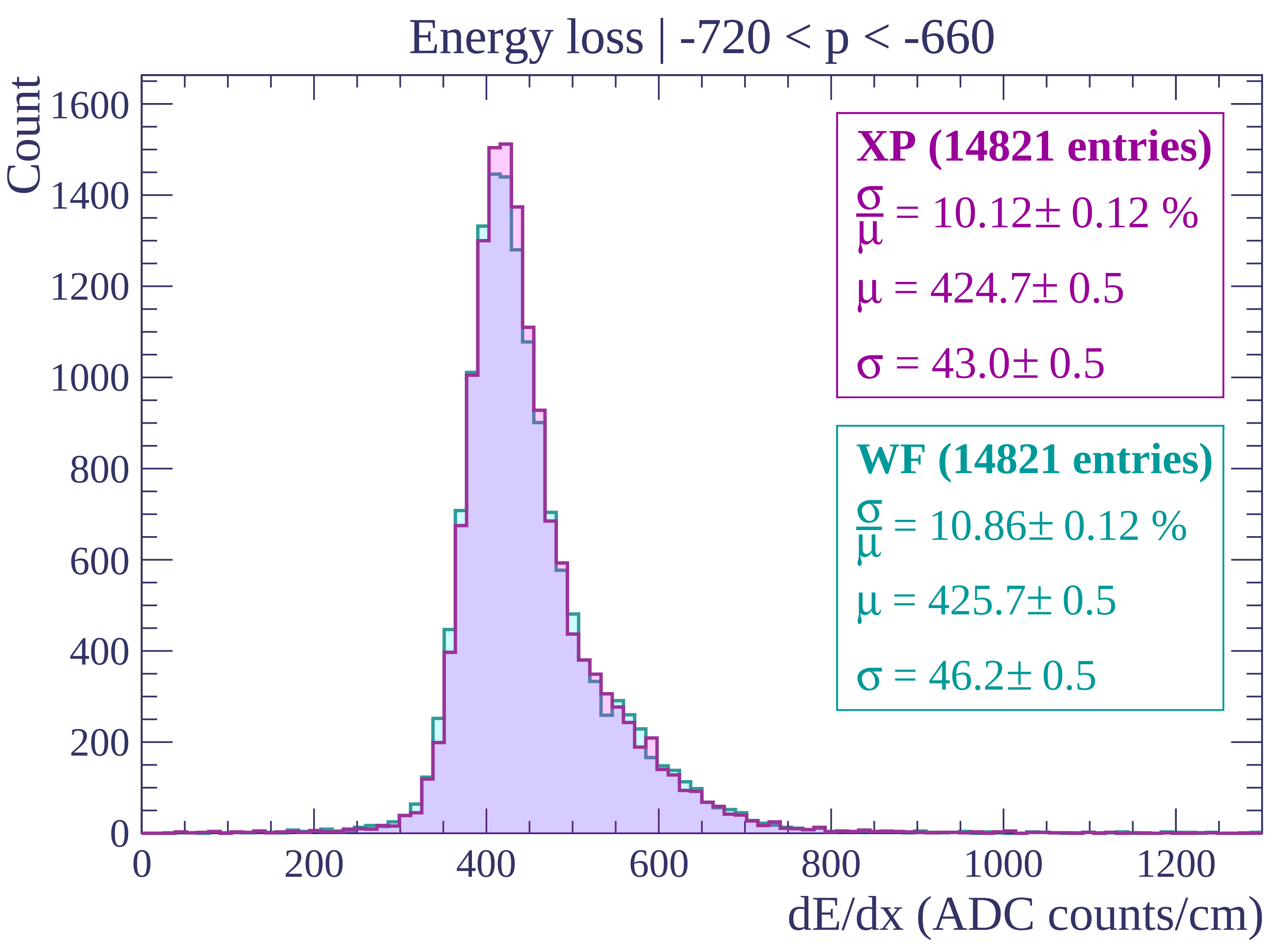

Energy loss | -960 < p < -900
Count
dE/dx (ADC counts/cm)
XP (12930 entries)
$\frac{\sigma}{\mu}$ = 10.77± 0.14 %
μ = 442.9± 0.6
σ = 47.7± 0.6
WF (12930 entries)
$\frac{\sigma}{\mu}$ = 11.30± 0.14 %
μ = 444.5± 0.6
σ = 50.2± 0.5


Energy loss | -900 < p < -840
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (13322 entries)
$\frac{\sigma}{\mu}$ = 10.55± 0.12 %
μ = 439.5± 0.5
σ = 46.4± 0.5
WF (13322 entries)
$\frac{\sigma}{\mu}$ = 10.93± 0.14 %
μ = 438.9± 0.6
σ = 48.0± 0.5

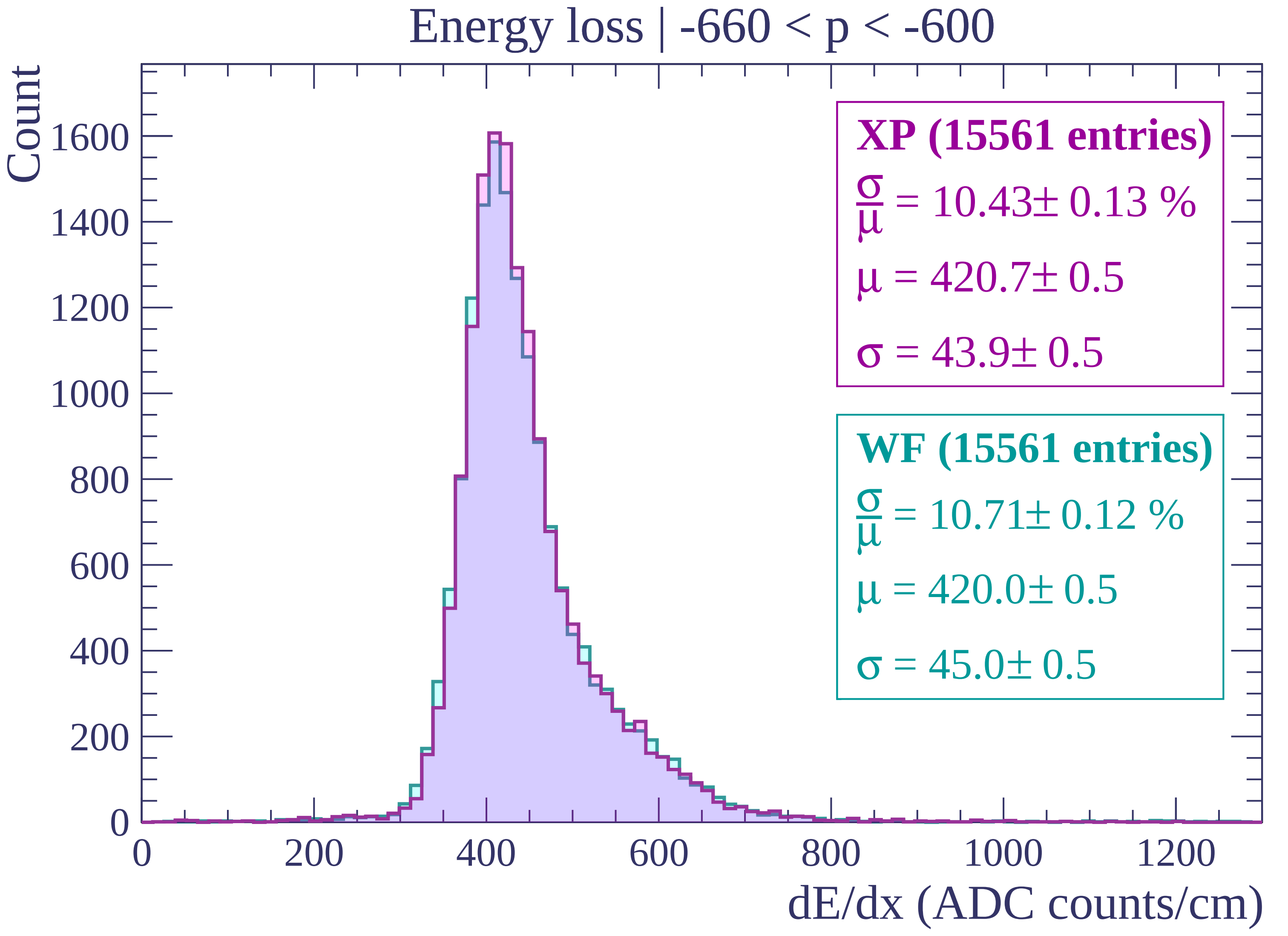

Energy loss | -840 < p < -780
Count
dE/dx (ADC counts/cm)
XP (13950 entries)
σ/μ = 11.07± 0.14 %
μ = 436.1± 0.6
σ = 48.3± 0.5
WF (13950 entries)
σ/μ = 11.50± 0.14 %
μ = 435.0± 0.6
σ = 50.0± 0.6
0
200
400
600
800
1000
1200
1400

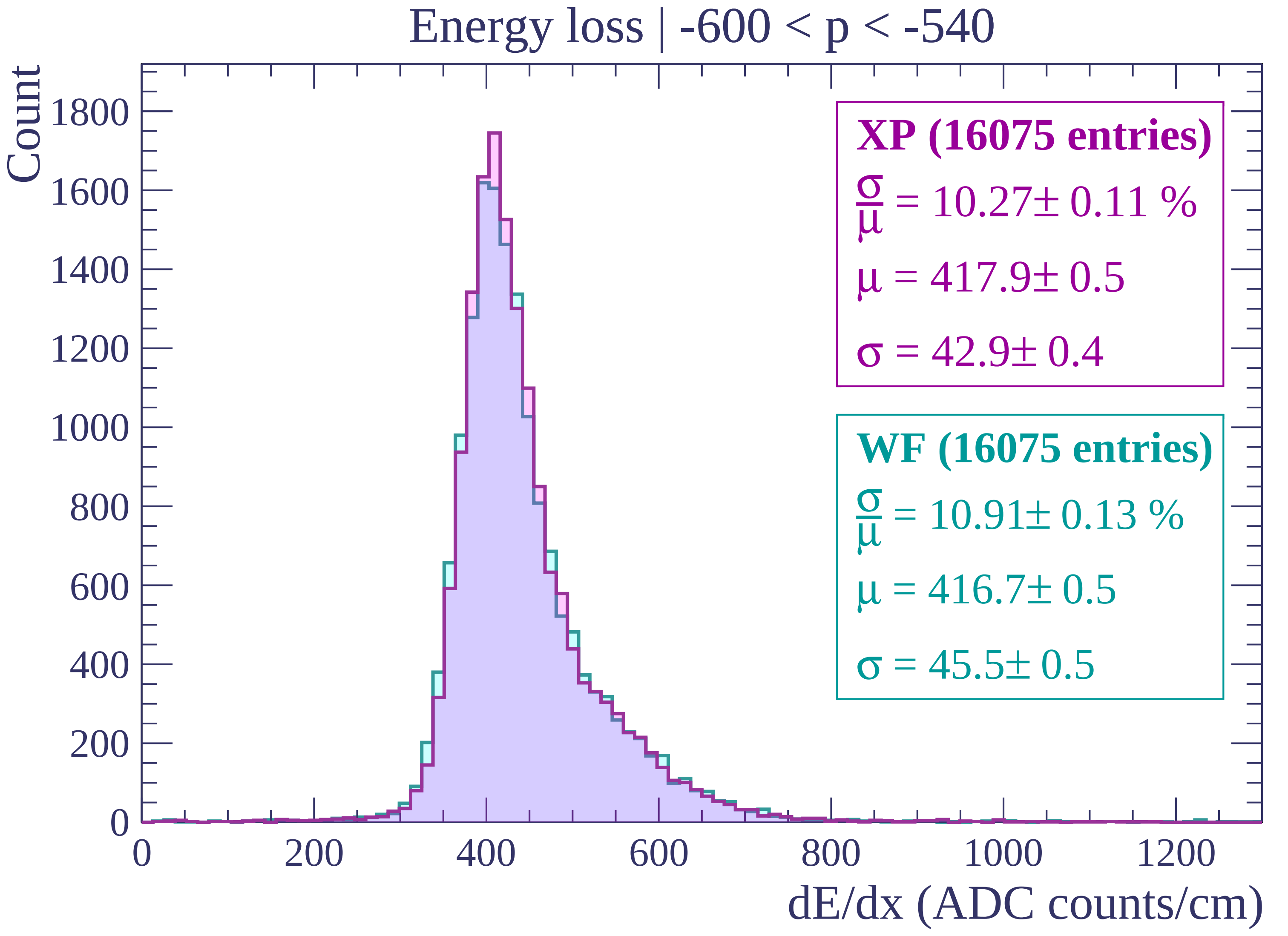


Energy loss | -780 < p < -720
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
XP (14607 entries)
σ/μ = 10.52± 0.13 %
μ = 431.2± 0.5
σ = 45.4± 0.5
WF (14607 entries)
σ/μ = 11.14± 0.14 %
μ = 430.1± 0.5
σ = 47.9± 0.6

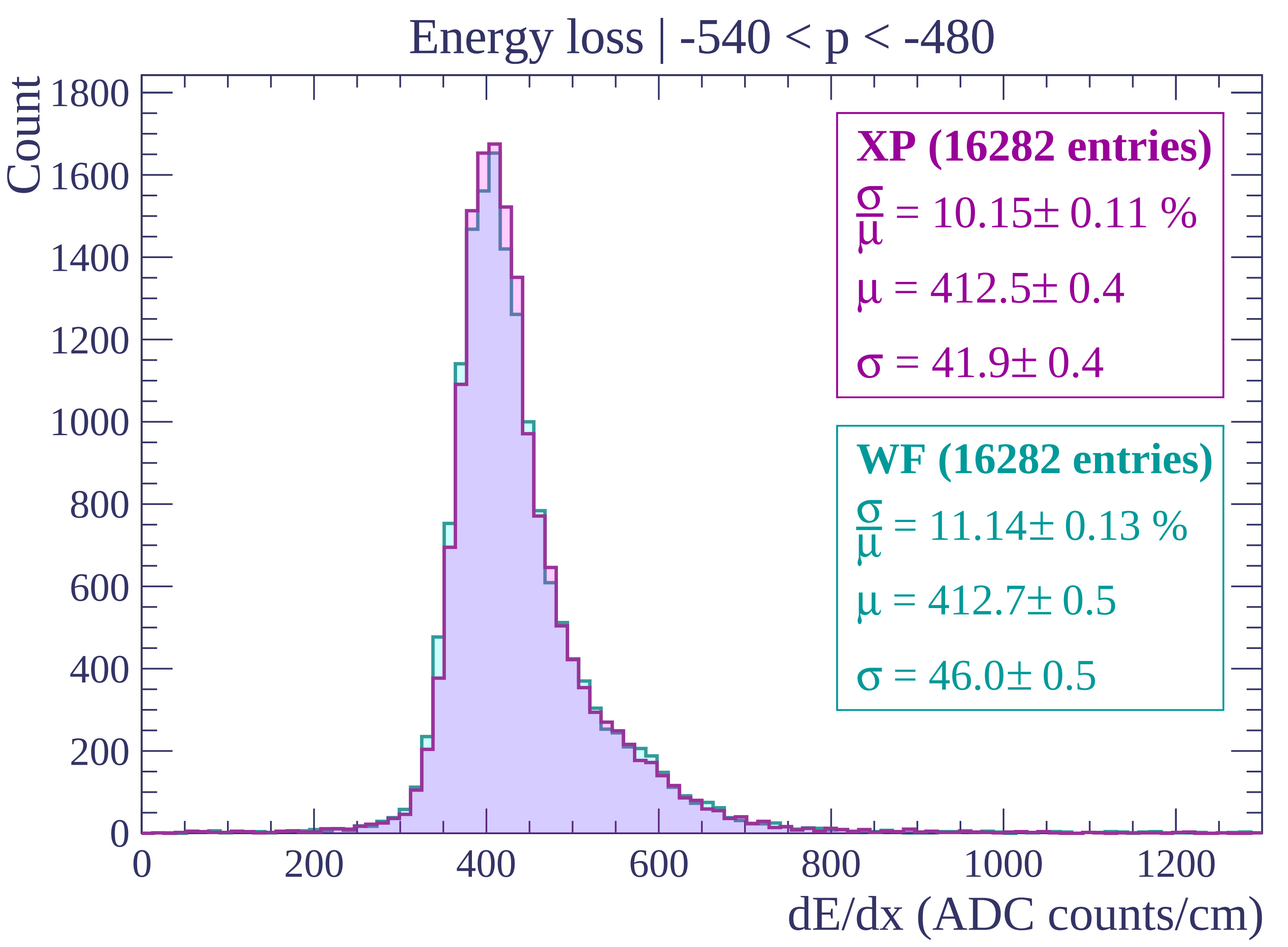


Energy loss | -720 < p < -660
Count
dE/dx (ADC counts/cm)
XP (14821 entries)
σ/μ = 10.12± 0.12 %
μ = 424.7± 0.5
σ = 43.0± 0.5
WF (14821 entries)
σ/μ = 10.86± 0.12 %
μ = 425.7± 0.5
σ = 46.2± 0.5
0
200
400
600
800
1000
1200
1400
1600

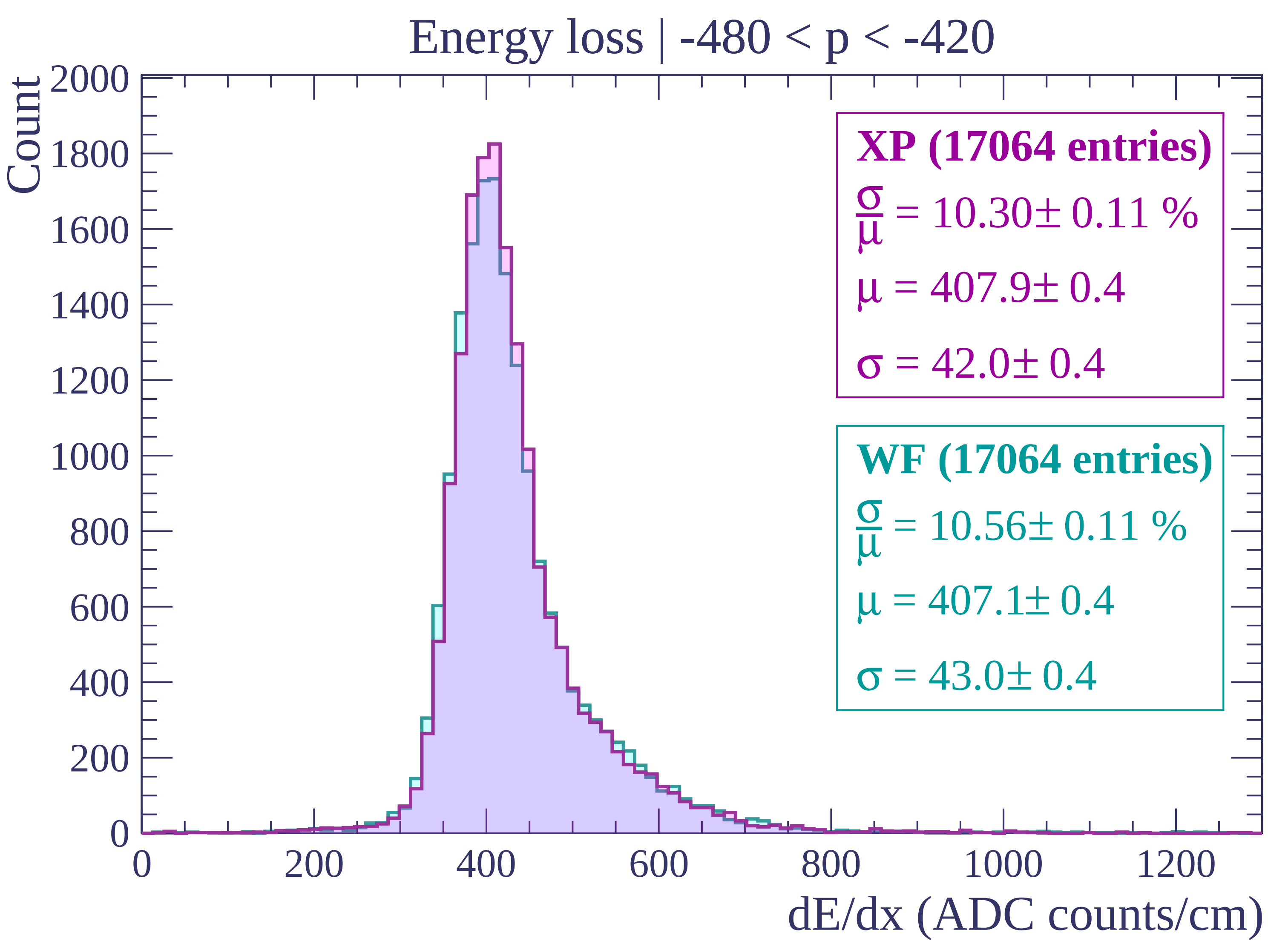

Energy loss | -660 < p < -600
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
0
200
400
600
800
1000
1200
XP (15561 entries)
σ/μ = 10.43± 0.13 %
μ = 420.7± 0.5
σ = 43.9± 0.5
WF (15561 entries)
σ/μ = 10.71± 0.12 %
μ = 420.0± 0.5
σ = 45.0± 0.5



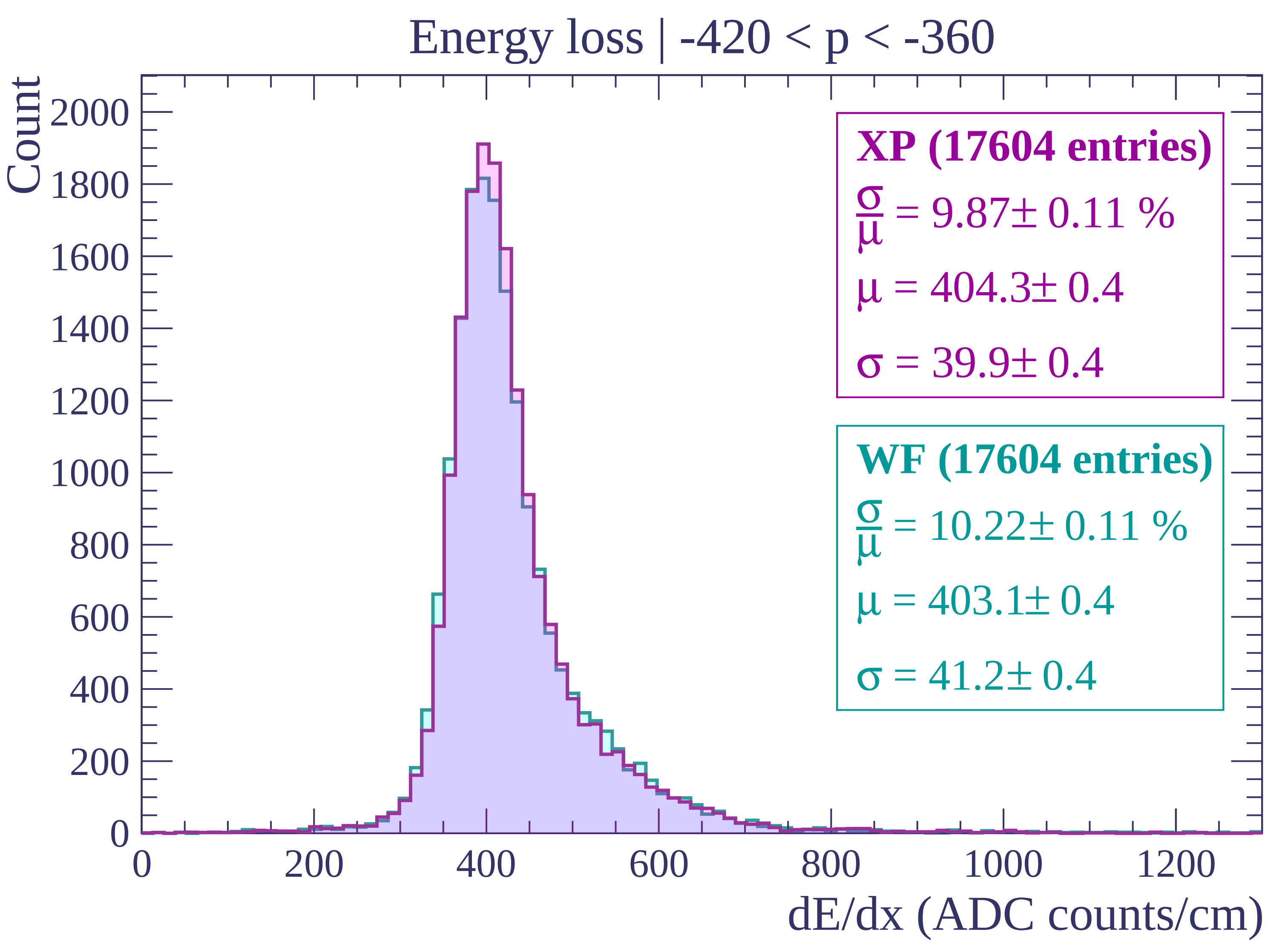
Energy loss | -600 < p < -540
Count
dE/dx (ADC counts/cm)
XP (16075 entries)
$\frac{\sigma}{\mu}$ = 10.27± 0.11 %
μ = 417.9± 0.5
σ = 42.9± 0.4
WF (16075 entries)
$\frac{\sigma}{\mu}$ = 10.91± 0.13 %
μ = 416.7± 0.5
σ = 45.5± 0.5
0
200
400
600
800
1000
1200
1400
1600
1800



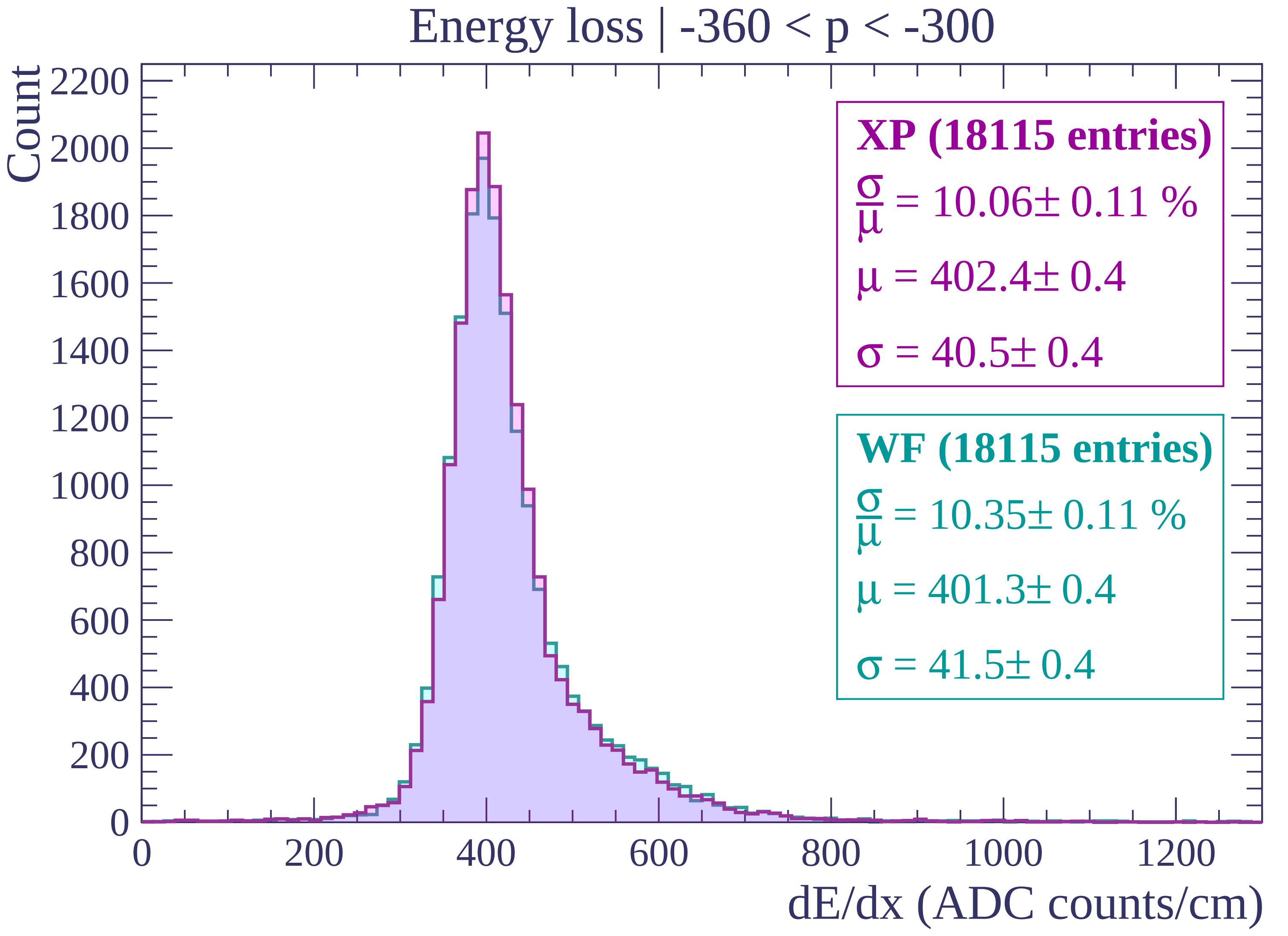
Energy loss | -540 < p < -480
Count
dE/dx (ADC counts/cm)
XP (16282 entries)
σ/μ = 10.15± 0.11 %
μ = 412.5± 0.4
σ = 41.9± 0.4
WF (16282 entries)
σ/μ = 11.14± 0.13 %
μ = 412.7± 0.5
σ = 46.0± 0.5



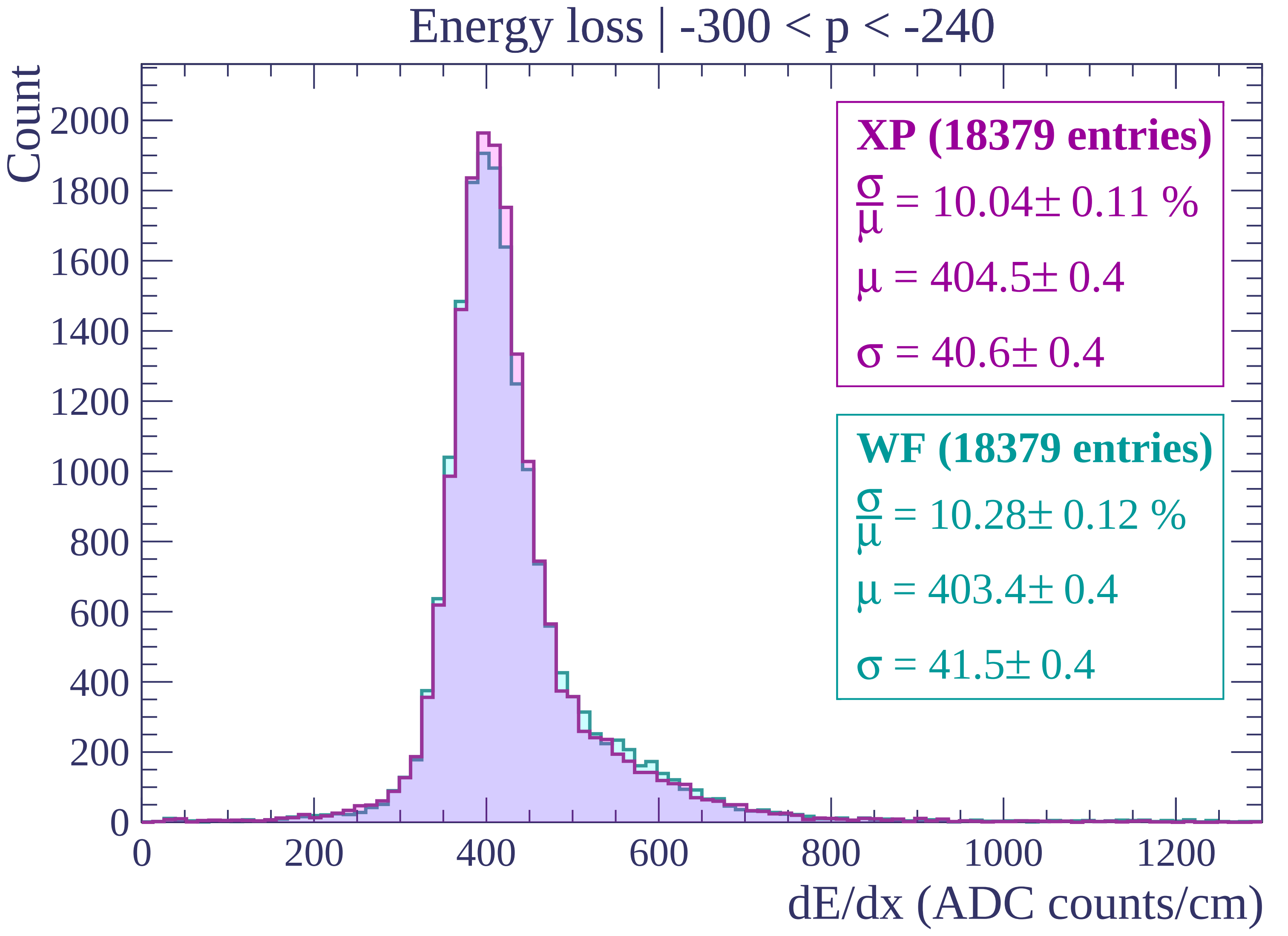
Energy loss | -480 < p < -420
Count
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (17064 entries)
σ/μ = 10.30± 0.11 %
μ = 407.9± 0.4
σ = 42.0± 0.4
WF (17064 entries)
σ/μ = 10.56± 0.11 %
μ = 407.1± 0.4
σ = 43.0± 0.4

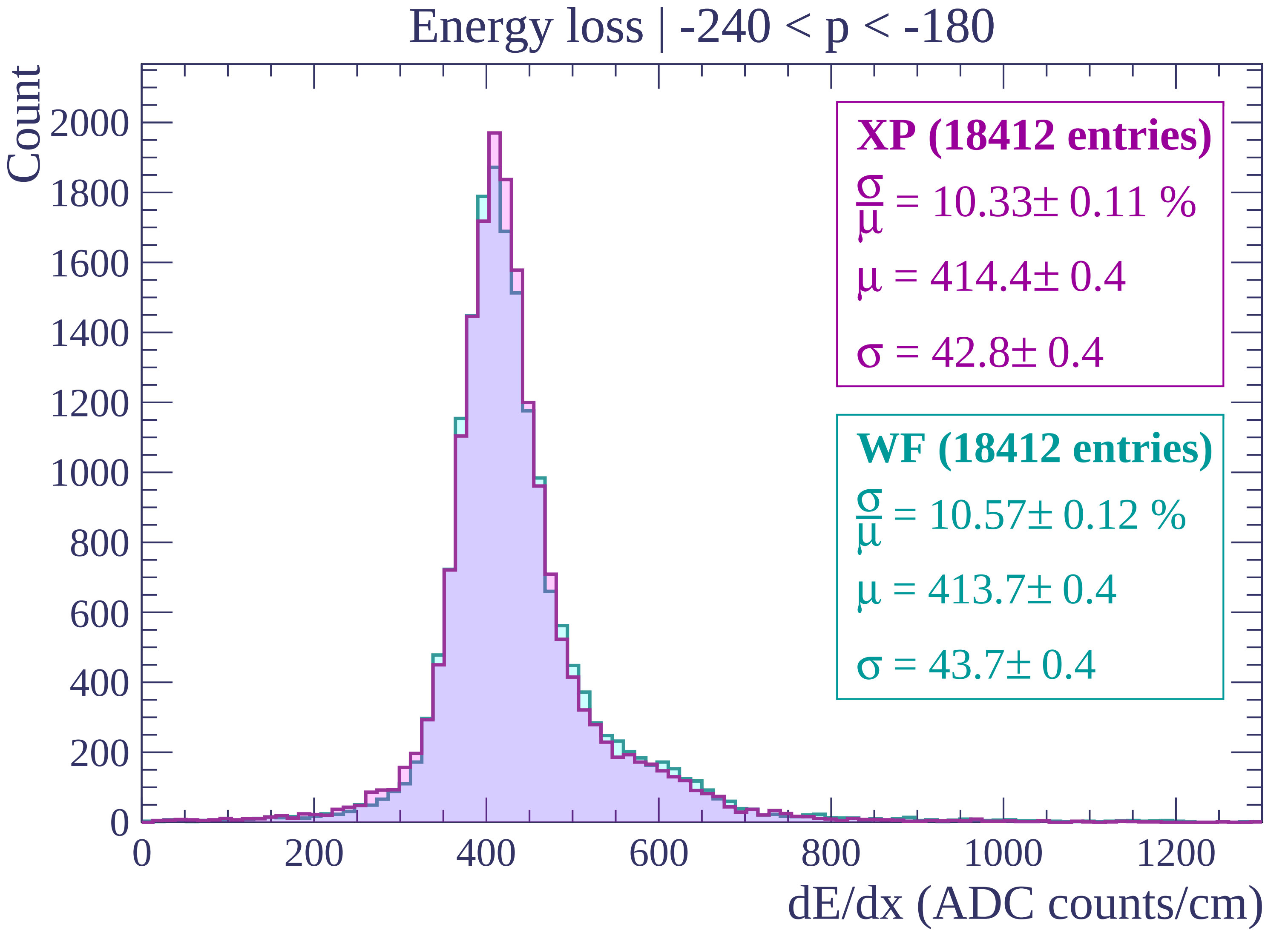


Energy loss | -420 < p < -360
Count
dE/dx (ADC counts/cm)
XP (17604 entries)
$\frac{\sigma}{\mu}$ = 9.87± 0.11 %
μ = 404.3± 0.4
σ = 39.9± 0.4
WF (17604 entries)
$\frac{\sigma}{\mu}$ = 10.22± 0.11 %
μ = 403.1± 0.4
σ = 41.2± 0.4
0
200
400
600
800
1000
1200
1400
1600
1800
2000

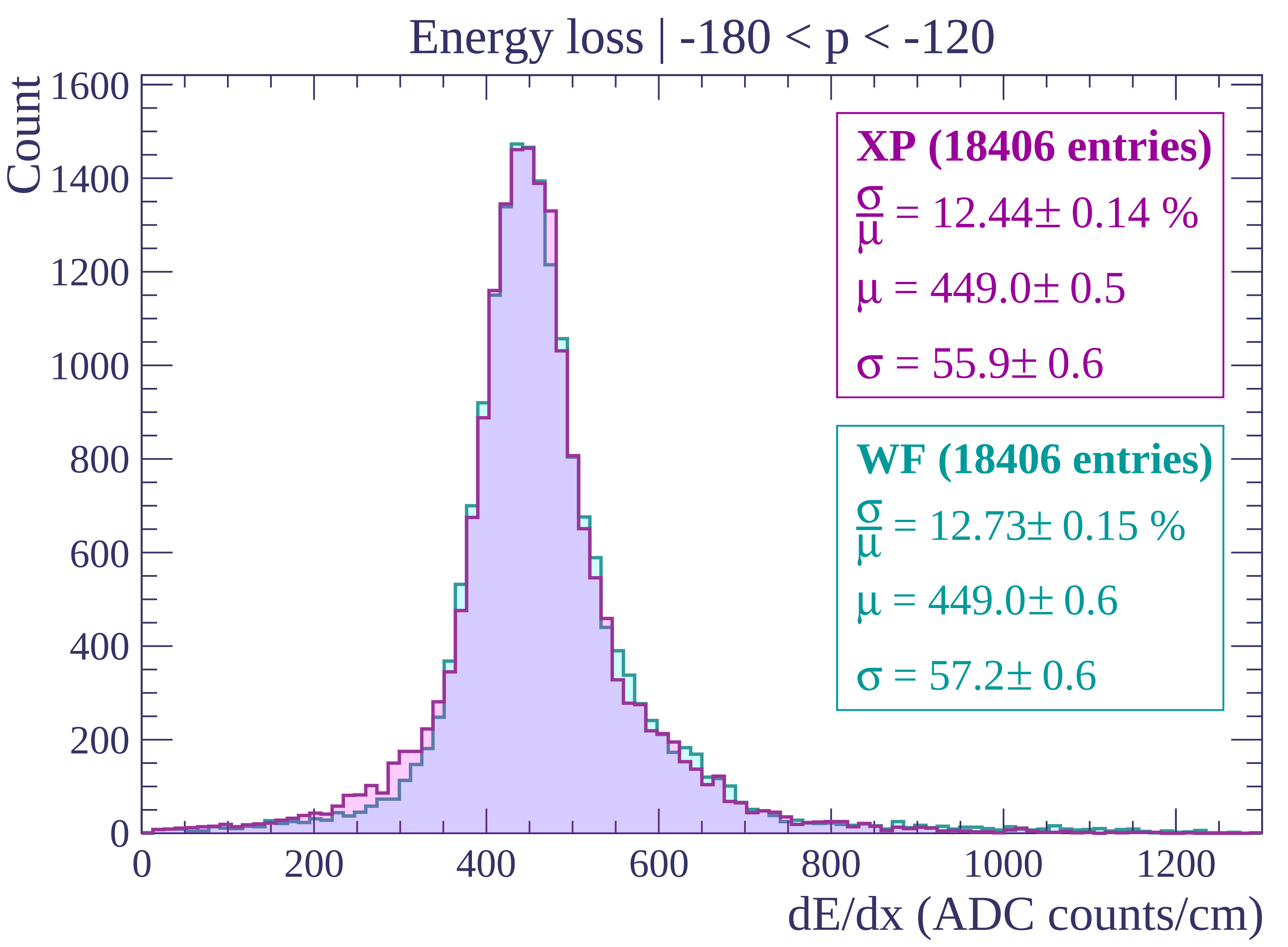


Energy loss | -360 < p < -300
Count
dE/dx (ADC counts/cm)
XP (18115 entries)
$\frac{\sigma}{\mu}$ = 10.06± 0.11 %
μ = 402.4± 0.4
σ = 40.5± 0.4
WF (18115 entries)
$\frac{\sigma}{\mu}$ = 10.35± 0.11 %
μ = 401.3± 0.4
σ = 41.5± 0.4
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200
0
200
400
600
800
1000
1200

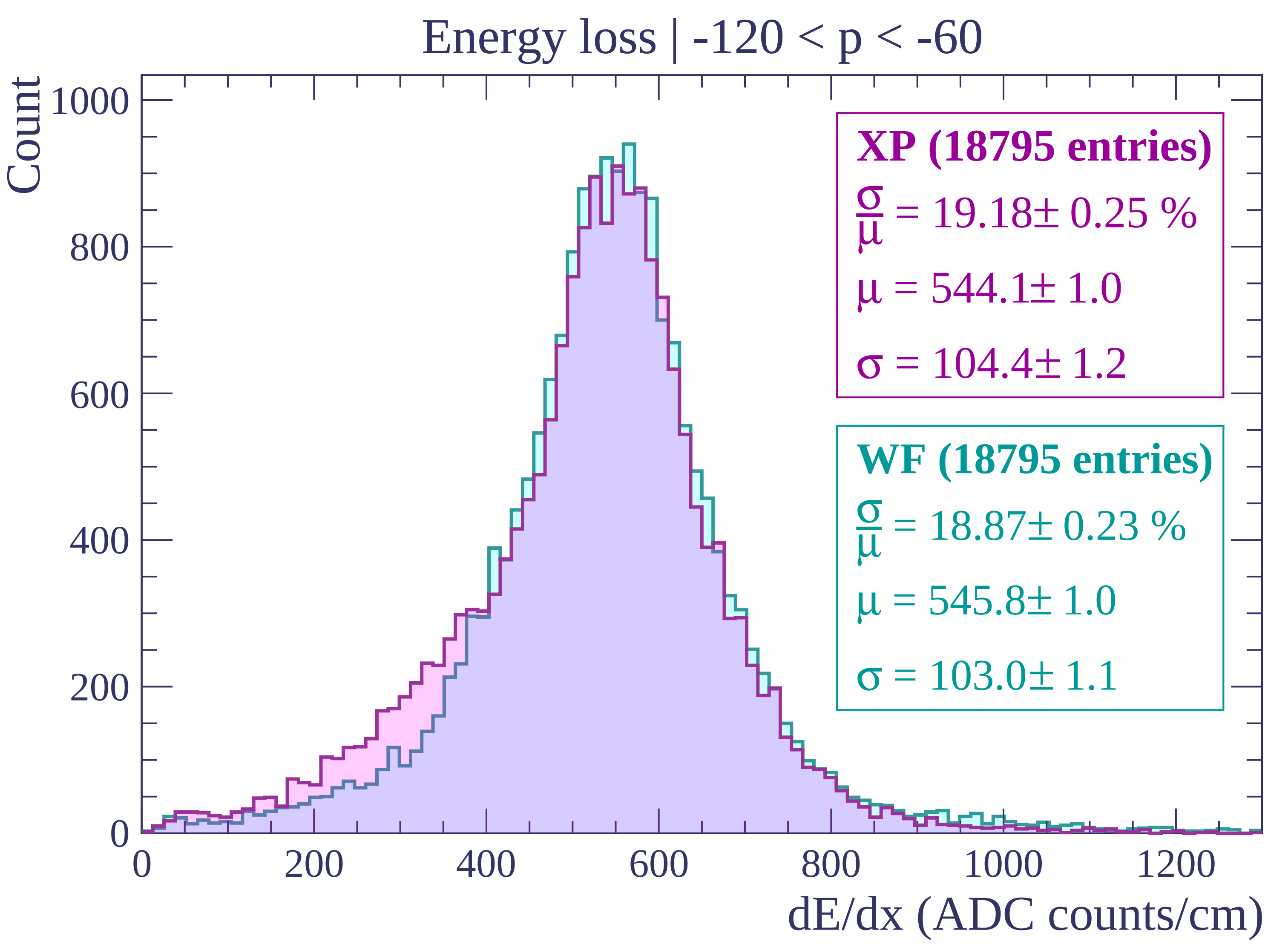

Energy loss | -300 < p < -240
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
0
200
400
600
800
1000
1200
XP (18379 entries)
σ/μ = 10.04± 0.11 %
μ = 404.5± 0.4
σ = 40.6± 0.4
WF (18379 entries)
σ/μ = 10.28± 0.12 %
μ = 403.4± 0.4
σ = 41.5± 0.4

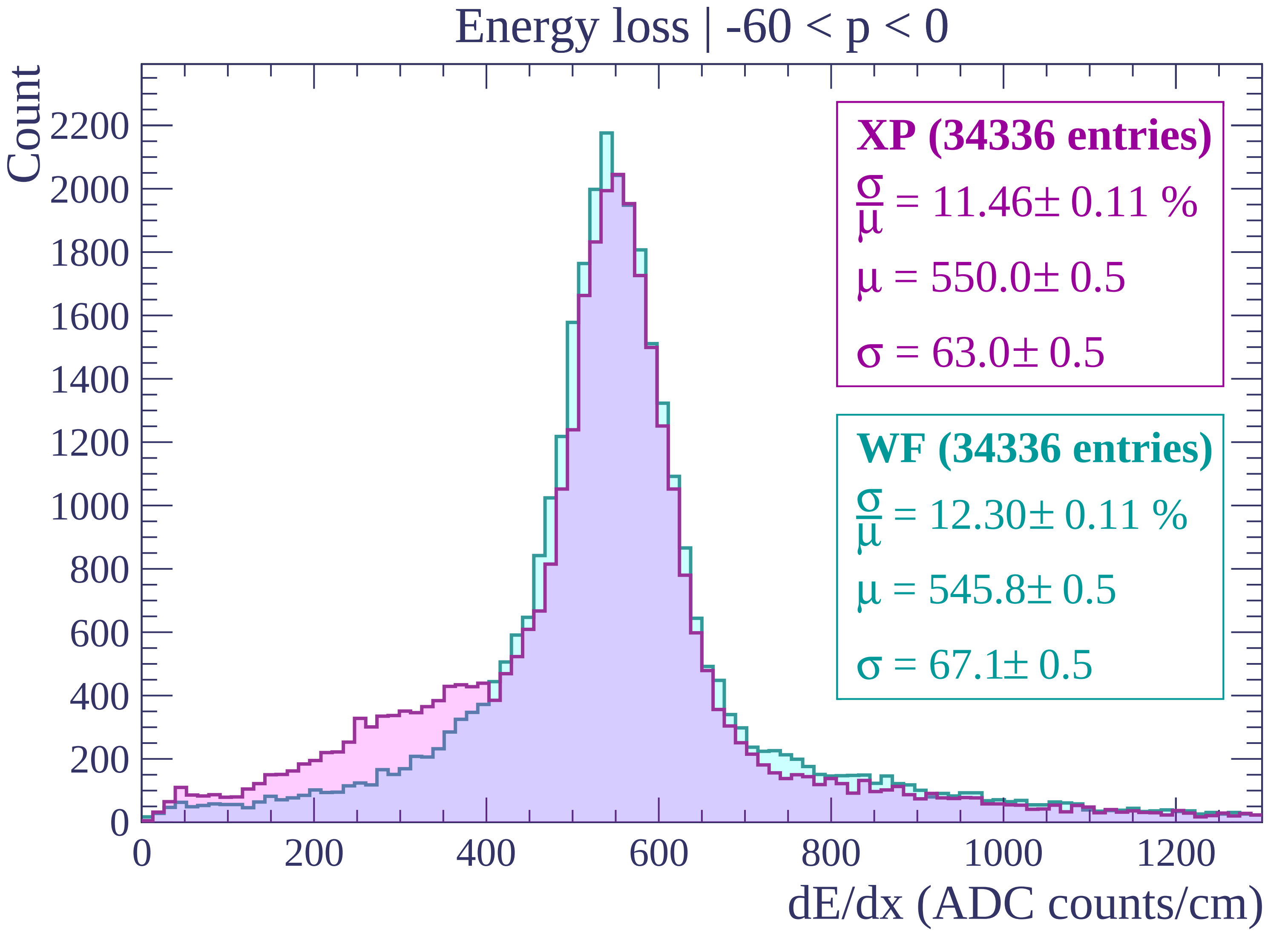


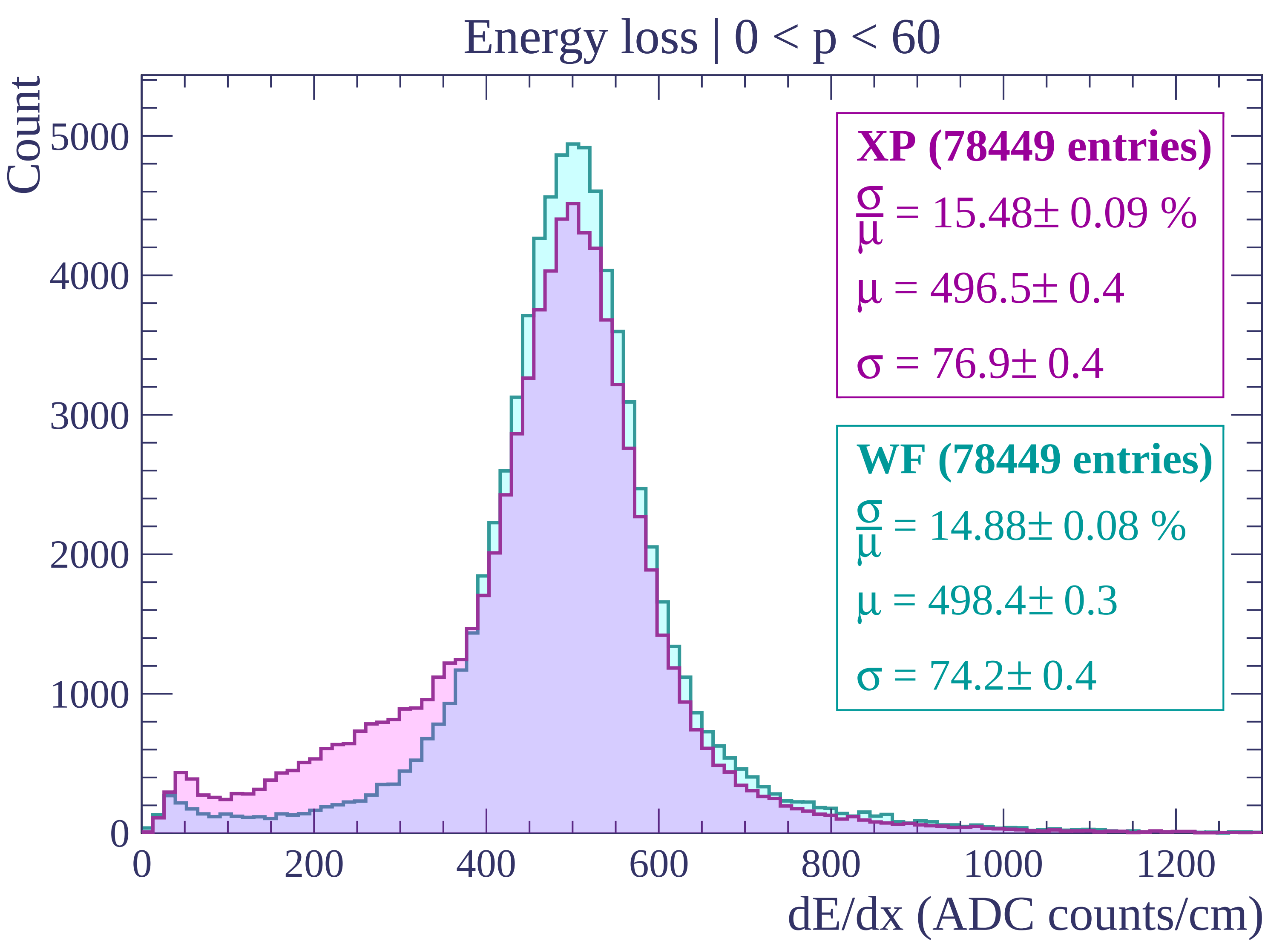
Energy loss | -240 < p < -180
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
0
200
400
600
800
1000
1200
XP (18412 entries)
σ/μ = 10.33± 0.11 %
μ = 414.4± 0.4
σ = 42.8± 0.4
WF (18412 entries)
σ/μ = 10.57± 0.12 %
μ = 413.7± 0.4
σ = 43.7± 0.4

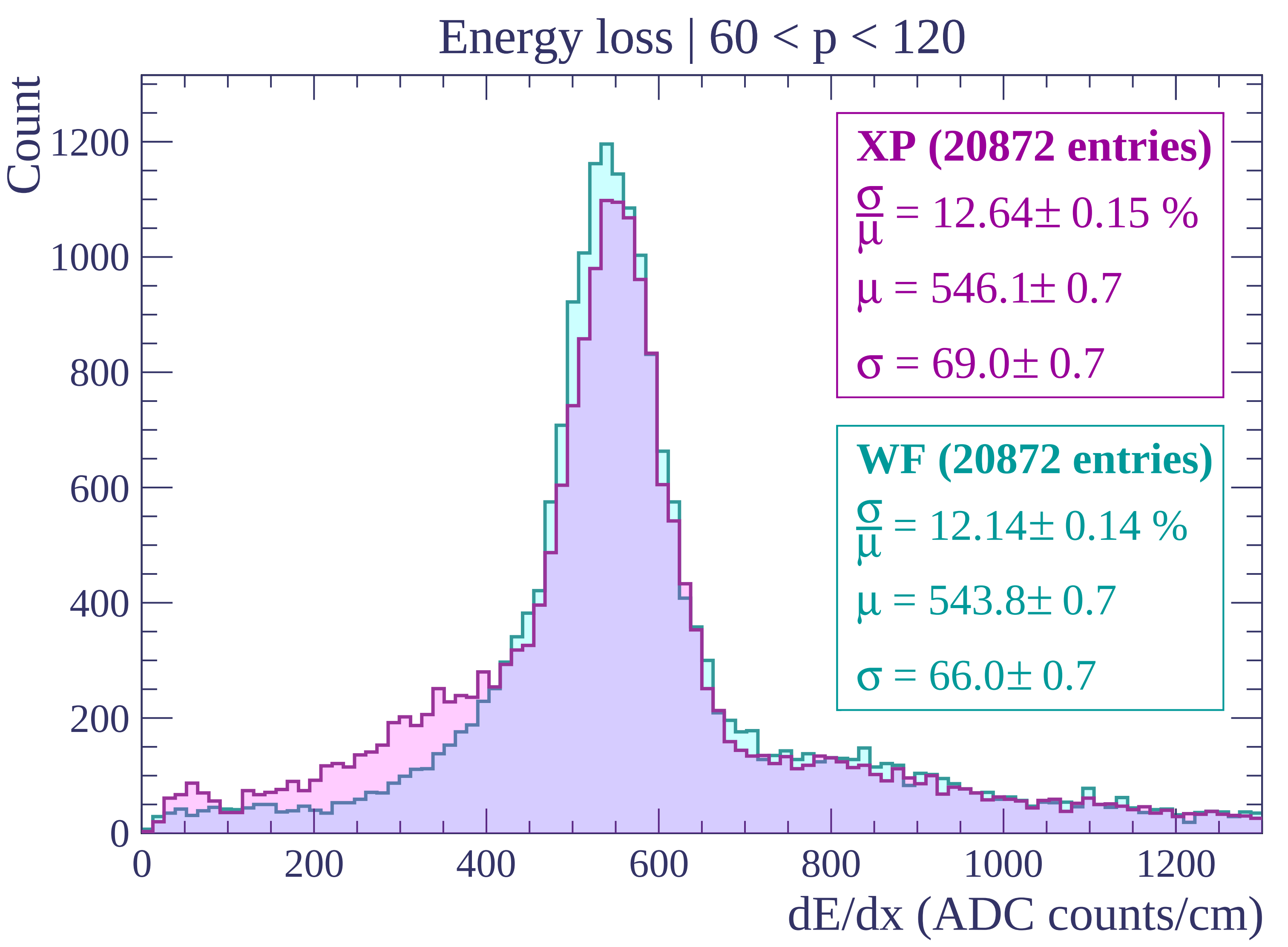


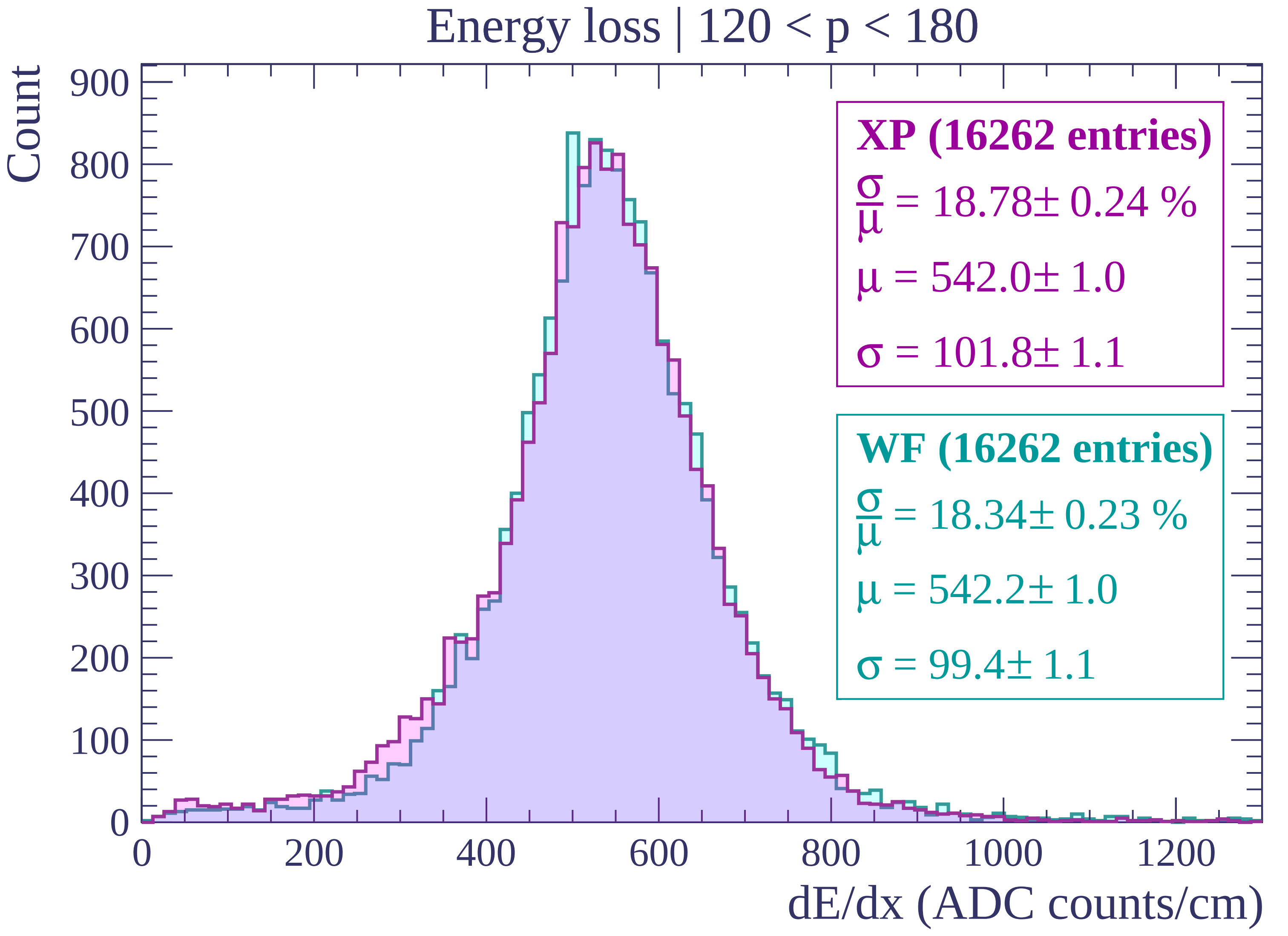
Energy loss | -180 < p < -120
Count
0
200
400
600
800
1000
1200
1400
1600
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (18406 entries)
σ/μ = 12.44± 0.14 %
μ = 449.0± 0.5
σ = 55.9± 0.6
WF (18406 entries)
σ/μ = 12.73± 0.15 %
μ = 449.0± 0.6
σ = 57.2± 0.6


Energy loss | -120 < p < -60
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (18795 entries)
$\frac{\sigma}{\mu}$ = 19.18± 0.25 %
μ = 544.1± 1.0
σ = 104.4± 1.2
WF (18795 entries)
$\frac{\sigma}{\mu}$ = 18.87± 0.23 %
μ = 545.8± 1.0
σ = 103.0± 1.1

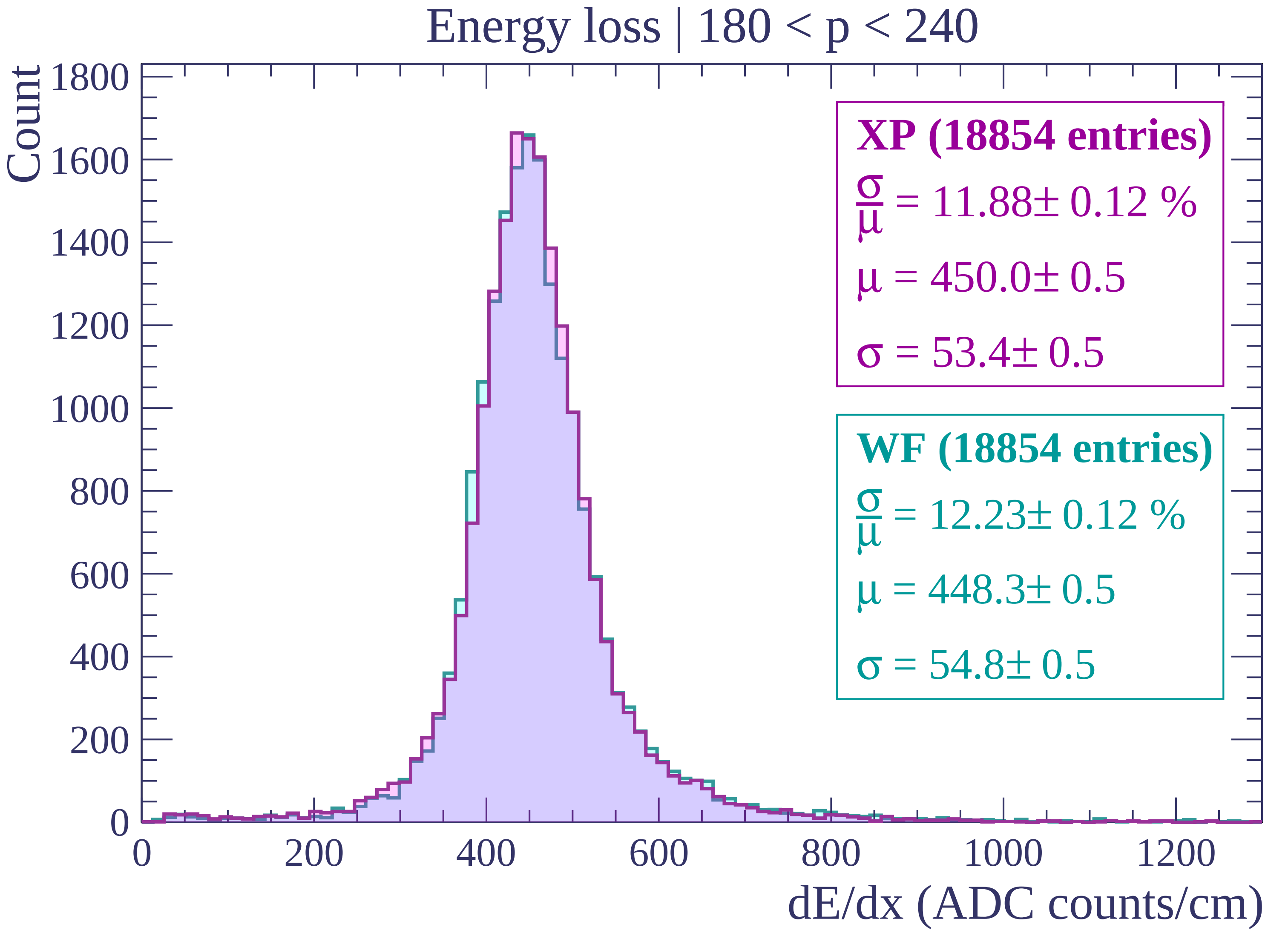

Energy loss | -60 < p < 0
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200
0
200
400
600
800
1000
1200
XP (34336 entries)
$\frac{\sigma}{\mu}$ = 11.46± 0.11 %
μ = 550.0± 0.5
σ = 63.0± 0.5
WF (34336 entries)
$\frac{\sigma}{\mu}$ = 12.30± 0.11 %
μ = 545.8± 0.5
σ = 67.1± 0.5


Energy loss | 0 < p < 60
Count
5000
4000
3000
2000
1000
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (78449 entries)
σ/μ = 15.48± 0.09 %
μ = 496.5± 0.4
σ = 76.9± 0.4
WF (78449 entries)
σ/μ = 14.88± 0.08 %
μ = 498.4± 0.3
σ = 74.2± 0.4

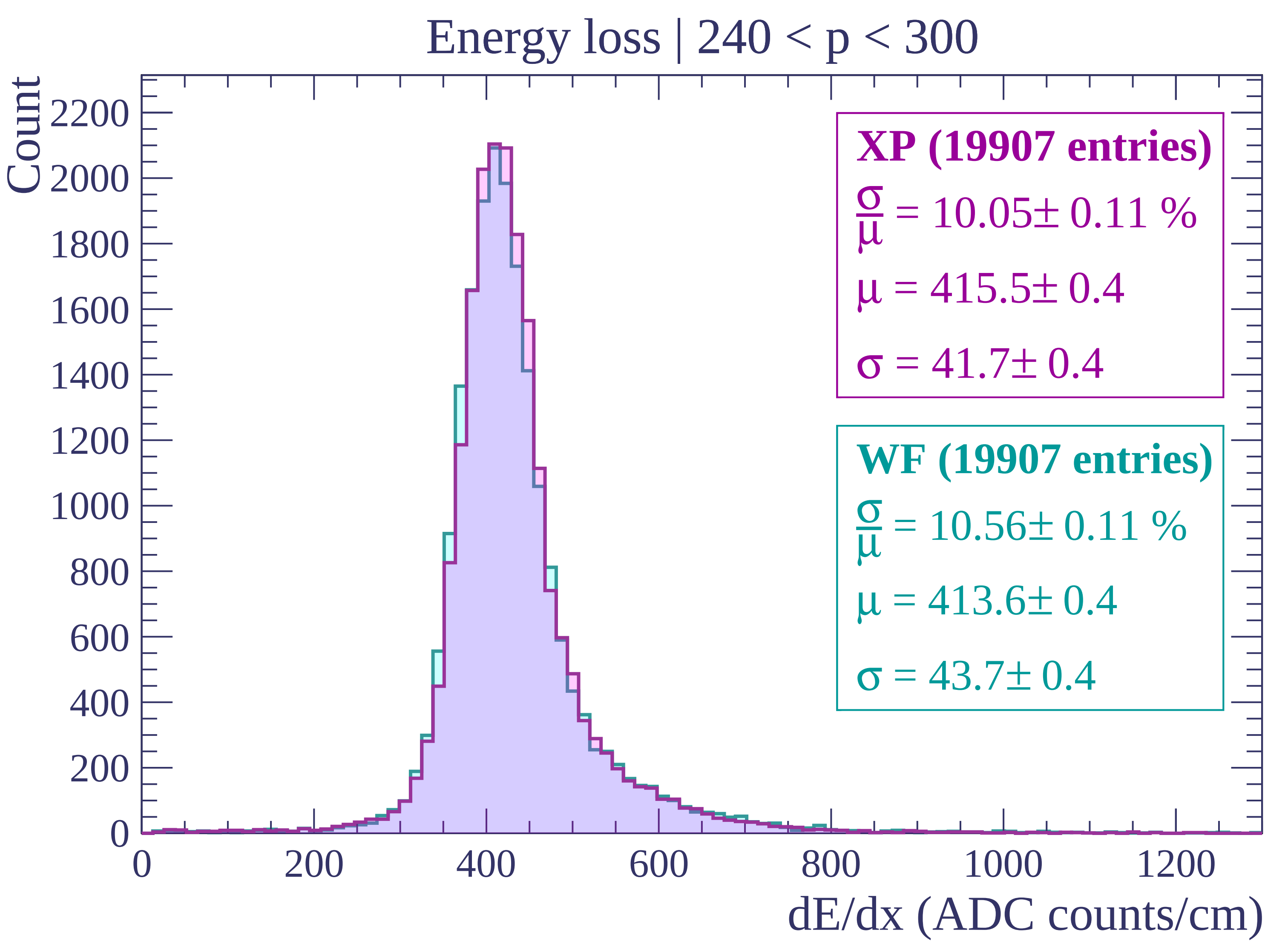

Energy loss | 60 < p < 120
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (20872 entries)
σ/μ = 12.64± 0.15 %
μ = 546.1± 0.7
σ = 69.0± 0.7
WF (20872 entries)
σ/μ = 12.14± 0.14 %
μ = 543.8± 0.7
σ = 66.0± 0.7



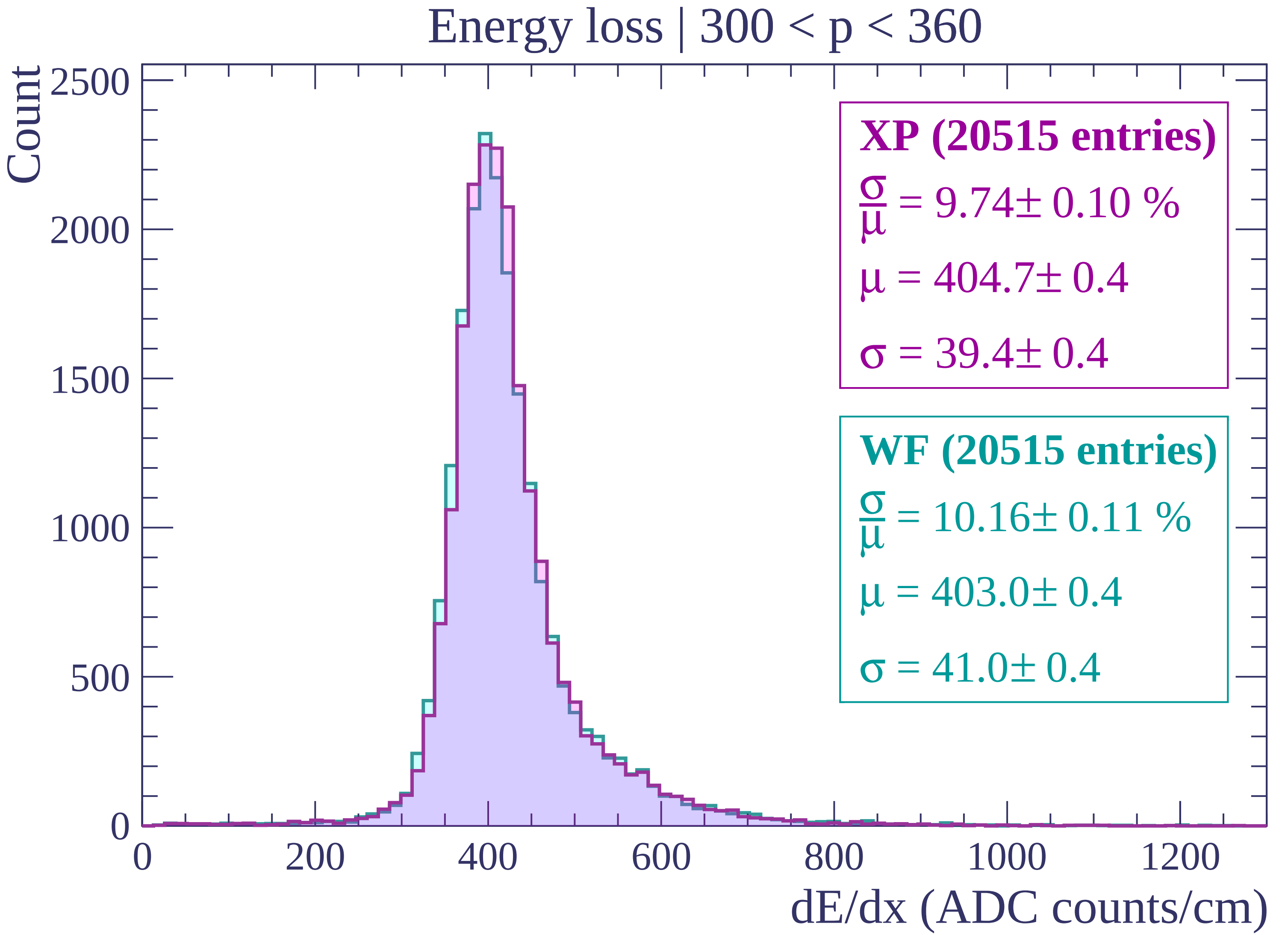
Energy loss | 120 < p < 180
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
700
900
XP (16262 entries)
σ/μ = 18.78± 0.24 %
μ = 542.0± 1.0
σ = 101.8± 1.1
WF (16262 entries)
σ/μ = 18.34± 0.23 %
μ = 542.2± 1.0
σ = 99.4± 1.1



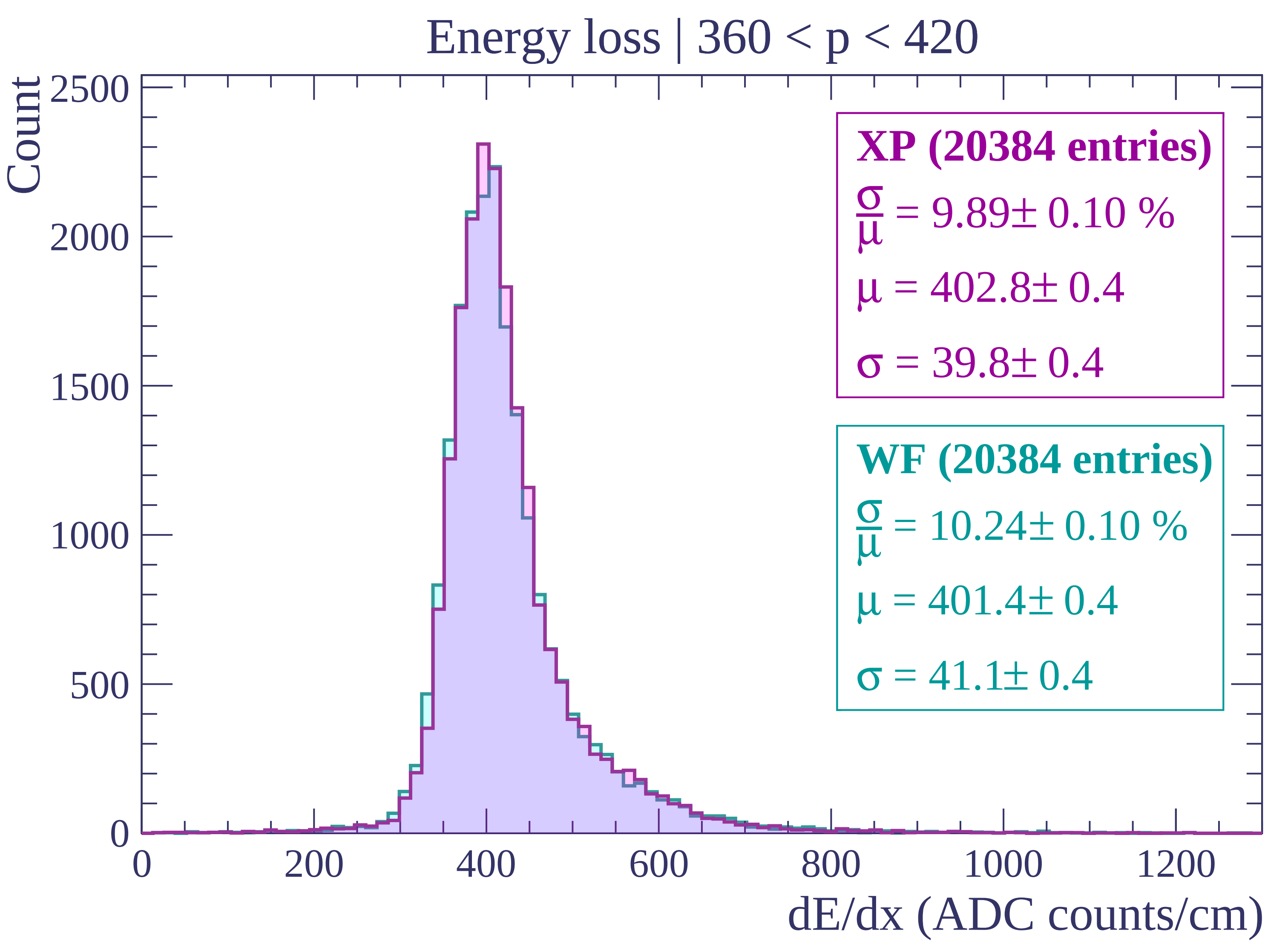
Energy loss | 180 < p < 240
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
0
200
400
600
800
1000
1200
XP (18854 entries)
σ/μ = 11.88± 0.12 %
μ = 450.0± 0.5
σ = 53.4± 0.5
WF (18854 entries)
σ/μ = 12.23± 0.12 %
μ = 448.3± 0.5
σ = 54.8± 0.5



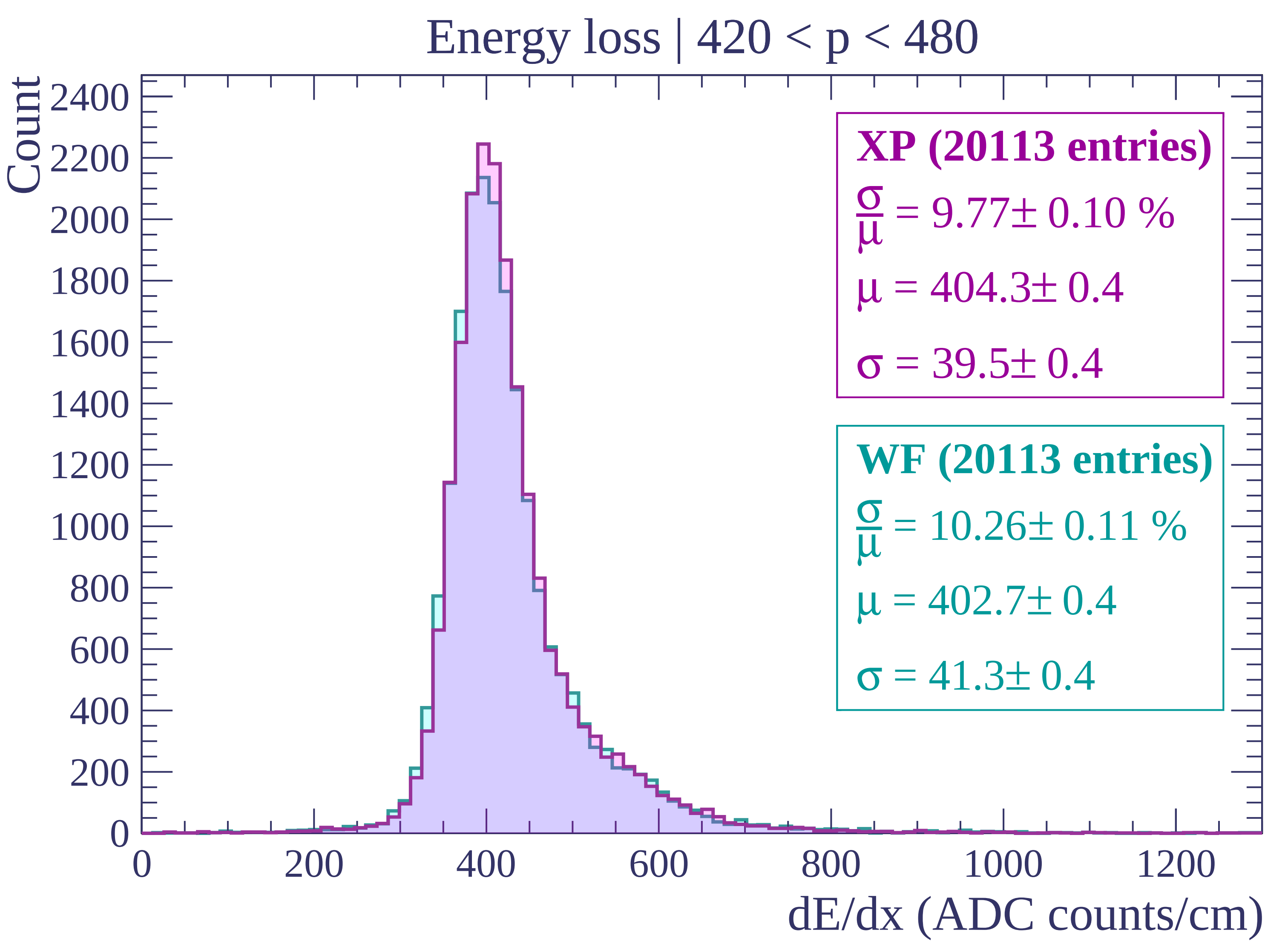
Energy loss | 240 < p < 300
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200
XP (19907 entries)
$\frac{\sigma}{\mu}$ = 10.05± 0.11 %
μ = 415.5± 0.4
σ = 41.7± 0.4
WF (19907 entries)
$\frac{\sigma}{\mu}$ = 10.56± 0.11 %
μ = 413.6± 0.4
σ = 43.7± 0.4

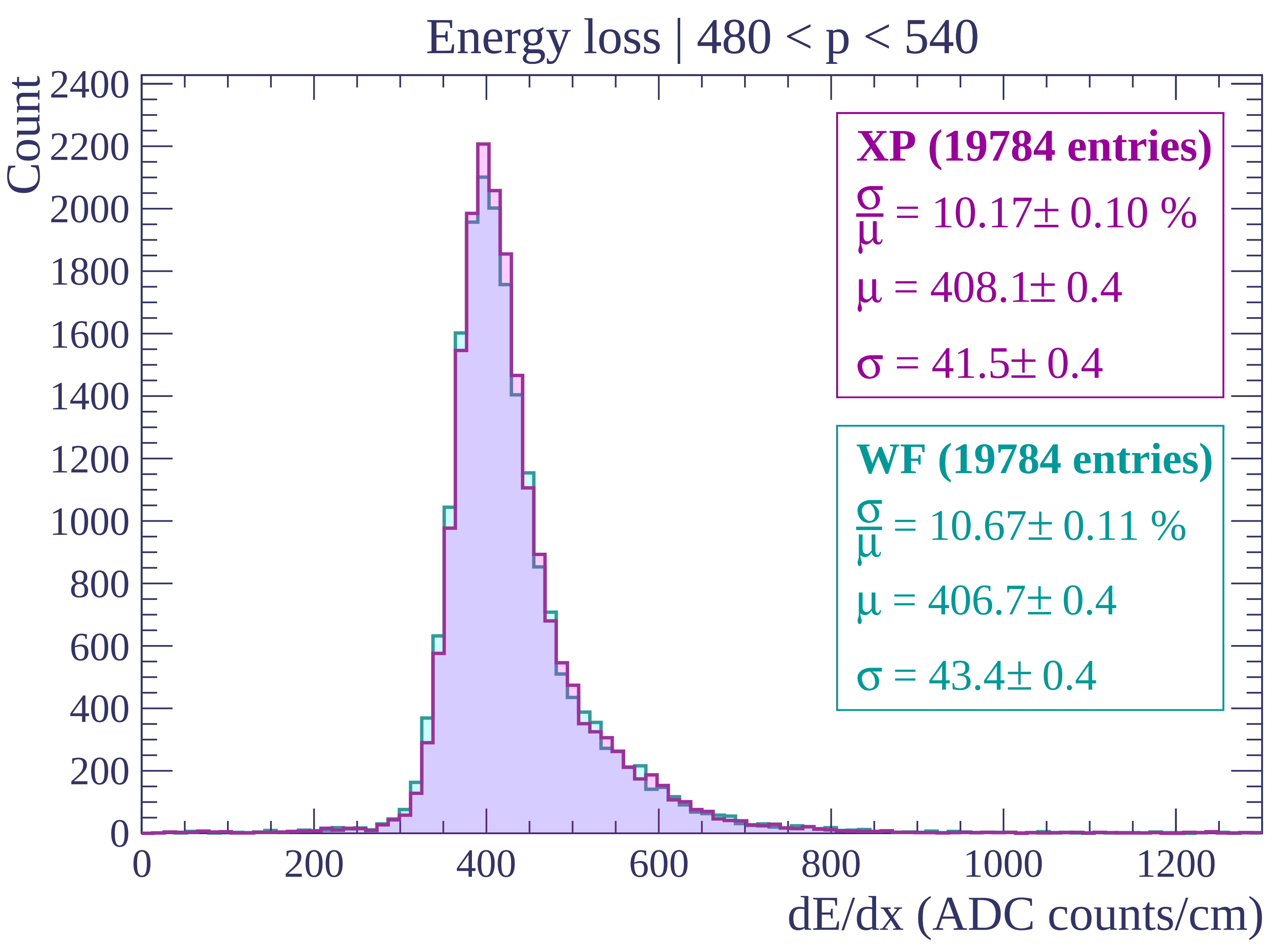


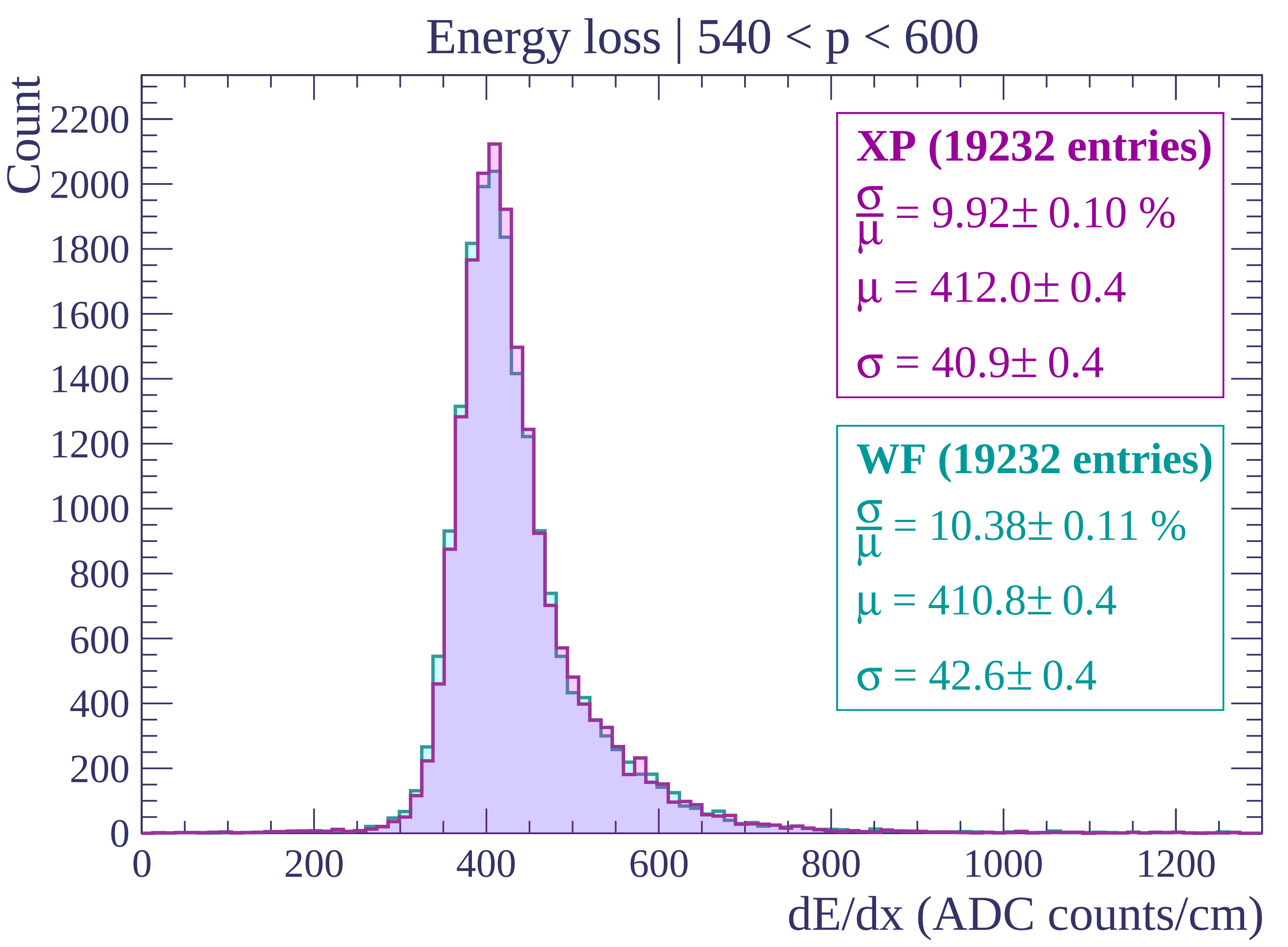
Energy loss | 300 < p < 360
Count
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (20515 entries)
$\frac{\sigma}{\mu}$ = 9.74± 0.10 %
μ = 404.7± 0.4
σ = 39.4± 0.4
WF (20515 entries)
$\frac{\sigma}{\mu}$ = 10.16± 0.11 %
μ = 403.0± 0.4
σ = 41.0± 0.4



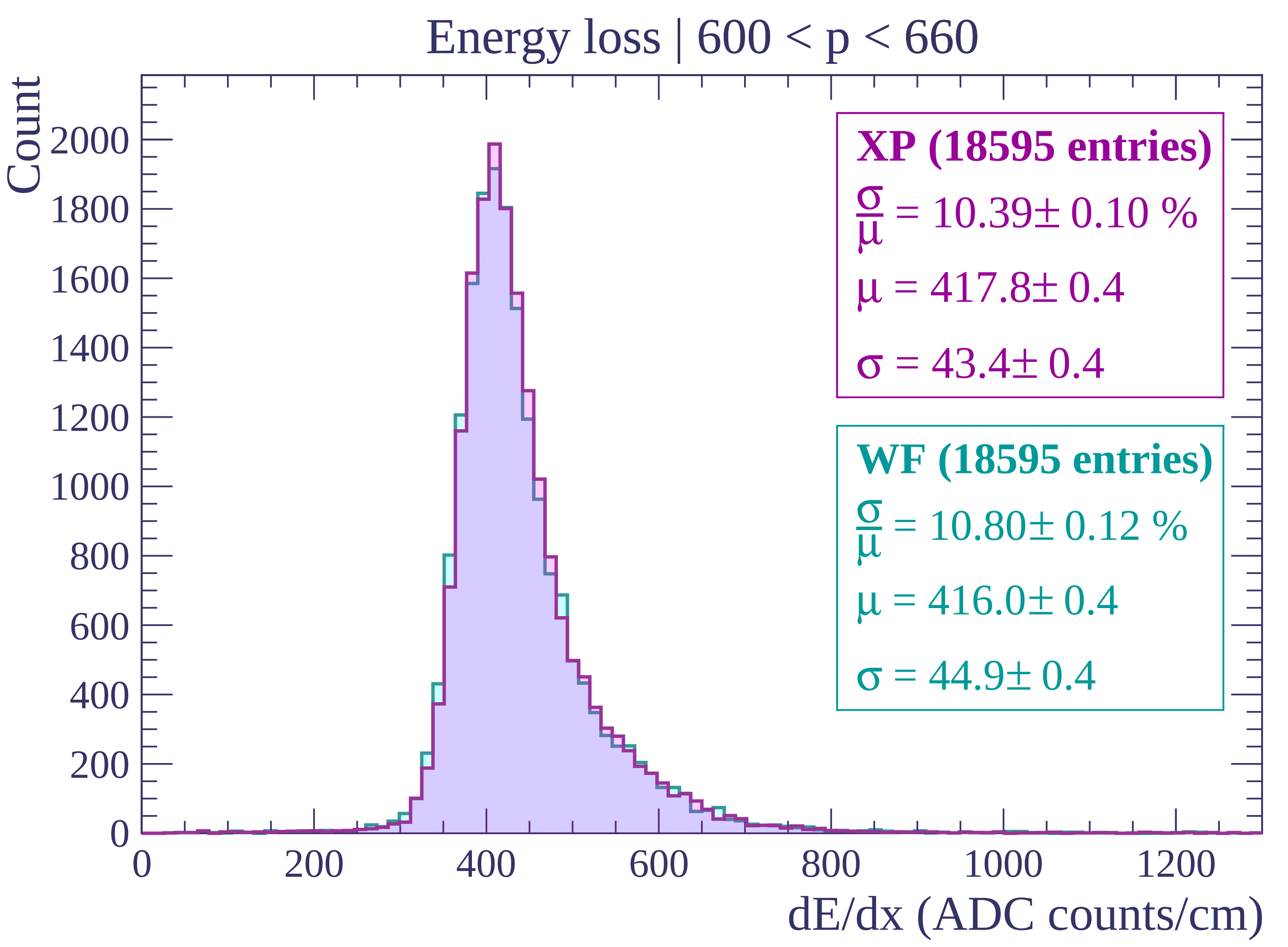
Energy loss | 360 < p < 420
Count
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (20384 entries)
$\frac{\sigma}{\mu}$ = 9.89± 0.10 %
μ = 402.8± 0.4
σ = 39.8± 0.4
WF (20384 entries)
$\frac{\sigma}{\mu}$ = 10.24± 0.10 %
μ = 401.4± 0.4
σ = 41.1± 0.4


Energy loss | 420 < p < 480
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200
2400
0
200
400
600
800
1000
1200
XP (20113 entries)
σ/μ = 9.77± 0.10 %
μ = 404.3± 0.4
σ = 39.5± 0.4
WF (20113 entries)
σ/μ = 10.26± 0.11 %
μ = 402.7± 0.4
σ = 41.3± 0.4

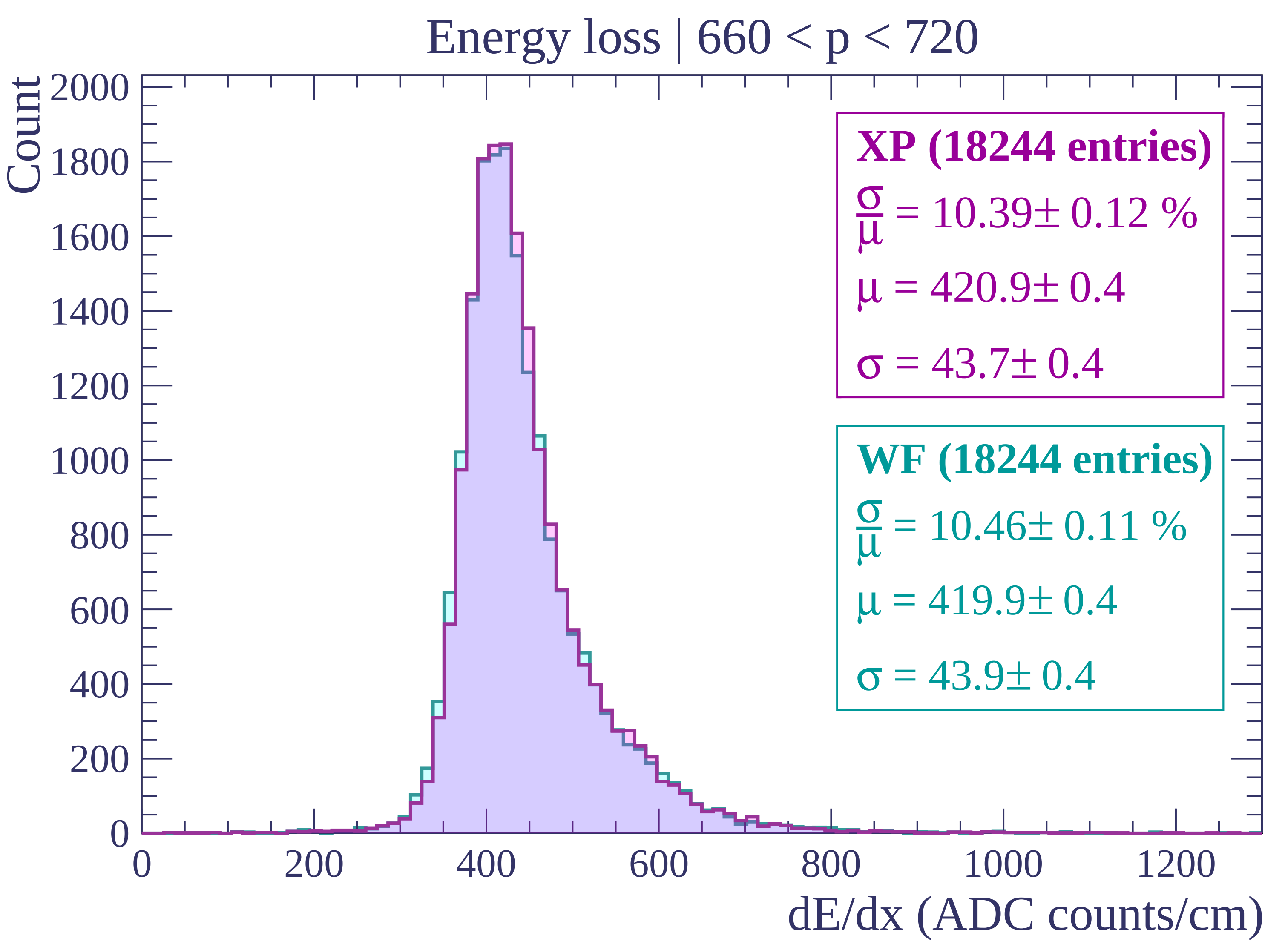

Energy loss | 480 < p < 540
Count
dE/dx (ADC counts/cm)
XP (19784 entries)
σ/μ = 10.17± 0.10 %
μ = 408.1± 0.4
σ = 41.5± 0.4
WF (19784 entries)
σ/μ = 10.67± 0.11 %
μ = 406.7± 0.4
σ = 43.4± 0.4
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200
2400



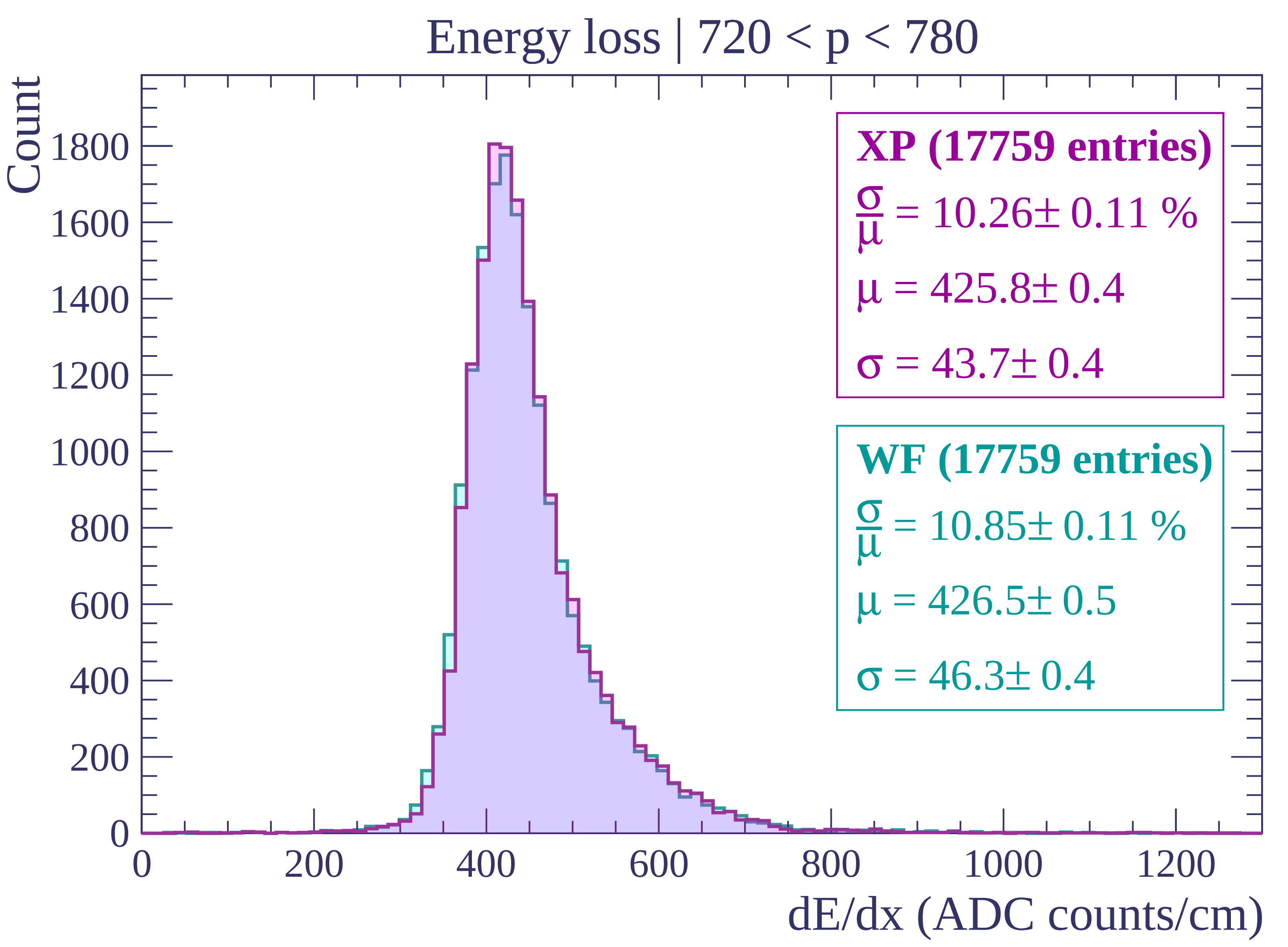
Energy loss | 540 < p < 600
Count
dE/dx (ADC counts/cm)
XP (19232 entries)
$\frac{\sigma}{\mu}$ = 9.92± 0.10 %
μ = 412.0± 0.4
σ = 40.9± 0.4
WF (19232 entries)
$\frac{\sigma}{\mu}$ = 10.38± 0.11 %
μ = 410.8± 0.4
σ = 42.6± 0.4
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200

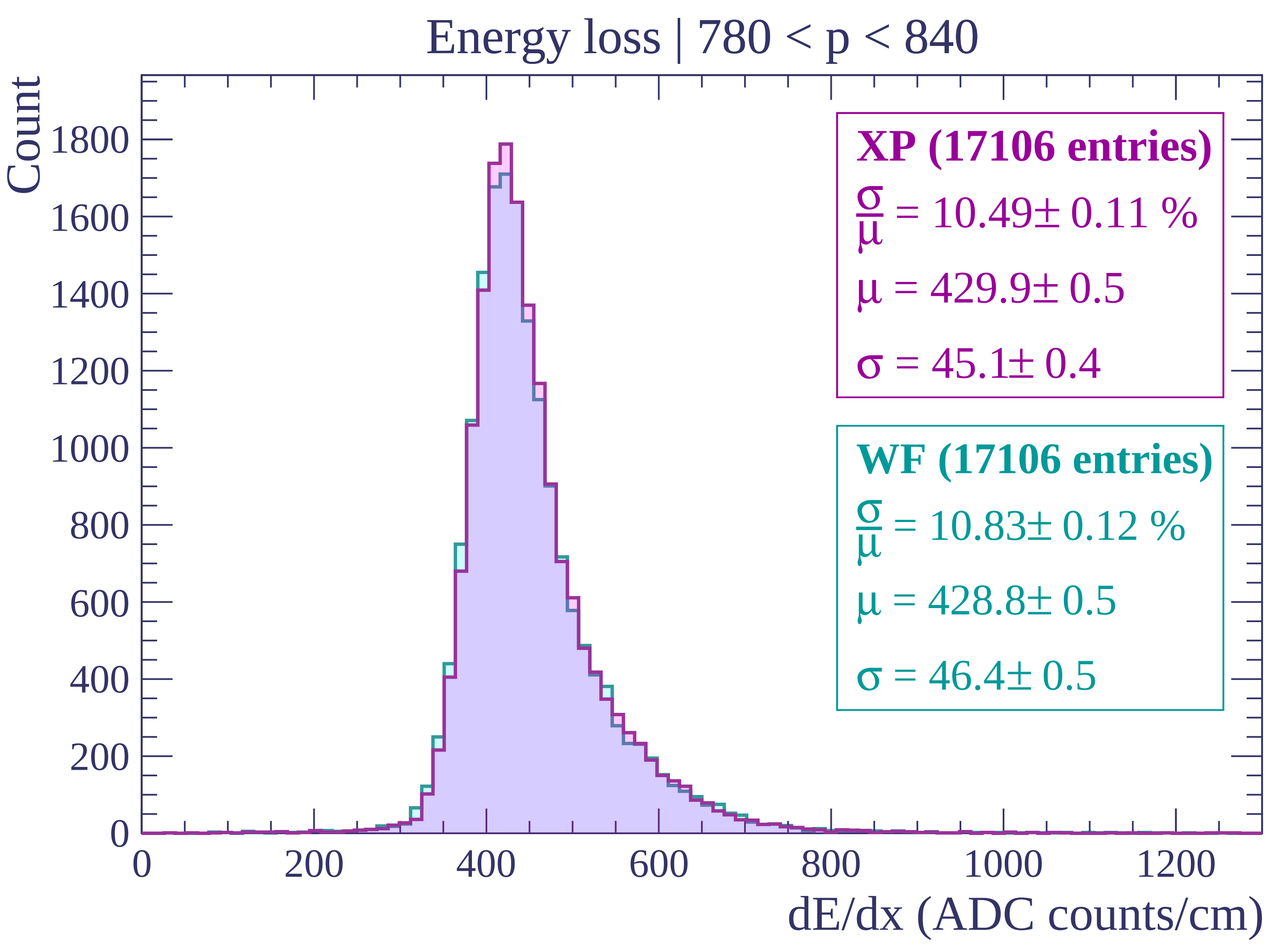


Energy loss | 600 < p < 660
Count
dE/dx (ADC counts/cm)
XP (18595 entries)
$\frac{\sigma}{\mu}$ = 10.39± 0.10 %
μ = 417.8± 0.4
σ = 43.4± 0.4
WF (18595 entries)
$\frac{\sigma}{\mu}$ = 10.80± 0.12 %
μ = 416.0± 0.4
σ = 44.9± 0.4

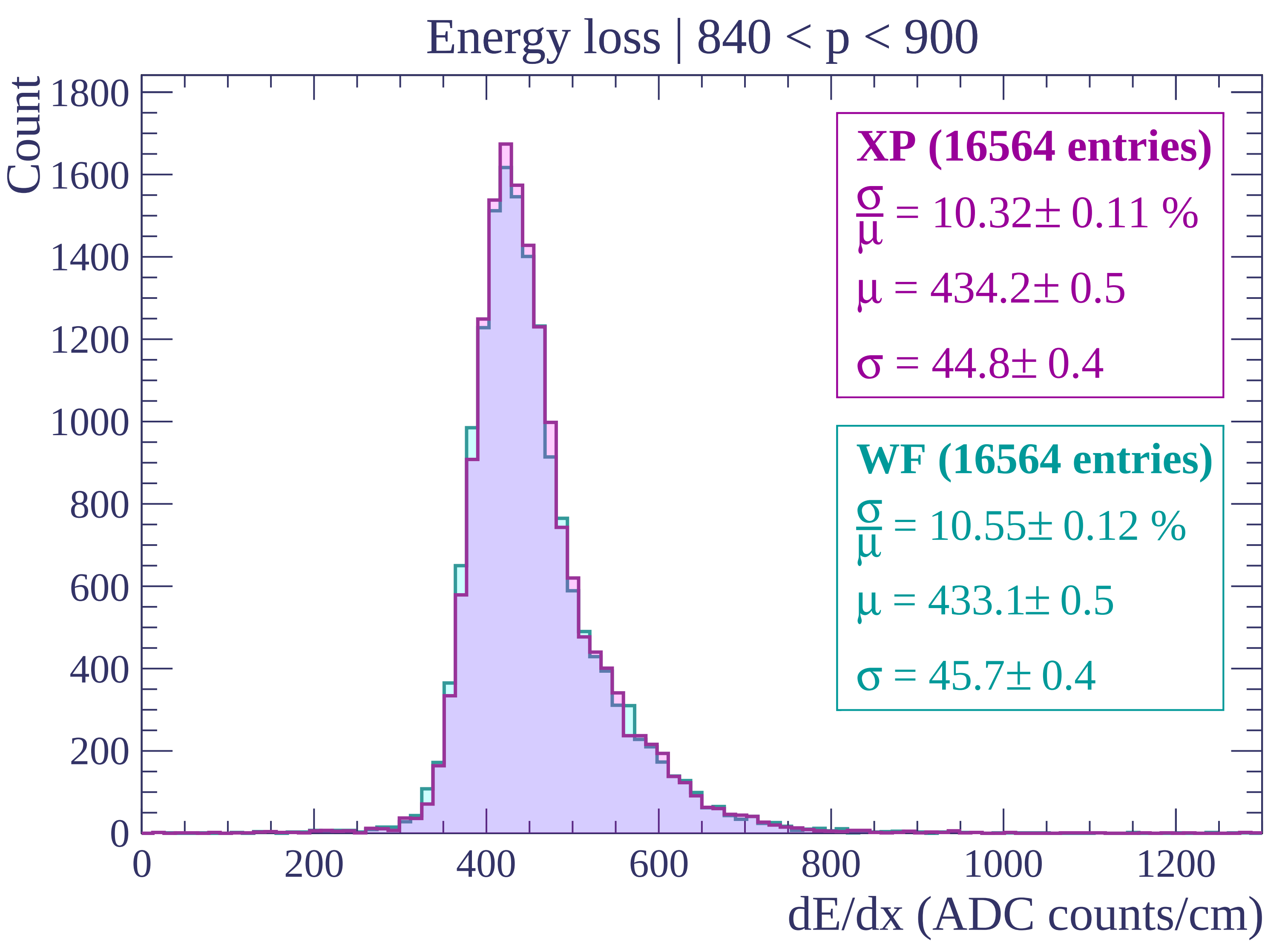


Energy loss | 660 < p < 720
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
0
200
400
600
800
1000
1200
XP (18244 entries)
$\frac{\sigma}{\mu}$ = 10.39± 0.12 %
μ = 420.9± 0.4
σ = 43.7± 0.4
WF (18244 entries)
$\frac{\sigma}{\mu}$ = 10.46± 0.11 %
μ = 419.9± 0.4
σ = 43.9± 0.4

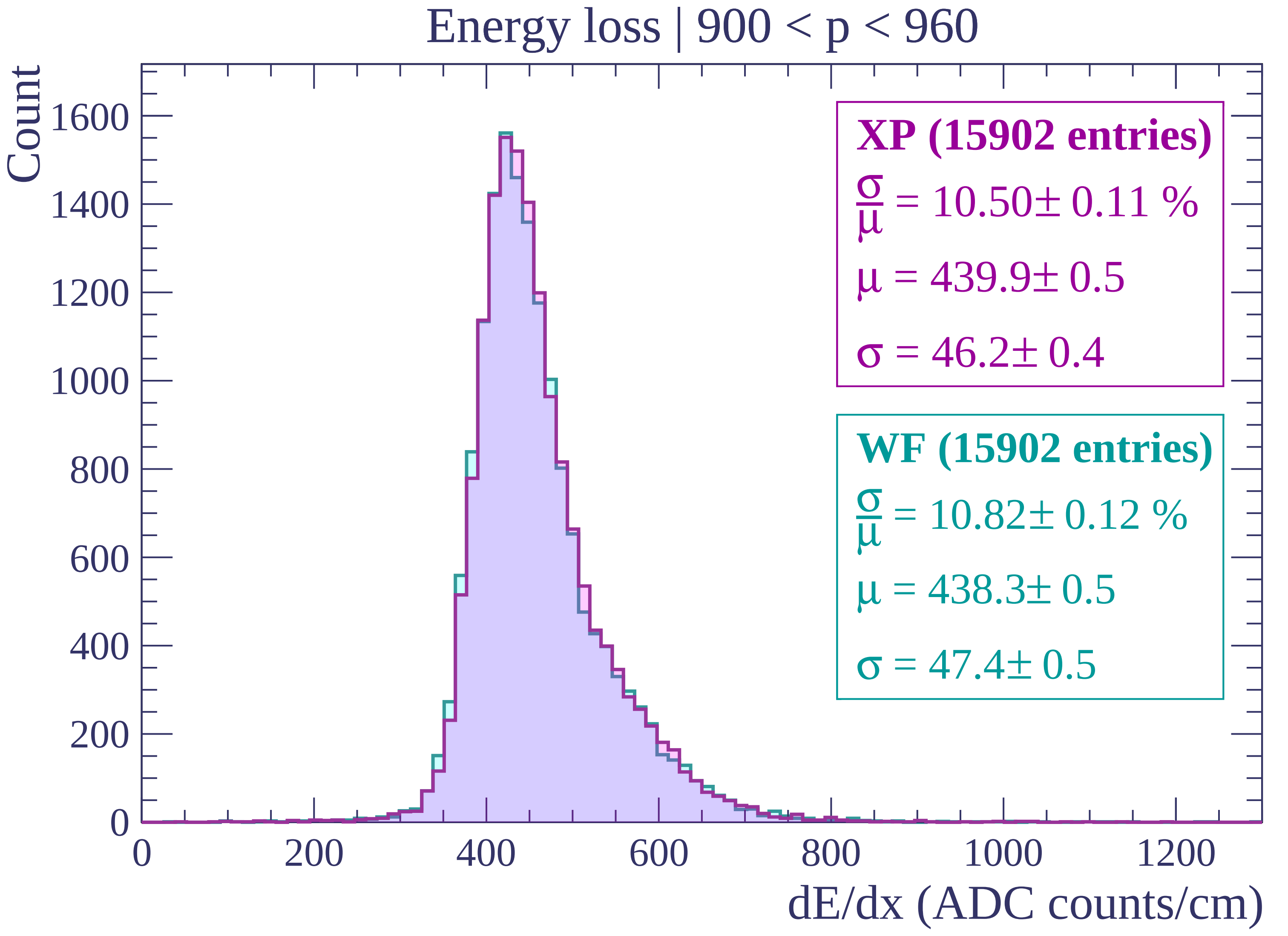

Energy loss | 720 < p < 780
Count
1800
1600
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (17759 entries)
$\frac{\sigma}{\mu}$ = 10.26± 0.11 %
μ = 425.8± 0.4
σ = 43.7± 0.4
WF (17759 entries)
$\frac{\sigma}{\mu}$ = 10.85± 0.11 %
μ = 426.5± 0.5
σ = 46.3± 0.4

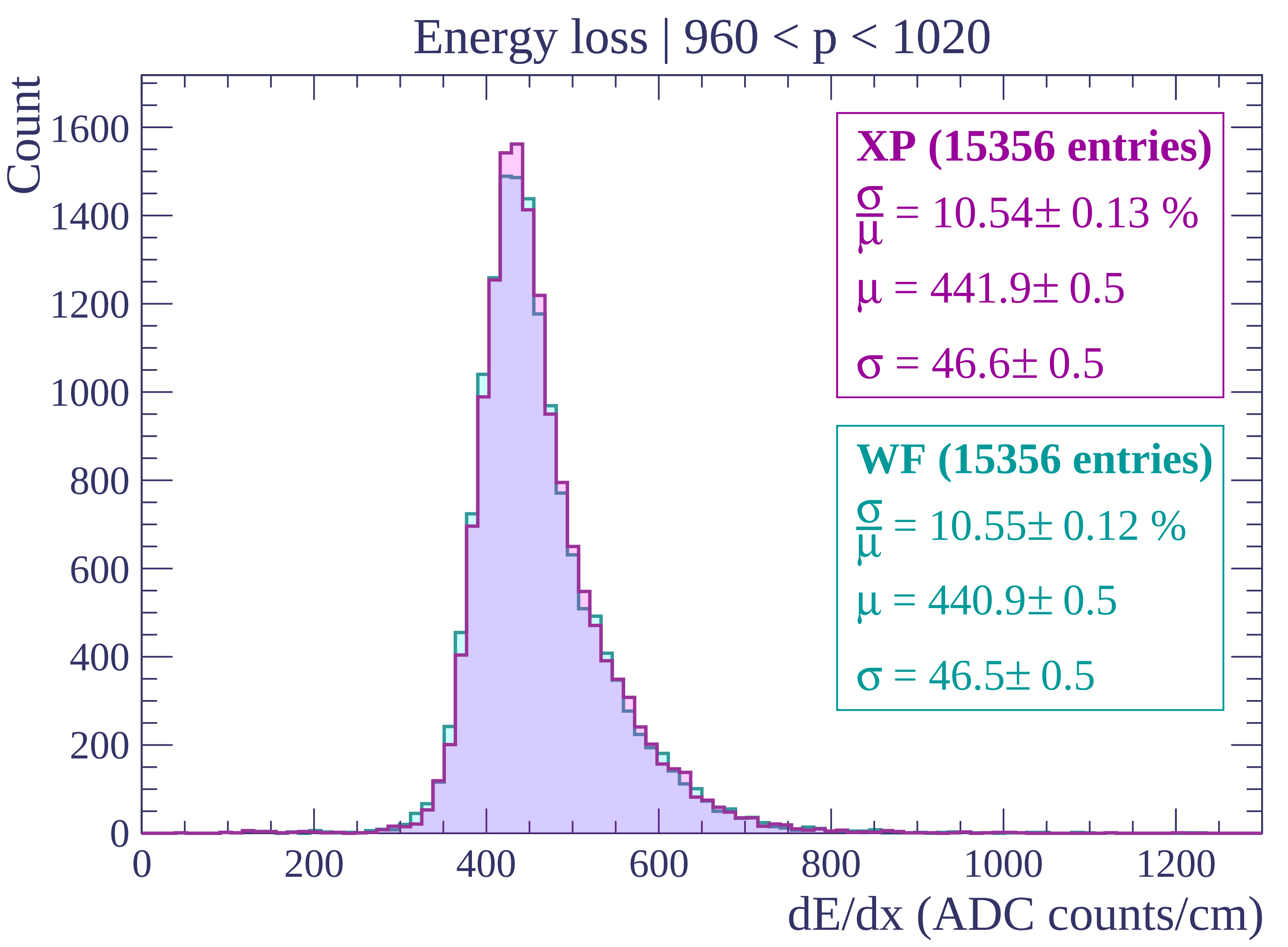


Energy loss | 780 < p < 840
Count
0
200
400
600
800
1000
1200
1400
1600
1800
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (17106 entries)
σ/μ = 10.49± 0.11 %
μ = 429.9± 0.5
σ = 45.1± 0.4
WF (17106 entries)
σ/μ = 10.83± 0.12 %
μ = 428.8± 0.5
σ = 46.4± 0.5

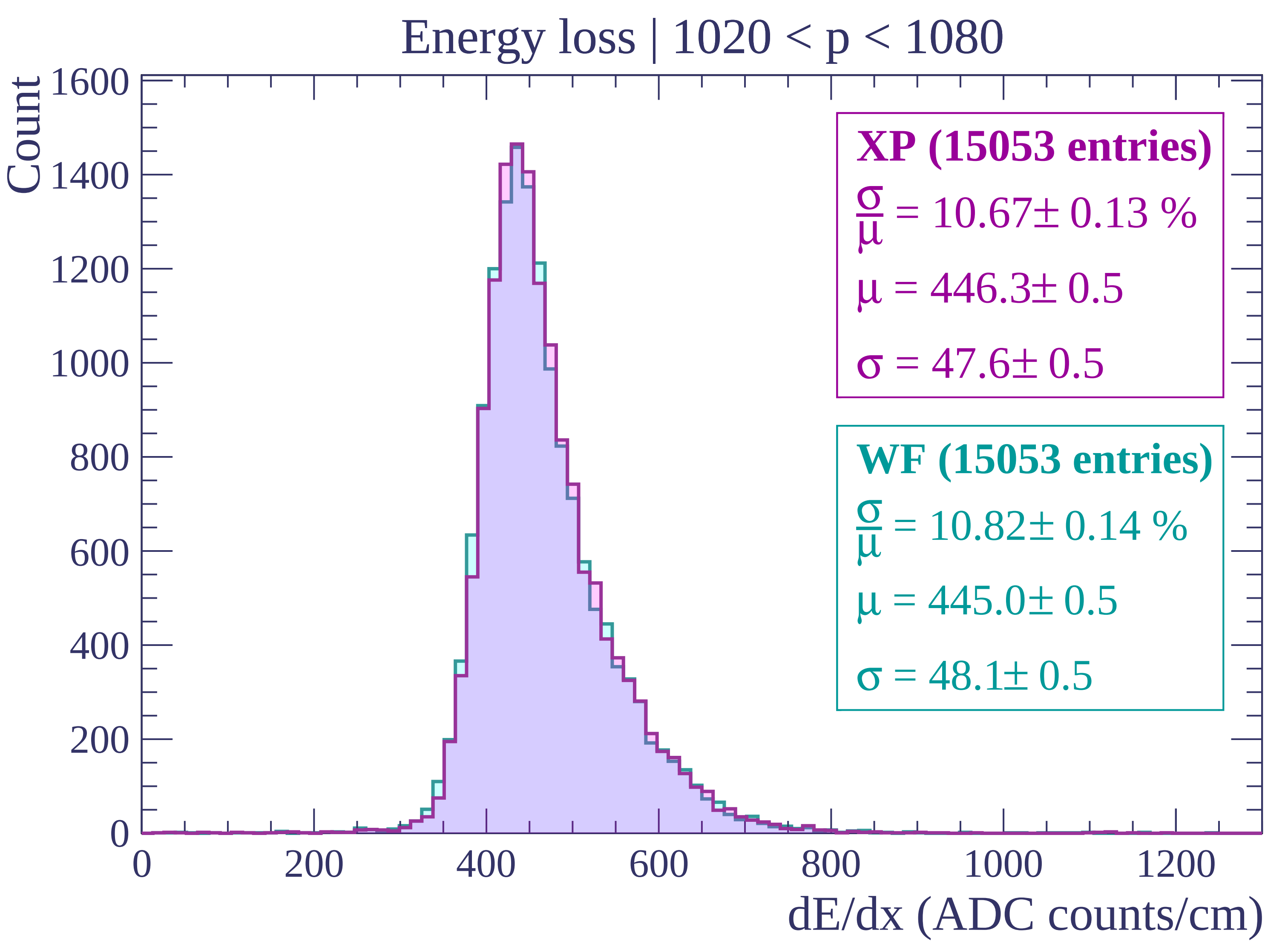


Energy loss | 840 < p < 900
Count
dE/dx (ADC counts/cm)
XP (16564 entries)
$\frac{\sigma}{\mu}$ = 10.32± 0.11 %
μ = 434.2± 0.5
σ = 44.8± 0.4
WF (16564 entries)
$\frac{\sigma}{\mu}$ = 10.55± 0.12 %
μ = 433.1± 0.5
σ = 45.7± 0.4

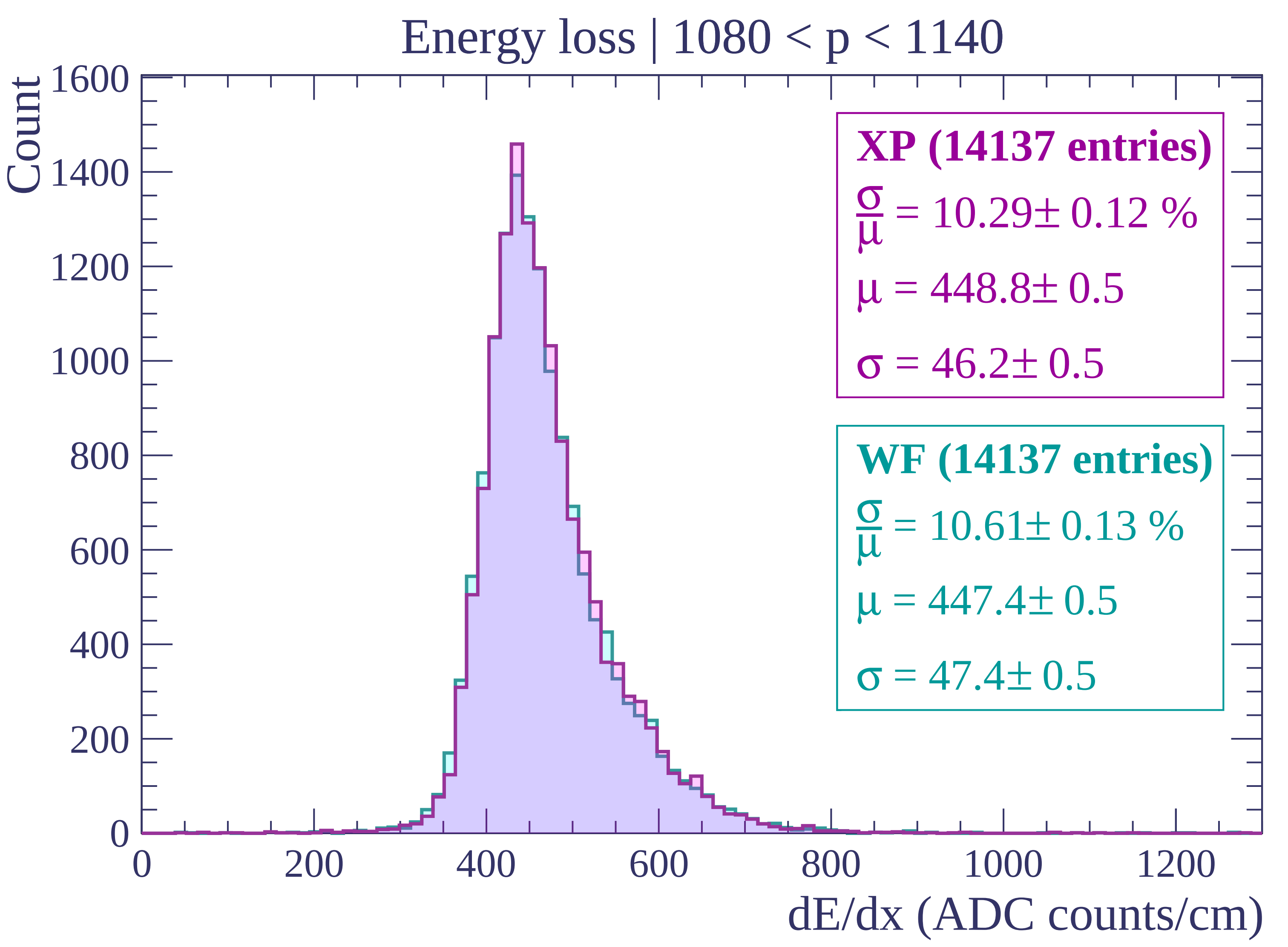

Energy loss | 900 < p < 960
Count
dE/dx (ADC counts/cm)
XP (15902 entries)
$\frac{\sigma}{\mu}$ = 10.50± 0.11 %
μ = 439.9± 0.5
σ = 46.2± 0.4
WF (15902 entries)
$\frac{\sigma}{\mu}$ = 10.82± 0.12 %
μ = 438.3± 0.5
σ = 47.4± 0.5



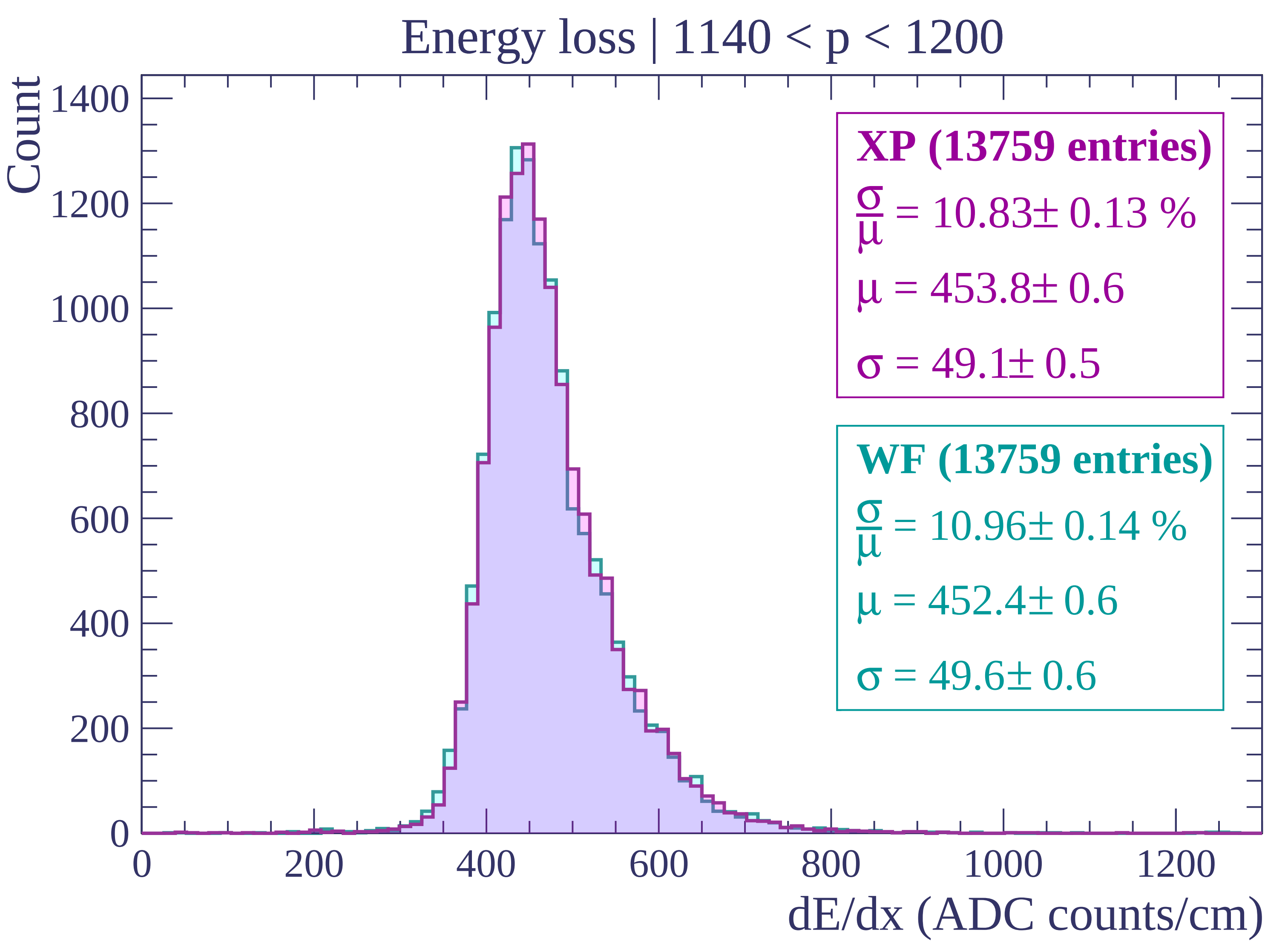
Energy loss | 960 < p < 1020
Count
dE/dx (ADC counts/cm)
XP (15356 entries)
σ/μ = 10.54± 0.13 %
μ = 441.9± 0.5
σ = 46.6± 0.5
WF (15356 entries)
σ/μ = 10.55± 0.12 %
μ = 440.9± 0.5
σ = 46.5± 0.5
0
200
400
600
800
1000
1200
1400
1600



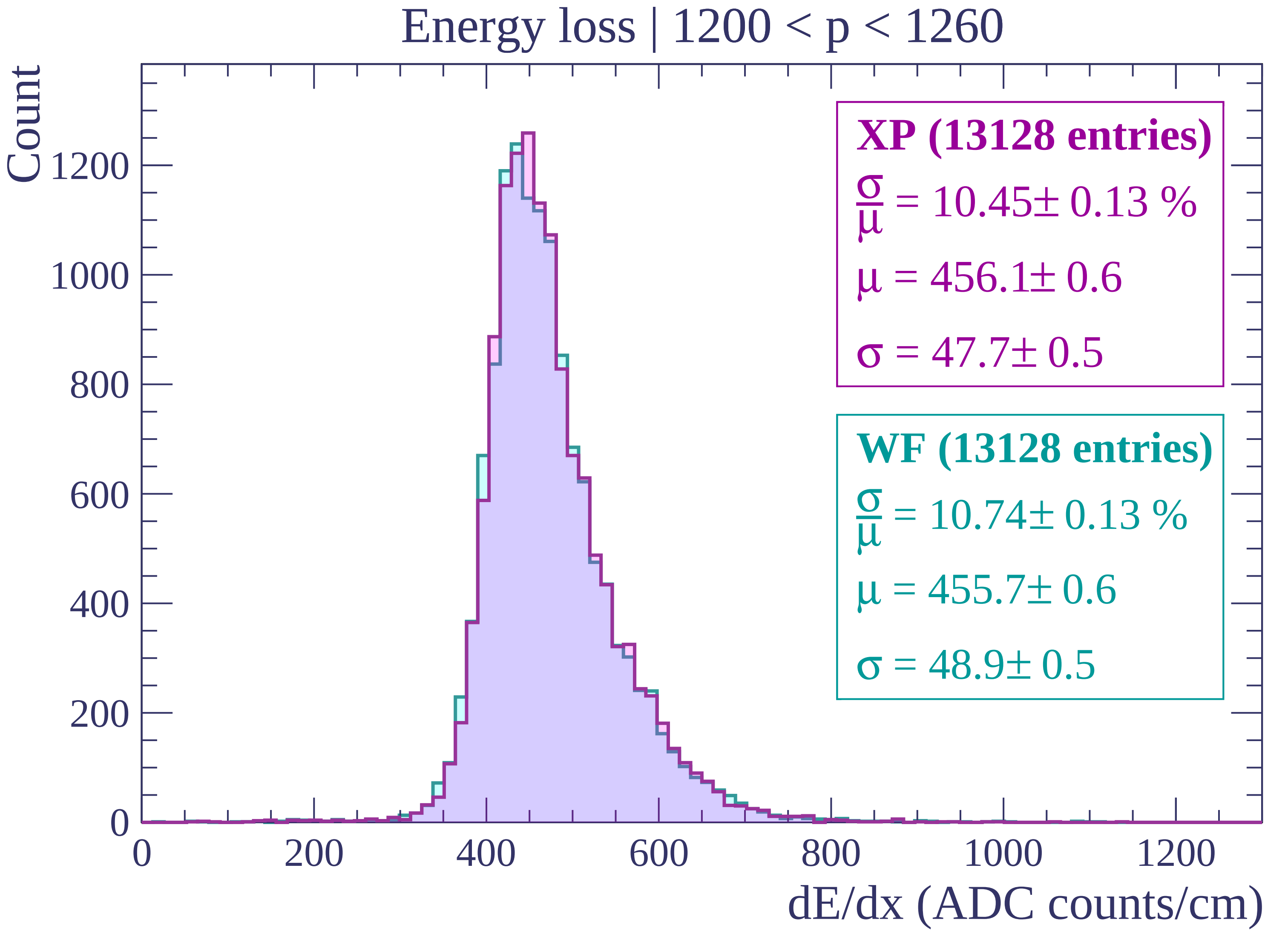
Energy loss | 1020 < p < 1080
Count
dE/dx (ADC counts/cm)
XP (15053 entries)
σ/μ = 10.67± 0.13 %
μ = 446.3± 0.5
σ = 47.6± 0.5
WF (15053 entries)
σ/μ = 10.82± 0.14 %
μ = 445.0± 0.5
σ = 48.1± 0.5

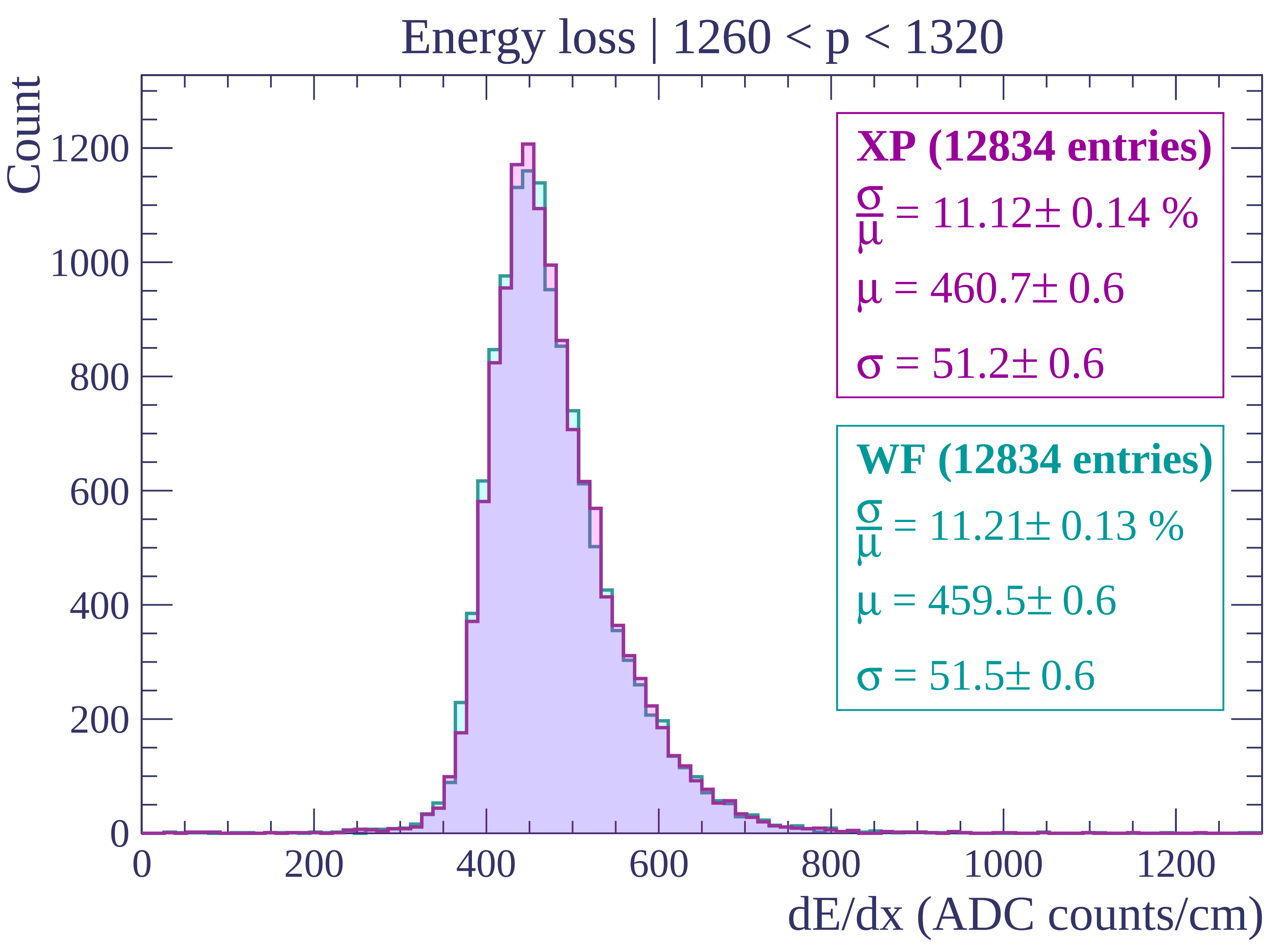


Energy loss | 1080 < p < 1140
Count
dE/dx (ADC counts/cm)
XP (14137 entries)
$\frac{\sigma}{\mu}$ = 10.29± 0.12 %
μ = 448.8± 0.5
σ = 46.2± 0.5
WF (14137 entries)
$\frac{\sigma}{\mu}$ = 10.61± 0.13 %
μ = 447.4± 0.5
σ = 47.4± 0.5

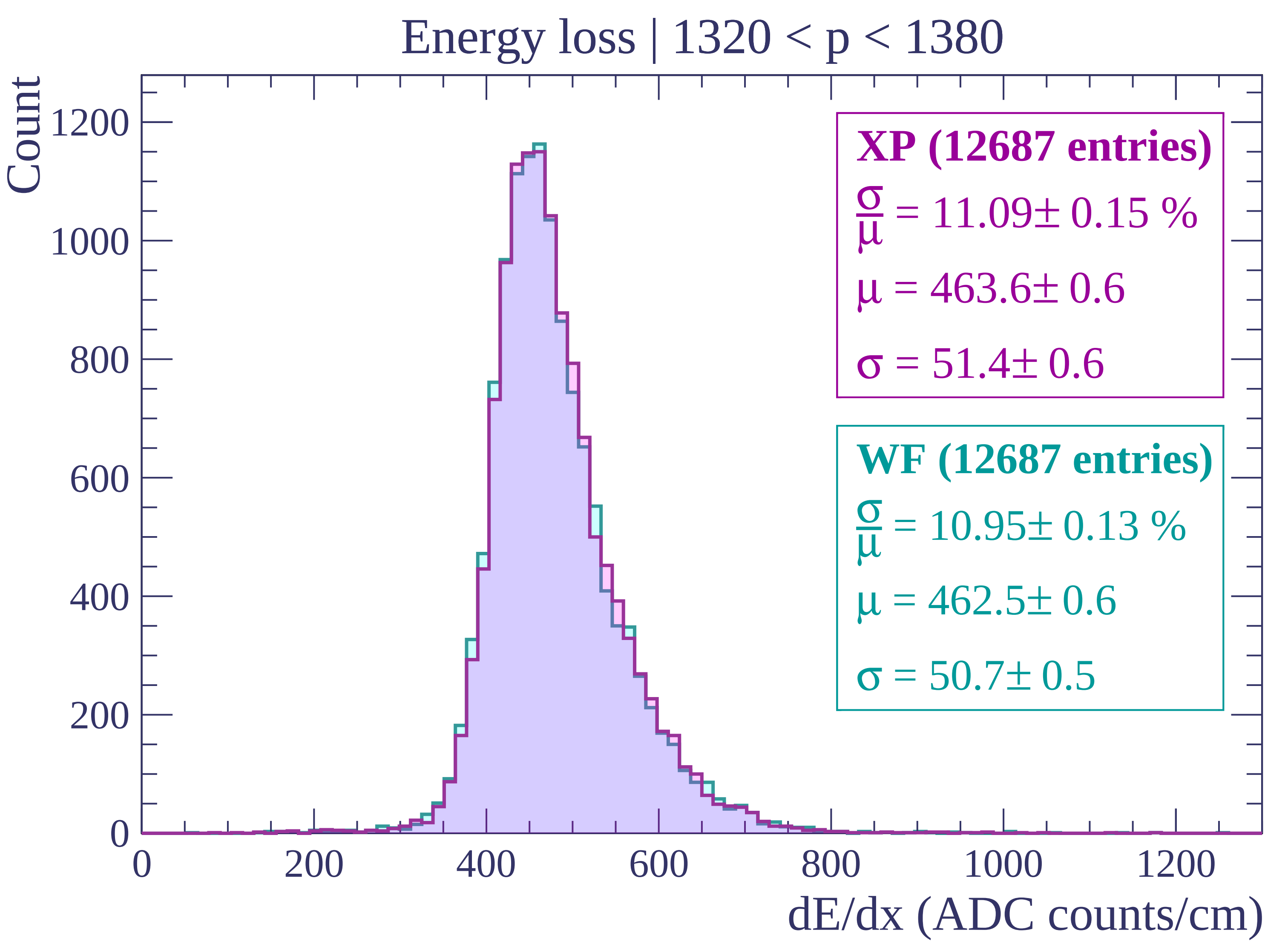


Energy loss | 1140 < p < 1200
Count
dE/dx (ADC counts/cm)
XP (13759 entries)
$\frac{\sigma}{\mu}$ = 10.83± 0.13 %
μ = 453.8± 0.6
σ = 49.1± 0.5
WF (13759 entries)
$\frac{\sigma}{\mu}$ = 10.96± 0.14 %
μ = 452.4± 0.6
σ = 49.6± 0.6

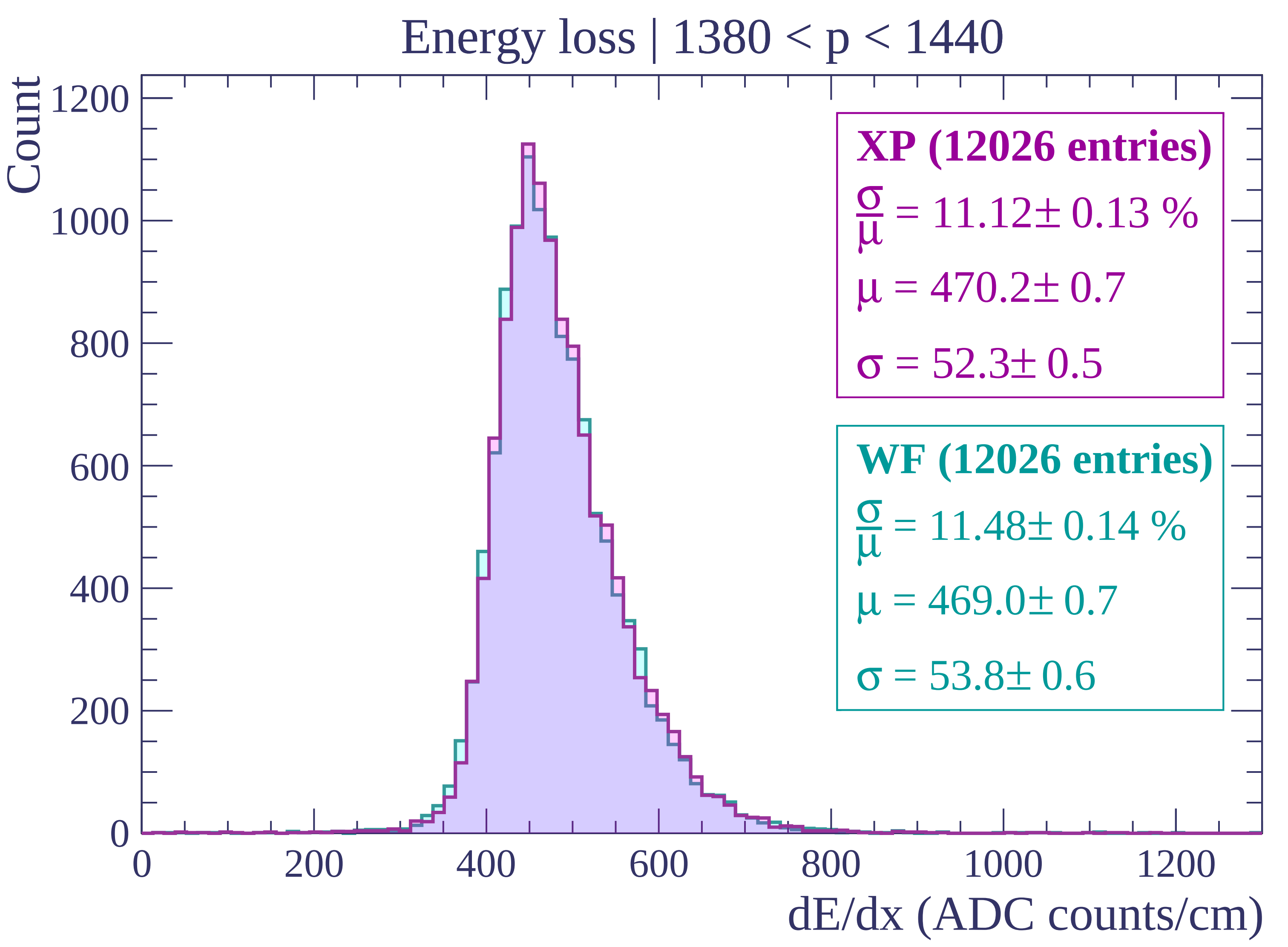

Energy loss | 1200 < p < 1260
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (13128 entries)
$\frac{\sigma}{\mu}$ = 10.45± 0.13 %
μ = 456.1± 0.6
σ = 47.7± 0.5
WF (13128 entries)
$\frac{\sigma}{\mu}$ = 10.74± 0.13 %
μ = 455.7± 0.6
σ = 48.9± 0.5



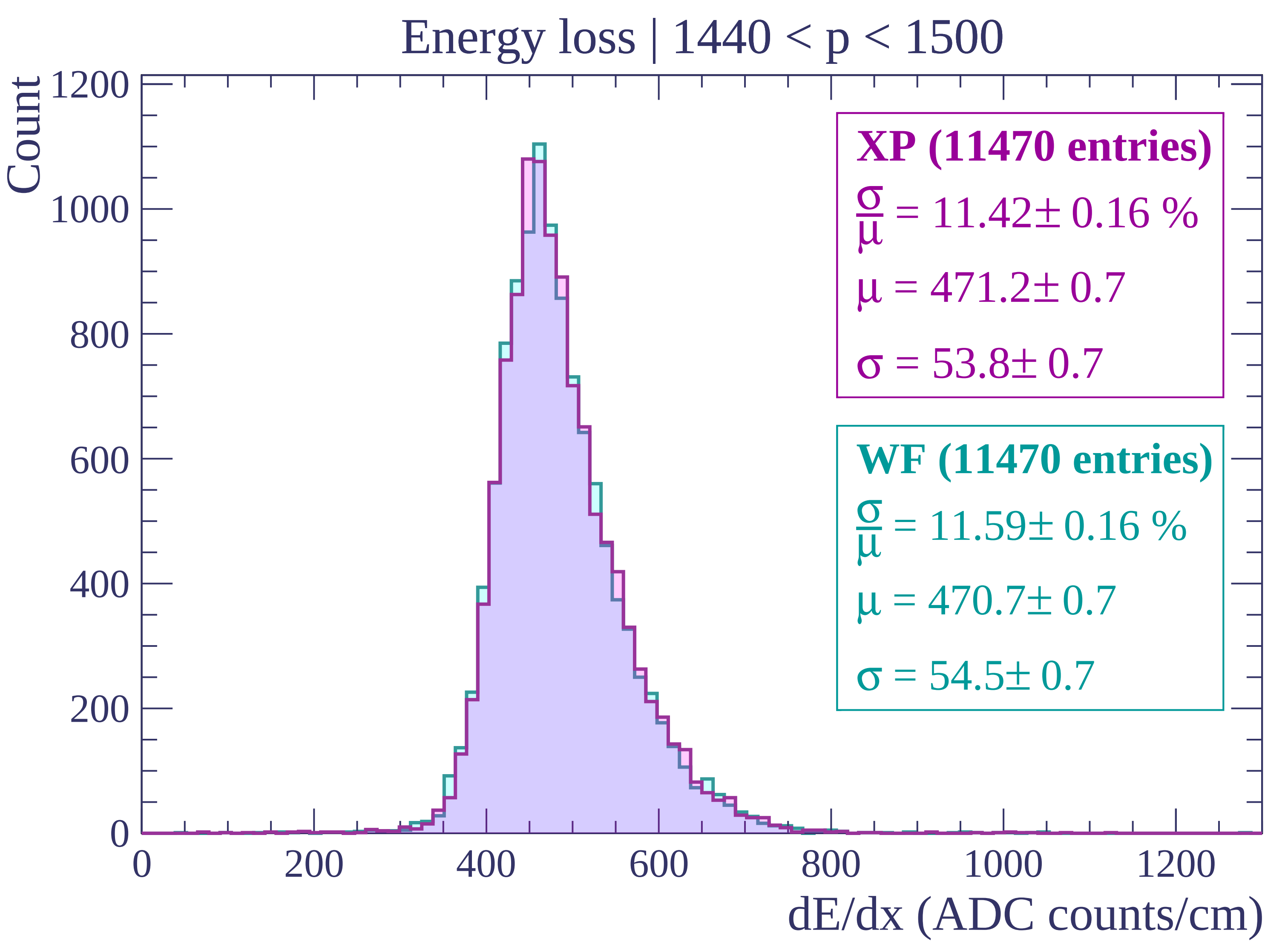
Energy loss | 1260 < p < 1320
Count
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (12834 entries)
σ/μ = 11.12± 0.14 %
μ = 460.7± 0.6
σ = 51.2± 0.6
WF (12834 entries)
σ/μ = 11.21± 0.13 %
μ = 459.5± 0.6
σ = 51.5± 0.6



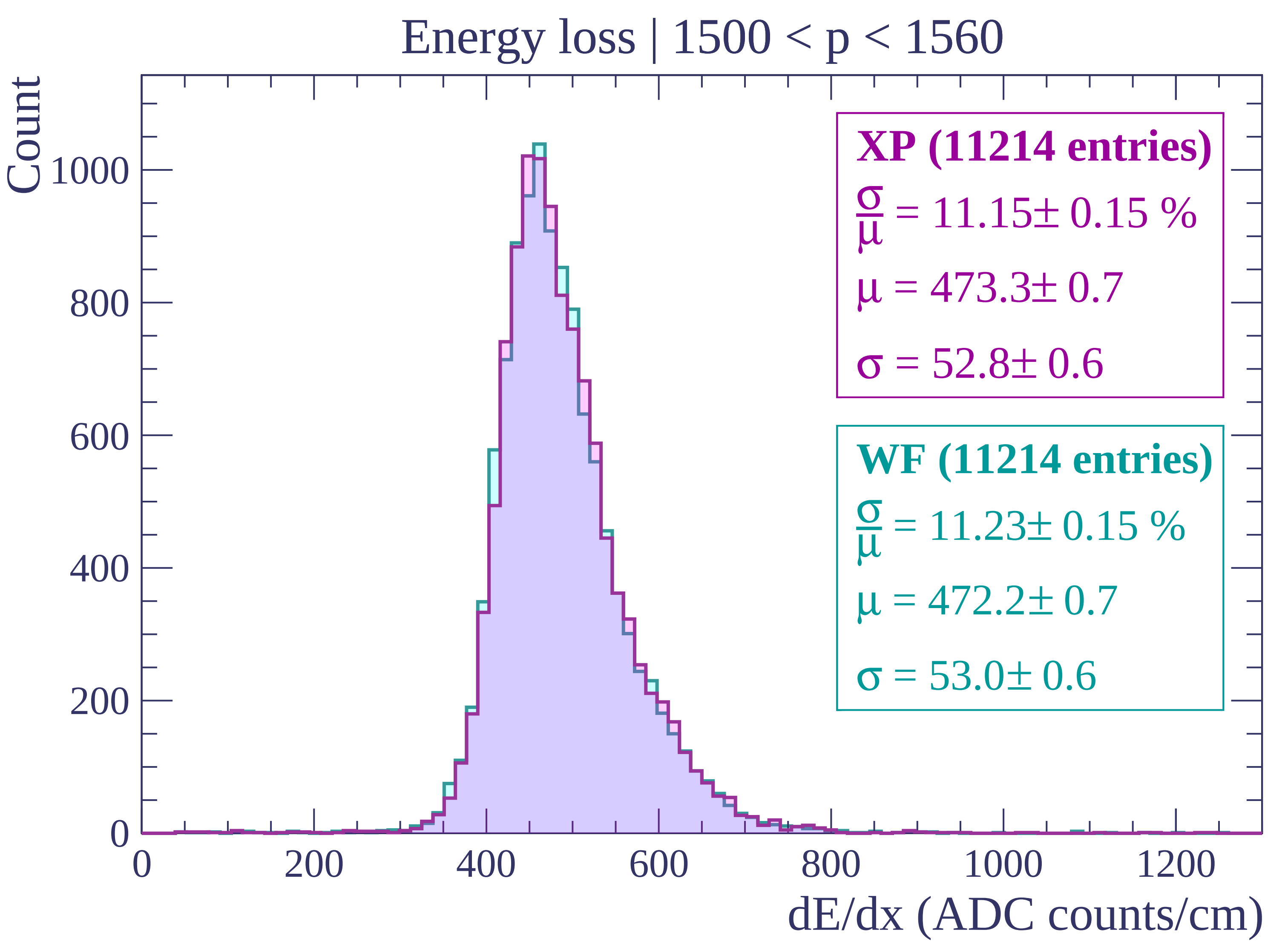
Energy loss | 1320 < p < 1380
Count
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (12687 entries)
$\frac{\sigma}{\mu}$ = 11.09± 0.15 %
μ = 463.6± 0.6
σ = 51.4± 0.6
WF (12687 entries)
$\frac{\sigma}{\mu}$ = 10.95± 0.13 %
μ = 462.5± 0.6
σ = 50.7± 0.5



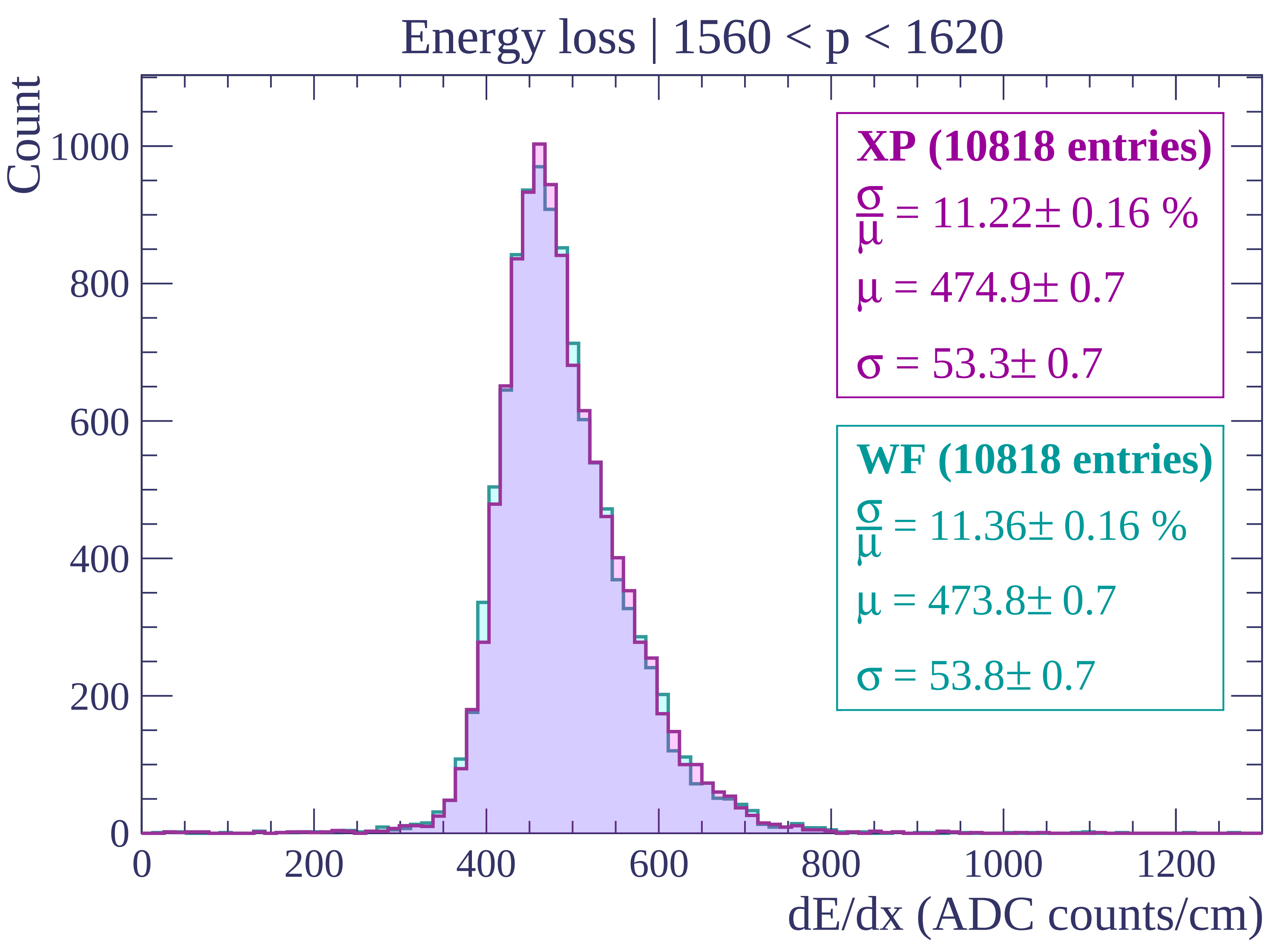
Energy loss | 1380 < p < 1440
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (12026 entries)
σ/μ = 11.12± 0.13 %
μ = 470.2± 0.7
σ = 52.3± 0.5
WF (12026 entries)
σ/μ = 11.48± 0.14 %
μ = 469.0± 0.7
σ = 53.8± 0.6

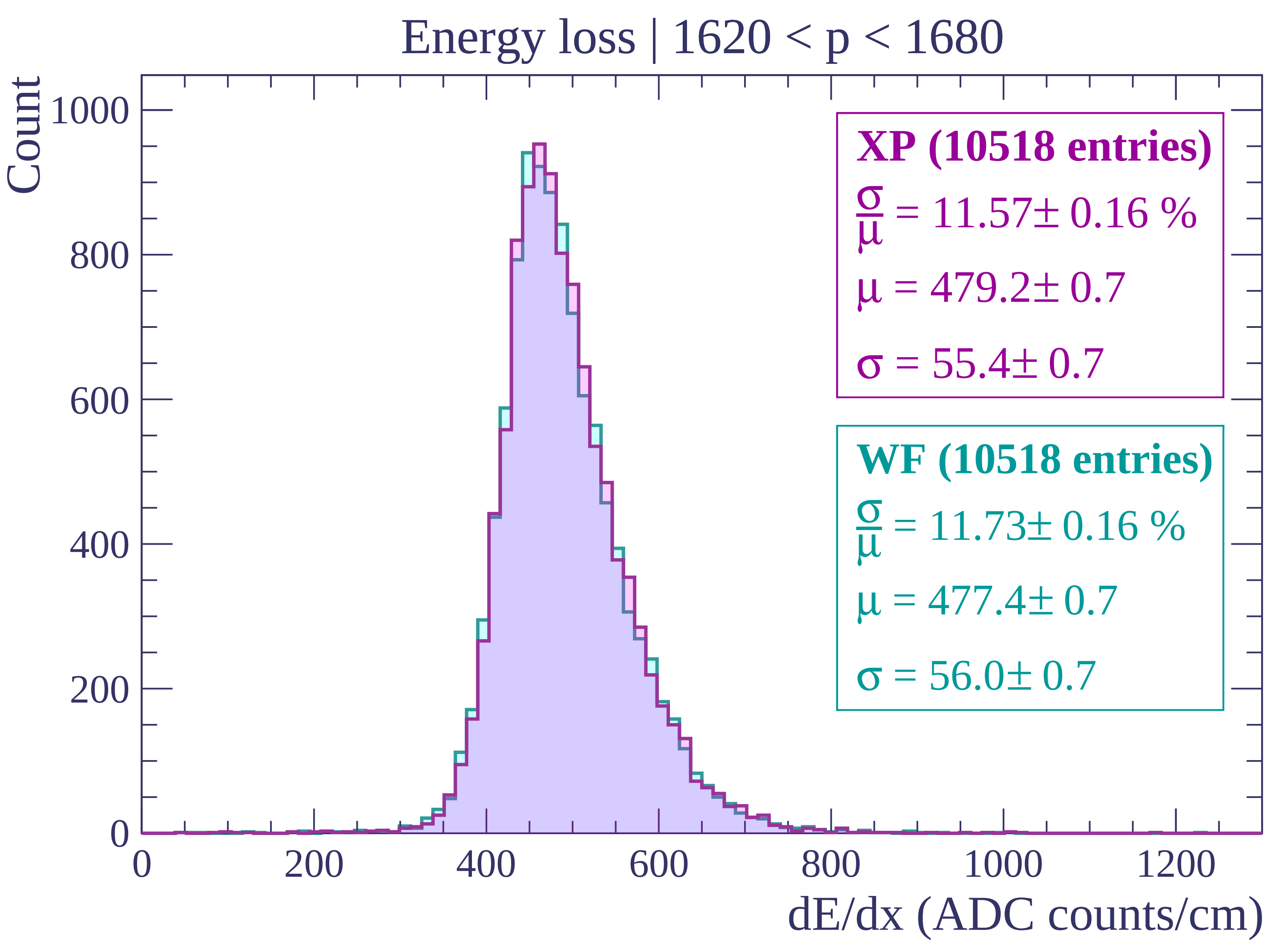


Energy loss | 1440 < p < 1500
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (11470 entries)
$\frac{\sigma}{\mu}$ = 11.42± 0.16 %
μ = 471.2± 0.7
σ = 53.8± 0.7
WF (11470 entries)
$\frac{\sigma}{\mu}$ = 11.59± 0.16 %
μ = 470.7± 0.7
σ = 54.5± 0.7

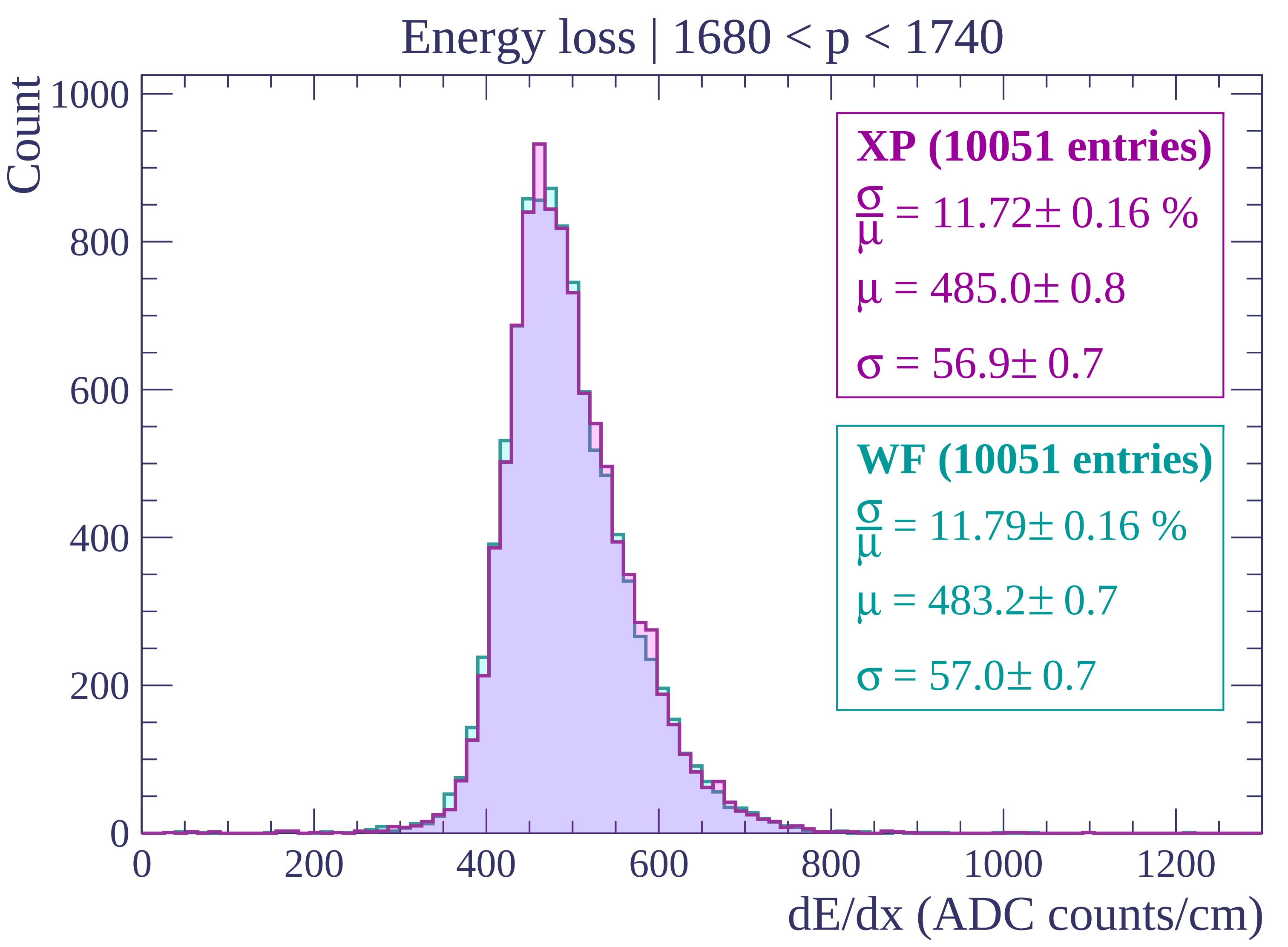


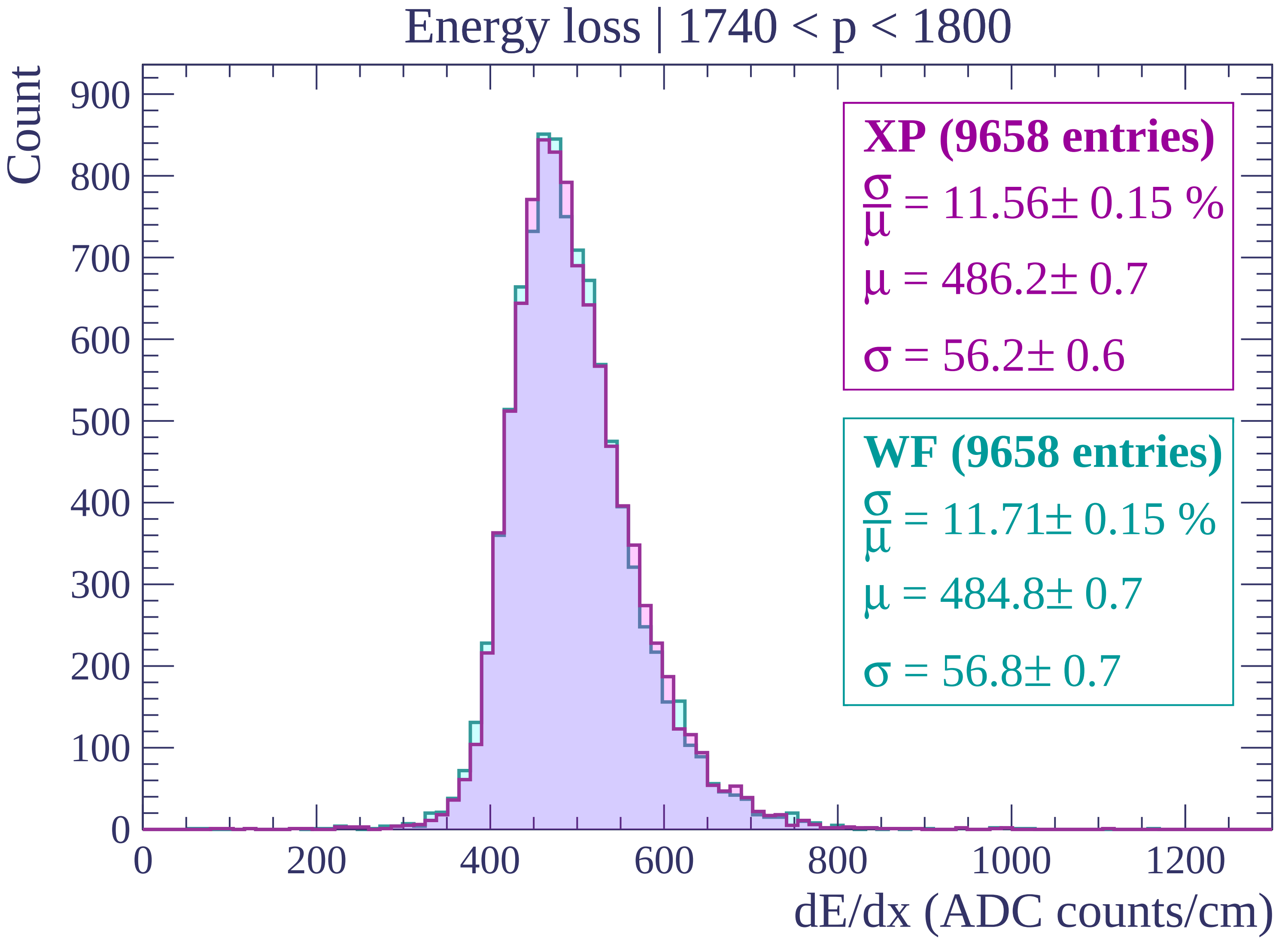
Energy loss | 1500 < p < 1560
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (11214 entries)
$\frac{\sigma}{\mu}$ = 11.15± 0.15 %
μ = 473.3± 0.7
σ = 52.8± 0.6
WF (11214 entries)
$\frac{\sigma}{\mu}$ = 11.23± 0.15 %
μ = 472.2± 0.7
σ = 53.0± 0.6

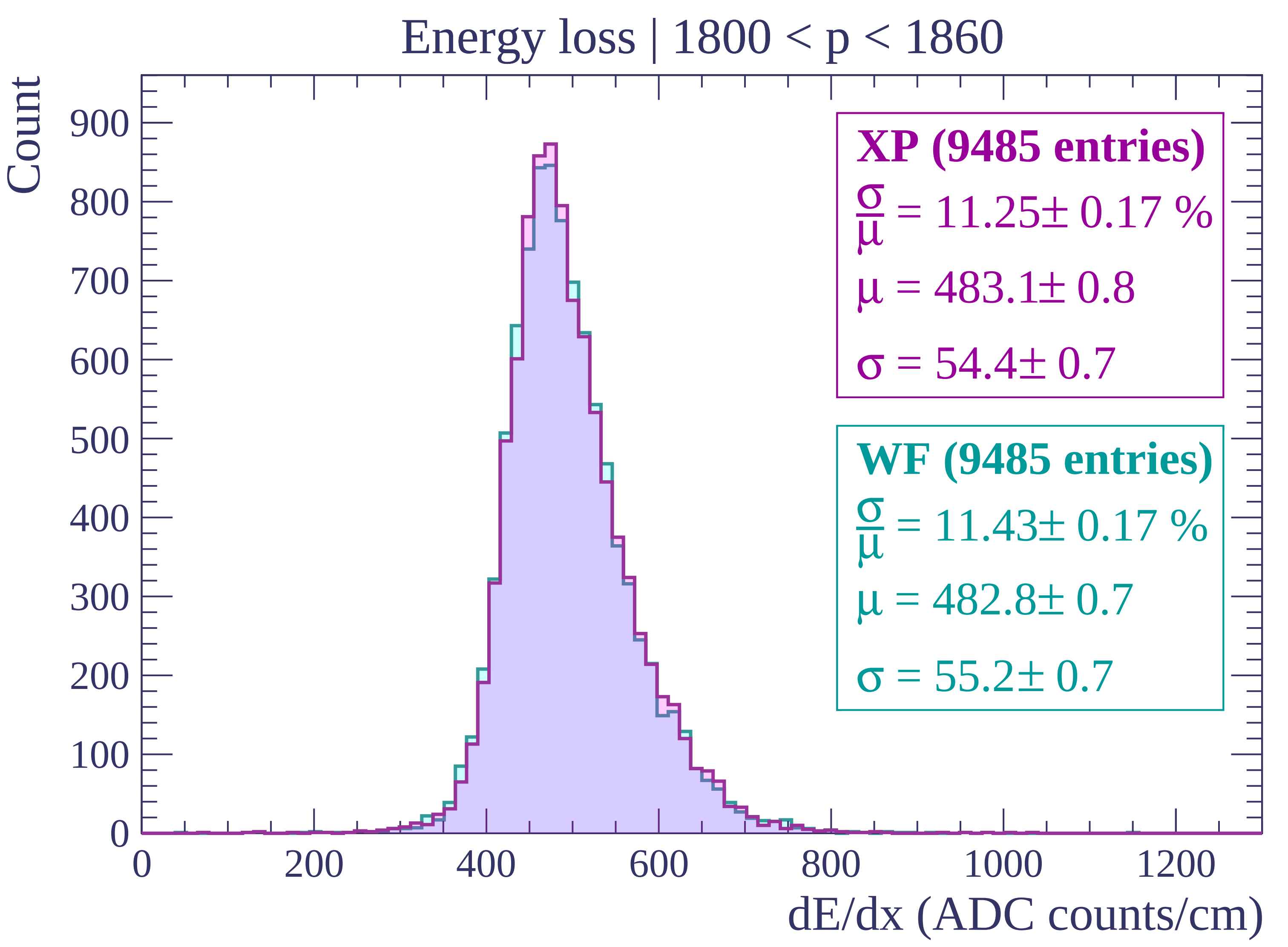

Energy loss | 1560 < p < 1620
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (10818 entries)
σ/μ = 11.22± 0.16 %
μ = 474.9± 0.7
σ = 53.3± 0.7
WF (10818 entries)
σ/μ = 11.36± 0.16 %
μ = 473.8± 0.7
σ = 53.8± 0.7

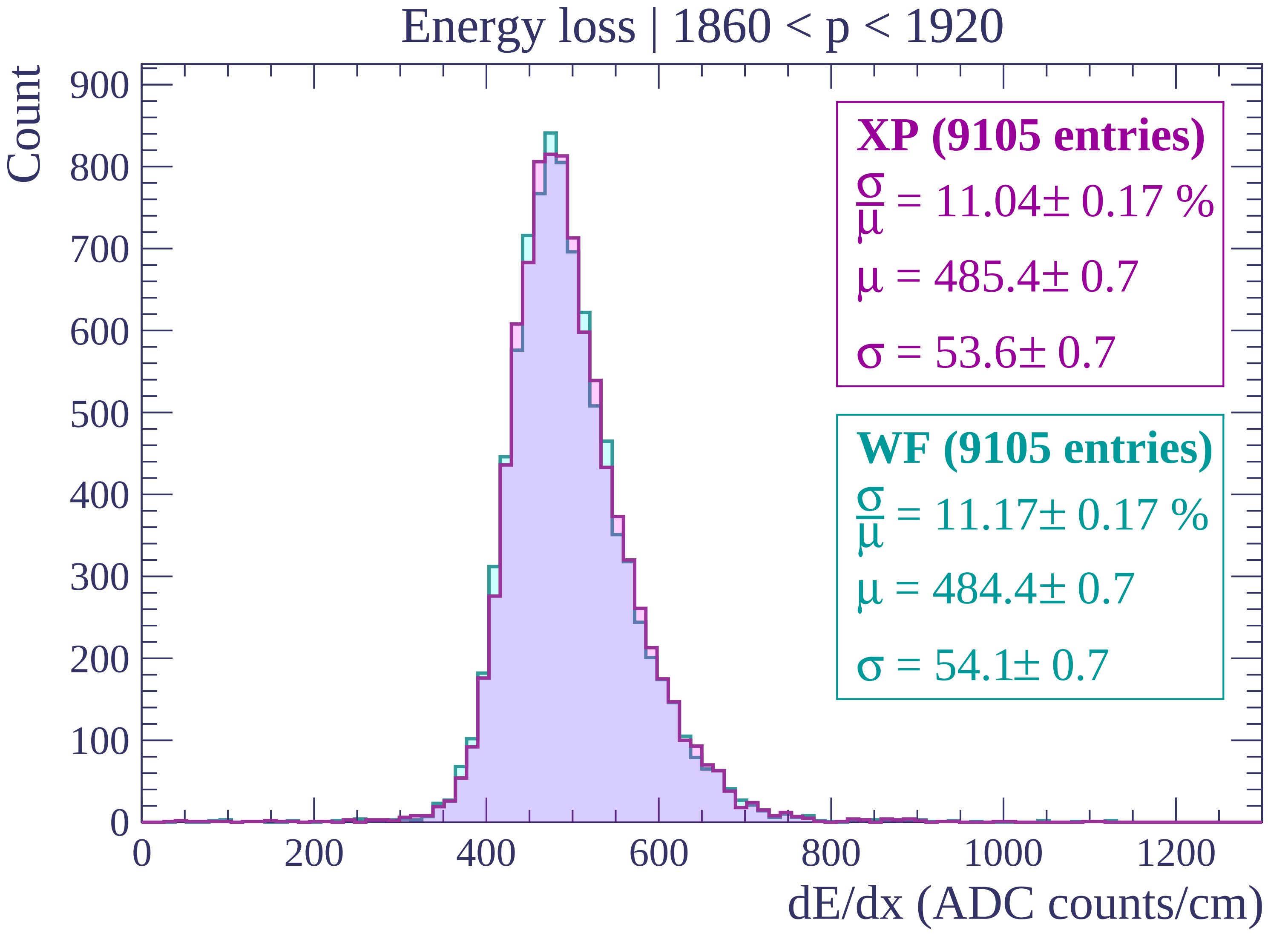


Energy loss | 1620 < p < 1680
Count
dE/dx (ADC counts/cm)
XP (10518 entries)
σ/μ = 11.57± 0.16 %
μ = 479.2± 0.7
σ = 55.4± 0.7
WF (10518 entries)
σ/μ = 11.73± 0.16 %
μ = 477.4± 0.7
σ = 56.0± 0.7

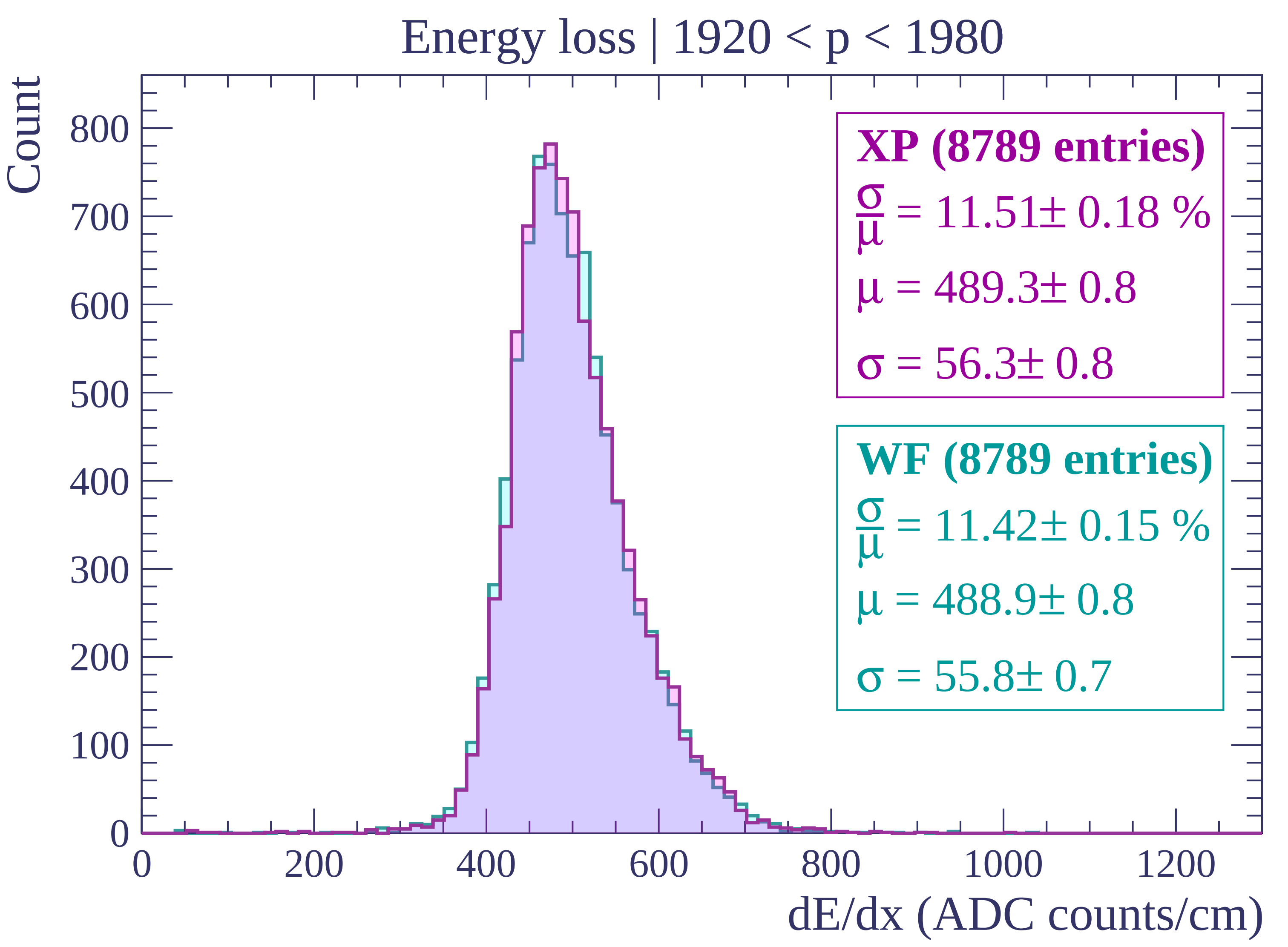

Energy loss | 1680 < p < 1740
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (10051 entries)
$\frac{\sigma}{\mu}$ = 11.72± 0.16 %
μ = 485.0± 0.8
σ = 56.9± 0.7
WF (10051 entries)
$\frac{\sigma}{\mu}$ = 11.79± 0.16 %
μ = 483.2± 0.7
σ = 57.0± 0.7


Energy loss | 1740 < p < 1800
Count
dE/dx (ADC counts/cm)
XP (9658 entries)
$\frac{\sigma}{\mu}$ = 11.56± 0.15 %
μ = 486.2± 0.7
σ = 56.2± 0.6
WF (9658 entries)
$\frac{\sigma}{\mu}$ = 11.71± 0.15 %
μ = 484.8± 0.7
σ = 56.8± 0.7
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200

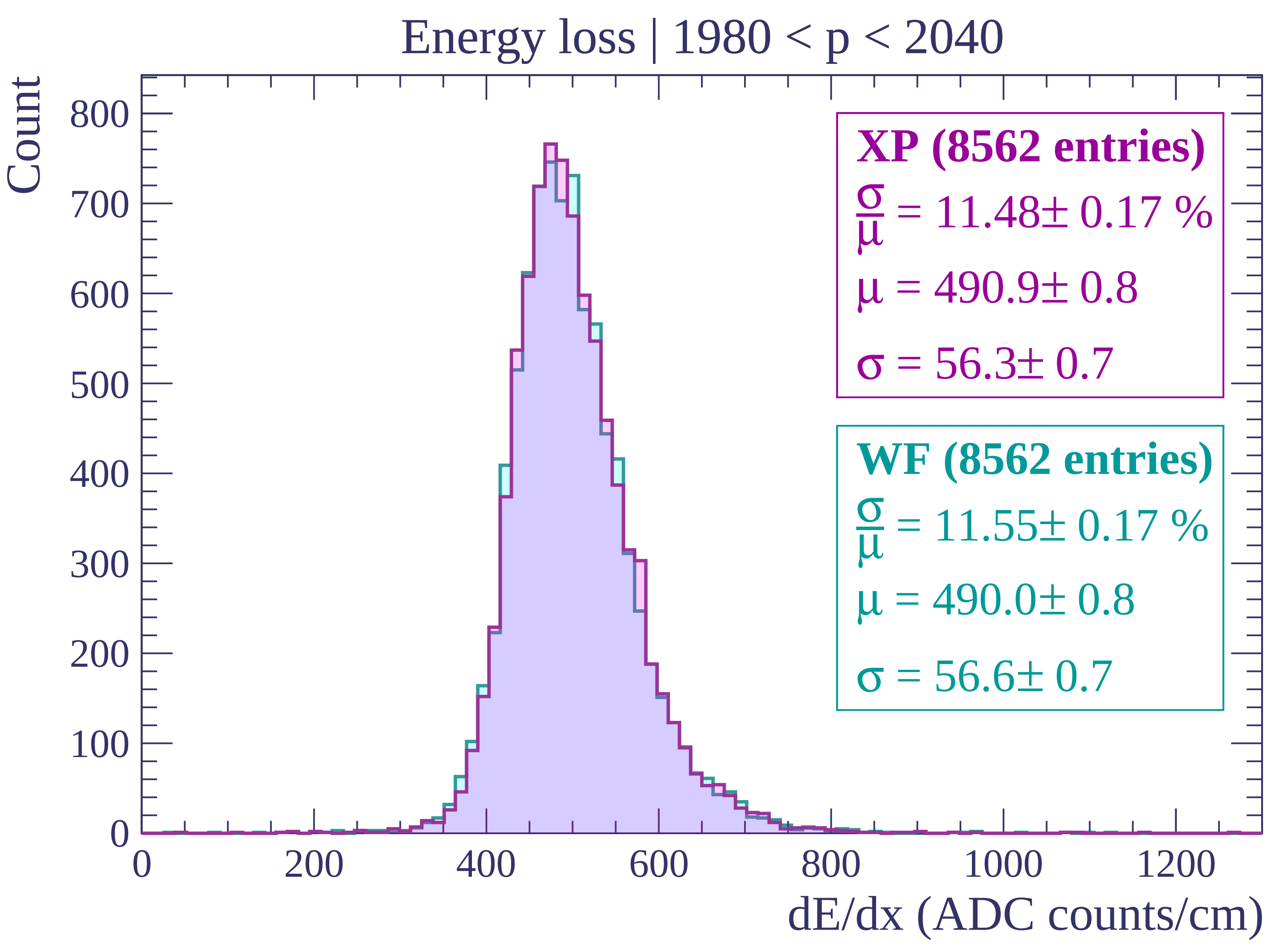

Energy loss | 1800 < p < 1860
Count
dE/dx (ADC counts/cm)
XP (9485 entries)
$\frac{\sigma}{\mu}$ = 11.25± 0.17 %
μ = 483.1± 0.8
σ = 54.4± 0.7
WF (9485 entries)
$\frac{\sigma}{\mu}$ = 11.43± 0.17 %
μ = 482.8± 0.7
σ = 55.2± 0.7

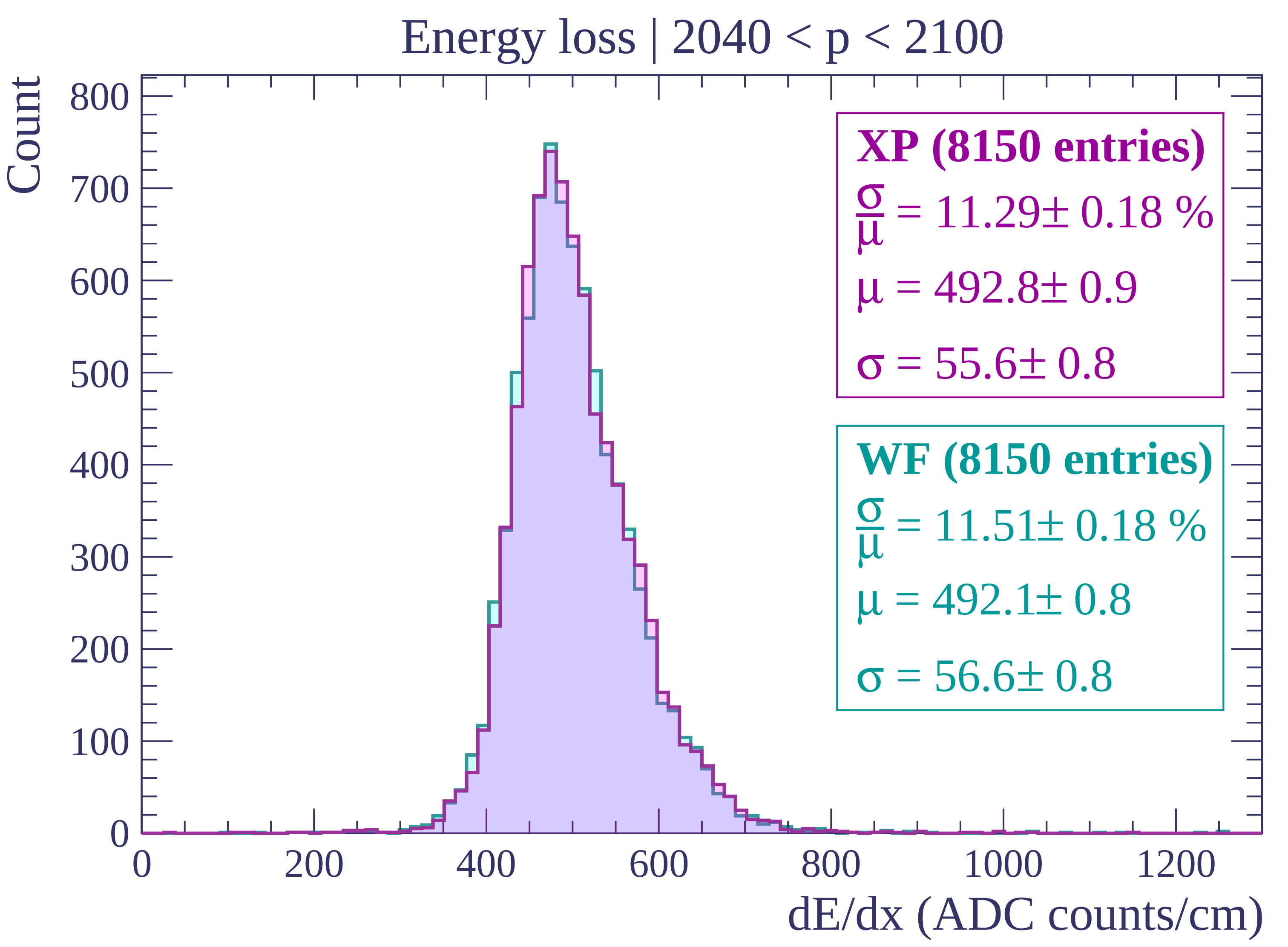


Energy loss | 1860 < p < 1920
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
700
900
XP (9105 entries)
$\frac{\sigma}{\mu}$ = 11.04± 0.17 %
μ = 485.4± 0.7
σ = 53.6± 0.7
WF (9105 entries)
$\frac{\sigma}{\mu}$ = 11.17± 0.17 %
μ = 484.4± 0.7
σ = 54.1± 0.7

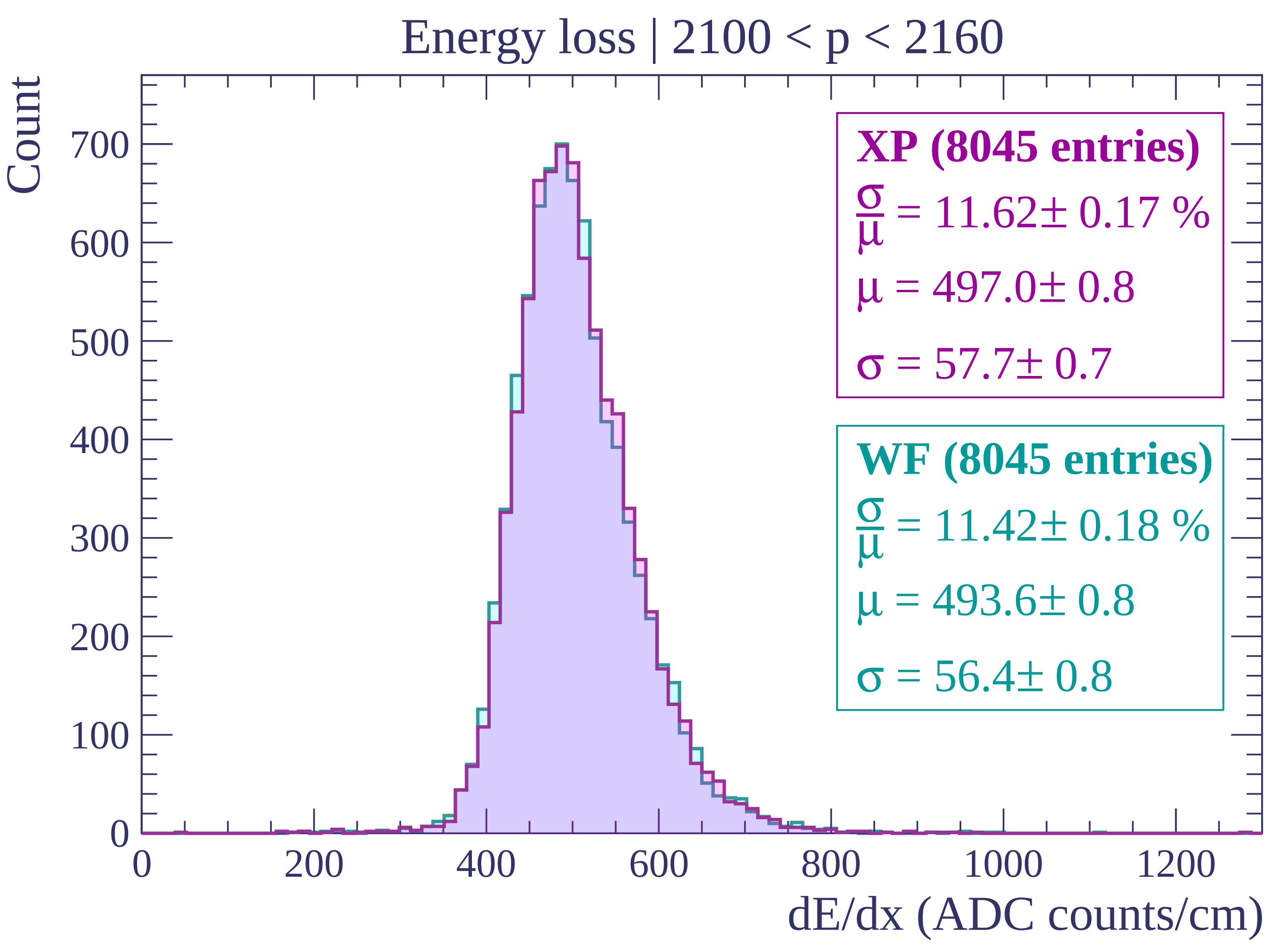


Energy loss | 1920 < p < 1980
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP (8789 entries)
σ/μ = 11.51± 0.18 %
μ = 489.3± 0.8
σ = 56.3± 0.8
WF (8789 entries)
σ/μ = 11.42± 0.15 %
μ = 488.9± 0.8
σ = 55.8± 0.7

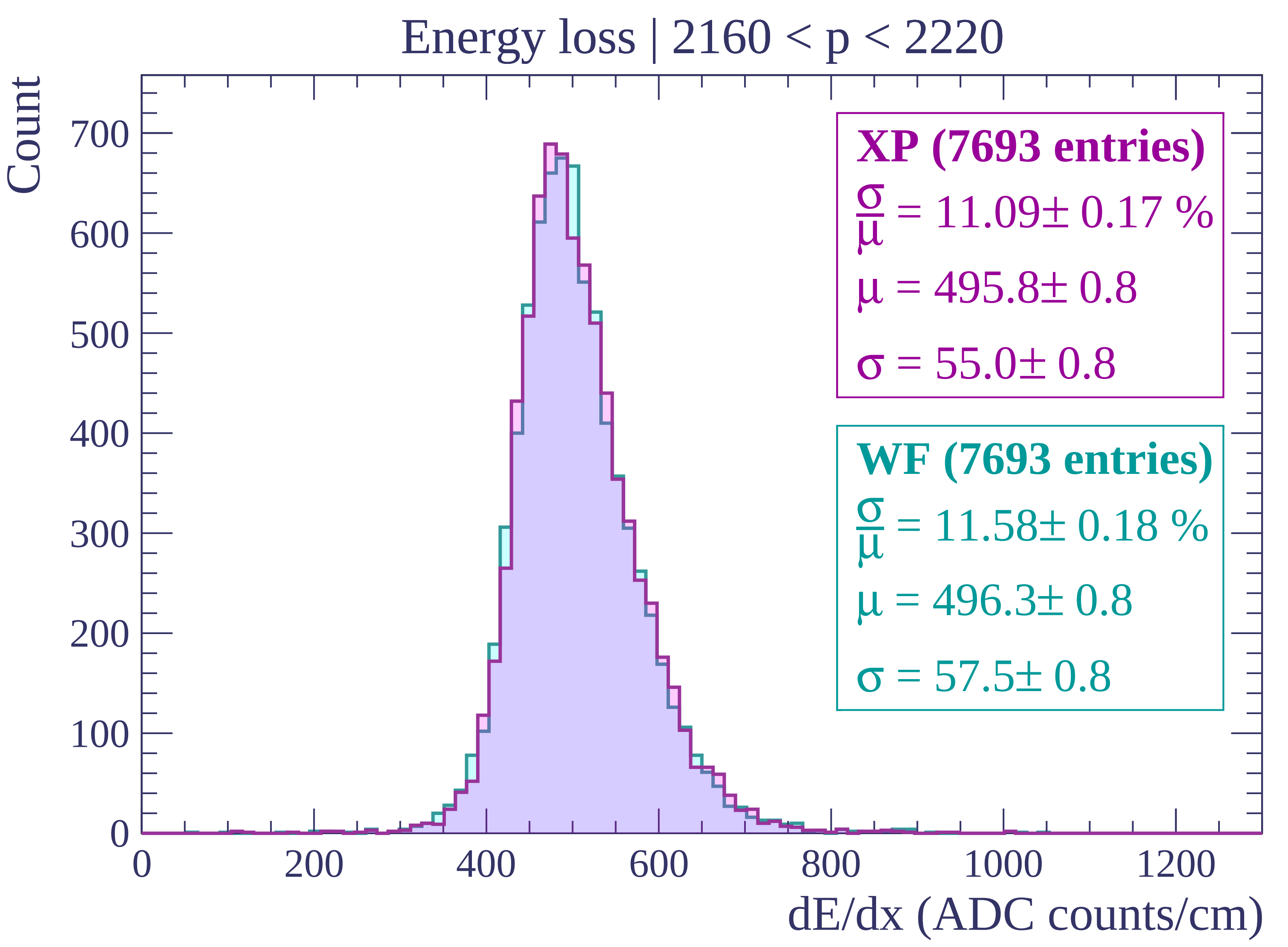

Energy loss | 1980 < p < 2040
Count
dE/dx (ADC counts/cm)
XP (8562 entries)
σ/μ = 11.48± 0.17 %
μ = 490.9± 0.8
σ = 56.3± 0.7
WF (8562 entries)
σ/μ = 11.55± 0.17 %
μ = 490.0± 0.8
σ = 56.6± 0.7
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200

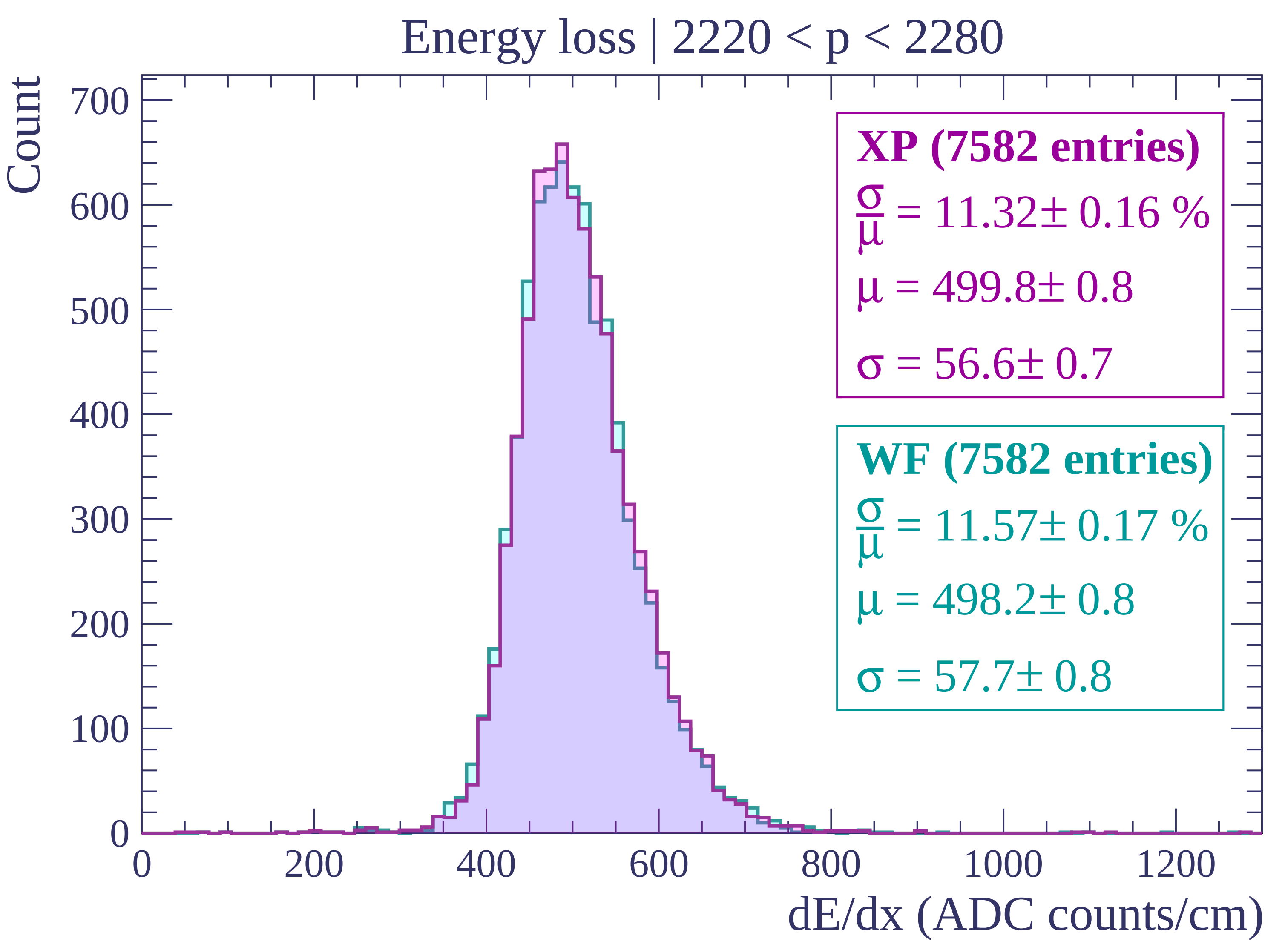


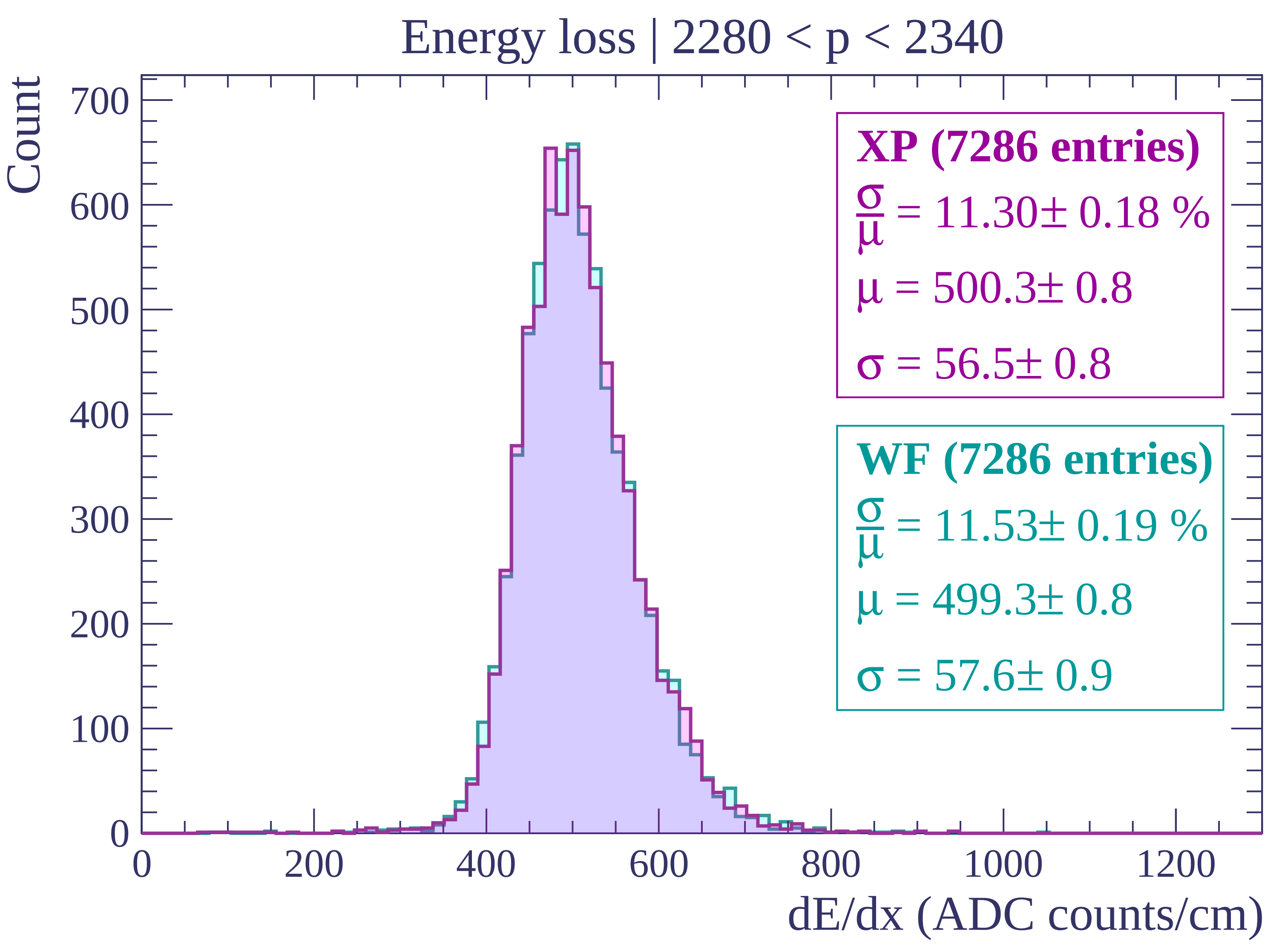
Energy loss | 2040 < p < 2100
Count
dE/dx (ADC counts/cm)
XP (8150 entries)
σ/μ = 11.29± 0.18 %
μ = 492.8± 0.9
σ = 55.6± 0.8
WF (8150 entries)
σ/μ = 11.51± 0.18 %
μ = 492.1± 0.8
σ = 56.6± 0.8

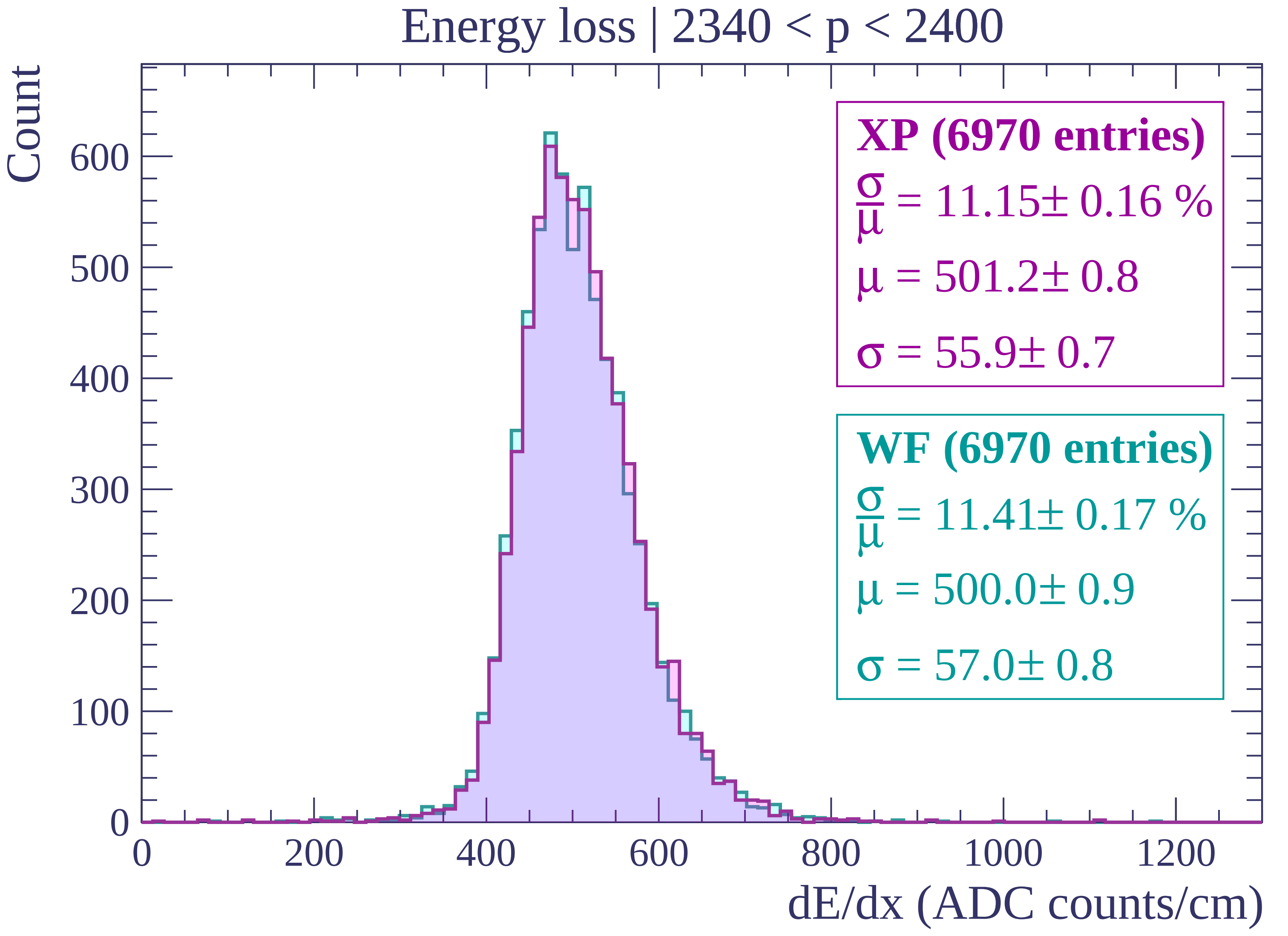


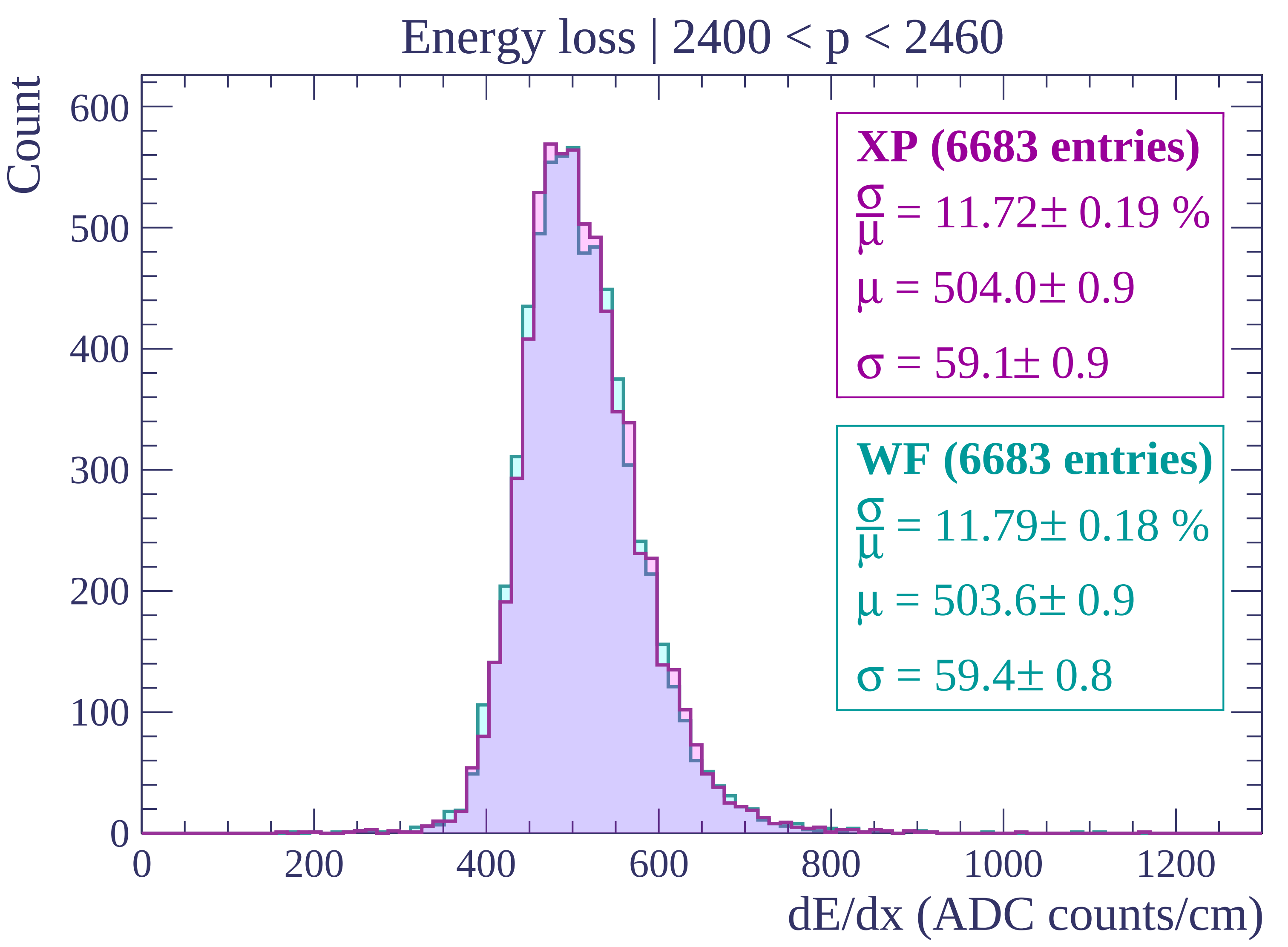
Energy loss | 2100 < p < 2160
Count
dE/dx (ADC counts/cm)
XP (8045 entries)
σ/μ = 11.62± 0.17 %
μ = 497.0± 0.8
σ = 57.7± 0.7
WF (8045 entries)
σ/μ = 11.42± 0.18 %
μ = 493.6± 0.8
σ = 56.4± 0.8
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200

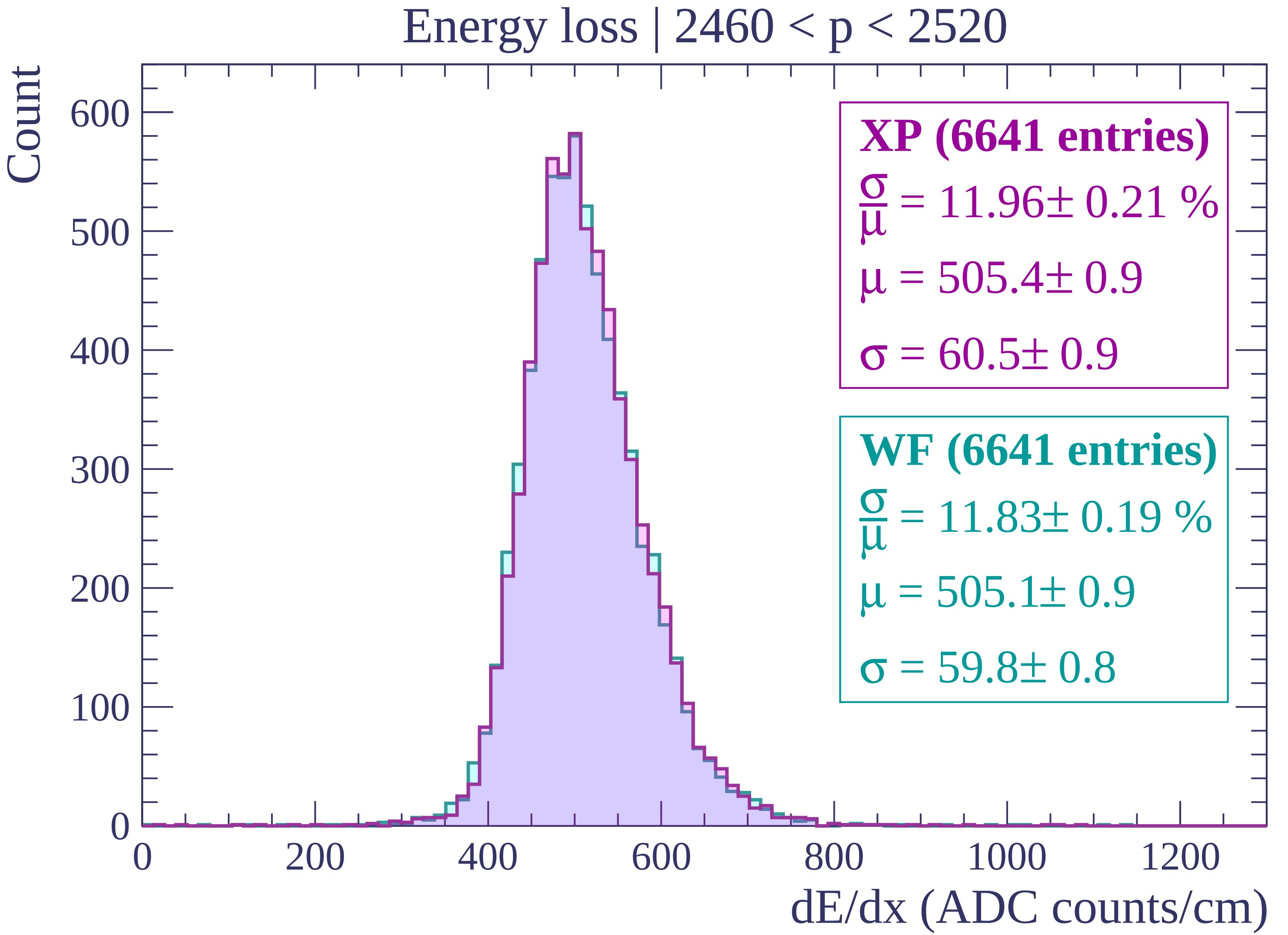

Energy loss | 2160 < p < 2220
Count
dE/dx (ADC counts/cm)
XP (7693 entries)
σ/μ = 11.09± 0.17 %
μ = 495.8± 0.8
σ = 55.0± 0.8
WF (7693 entries)
σ/μ = 11.58± 0.18 %
μ = 496.3± 0.8
σ = 57.5± 0.8
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200


Energy loss | 2220 < p < 2280
Count
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (7582 entries)
σ/μ = 11.32± 0.16 %
μ = 499.8± 0.8
σ = 56.6± 0.7
WF (7582 entries)
σ/μ = 11.57± 0.17 %
μ = 498.2± 0.8
σ = 57.7± 0.8

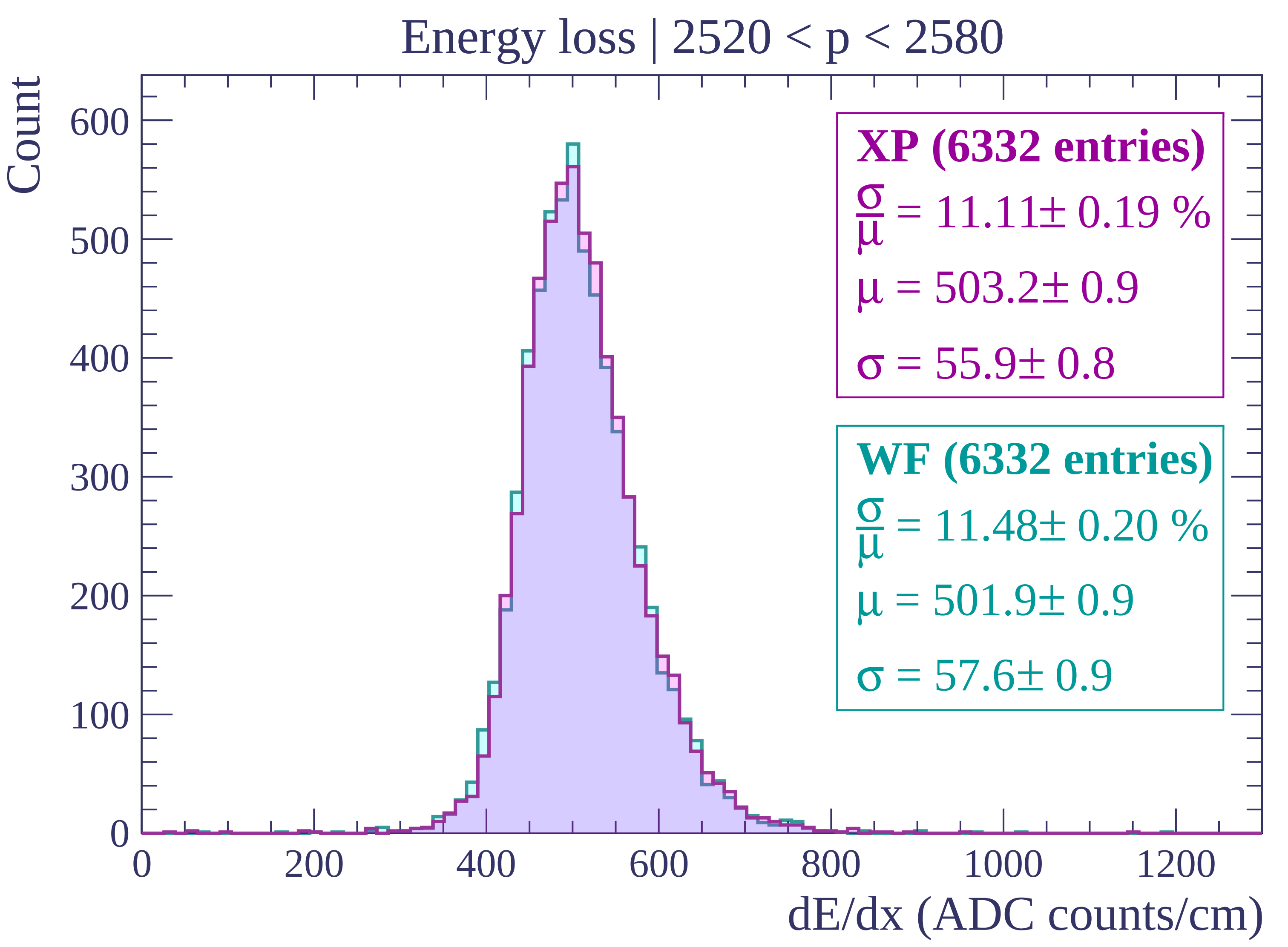

Energy loss | 2280 < p < 2340
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (7286 entries)
$\frac{\sigma}{\mu}$ = 11.30± 0.18 %
μ = 500.3± 0.8
σ = 56.5± 0.8
WF (7286 entries)
$\frac{\sigma}{\mu}$ = 11.53± 0.19 %
μ = 499.3± 0.8
σ = 57.6± 0.9


Energy loss | 2340 < p < 2400
Count
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6970 entries)
σ/μ = 11.15± 0.16 %
μ = 501.2± 0.8
σ = 55.9± 0.7
WF (6970 entries)
σ/μ = 11.41± 0.17 %
μ = 500.0± 0.9
σ = 57.0± 0.8

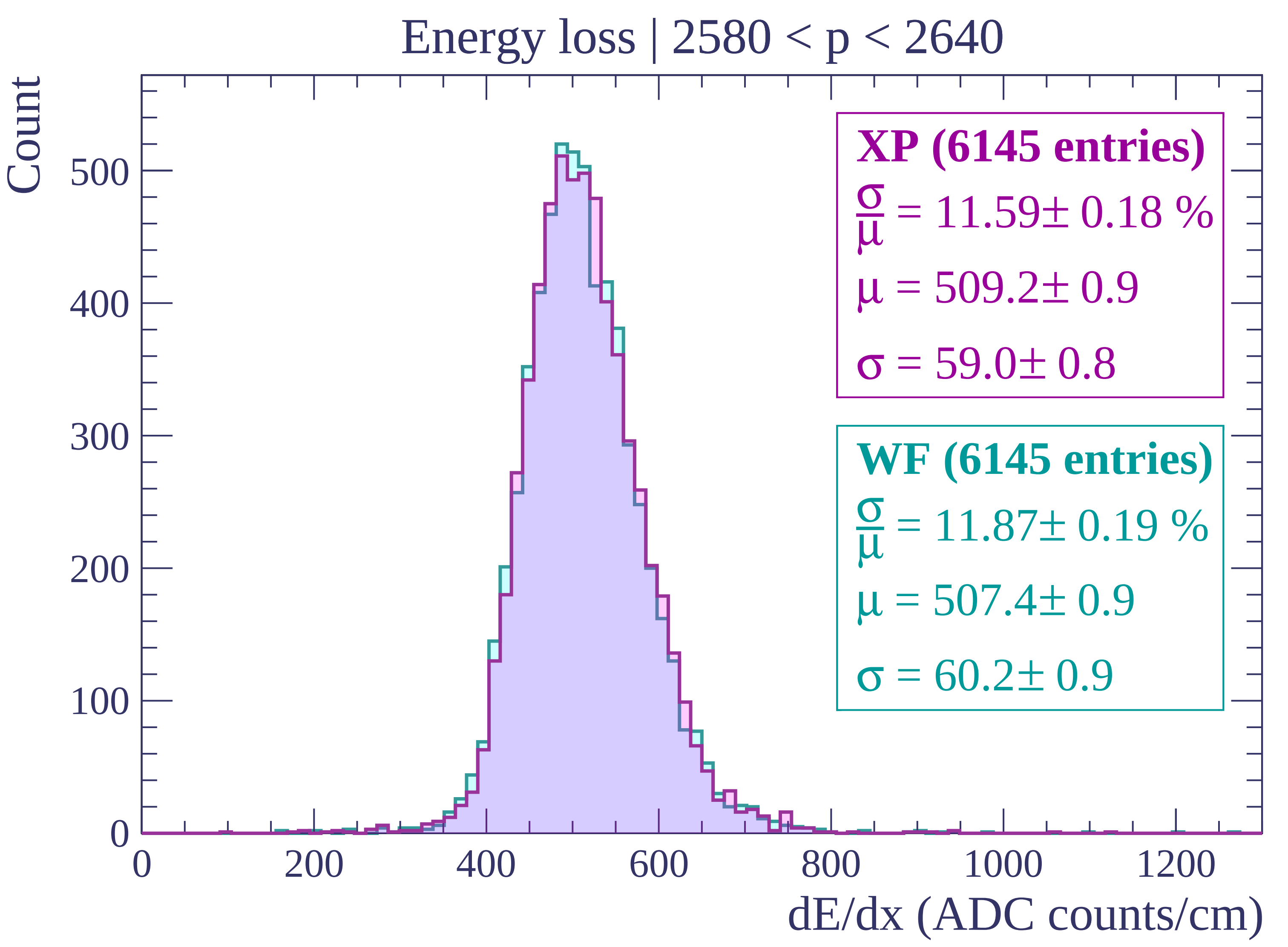

Energy loss | 2400 < p < 2460
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
XP (6683 entries)
$\frac{\sigma}{\mu}$ = 11.72± 0.19 %
μ = 504.0± 0.9
σ = 59.1± 0.9
WF (6683 entries)
$\frac{\sigma}{\mu}$ = 11.79± 0.18 %
μ = 503.6± 0.9
σ = 59.4± 0.8



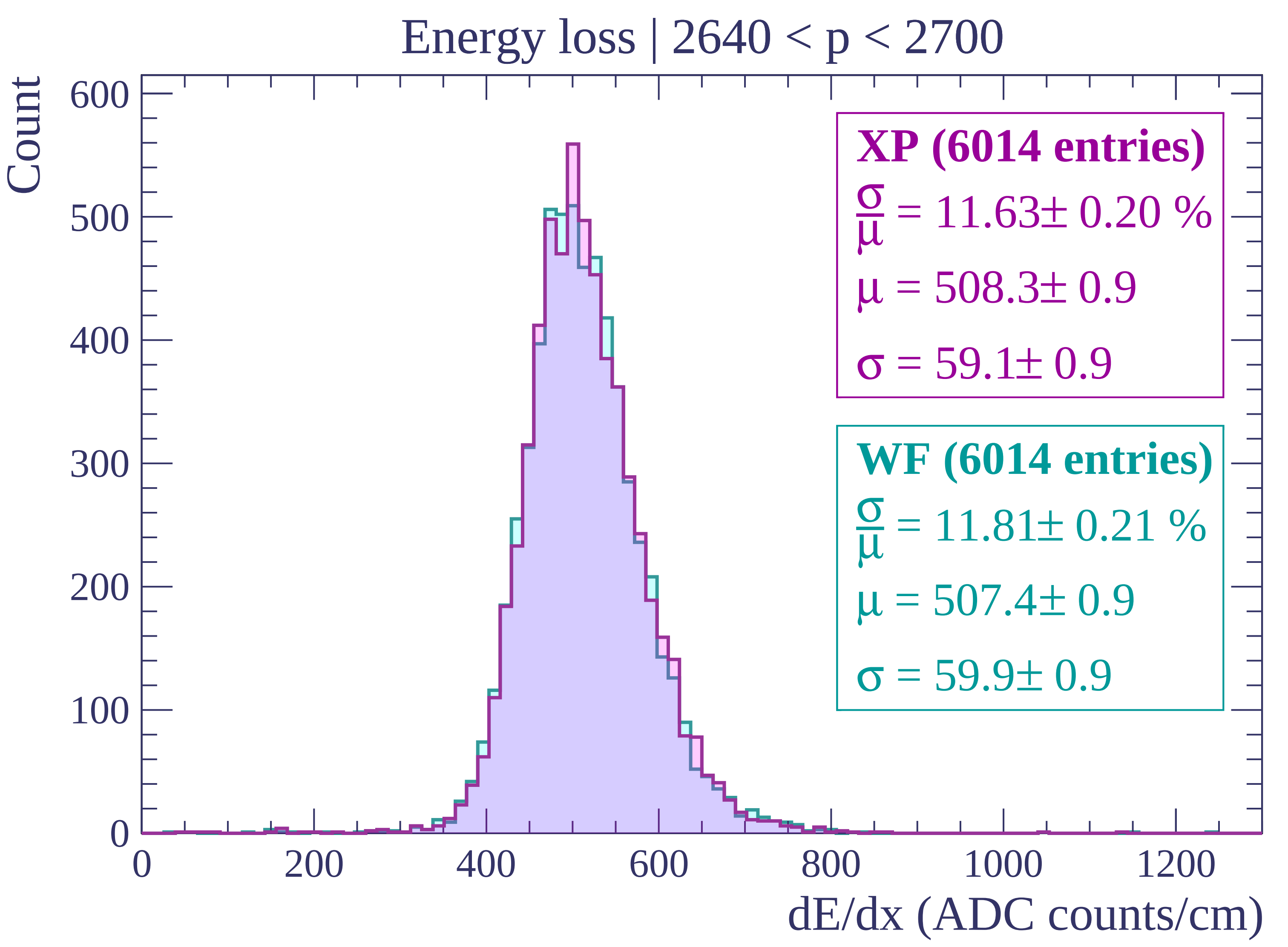
Energy loss | 2460 < p < 2520
Count
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6641 entries)
$\frac{\sigma}{\mu}$ = 11.96± 0.21 %
μ = 505.4± 0.9
σ = 60.5± 0.9
WF (6641 entries)
$\frac{\sigma}{\mu}$ = 11.83± 0.19 %
μ = 505.1± 0.9
σ = 59.8± 0.8



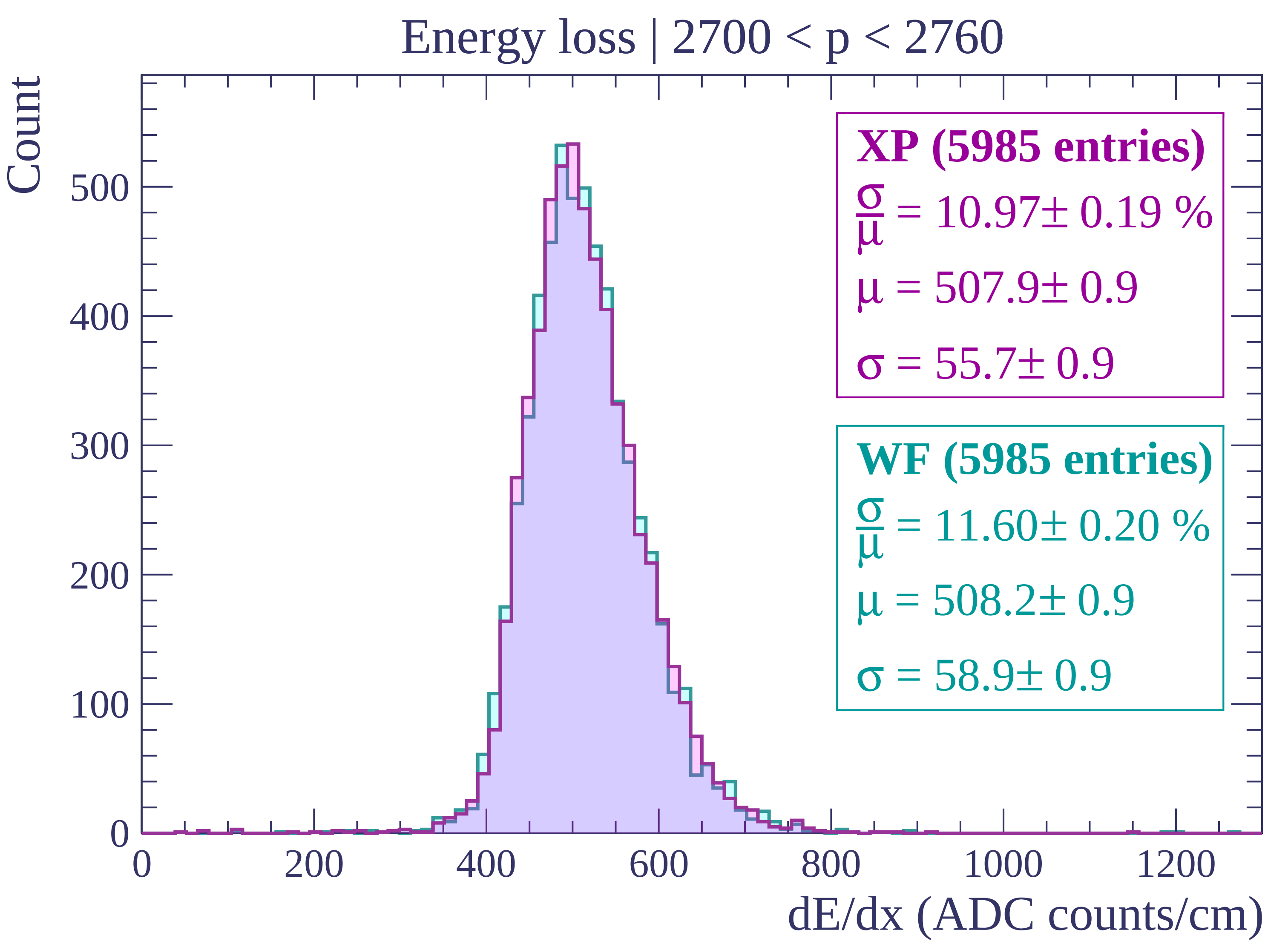
Energy loss | 2520 < p < 2580
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6332 entries)
σ/μ = 11.11± 0.19 %
μ = 503.2± 0.9
σ = 55.9± 0.8
WF (6332 entries)
σ/μ = 11.48± 0.20 %
μ = 501.9± 0.9
σ = 57.6± 0.9



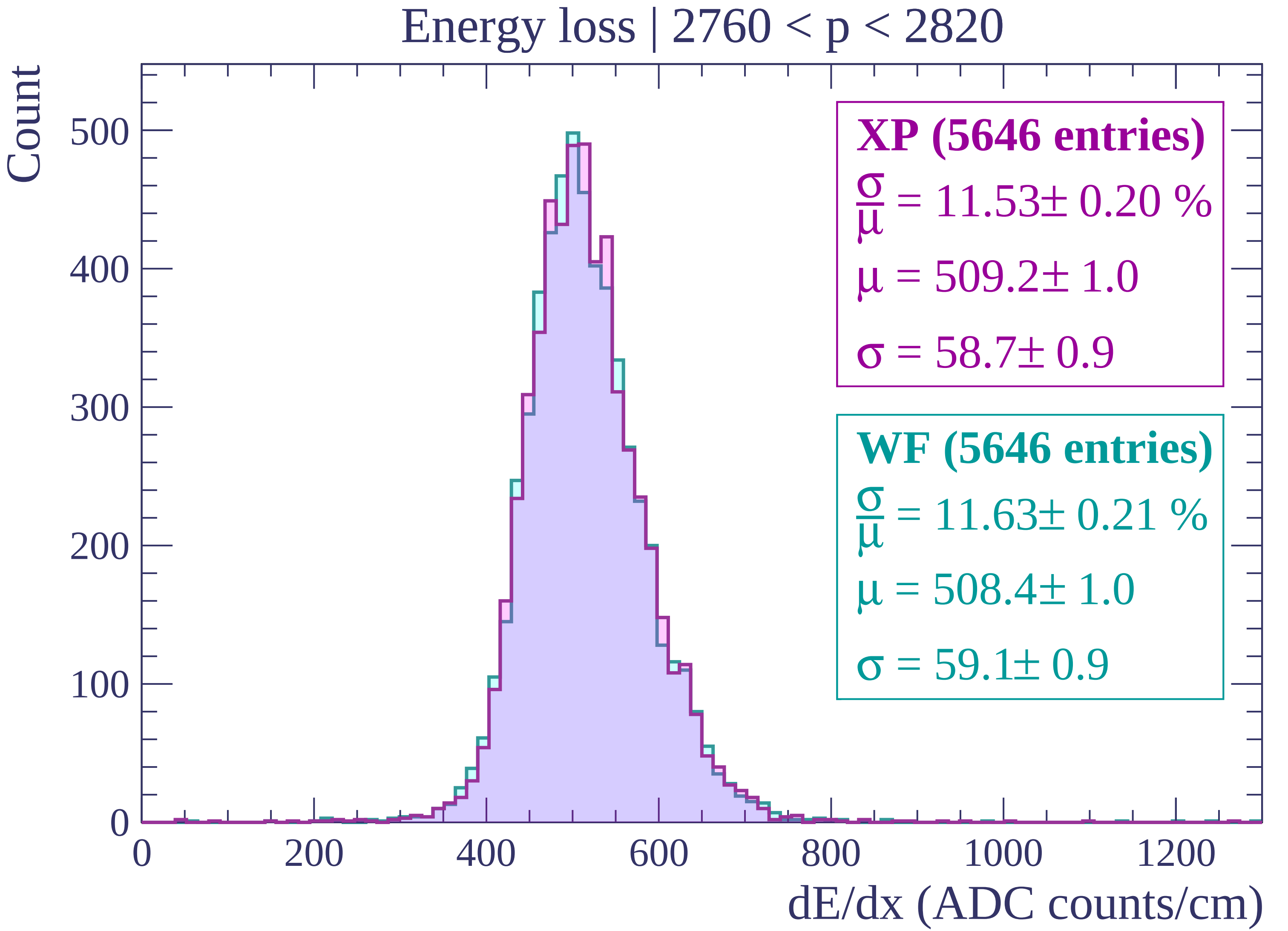
Energy loss | 2580 < p < 2640
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6145 entries)
σ/μ = 11.59± 0.18 %
μ = 509.2± 0.9
σ = 59.0± 0.8
WF (6145 entries)
σ/μ = 11.87± 0.19 %
μ = 507.4± 0.9
σ = 60.2± 0.9



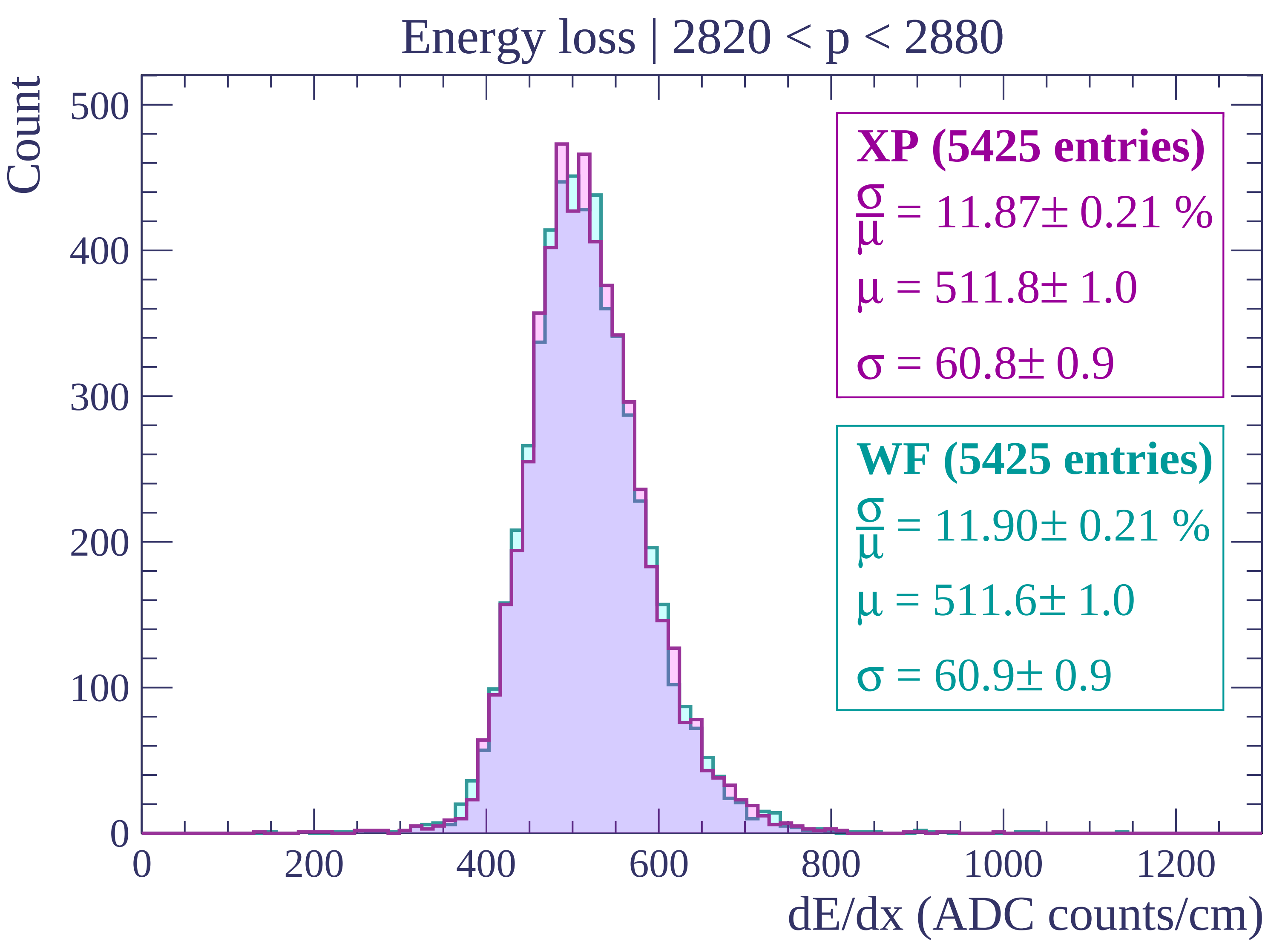
Energy loss | 2640 < p < 2700
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (6014 entries)
$\frac{\sigma}{\mu}$ = 11.63± 0.20 %
μ = 508.3± 0.9
σ = 59.1± 0.9
WF (6014 entries)
$\frac{\sigma}{\mu}$ = 11.81± 0.21 %
μ = 507.4± 0.9
σ = 59.9± 0.9

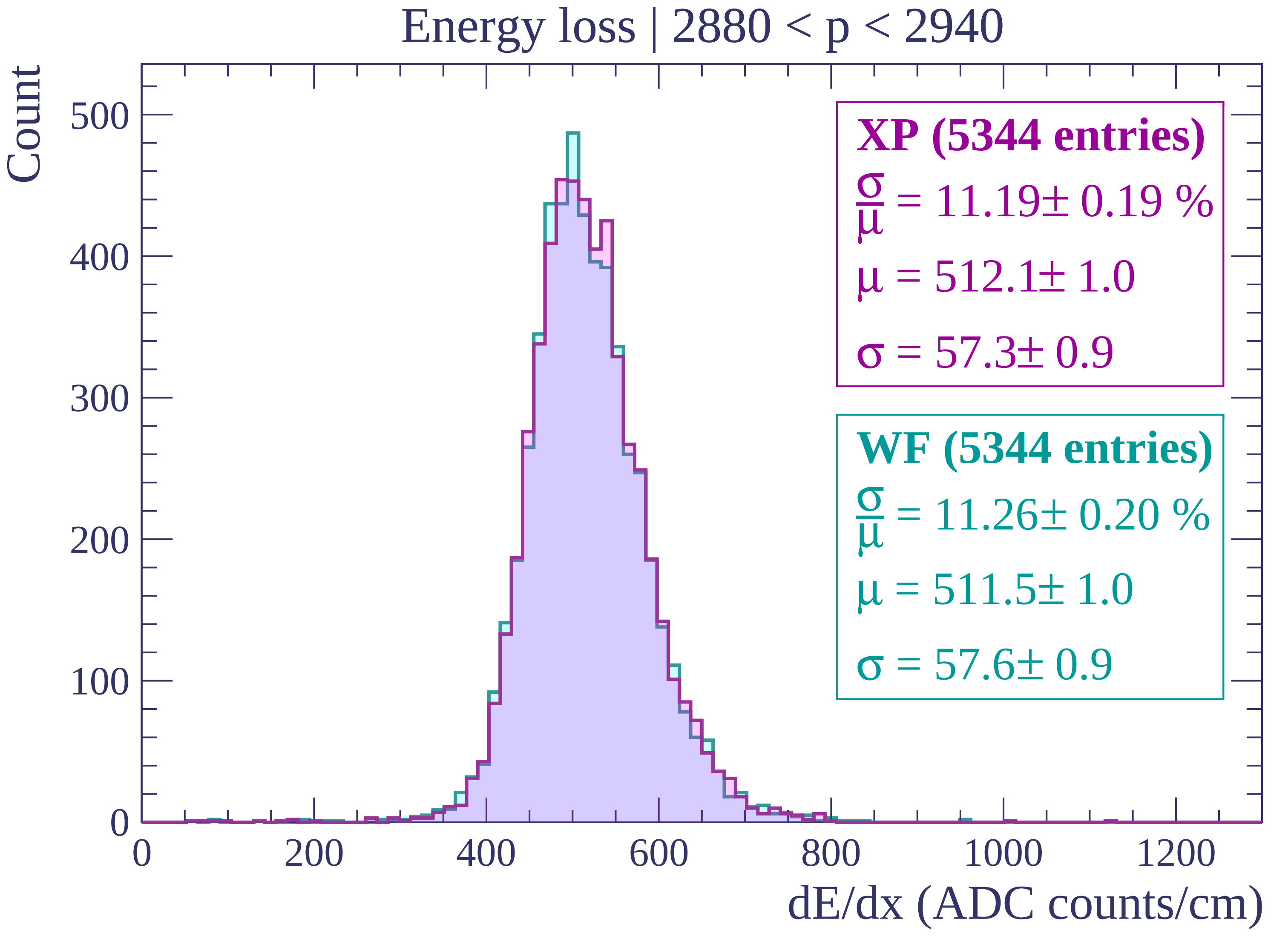


Energy loss | 2700 < p < 2760
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (5985 entries)
σ/μ = 10.97± 0.19 %
μ = 507.9± 0.9
σ = 55.7± 0.9
WF (5985 entries)
σ/μ = 11.60± 0.20 %
μ = 508.2± 0.9
σ = 58.9± 0.9

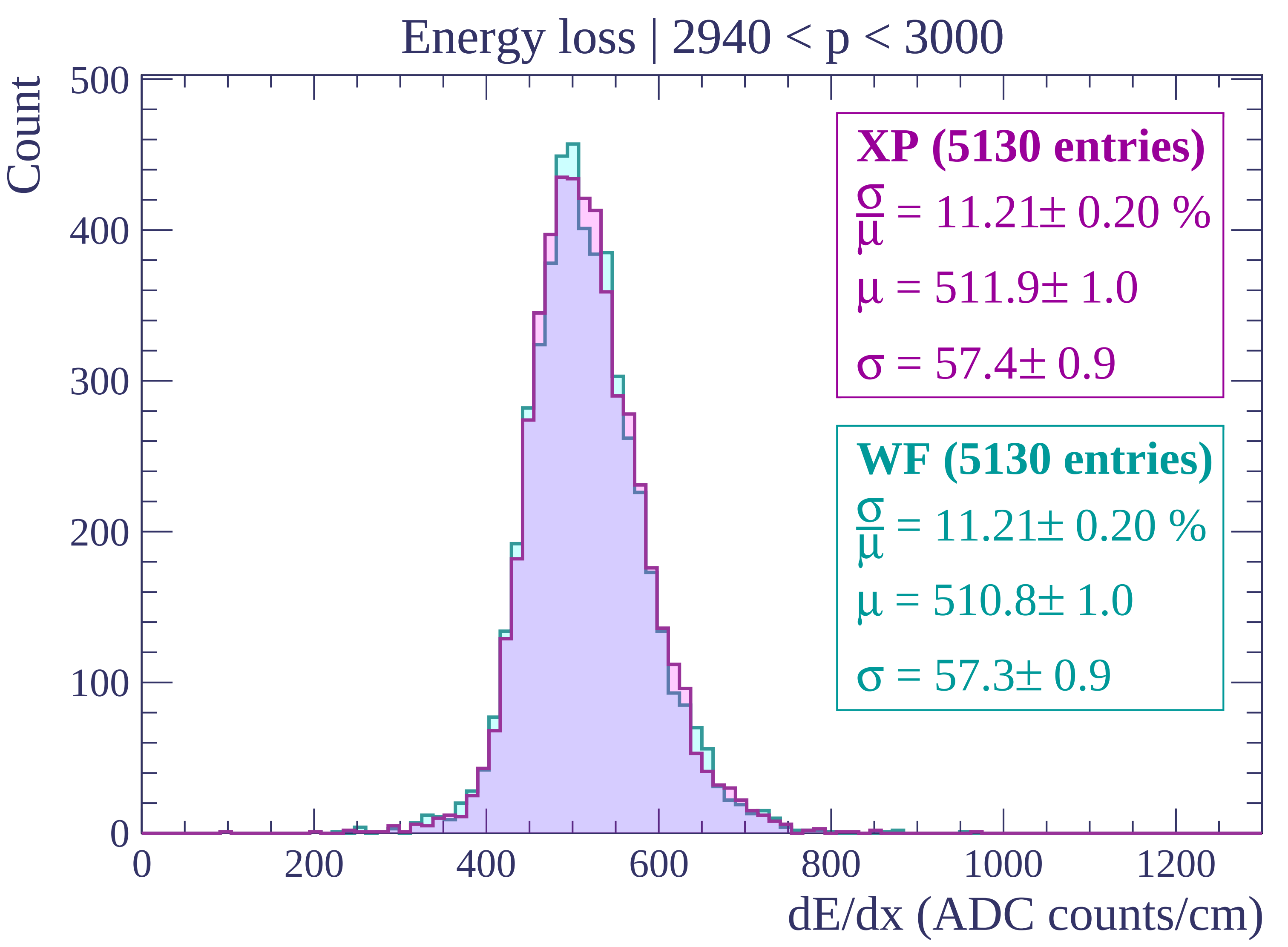


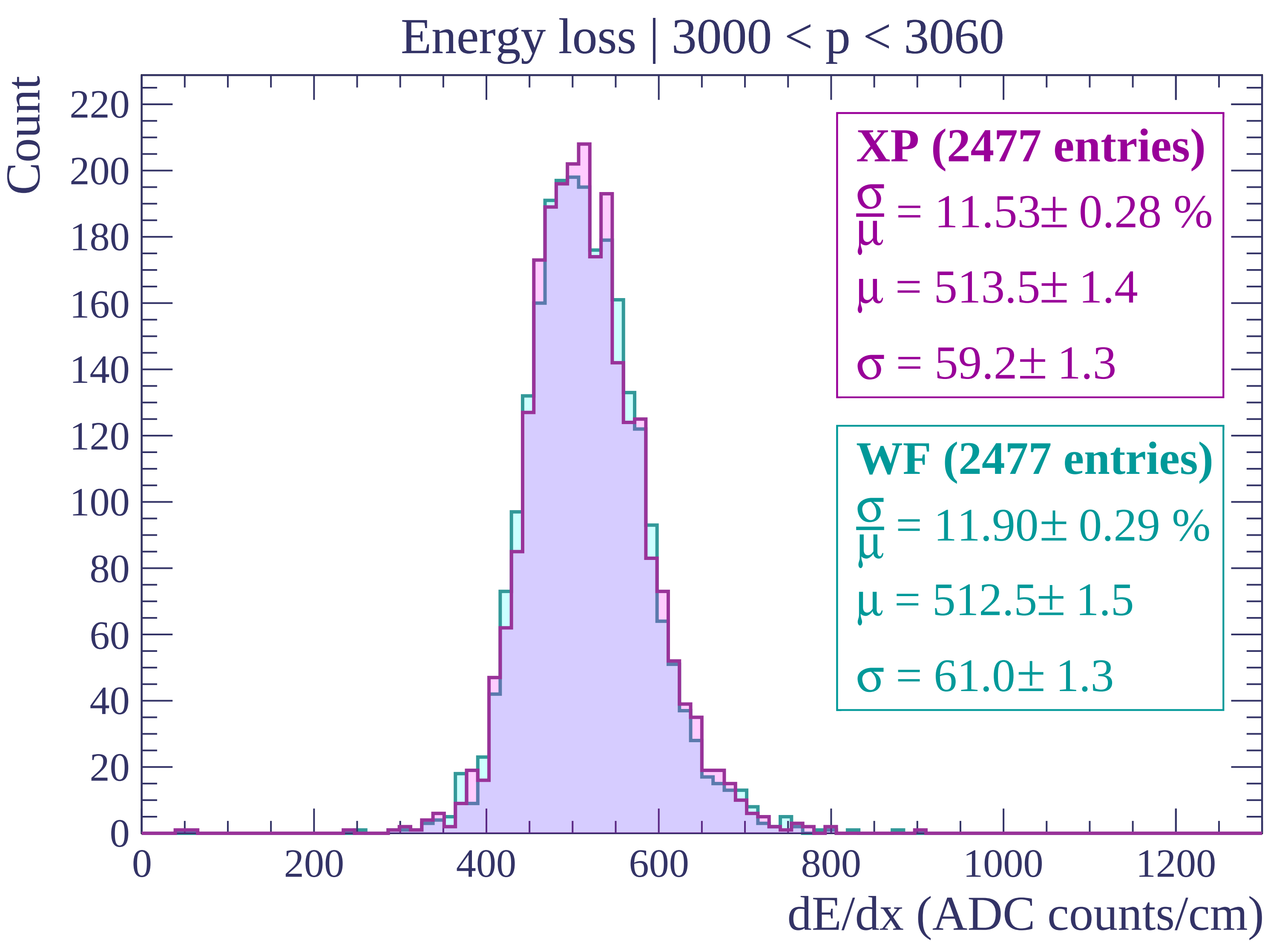
Energy loss | 2760 < p < 2820
Count
dE/dx (ADC counts/cm)
XP (5646 entries)
σ/μ = 11.53± 0.20 %
μ = 509.2± 1.0
σ = 58.7± 0.9
WF (5646 entries)
σ/μ = 11.63± 0.21 %
μ = 508.4± 1.0
σ = 59.1± 0.9
0
100
200
300
400
500
0
200
400
600
800
1000
1200

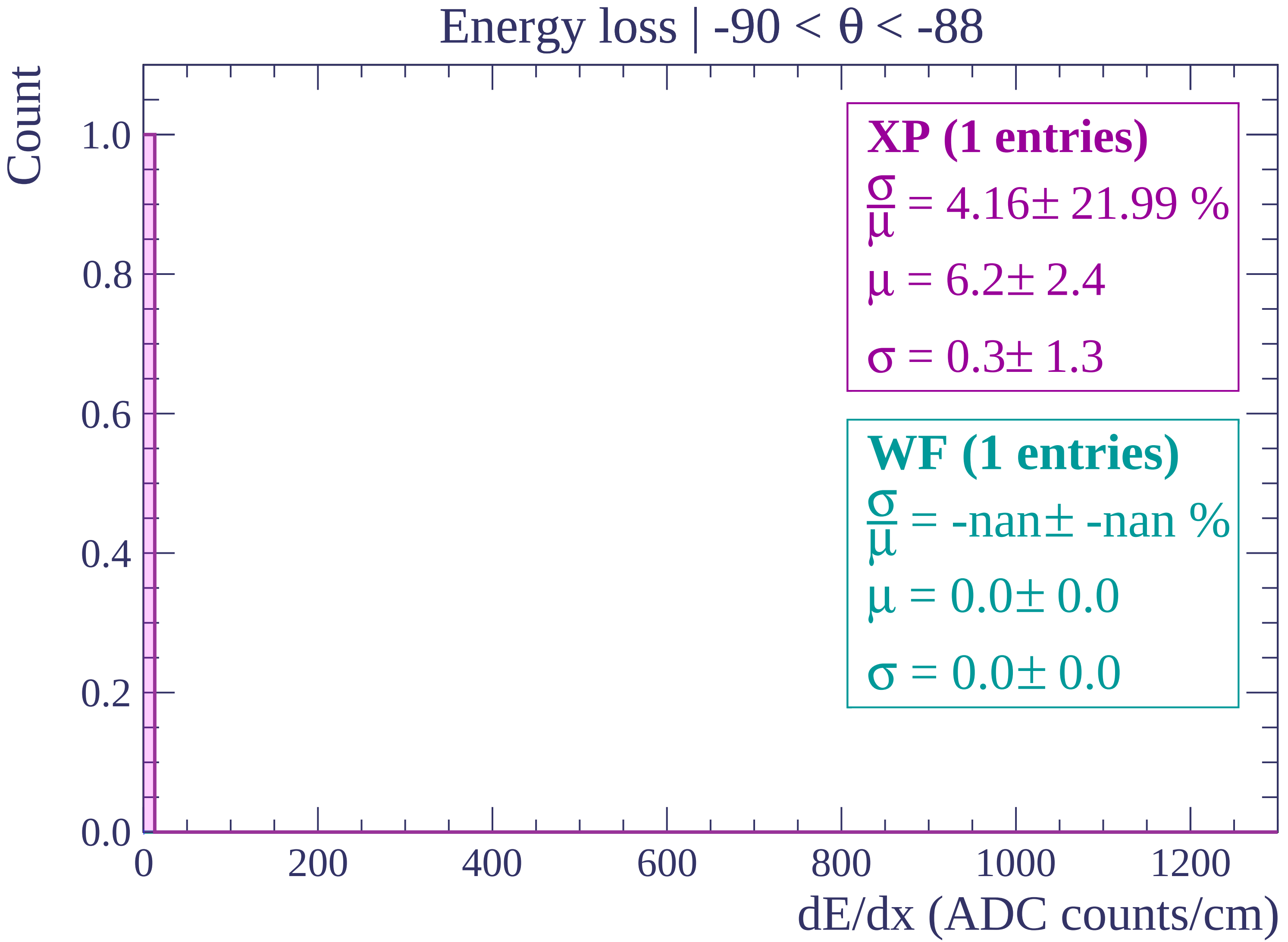

Energy loss | 2820 < p < 2880
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
XP (5425 entries)
σ/μ = 11.87± 0.21 %
μ = 511.8± 1.0
σ = 60.8± 0.9
WF (5425 entries)
σ/μ = 11.90± 0.21 %
μ = 511.6± 1.0
σ = 60.9± 0.9

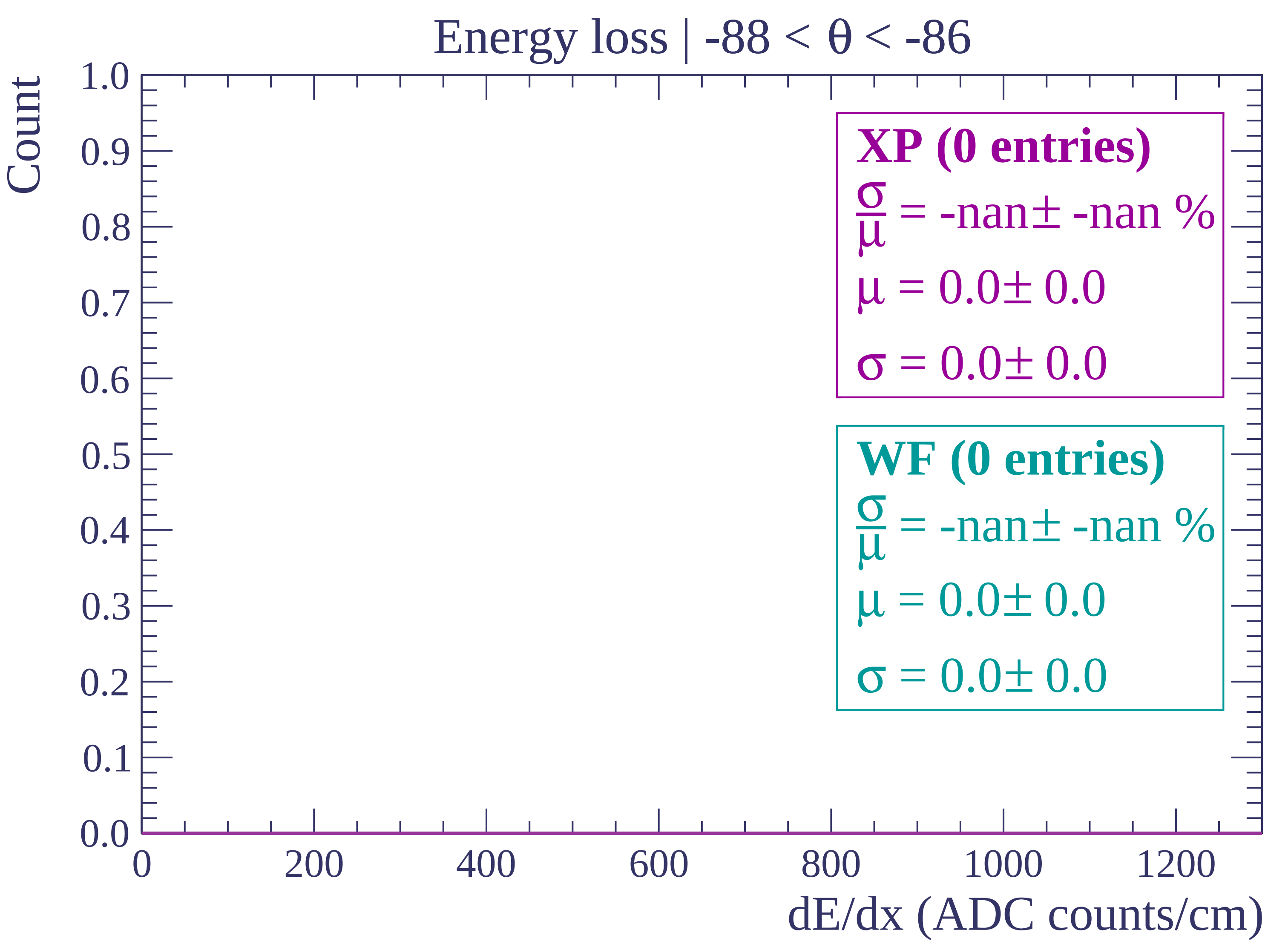


Energy loss | 2880 < p < 2940
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5344 entries)
σ/μ = 11.19± 0.19 %
μ = 512.1± 1.0
σ = 57.3± 0.9
WF (5344 entries)
σ/μ = 11.26± 0.20 %
μ = 511.5± 1.0
σ = 57.6± 0.9

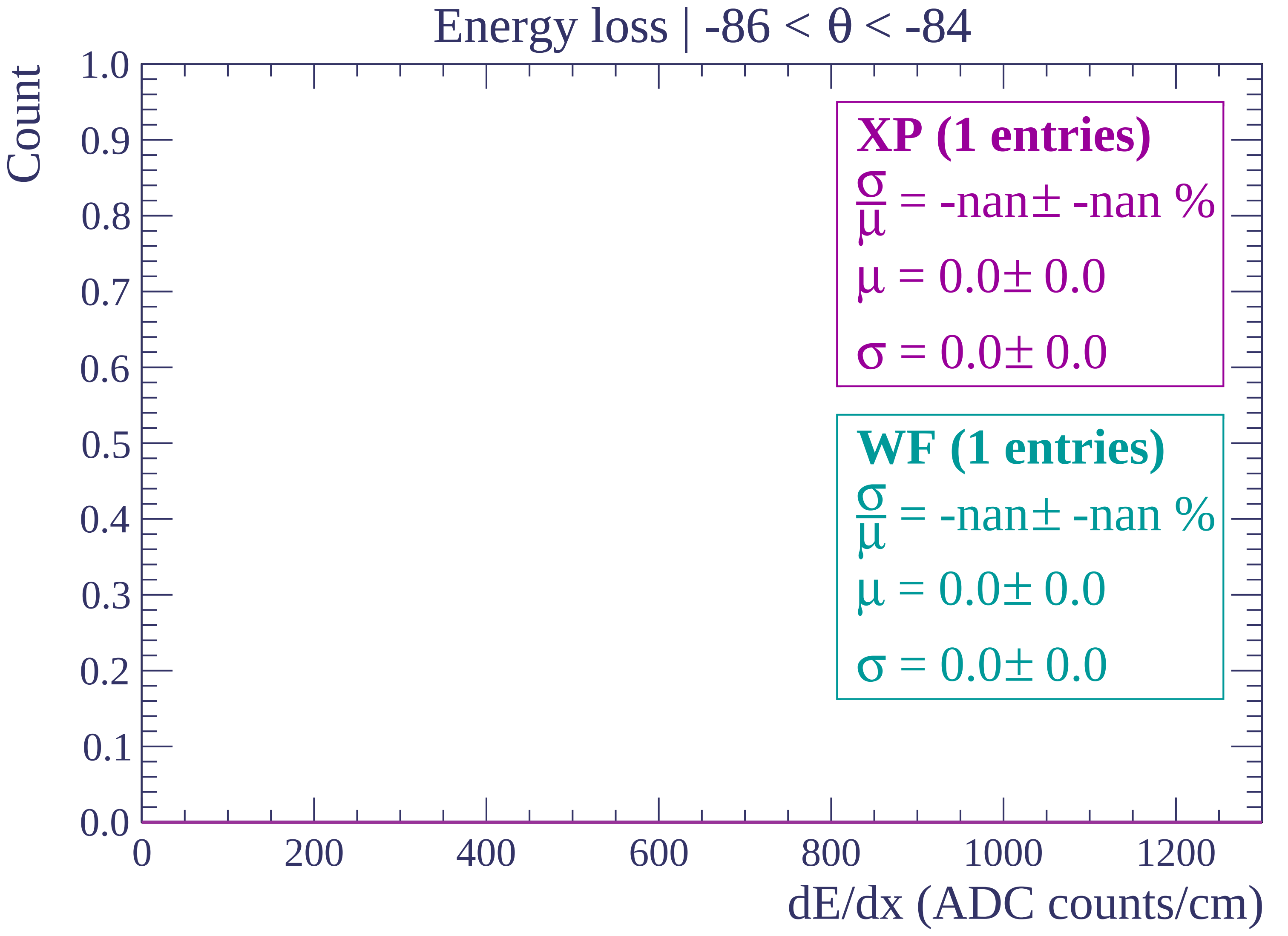

Energy loss | 2940 < p < 3000
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (5130 entries)
σ/μ = 11.21± 0.20 %
μ = 511.9± 1.0
σ = 57.4± 0.9
WF (5130 entries)
σ/μ = 11.21± 0.20 %
μ = 510.8± 1.0
σ = 57.3± 0.9


Energy loss | 3000 < p < 3060
Count
dE/dx (ADC counts/cm)
XP (2477 entries)
$\frac{\sigma}{\mu}$ = 11.53± 0.28 %
μ = 513.5± 1.4
σ = 59.2± 1.3
WF (2477 entries)
$\frac{\sigma}{\mu}$ = 11.90± 0.29 %
μ = 512.5± 1.5
σ = 61.0± 1.3
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200

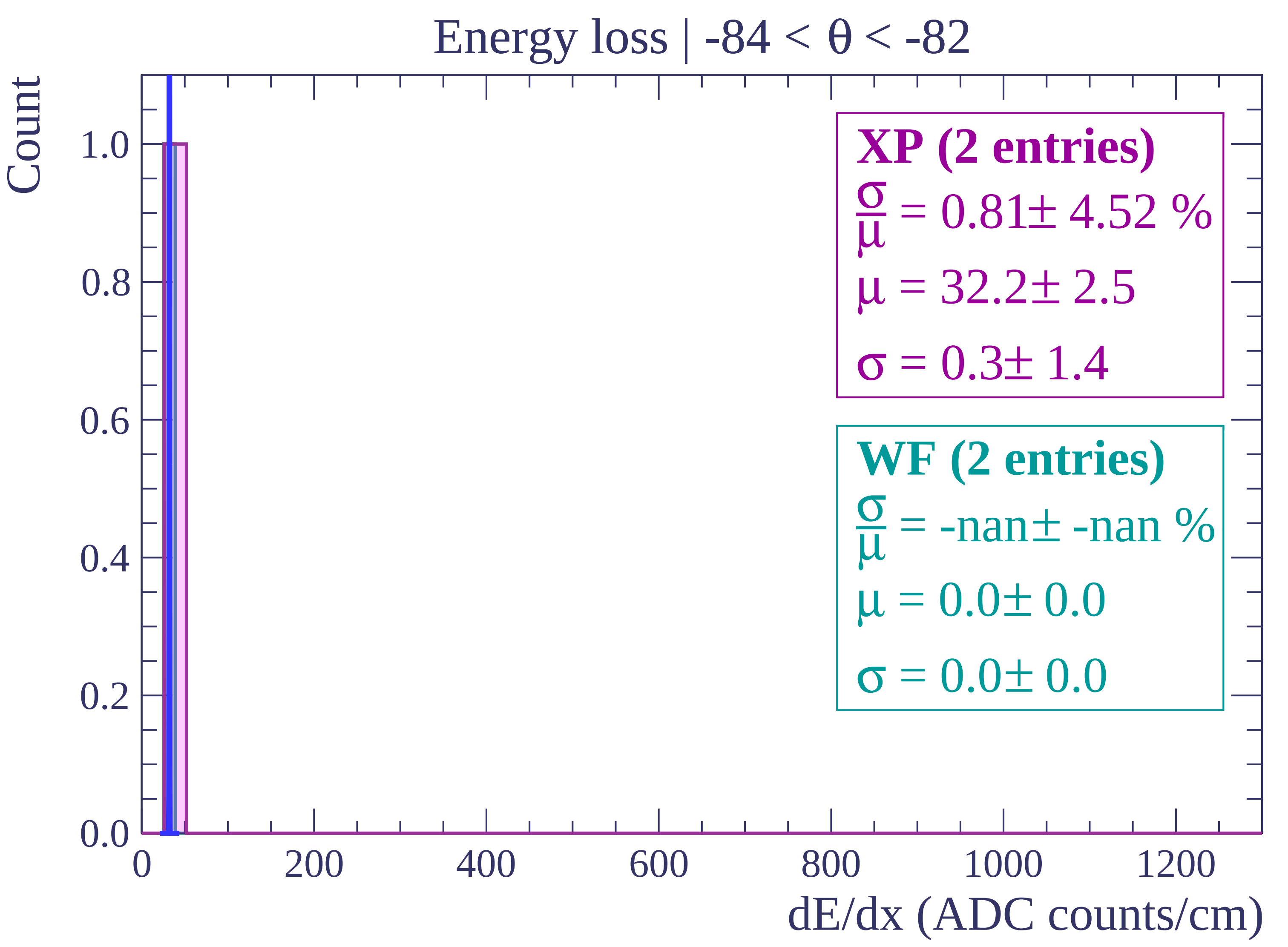

Energy loss | -90 < θ < -88
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1 entries)
σ/μ = 4.16± 21.99 %
μ = 6.2± 2.4
σ = 0.3± 1.3
WF (1 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



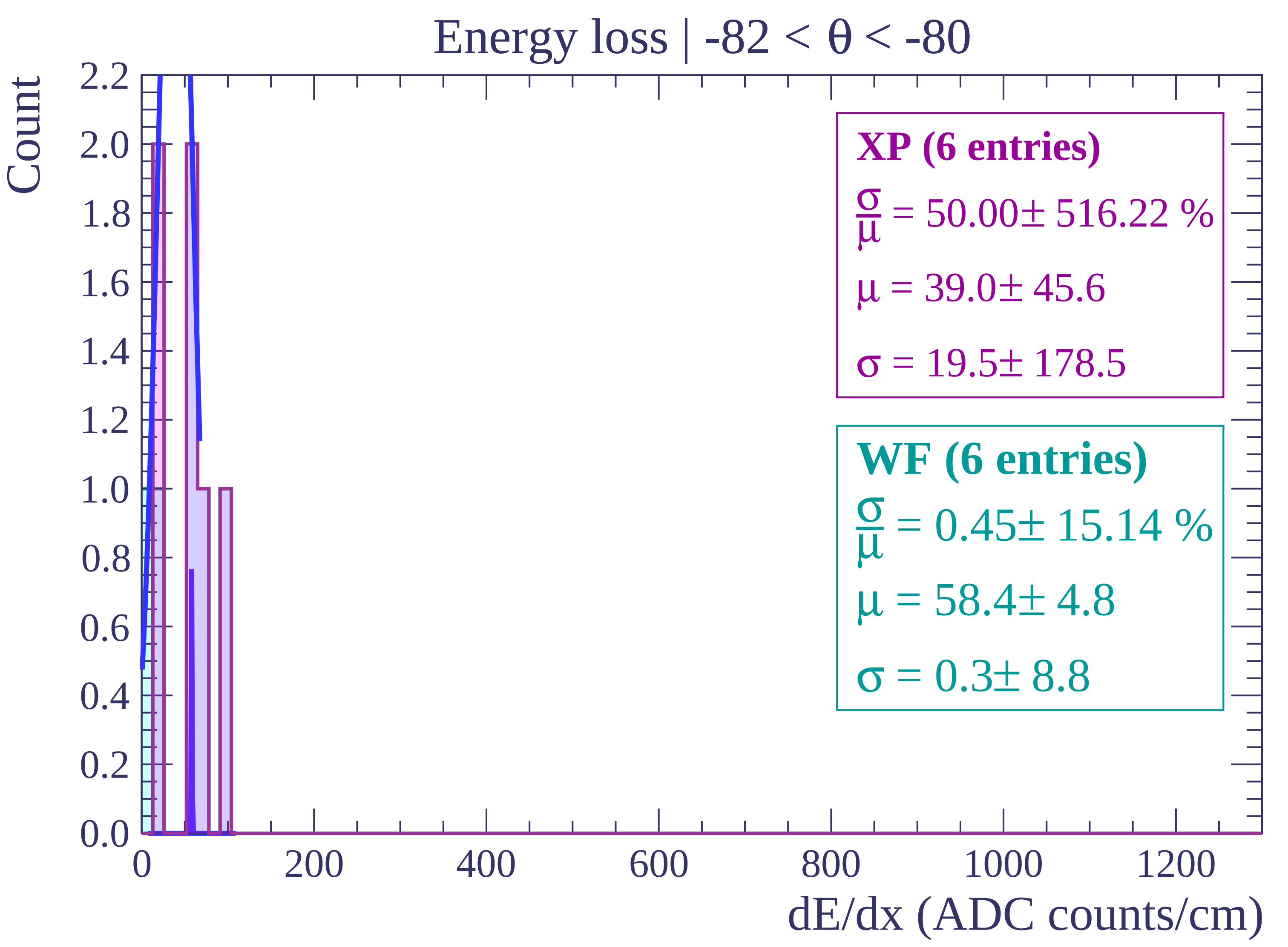
Energy loss | -88 < θ < -86
Count
dE/dx (ADC counts/cm)
XP (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



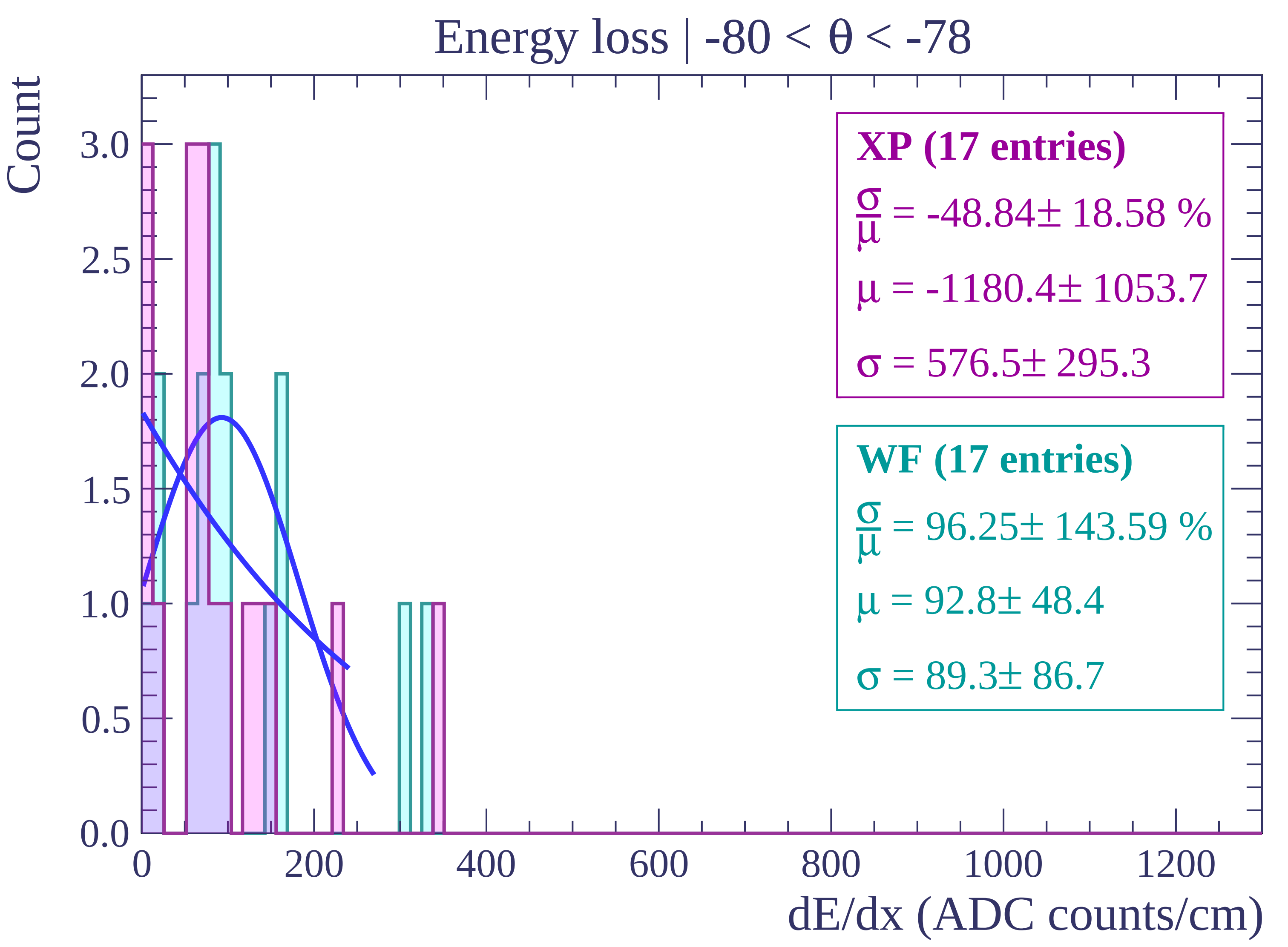
Energy loss | -86 < θ < -84
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (1 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



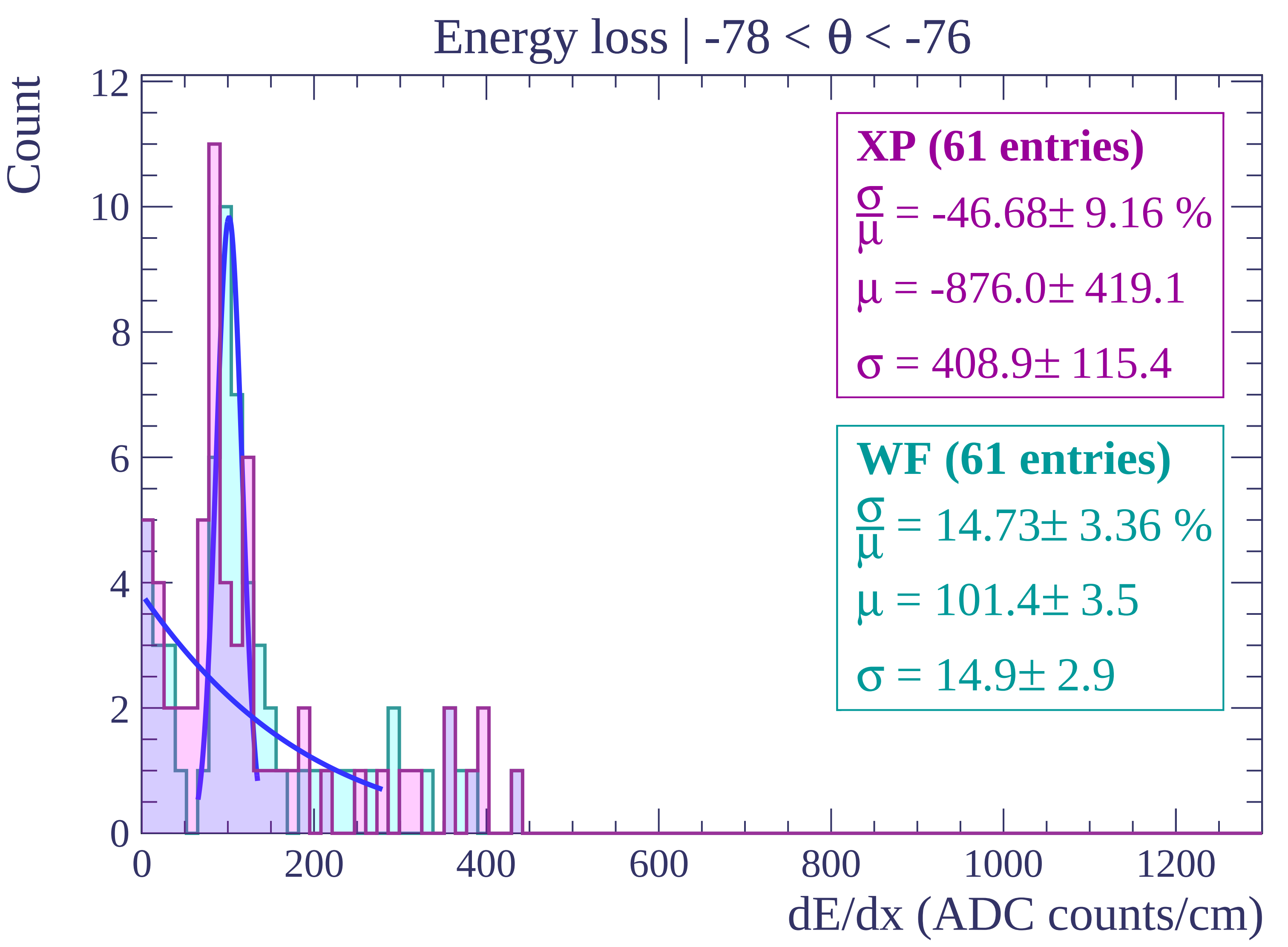
Energy loss | -84 < θ < -82
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2 entries)
σ/μ = 0.81± 4.52 %
μ = 32.2± 2.5
σ = 0.3± 1.4
WF (2 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



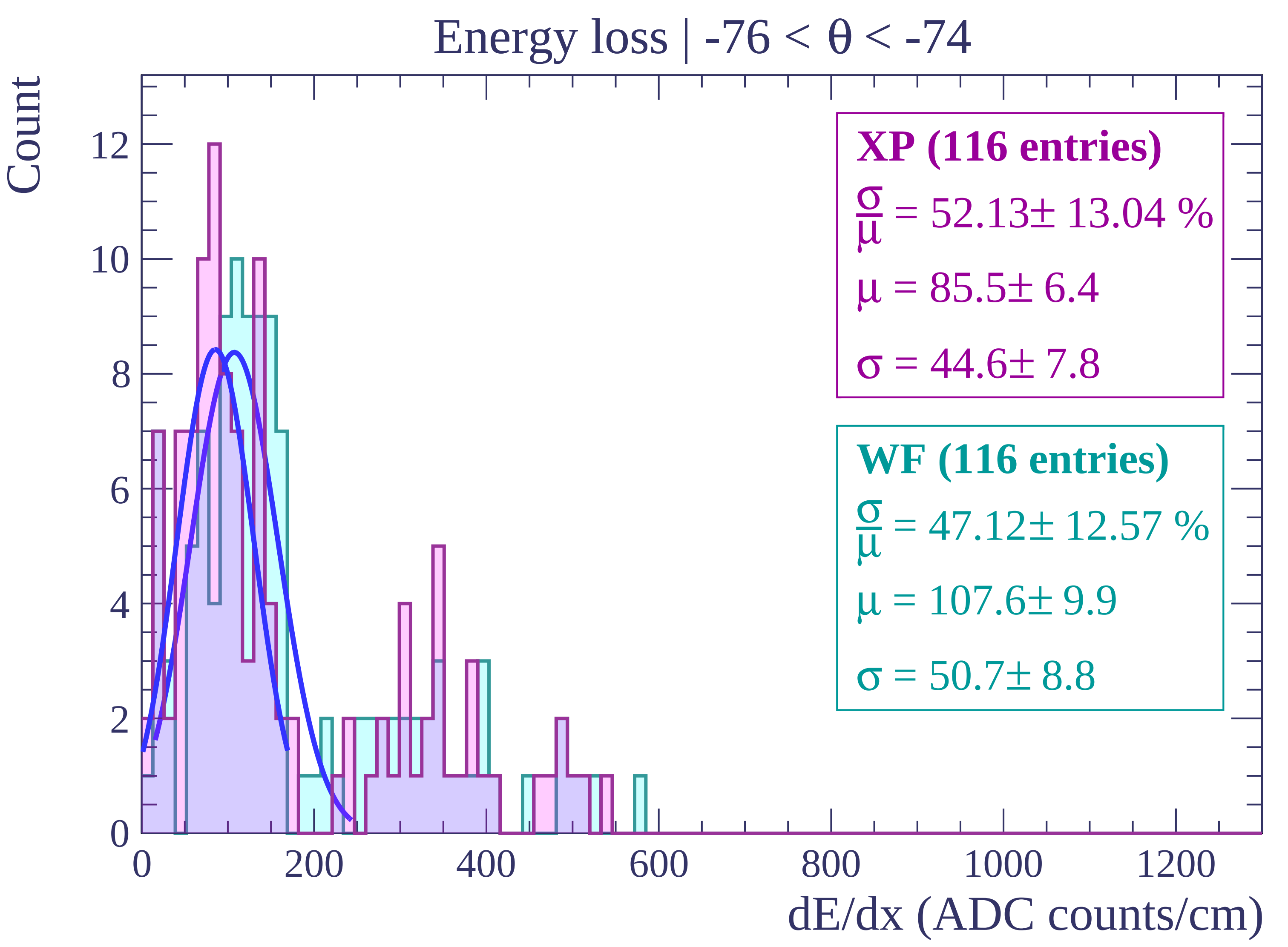
Energy loss | -82 < θ < -80
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
XP (6 entries)
$\frac{\sigma}{\mu}$ = 50.00± 516.22 %
μ = 39.0± 45.6
σ = 19.5± 178.5
WF (6 entries)
$\frac{\sigma}{\mu}$ = 0.45± 15.14 %
μ = 58.4± 4.8
σ = 0.3± 8.8



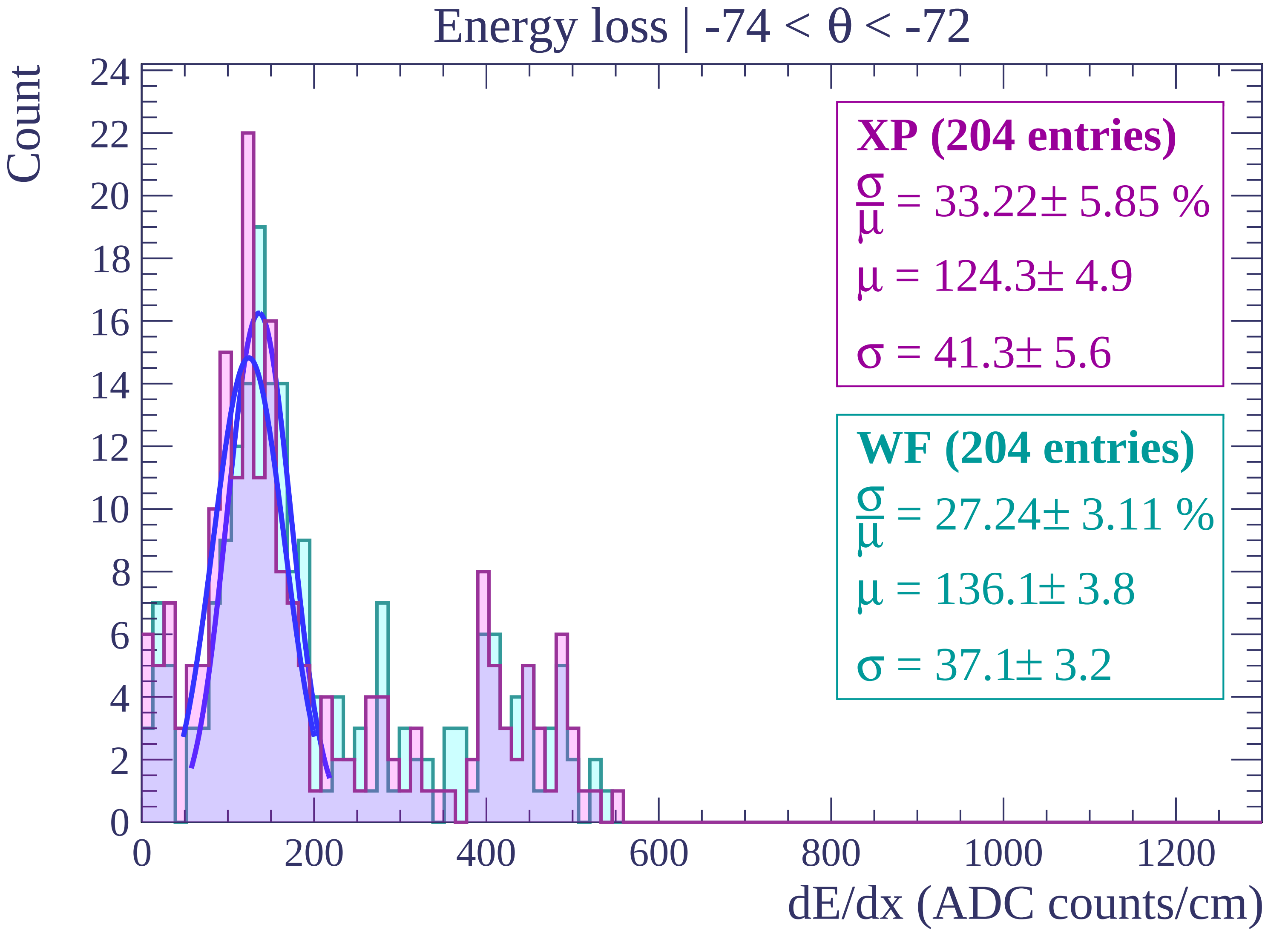
Energy loss | -80 < θ < -78
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
200
400
600
800
1000
1200
XP (17 entries)
σ/μ = -48.84± 18.58 %
μ = -1180.4± 1053.7
σ = 576.5± 295.3
WF (17 entries)
σ/μ = 96.25± 143.59 %
μ = 92.8± 48.4
σ = 89.3± 86.7



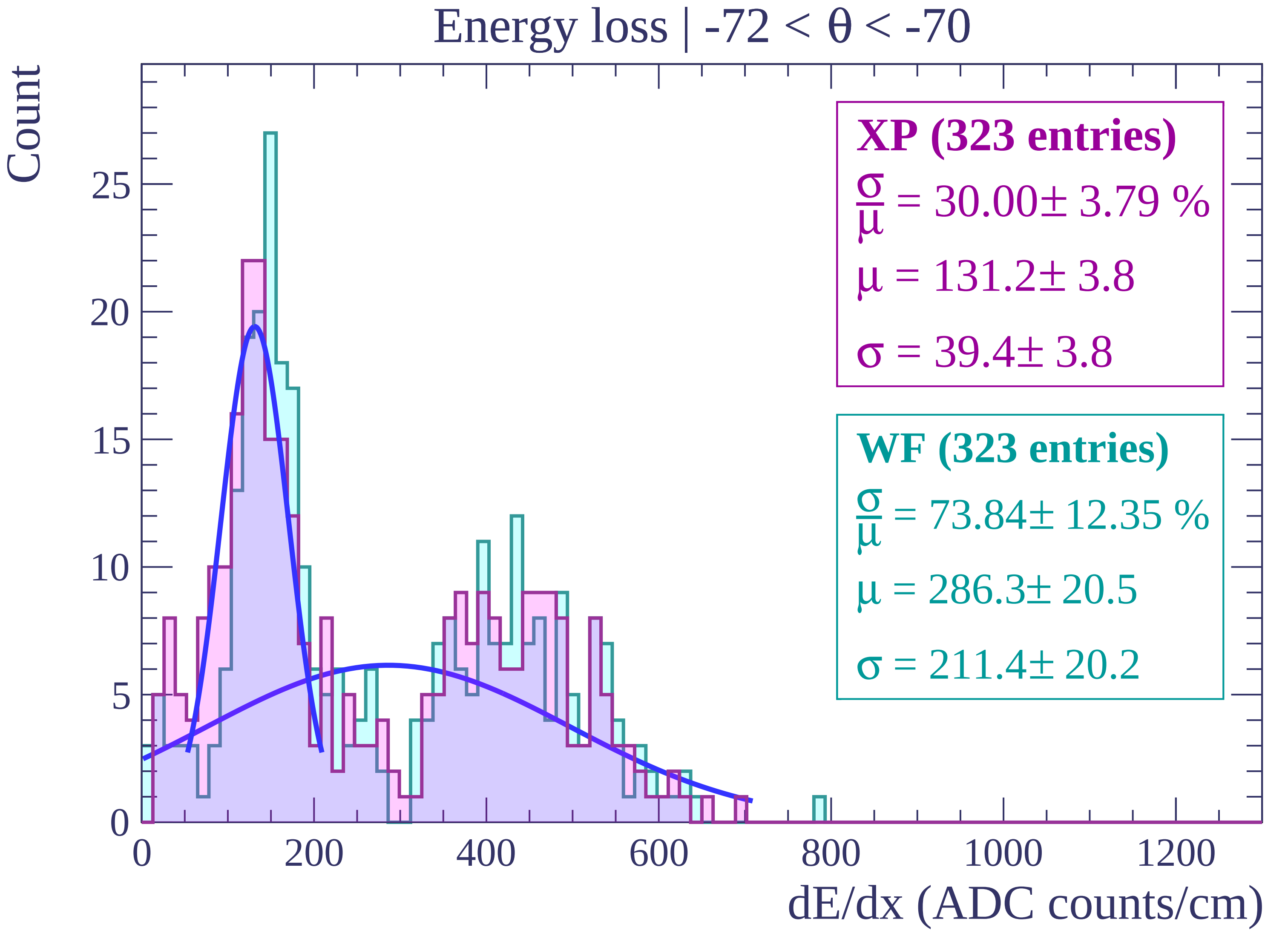
Energy loss | -78 < θ < -76
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
XP (61 entries)
σ/μ = -46.68± 9.16 %
μ = -876.0± 419.1
σ = 408.9± 115.4
WF (61 entries)
σ/μ = 14.73± 3.36 %
μ = 101.4± 3.5
σ = 14.9± 2.9



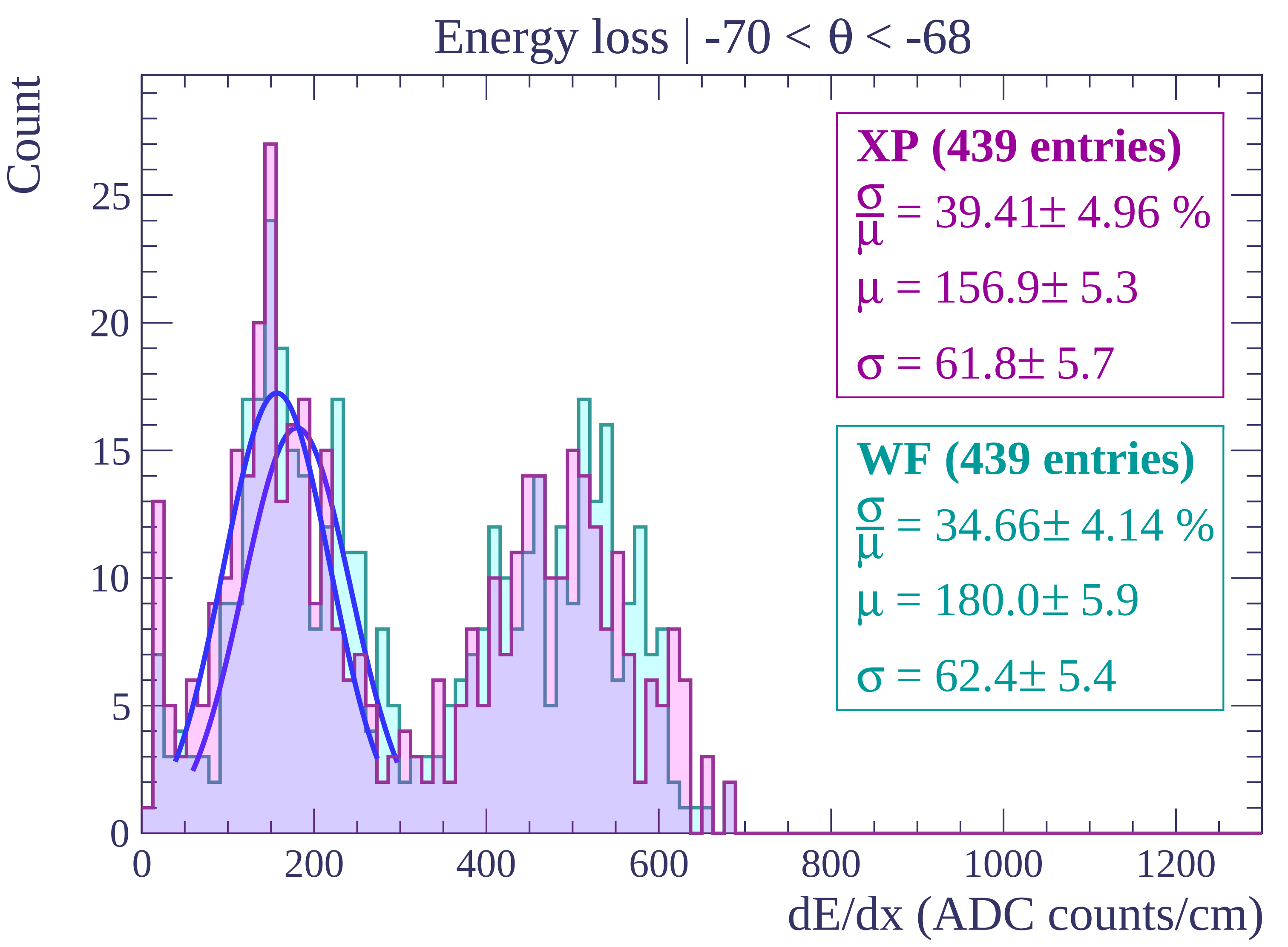
Energy loss | -76 < θ < -74
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
XP (116 entries)
σ/μ = 52.13± 13.04 %
μ = 85.5± 6.4
σ = 44.6± 7.8
WF (116 entries)
σ/μ = 47.12± 12.57 %
μ = 107.6± 9.9
σ = 50.7± 8.8



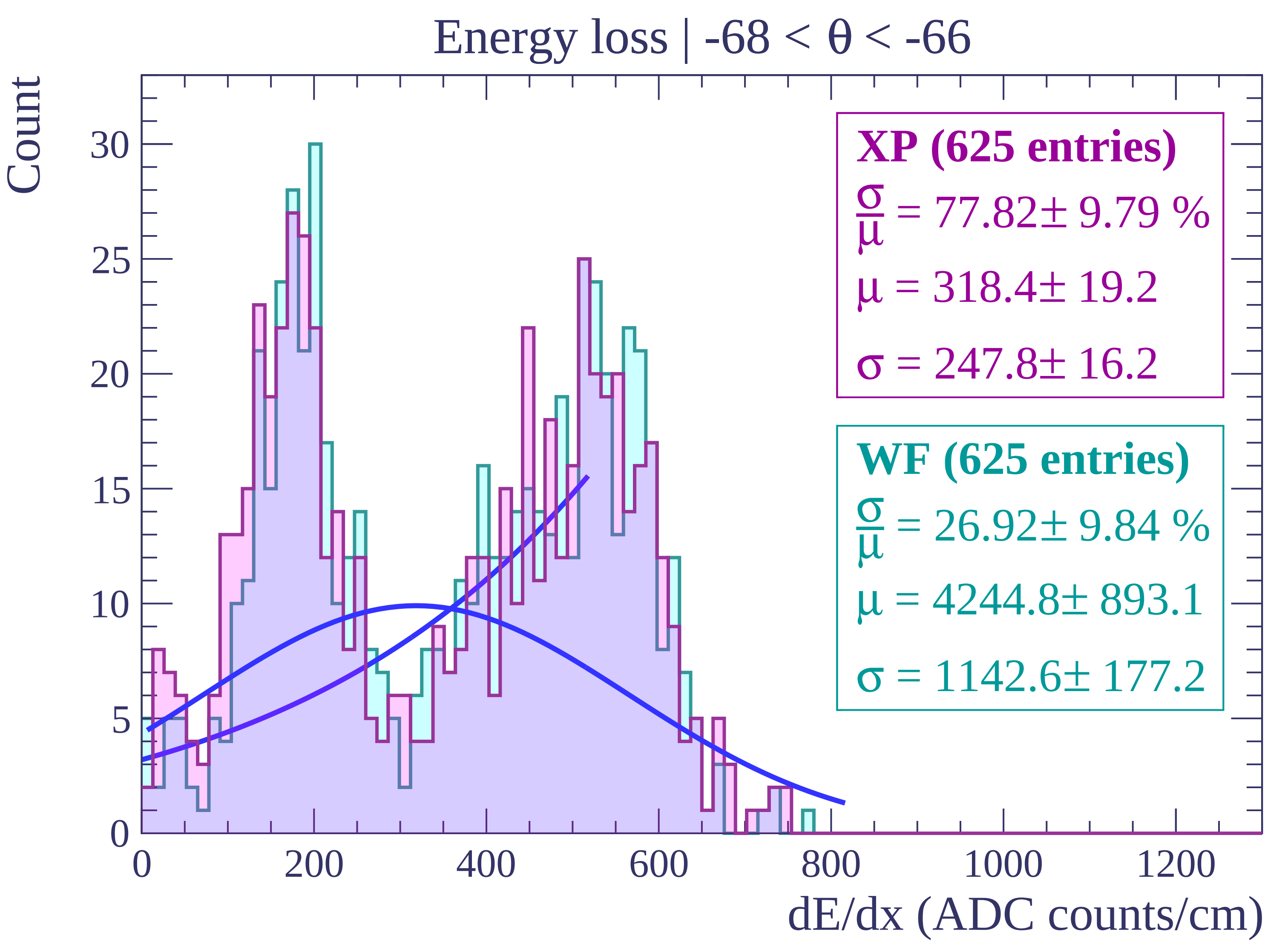
Energy loss | -74 < θ < -72
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
18
20
22
24
0
200
400
600
800
1000
1200
XP (204 entries)
σ/μ = 33.22± 5.85 %
μ = 124.3± 4.9
σ = 41.3± 5.6
WF (204 entries)
σ/μ = 27.24± 3.11 %
μ = 136.1± 3.8
σ = 37.1± 3.2

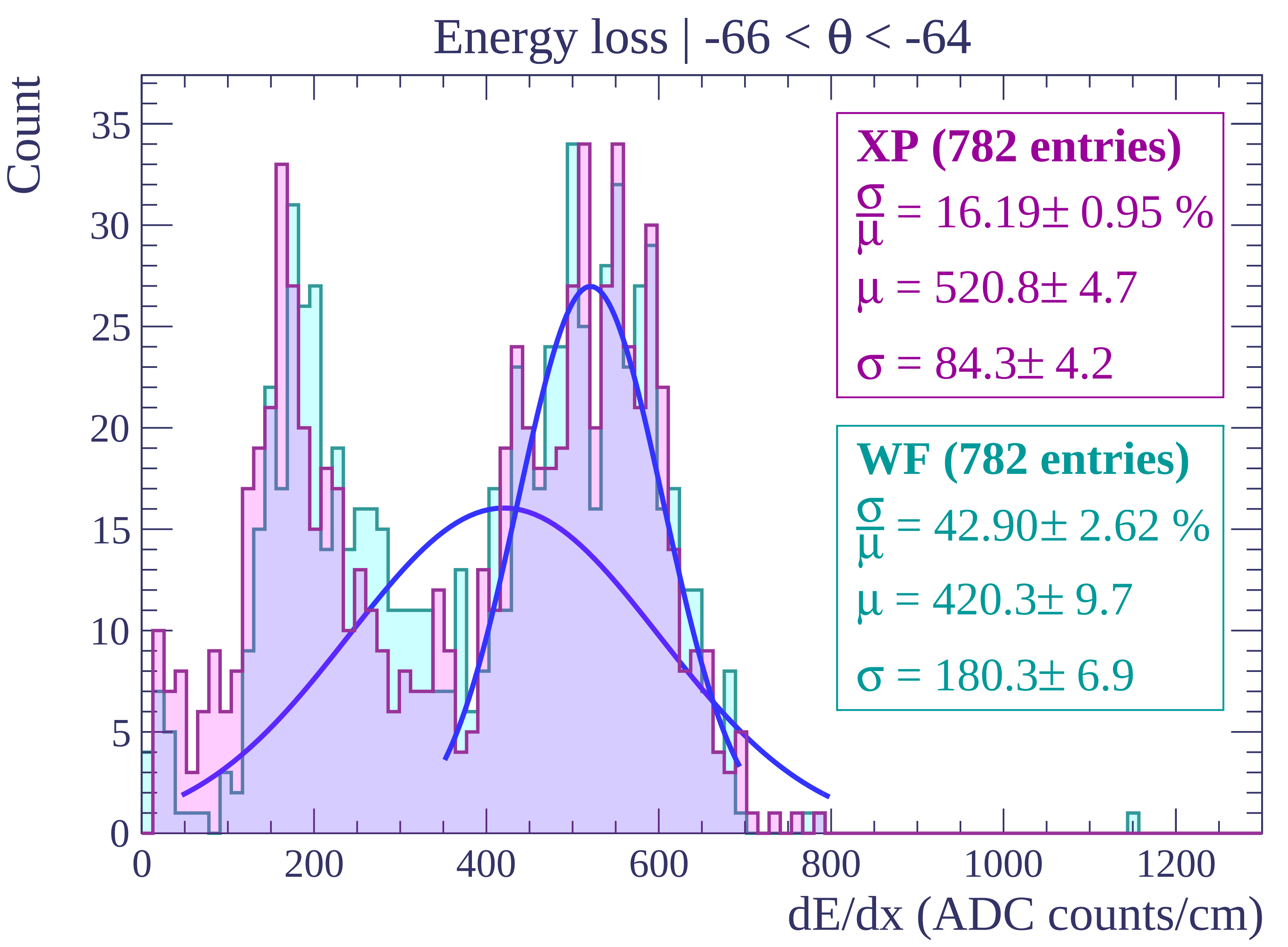


Energy loss | -72 < θ < -70
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
XP (323 entries)
$\frac{\sigma}{\mu}$ = 30.00± 3.79 %
μ = 131.2± 3.8
σ = 39.4± 3.8
WF (323 entries)
$\frac{\sigma}{\mu}$ = 73.84± 12.35 %
μ = 286.3± 20.5
σ = 211.4± 20.2

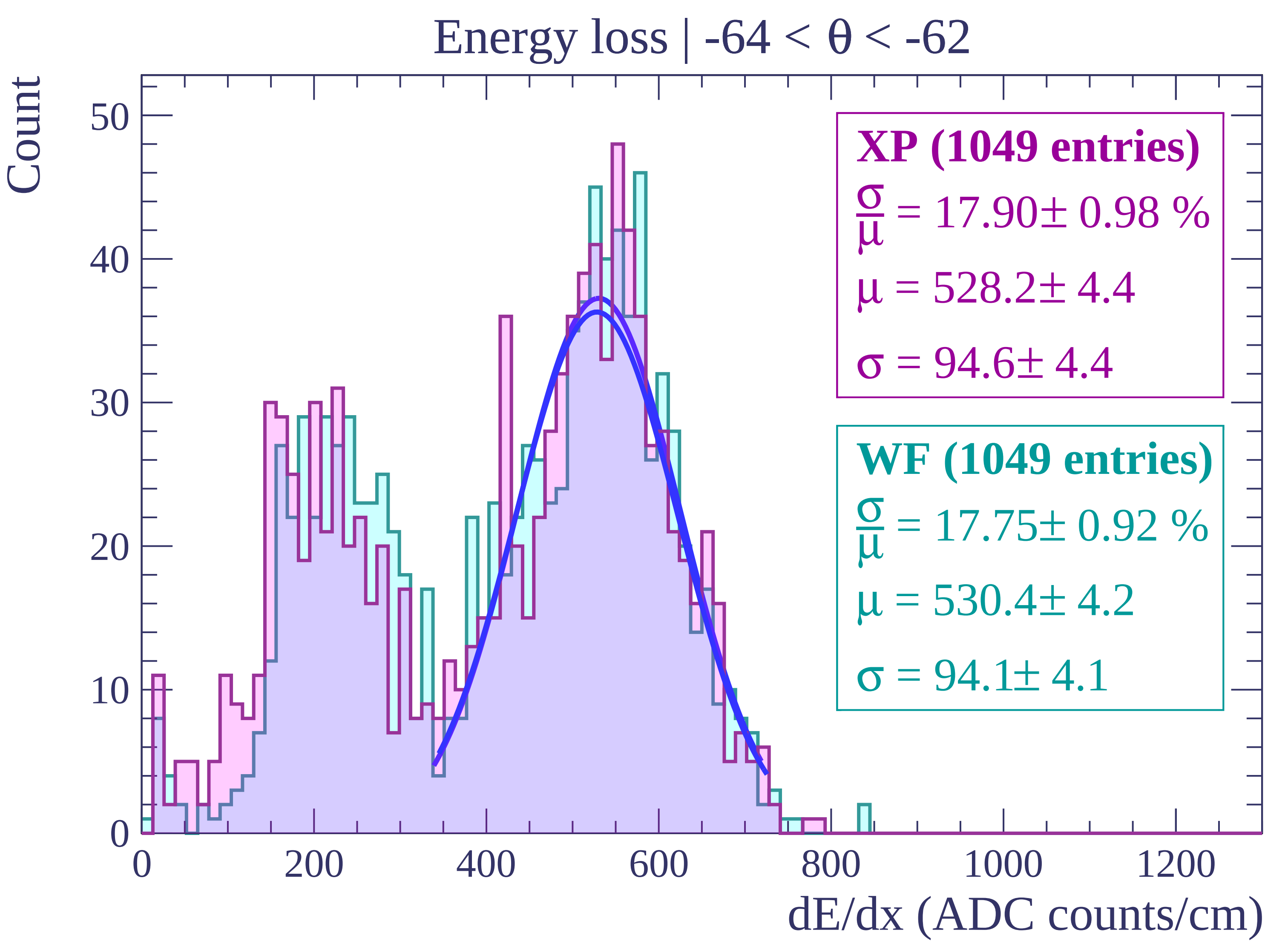


Energy loss | -70 < θ < -68
Count
XP (439 entries)
$\frac{\sigma}{\mu}$ = 39.41± 4.96 %
μ = 156.9± 5.3
σ = 61.8± 5.7
WF (439 entries)
$\frac{\sigma}{\mu}$ = 34.66± 4.14 %
μ = 180.0± 5.9
σ = 62.4± 5.4
0
5
10
15
20
25
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)

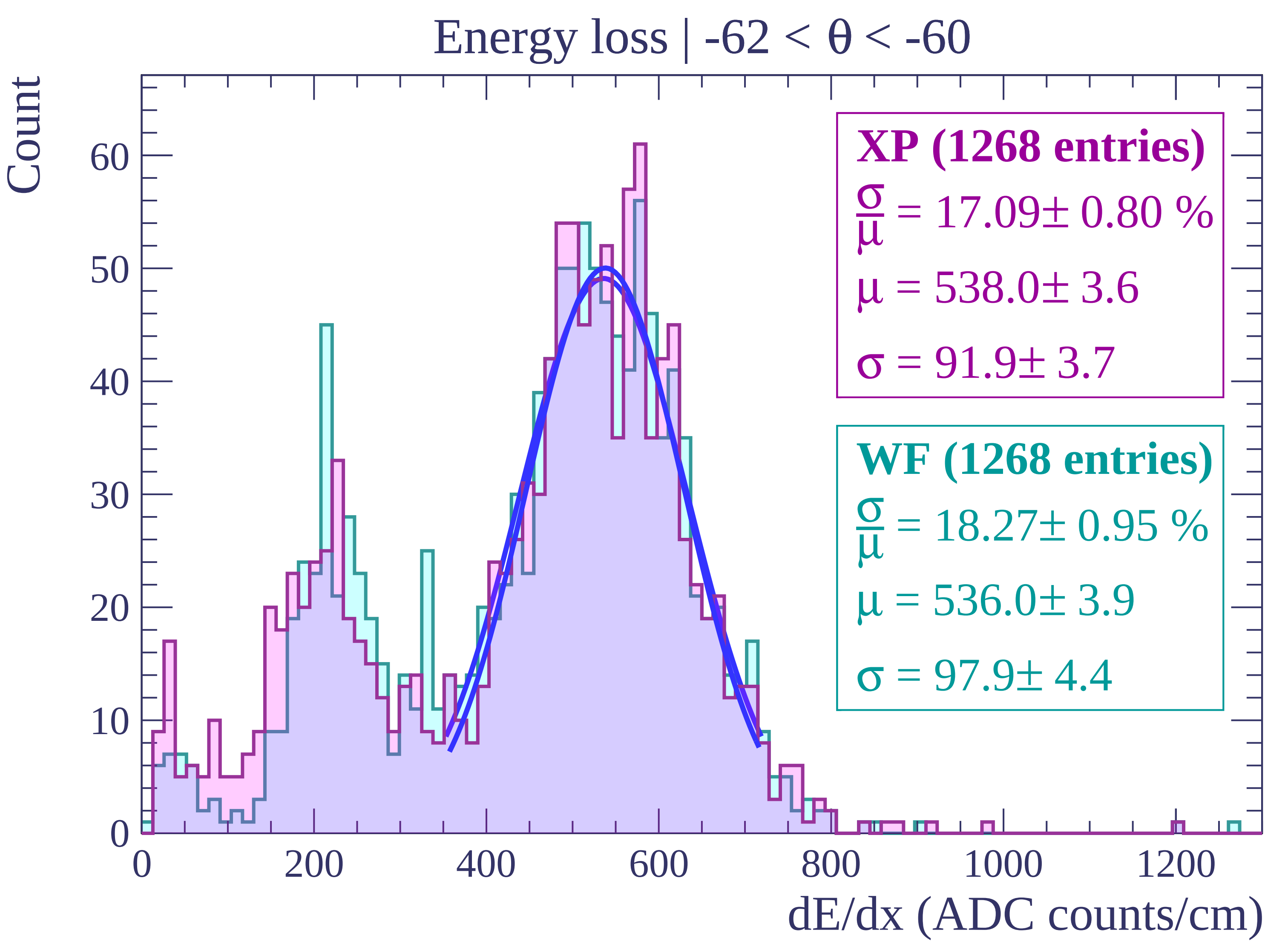

Energy loss | -68 < θ < -66
Count
dE/dx (ADC counts/cm)
XP (625 entries)
σ/μ = 77.82± 9.79 %
μ = 318.4± 19.2
σ = 247.8± 16.2
WF (625 entries)
σ/μ = 26.92± 9.84 %
μ = 4244.8± 893.1
σ = 1142.6± 177.2
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200

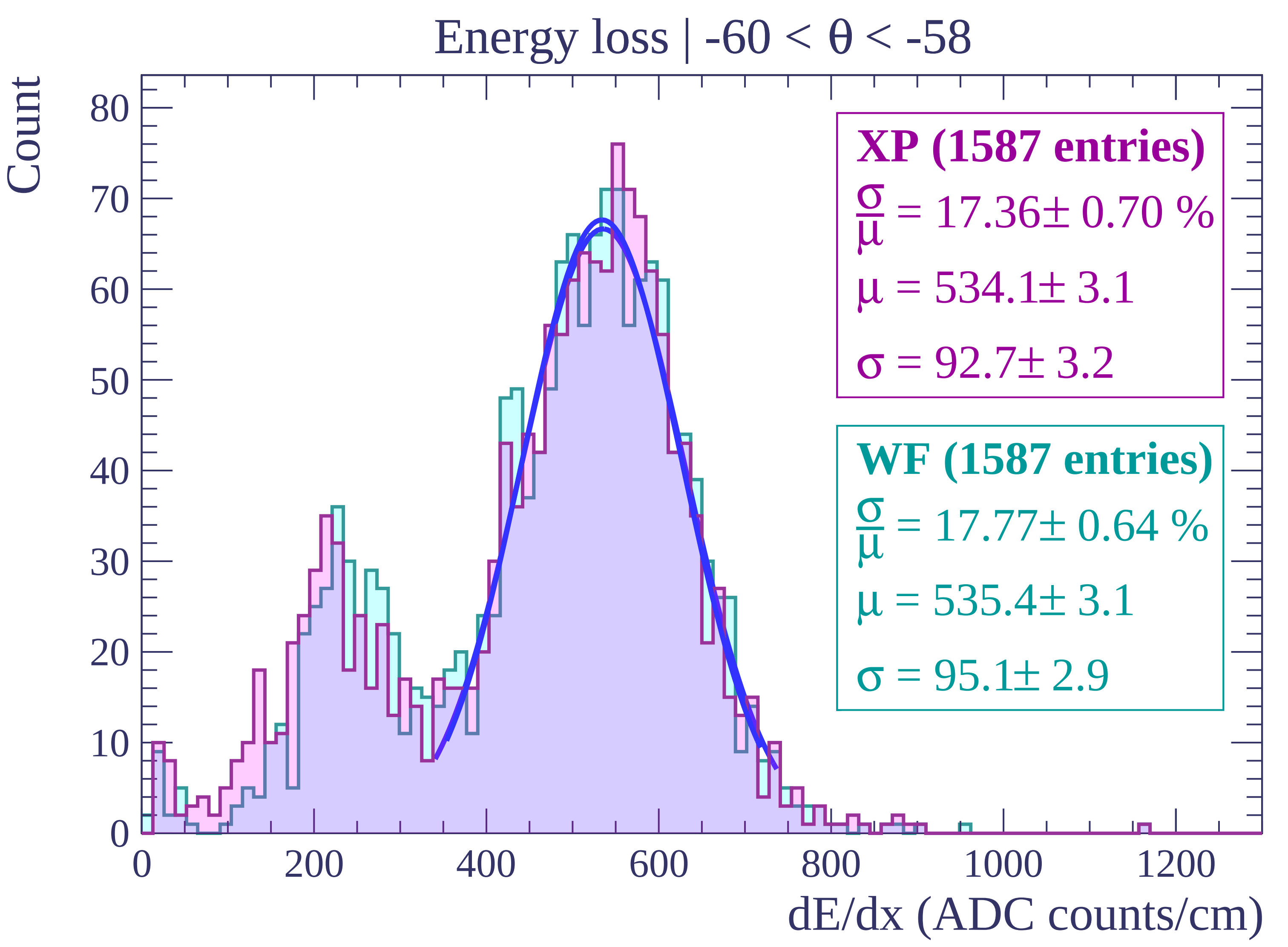


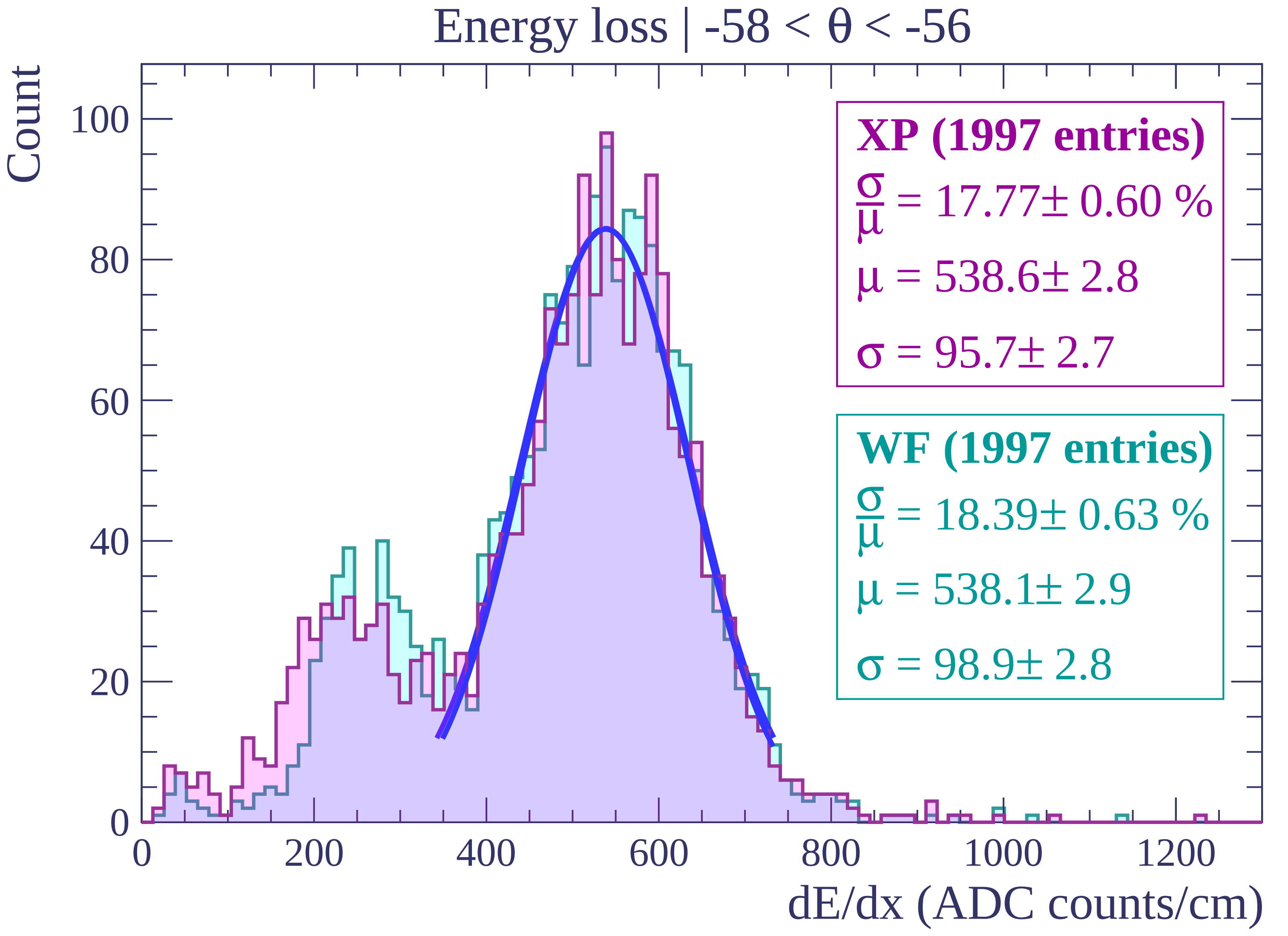
Energy loss | -66 < θ < -64
Count
dE/dx (ADC counts/cm)
XP (782 entries)
σ/μ = 16.19± 0.95 %
μ = 520.8± 4.7
σ = 84.3± 4.2
WF (782 entries)
σ/μ = 42.90± 2.62 %
μ = 420.3± 9.7
σ = 180.3± 6.9
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200

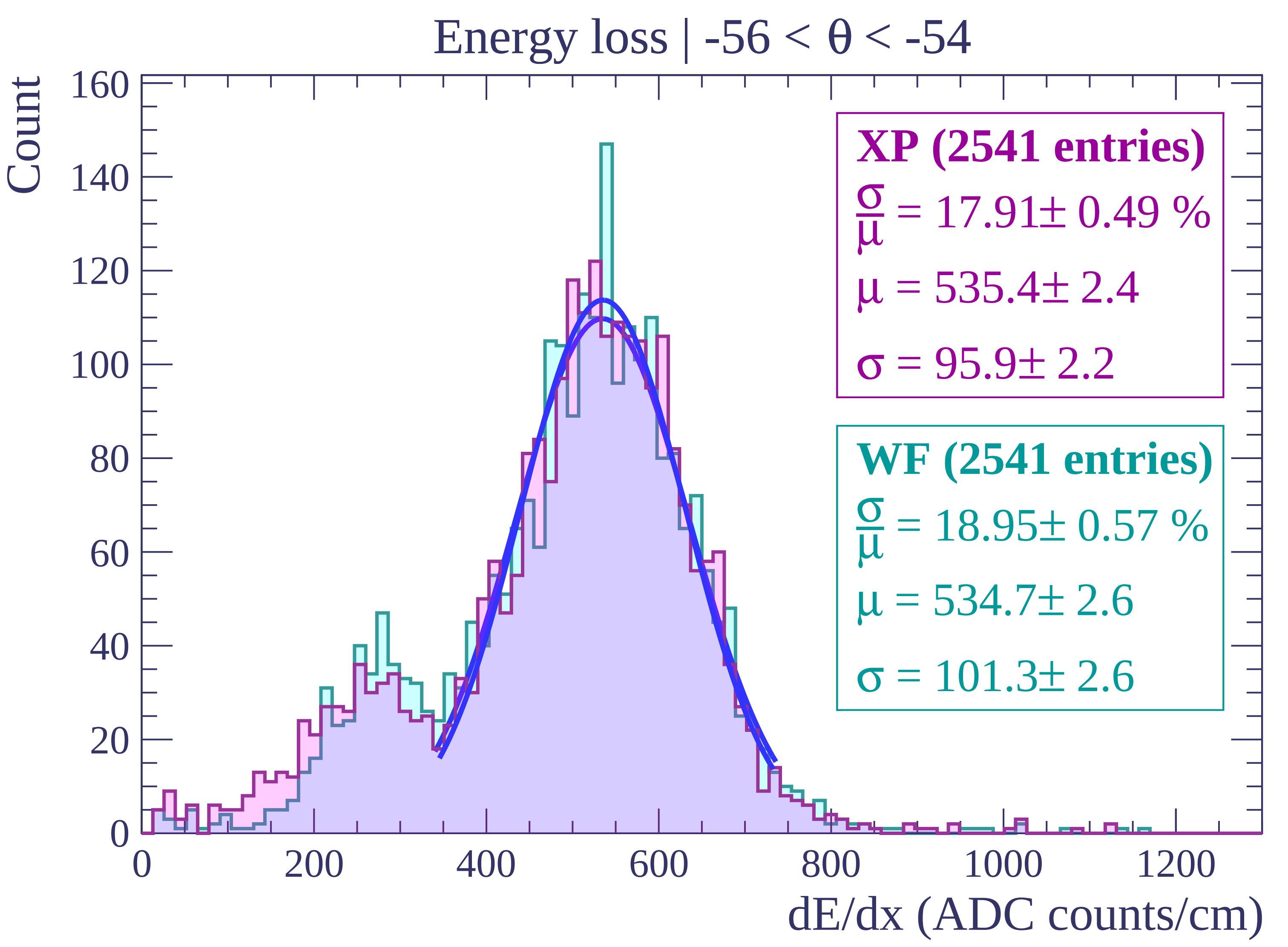


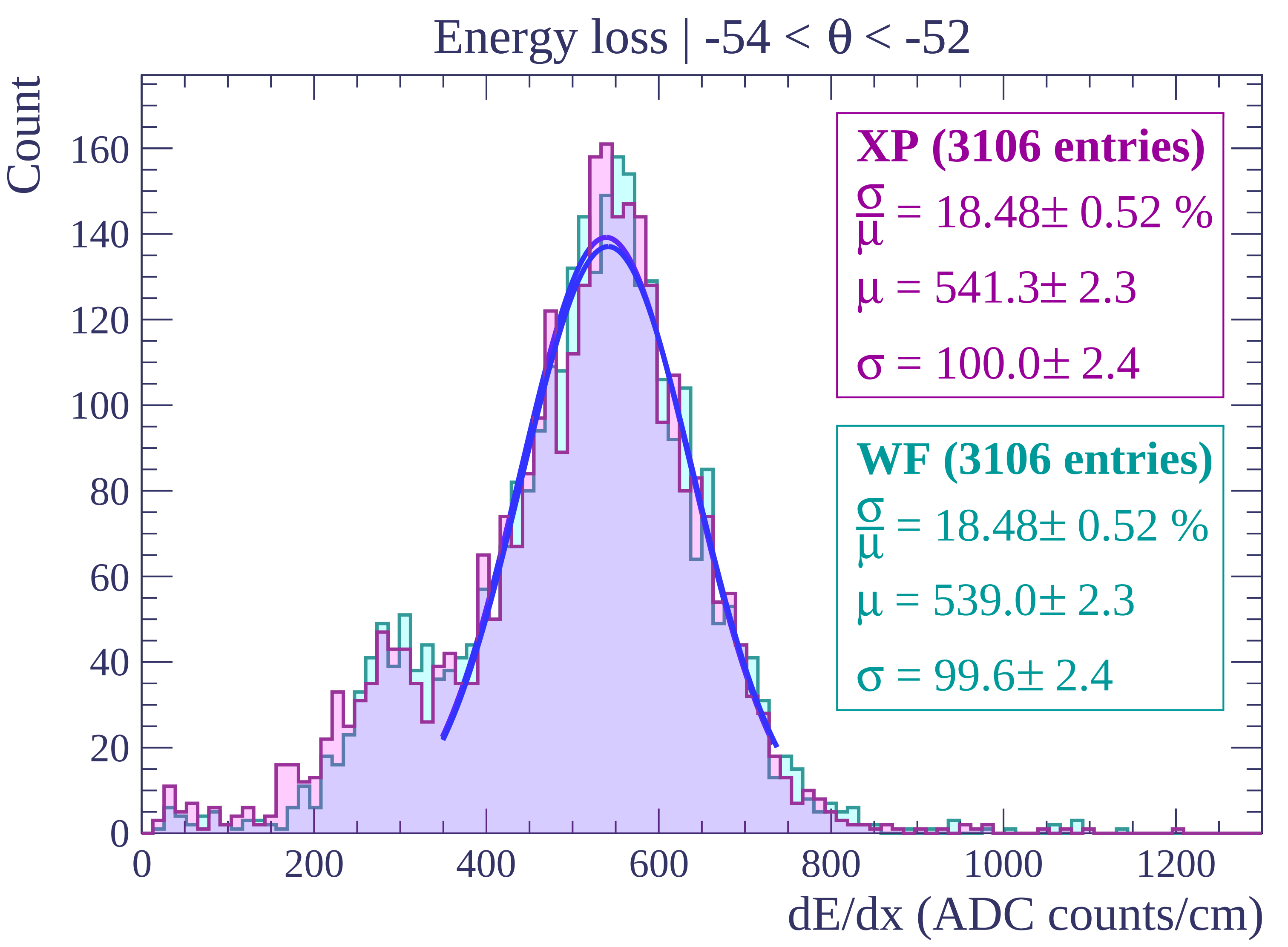
Energy loss | -64 < θ < -62
Count
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1049 entries)
σ/μ = 17.90± 0.98 %
μ = 528.2± 4.4
σ = 94.6± 4.4
WF (1049 entries)
σ/μ = 17.75± 0.92 %
μ = 530.4± 4.2
σ = 94.1± 4.1

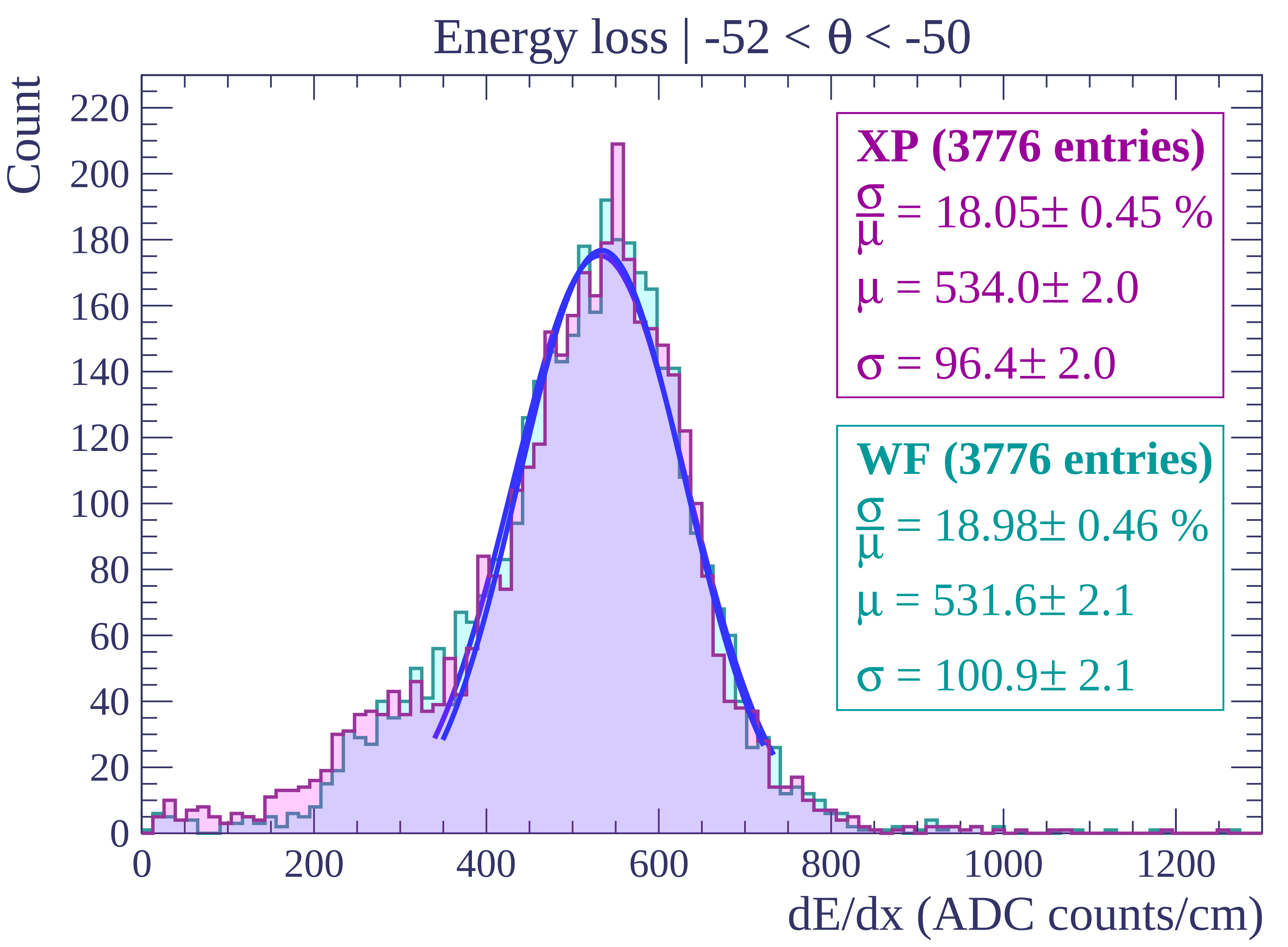

Energy loss | -62 < θ < -60
Count
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1268 entries)
σ/μ = 17.09± 0.80 %
μ = 538.0± 3.6
σ = 91.9± 3.7
WF (1268 entries)
σ/μ = 18.27± 0.95 %
μ = 536.0± 3.9
σ = 97.9± 4.4


Energy loss | -60 < θ < -58
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP (1587 entries)
σ/μ = 17.36± 0.70 %
μ = 534.1± 3.1
σ = 92.7± 3.2
WF (1587 entries)
σ/μ = 17.77± 0.64 %
μ = 535.4± 3.1
σ = 95.1± 2.9

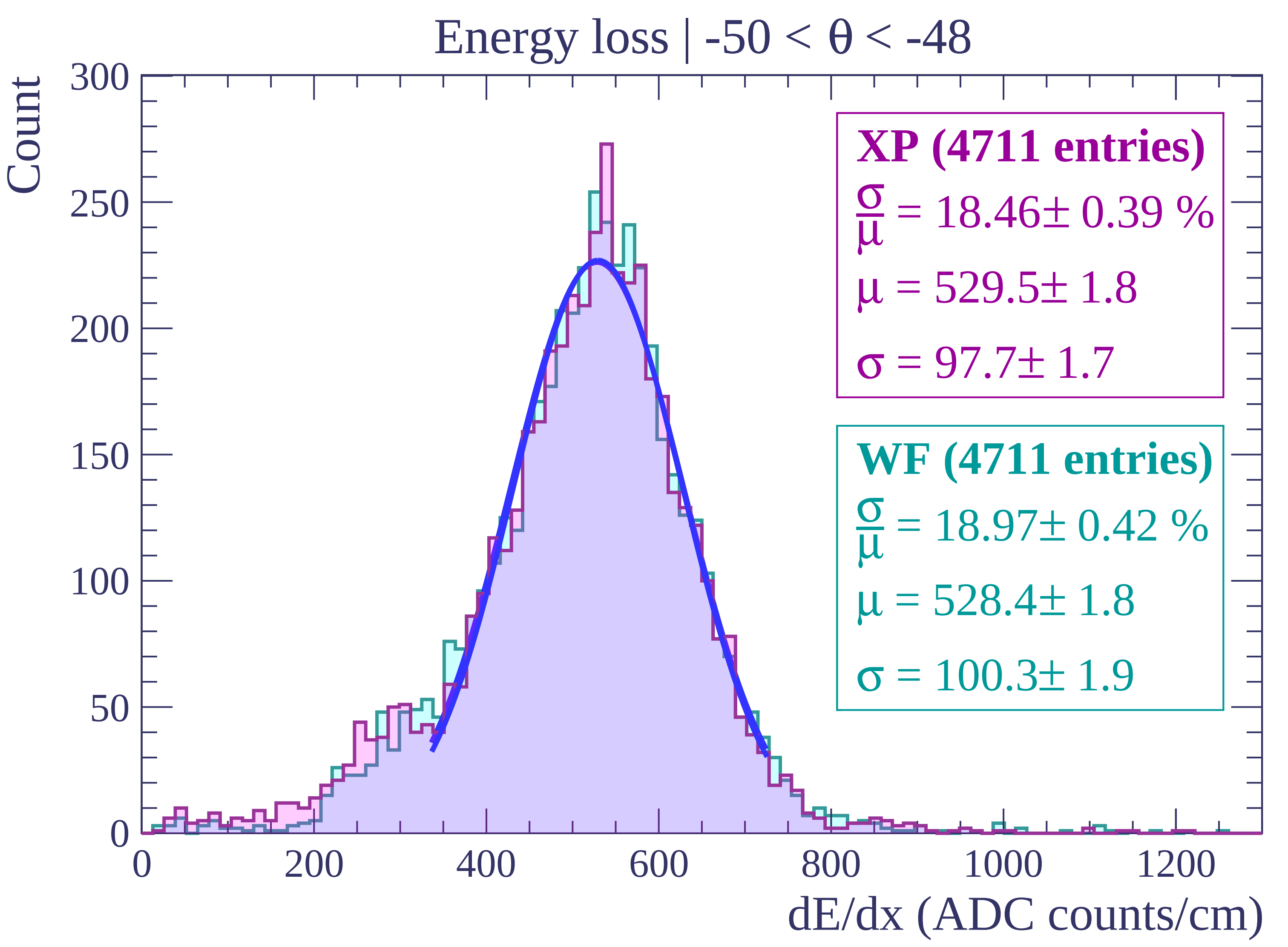

Energy loss | -58 < θ < -56
Count
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1997 entries)
σ/μ = 17.77± 0.60 %
μ = 538.6± 2.8
σ = 95.7± 2.7
WF (1997 entries)
σ/μ = 18.39± 0.63 %
μ = 538.1± 2.9
σ = 98.9± 2.8


Energy loss | -56 < θ < -54
Count
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2541 entries)
σ/μ = 17.91± 0.49 %
μ = 535.4± 2.4
σ = 95.9± 2.2
WF (2541 entries)
σ/μ = 18.95± 0.57 %
μ = 534.7± 2.6
σ = 101.3± 2.6

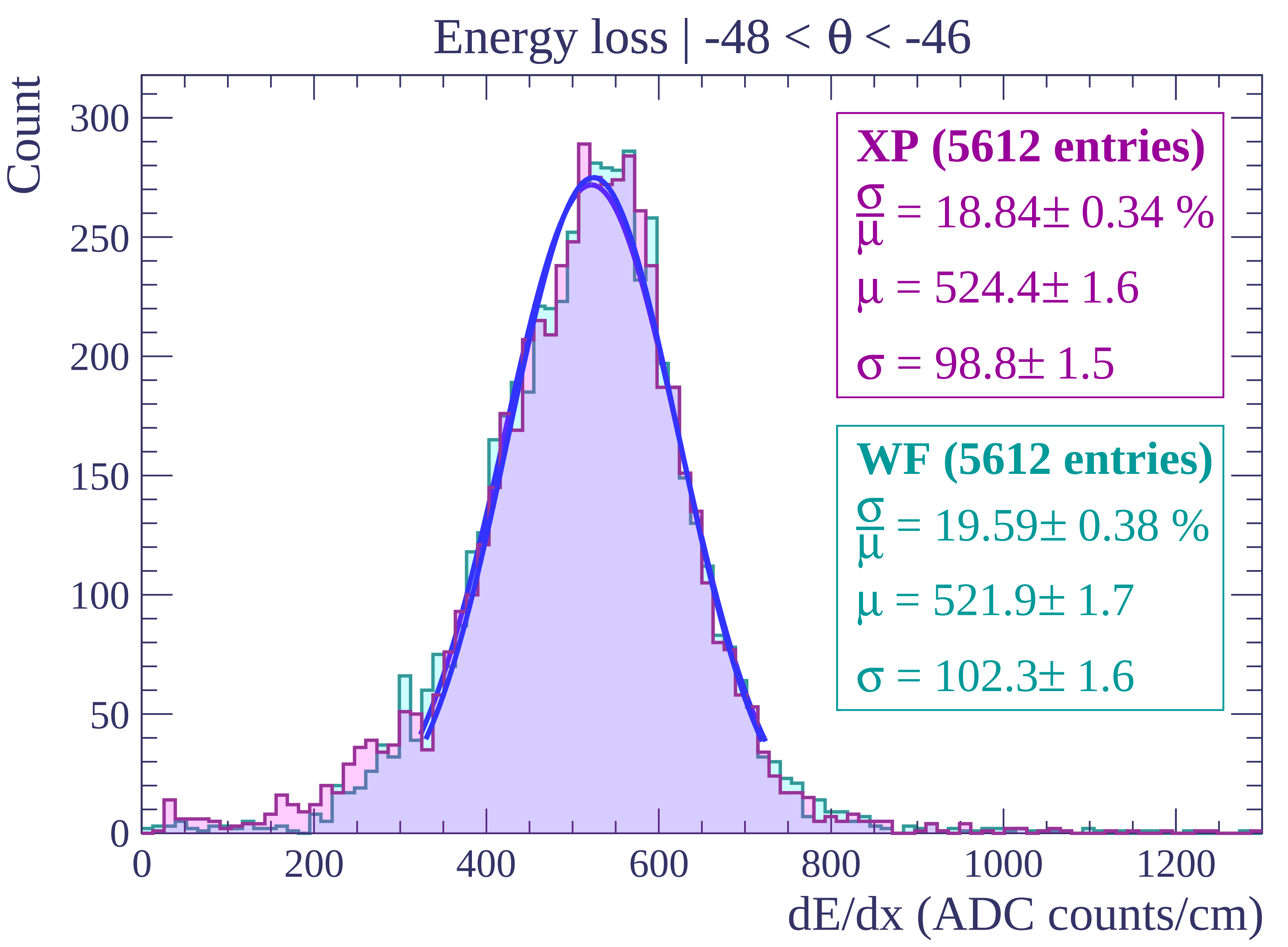

Energy loss | -54 < θ < -52
Count
dE/dx (ADC counts/cm)
XP (3106 entries)
σ/μ = 18.48± 0.52 %
μ = 541.3± 2.3
σ = 100.0± 2.4
WF (3106 entries)
σ/μ = 18.48± 0.52 %
μ = 539.0± 2.3
σ = 99.6± 2.4
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200



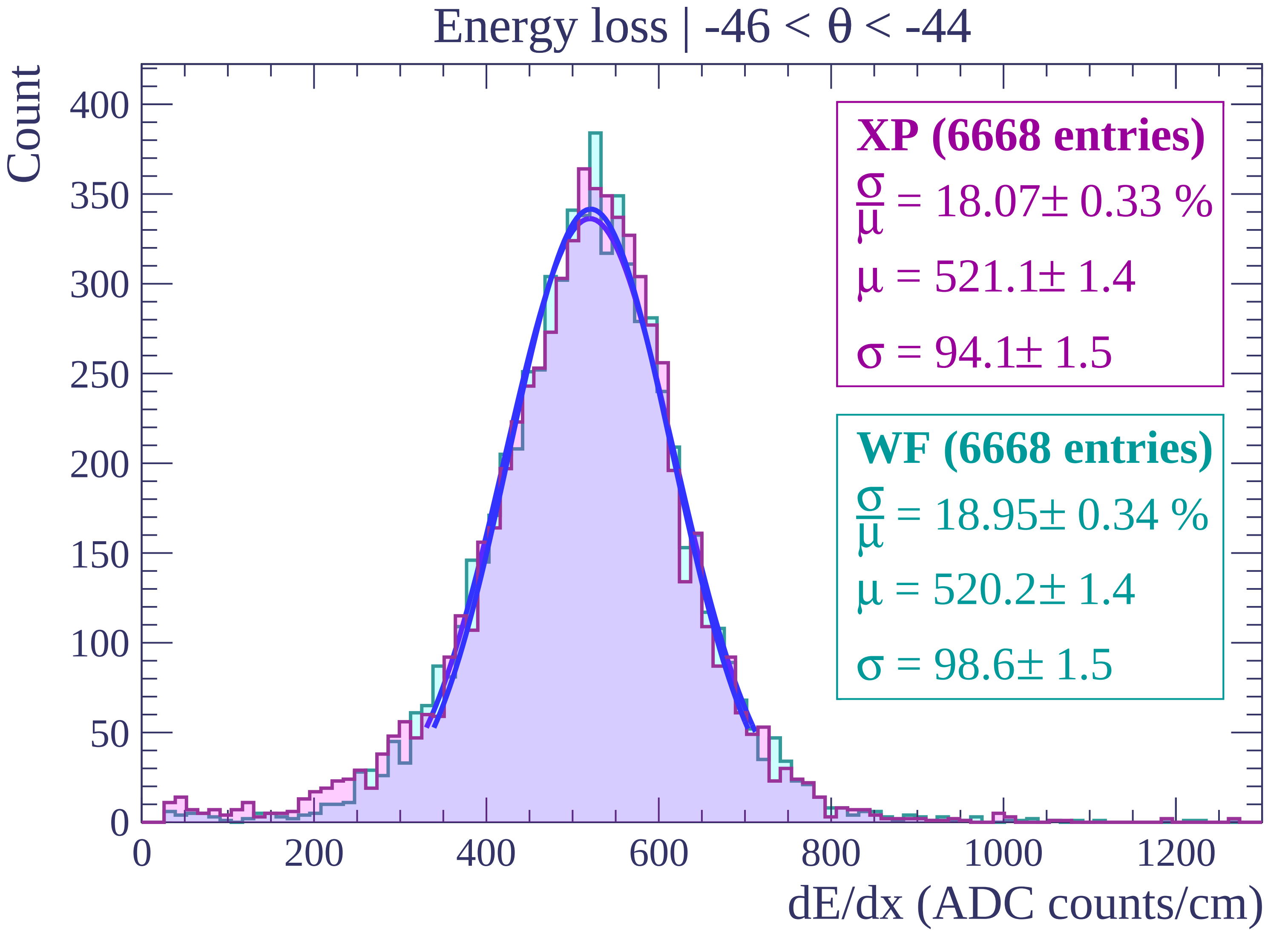
Energy loss | -52 < θ < -50
Count
dE/dx (ADC counts/cm)
XP (3776 entries)
$\frac{\sigma}{\mu}$ = 18.05± 0.45 %
μ = 534.0± 2.0
σ = 96.4± 2.0
WF (3776 entries)
$\frac{\sigma}{\mu}$ = 18.98± 0.46 %
μ = 531.6± 2.1
σ = 100.9± 2.1



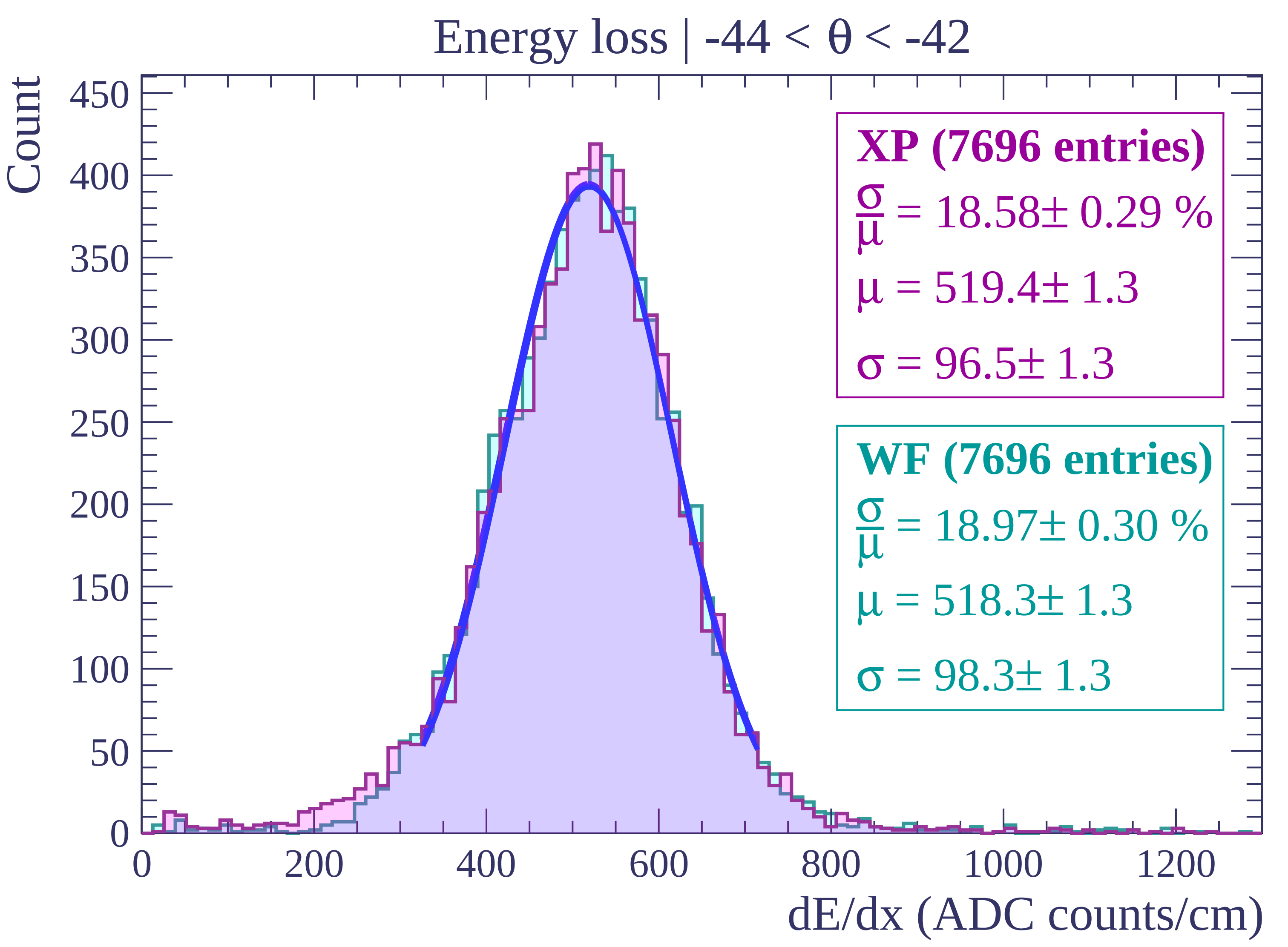
Energy loss | -50 < θ < -48
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (4711 entries)
σ/μ = 18.46± 0.39 %
μ = 529.5± 1.8
σ = 97.7± 1.7
WF (4711 entries)
σ/μ = 18.97± 0.42 %
μ = 528.4± 1.8
σ = 100.3± 1.9

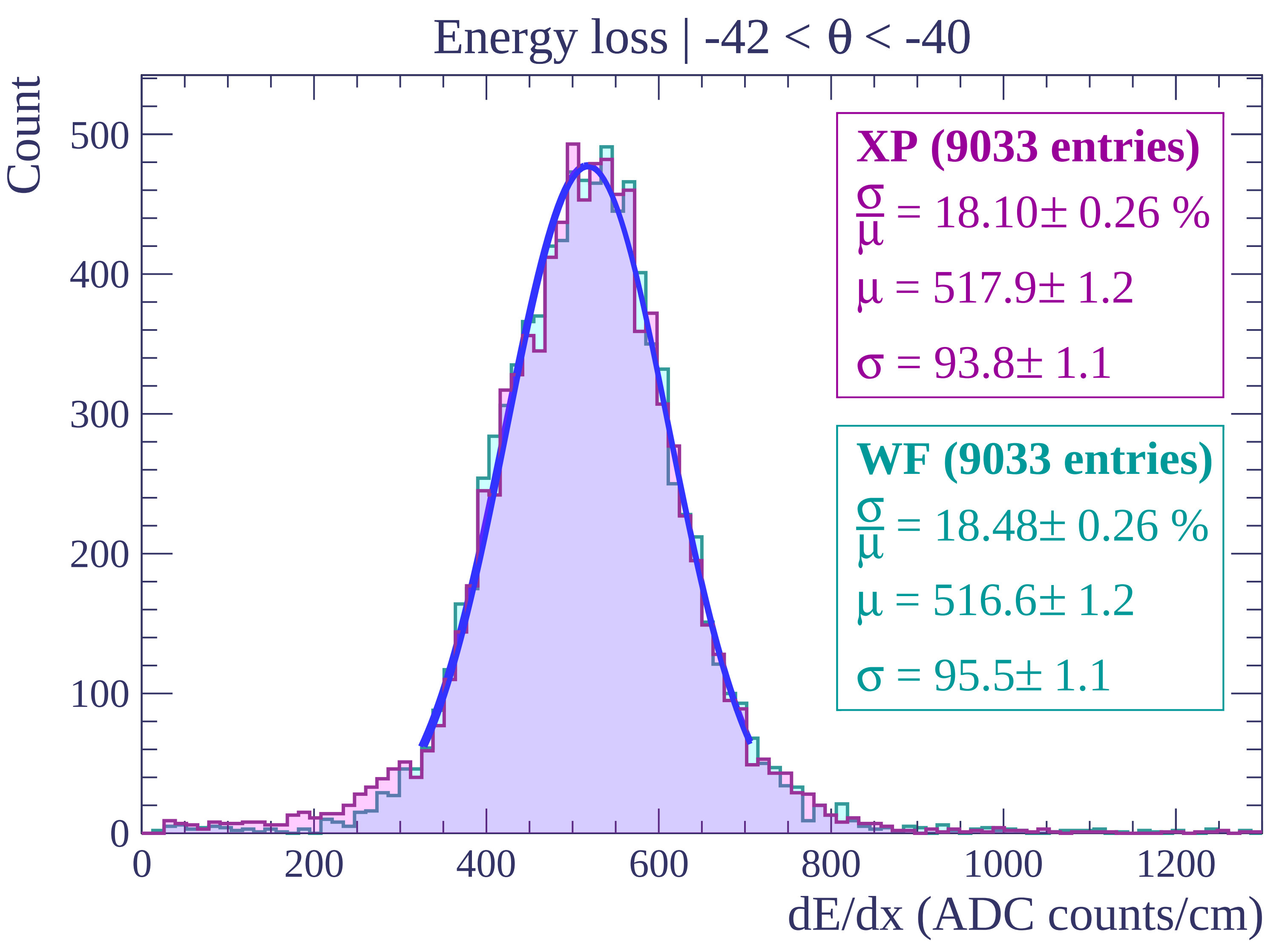


Energy loss | -48 < θ < -46
Count
XP (5612 entries)
σ/μ = 18.84± 0.34 %
μ = 524.4± 1.6
σ = 98.8± 1.5
WF (5612 entries)
σ/μ = 19.59± 0.38 %
μ = 521.9± 1.7
σ = 102.3± 1.6
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)

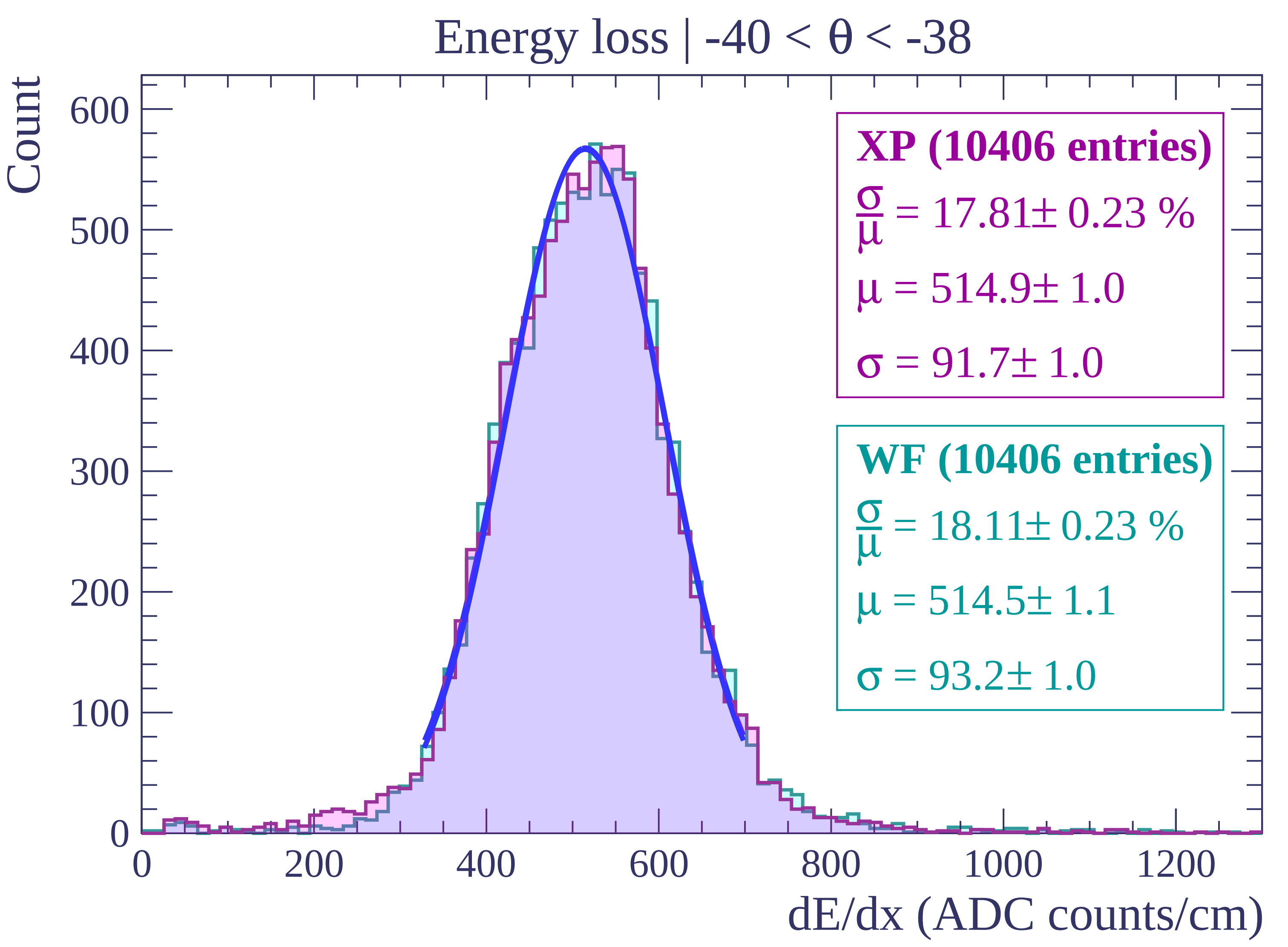


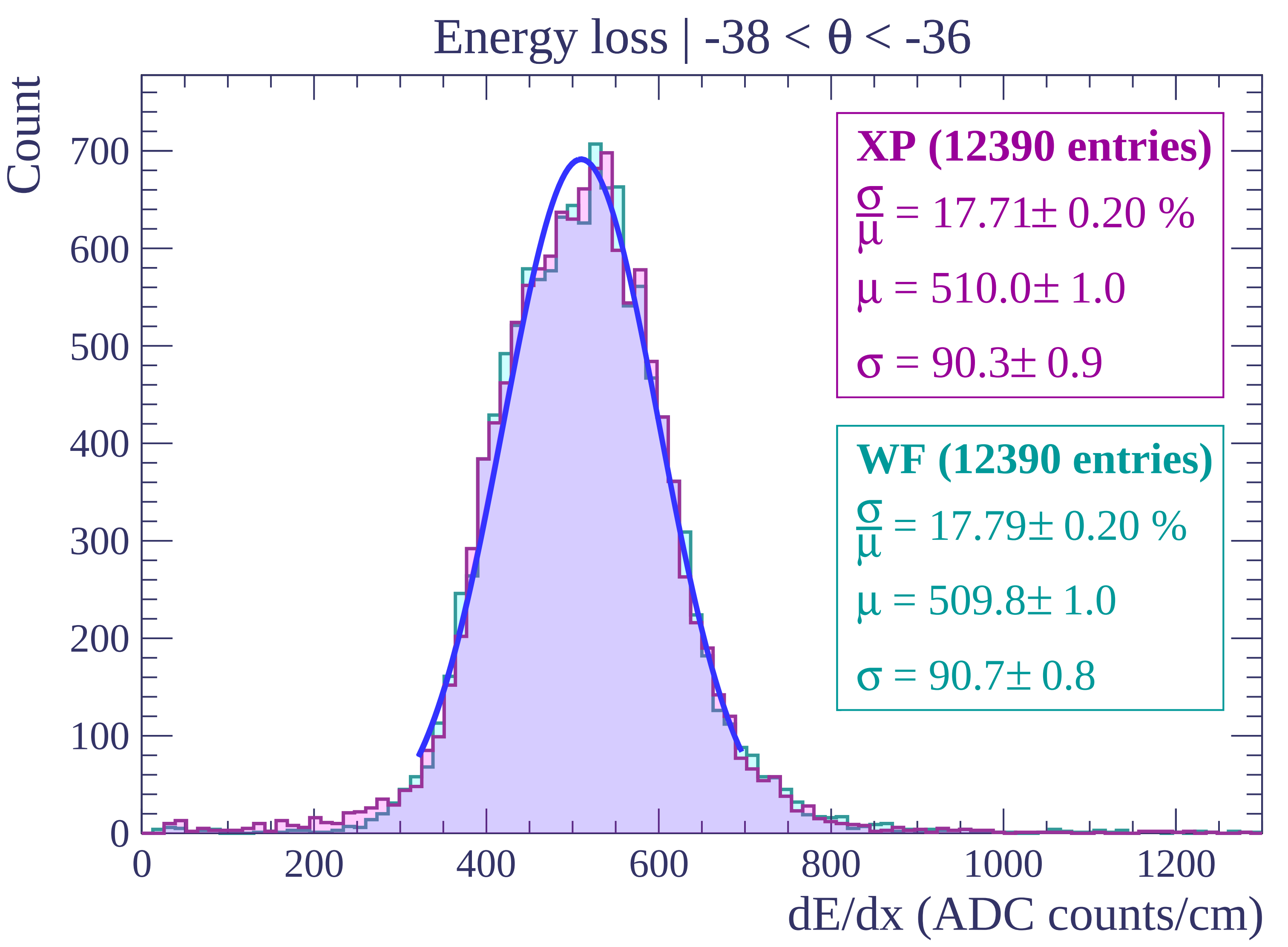
Energy loss | -46 < θ < -44
Count
dE/dx (ADC counts/cm)
XP (6668 entries)
σ/μ = 18.07± 0.33 %
μ = 521.1± 1.4
σ = 94.1± 1.5
WF (6668 entries)
σ/μ = 18.95± 0.34 %
μ = 520.2± 1.4
σ = 98.6± 1.5
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200


Energy loss | -44 < θ < -42
Count
dE/dx (ADC counts/cm)
XP (7696 entries)
σ/μ = 18.58± 0.29 %
μ = 519.4± 1.3
σ = 96.5± 1.3
WF (7696 entries)
σ/μ = 18.97± 0.30 %
μ = 518.3± 1.3
σ = 98.3± 1.3

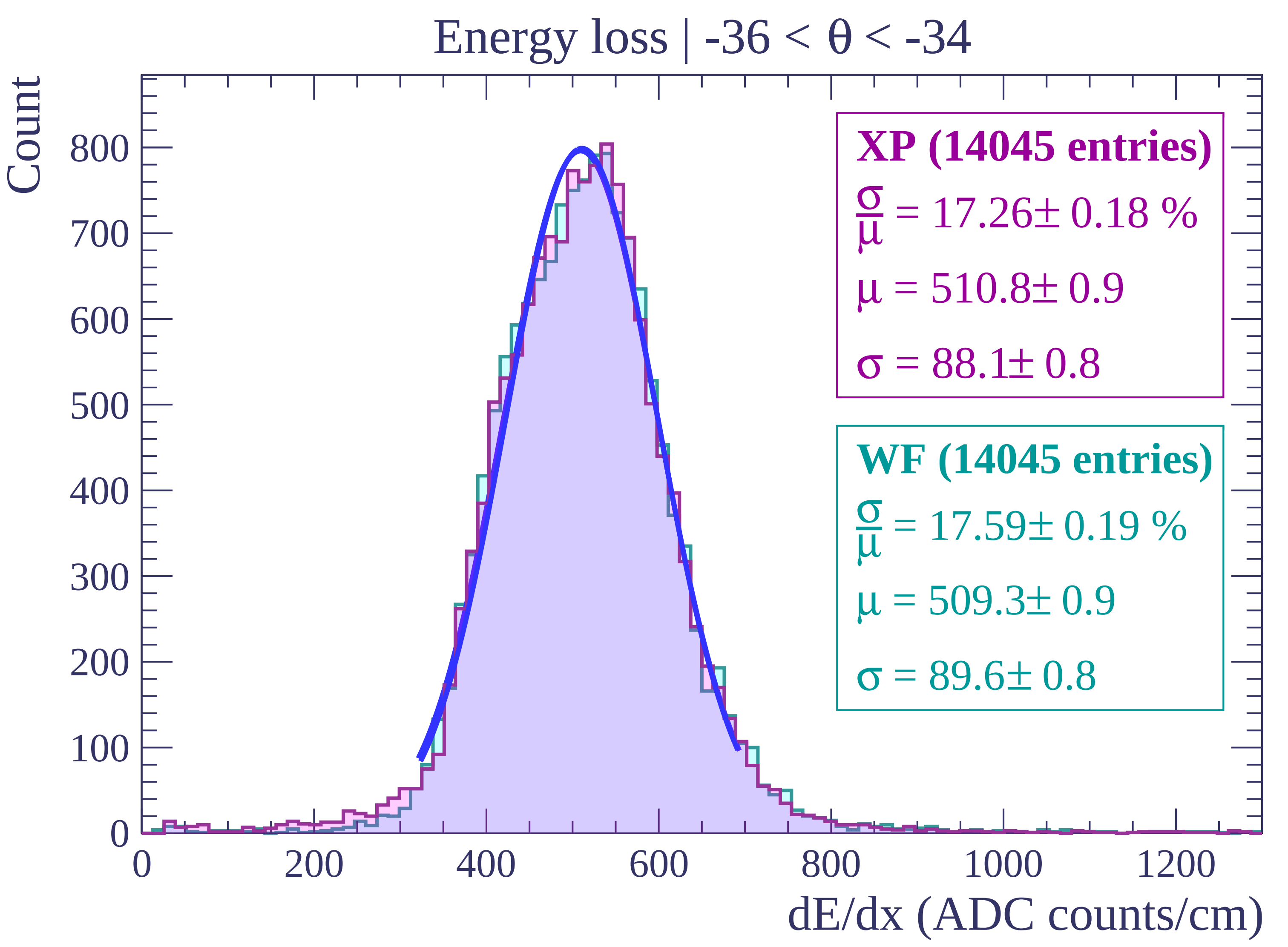


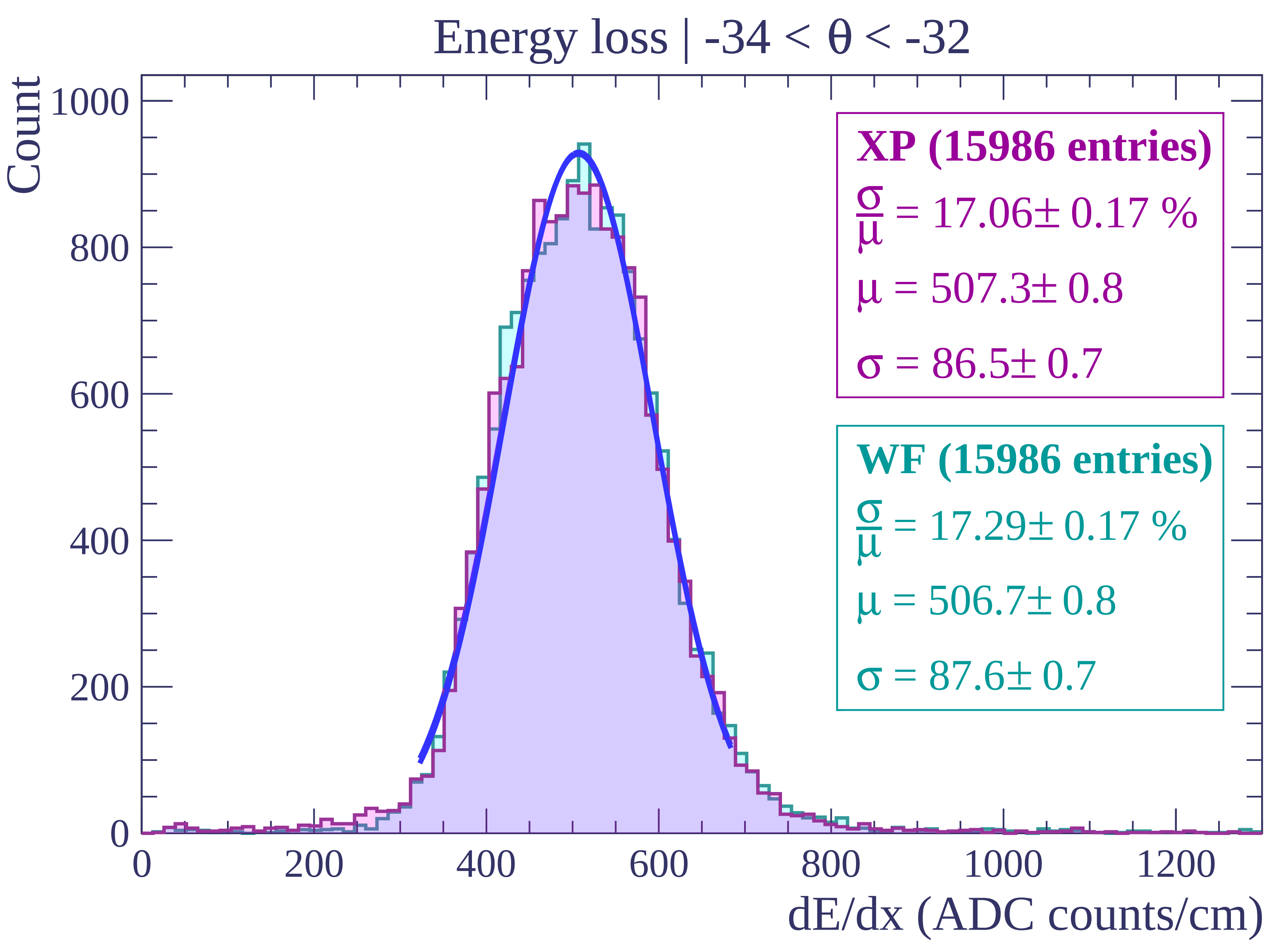
Energy loss | -42 < θ < -40
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
XP (9033 entries)
σ/μ = 18.10± 0.26 %
μ = 517.9± 1.2
σ = 93.8± 1.1
WF (9033 entries)
σ/μ = 18.48± 0.26 %
μ = 516.6± 1.2
σ = 95.5± 1.1

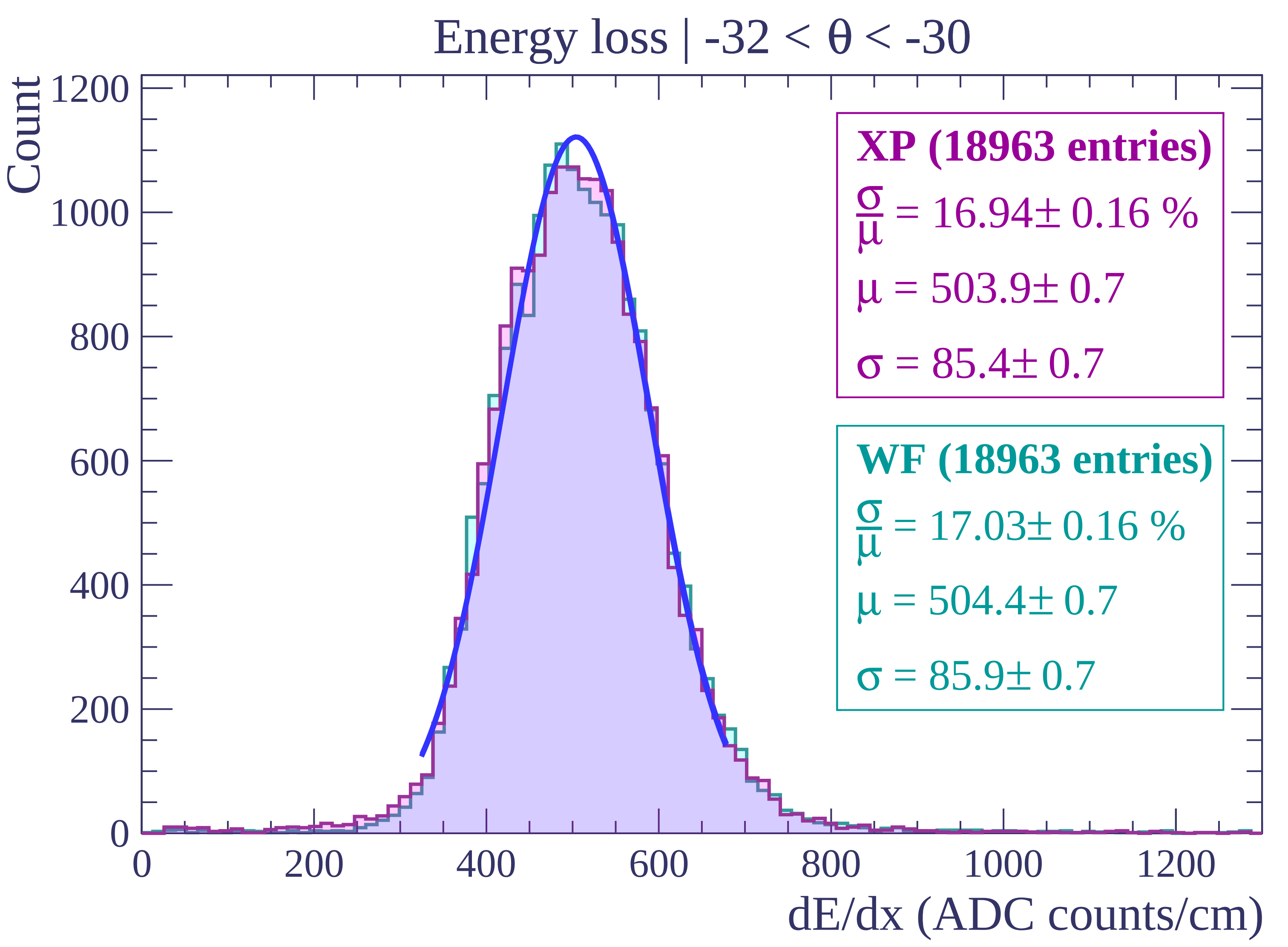

Energy loss | -40 < θ < -38
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (10406 entries)
σ/μ = 17.81± 0.23 %
μ = 514.9± 1.0
σ = 91.7± 1.0
WF (10406 entries)
σ/μ = 18.11± 0.23 %
μ = 514.5± 1.1
σ = 93.2± 1.0

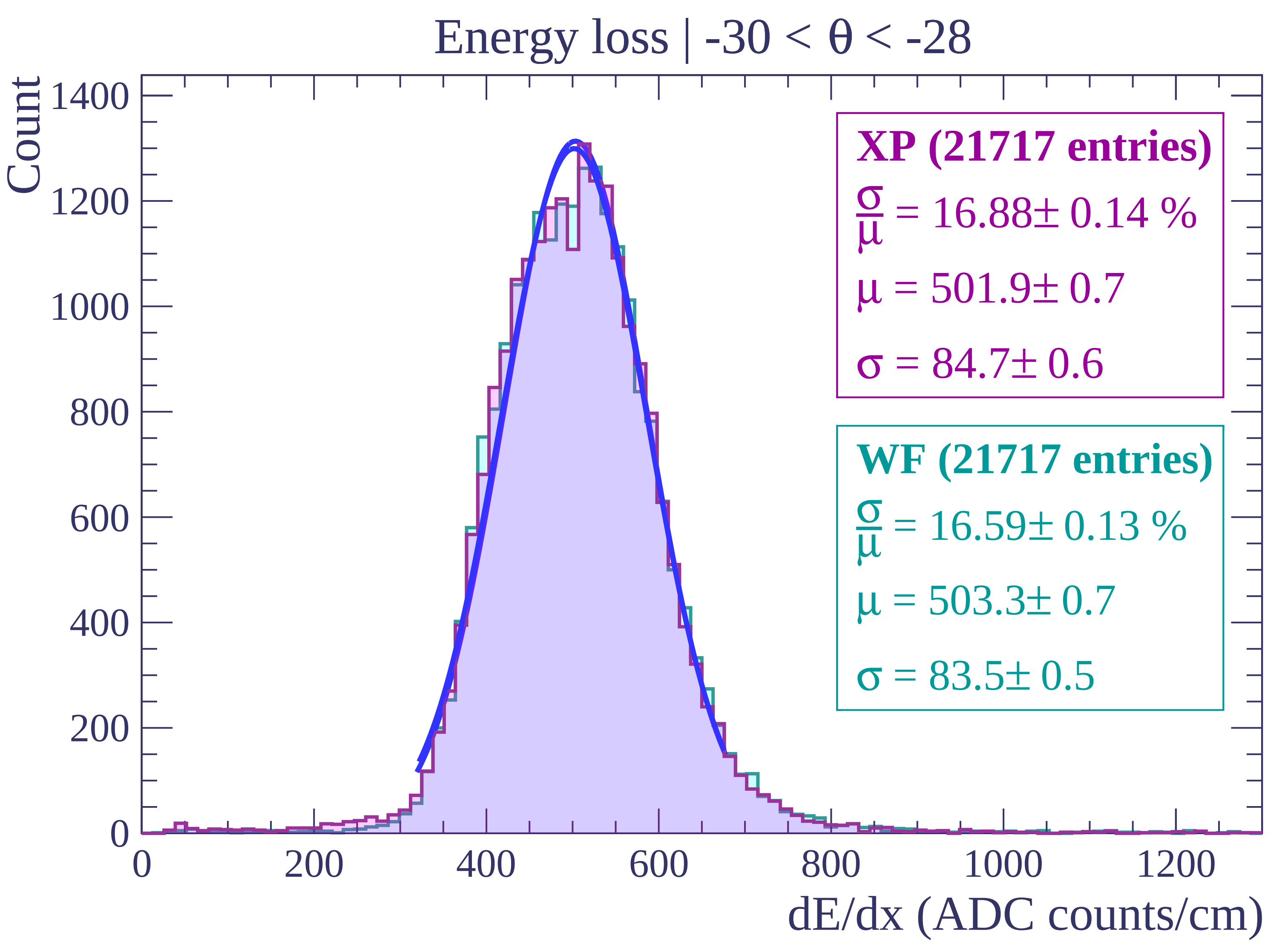


Energy loss | -38 < θ < -36
Count
dE/dx (ADC counts/cm)
XP (12390 entries)
$\frac{\sigma}{\mu}$ = 17.71± 0.20 %
μ = 510.0± 1.0
σ = 90.3± 0.9
WF (12390 entries)
$\frac{\sigma}{\mu}$ = 17.79± 0.20 %
μ = 509.8± 1.0
σ = 90.7± 0.8
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200

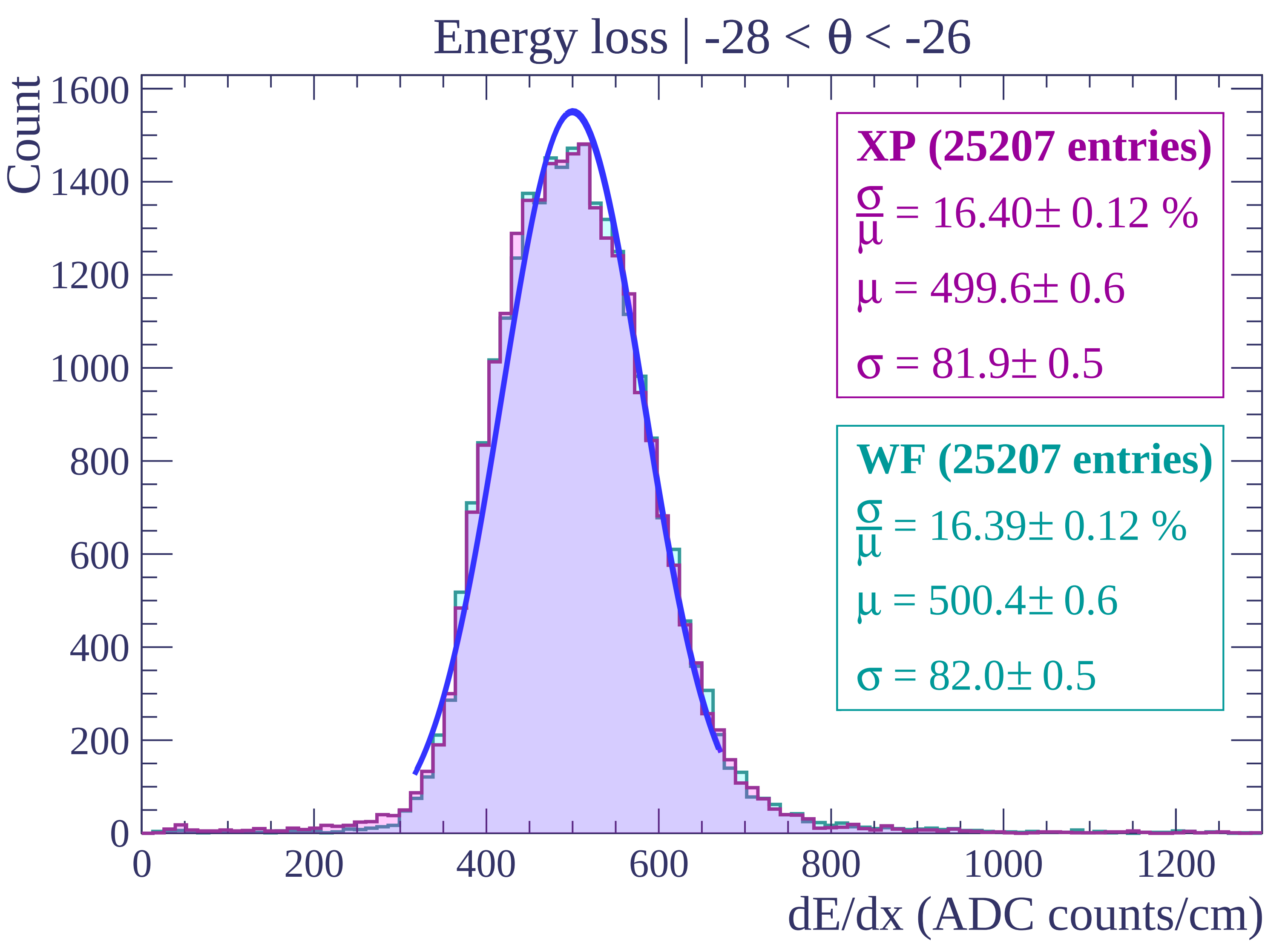

Energy loss | -36 < θ < -34
Count
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (14045 entries)
$\frac{\sigma}{\mu}$ = 17.26± 0.18 %
μ = 510.8± 0.9
σ = 88.1± 0.8
WF (14045 entries)
$\frac{\sigma}{\mu}$ = 17.59± 0.19 %
μ = 509.3± 0.9
σ = 89.6± 0.8


Energy loss | -34 < θ < -32
Count
dE/dx (ADC counts/cm)
XP (15986 entries)
$\frac{\sigma}{\mu}$ = 17.06± 0.17 %
μ = 507.3± 0.8
σ = 86.5± 0.7
WF (15986 entries)
$\frac{\sigma}{\mu}$ = 17.29± 0.17 %
μ = 506.7± 0.8
σ = 87.6± 0.7
0
200
400
600
800
1000
1200

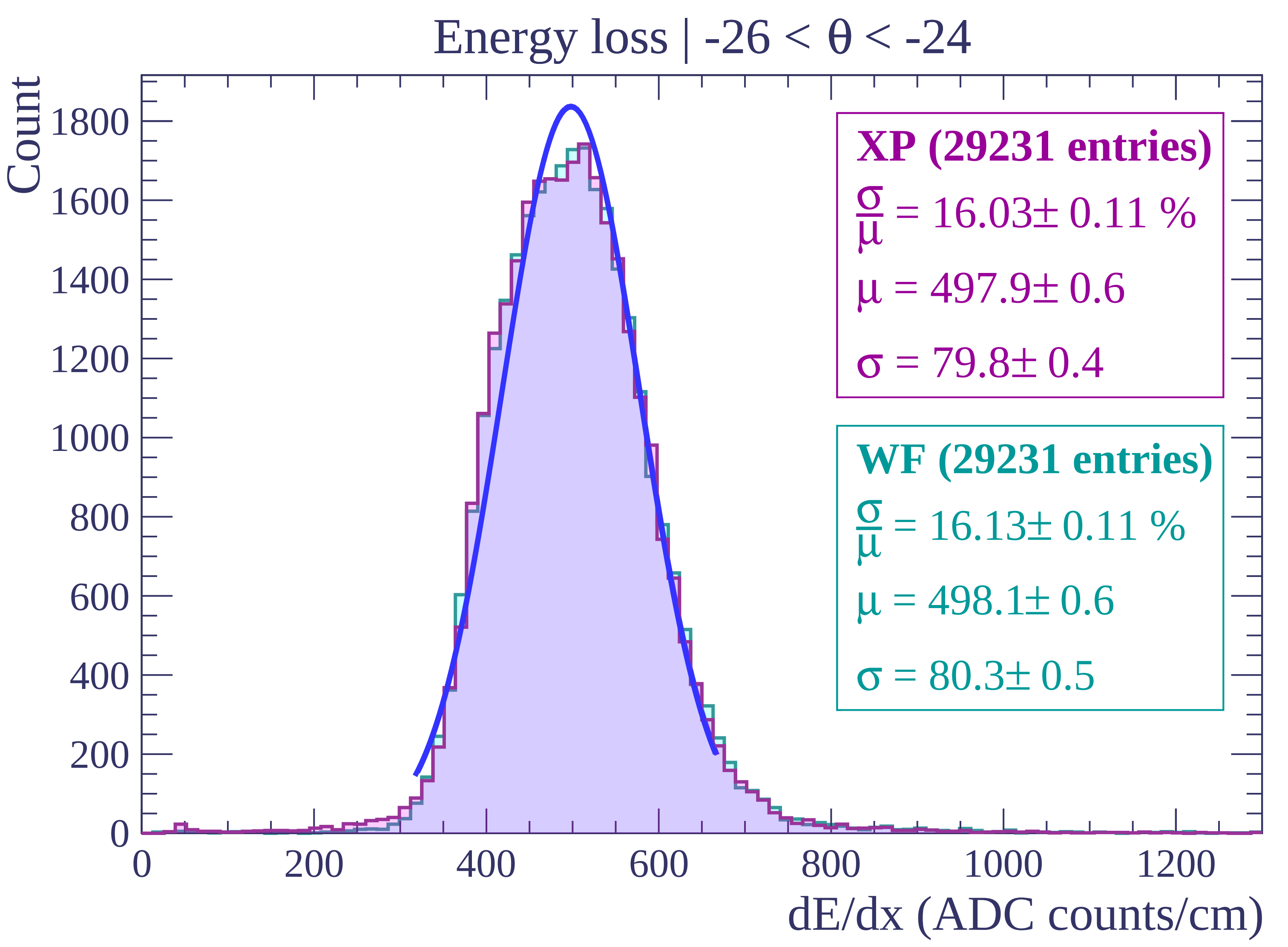

Energy loss | -32 < θ < -30
Count
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (18963 entries)
σ/μ = 16.94± 0.16 %
μ = 503.9± 0.7
σ = 85.4± 0.7
WF (18963 entries)
σ/μ = 17.03± 0.16 %
μ = 504.4± 0.7
σ = 85.9± 0.7



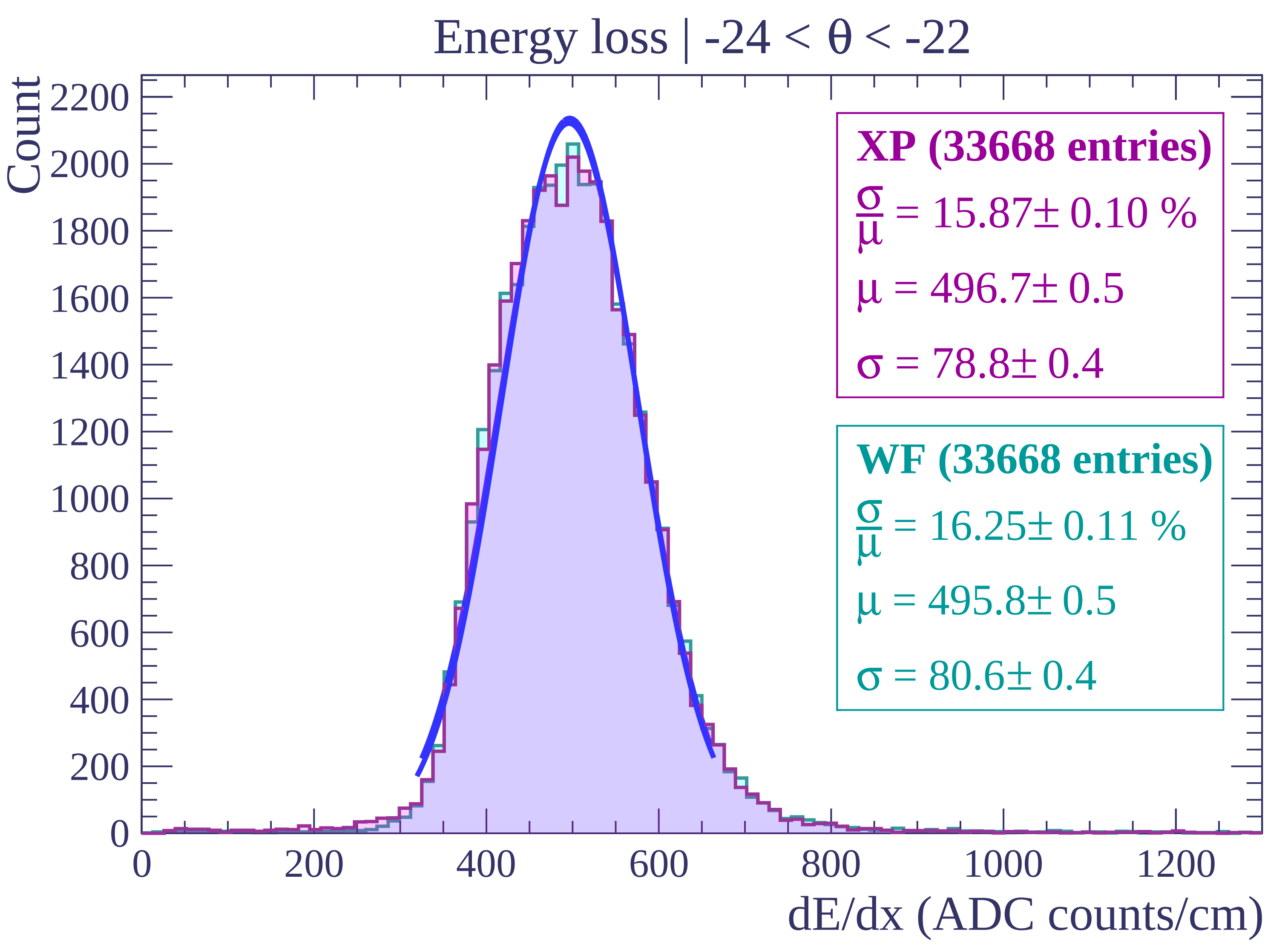
Energy loss | -30 < θ < -28
Count
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (21717 entries)
σ/μ = 16.88± 0.14 %
μ = 501.9± 0.7
σ = 84.7± 0.6
WF (21717 entries)
σ/μ = 16.59± 0.13 %
μ = 503.3± 0.7
σ = 83.5± 0.5

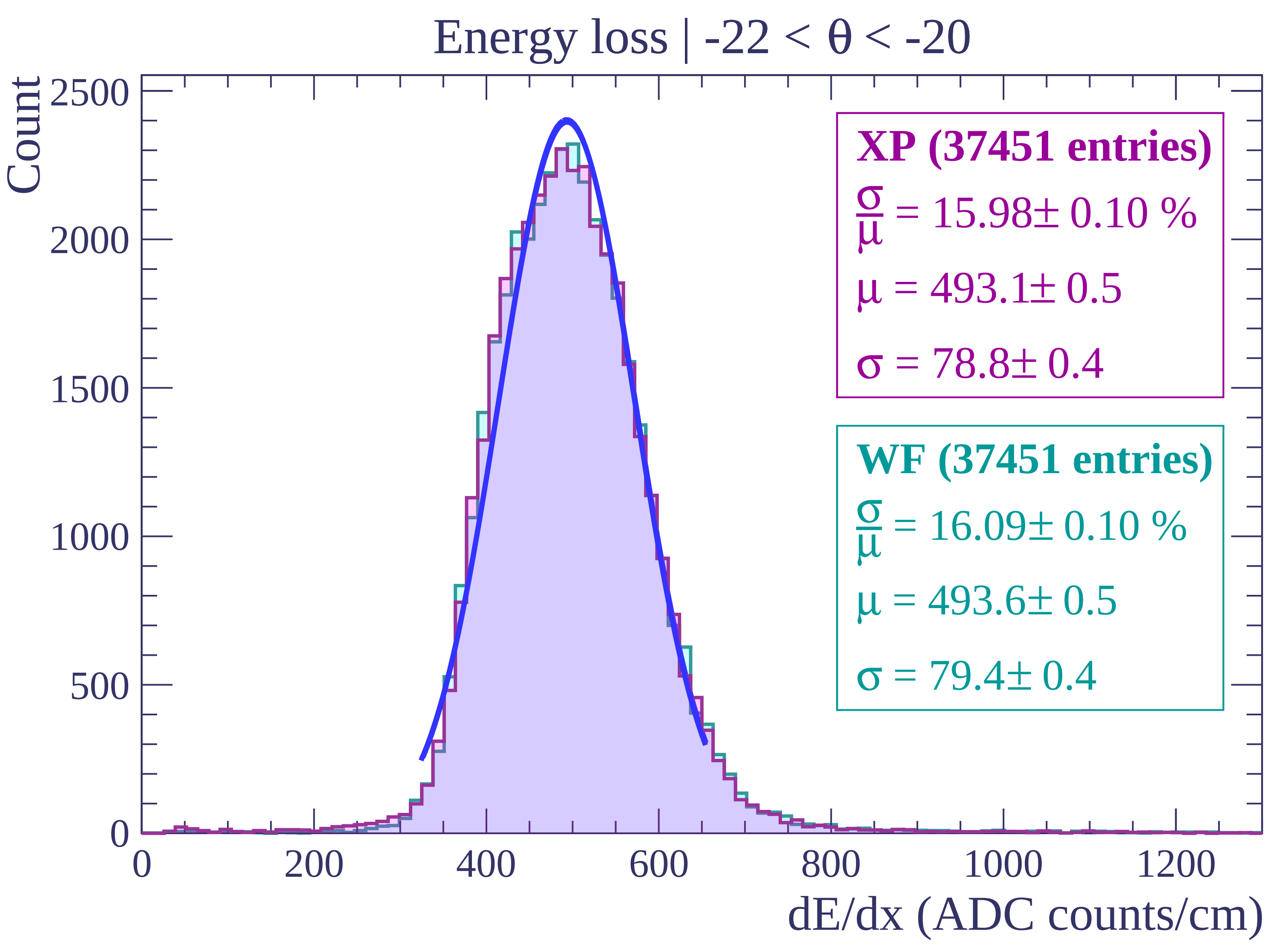


Energy loss | -28 < θ < -26
Count
1600
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (25207 entries)
$\frac{\sigma}{\mu}$ = 16.40± 0.12 %
μ = 499.6± 0.6
σ = 81.9± 0.5
WF (25207 entries)
$\frac{\sigma}{\mu}$ = 16.39± 0.12 %
μ = 500.4± 0.6
σ = 82.0± 0.5

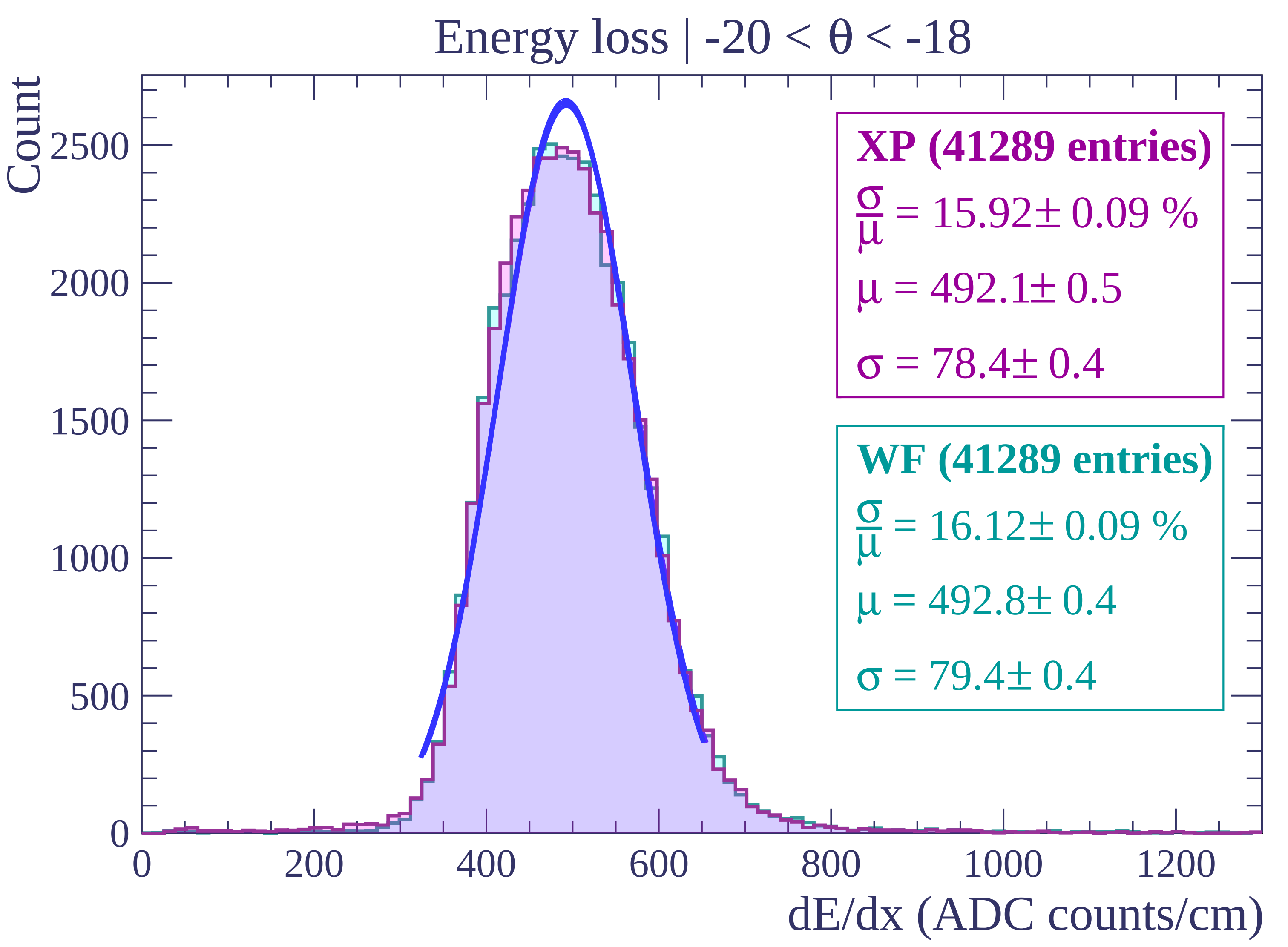


Energy loss | -26 < θ < -24
Count
dE/dx (ADC counts/cm)
XP (29231 entries)
σ/μ = 16.03± 0.11 %
μ = 497.9± 0.6
σ = 79.8± 0.4
WF (29231 entries)
σ/μ = 16.13± 0.11 %
μ = 498.1± 0.6
σ = 80.3± 0.5
0
200
400
600
800
1000
1200
1400
1600
1800

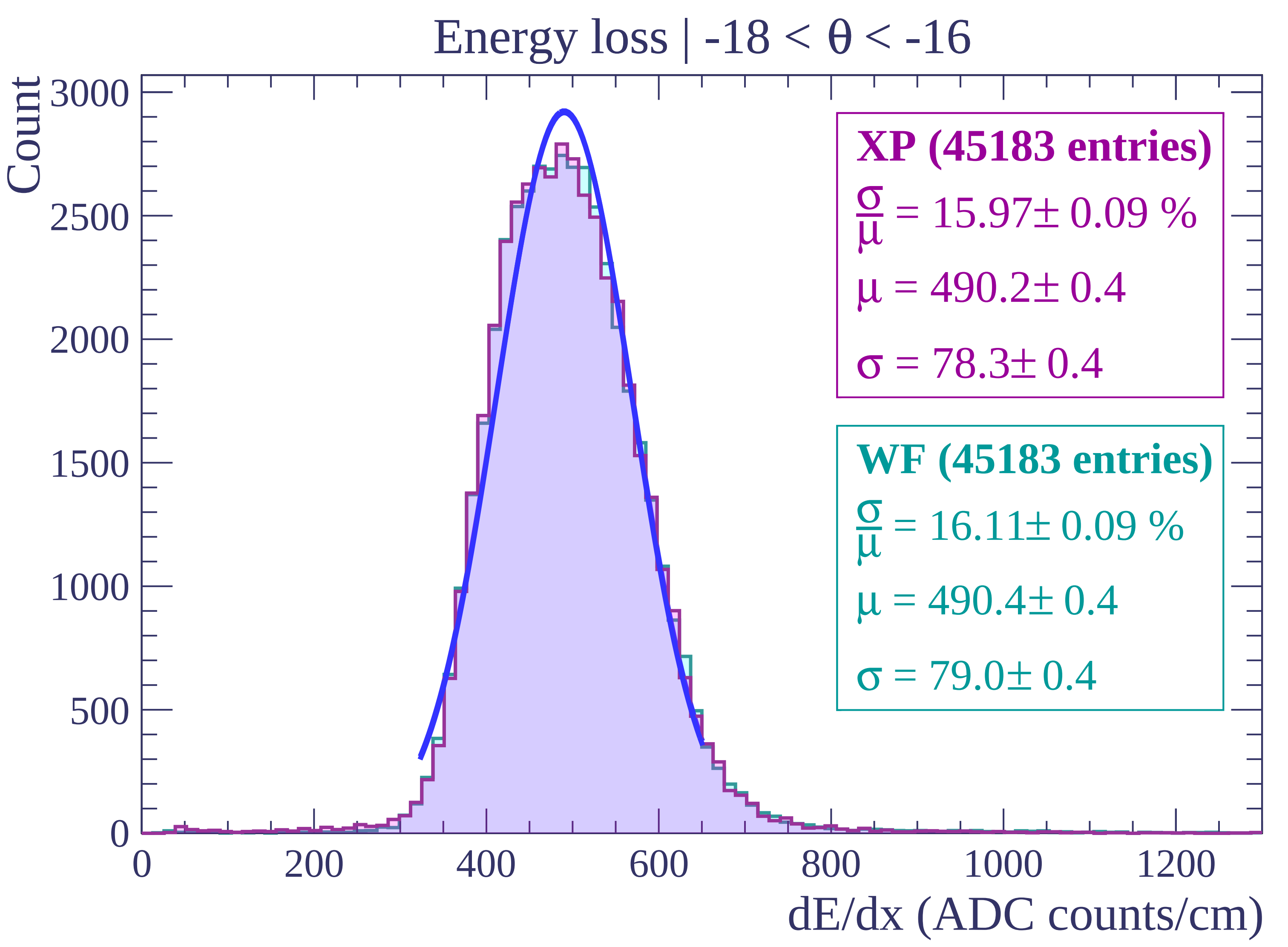

Energy loss | -24 < θ < -22
Count
dE/dx (ADC counts/cm)
XP (33668 entries)
σ/μ = 15.87± 0.10 %
μ = 496.7± 0.5
σ = 78.8± 0.4
WF (33668 entries)
σ/μ = 16.25± 0.11 %
μ = 495.8± 0.5
σ = 80.6± 0.4
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200
0
200
400
600
800
1000
1200



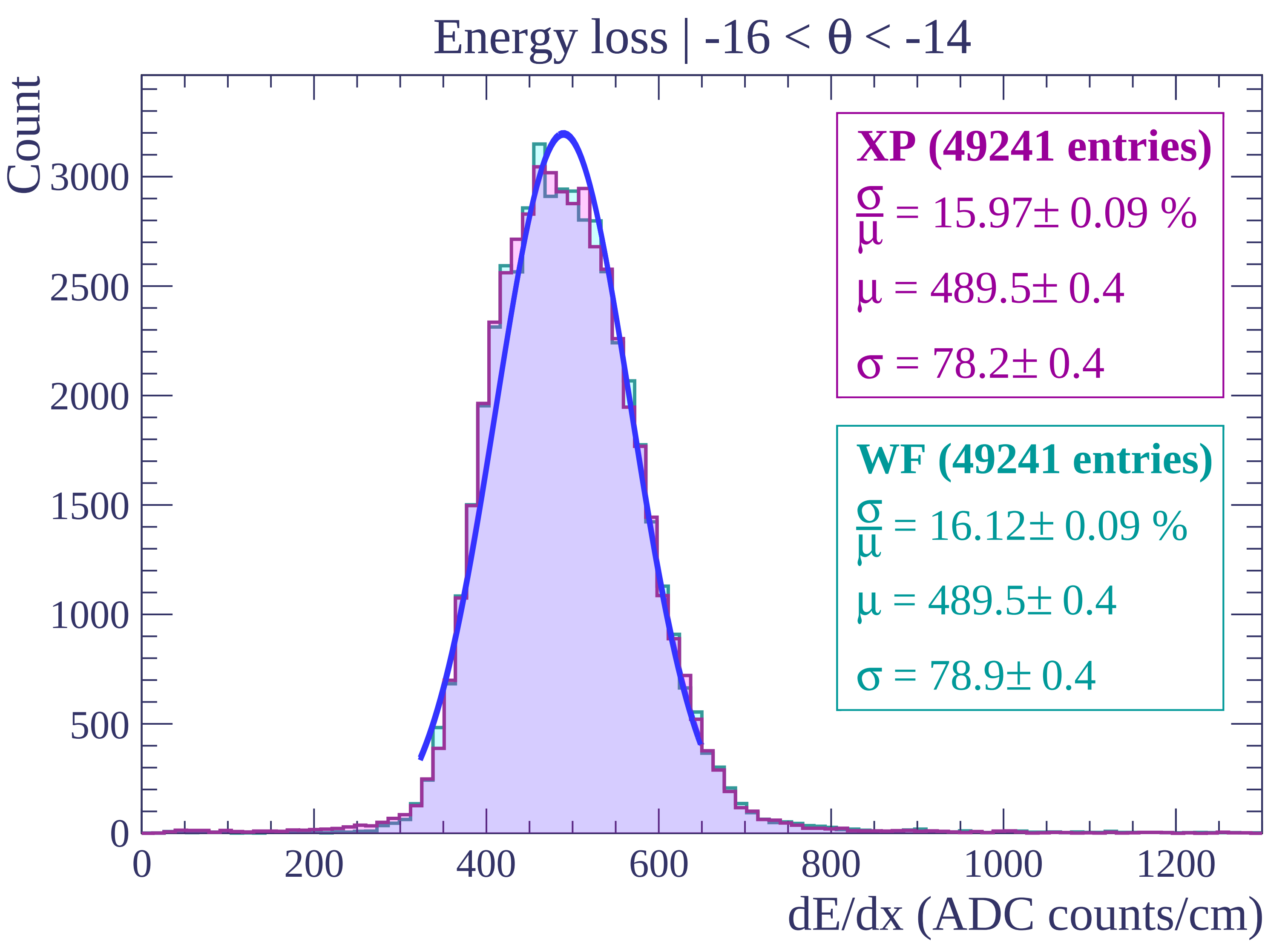
Energy loss | -22 < θ < -20
Count
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (37451 entries)
σ/μ = 15.98± 0.10 %
μ = 493.1± 0.5
σ = 78.8± 0.4
WF (37451 entries)
σ/μ = 16.09± 0.10 %
μ = 493.6± 0.5
σ = 79.4± 0.4



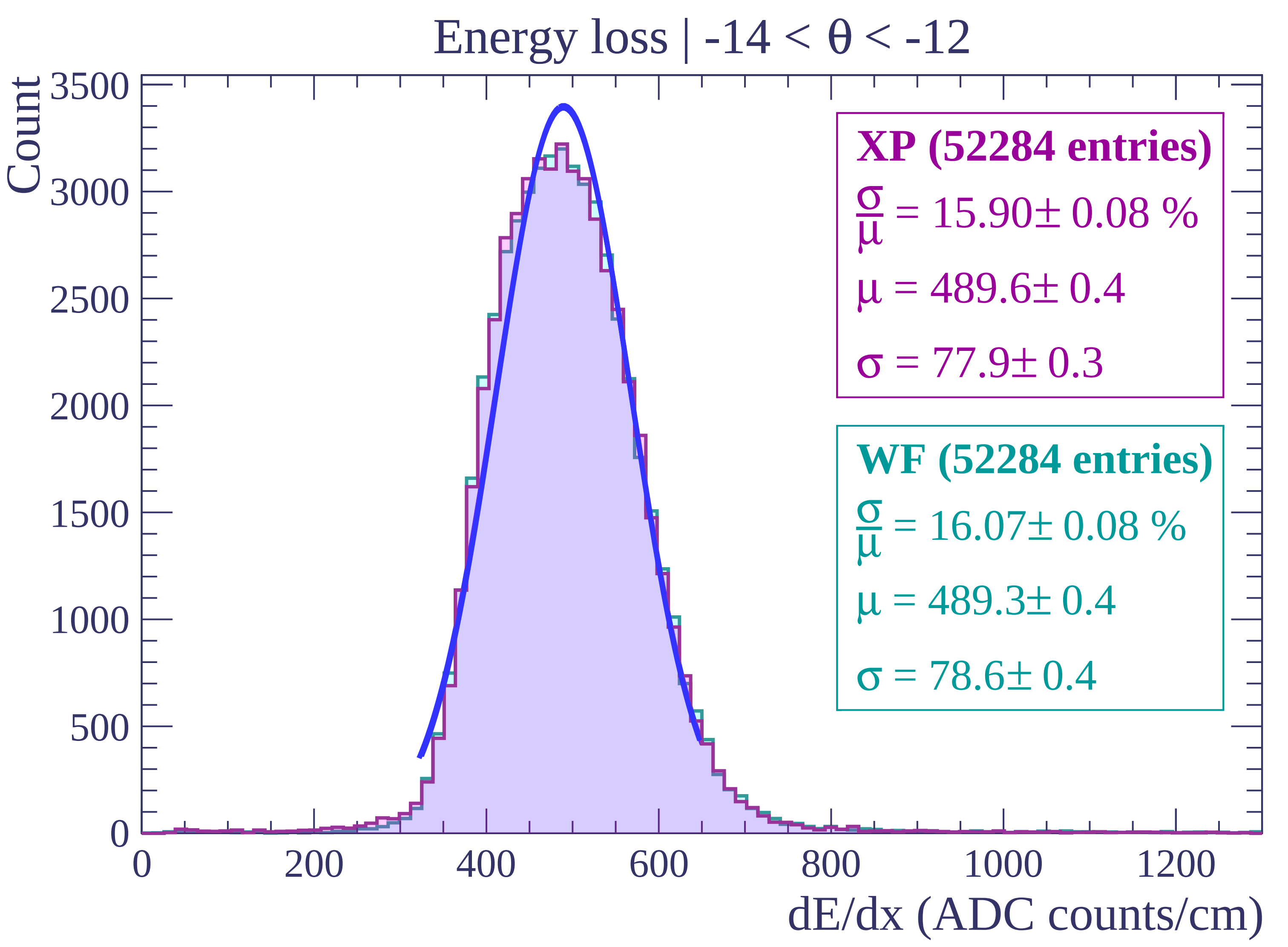
Energy loss | -20 < θ < -18
Count
2500
2000
1500
1000
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (41289 entries)
σ/μ = 15.92± 0.09 %
μ = 492.1± 0.5
σ = 78.4± 0.4
WF (41289 entries)
σ/μ = 16.12± 0.09 %
μ = 492.8± 0.4
σ = 79.4± 0.4



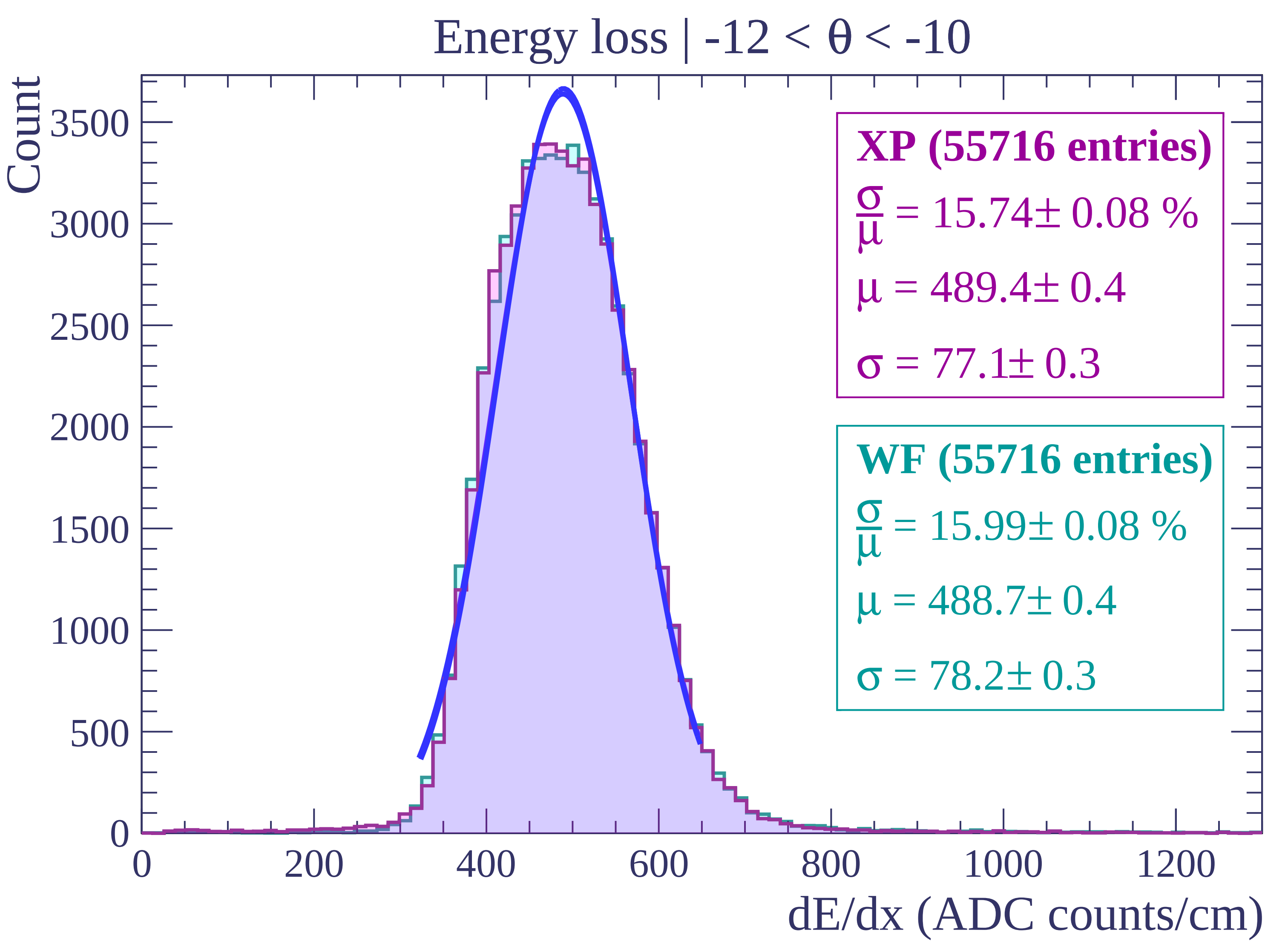
Energy loss | -18 < θ < -16
Count
3000
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (45183 entries)
σ/μ = 15.97± 0.09 %
μ = 490.2± 0.4
σ = 78.3± 0.4
WF (45183 entries)
σ/μ = 16.11± 0.09 %
μ = 490.4± 0.4
σ = 79.0± 0.4



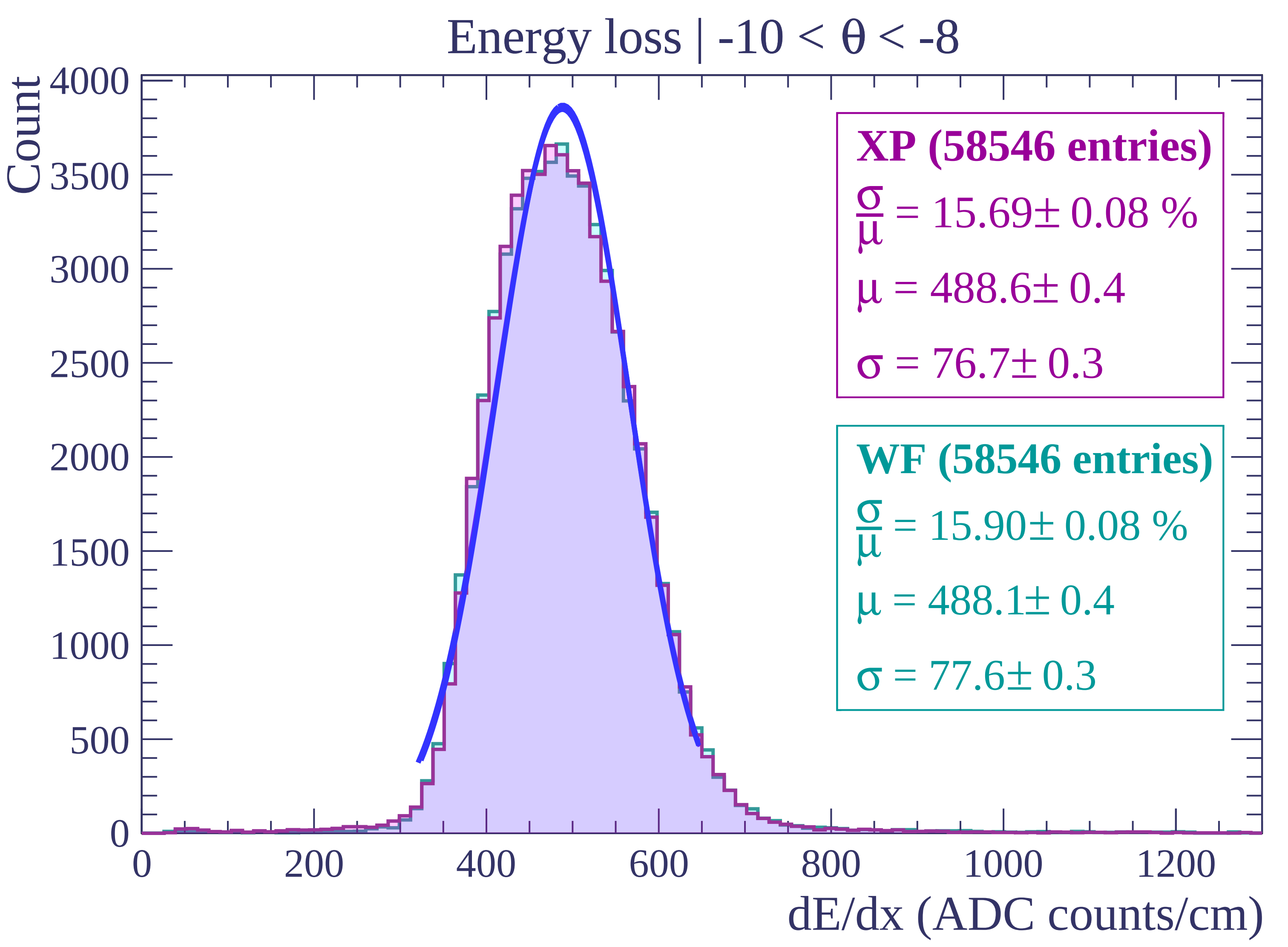
Energy loss | -16 < θ < -14
Count
0
500
1000
1500
2000
2500
3000
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (49241 entries)
σ/μ = 15.97± 0.09 %
μ = 489.5± 0.4
σ = 78.2± 0.4
WF (49241 entries)
σ/μ = 16.12± 0.09 %
μ = 489.5± 0.4
σ = 78.9± 0.4

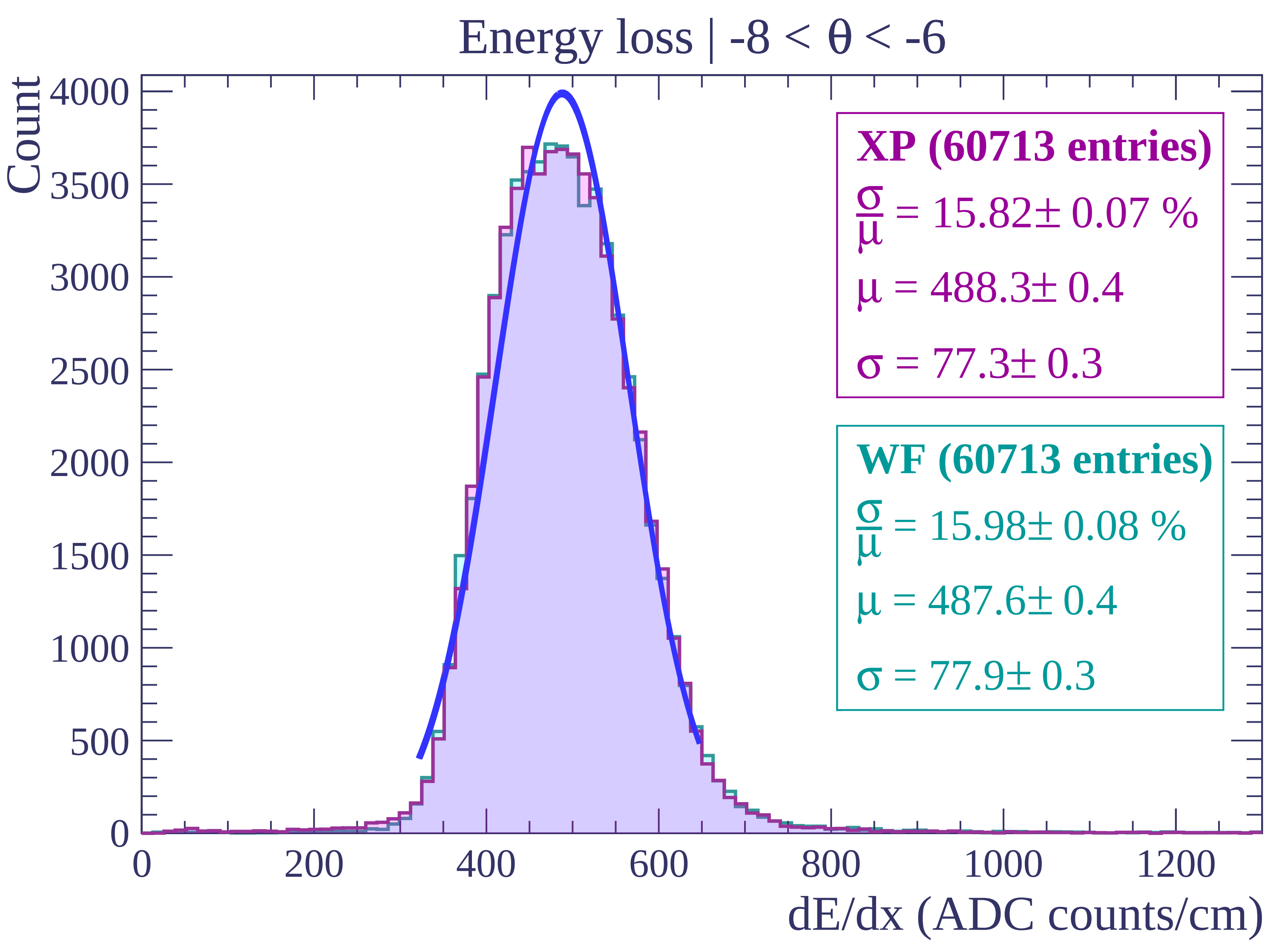


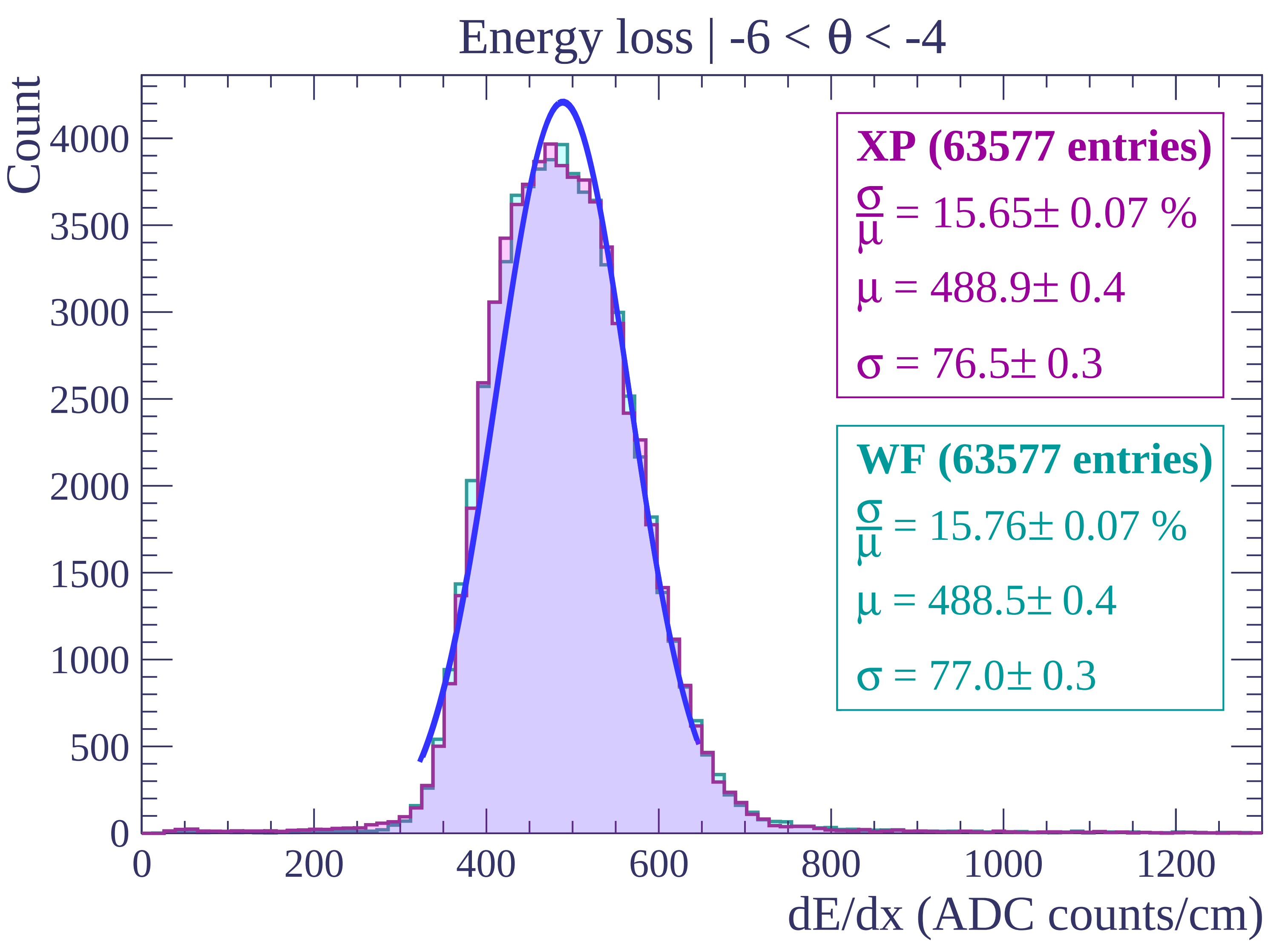
Energy loss | -14 < θ < -12
Count
dE/dx (ADC counts/cm)
XP (52284 entries)
σ/μ = 15.90± 0.08 %
μ = 489.6± 0.4
σ = 77.9± 0.3
WF (52284 entries)
σ/μ = 16.07± 0.08 %
μ = 489.3± 0.4
σ = 78.6± 0.4

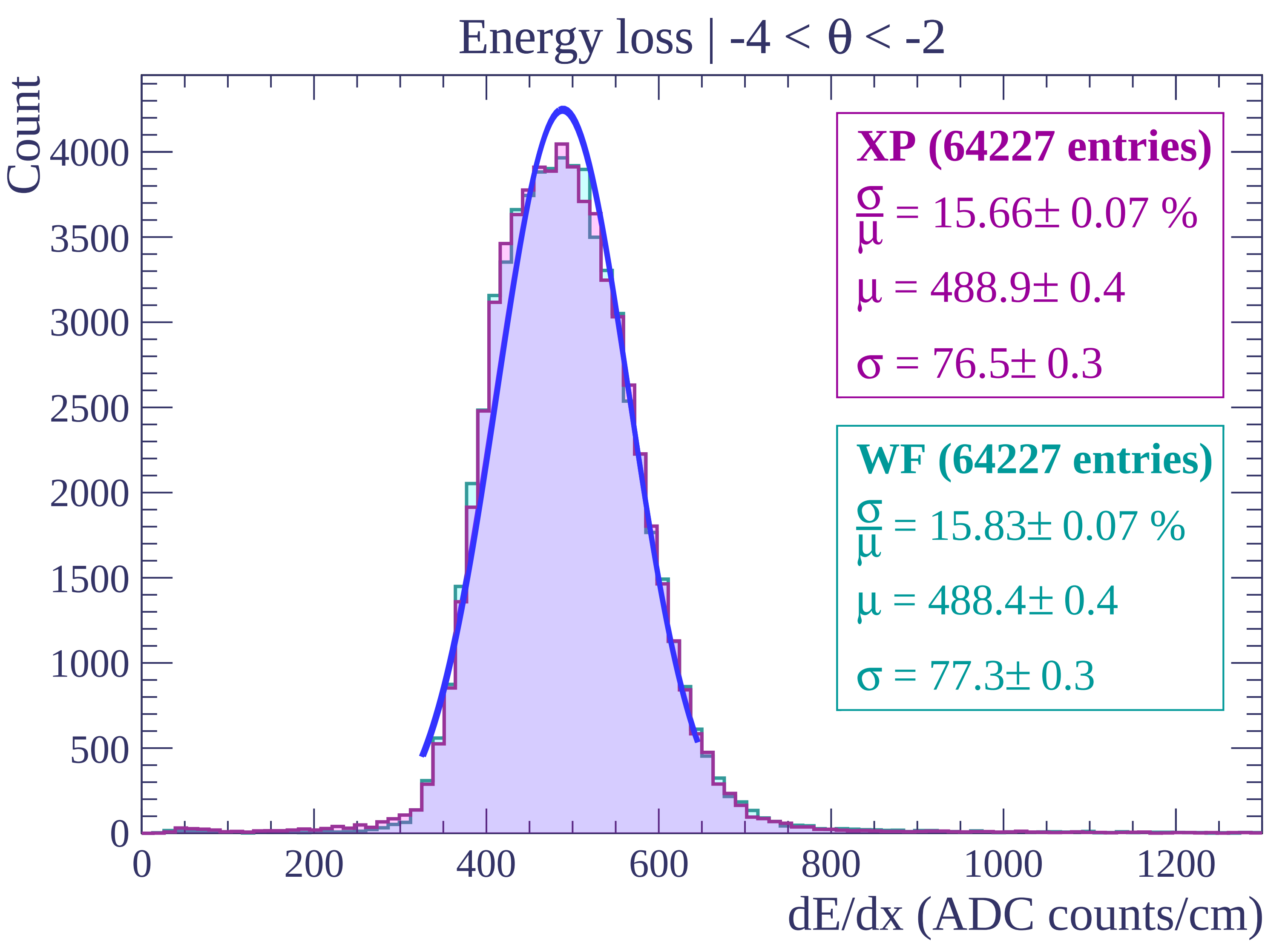


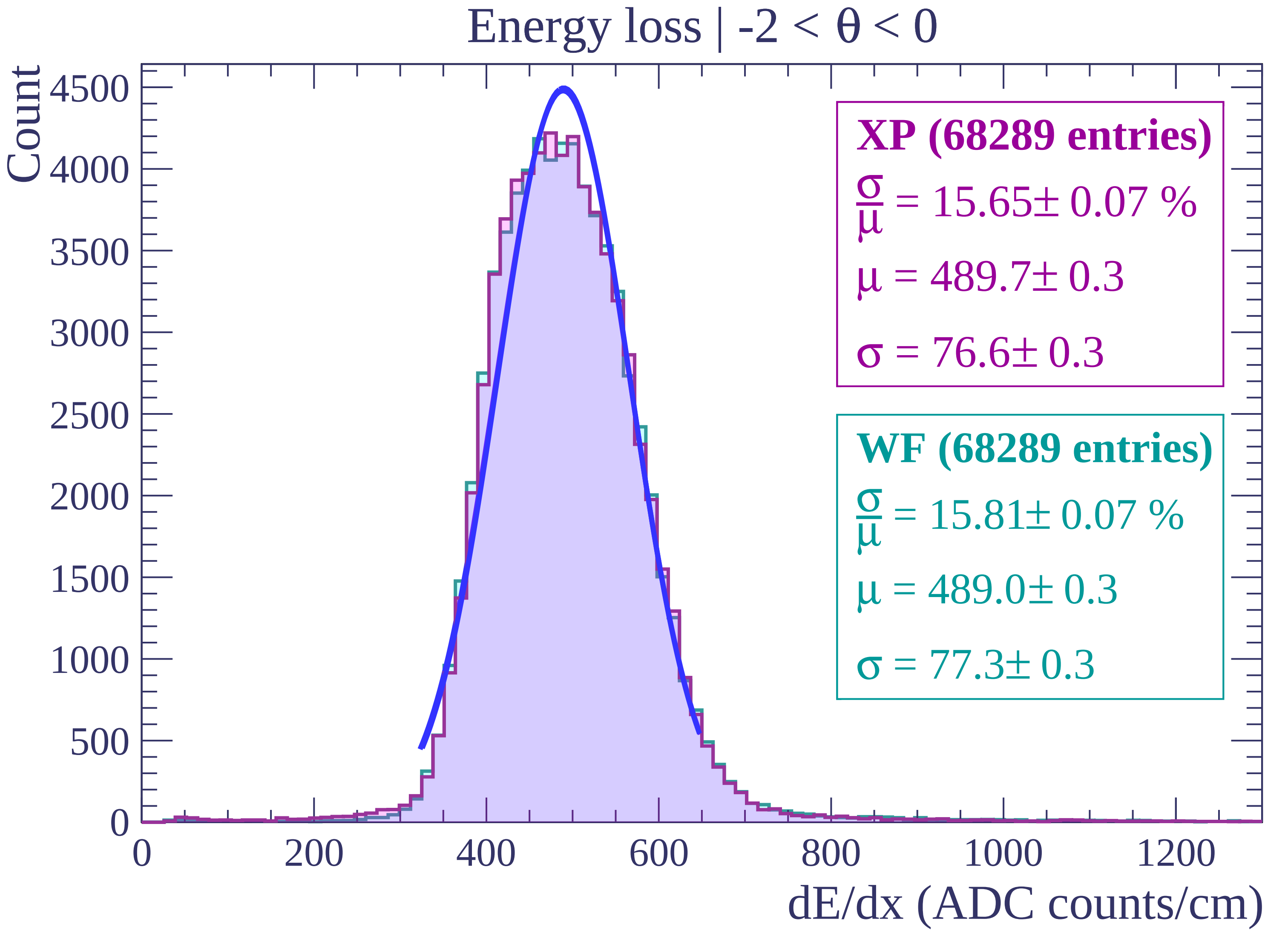
Energy loss | -12 < θ < -10
Count
3500
3000
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (55716 entries)
σ/μ = 15.74± 0.08 %
μ = 489.4± 0.4
σ = 77.1± 0.3
WF (55716 entries)
σ/μ = 15.99± 0.08 %
μ = 488.7± 0.4
σ = 78.2± 0.3

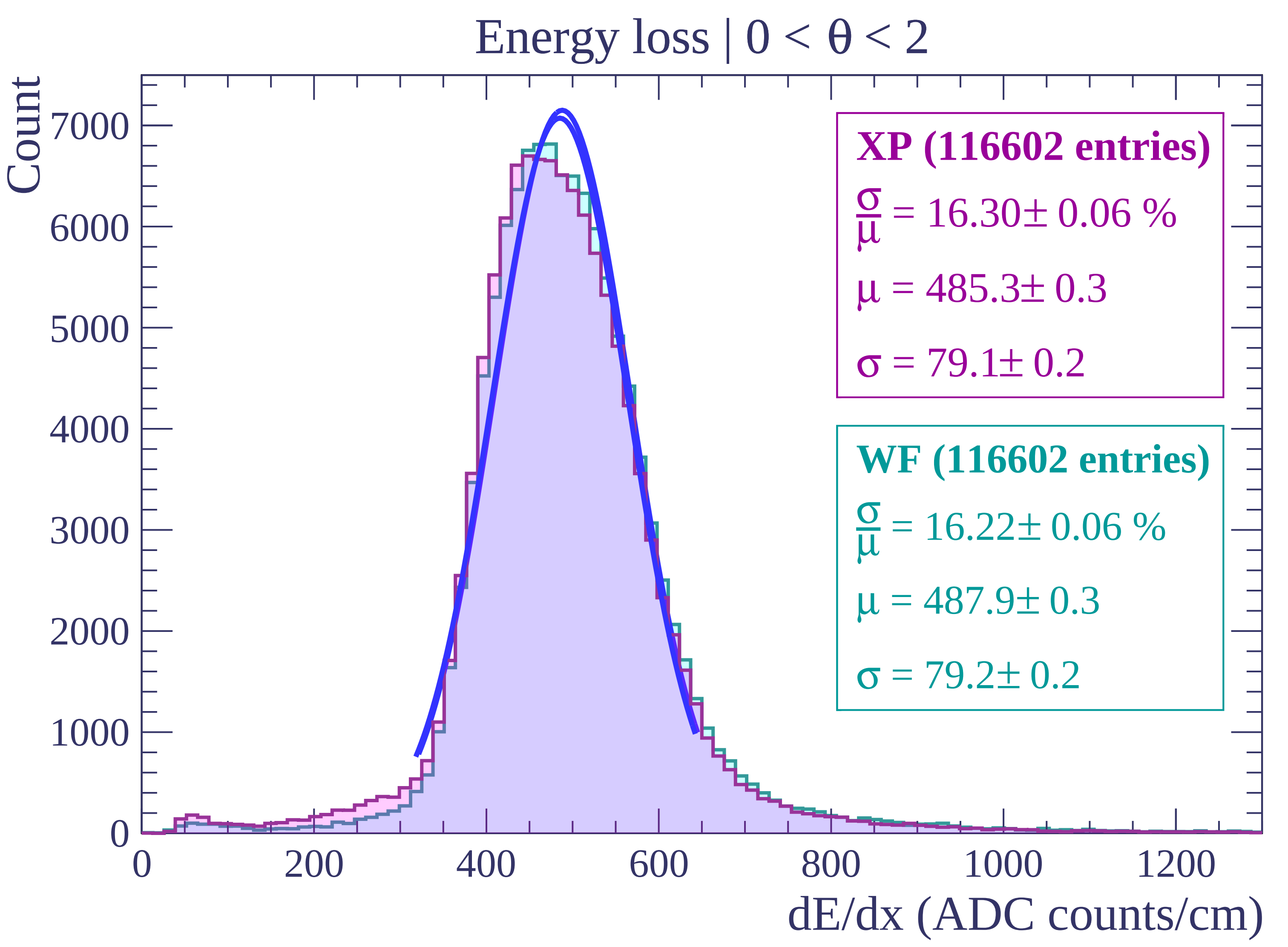

Energy loss | -10 < θ < -8
Count
4000
3500
3000
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (58546 entries)
σ/μ = 15.69± 0.08 %
μ = 488.6± 0.4
σ = 76.7± 0.3
WF (58546 entries)
σ/μ = 15.90± 0.08 %
μ = 488.1± 0.4
σ = 77.6± 0.3


Energy loss | -8 < θ < -6
Count
dE/dx (ADC counts/cm)
0
500
1000
1500
2000
2500
3000
3500
4000
0
200
400
600
800
1000
1200
XP (60713 entries)
σ/μ = 15.82± 0.07 %
μ = 488.3± 0.4
σ = 77.3± 0.3
WF (60713 entries)
σ/μ = 15.98± 0.08 %
μ = 487.6± 0.4
σ = 77.9± 0.3

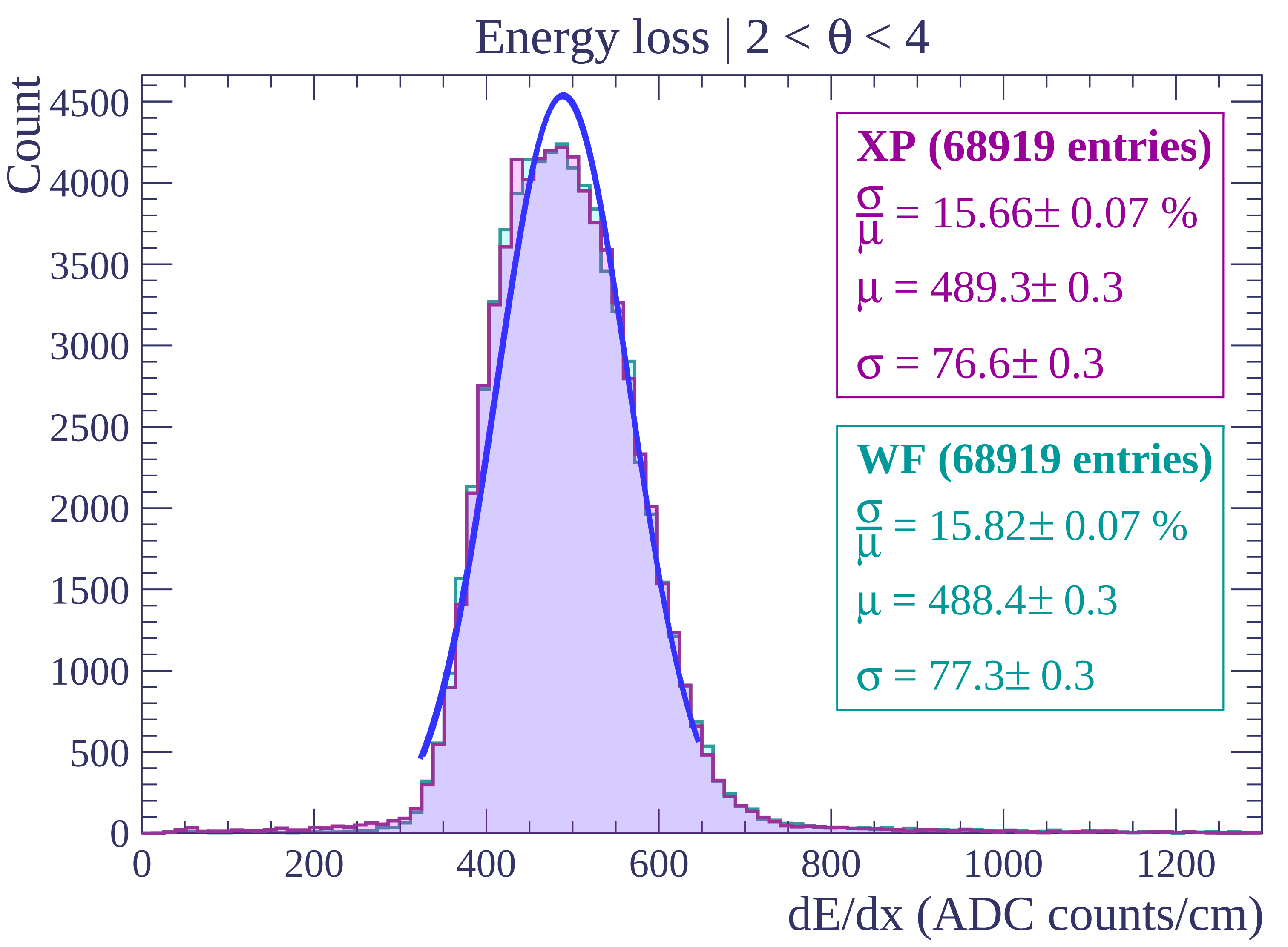

Energy loss | -6 < θ < -4
Count
dE/dx (ADC counts/cm)
XP (63577 entries)
σ/μ = 15.65± 0.07 %
μ = 488.9± 0.4
σ = 76.5± 0.3
WF (63577 entries)
σ/μ = 15.76± 0.07 %
μ = 488.5± 0.4
σ = 77.0± 0.3
0
500
1000
1500
2000
2500
3000
3500
4000
0
200
400
600
800
1000
1200


Energy loss | -4 < θ < -2
Count
dE/dx (ADC counts/cm)
0
500
1000
1500
2000
2500
3000
3500
4000
0
200
400
600
800
1000
1200
XP (64227 entries)
σ/μ = 15.66± 0.07 %
μ = 488.9± 0.4
σ = 76.5± 0.3
WF (64227 entries)
σ/μ = 15.83± 0.07 %
μ = 488.4± 0.4
σ = 77.3± 0.3

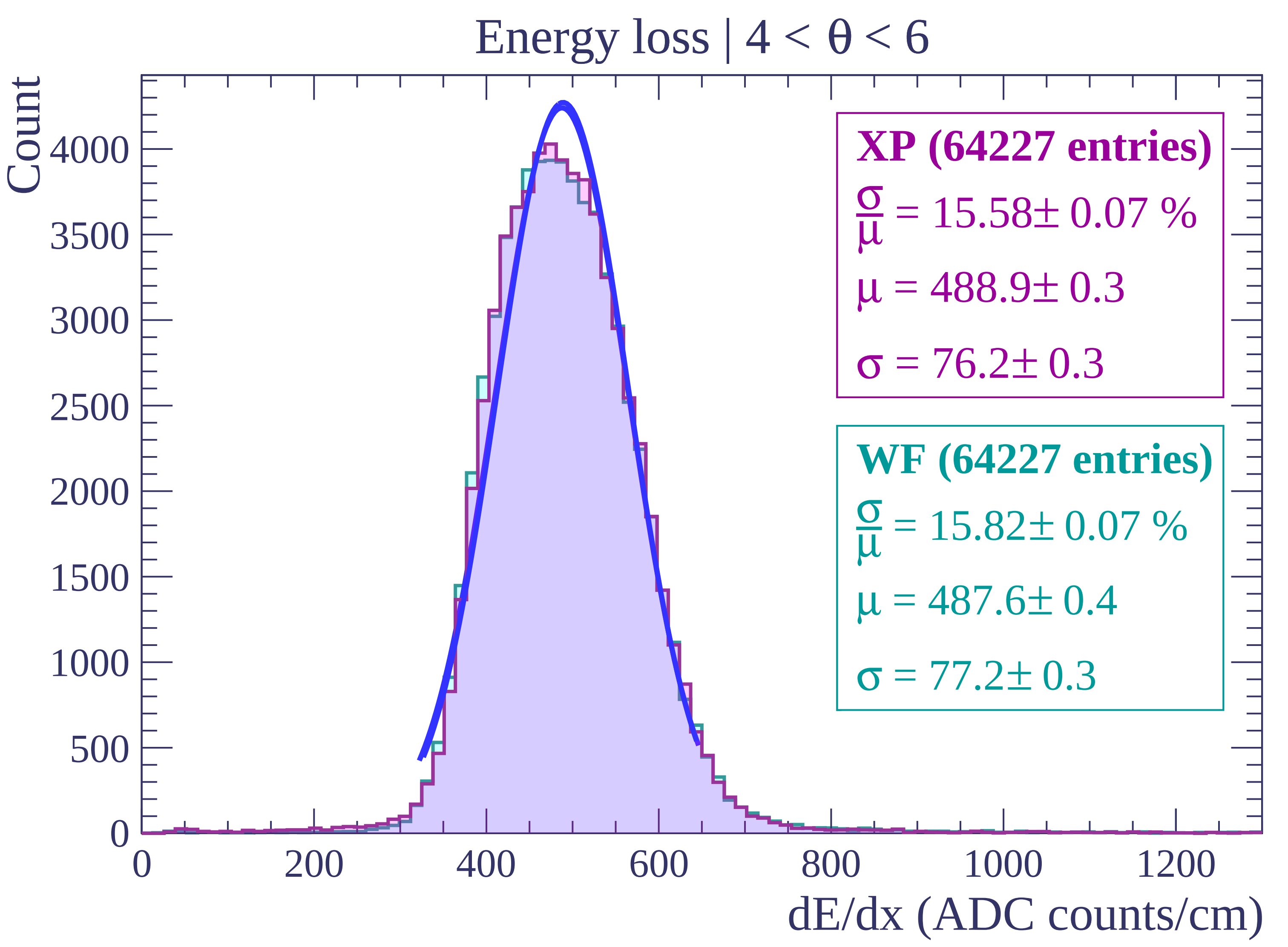

Energy loss | -2 < θ < 0
Count
dE/dx (ADC counts/cm)
XP (68289 entries)
σ/μ = 15.65± 0.07 %
μ = 489.7± 0.3
σ = 76.6± 0.3
WF (68289 entries)
σ/μ = 15.81± 0.07 %
μ = 489.0± 0.3
σ = 77.3± 0.3
0
500
1000
1500
2000
2500
3000
3500
4000
4500
0
200
400
600
800
1000
1200

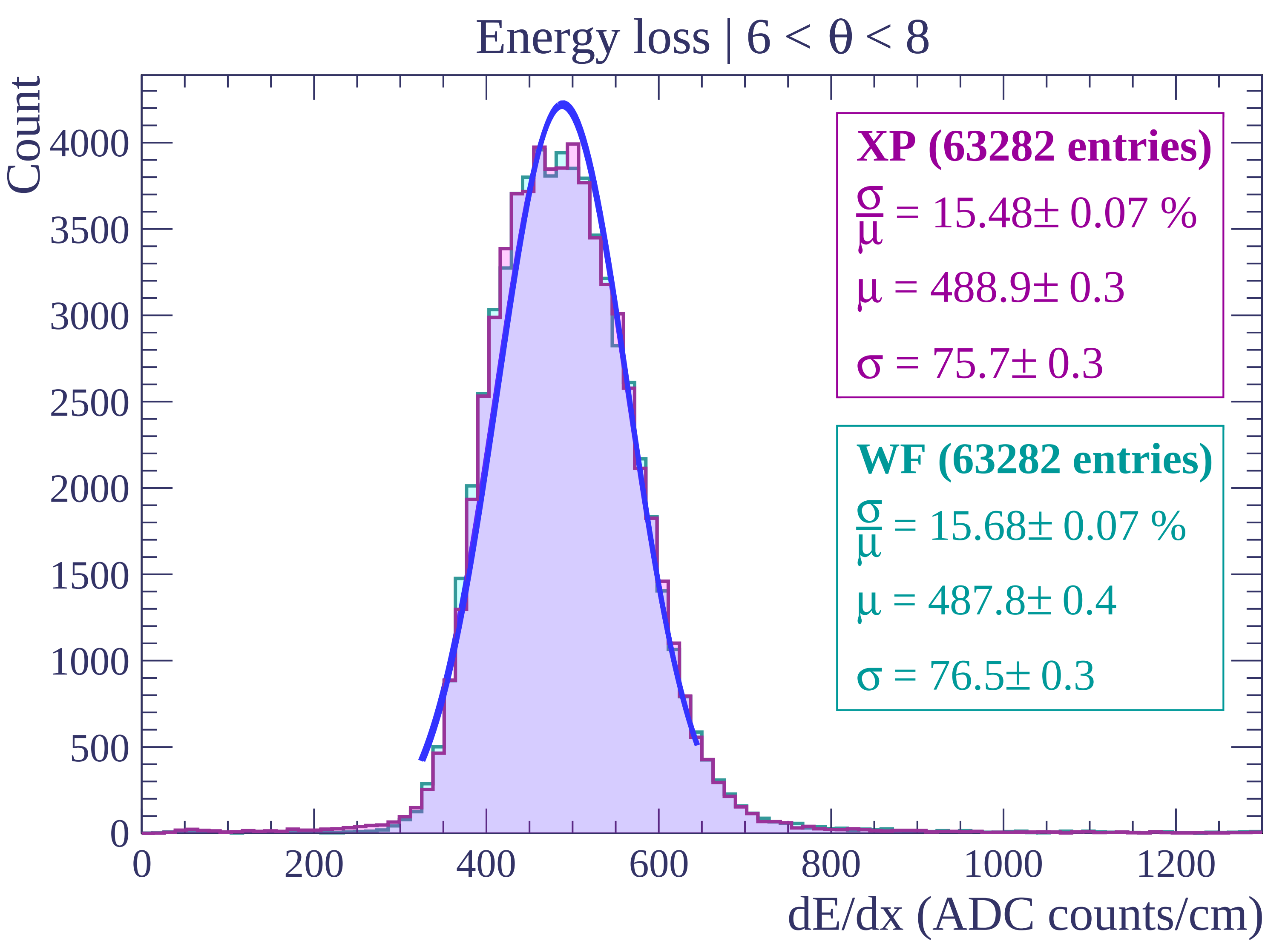


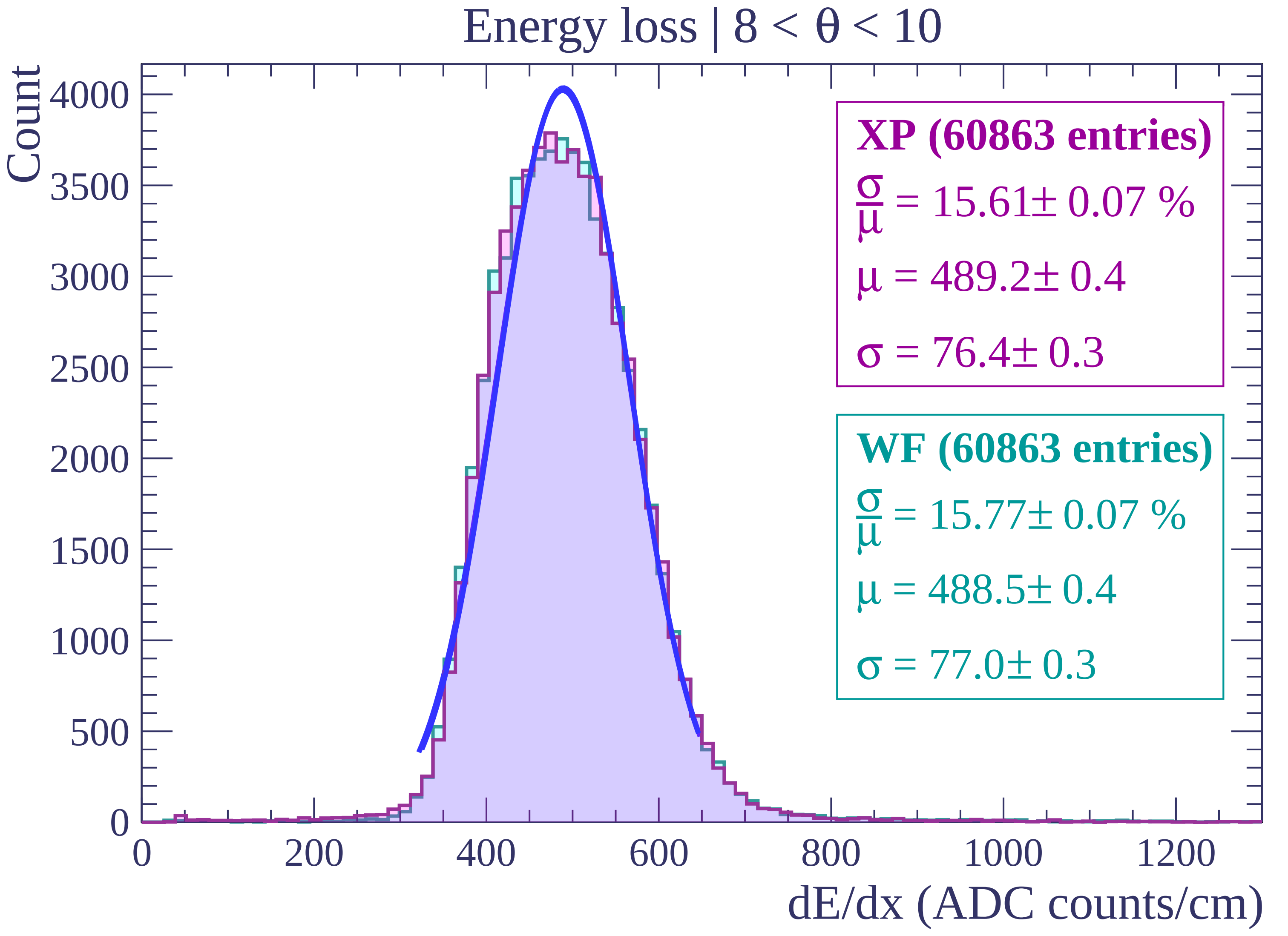
Energy loss | 0 < θ < 2
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1000
2000
3000
4000
5000
6000
7000
XP (116602 entries)
σ/μ = 16.30± 0.06 %
μ = 485.3± 0.3
σ = 79.1± 0.2
WF (116602 entries)
σ/μ = 16.22± 0.06 %
μ = 487.9± 0.3
σ = 79.2± 0.2

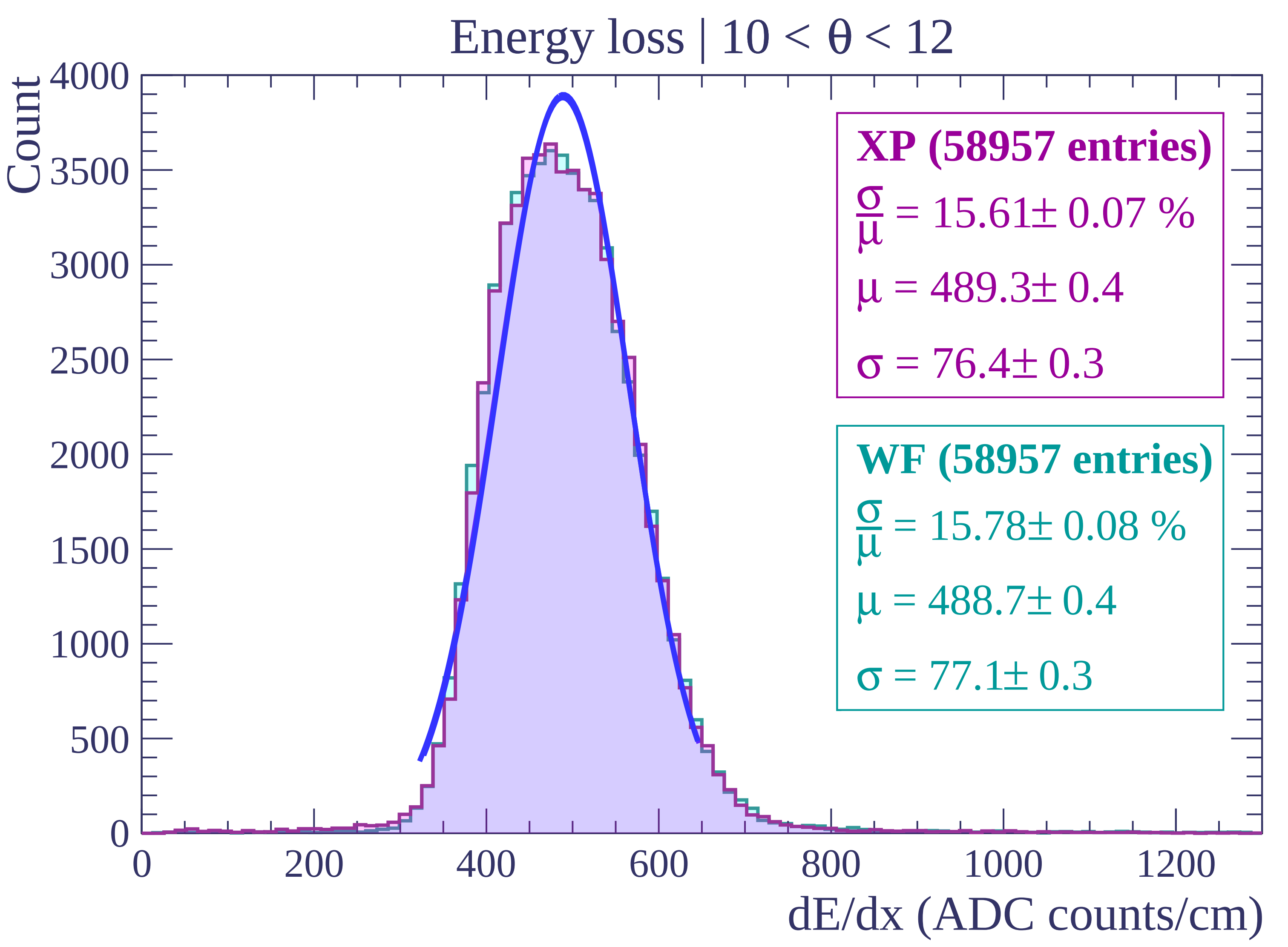


Energy loss | 2 < θ < 4
Count
dE/dx (ADC counts/cm)
XP (68919 entries)
σ/μ = 15.66± 0.07 %
μ = 489.3± 0.3
σ = 76.6± 0.3
WF (68919 entries)
σ/μ = 15.82± 0.07 %
μ = 488.4± 0.3
σ = 77.3± 0.3
0
500
1000
1500
2000
2500
3000
3500
4000
4500
0
200
400
600
800
1000
1200

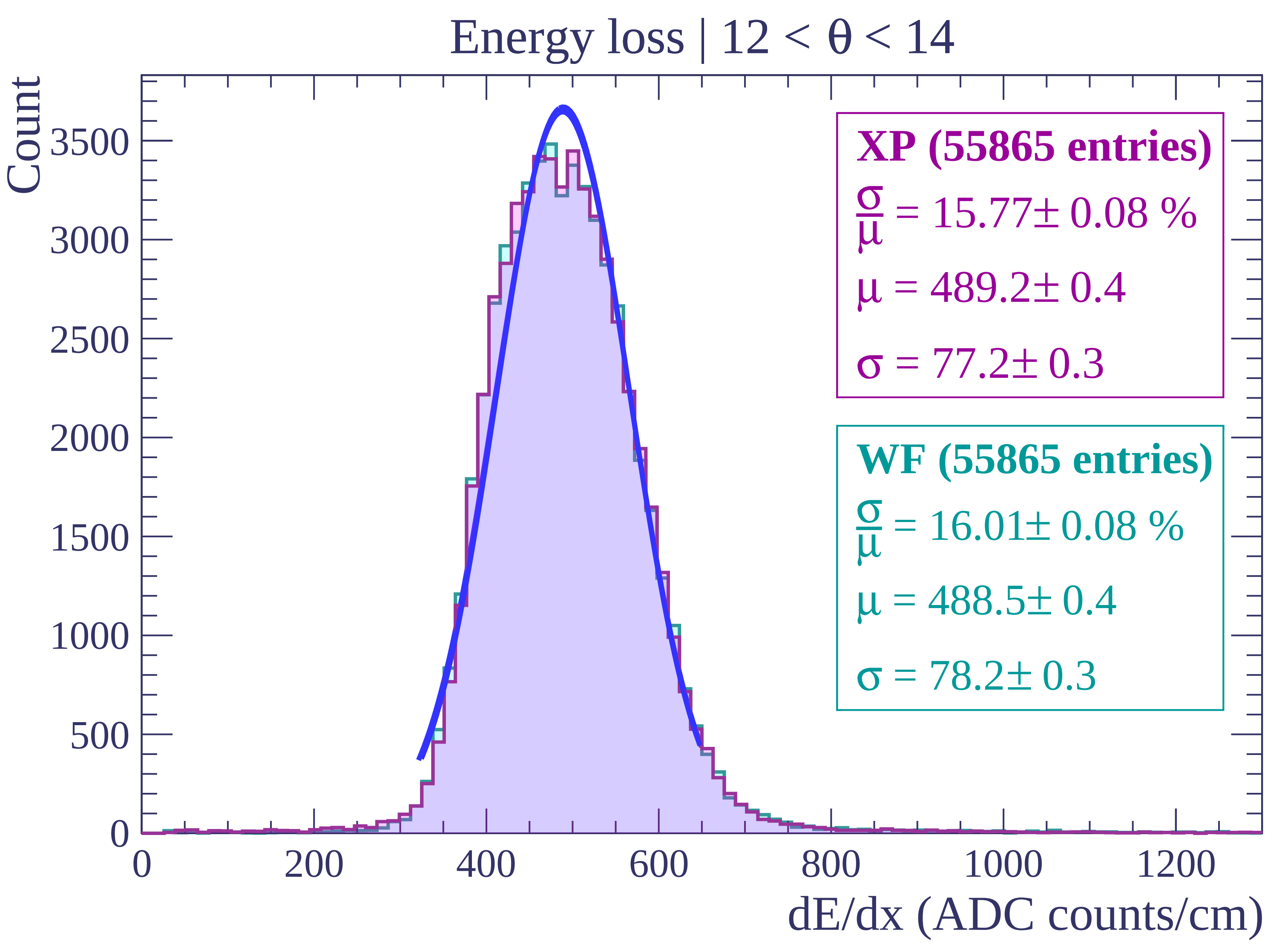

Energy loss | 4 < θ < 6
Count
dE/dx (ADC counts/cm)
XP (64227 entries)
σ/μ = 15.58± 0.07 %
μ = 488.9± 0.3
σ = 76.2± 0.3
WF (64227 entries)
σ/μ = 15.82± 0.07 %
μ = 487.6± 0.4
σ = 77.2± 0.3
0
500
1000
1500
2000
2500
3000
3500
4000
0
200
400
600
800
1000
1200


Energy loss | 6 < θ < 8
Count
4000
3500
3000
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (63282 entries)
σ/μ = 15.48± 0.07 %
μ = 488.9± 0.3
σ = 75.7± 0.3
WF (63282 entries)
σ/μ = 15.68± 0.07 %
μ = 487.8± 0.4
σ = 76.5± 0.3

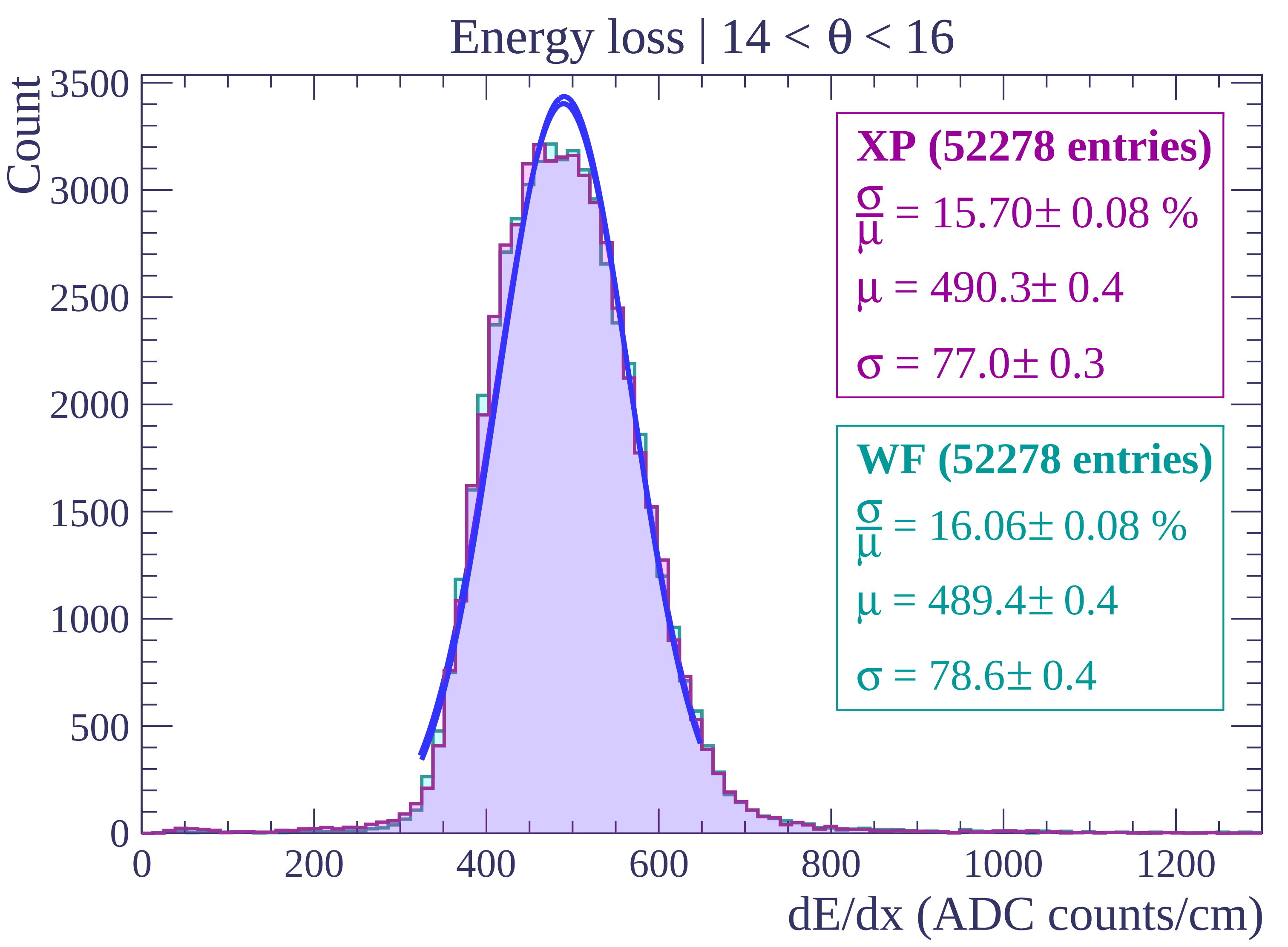

Energy loss | 8 < θ < 10
Count
4000
3500
3000
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (60863 entries)
σ/μ = 15.61 ± 0.07 %
μ = 489.2 ± 0.4
σ = 76.4 ± 0.3
WF (60863 entries)
σ/μ = 15.77 ± 0.07 %
μ = 488.5 ± 0.4
σ = 77.0 ± 0.3



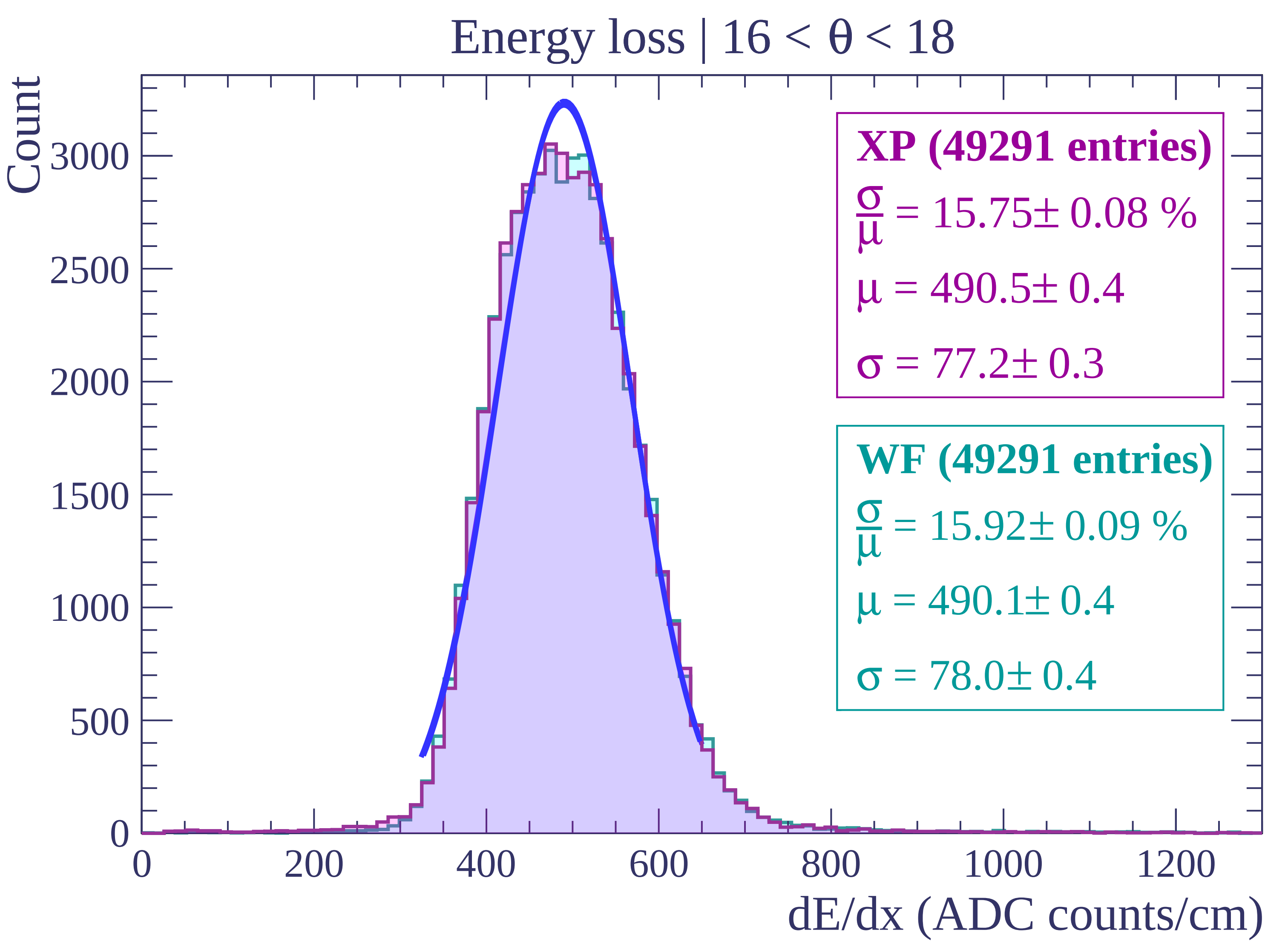
Energy loss | 10 < θ < 12
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
500
1000
1500
2000
2500
3000
3500
4000
XP (58957 entries)
σ/μ = 15.61± 0.07 %
μ = 489.3± 0.4
σ = 76.4± 0.3
WF (58957 entries)
σ/μ = 15.78± 0.08 %
μ = 488.7± 0.4
σ = 77.1± 0.3



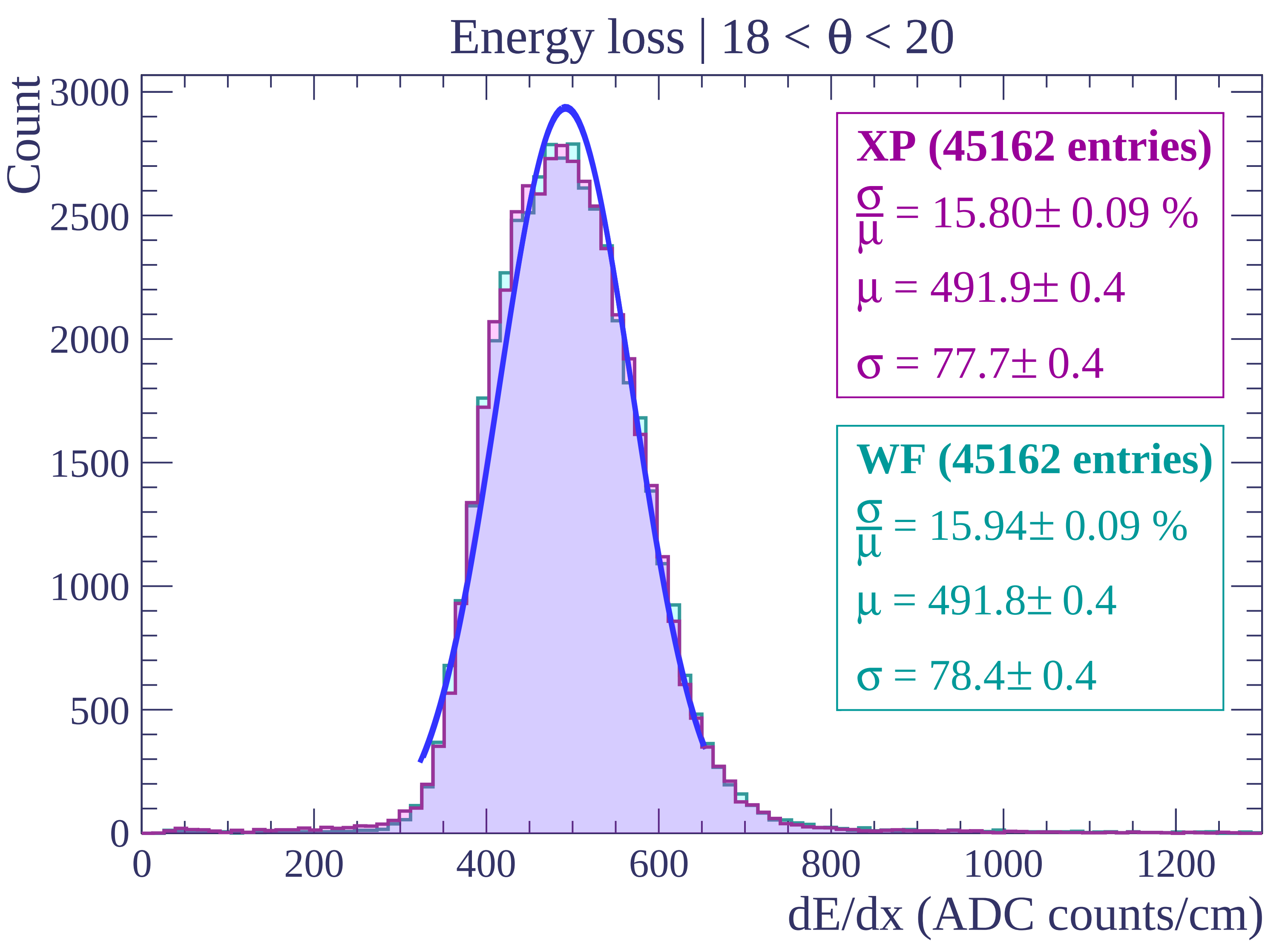
Energy loss | 12 < θ < 14
Count
dE/dx (ADC counts/cm)
XP (55865 entries)
σ/μ = 15.77± 0.08 %
μ = 489.2± 0.4
σ = 77.2± 0.3
WF (55865 entries)
σ/μ = 16.01± 0.08 %
μ = 488.5± 0.4
σ = 78.2± 0.3

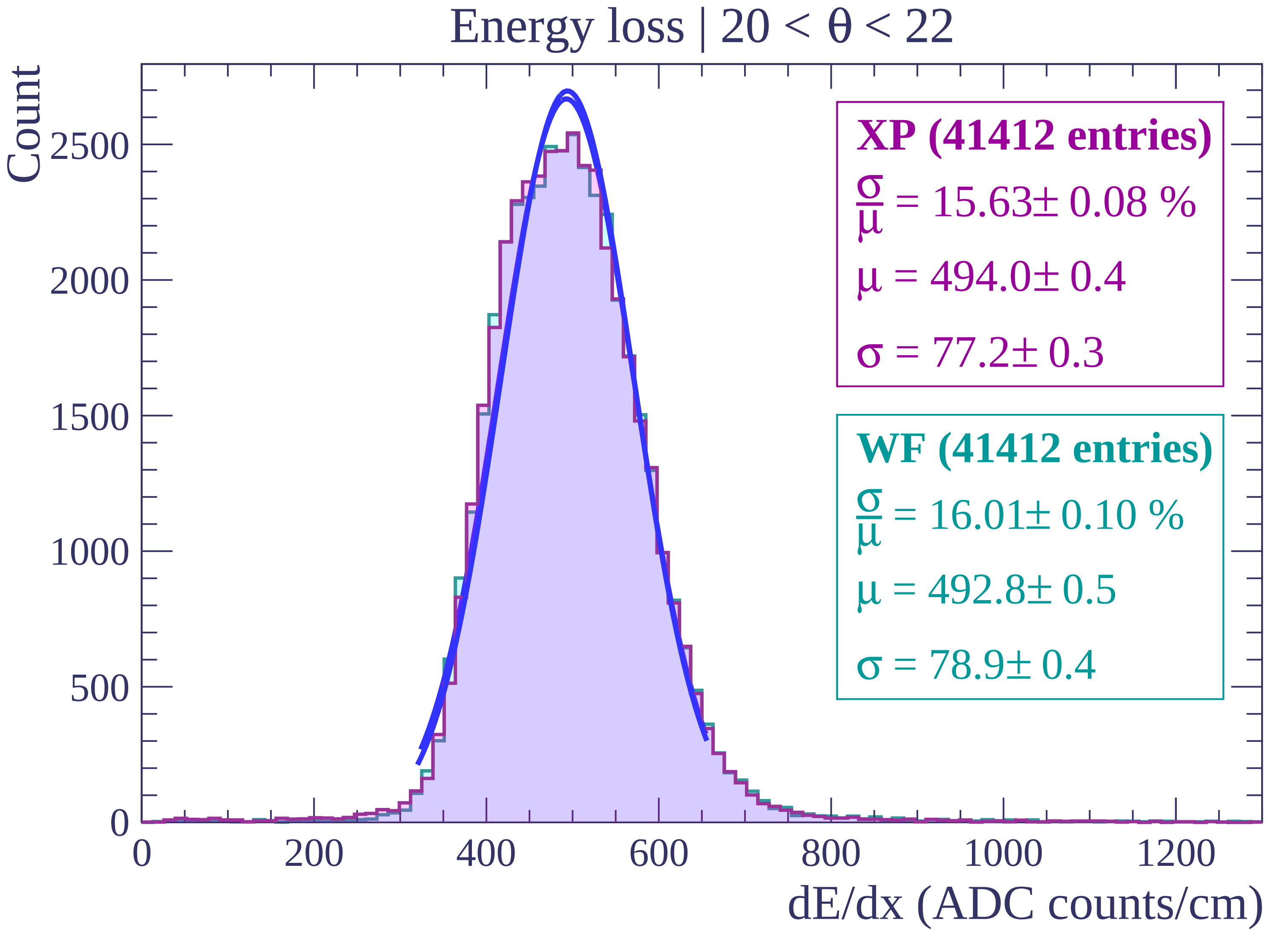


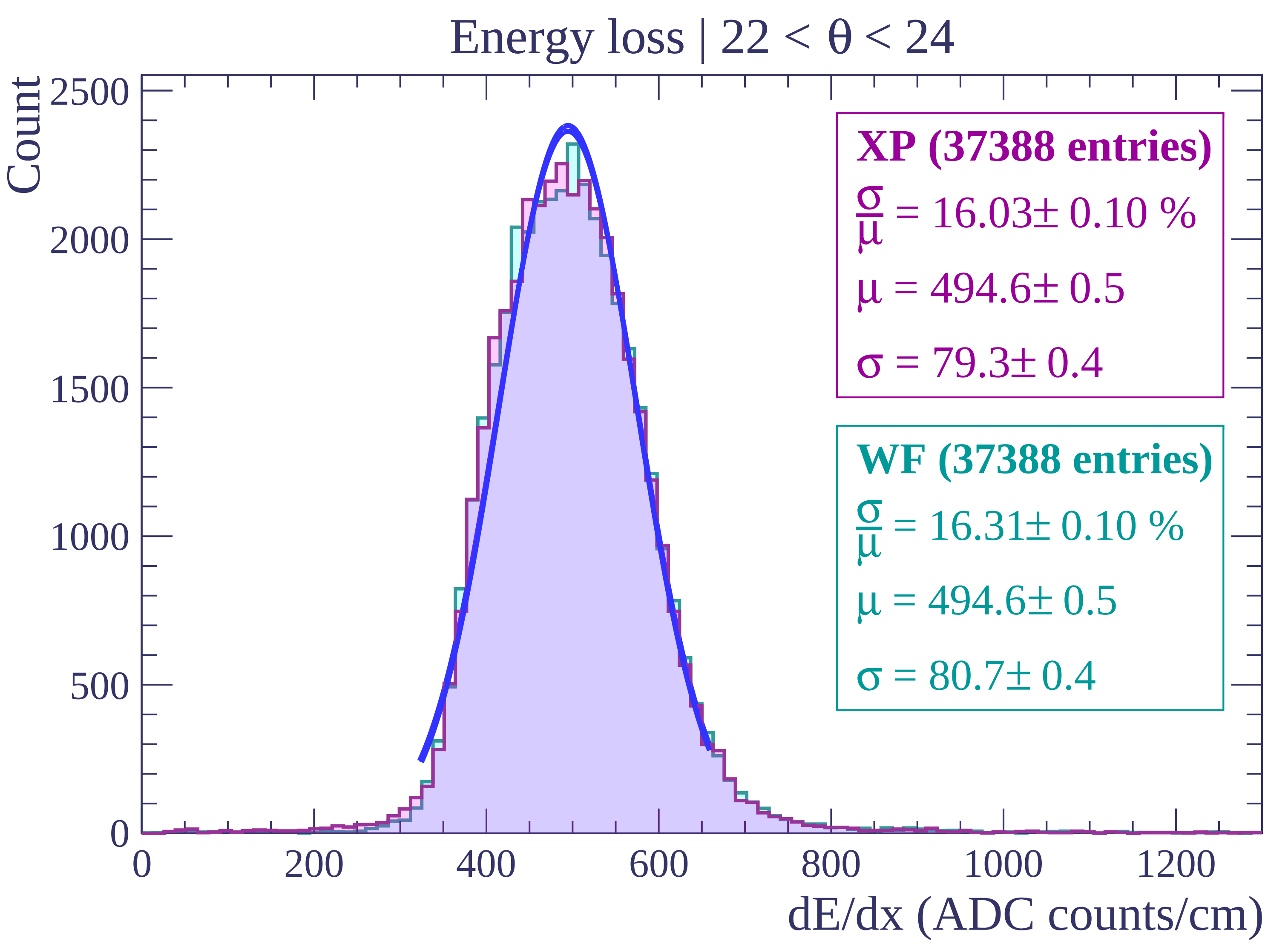
Energy loss | 14 < θ < 16
Count
dE/dx (ADC counts/cm)
XP (52278 entries)
σ/μ = 15.70± 0.08 %
μ = 490.3± 0.4
σ = 77.0± 0.3
WF (52278 entries)
σ/μ = 16.06± 0.08 %
μ = 489.4± 0.4
σ = 78.6± 0.4



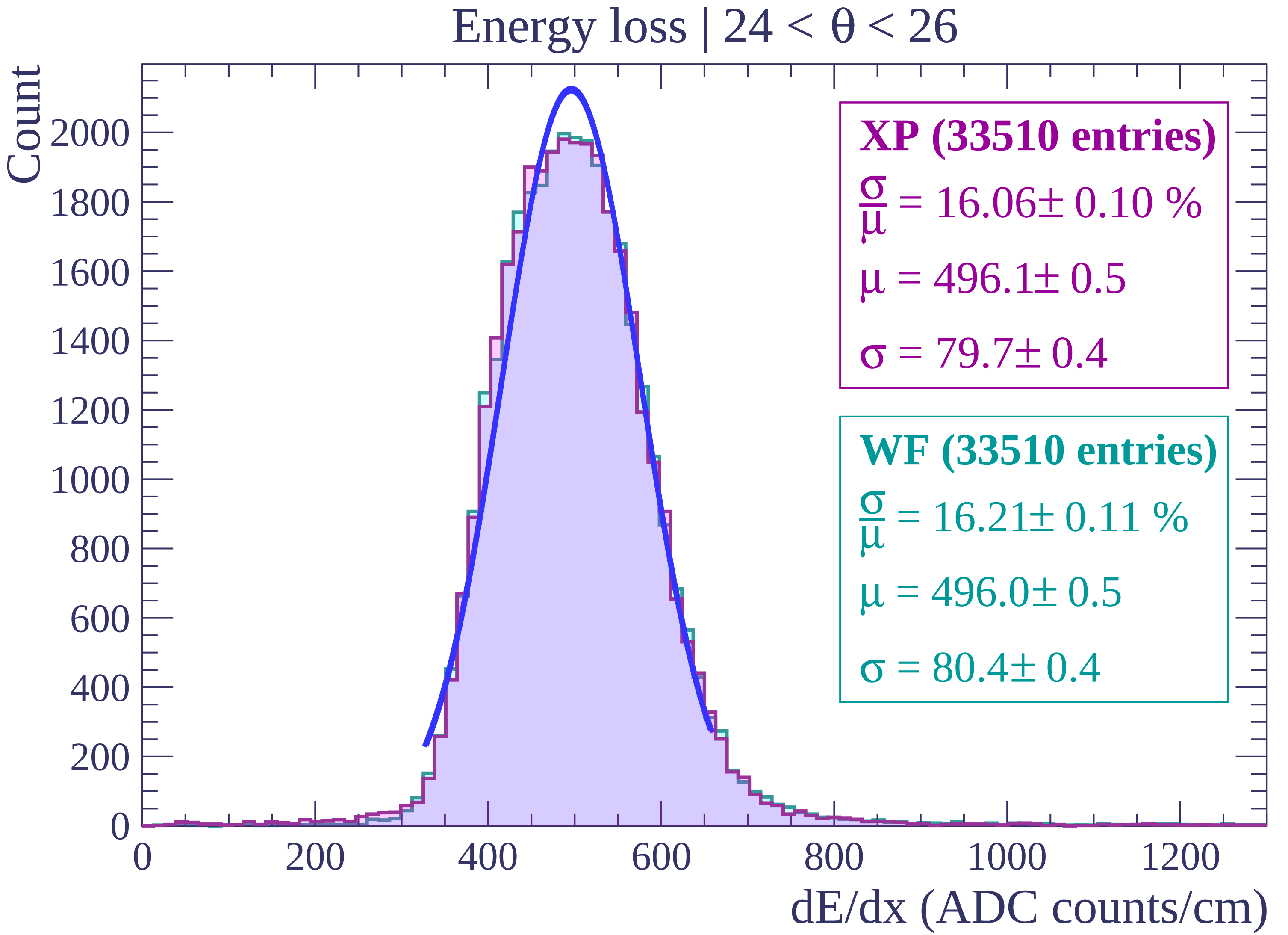
Energy loss | 16 < θ < 18
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
500
1000
1500
2000
2500
3000
XP (49291 entries)
$\frac{\sigma}{\mu}$ = 15.75± 0.08 %
μ = 490.5± 0.4
σ = 77.2± 0.3
WF (49291 entries)
$\frac{\sigma}{\mu}$ = 15.92± 0.09 %
μ = 490.1± 0.4
σ = 78.0± 0.4

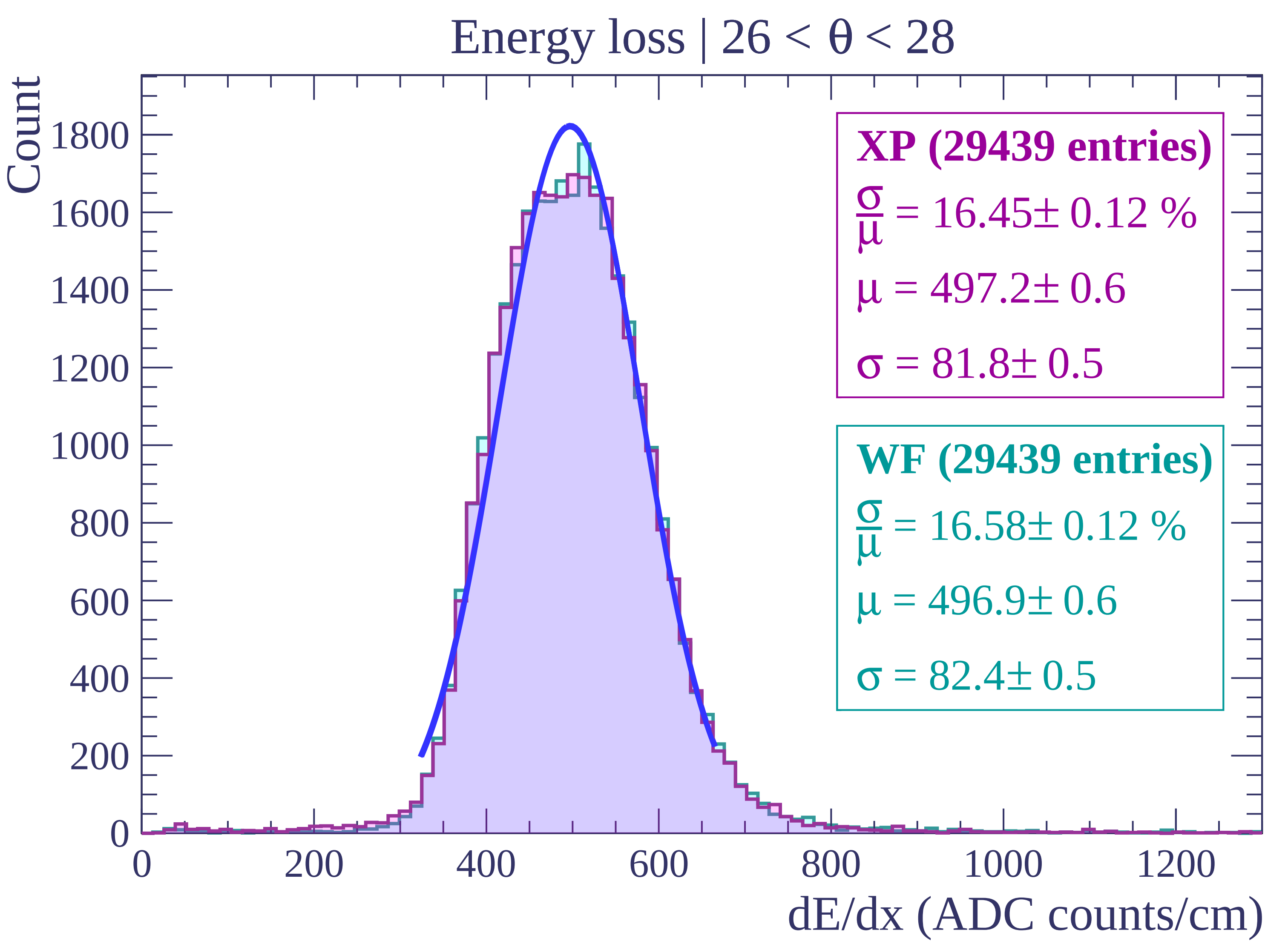

Energy loss | 18 < θ < 20
Count
3000
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (45162 entries)
σ/μ = 15.80± 0.09 %
μ = 491.9± 0.4
σ = 77.7± 0.4
WF (45162 entries)
σ/μ = 15.94± 0.09 %
μ = 491.8± 0.4
σ = 78.4± 0.4


Energy loss | 20 < θ < 22
Count
dE/dx (ADC counts/cm)
0
500
1000
1500
2000
2500
0
200
400
600
800
1000
1200
XP (41412 entries)
σ/μ = 15.63± 0.08 %
μ = 494.0± 0.4
σ = 77.2± 0.3
WF (41412 entries)
σ/μ = 16.01± 0.10 %
μ = 492.8± 0.5
σ = 78.9± 0.4

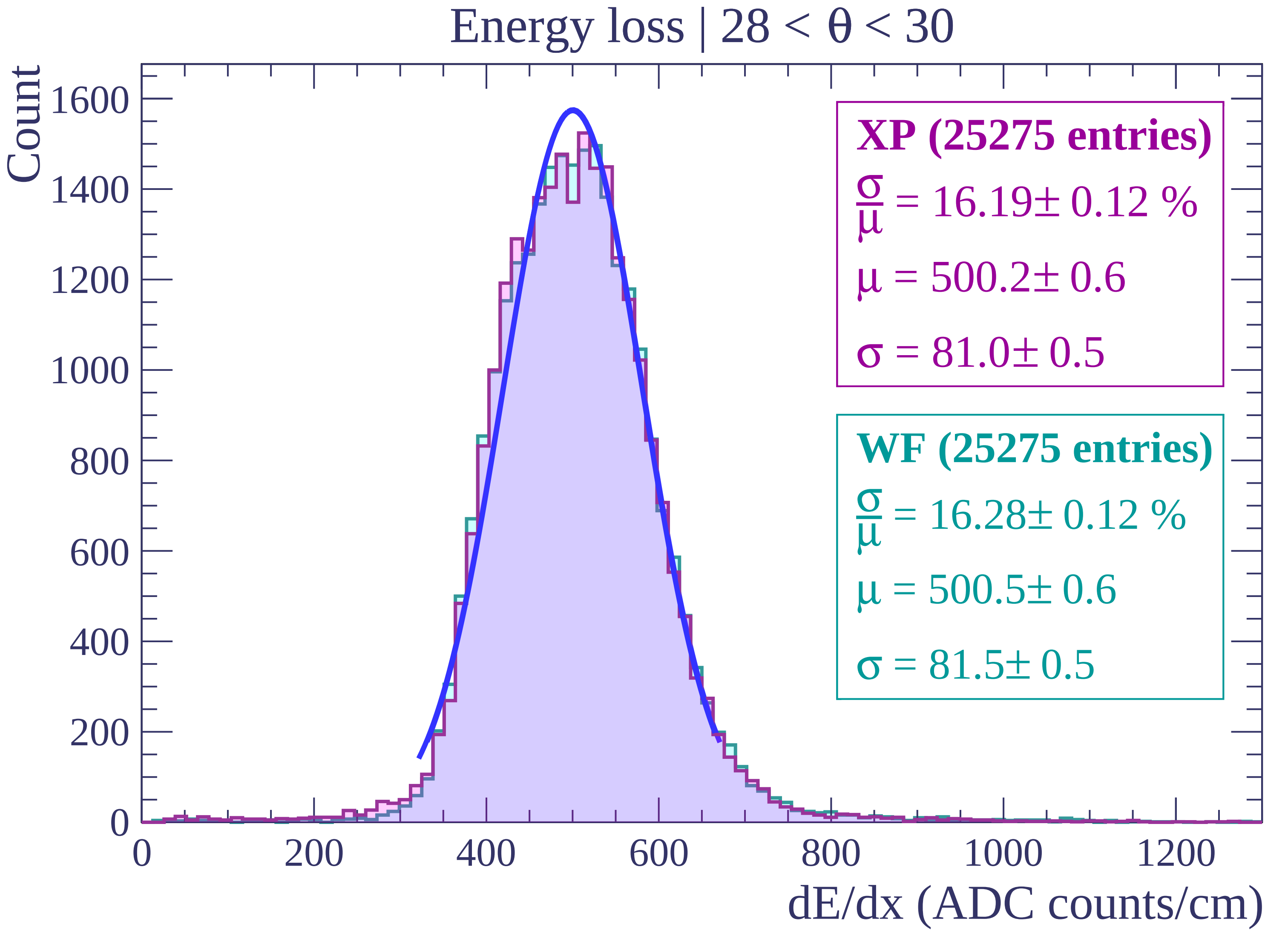


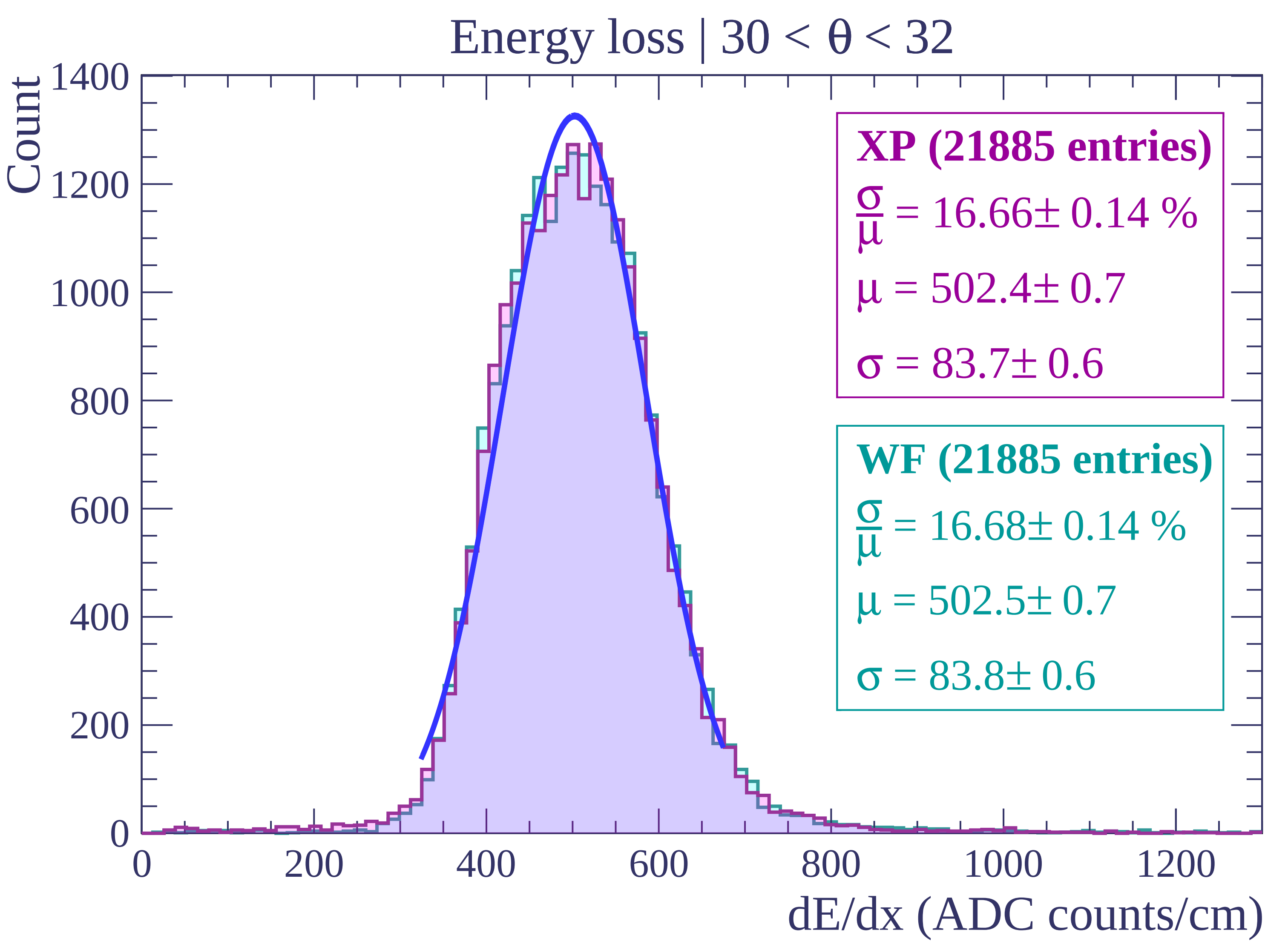
Energy loss | 22 < θ < 24
Count
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (37388 entries)
σ/μ = 16.03± 0.10 %
μ = 494.6± 0.5
σ = 79.3± 0.4
WF (37388 entries)
σ/μ = 16.31± 0.10 %
μ = 494.6± 0.5
σ = 80.7± 0.4


Energy loss | 24 < θ < 26
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
0
200
400
600
800
1000
1200
XP (33510 entries)
σ/μ = 16.06± 0.10 %
μ = 496.1± 0.5
σ = 79.7± 0.4
WF (33510 entries)
σ/μ = 16.21± 0.11 %
μ = 496.0± 0.5
σ = 80.4± 0.4

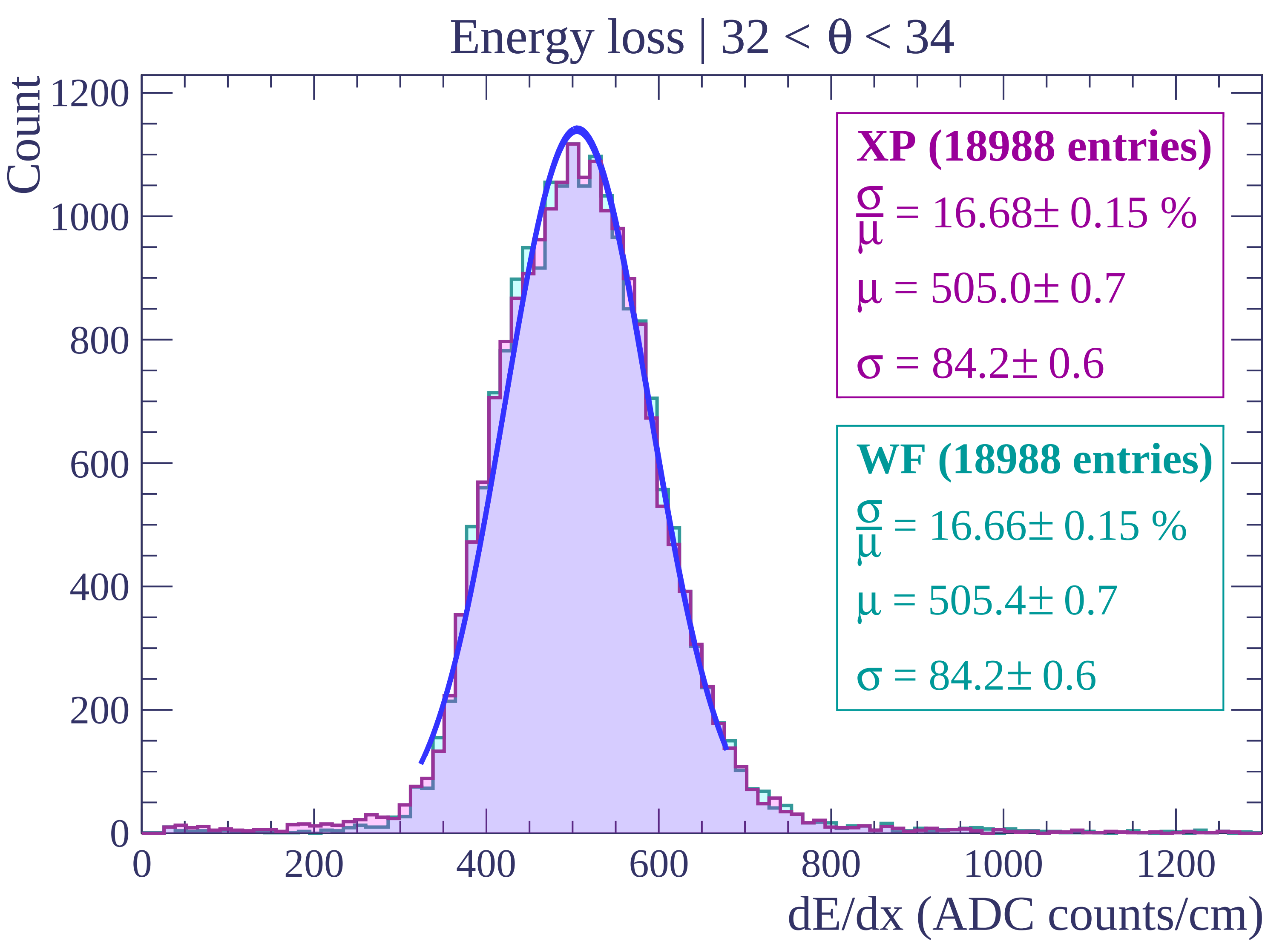


Energy loss | 26 < θ < 28
Count
dE/dx (ADC counts/cm)
XP (29439 entries)
σ/μ = 16.45± 0.12 %
μ = 497.2± 0.6
σ = 81.8± 0.5
WF (29439 entries)
σ/μ = 16.58± 0.12 %
μ = 496.9± 0.6
σ = 82.4± 0.5
0
200
400
600
800
1000
1200
1400
1600
1800

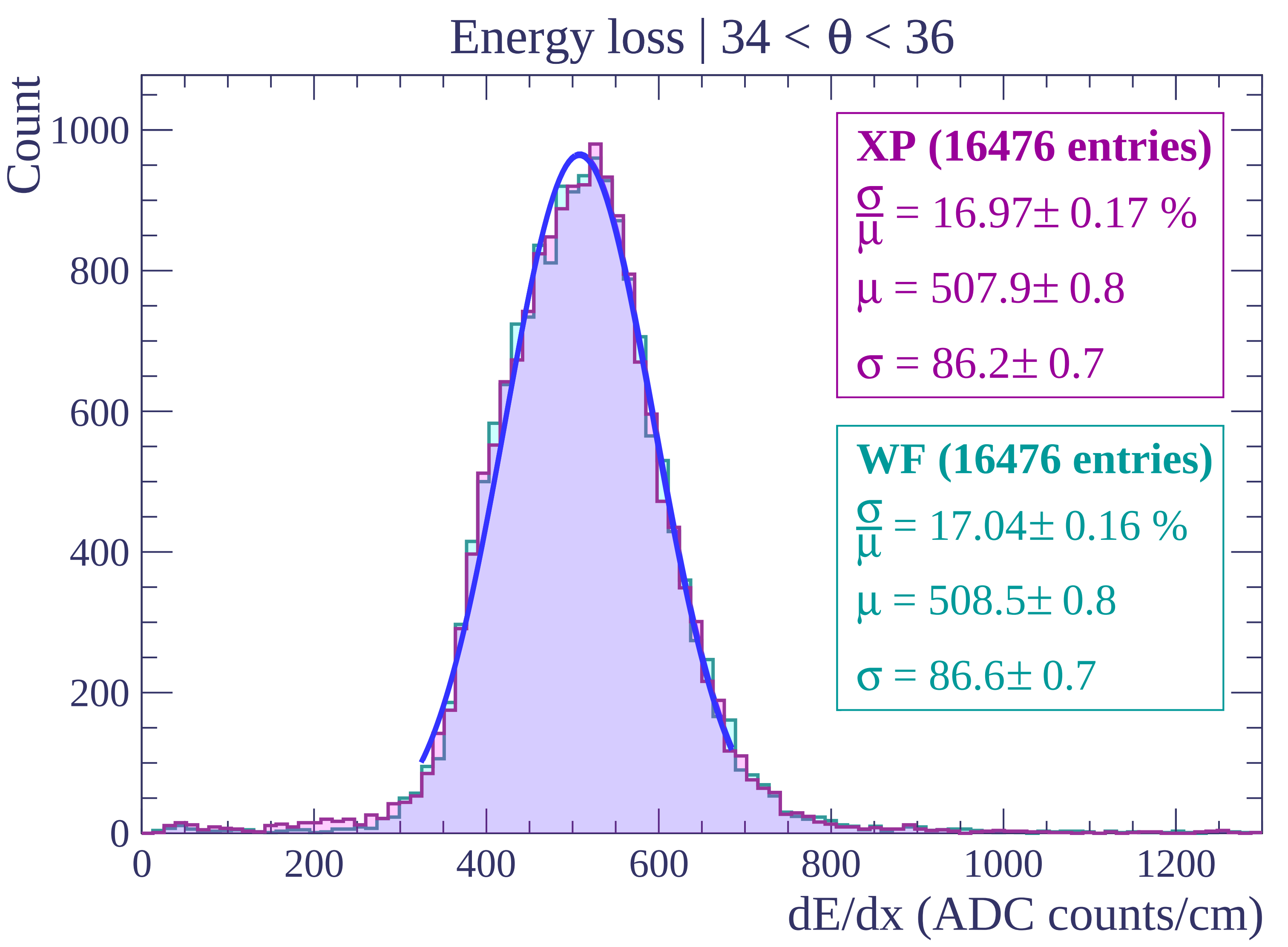

Energy loss | 28 < θ < 30
Count
1600
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (25275 entries)
σ/μ = 16.19± 0.12 %
μ = 500.2± 0.6
σ = 81.0± 0.5
WF (25275 entries)
σ/μ = 16.28± 0.12 %
μ = 500.5± 0.6
σ = 81.5± 0.5

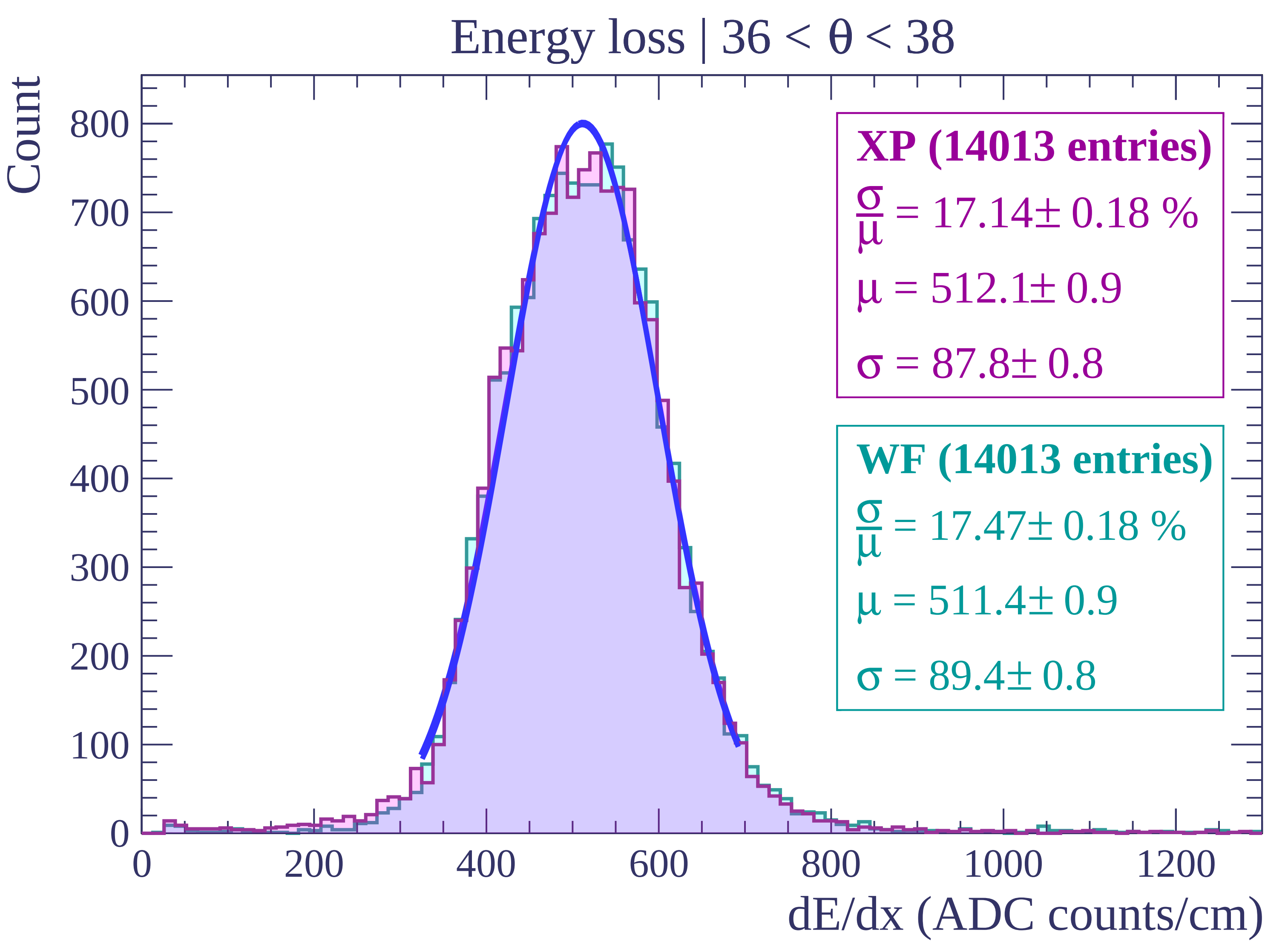


Energy loss | 30 < θ < 32
Count
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (21885 entries)
σ/μ = 16.66± 0.14 %
μ = 502.4± 0.7
σ = 83.7± 0.6
WF (21885 entries)
σ/μ = 16.68± 0.14 %
μ = 502.5± 0.7
σ = 83.8± 0.6

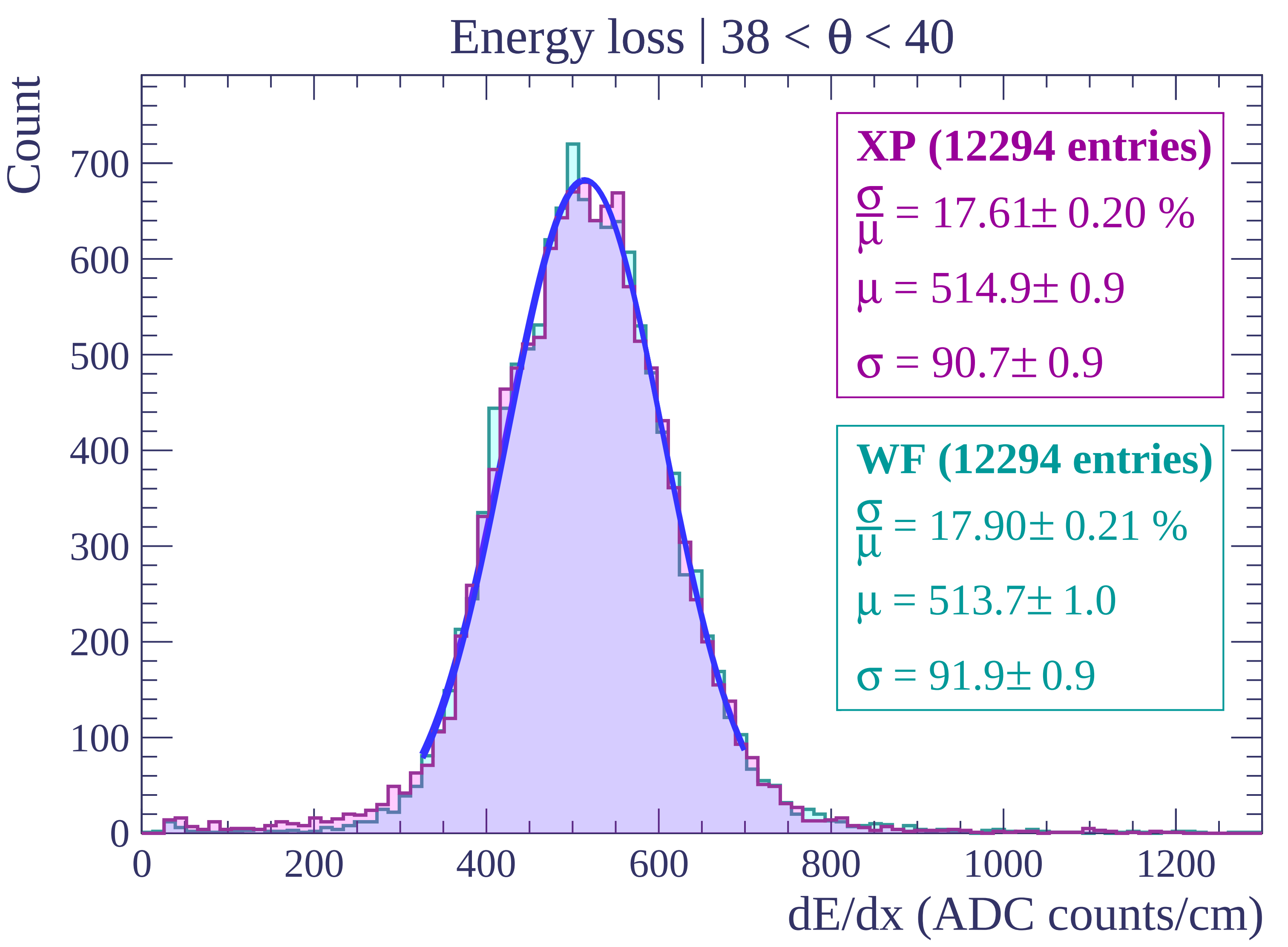


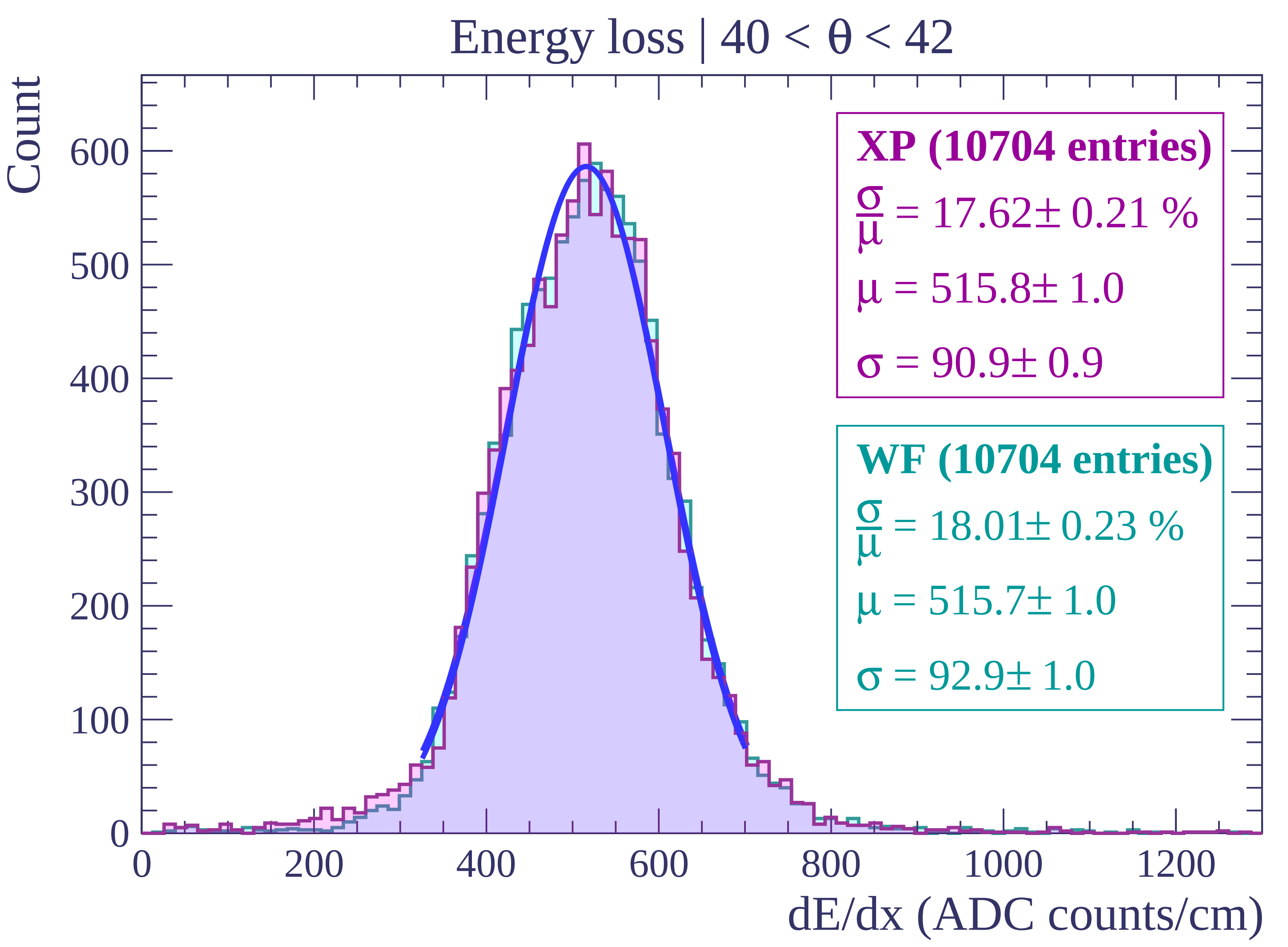
Energy loss | 32 < θ < 34
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (18988 entries)
σ/μ = 16.68± 0.15 %
μ = 505.0± 0.7
σ = 84.2± 0.6
WF (18988 entries)
σ/μ = 16.66± 0.15 %
μ = 505.4± 0.7
σ = 84.2± 0.6


Energy loss | 34 < θ < 36
Count
dE/dx (ADC counts/cm)
XP (16476 entries)
σ/μ = 16.97± 0.17 %
μ = 507.9± 0.8
σ = 86.2± 0.7
WF (16476 entries)
σ/μ = 17.04± 0.16 %
μ = 508.5± 0.8
σ = 86.6± 0.7
0
200
400
600
800
1000
1200

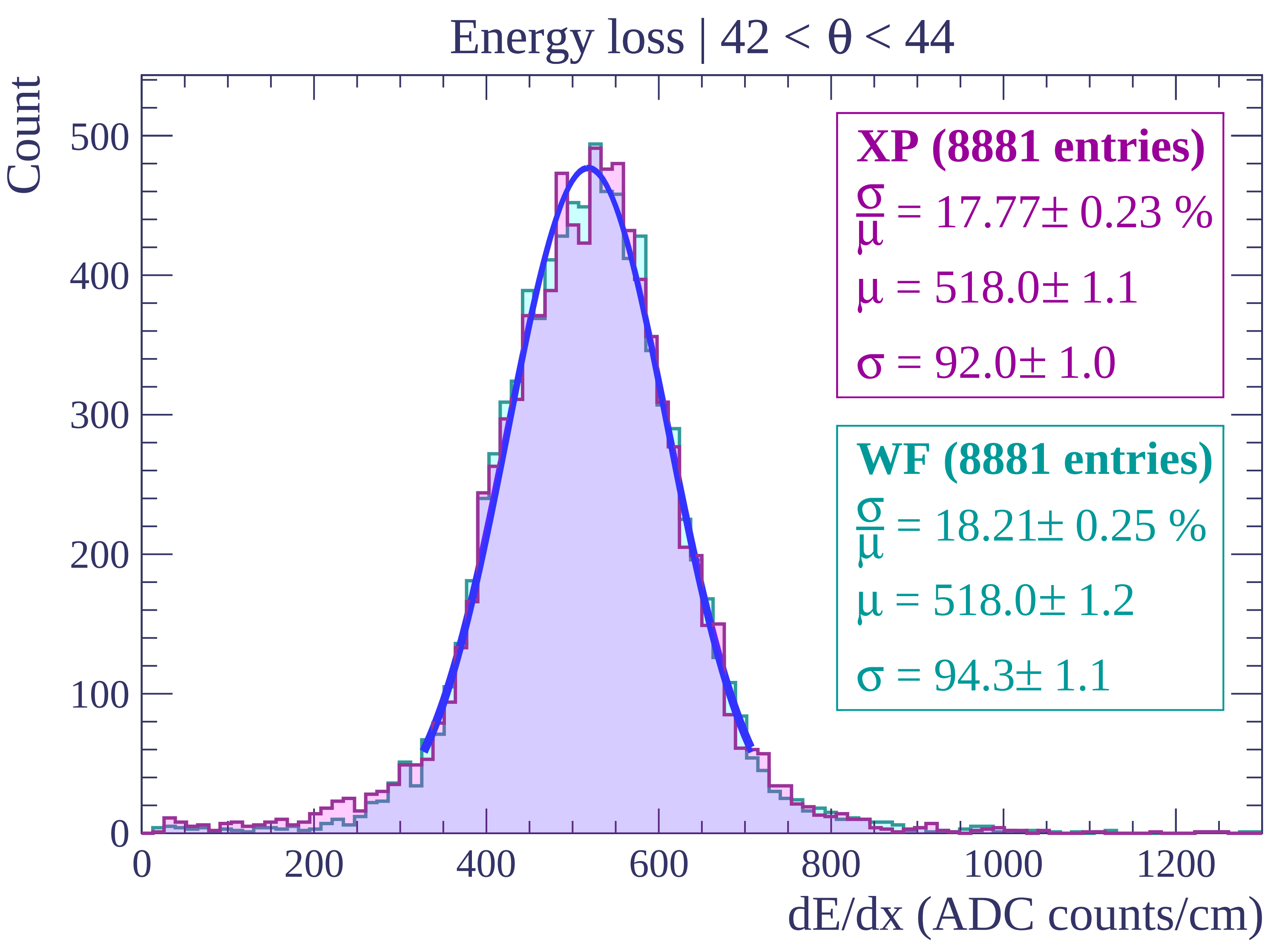


Energy loss | 36 < θ < 38
Count
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (14013 entries)
σ/μ = 17.14± 0.18 %
μ = 512.1± 0.9
σ = 87.8± 0.8
WF (14013 entries)
σ/μ = 17.47± 0.18 %
μ = 511.4± 0.9
σ = 89.4± 0.8

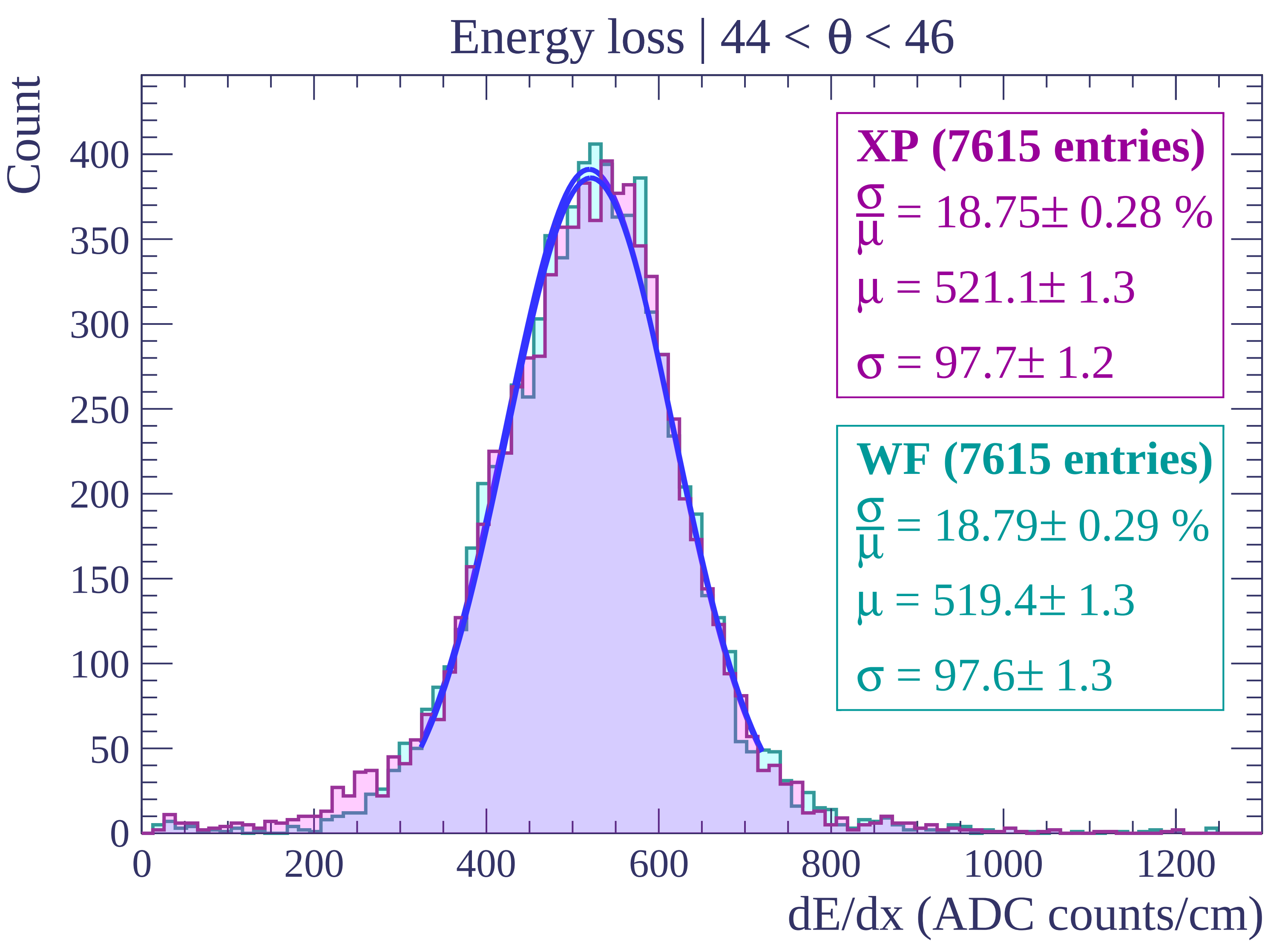

Energy loss | 38 < θ < 40
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (12294 entries)
σ/μ = 17.61± 0.20 %
μ = 514.9± 0.9
σ = 90.7± 0.9
WF (12294 entries)
σ/μ = 17.90± 0.21 %
μ = 513.7± 1.0
σ = 91.9± 0.9



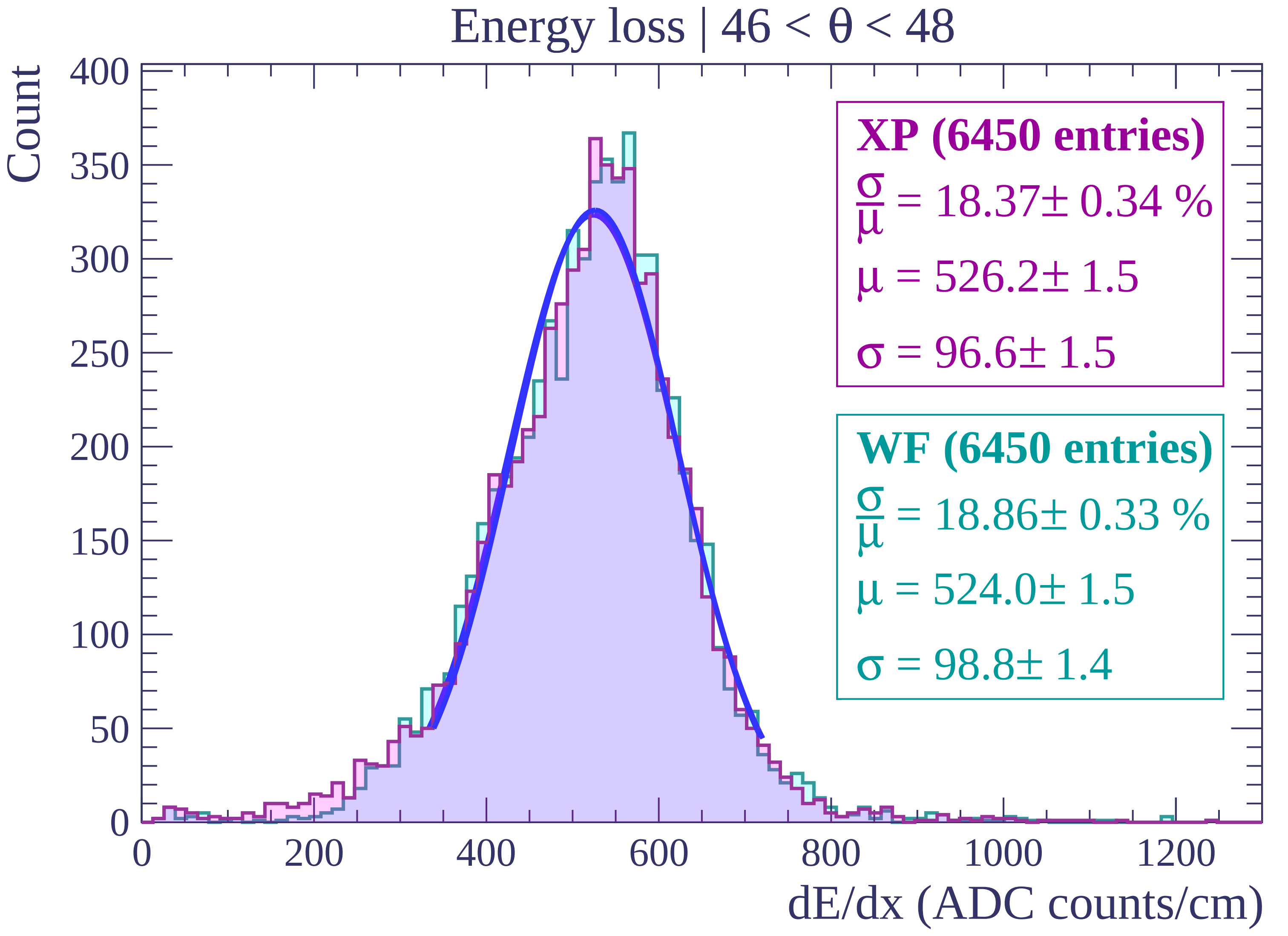
Energy loss | 40 < θ < 42
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (10704 entries)
σ/μ = 17.62± 0.21 %
μ = 515.8± 1.0
σ = 90.9± 0.9
WF (10704 entries)
σ/μ = 18.01± 0.23 %
μ = 515.7± 1.0
σ = 92.9± 1.0



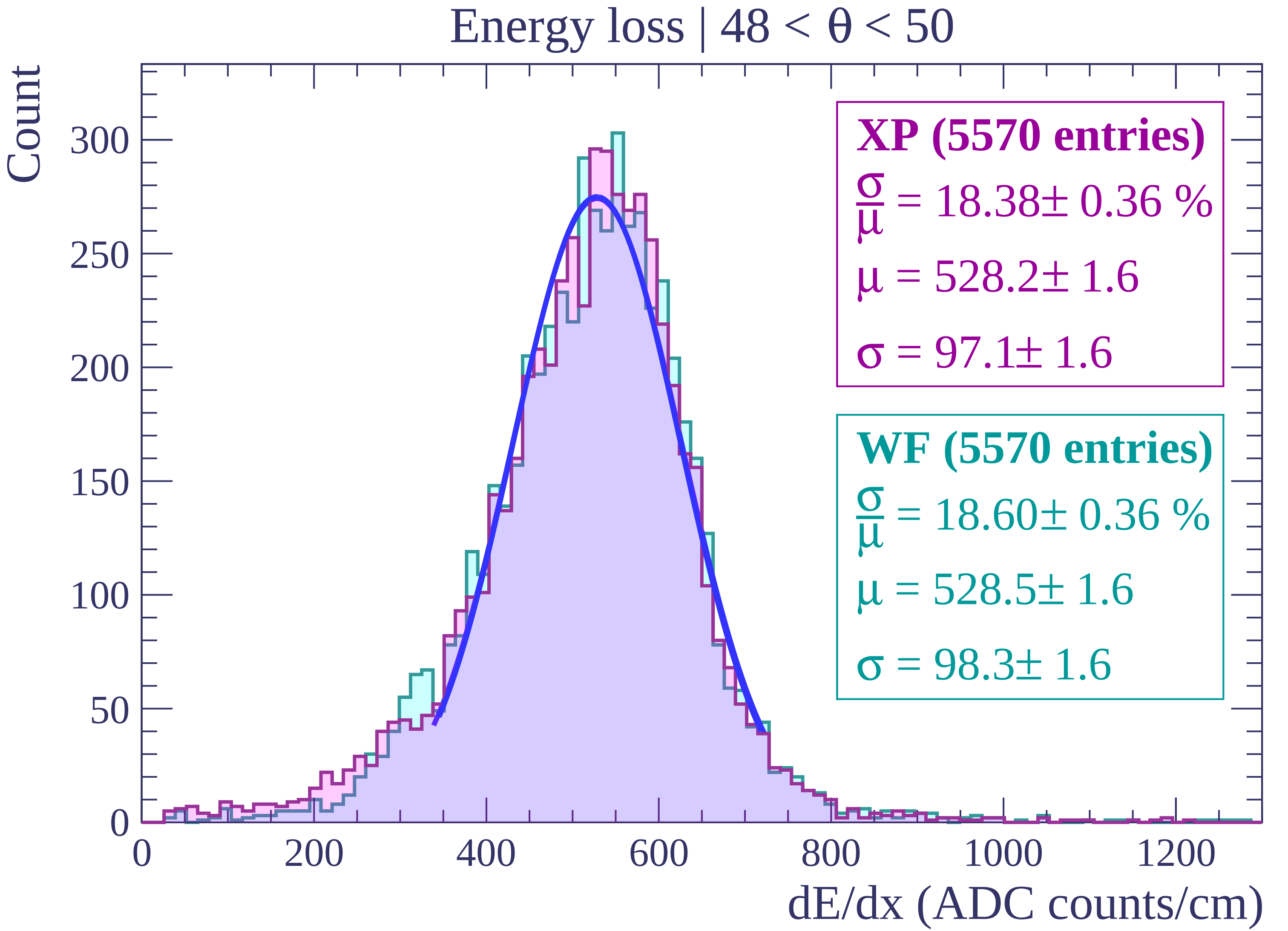
Energy loss | 42 < θ < 44
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (8881 entries)
σ/μ = 17.77± 0.23 %
μ = 518.0± 1.1
σ = 92.0± 1.0
WF (8881 entries)
σ/μ = 18.21± 0.25 %
μ = 518.0± 1.2
σ = 94.3± 1.1

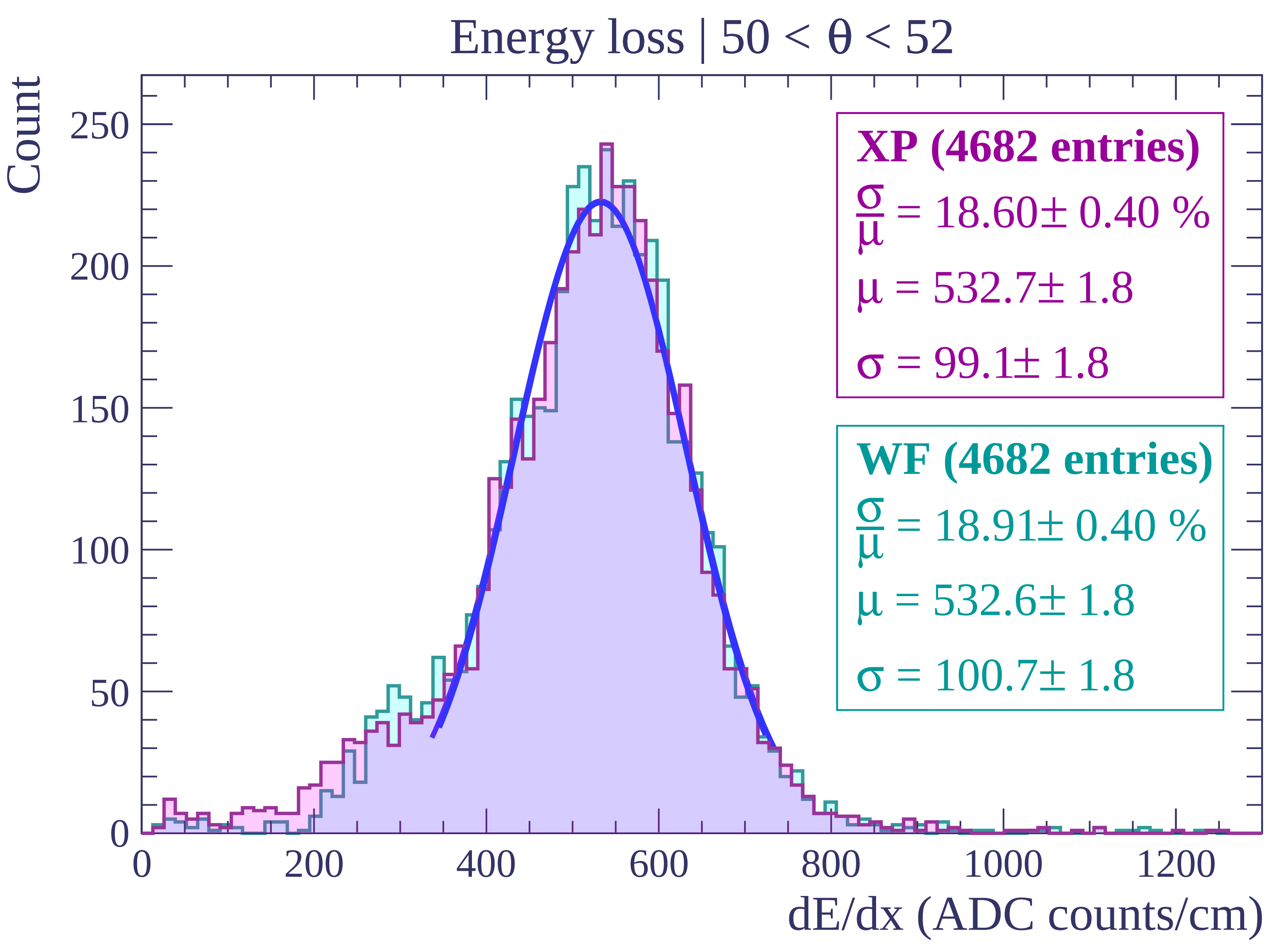


Energy loss | 44 < θ < 46
Count
XP (7615 entries)
σ/μ = 18.75± 0.28 %
μ = 521.1± 1.3
σ = 97.7± 1.2
WF (7615 entries)
σ/μ = 18.79± 0.29 %
μ = 519.4± 1.3
σ = 97.6± 1.3
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)

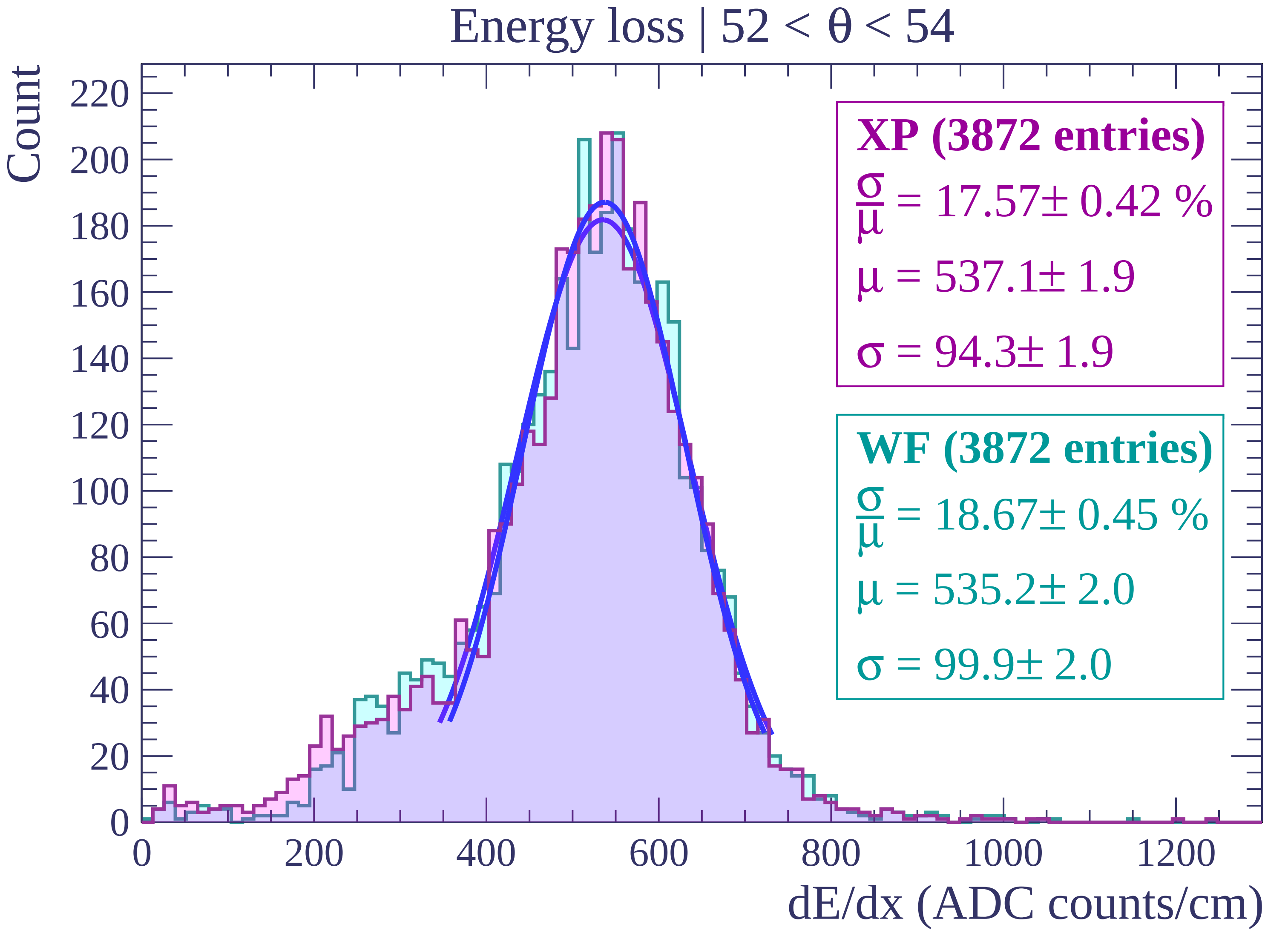


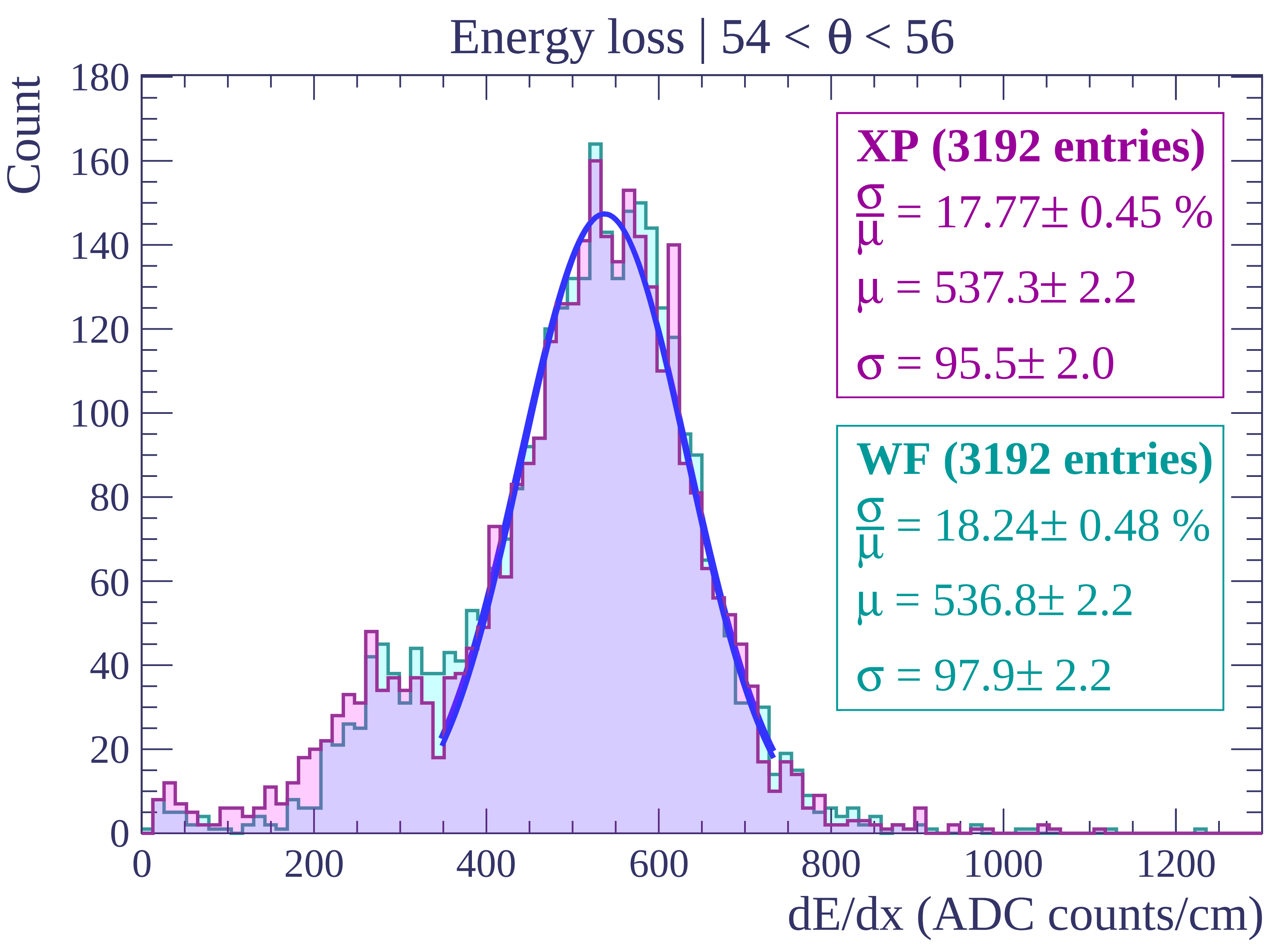
Energy loss | 46 < θ < 48
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
400
XP (6450 entries)
σ/μ = 18.37± 0.34 %
μ = 526.2± 1.5
σ = 96.6± 1.5
WF (6450 entries)
σ/μ = 18.86± 0.33 %
μ = 524.0± 1.5
σ = 98.8± 1.4

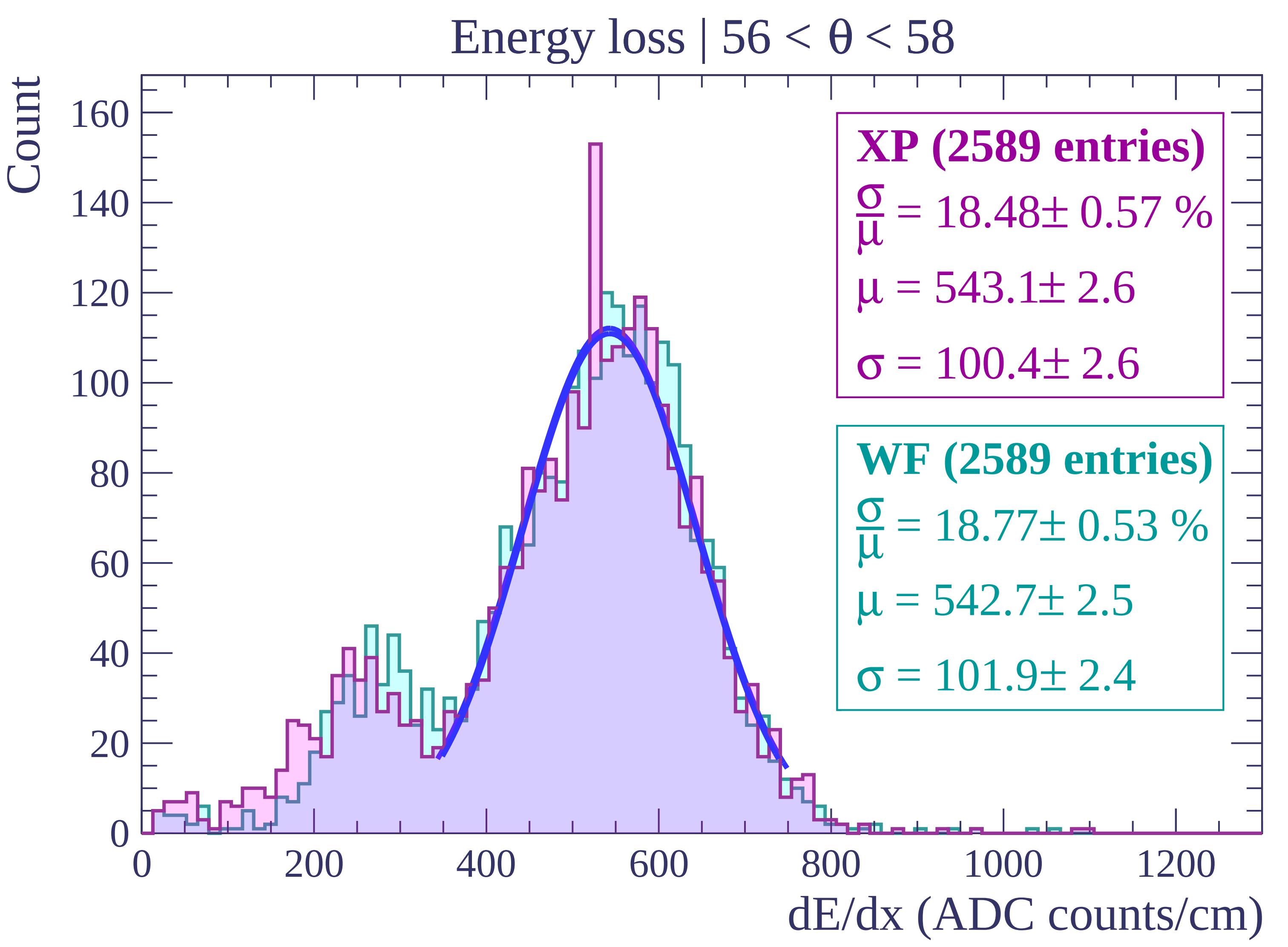

Energy loss | 48 < θ < 50
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5570 entries)
σ/μ = 18.38± 0.36 %
μ = 528.2± 1.6
σ = 97.1± 1.6
WF (5570 entries)
σ/μ = 18.60± 0.36 %
μ = 528.5± 1.6
σ = 98.3± 1.6

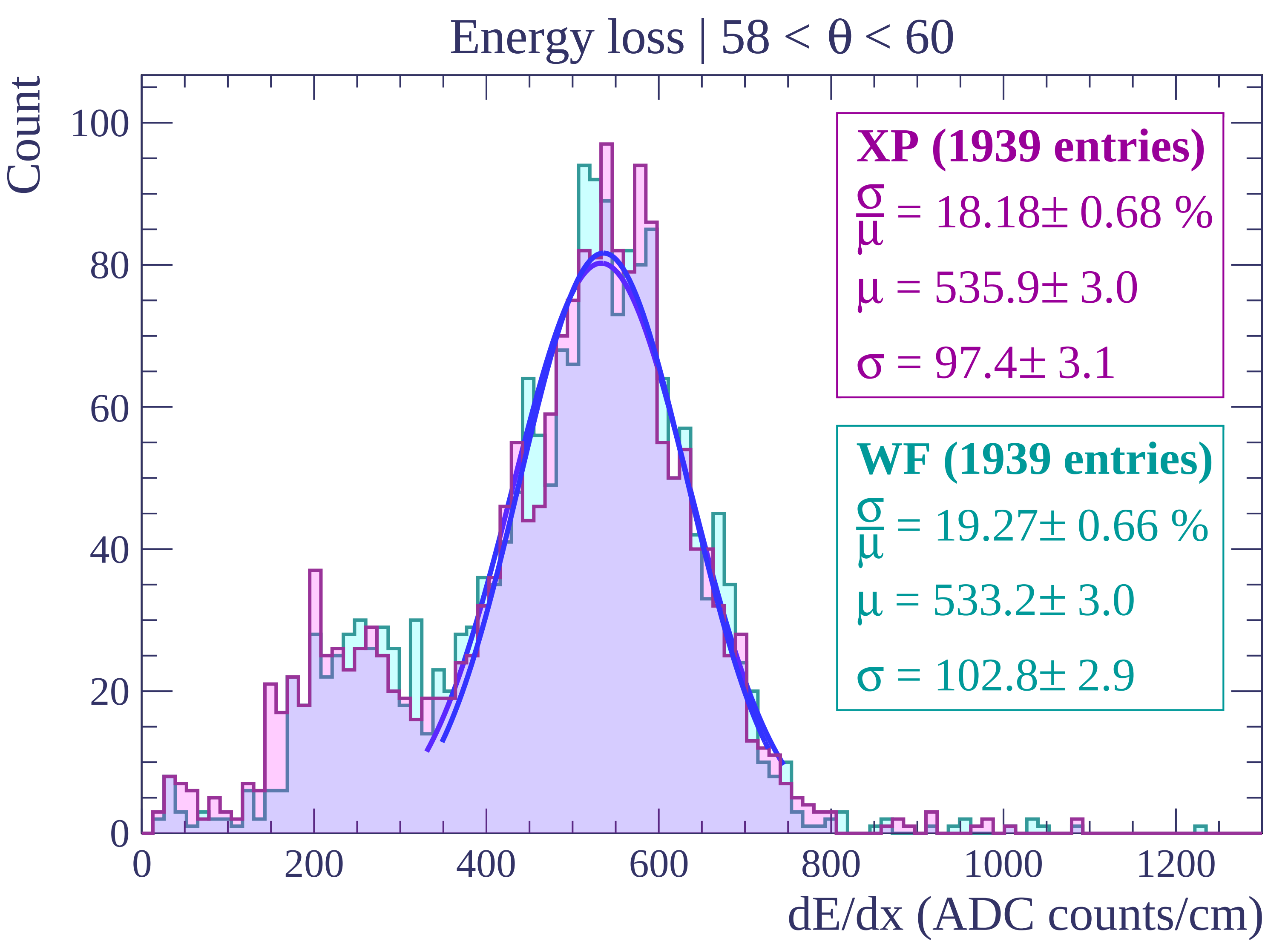


Energy loss | 50 < θ < 52
Count
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4682 entries)
σ/μ = 18.60± 0.40 %
μ = 532.7± 1.8
σ = 99.1± 1.8
WF (4682 entries)
σ/μ = 18.91± 0.40 %
μ = 532.6± 1.8
σ = 100.7± 1.8

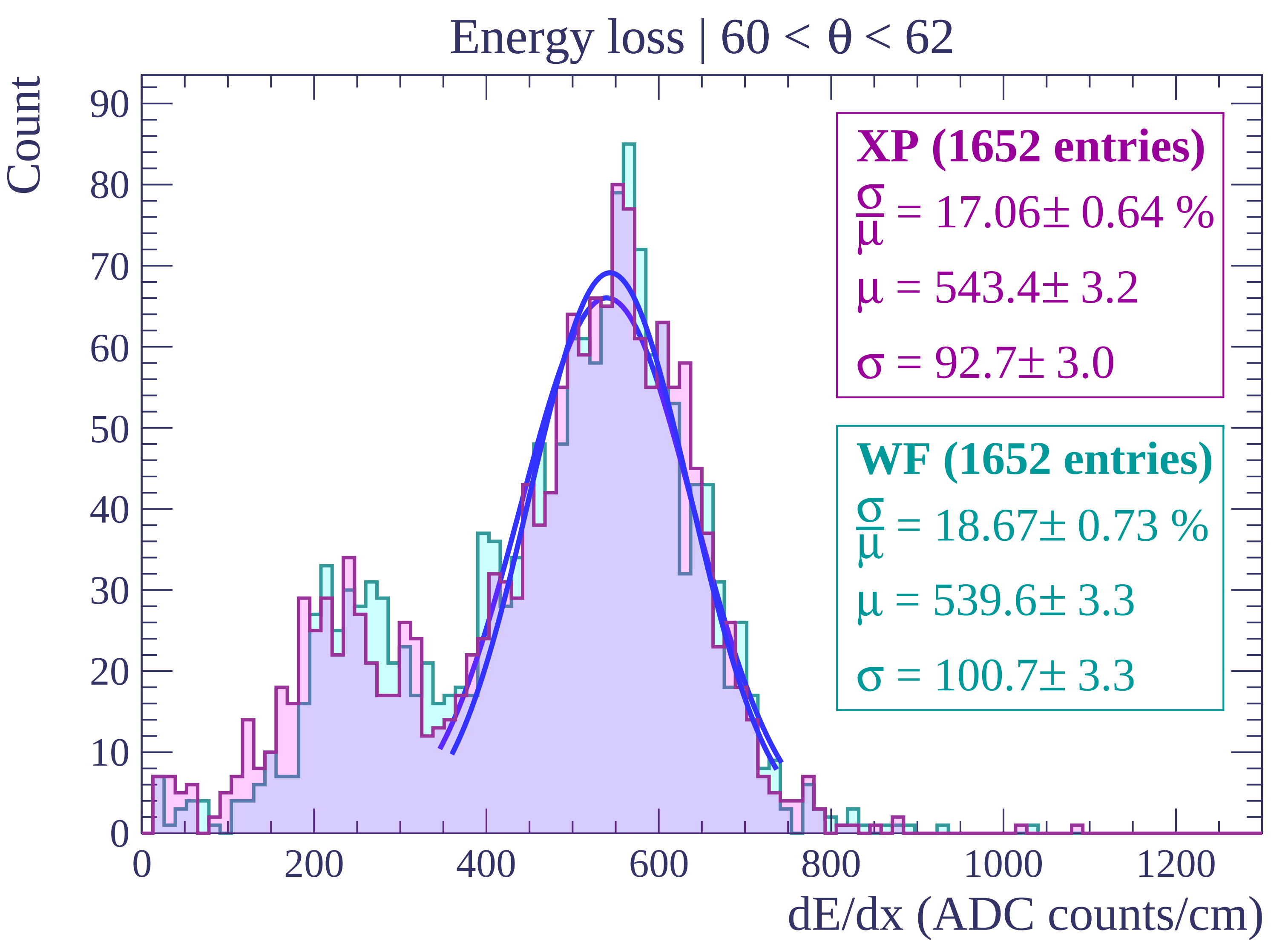

Energy loss | 52 < θ < 54
Count
dE/dx (ADC counts/cm)
XP (3872 entries)
σ/μ = 17.57± 0.42 %
μ = 537.1± 1.9
σ = 94.3± 1.9
WF (3872 entries)
σ/μ = 18.67± 0.45 %
μ = 535.2± 2.0
σ = 99.9± 2.0


Energy loss | 54 < θ < 56
Count
dE/dx (ADC counts/cm)
XP (3192 entries)
σ/μ = 17.77± 0.45 %
μ = 537.3± 2.2
σ = 95.5± 2.0
WF (3192 entries)
σ/μ = 18.24± 0.48 %
μ = 536.8± 2.2
σ = 97.9± 2.2
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200

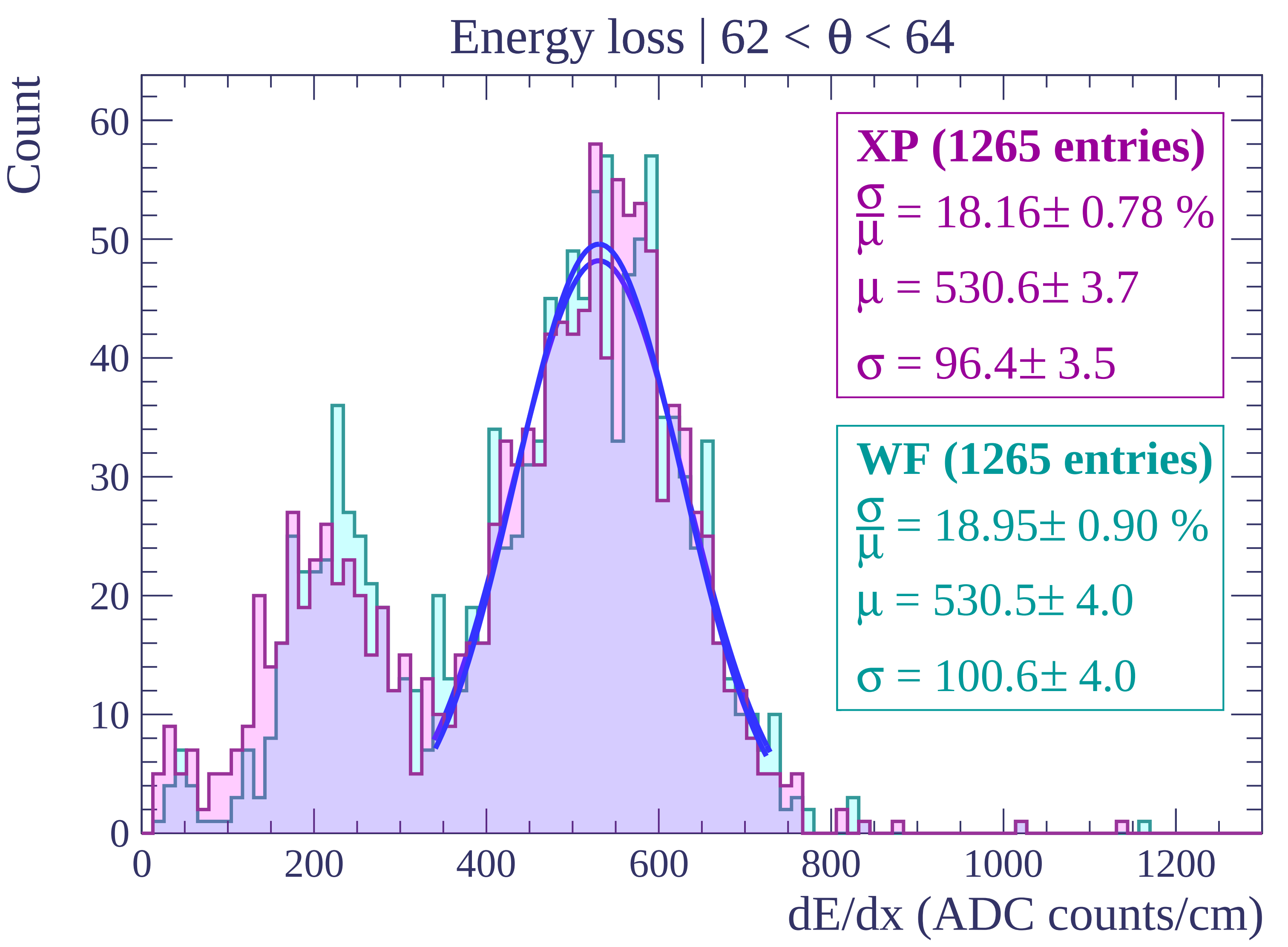

Energy loss | 56 < θ < 58
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP (2589 entries)
σ/μ = 18.48± 0.57 %
μ = 543.1± 2.6
σ = 100.4± 2.6
WF (2589 entries)
σ/μ = 18.77± 0.53 %
μ = 542.7± 2.5
σ = 101.9± 2.4



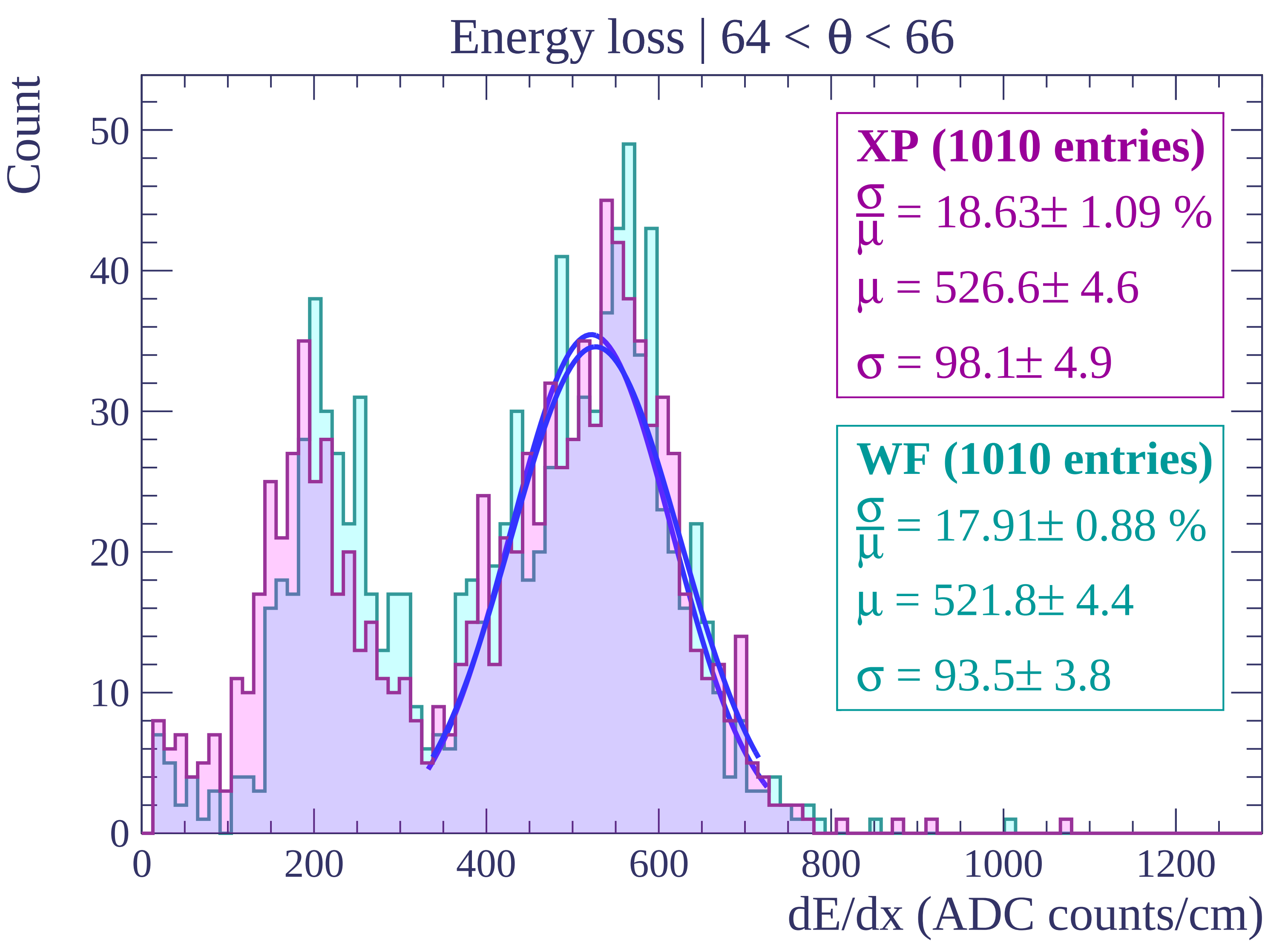
Energy loss | 58 < θ < 60
Count
dE/dx (ADC counts/cm)
XP (1939 entries)
σ/μ = 18.18± 0.68 %
μ = 535.9± 3.0
σ = 97.4± 3.1
WF (1939 entries)
σ/μ = 19.27± 0.66 %
μ = 533.2± 3.0
σ = 102.8± 2.9

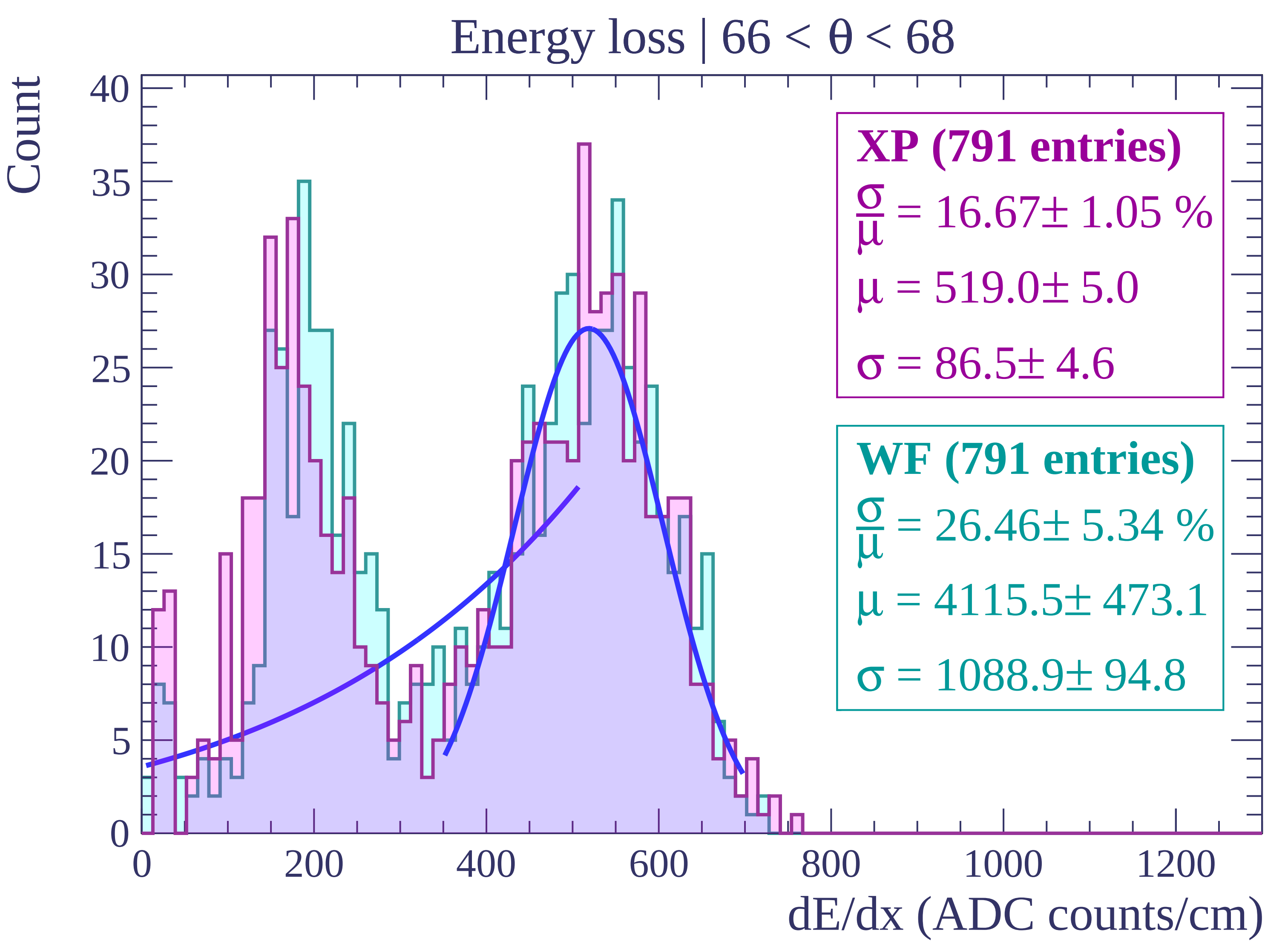


Energy loss | 60 < θ < 62
Count
dE/dx (ADC counts/cm)
XP (1652 entries)
σ/μ = 17.06± 0.64 %
μ = 543.4± 3.2
σ = 92.7± 3.0
WF (1652 entries)
σ/μ = 18.67± 0.73 %
μ = 539.6± 3.3
σ = 100.7± 3.3
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200

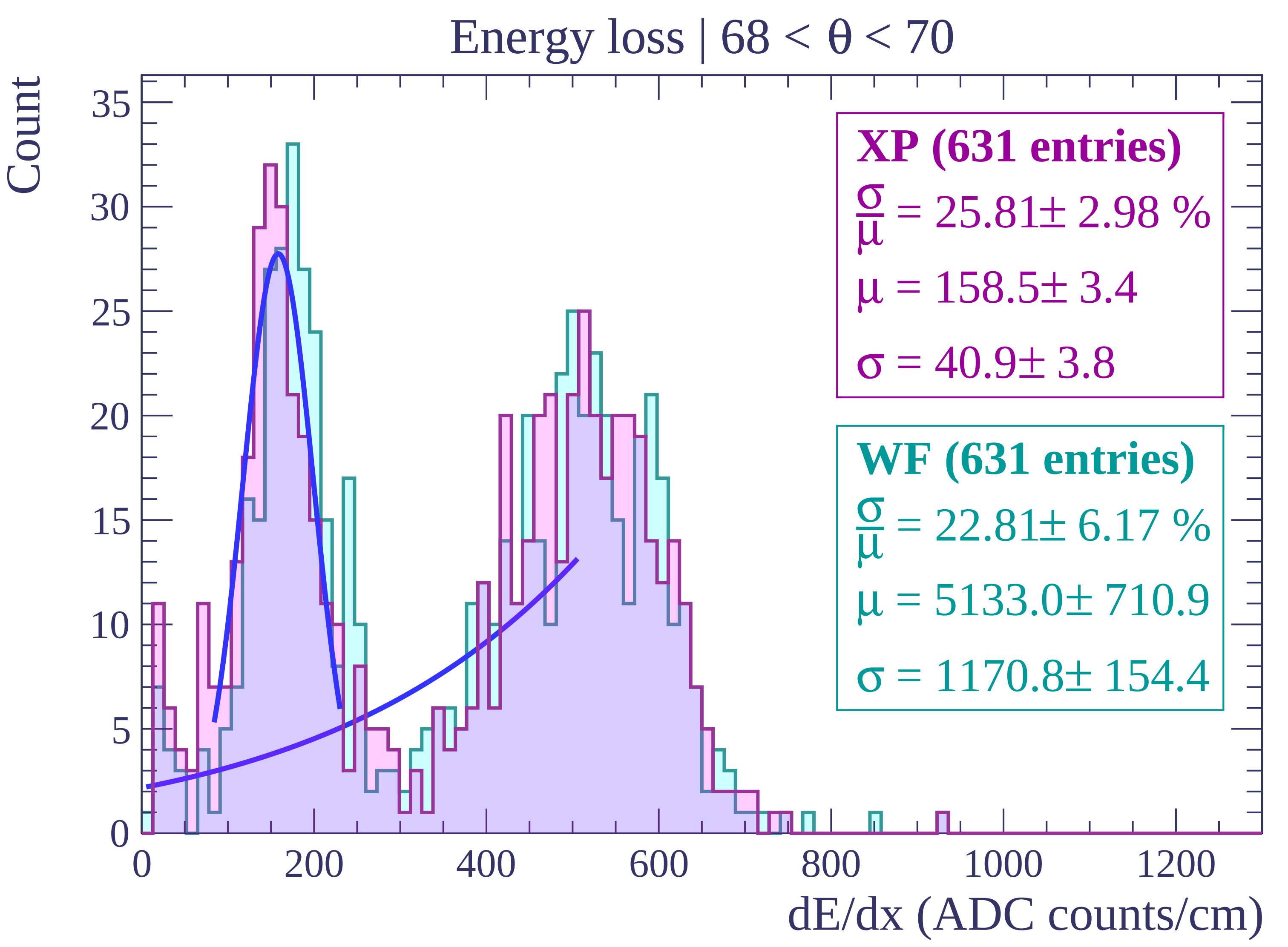


Energy loss | 62 < θ < 64
Count
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1265 entries)
σ/μ = 18.16± 0.78 %
μ = 530.6± 3.7
σ = 96.4± 3.5
WF (1265 entries)
σ/μ = 18.95± 0.90 %
μ = 530.5± 4.0
σ = 100.6± 4.0

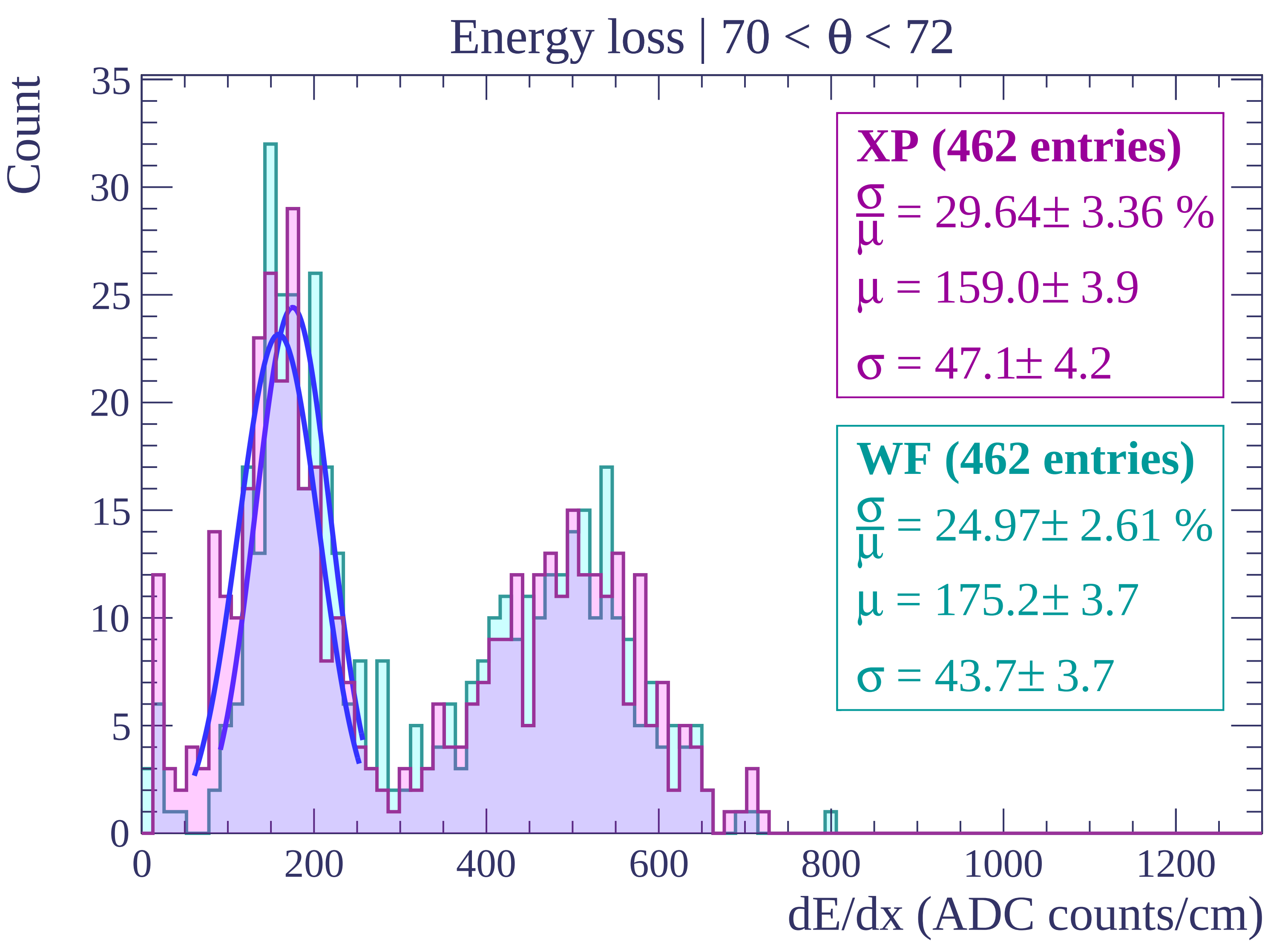

Energy loss | 64 < θ < 66
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP (1010 entries)
σ/μ = 18.63 ± 1.09 %
μ = 526.6 ± 4.6
σ = 98.1 ± 4.9
WF (1010 entries)
σ/μ = 17.91 ± 0.88 %
μ = 521.8 ± 4.4
σ = 93.5 ± 3.8



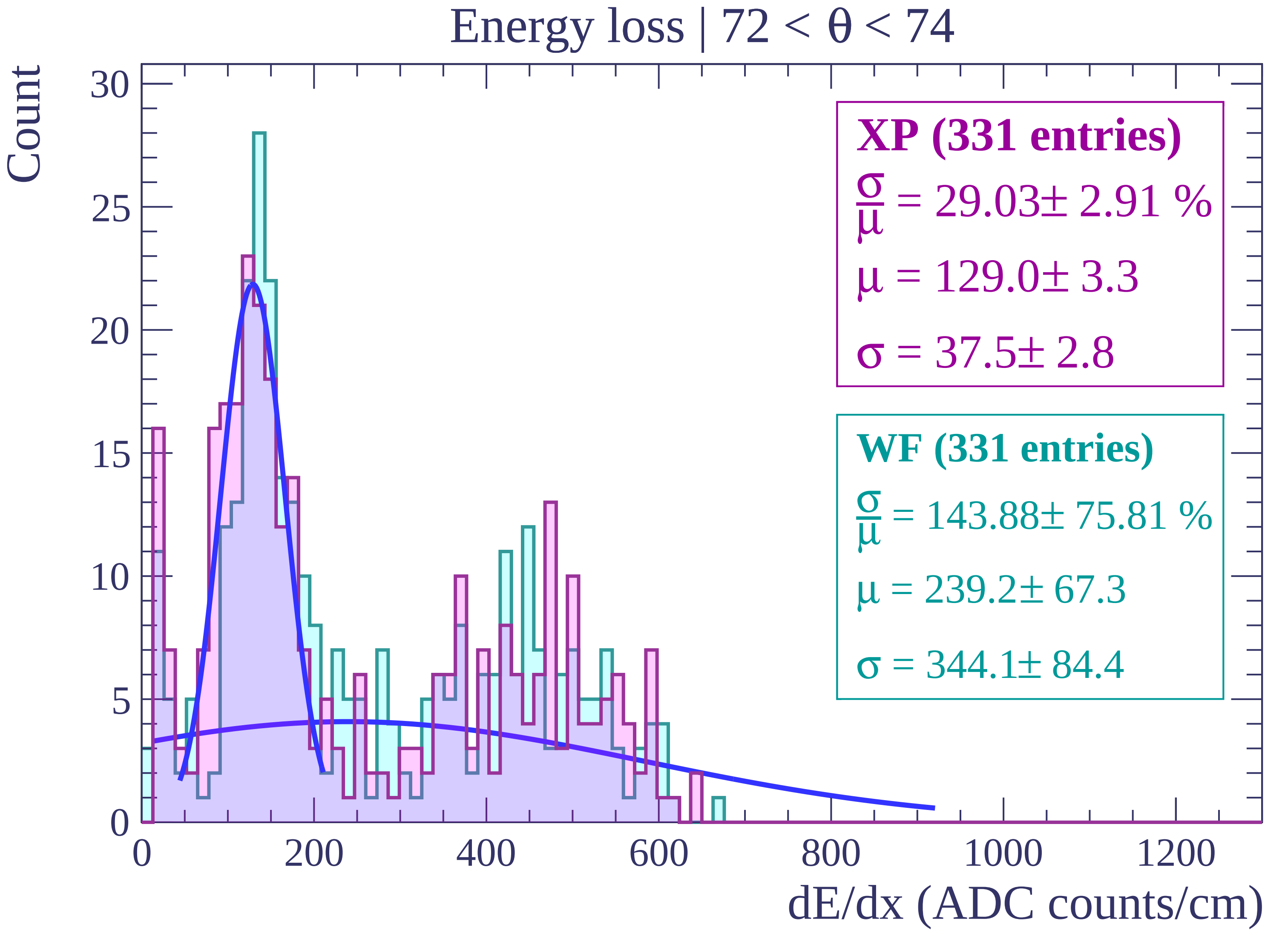
Energy loss | 66 < θ < 68
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
40
XP (791 entries)
σ/μ = 16.67± 1.05 %
μ = 519.0± 5.0
σ = 86.5± 4.6
WF (791 entries)
σ/μ = 26.46± 5.34 %
μ = 4115.5± 473.1
σ = 1088.9± 94.8



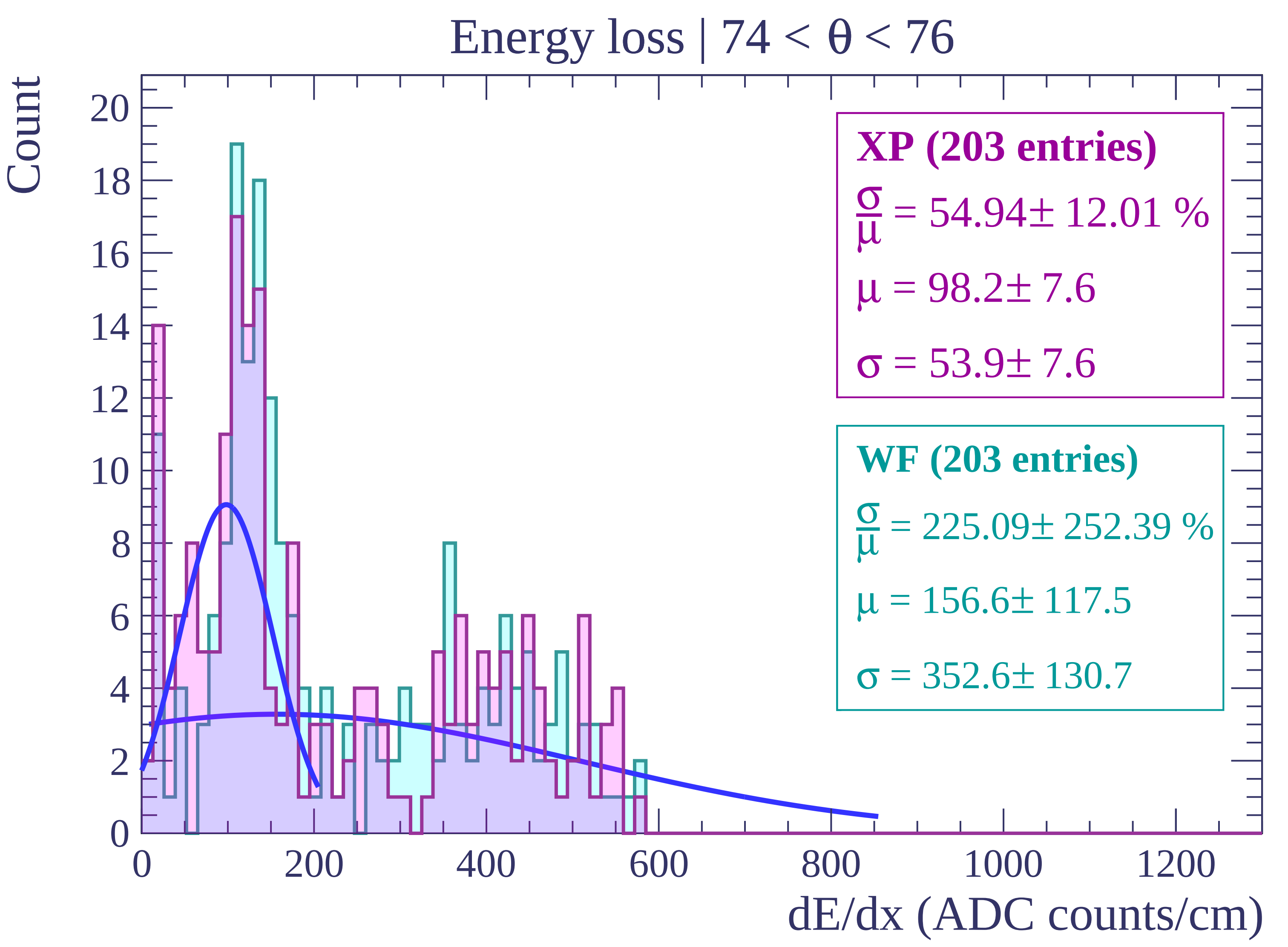
Energy loss | 68 < θ < 70
Count
dE/dx (ADC counts/cm)
XP (631 entries)
σ/μ = 25.81± 2.98 %
μ = 158.5± 3.4
σ = 40.9± 3.8
WF (631 entries)
σ/μ = 22.81± 6.17 %
μ = 5133.0± 710.9
σ = 1170.8± 154.4



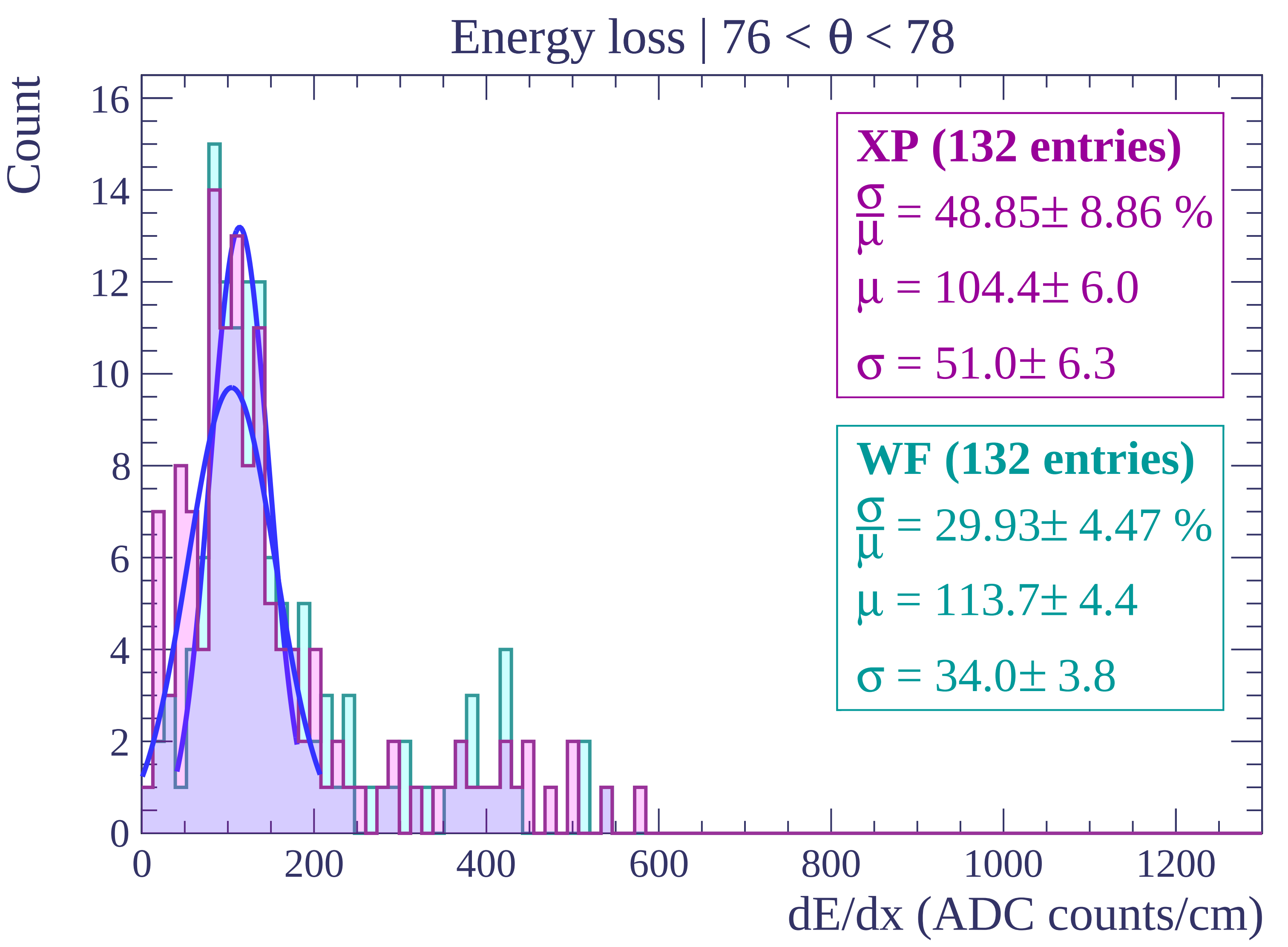
Energy loss | 70 < θ < 72
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200
XP (462 entries)
σ/μ = 29.64± 3.36 %
μ = 159.0± 3.9
σ = 47.1± 4.2
WF (462 entries)
σ/μ = 24.97± 2.61 %
μ = 175.2± 3.7
σ = 43.7± 3.7



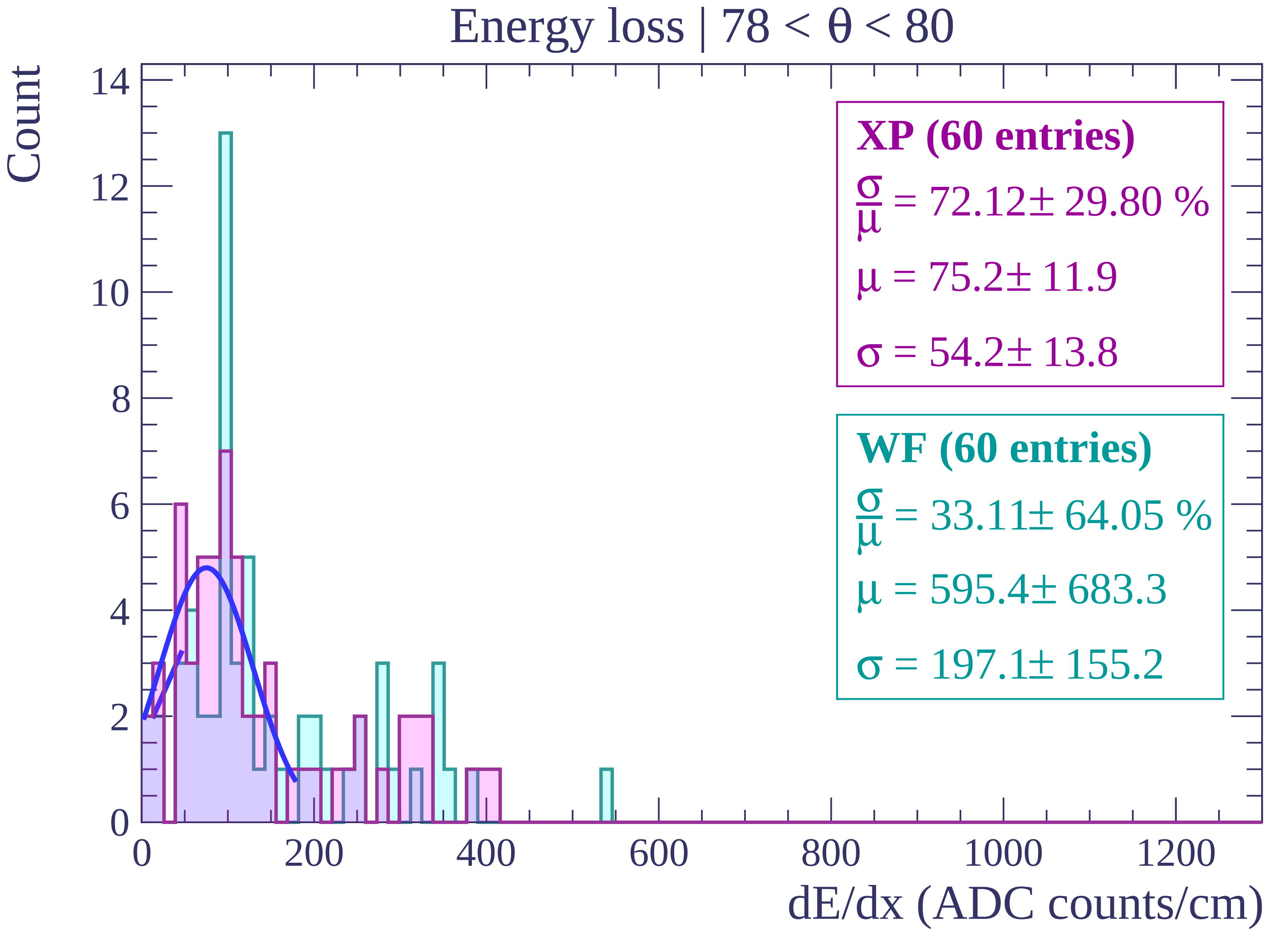
Energy loss | 72 < θ < 74
Count
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (331 entries)
σ/μ = 29.03± 2.91 %
μ = 129.0± 3.3
σ = 37.5± 2.8
WF (331 entries)
σ/μ = 143.88± 75.81 %
μ = 239.2± 67.3
σ = 344.1± 84.4

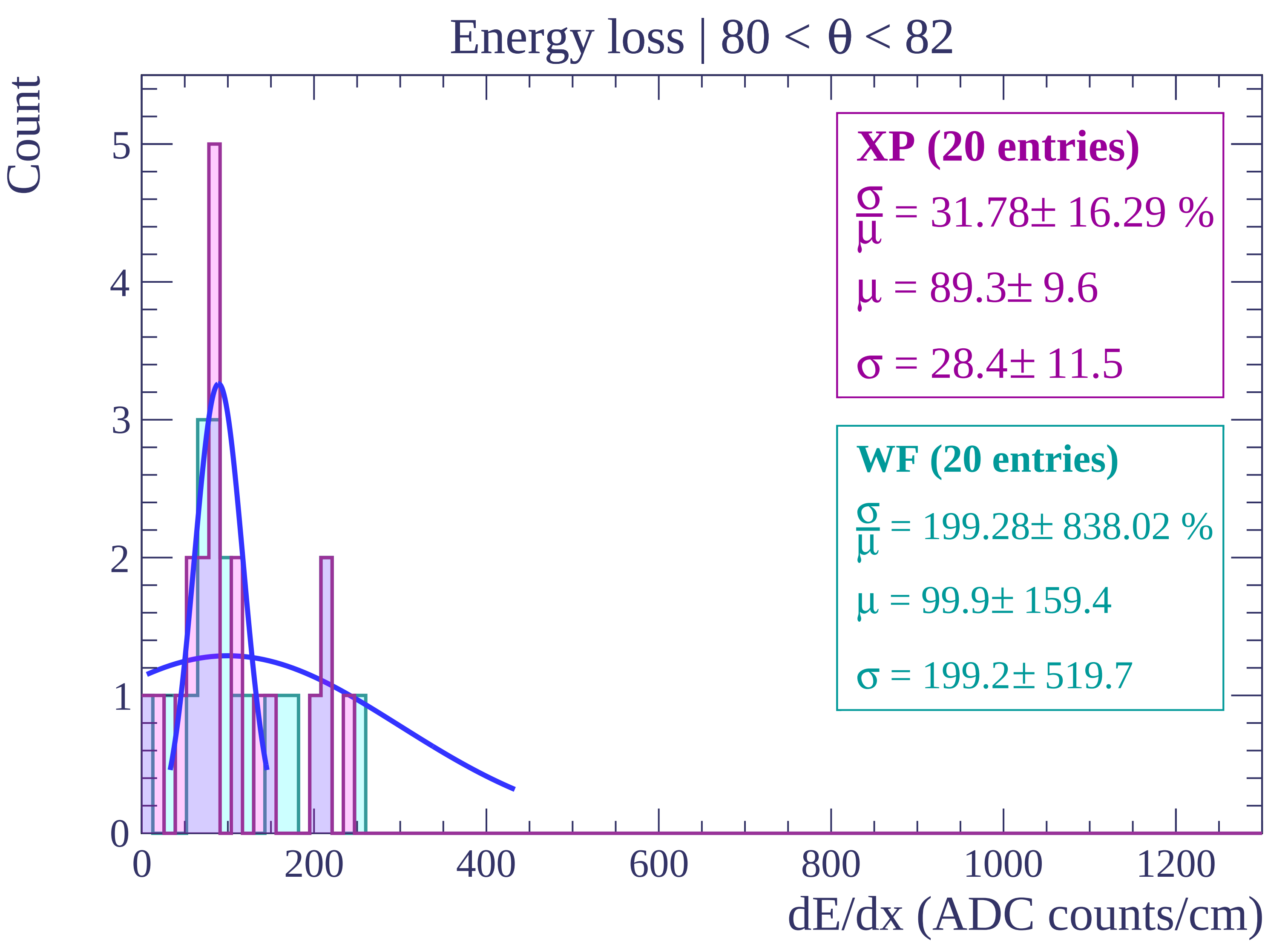


Energy loss | 74 < θ < 76
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
18
20
0
200
400
600
800
1000
1200
XP (203 entries)
$\frac{\sigma}{\mu}$ = 54.94± 12.01 %
μ = 98.2± 7.6
σ = 53.9± 7.6
WF (203 entries)
$\frac{\sigma}{\mu}$ = 225.09± 252.39 %
μ = 156.6± 117.5
σ = 352.6± 130.7

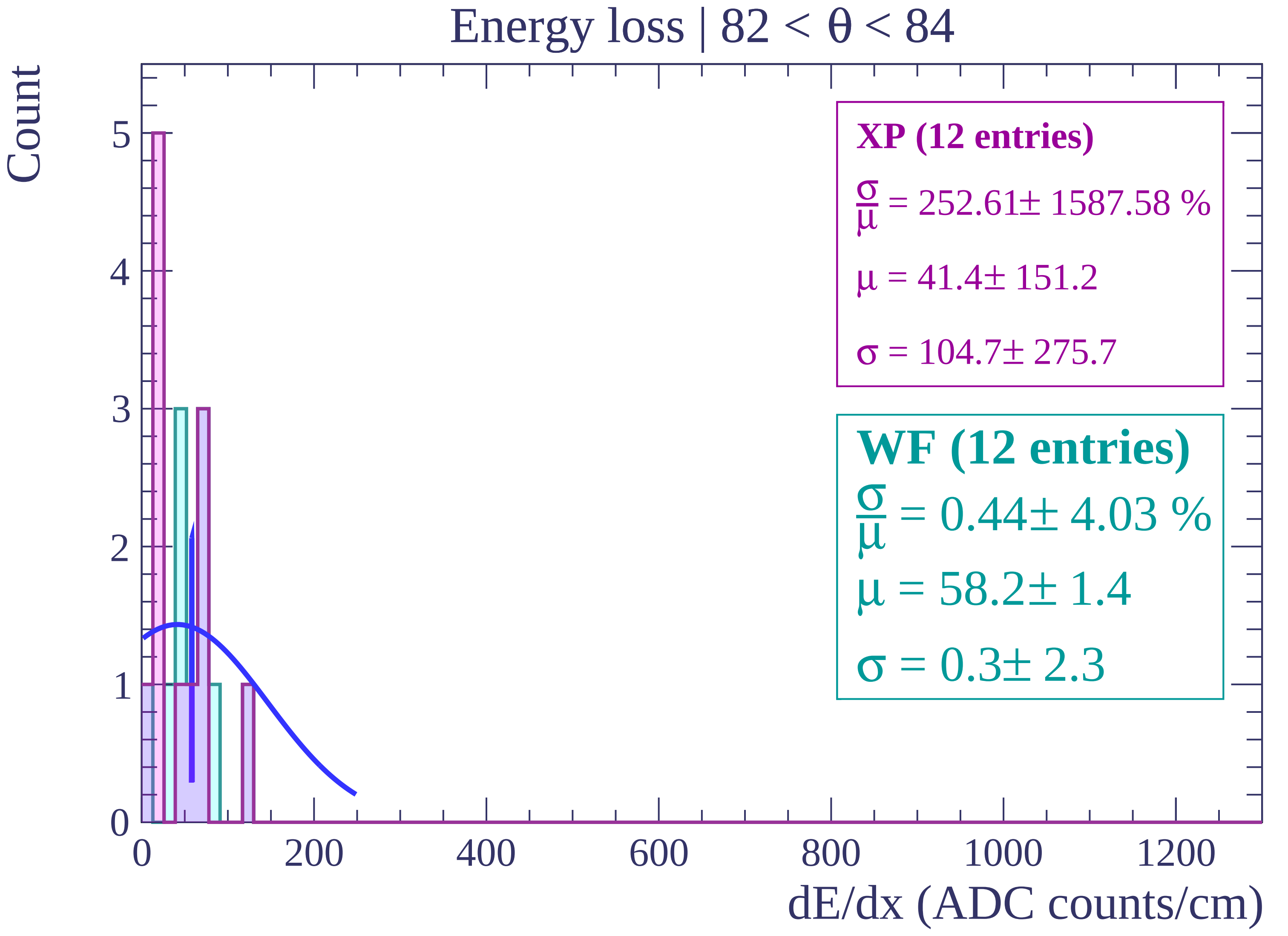


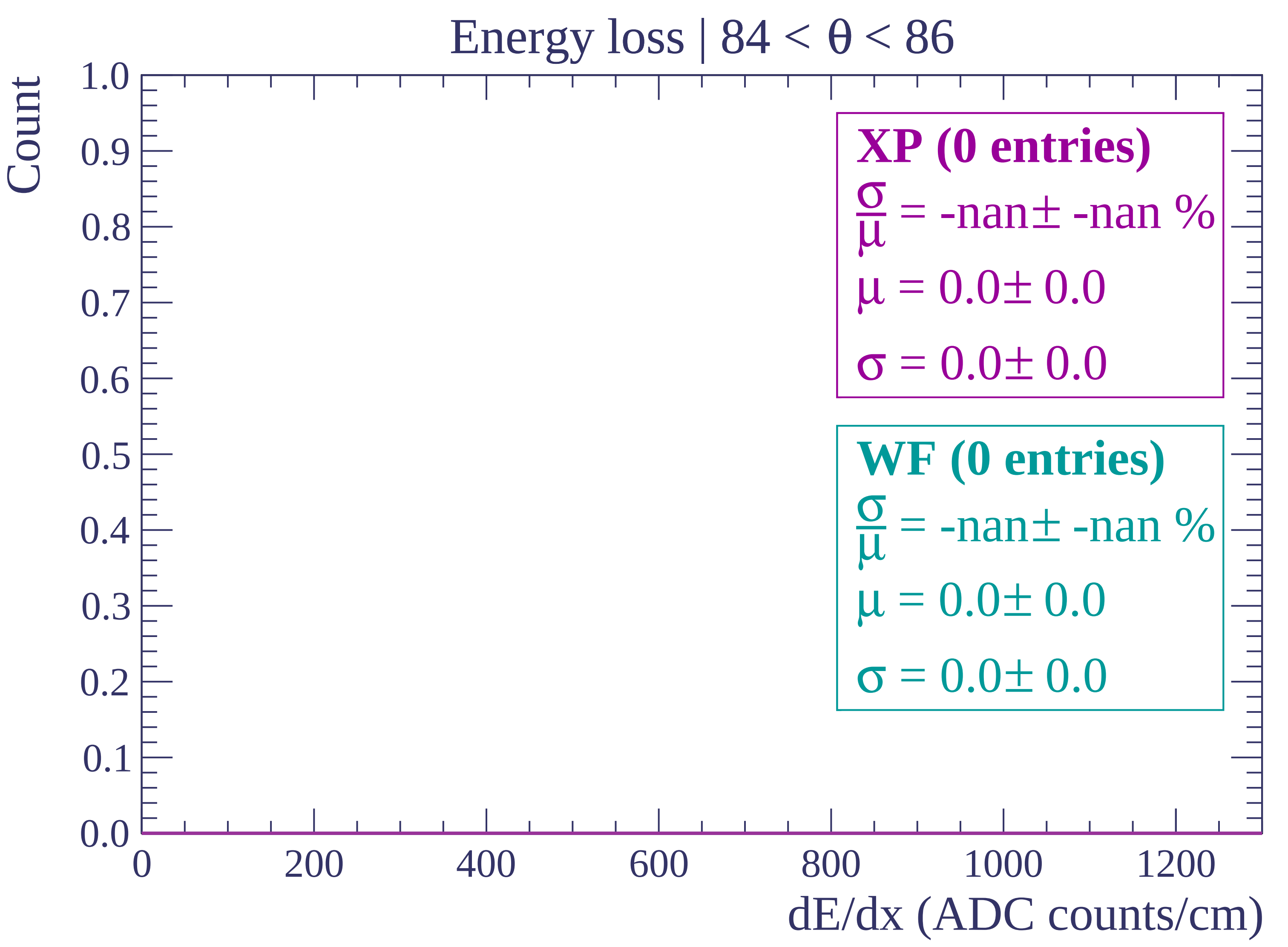
Energy loss | 76 < θ < 78
Count
dE/dx (ADC counts/cm)
XP (132 entries)
σ/μ = 48.85± 8.86 %
μ = 104.4± 6.0
σ = 51.0± 6.3
WF (132 entries)
σ/μ = 29.93± 4.47 %
μ = 113.7± 4.4
σ = 34.0± 3.8


Energy loss | 78 < θ < 80
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
XP (60 entries)
σ/μ = 72.12± 29.80 %
μ = 75.2± 11.9
σ = 54.2± 13.8
WF (60 entries)
σ/μ = 33.11± 64.05 %
μ = 595.4± 683.3
σ = 197.1± 155.2

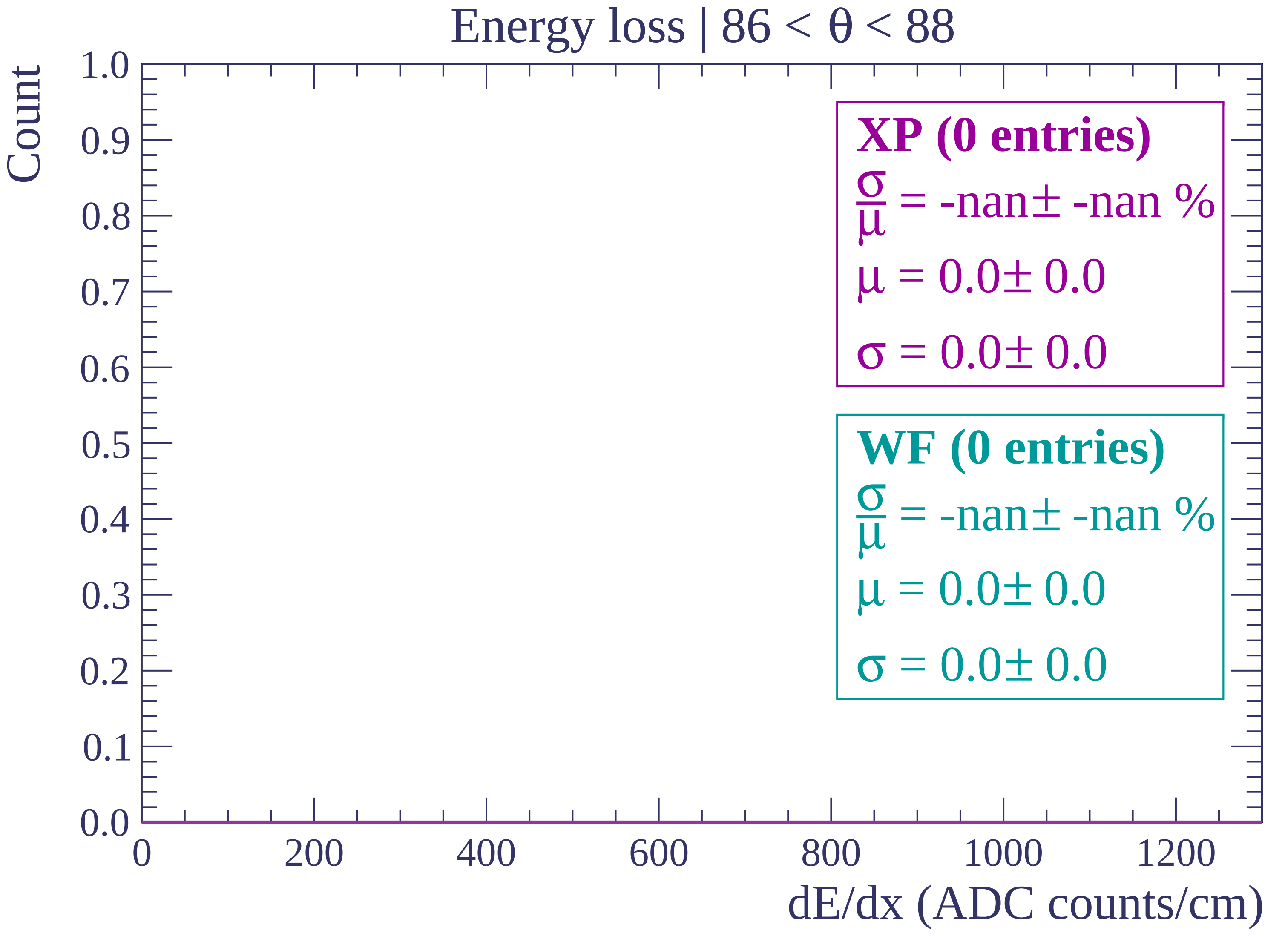


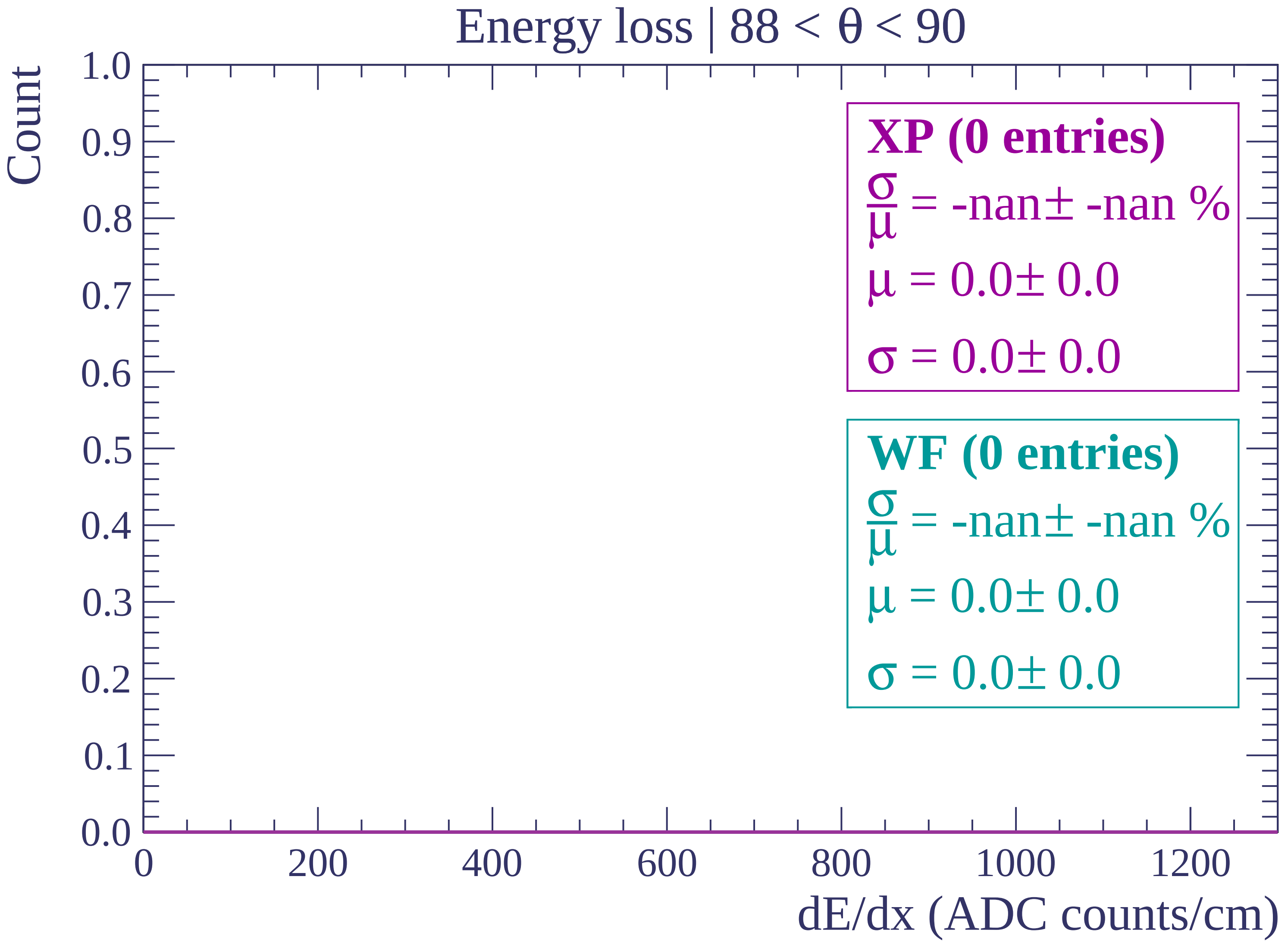
Energy loss | 80 < θ < 82
Count
0
1
2
3
4
5
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (20 entries)
σ/μ = 31.78± 16.29 %
μ = 89.3± 9.6
σ = 28.4± 11.5
WF (20 entries)
σ/μ = 199.28± 838.02 %
μ = 99.9± 159.4
σ = 199.2± 519.7


Energy loss | 82 < θ < 84
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
XP (12 entries)
σ/μ = 252.61± 1587.58 %
μ = 41.4± 151.2
σ = 104.7± 275.7
WF (12 entries)
σ/μ = 0.44± 4.03 %
μ = 58.2± 1.4
σ = 0.3± 2.3

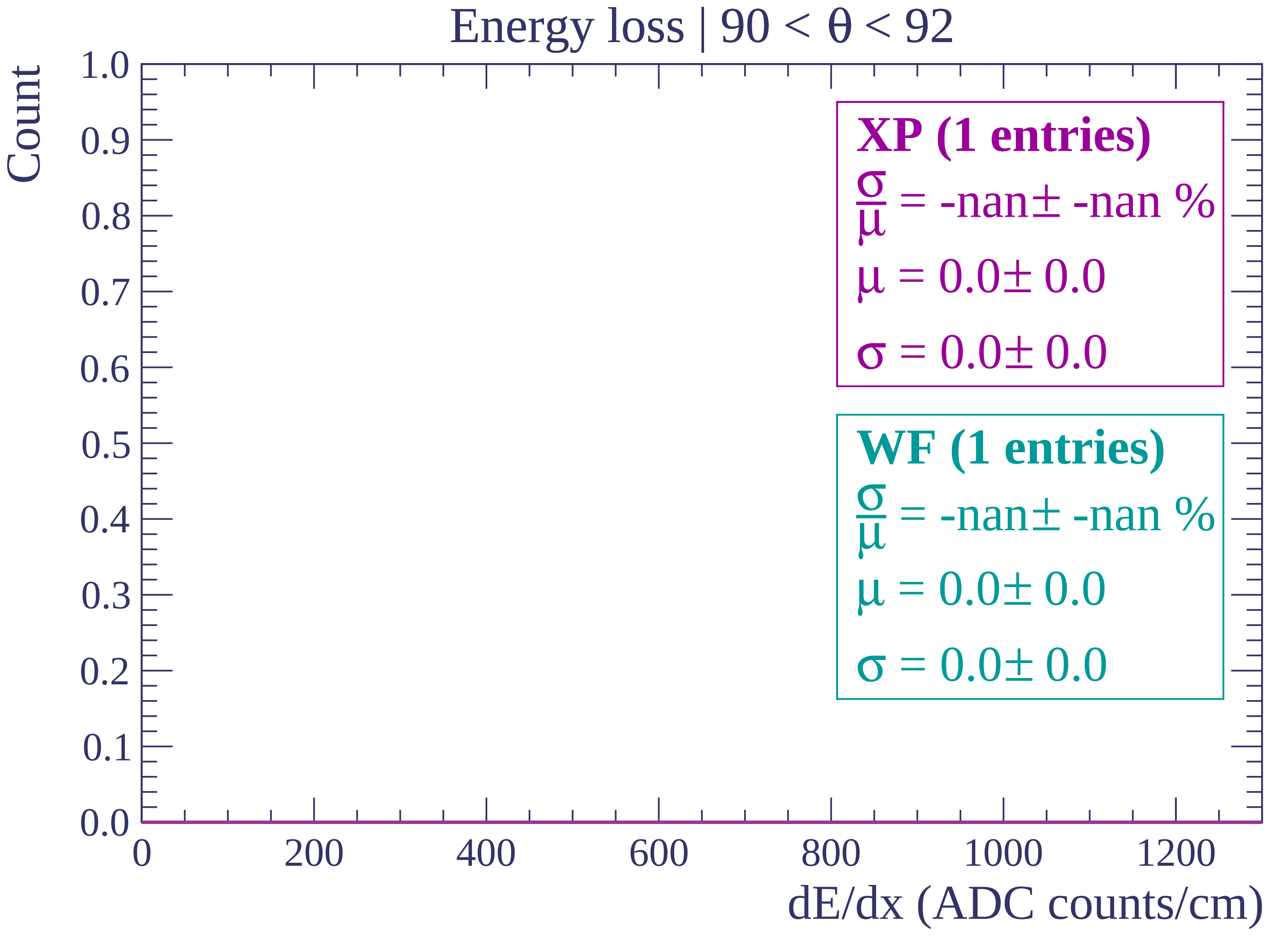


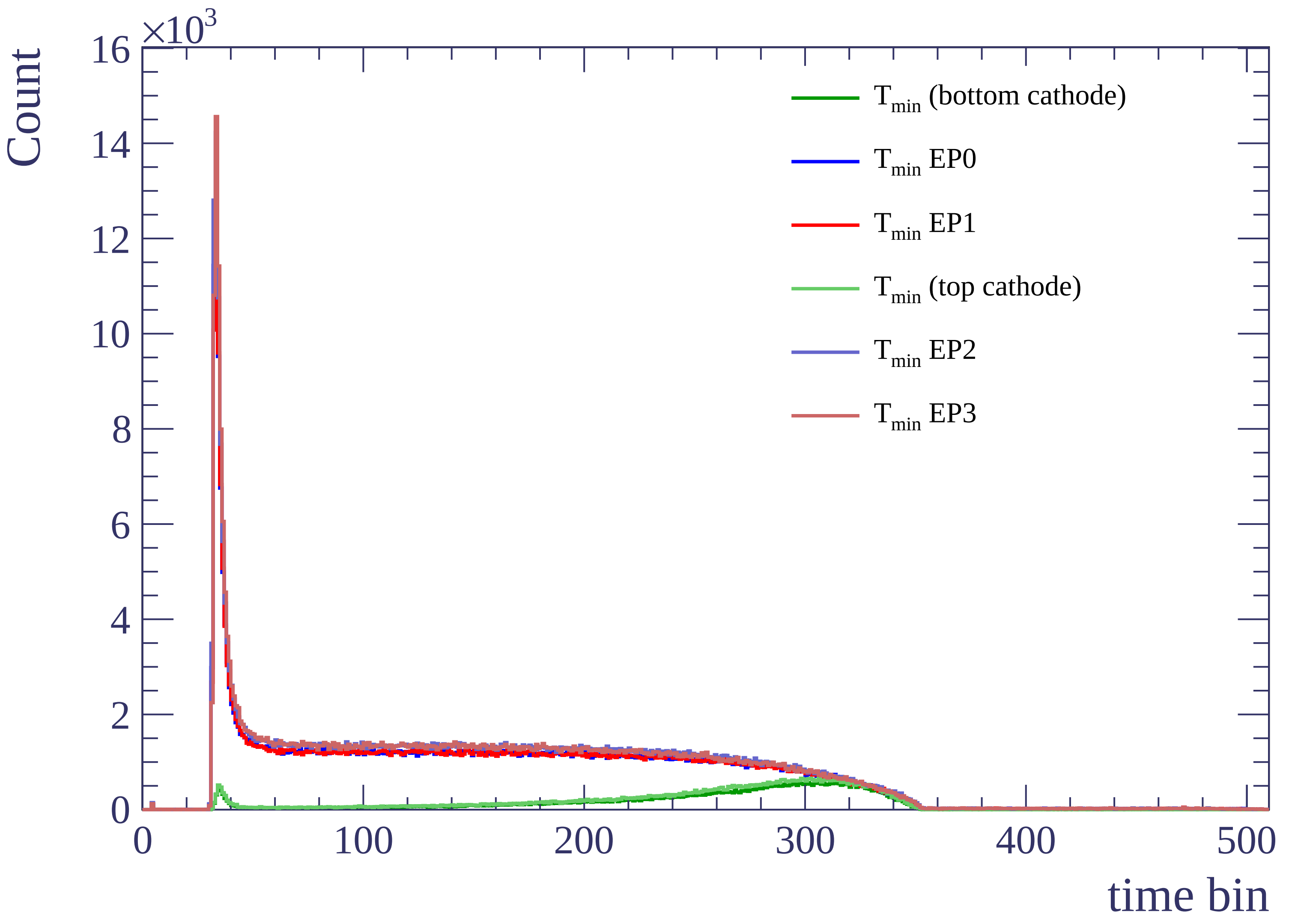
Energy loss | 84 < θ < 86
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 86 < θ < 88
Count
dE/dx (ADC counts/cm)
XP (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200

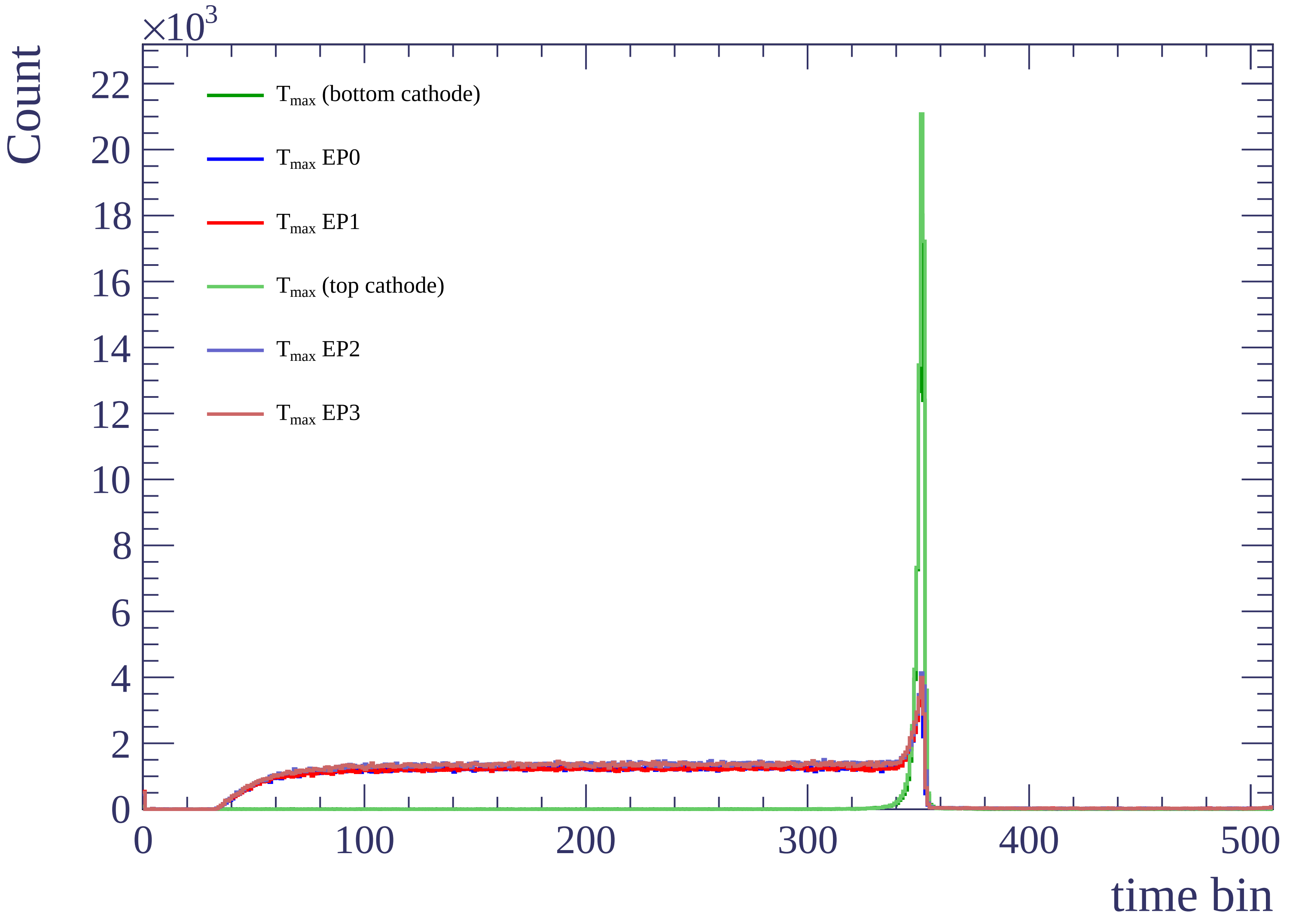


Energy loss | 88 < θ < 90
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 90 < θ < 92
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (1 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



$\times 10^3$
Count
0
2
4
6
8
10
12
14
16
0
100
200
300
400
500
time bin
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3



$\times 10^3$
Count
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3
0
2
4
6
8
10
12
14
16
18
20
22
0
100
200
300
400
500
time bin